

Université Paris Cité
UFR Linguistique

Form-meaning mismatches in Nanosyntax

HABILITATION

un recueil d'articles

avec une introduction résumée

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¹This online version differs slightly from the original submitted version. Minor typographical errors have been corrected, and a small number of minor content-related revisions have been made. An updated version of Chapter 1 in Part II has also been included. The papers in Part II are reproduced in their original published versions wherever copyright permissions allow; where this was not possible, the final author version is included instead. Part III, About the Author, is not included in the online version.

“You only grow by coming to the
end of something and by beginning
something else.”

John Irving

The World According to Garp

Acknowledgement

Writing this summary of my work, a bit more than 10 years after I defended my PhD, was an intense experience. But having completed the work I realise once more how incredibly grateful I am for the opportunities that I have ; for this gift of being able to do research as a daily activity ; to have a job whose essence is to think and learn and share knowledge. Of course, working in an academic environment is not all sunshine and rainbows : I've had my fair share of frustration and imposter syndrome as well over the past ten years, but in the end the overall balance is positive.

In many ways this is due to the fact that I ended up having a permanent position as a CNRS researcher at the *Laboratoire de Linguistique Formelle* (U Paris Cité). There is something about this place, and the people who work there, that gives energy and inspiration. From the very first time when I was visiting the lab in 2019 as a CNRS applicant, the vibrant open atmosphere struck me and I knew that I wanted to work there. Fortunately, this came true in 2020.

The role of Olivier Bonami in how I feel about my current position and at the lab cannot be underestimated. From day one he has been a supportive, loyal, patient, and enthusiastic mentor, who guided me through the CNRS application process - always willing to answer my questions - who welcomed me to the lab in the most difficult time of covid confinement, who encouraged me to do the HDR and who has -ever since - been an incredibly nice colleague whom I want to wholeheartedly thank. Special thanks also go to Caterina Donati who has also been there for me from my very first days at the lab and whose intelligent advice and warm support are invaluable to me.

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Ghent, November 2023

Karen De Clercq

Table des matières

I Summary of previous research 3

Form-meaning mismatches in Nanosyntax

K. De Clercq	5
1.1 Introduction	5
1.2 One Form-Many Meanings	18
1.2.1 From syncretism to portmanteaus	18
1.2.2 Fusion	23
1.2.2.1 False vs. untrue	23
1.2.2.2 Korean negative and honorific portmanteaus	31
1.2.2.3 Bambara	39
1.3 0-to-M	52
1.4 Many Forms to One Meaning (F^n -to- M)	60
1.4.1 Multiple exponence	60
1.4.2 Doubling	66
1.4.2.1 Negative allomorphy	66
1.4.2.2 Negative concord	68
1.5 Form-to-No Meaning	75
1.6 Conclusion	81

II Collection of papers 95

Nanosyntax: state of the art

K. De Clercq, P. Caha, M. Starke and G. Vanden Wyngaerd	97
---	----

Zero morphology in change-of-state verbs

P. Caha, K. De Clercq, and G. Vanden Wyngaerd	159
---	-----

Negation and the functional sequence

K. De Clercq and G. Vanden Wyngaerd	179
---	-----

Late Insertion and Root Suppletion

G. Vanden Wyngaerd, K. De Clercq and P. Caha	223
--	-----

Tense and sentential negation	
K. De Clercq	267
The Addressee in the present imperative in Latin	
A.-L. Demonie and K. De Clercq	287
The Fine structure of the comparative	
P. Caha, K. De Clercq and G. Vanden Wyngaerd	313
Negative Concord and TAM	
K. De Clercq	417
How to be positive	
G. Vanden Wyngaerd, M. Starke, K. De Clercq and P. Caha	447
 III About the author	 483

Preface

The synthesising paper in chapter 1 of Part I below looks at the topic of form-meaning mismatches that runs as a recurring theme through the publications from my PhD in 2013 to the present. However, it was only when I was invited to teach a summer school in Göttingen (upcoming, August 2024) within the context of the DFG funded project ‘Form meaning mismatches’ (DFG Research Training Group 2021) that I started thinking about my own work in these terms. The articles that I mainly draw upon are included in the different chapters of Part II. Each chapter in Part II brings key articles to the fore linked to one of the sections in the synthesising chapter of Part I. My other publications, even though not explicitly included here, also contributed to the conceptualisation of this overarching theme. Most of my publications can be downloaded via open access or in prefinal version from the LLF website, Lingbuzz or HAL.

Part I

Summary of previous research

1 Form-meaning mismatches in Nanosyntax

K. De Clercq

1.1 Introduction

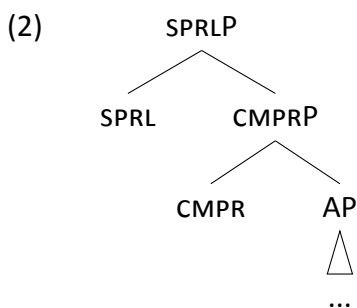
The phrase *form-meaning mismatches* comes with the presupposition that there is usually a match between form and meaning. Indeed, traditional linguistic theory—going back at least to Saussure and beyond—took the stance that “morphemes are linked bidirectionally with meanings” (A. C. Harris 2009: 271). While this is a debatable thing for many reasons, as we will see in the discussion of several empirical phenomena in this chapter, this idea has been around, both in philosophy and in linguistics. De Saussure (1916) for instance considered the association of *signifiant* (form) and *signifié* (meaning) as the corner-stone of linguistic theory (Koutsoukos, Van Goethem & De Smet 2016). The related principle of (semantic) compositionality, also (incorrectly, see Janssen 2001) dubbed Frege’s Principle, comes with the inherent idea that “the meaning of a compound expression is a function of the meaning of its parts and of the syntactic rule by which they are combined” (Janssen 2001: 115). Inherent to this principle is the idea that the “parts” (i) can be easily distinguished and (ii) have a straightforward meaning, i.e. there is a match between form and meaning, so that the meaning of the whole can be computed. Bybee (1985: 3) refers to this as the ‘ideal of a one-to-one relationship’. However, it is a fairly well-accepted fact in linguistics that this ideal is not reality and that there is lots of deviation from the ideal.

In this introductory section, we will outline four logical form-meaning mismatches (see also DFG Research Training Group 2021; A. C. Harris 2009; Crysmann & Sailer 2021; Yip 1998), and illustrate them with an example.

1. One-Form-to-Many-Meanings ($F\text{-to-}M^n$)
2. No-Form-to-Meaning ($0\text{-to-}M$)
3. Many-Forms-to-One-Meaning ($F^n\text{-to-}M$)
4. Form-to-No-Meaning ($F\text{-to-}0$)

In the subsequent sections (starting with section 1.2), these four types of mismatches will be discussed in greater detail by zooming in on one, two or three case studies per mismatch. The core idea in all these sections will be that the traditional notion of a form-meaning mismatch is rooted in the illusion of a one-to-one correspondence between form and meaning. If we make the ‘parts’ smaller by cutting open the morpheme to look at the underlying distinctive morpho-syntactic features, then it becomes clear that these mismatches do not require any

the functional heads CMPR and SPRL for the syntactic representation of the comparative and superlative respectively, the *ABA generalisation, established by Bobaljik (2012) on the basis of (the absence of) syncretisms between root forms, can be naturally accounted for by the containment structure in (2).



Additional support for this structure comes from containment data, as in Table 1.1 from Bobaljik (2012: 50), where we see that the form of the superlative in a variety of languages contains the form of the comparative.

Table 1.1: Morphological containment relations

		CMPR	SPRL
a.	Persian	X- tær	X- tær-in
b.	Lithuanian	X- iau	X- iau-sia
c.	Cimbrian German	X- ar	X- ar-ste
d.	Batsbi	X- vx	X- vx-č
e.	Latvian	X- âk	vis -X- âk
f.	Czech	X- ší	nej -X- ší
g.	Hungarian	X- bb	leg -X- bb
h.	Chukchi	X- əŋ	ənan -X- əŋ
i.	Cherokee	X- ka/ya/...	w -X- kǎʔi/yǎʔi/...
j.	Ubykh	ç'a -X	a-ç'a -X

What this example from degree morphology shows is that forms with many meanings, like Armenian *partsrahasag*, shed light on relations between morphosyntactic features. In this particular case, the form shows that there is a very close relation between the features responsible for the positive form of the adjective, the comparative and the superlative. This idea is not new and the work by Bresnan (1973); Corver (1997) already proposed to treat comparative and superlative morphology in the extended projection line of the AP, in the form of a functional projection DEGP. However, Bobaljik's study of root suppletion, and more in particular the study of syncretisms in this domain and the *ABA generalisation, as well as the containment data in table 1.1, gave greater urgency to the idea that the features corresponding to the positive, generalisation in this chapter, but we refer the reader to Bobaljik (2012) and De Clercq & Vanden Wyngaerd (2017a), De Clercq, Caha et al. (2023) for discussion.

comparative and superlative can be ordered hierarchically as in (2), and that a principled decomposition of the DEGP layer (Corver 1997), into CMPR and SPRL, is warranted.

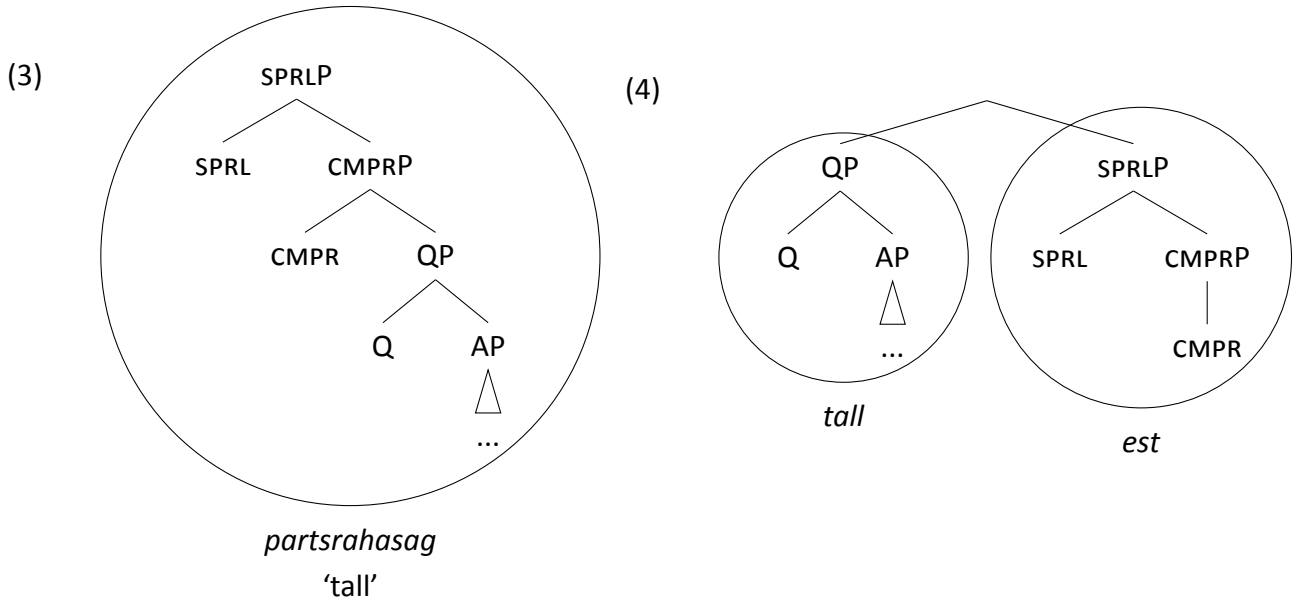
In Nanosyntax as well, the study of syncretism, *ABA and containment has been a key tool to probe the structure of a morpheme, and to decompose.² This has led to very fine-grained decompositions of case (Caha 2009; 2022; Taraldsen 2010; Starke 2017; Bergsma 2019; Kloudová 2020; Türk & Caha 2021; Janků 2022), directional expressions (Pantcheva 2011), negation (De Clercq 2013; 2020b; Baunaz & Lander 2023), complementizers (Baunaz 2018; Baunaz & Lander 2018b; c), demonstratives (Lander 2016; Lander & Haegeman 2018), pronominals (Vanden Wyngaerd 2018), cross-categorical syncretisms (Wiland 2019), inflectional and derivational morphology of verbs (Taraldsen Medová & Wiland 2018; Baunaz & Lander 2019; Starke & Cortiula 2021; Cortiula 2023), degrees (Caha, De Clercq & Vanden Wyngaerd 2019; Vyshnevskaya 2022; Wiland 2023), and indefinites (Dekier 2021). All of this work has taken syncretism and *ABA as a means to probe the underlying feature structure of the relevant morphemes, and hence has helped in understanding the order of syntactico-semantic features inside markers for case, negation, complementizers, etc. Consequently, these studies have helped to get closer to getting a picture of what is referred to as “the functional sequence” (henceforth FSEQ), i.e. the sequence of features that syntax uses to build structure.³ The consequence of this type of approach is that almost all morphemes, the so-called smallest units of meaning of natural language, actually turn out to be nonatomic, and to lexicalise many or several features. In other words, even small pieces of morphology are actually portmanteaus or, to use a term from the typological literature, instances of *polyexponence*, i.e. “formatives which simultaneously code more than one category” (Bickel & Nicol 2013), or of (*morphological*) *fusion* (Bybee 1985; Rathi, Hahn & Futrell 2022).⁴ This naturally brings us to one of the core ideas in Nanosyntax, namely phrasal lexicalisation. Let us have a look at what this means exactly.

²Bybee (1985: 13) also considers the degree of fusion indicative of the relevance that particular ‘meaning elements’ have with respect to each other, and that they must be closer to each other.

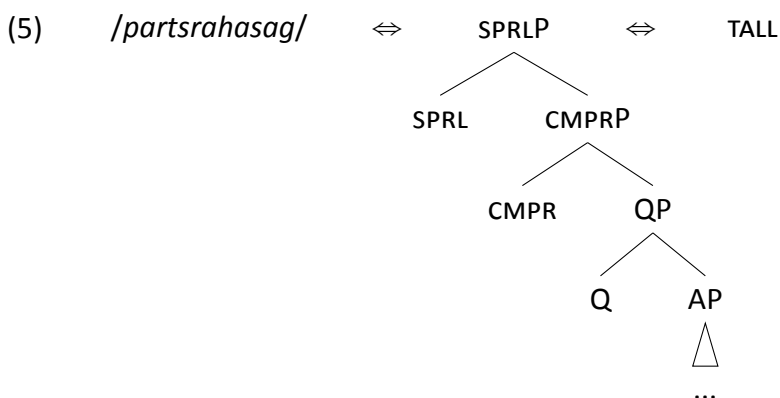
³One could argue that the functional sequence in Nanosyntax corresponds to the morpho-syntactic structure whose feature ordering allows for a maximally processable structure. According to Hahn, Matthew & Degen (2021) universal patterns in affix order can be accounted for as the consequence of a trade-off between memory and surprisal.

⁴The term fusion is used in at least four different ways in the literature: in the typological literature, fusion refers to the “the degree to which grammatical markers [...] are phonologically connected to a host word or stem” (Bickel & Nicol 2013). The term is also used to refer to *phonological fusion*, in which case two morphemes are conflated to one. A third usage is the one used in the main body of the text, i.e. *morphological fusion*, referring to a situation of polyexponence, in which one morpheme lexicalizes multiple features (Bybee 1985; Rathi, Hahn & Futrell 2022). Finally, fusion can also refer to a post-syntactic operation in Distributed Morphology, that is responsible for fusing two syntactic features into one exponent (Harley & Noyer 1999).

As we said before, the Armenian root *partsrahasag* in (1) is used for the positive, comparative and superlative, and is hence syncretic for these categories. If we zoom in on its superlative use in (1c) and compare it to the English superlative form *tallest*, then we can deduce that *partsrahasag* can lexicalize the entire structure in (3), represented by means of a circle around the syntactic structure. We adopt Bobaljik’s containment structure, and add a QP layer (Bresnan 1973; Corver 1997; De Clercq 2013) to account for the fact that adjectives that are able to form comparative and superlative forms are gradable. The Q head can be lexicalised by an independent morpheme in a language like Czech (see Vanden Wyngaerd, Starke et al. 2020 and section 1.5 below). In English the comparative and superlative layers are lexicalized by *-est* and the AP and QP node are lexicalised by the root *tall*, as in (4).



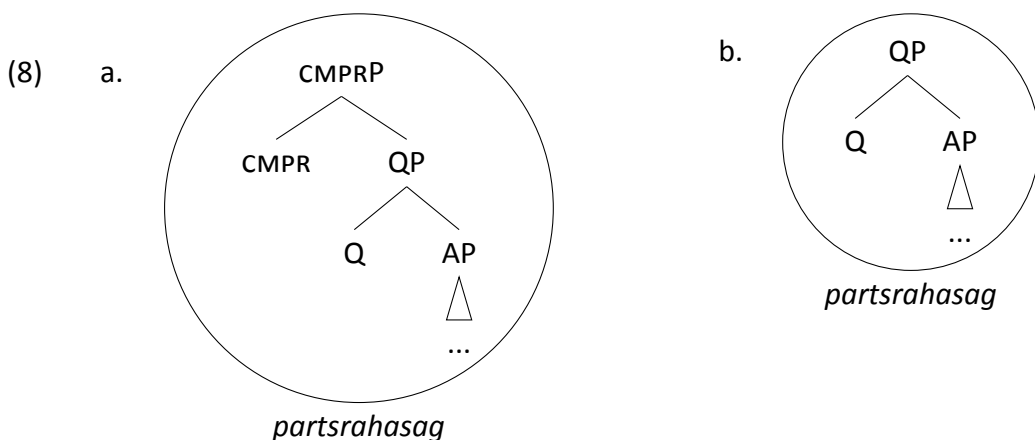
Differences between languages can thus be captured in terms of how much structure particular exponents can lexicalize (Starke 2014b): the structure of the root in Armenian is much bigger than the structure of the root in English, obviating the need for additional markers in the comparative and superlative. The place to store this information is the lexicon. As a consequence, lexical items in Nanosyntax are stored with a syntactic structure, conceptual information and phonological information. As such, the lexical item for *partsrahasag* for instance is stored in the lexicon with the features of the positive, comparative and superlative degree, and with the conceptual and phonological information as in (5) (see also De Clercq, Caha et al. 2023; Caha, De Clercq & Vanden Wyngaerd 2023c).



This allows (5) to lexicalise the structure in (3), but thanks to the matching condition, (6) (see also Caha, De Clercq & Vanden Wyngaerd 2023c), and its concomitant Superset Effect, (7), the lexical item in (5) can also lexicalise the structure in (8a) (the comparative) and (8b) (the positive).⁵

(6) Condition on matching: A lexically stored constituent L matches a syntactic phrase S iff S is identical to L.

(7) Superset Effect: A lexically stored tree matches a syntactic node iff the lexically stored tree contains the syntactic node.

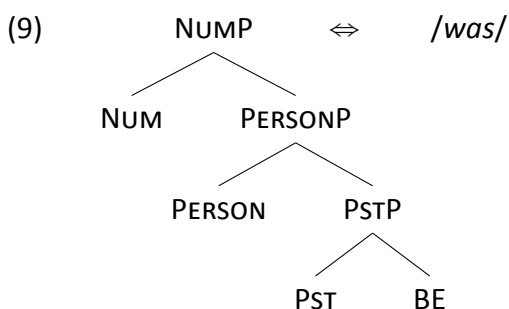


Summarizing, the Armenian example provided us not only with an ideal example of a *F-to-Mⁿ* mismatch, but it also allowed us to introduce the basics of Nanosyntax, namely phrasal lexicalisation and the matching condition (see also Starke 2009; Caha 2009; Baunaz & Lander 2018a; De Clercq, Caha et al. 2025). The discussion has revealed that the form-meaning mismatch of Armenian can be explained once the idea of phrasal lexicalisation is paired with the

⁵The Superset Effect has often been informally referred to as the Superset Principle (Starke 2009: 3). However, since the Superset Effect actually follows from the condition on matching, it is an effect or theorem and not a principle. See also De Clercq, Caha et al. (2025) in chapter 2 for discussion.

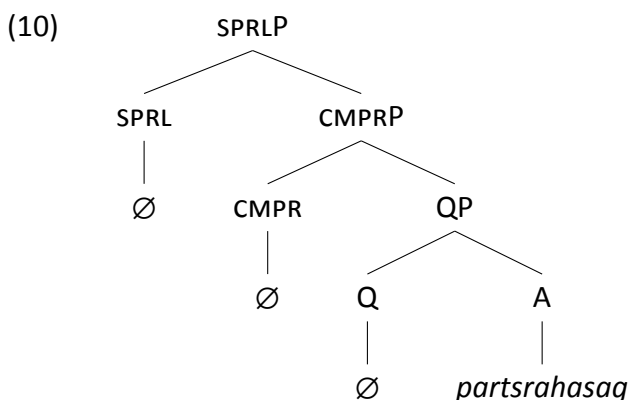
identity condition on matching between syntactic trees and constituents in the lexicon. We will discuss another instance of syncretism by means of a study of conversion in section 1.2.1, focussing on the papers by Caha, De Clercq & Vanden Wyngaerd 2023c and Vanden Wyngaerd, De Clercq & Caha 2022.

Before we continue to the next type of mismatch, we want to draw the attention to the fact that not all instances of fusion also involve syncretism. Actually, the prototypical case of fusion does not involve syncretism, but involves a unique form that seems indivisible and that lexicalizes multiple features. Take for instance the irregular simple past form of *be*: *was*. This is a typical illustration of what is traditionally referred to as morphological fusion: the form not only expresses the lexical semantics of BE, but also the past tense PST, singular number NUM, and 3rd person (captured by the PERSON feature (Harbour 2016; Ackema & Neeleman 2018; Vanden Wyngaerd 2018) in (9). The exponent *was* can be informally (and minimally) represented as lexicalizing the structure in (9) (cf. De Clercq 2020b: 177-178).



In this case, we are dealing with one unique form that corresponds to at least four different meaning components: BE, PST, PERSON and NUMBER. All these meaning components need to be activated to give rise to the exponent *was*. If we were to take away PST, the structure would be lexicalized as *is*, a non-syncretic form. In section 1.2.2, we explore three different types of fusion, which also instantiate cases of suppletion (Fertig 1998; G. G. Corbett 2007). We first discuss a case of suppletion in the lexical domain, and then two cases in the inflectional domain. As with the portmanteaus that arise due to syncretism, we will illustrate how the form-meaning mismatch that fusion seems to give rise to finds a natural explanation under phrasal lexicalisation.

Another mismatch that gets an immediate and natural explanation in Nanosyntax thanks to phrasal lexicalisation is the *0-to-M* mismatch. If we take the Armenian example again, then under head insertion the structure in (3), modified after Bobaljik (2012) with the additional layer for gradability, would have *partsrahasag* inserted under the root, as in (10), and then zeroes would need to be inserted in all the other head positions to get the meaning of the positive, the comparative, and the superlative.



However, thanks to phrasal lexicalisation these zeroes are no longer needed, and the mismatch also disappears, as we discussed above. In section 1.3 we will zoom in on some recent joint research of mine on Latin imperatives (Demonie & Clercq 2023). The absence of number or person morphology on second person singular imperative forms (e.g. *veni come!*) makes it a clear *O-to-M* mismatch at first sight (see also DFG Research Training Group 2021). We will discuss data from - amongst others - the future imperative to argue that there is more underlying featural structure in morphologically poor forms than meets the eye, thus showing how decomposition and phrasal lexicalisation challenge the illusion of a one-to-one correspondence between form and meaning.

I will now proceed to discuss two other traditional mismatches: *Fⁿ-to-M* and *F-to-O*. We will discuss the *Fⁿ-to-M* mismatch first. A typical example of that type comes from doubling.⁶ Doubling phenomena come in different types, and not all of them are mismatches (van Riemsdijk 2012). Some types of doubling reveal a form-meaning match. For instance, negations with different scopal properties, regardless of whether they are homonymous or not, can give rise to a double negation at the interpretative level, as in (11).

- (11) a. John can't not hate the man. (Horn 2001)
 b. He sent me a not unfriendly letter. (Zimmer 1964)

The same is true for *wh*-words in sentences like *Who did what?* or *What did he say about what?*: the two *wh*-words each come with the semantics of a *wh*-word, i.e. as elements that introduce alternatives (Hamblin 1973).⁷ These phenomena lie outside of the scope of this chapter. We focus on instances of doubling that may give rise to a mismatch, at least at first sight.

Within this group, a distinction can be made between doubling that happens at the word level, or doubling that happens at the clause level. Doubling at the word-level involves reduplic-

⁶A *Fⁿ-to-M* mismatch that we do not discuss here is overabundance (Thornton 2019). We only discuss *Fⁿ-to-M* mismatches at the level of the clause/word. Overabundance is a mismatch situated at the level of the paradigm (or in the lexicon in Nanosyntactic terms).

⁷I abstract away here from the single pair vs. pair list readings, and echo readings that multiple questions can trigger. I refer the reader to Kotek (2016) and the references cited there.

ation (see Urbanczyk 2017 for an overview) and multiple exponence (see P. Matthews 1972: 82 for coinage of the term) for instance, while doubling at the clause level involves concord or agreement (Cabredo Hofherr 2020). For this overview we will focus on one type of word-level doubling, i.e. multiple exponence, and two related clause-level doubling phenomena, i.e. bipartite negation and negative concord.

Multiple (or exuberant, or extended) exponence (A. C. Harris 2009; Caballero & A. C. Harris 2012; A. Harris 2017) refers to the phenomenon whereby two or more morphemes in a single word realize the same grammatical information. The data mentioned in the literature vary in complexity. On the one hand, very complex data as in (12) from Batsbi (Tsova-Tush), a language of the Nakh group of the Nakh-Dagest (A. C. Harris 2009: 268, Crysmann 2021) have been classified as instances of extended or exuberant exponence. (12) shows three repetitions of a class/gender marker, **d-**, that agrees with the nominal *c'a* 'house', within one grammatical word. Up to five of these markers can be present (A. C. Harris 2009).

- (12) *tiš'iⁿ c'a daħ d-ex-d-o-d-an-iš*
 Old house(d/d).ABS PV CM-destroy-CM-PRES-CM-EVIDI-2PL.ERG
 'Y'all are evidently tearing down the old house.'

While I have not yet discussed this type of multiple exponence in my work (but see Caha 2022 for a Nanosyntactic approach to agreement), a less radical type of multiple exponence has figured in my work. These less radical types have also been discussed by P. Matthews (1972); P. H. Matthews (1974). A case in point is for instance English *told*: in this verbal form the past tense is not only marked by the vowel change (as in *sank*), but also by means of the presence of *-d* (as in *walk-ed*). A similar example comes from German *Hälse* 'necks', where the plural is both marked by means of the umlaut and the plural morpheme *-e*. Another typical example comes from the domain of degree morphology in English: *bett-er* for instance can also be regarded as a case of multiple exponence. Both the root and the suffix give rise to the meaning of the comparative. In my work with Pavel Caha and Guido Vanden Wyngaerd, I tackled this type of multiple exponence, both in English and in Czech. We will discuss these data in section 1.4.1.

With respect to clausal doubling phenomena, a well-known example comes from *wh*-doubling as described for Canton Uri in Switzerland, illustrated in (13a). The exact same *wh*-word *was* can appear in clause initial and clause final position, giving rise to only one interpretation for the *wh*-words. The interpretation of (13a) is that of a traditional information seeking question (van Riemsdijk 2012), and hence does not differ semantically from (13b), the same sentence without doubling.

- (13) a. **Was** mach-ä-mer moorä **was**?
 what do-PL-we tomorrow what
 'What do we do tomorrow?'

- b. **Was** mach-ä-mer moorä?
 what do-PL-we tomorrow what
 'What do we do tomorrow?' (Glaser & Frey n.d.: 8)

Many different analyses can be and have been entertained for this type of doubling. van Riemsdijk (2012) proposes an analysis in terms of AGREE (Chomsky 2000), and basically treats the doubling as some type of agreement. For other types of clausal doubling, like bipartite negation, AGREE inspired analyses have also been proposed.

Bipartite negation is a doubling phenomenon in which the standard negator consists of two elements, of which one usually precedes the verb and another follows it (Miestamo 2005). For this reason, bipartite negation is also referred to as discontinuous negation. Crucially, the two particles together can give rise to one semantic negation, and hence this phenomenon totally matches the type of mismatches we are interested in in this chapter. A famous example to illustrate bipartite negation comes from *ne ...pas* in *le bon usage* French (14a) (see amongst many others Pollock (1989); Haegeman (1995); Rowlett (1998); Hansen (2013); Willis, Lucas & Breitbarth (2013); De Clercq (2019; 2020b), Abeillé & Godard (2021: ch10)). But the phenomenon is widespread and can be found in many languages, like for instance also in Sgaw Karen (14b) (Sino-Tibetan, Tilleson 2019: 5), Afrikaans (14c) (Biberauer & Zeijlstra 2012), or Babanki (14d) (Bantu, Northwest Cameroon) (Akumbu 2016: 151) (see Dryer (2013) for a typological overview).

- (14) a. Je (**ne**) sais **pas**.
 I NEG know NEG
 'I don't know.'
- b. jə1 tə1 nə2 pi2 (**bə5**)
 I NEG understand NEG
 I don't understand.
- c. Hij is **nie** moeg **nie**.
 he is NEG tired NEG
 'He is not tired.'
- d. fà-nín fá **kó'** fəŋ-ə **bwen**
 C19-bird SM NEG fall-PROG NEG
 'The bird is not falling.'

Very often one of the two elements is optional, as is also the case for the wh-words in Canton Uri Swiss, or can disappear in particular contexts. With respect to bipartite negation this type of optionality is usually understood as part of the diachronic development that negators seem prone to undergo, as described by Jespersen (1917: 4):

'The history of negative expressions in various languages makes us witness the following curious fluctuation: the original negative adverb is first weakened, then found insufficient and therefore strengthened, generally through some additional

word, and this in its turn may be felt as the negative proper and may then in course of time be subject to the same development as the original word.'

Since Dahl (1979), this succession of diachronic stages has been referred to as Jespersen's cycle. If all the data in (14) are indeed undergoing some type of Jespersen Cycle, then the examples all illustrate the stage in which the original negator and the new negator are both present, the so-called Stage II in the traditional three-stage depiction (15) (from De Swart 2010: 114).

- (15)
- a. Stage I: Preverbal expression of sentential negation.
 - b. Stage II: Discontinuous expression of sentential negation.
 - c. Stage III: Postverbal expression of sentential negation.

Crucial for all four languages in (14) is that the two forms, if they co-occur, convey only one semantic negation together. Many different analyses have been proposed in the literature, but most formal accounts are asymmetric (Breitbarth, Anne 2009), which boils down to the fact that essentially only one of the elements is really negative, with the other element being an NPI (see Zeijlstra 2009 for discussion of *ne*), or a semantically non-negative element that enters into an AGREE relation with the real negator (see Rowlett 1998: 28, Rizzi & Roberts 1996: 76, Roberts & Roussou 2003: 154-155, Roberts 2007: 64-81, see De Clercq 2020b: 192-195 for discussion). However, what these accounts cannot explain is why the so-called non-negative or NPI-like element is in some languages also really negative in particular environments, as for instance in the French example (16).

- (16)
- Julie ne boit plus.
Julie ne drinks more
'Julie no longer drinks.'

This example shows that *ne* can sometimes really carry negative force, since without *ne* the sentence gets the opposite affirmative meaning, i.e. *Julie drinks more*. What the asymmetric accounts can also not explain is why the non-negative or NPI-like element is compulsory in some languages, as for instance in certain stages of the French language (and Jespersen's Stage II in general), in which the co-occurrence of *ne* with *pas* was required for a sentence to pass as grammatical.

A similar type of problem arises with a related doubling phenomenon, more in particular with the phenomenon of negative concord, illustrated for French in (17a) (Zeijlstra 2004: 62), and for West Flemish in (17b) (Haegeman 1995: 116).

- (17)
- a. **Personne ne** mange **rien**
NEG.body NEG eats NEG.thing
'Nobody eats anything'

- b. da Valere woarschijnlijk an **niemand niets** **nie (en)**-zegt
 that Valere probably to nobody nothing not en-says
 'that Valere probably does not say anything to anyone'

For French in (17a) there are three negatively marked elements in the clause of which *personne* and *rien* are Negative Concord Items (NCIs). For West Flemish, in (17b), there are four negatively marked words in the clause and *niemand* and *niets* are NCIs. In both examples there is only one negative meaning expressed in spite of the multitude of negative forms. As with bipartite negation, there are different ways in the literature to deal with this form-meaning mismatch, which we will discuss briefly in section 1.4.2.2. Crucially, in that section, we will focus on Berber data to explain how negative concord, a clear form-meaning mismatch of the F-to-Mⁿ type, can be captured in Nanosyntax.

While looking at the Berber data, we will zoom in on another doubling phenomenon, namely TAM allomorphy conditioned by negation, across word boundaries (see section 1.4.2.1). In this case there is a negation specific verb form that co-occurs with the negator, hence giving rise to a Fⁿ-to-M mismatch. We discuss this in section 1.4.2.1.

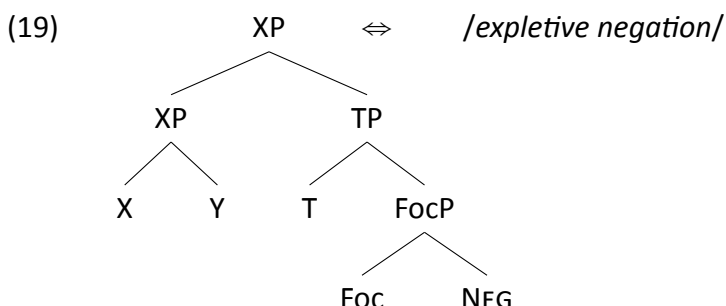
A form meaning mismatch that we have not yet discussed is one in which a particular form does not seem to contribute a meaning, a F-to-O correspondence as it were. A famous example of that type is for instance the case of expletive negation, in (18) (Delfitto & Fiorin 2014).

- (18) a. Dubito que **no** mengi.
 I.doubt that NEG eat.3sg
 'I doubt that he eats.'
- b. Che cosa **non** ha fatto Gianni!
 what thing neg has done Gianni
 'The things Gianni did!'

Expletive negation (henceforth EN) involves the use of a morphological negator, similar or different from the standard negator, that is used in an embedded clause selected by a verb or noun “expressing fear, prohibition, hindering, avoidance, denial, or doubt,” or in *before* and *until* clauses, in the standard of comparatives, and exclamations (Delfitto 2020). Several proposals have been made in the literature for this type of mismatch. An influential approach, introduced by Potts (2005), claims that the EN element is truth-conditionally vacuous, but triggers an evaluative sense, i.e. the EN “should be understood as an utterance modifier, in the form of a conventionalized implicature” (Delfitto 2020). For the sentence in (18) this means that the speaker evaluates it as unlikely that he eats. In some syntactic approaches (Espinal 1992; 1997; 2007), the element is considered negative in so far that it enters into a concord relation with a negative feature present in the superordinate predicate. Something along these lines was also suggested by Jespersen (1917); van der Wouden (1994), who refer to EN as paratactic negation. Delfitto (2020) on the other hand proposes an analysis in which the EN keeps

its negative semantics, but in which the domain of application is not the proposition but the conversational implicature. Greco (2020b; a) proposes that the different readings for negation arise as a consequence of the fact that EN is generated in the CP domain, and not in the vP phase. Only negation that is merged in the vP phase can reverse the truth conditions of the sentence. The problem with this type of approach is that it remains unexplained why a negative marker would not be able to have real negative scope from this position. In English for instance negative DPs and PPs move to CP for extra negative force (Haegeman 2000; De Clercq, Haegeman & Lohndal 2012), and in Jewish Babylonian Aramaic (Bar-Asher Siegal 2015; Bar-Asher Siegal & De Clercq 2019) a dedicated marker in the left periphery gives rise to external negation.

A way to approach expletive negation in Nanosyntax would be by means of complex lexical items, i.e. the type of lexical items adopted by Blix (2021; 2022), which consist of an additional complex left branch. Let me illustrate this with a hypothetical example. Imagine that some negators are stored in the lexicon with a structure as in (19).



This structure would force them to be really negative whenever features on the right branch are lexicalised. However, the structure of the lexical item also suggests that the same exponent could be used to also lexicalize just the left branch, as well as both branches together (i.e. the situation where there is a perfect match). Whenever there is a context in which just the left branch gets lexicalised, we are dealing with a situation that is referred to as expletive negation: the syntax of that structure will not have a NEG-feature, but the lexical item that lexicalises this structure will contain of a NEG-feature.

Without going into any details, and keeping this proposal at an informal level, it would allow us to capture the hybrid behaviour of expletive negation, without having to postulate different lexical items in the lexicon. However, expletive negation is not what we will zoom in on as a F-to-O mismatch. Instead, we will focus (in section 1.5) on what looks like a F-to-O mismatch in the domain of Czech adjectives. The positive form of some Czech adjectives comes with the marker *k* or the marker *n*. At first sight these markers do not seem to contribute any meaning. However, upon closer consideration we will see that it makes sense to think of them as lexicalisations of the internal feature structure of gradable adjectives. Section 1.5 presents unpublished work with Pavel Caha (Masaryk University, Brno) and Guido Vanden Wyngaerd

(CRISSP/KU Leuven) (Caha, De Clercq & Vanden Wyngaerd 2023a; b). This research draws in part on Vanden Wyngaerd, Starke et al. 2020, see chapter 10, but it also goes beyond the analysis presented there.

In the remainder of this chapter, I will discuss the four form meaning mismatches introduced in this section in more detail, focussing on some of my past research projects. The upshot of all investigations mentioned in this first chapter is that when we make the basic unit of investigation smaller, i.e. when we move to the Nanolevel of a morpheme, form-meaning mismatches turn out to be nothing exceptional, but rather a natural consequence of phrasal lexicalisation on the one hand and the size/structure of stored lexical items on the other hand.

1.2 One Form-Many Meanings

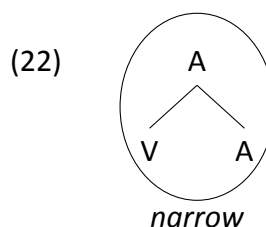
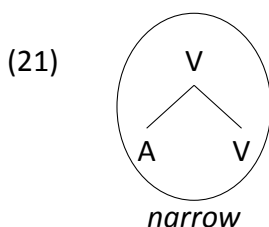
In this section we discuss two subtypes of the One Form-Many Meanings mismatch, i.e. F-to-Mⁿ. In the first subsection, 1.2.1, we discuss **syncretism**, in which one form lexicalizes different meanings. We analyse these forms as consisting of different underlying syntactic-semantic features, hence accounting for why they can lexicalize different meanings. Crucially, if one or several features are pruned in the syntax, the forms discussed in section 1.2.1 will continue to lexicalize that set of features. In other words, these forms consist of features that are in a subset-superset relation, and of which different portions can be lexicalized by the same form, giving rise to syncretism. In section 1.2.2 on the other hand, I discuss **fusion**, which I use here to refer to the situation in which one unique form lexicalizes several syntactico-semantic features (Rathi, Hahn & Futrell 2022). If one of these features is not present in the syntax, that particular form will no longer be available (see also section 1.1). Fusion can coincide with suppletion (G. G. Corbett 2007), the situation in which “regular semantic and/or grammatical relations are encoded by unpredictable formal patterns” (Veselinova 2006; 2017). Two cases of fusion that we will tackle, i.e. negative adjectives (section 1.2.2.1) and Korean portmanteaus (section 1.2.2.2), could also be regarded as cases of suppletion, because in both cases a regular one-to-one pattern is also available in the language. The case of fusion discussed for Bambara (section 1.2.2.3), is a case of pure polyexponence or cumulative exponence. Before we dive into fusion, we tackle syncretism.

1.2.1 From syncretism to portmanteaus

Caha, De Clercq & Vanden Wyngaerd (2023c) (chapter 3 in part II), discuss a clear-cut form-meaning mismatch: conversion. An example from English is given in (20): *narrow* is ambiguous between being an adjective (20a) and a verb, and as a verb it can give rise to an inchoative (20b) and a causative (20c) reading.

- (20) a. The road is narrow.
 b. The road narrow-ed.
 c. The workers narrow-ed the road.

One form thus seems to be able to express multiple meanings, a situation that reminds us of what we discussed in section 1.1 with respect to the Armenian adjective *partsrahasag*. Without closer inspection, one could either assume that the verb is derived from the adjective or the adjective from the verb, and hence both structures in (21)-(22) would be plausible (see Haspel-math 1993).



However, on the basis of English morphology shown in Table 1.2 (Caha, De Clercq & Vanden Wyngaerd 2023c: 37), and on the basis of the overt morphology in languages like Czech and Turkish, we can deduce that the adjective is contained in the verb, and hence that the structure in (21) is on the right track.

Table 1.2: Deadjectival verbs in English

Ø	-en	-ify	-ise	en-
cool	tighten	solidify	generalise	enlarge
narrow	widen	prettify	formalise	enfeeble
open	shorten	simplify	americanise	enrich
thin	sharpen	humidify	sexualise	
dim	slacken	acidify	christianise	
tame	brighten	fluidify	commercialise	
blind	cheapen	falsify	conceptualise	
warm	coarsen	Frenchify	actualise	
clean	dampen	intensify	annualise	
empty	darken	uglify	grammaticalise	
clear	deaden	diversify	brutalise	
dry	deafen	greenify	centralise	

Moreover, the semantics also seems to back up the morphology in that way. Take for instance *to narrow*, which can be intuitively paraphrased to mean *to make narrow* or *to become narrow*, suggesting that the adjectival meaning (the property) is contained in the verbal meaning (the event). With respect to the second ambiguity, the causative and inchoative reading in (20c)-(20b), respectively, the morphology of English is not very helpful, i.e. there is no dedicated marker for the inchoative that is not also used for the causative, nor vice versa. Moreover,

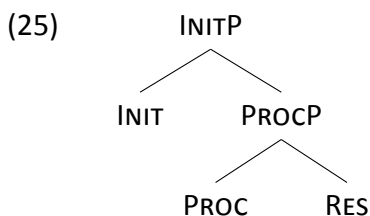
from a crosslinguistic perspective the morphology seems to go both ways (Haspelmath 1993): some languages use a causative that is morphologically more complex than the inchoative, but other languages do the opposite, i.e. they build the inchoative on top of the causative.⁸ Semantically however, the inchoative seems entailed by the causative: i.e. *to narrow* means *to cause to become narrow* which suggests that the inchoative meaning is contained in the meaning of the causative. Lundquist et al. (2016: 2) argue for a Causational Entailment, as formalized in (23):

(23) $\forall x \forall y [\text{CAUSE}(x, \text{INCH}(\text{Pred}(y))) \rightarrow \text{INCH}(\text{Pred}(y))]$

The fact that there is this causational entailment explains why (24) is infelicitous:

(24) #John broke the glass, but the glass didn't break.

The semantic factors are thus in this case decisive in choosing for a structure in which the inchoative is contained in the causative. Ramchand's (2008) work on the decomposition of the verb phrase supports this type of containment, in that she adopts the structure in (25) for causative verbs. INIT stands for initiation and is the feature that contributes the initiation event. PROC stands for Process and this is the feature that specifies the nature of the change or process, and RES stands for Result. INIT and RES are optional.⁹

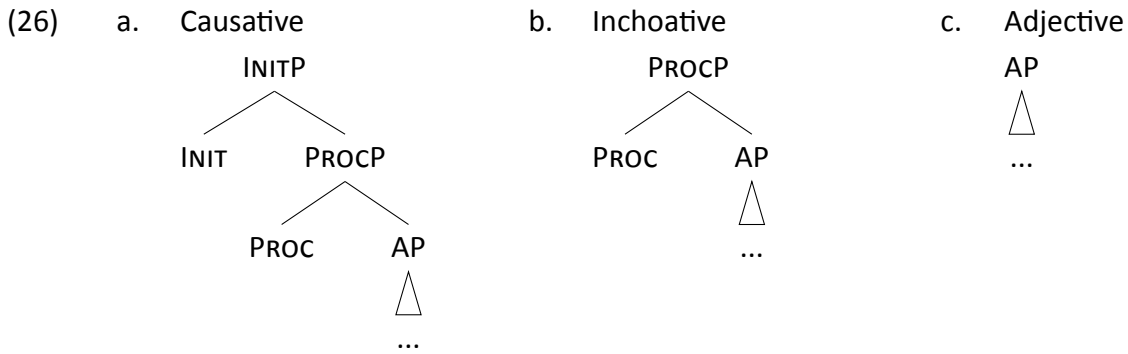


For causative deadjectival verbs Ramchand (2008) proposes the structure in (26a) with the AP contained in the verbal structure. For inchoatives the INIT feature is missing, as in (26b).¹⁰

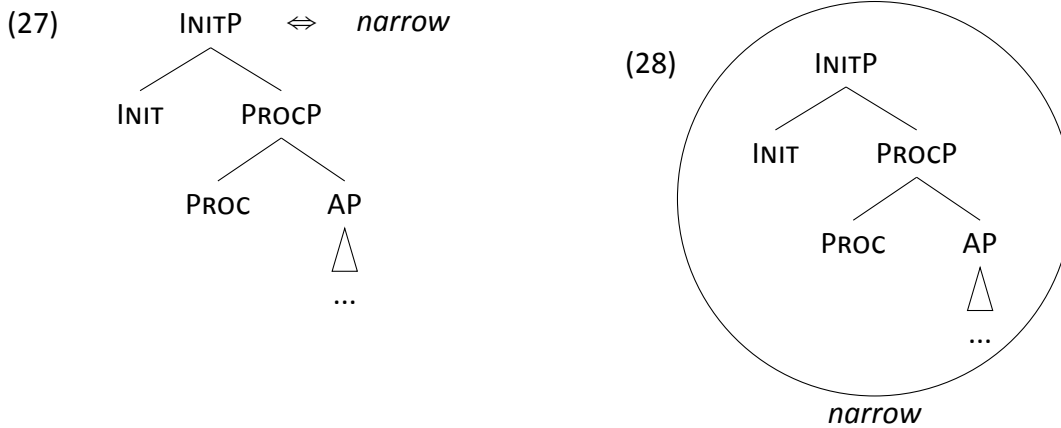
⁸This type of inchoative formation is also referred to as the anticausative, see Caha, De Clercq & Vanden Wyngaerd (2023c: 56-60) for a possible way to analyse anticausatives within the current approach.

⁹For stative verbs Ramchand (2008: 55) proposes that there is only an INIT feature and no PROC feature. The absence of PROC gives rise to the stative interpretation, i.e. there is no process for the initiator to work upon.

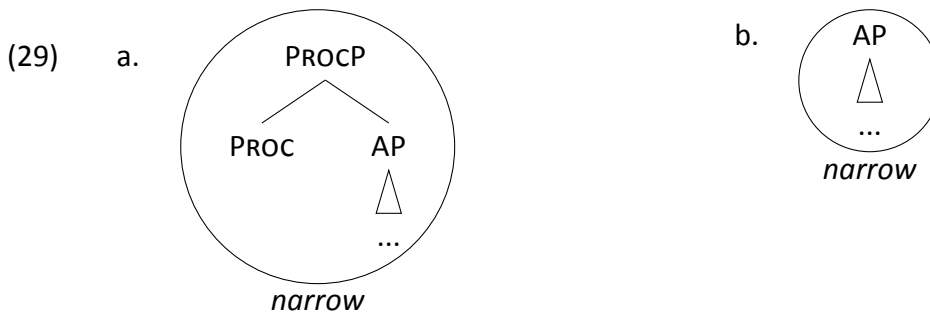
¹⁰Ramchand proposes that RES is absent from the structure of deadjectival verbs, but see Vanden Wyngaerd, De Clercq & Caha (2022) for the inclusion of RES in the derivation of Dutch deadjectival verbs.



In line with the Nanosyntactic ideas we developed in section 1.1, the lexical structure for a verb like *narrow* is as in (27), and the syntax of the causative verb that it can lexicalize is in (28).



Thanks to the lexical structure in (27), and thanks to the Superset Effect which is an inherent consequence of the matching condition (see definition (6) in section 1.1), (27) can also lexicalize the syntax of an inchoative verb (29a), and of a regular adjective (29b).

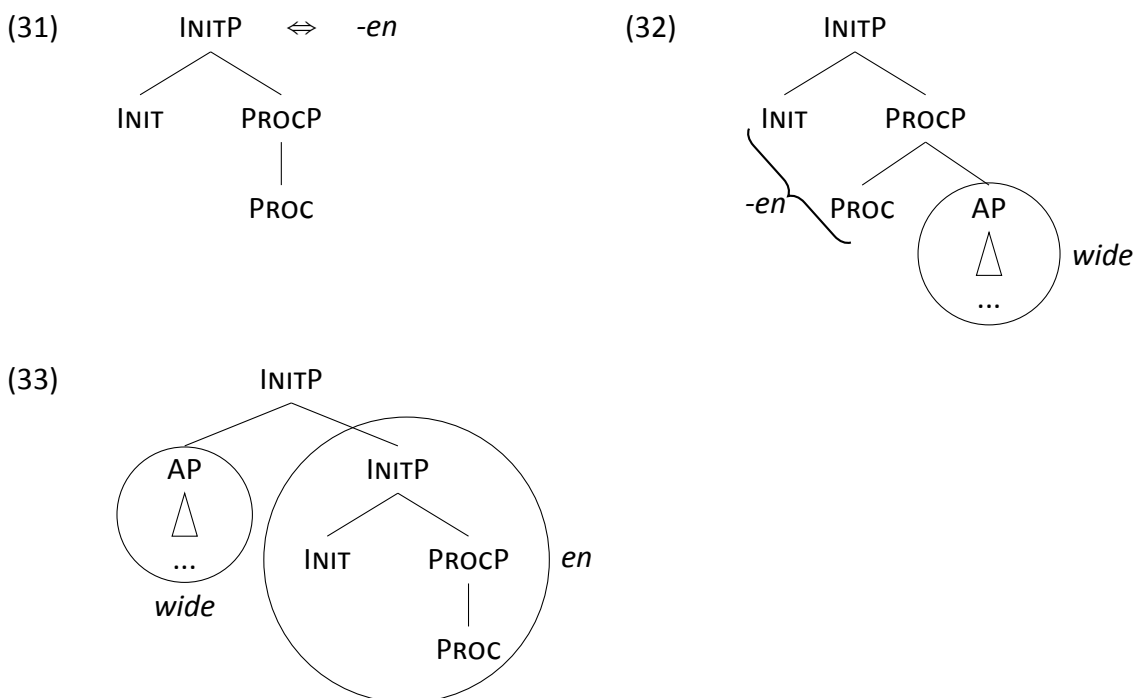


At this point, we already demonstrated how syncretism, or conversion in this case, can be accounted for in Nanosyntax: the mismatch between form and meaning can be explained as the consequence of the fact that the lexical structure of an element that is syncretic across different functions is structurally big and consists of all the relevant syntactico-semantic features that give rise to these different functions. Thanks to phrasal lexicalisation, and the Superset Effect, this lexically stored structure can match the different syntactic structures that each take care of a different function/meaning.

In what follows I elaborate on the same point by looking at the adjective-verb pair *wide-widen*. This case is different, because here it is an affix that gives rise to the ambiguity between the inchoative and causative reading, as shown in (30).

- (30) a. The workers widened the road. [CAUS]
 b. The road widened. [INCH]

Consequently, we deduce that *-en* lexicalises the features PROC and INIT, and hence we propose the lexical item in (31) for *-en*. The structure in (32) is an informal representation of what the lexicalisation looks like. Actually, the lexical item of *-en* cannot lexicalize the feature PROC and INIT in (32), since the tree in (32) is not identical to (or a subconstituent of) the lexical tree in (31). In order to lexicalise, syntax will perform a rescue operation, a so-called lexicalisation-driven movement, resulting in the structure in (33), which will now have a right branch that has not yet been lexicalised, and which is a perfect match for the lexical structure in (31). This movement is part of a set of rescue operations, the lexicalisation algorithm, which the syntax can perform in order to get a lexicalisation. We will discuss this in more detail in section 1.2.2.3 (or see De Clercq, Caha et al. (2025) in chapter 2).



We discussed the zero-derived verb *narrow* in this section, and the verbalising affix *-en*. Both case studies illustrated how syncretism, a form-meaning mismatch, can be captured in Nanosyntax as a consequence of stored lexical trees, phrasal lexicalisation and the matching condition.

1.2.2 Fusion

In this section I discuss three different types of fusion that are all F-to-Mⁿ mismatches. The first two cases in this section concern fusion that also involves suppletion, i.e. an irregular form is used while the language also has a regular pattern that is more of a form-meaning match. The last case of fusion concerns inflectional morphemes in Bambara which are fusional but not suppletive, i.e. they are not irregular in any possible way.

1.2.2.1 False vs. untrue

The first type of fusion I want to discuss involves adjectives like *false* or *sad*, in comparison to the quasi-synonymous adjectives *untrue* and *unhappy*. One could regard adjectives like *false* or *sad* as instances of the highest degree of fusion, on a par with what Bybee (1985: 12) argued for *kill* ‘cause to die’ (as opposed to *die*). In both cases ‘two or more semantic elements are expressed by one monomorphemic lexical expression’. While *un-true* and *un-happy* are instances of a perfect form-meaning match, adjectives like *false* or *sad* have a negative meaning, without consisting of an overt dedicated negative morpheme. De Clercq & Vanden Wyngaerd (2019a) (chapter 4 in Part II), argue that the adjective *false* is a portmanteau, a case of morphological fusion, that lexicalizes the features of positive gradable adjectives with an additional NEG feature.

The empirical argument for this claim starts from the well-known observation, made for English *un-* in Zimmer (1964); Reece Allen (1978); Siegel (1978), Horn (2001: 275) and dating back to Jespersen (1942: 466), to the effect that morphological negation with the negative prefix *un-* in English cannot combine with negative adjectives. This restriction is not a restriction on the combination of negation and a negative adjective as such, given the facts in (34c).

(34)	a.	unhappy	b.	*unsad	c.	not sad
		unwise		*unfoolish		not foolish
		unclean		*undirty		not dirty
		unfriendly		*unhostile		not hostile
		unhealthy		*unsick		not sick
		unkind		*unrude		not rude
		untrue		*unfalse		not false
		uneasy		*undifficult		not difficult

The restriction extends beyond *un-* prefixation and is relevant for all kinds of negative prefixes, and also for the negative caritive suffix *-less*, as shown in (35).

(35)

*un-dis-	*undishonest *undiscourteous	not dishonest not discourteous
*un-iN-	*unimpossible *unillogical	not impossible not illogical
*un-a-	*unabnormal *unatypical	not abnormal not atypical
*un-un-	*ununhappy *ununtrue	not unhappy not untrue
*dis-dis-	*disdishonest *disdisabled	not dishonest not disabled
*un-X-less	*unbreathless *unsenseless	not breathless not senseless

De Clercq & Vanden Wyngaerd (2019a: 426) provide support for the absence of these patterns by doing a corpus research in the BNC and COCA, which indeed shows that this combination is not attested in these corpora. At first sight, these type of data seem to support the morphology-syntax divide, in the sense that the constraint seems a constraint on morphological negation, since negation by means of the so-called syntactic negator *not* is totally acceptable. However, when we extend our scope from English to Dutch, we first of all see that exactly the same pattern holds as in English: negative adjectives cannot be combined with negative affixes, (36).

(36)

<i>on</i> +A _{POS}	<i>on</i> +A _{NEG}	
on-gelukkig 'unhappy'	*on-droef, *on-triest 'unsad'	niet droef/triest not sad
on-verstandig, on-wijs 'unwise'	*on-dom 'unfoolish'	niet dom 'not dumb'
on-gezond, on-wel 'unhealthy, unwell'	*on-ziek 'unsick'	niet ziek 'not sick'
zin-loos sense-less	*on-zinloos un-senseless	niet zinloos not senseless

However, crucially, the same restriction holds for the negative adjectival modifier *weinig* 'little', which, like *on-*, cannot modify negative adjectives, but which, unlike the prefix *on-* 'un-', is a syntactic negator in the traditional sense of the word:

- (37)
- a. weinig actief/*passief
little active/passive
 - b. weinig gezond/*ziek
little healthy/sick
 - c. weinig correct/*fout, *incorrect
little correct/wrong
 - d. weinig verstandig/*dom, *onverstandig
little clear/confused
 - e. weinig zinvol/*zinloos
little meaningful/senseless
 - f. weinig nuttig/*nutteloos
little useful/useless

It thus seems that the distinction between a morphological negator and syntactic negator is not the relevant distinction, and the Dutch data point to the fact that the constraint at work here cuts across the morphology-syntax divide. This is confirmed by the fact that the French adjectival modifier *peu* has a distribution similar to Dutch *weinig* 'little', e.g. *peu actif/*passif* 'little active/passive', *peu probable/*improbable* 'little likely/*unlikely' (see De Clercq & Vanden Wyngaerd 2017b for discussion). A possible reaction from semantic side could at this point be that the pattern can be explained in terms of the generalisation in (38): contrary negations cannot be stacked, but contradictory negations can (see also De Clercq & Vanden Wyngaerd 2018), as in (38).

- (38)
- a. $P(x)$
 - b. $\neg P(x)$
 - c. $\neg \neg P(x)$ (= (38a))
 - d. ...

Let us consider for a moment what contradictory and contrary negators are. Two propositions are contradictory if they cannot be true together, nor false together. An example would be the one in (39), with *n't/not* giving rise to contradictory negation.

- (39)
- a. Alexandra went to university.
 - b. Alexandra didn't go to university.

Two propositions are contrary if they cannot be true together but can be false together (see Horn 2001 for extensive discussion). An example would be the pair in (40), with *un-* giving rise to contrary negation.

- (40)
- a. Wendy is happy.
 - b. Wendy is **un**happy.

However, if this were indeed the relevant generalisation then it remains hard to understand why two contradictory negators cannot be stacked in particular contexts, as those mentioned by Collins (2018) in (41)-(46):

- (41) a. Not everybody was there.
b. *Not not everybody was there.
- (42) a. Not many people were there.
b. *Not not many people were there.
- (43) a. Not {even/only} John was there.
b. *Not not {even/only} John was there.
- (44) a. Not {more than three/less than three/a lot of} people were there.
b. *Not not {more than three/less than three/a lot of} people were there.
- (45) a. Not often do I manage to go on fall foliage tours.
b. *Not not often (do) I manage to go on fall foliage tours.
- (46) a. I persuaded John not to like Clinton.
b. *I persuaded John not not to like Clinton.

On the basis of this, we argue that a semantic account of these facts would undergenerate, and we propose to capture the stacking facts by adopting a *NEG NEG constraint in syntax, which is a concrete implementation of a more general syntactic constraint on permissible functional sequences in (47):

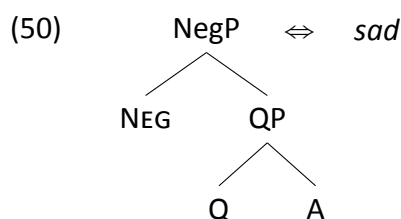
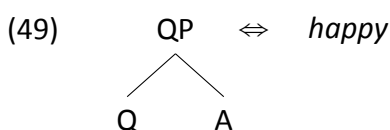
- (47) *<X, X>
The functional sequence must not contain two immediately consecutive identical projections.

The constraint boils down to the claim that syntax does not allow for two contiguous identical negative features, a phenomenon we can observe beyond negation, i.e. no functional head of the same type selects exactly the same functional head. Even if there are two or more positions for topic, these topics will be of a different kind, or if there is more than one tense position, the tense will be of a different type, etc The prediction of this is that whenever one can see negative markers stacked in a language giving rise to double negation, the negative markers will be featurally/structurally different, even if they are homonymous. The reason they can co-occur is that there will be intervening layers of structure between them. The upshot of this proposal is that the negative prefixes of the contrary type (*un-*, *dis-*, *iN-* in English and *on-* in Dutch) and the suffix *-less*, *-loos* must be of the same type, hence the ungrammaticality when they co-occur. In the same vein, negative adjectives like *false* or *sad* must consist of an internal NEG feature, since they give rise to unacceptability in the same way as their negatively sisters.

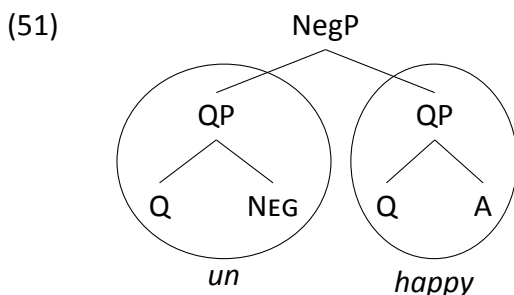
And indeed, negative adjectives give rise to contrary opposition, in the same way as *un-* does in (40), as one can observe in (48), i.e. it is possible that Rachel is neither tall nor short given a particular contextual standard (see De Clercq & Vanden Wyngaerd 2018 for a semantic proposal in terms of extent semantics).

- (48) a. Rachel is **tall**.
b. Rachel is **short**.

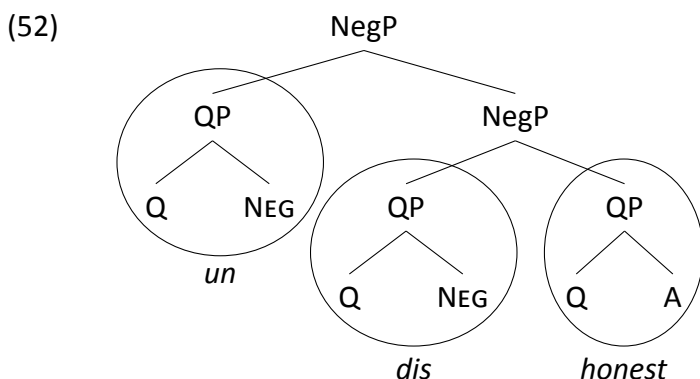
As a consequence of this, and adopting the internal structure of gradable adjectives discussed in section 1.1, we propose that positive adjectives have a lexical structure as in (49), and negative adjectives must have a lexical structure as in (50).



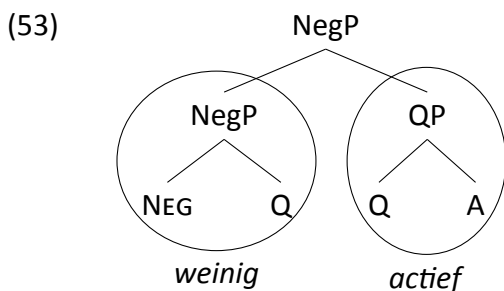
For negation we follow the decomposition of negative markers worked out in De Clercq (2013; 2018; 2020b), which boils down to the idea that structurally different negative markers take scope in different positions in the clause due to their internal structure. The idea that there are different positions for negation across the clausal domain is not new (Cormack2002; Zanuttini 1997; Poletto 2008; 2017), but the claim that there are positions for negation in domains that are normally covered by morphology was new. Especially the idea that these low (morphological) positions are connected with the high sentential (syntactic) positions via the internal structure of negation was a new way of thinking about negation. The essence of the proposal was that there are at least four relevant types of negation to distinguish: Quantity-negation, Classifier-negation, Focus-negation and Tense-negation. The labels are related to positions where the negators usually surface, i.e. QP in the extended projection of AP, ClassP in the extended projection of the NP, FocP in the extended projection of vP and the left periphery, and TP in the inflectional domain. Crucially, the structure for a negatively prefixed adjective would be as in (51). The Q-negator can project on top of the Q-layer of the AP thanks to its internal structure.



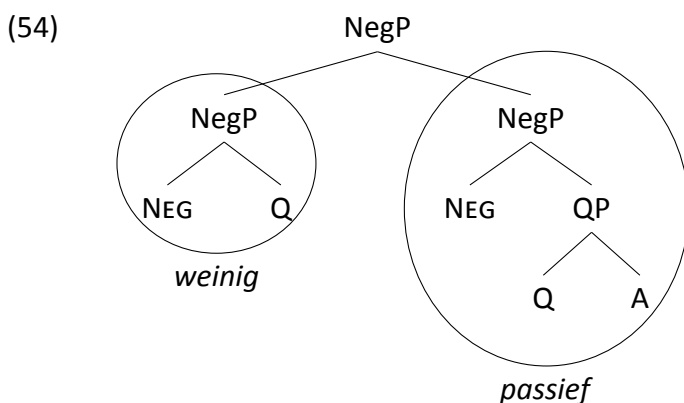
Since negative adjectives already contain a NEG feature, the same type of Q-related negation cannot project anymore, because that would violate the structure of the fseq, i.e. the idea that one cannot have two consecutive NEG features. As such, the structure in (52) is seen to violate (47).

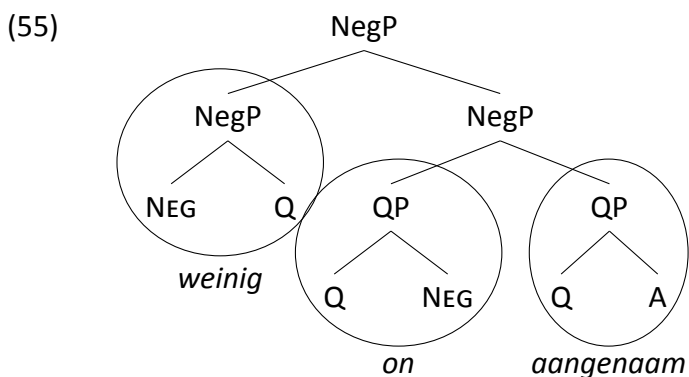


The crucial piece of evidence from Dutch, i.e. the fact that *weinig* ‘little’, a so-called ‘syntactic’ type of negation behaves in exactly the same way with respect to negative adjectives as the negative affixes, can be derived by adopting the structure in (53), which shows it modifying a positive adjective.



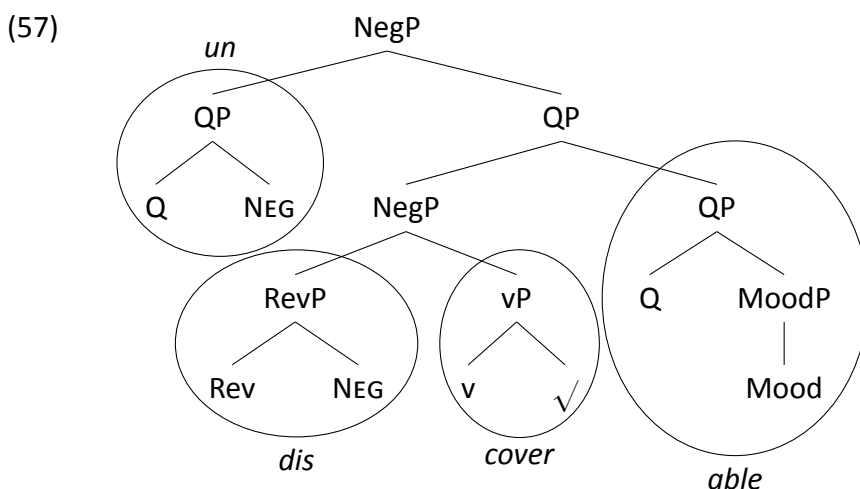
Since *weinig* ‘little’ has the structure of a Q-negator, it cannot combine with another Q-negation, hence the unacceptability of *weinig* with negative adjectives (54), as well as with negatively prefixed adjectives like *onaangenaam* ‘unpleasant’ (55).



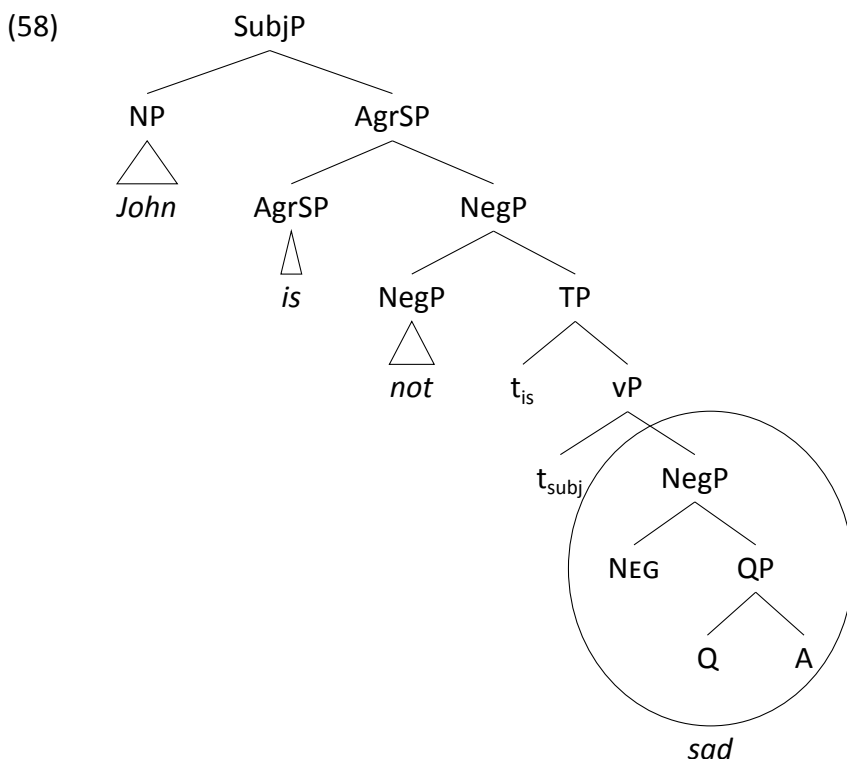


Some cases, like those in (56), look at first sight like counterexamples to the pattern discussed so far. However, upon closer consideration, all these examples can be accounted for in terms of structural nonadjacency, i.e. the two negative markers and their respective underlying negative features can be explained as being non-contiguous in the FSEQ due to intervening layers of structure, as illustrated for *undiscoverable* in (57).

- (56)
- | | |
|---------------------------|-------------------|
| un-dis -hearten-ed | un-scathe-d |
| un-dis -pute-d | un-defeated |
| un-dis -cover-able | un-blam-able |
| un-harm-ed | un-objection-able |
| in-vulner-able | ir-reproach-able |



The sentential negation is under the current approach entirely fine with all kinds of negative affixal markers due to the fact that sentential negators take scope in a position above TP or vP, (58).



The point of the discussion of this piece of research is that the data from Dutch (and by extension from French) show that ‘syntactic’ negators like Dutch *weinig* or French *peu* pattern exactly like the morphological negative prefixes *on-/loos*: neither of them can be combined with negative adjectives. De Clercq & Vanden Wyngaerd (2019a) take these data to mean that negative adjectives are instances of fusion, i.e. portmanteaus, lexicalizing not only features relevant to the adjectival domain, but also a NEG feature. As such, compared to negatively prefixed adjectives like *un-true*, which look like a perfect form-meaning match, *false* consists of the same relevant underlying features. However, due to the fact that negative adjectives are stored with a bigger structure in the lexicon, allowing them to realize a Q-related NEG feature, they can no longer combine with an additional NEG-feature of the same type, which rules out cases like **unfalse*. Positive adjectives on the other hand are smaller in size, and do not have lexical structures that consist of NEG. Hence, they can combine with a negative affix, as in *un-true*. Thanks to empirical studies as the one discussed in this section, the internal structure of what looks like an indivisible morpheme, can be cut open and a form-meaning mismatch can be understood as a consequence of the size of lexically stored trees.

1.2.2.2 Korean negative and honorific portmanteaus

In this section I zoom in on cases of fusion, more in particular on suppletive forms, in the verbal paradigm of Korean.¹¹ The suppletion arises under negation or honorification and gives rise to a paradox. The case is discussed in detail in Chung (2009) and Choi & Harley (2019), who offer conflicting solutions couched in Distributed Morphology. I will provide a solution that takes into account causative intervention facts (unlike Choi & Harley 2019) and refrains from drawing on post-syntactic operations at different levels (pre- and post-vocabulary insertion). In addition, the proposed account will also deal with suppletion in strictly local terms (unlike Chung 2009 and Choi & Harley (2019)). The proposal thus does not only offer a resolution of a form-meaning mismatch, but also adds to the discussion on the nature of suppletion (Veselinova 2017), and the role of locality in triggering it (Bobaljik 2000; 2012; Merchant 2015; Moskal 2015; Choi & Harley 2019). The data in this section are discussed in Vanden Wyngaerd, De Clercq & Caha (2021) (chapter 4 in Part II).

Before embarking on the relevant facts that will form the focus of this section, I want to draw attention to two empirical facts. First, sentence negation in Korean can be expressed by means of two different strategies. One strategy is referred to as short form negation (59), while the other form is referred to as long form negation. Long form negation combines the negative marker of the short form with the light verb *hata* ‘do’ in combination with *ci* marking on the lexical verb (60). Both long and short form negation can make use of the regular neutral negative marker *an(i)*, but also of the modal negator *mos*. In what follows, when a position for NEG is referred to, the position for short form negation is referred to, which is a position below TP and above vP, while long form negation takes scope in a position above TP (Hagstrom 1997; 2000; De Clercq 2020b).

- (59) a. eysute-ka **an(i)** ca-n-ta.
Esther-NOM NEG sleep-PRES-DECL
‘Esther doesn’t sleep/isn’t sleeping.’
b. eysute-ka **mos** ca-n-ta.
Esther-NOM NEG sleep-PRES-DECL
‘Esther cannot/is not allowed to sleep.’
- (60) a. eysute-ka ca-ci **an(i)** ha-n-ta.
Esther-NOM sleep-CI NEG do-PRES-DECL
‘Esther doesn’t sleep.’ (Chung 2007: 97)
b. eysute-ka ca-ci **mos** ha-n-ta.
Esther-NOM sleep-CI NEG do-PRES-DECL
‘Esther cannot/is not allowed to sleep.’ (Chung 2007: 98)

¹¹I want to thank Hyunjung Lee and Jaehoon Choi for their help with Korean data. All errors are of course mine.

Second, subject honorification in Korean is expressed by means of the nominative case marker *-kkeyse* on the subject (Schütze 2001). If a subject is marked like this, the honorific marker *-si* appears on the lexical verb, following the root and preceding tense and mood particles, as in (61a). With some inherently honorified subjects, like *ape* ‘father’ in (61b), the subject honorific marking can be absent, and hence also the marking on the verb can go missing (Kim & Sells 2007: 309). While subject honorification is usually treated as agreement (Ahn & Yoon 1989; Han 1993; Koopman 2005; Yun 1993), this is not always the case, as for instance in Kim & Sells (2007); L. Brown (2011).

- (61) a. *ape-nim-kkeyse mence ka-si-ess-ta*
 father-HON-HON.SUBJ first go-HON-PST-DECL
 ‘Father went first.’
 b. *ape-nim-i mence ka-(si)-ess-ta*
 father-HON-NOM first go-(HON-)PST-DECL
 ‘Father went first.’

Since we will be concerned with how suppletive forms can be conditioned in a honorific context triggered by *-si*, we will abstract away from what the right way is to capture the dependency between the subject honorific marking and the marking on the verb, a type of F^n -to- M in its own right. The pattern that we discuss in Vanden Wyngaerd, De Clercq & Caha (2021) is summarized in the table in (62).

(62)

	regular pattern	$\sqrt{\text{EXIST}}$	$\sqrt{\text{KNOW}}$
a.	$\sqrt{X}$	iss	al
b.	NEG $\sqrt{X}$	eps	molu
c.	$\sqrt{X}$ HON	kyey-si	al-si
d.	NEG $\sqrt{X}$ HON	ani/mos kyey-si	molu-si

It sketches a situation with regular verbs (shown as X in the first column), and two suppletive roots, *EXIST* and *KNOW*. These pattern differently with respect to the feature they are conditioned by. Line (a) in the table shows the bare roots of the two verbs without any conditioning factor, i.e. *iss* ‘exist’ and *al* ‘know’ respectively. In line (b) the same form with negation is shown: for *EXIST* this results in the suppletive form *eps*, and for *KNOW* this leads to the suppletive form *molu*. These forms are not compatible with the short form negation (see Chung 2009). If honorification is added to the root, which is depicted on line (c), then *EXIST* again gets a suppletive form, i.e. *kyey*, followed by the honorific marker *-(u)si*, while *KNOW* keeps its root form and just takes the honorific marker *-(u)si*.¹² When negation and honorification are both present, *EXIST*

¹²The co-occurrence of the suppletive honorific root *kyey* with the honorific suffix *si* is at first sight another form-meaning mismatch, i.e. a case of multiple exponence. See section 1.4.1 for discussion.)

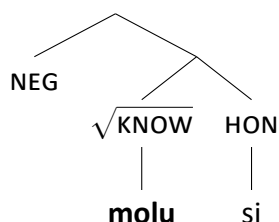
takes the form *kyey-si*, suggesting that honorification takes precedence over negation in conditioning the root, i.e. that HON is closer to the root than NEG. However, with KNOW the opposite pattern appears, with negation triggering the suppletive negative form *molu*, suggesting that NEG is closer to the root than HON.

The pattern in (62) hence presents us with a paradox, at least under the assumptions (i) that there is only one order of features for Korean, and (ii) that suppletion arises in a strictly local configuration. The EXIST verb suggests that the HON head is closer to the root, while the pattern displayed by KNOW suggests that NEG is closer to the root than HON.

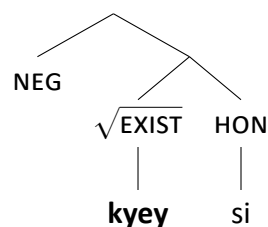
- (63) a. NEG > HON > $\sqrt{\text{EXIST}}$
 b. HON > NEG > $\sqrt{\text{KNOW}}$

However, while Chung (2009) and Choi & Harley (2019) agree on the existence of a paradox, they do not agree on the way to solve the paradox, nor on the underlying order of features. Choi & Harley (2019) argue for the structure in (64), while Chung (2009) argues for the structure in (65).

(64)



(65)



In order to account for suppletion under a terminal insertion system, Choi & Harley (2019) adopt the idea that HON conditions the root in a strictly local environment. However, if conditioning has not taken place in a strictly local configuration, as for instance in the case of KNOW, then it can happen from a distance (by negation for instance), as long as it happens within the complex X (see also Bobaljik 2012), hence supporting the idea that contextual allomorphy happens from the root-outwards (Bobaljik 2000). However, Vanden Wyngaerd, De Clercq & Caha (2021) adduce evidence from causative intervention that all conditioning in Korean must happen in a strictly local environment. The data in the table in (66) show that when the causative is present, in the form of the marker *-(l)i*, the conditioning by negation does not take place.

- | | | | | |
|------|----|-------------------------------|---------------|--------------------|
| (66) | a. | $\sqrt{\text{KNOW}}$ | al | 'know' |
| | b. | NEG $\sqrt{\text{KNOW}}$ | molu | 'not know' |
| | c. | $\sqrt{\text{KNOW}}$ CAUS | al-li | 'let know, inform' |
| | d. | NEG $\sqrt{\text{KNOW}}$ CAUS | ani/mos al-li | 'not inform' |

This suggests that the causative is closer to the root than the negation, as is confirmed by the semantics, which is 'not inform', rather than 'cause to not know'. Since the causative marker

triggers no suppletion in the root, we have exactly the same situation as in (64), and yet in this case the negation is unable to trigger the suppletive negative root.

There is also intervention from the causative with honorification, but this cannot be shown with the verb *EXIST*, since *EXIST* does not have a synthetic (“lexical”) causative form. However, we can show the pattern with some other verbs that take suppletive forms under honorification. The table in (67) illustrates that the root *EAT* and *SLEEP* take a suppletive form in the presence of the honorific marker *-(u)si-*.

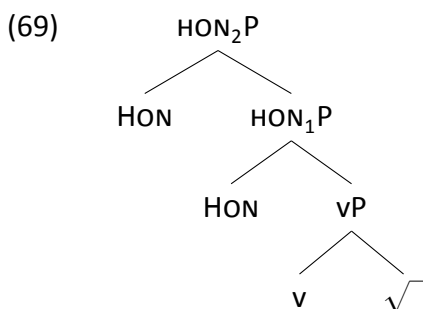
(67)	ROOT-DECL	ROOT-HON-DECL	ROOT-HON-DECL	
a.	mek-ta	*mek-usi-ta	caps -usi-ta	‘eat’
b.	ca-ta	*ca-si-ta	cwum -usi-ta	‘sleep’

When both the causative marker *-(l)i* and the honorific marker *-si* are present, they both occur on the same side of the root, suggesting that the underlying hierarchy is *HON > CAUS > ROOT*. In this case, the verb switches back to the nonsuppletive root *mek* (68d).

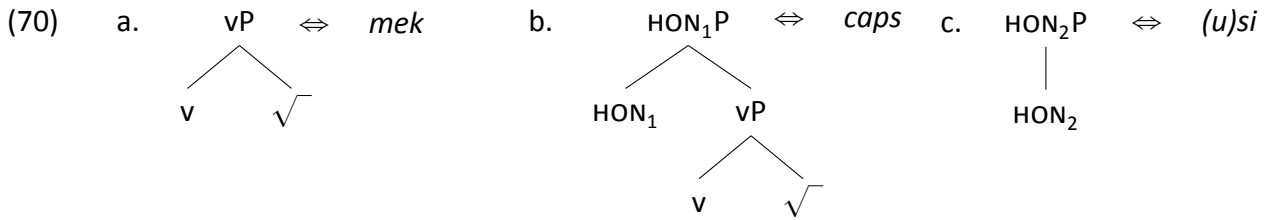
(68)	a.	$\sqrt{\text{EAT}}$	mek	‘eat’
	b.	$\sqrt{\text{EAT CAUS}}$	mek-i	‘let eat’
	c.	$\sqrt{\text{EAT HON}}$	caps -usi	‘eat’
	d.	$\sqrt{\text{EAT CAUS HON}}$	mek-i-si	‘let eat’

This is then a second case suggesting that the conditioning of root allomorphy needs to happen in a strictly local environment, and cannot be triggered across intervening heads like *HON* or *CAUS*.

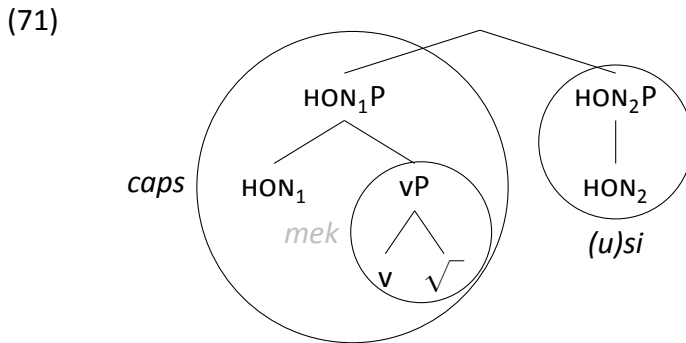
This obviously raises the question what is going on in (64), where *NEG* does seem to be able to trigger root suppletion across the intervening *HON*. The proposal advanced by Vanden Wyngaerd, De Clercq & Caha (2021) consists in a decomposition of honorification into two functional heads, as shown in (69).



We shall add negative and causative heads to this tree immediately, but for now we just look at honorification. We propose the lexical items for *mek* ‘eat’ in (70a), for the suppletive form *caps* in (70b), and the honorific marker *-(u)si* in (70c).



The suppletive honorific form *capsusi* is syntactically derived as in (71), with the circles indicating the different steps of phrasal lexicalisation.



After each merge step, at the phrasal level, different lexical items will be candidates for insertion. At the level of vP , the lexicon will be checked and *mek* in (70a) will be a candidate for insertion, and get inserted. When HON_1 is merged, the lexicon is checked again for a lexicalisation. The lexical item in (70b) will be a perfect match, and since it has matching conceptual information, it can be inserted and it will override the previous lexicalisation. When HON_2 is merged, there will first not be a lexicalisation, but then a lexicalisation-driven movement will be performed in accordance with the lexicalisation algorithm in (72) (modified after Starke 2018). This will move the complement of HON_2 , HON_1P (since there is no specifier) to seek compliance between the generated structure and the lexicon.¹³

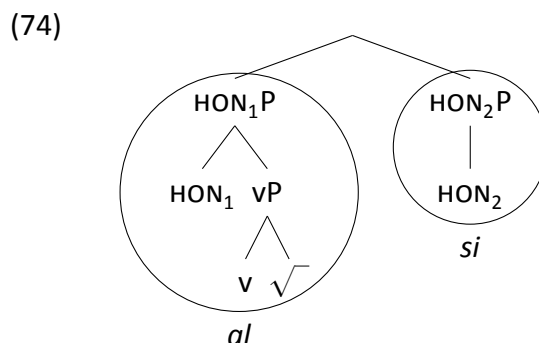
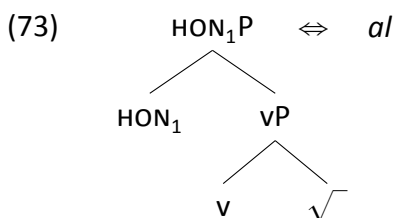
- (72)
- Merge-F and Lexicalise FP
 - If (a) fails, move the specifier of the complement of F, and retry (a)
 - If (b) fails, undo spec movement, move the complement of F, and retry (a)
 - If (c) also fails, attempt backtracking to the previous cycle and try next option for that cycle.
 - If merge-F has failed to spell out (even after backtracking), try to spawn a new derivation providing feature F and merge that with the current derivation, projecting feature F to the top node.

After complement movement, the right branch of (71) will be a match for the honorific suffix

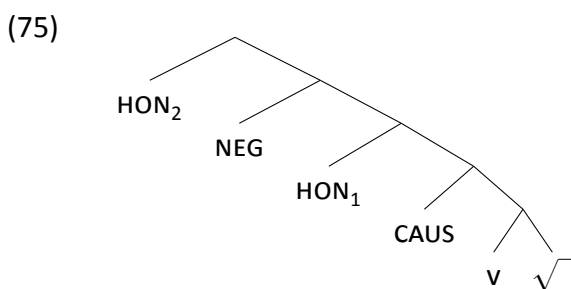
¹³See the new State-of-the-Art, i.e. De Clercq, Caha et al. (2025) in chapter 2 for more information on the lexicalisation algorithm, and for a recent update of the algorithm.

(70c), leading to the lexicalisation depicted in (71).

The lexical item for a regular verb like *al* ‘know’ will be as in (73). Since the root reaches up to HON₁, and there is no smaller root for the same concept, there will be no suppletion under honorification. The structure in (74) shows the syntax.

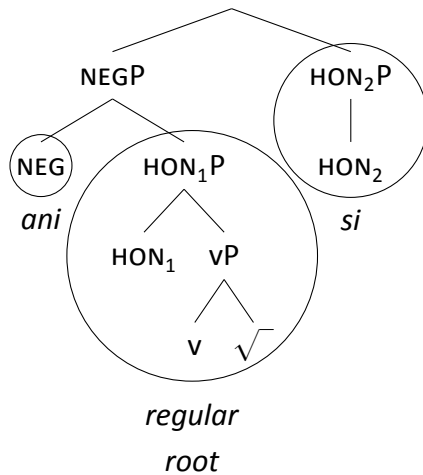


We now add the causative and negation to the structure. The causative is positioned closest to the root, accounting for the intervention effects that we showed the CAUS head triggered both with respect to honorific and negative suppletive roots. We put the negation in between the two honorific heads. As we will see below, this will allow us to capture the paradox.



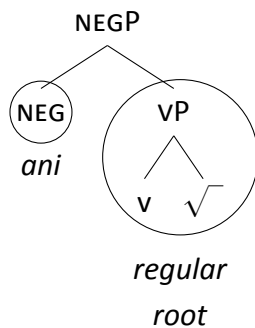
We are now almost in a position to explain the paradox that we presented in (62). Regular roots in Korean will be of the size HON₁, which means that they will combine with the regular negator and the regular honorification, as abstractly shown in (76).

(76)



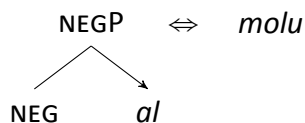
When honorification is not present in the structure, the same lexical items that contain HON₁ in their lexical structure can lexicalize syntactic structures without honorification thanks to the Superset Effect. This situation is depicted in (77).

(77)



Before we move on to continue the explanation as to how we derive the paradox, we present the lexical structure of *molu* 'not know' in (78).

(78)



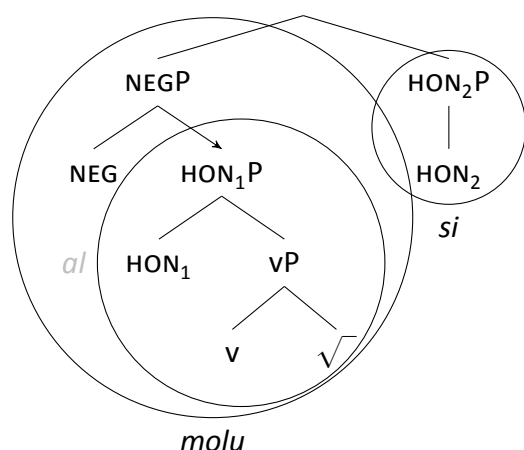
The lexical structure of *molu* 'not know' consists of negation and has a pointer to the lexical item for *al*- 'know'.¹⁴ This pointer is a means used in Nanosyntax to link two different lexical items with each other in the lexicon. The use of the pointer allows the lexical item that is pointed to

¹⁴Pointers in Nanosyntax are used to capture idioms (Starke 2011; 2014a), syncretisms in multidimensional paradigms (Caha & Pantcheva 2012; Vanden Wyngaerd 2018) and unproductive 'idiomatic' affixation (De Clercq & Vanden Wyngaerd 2019b). The mechanism of pointers can be somewhat compared to having a lexical sign in another sign's lexical entry to capture morphological dependencies, as in the HPSG analysis proposed for French suppletive verbal inflection by Bonami & Boyé (2002).

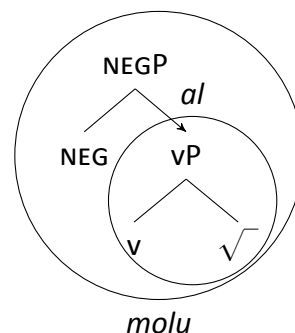
to be used in a subset of the environments for which the pointing lexical item is an identical match (i.e. in accordance with the matching condition).

Thanks to the way the lexical item in (78) is set up, *molu* can be inserted in the syntax that contains the projections for honorification (79a), and in a syntax that does not merge these projections (79b).

(79) a.

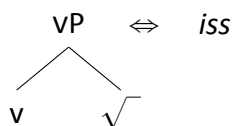


b.

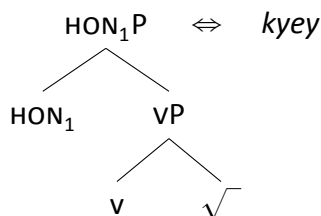


We now move to the next step in solving the paradox: the structure of the lexical items for EXIST. Crucially, there are three different conceptually related lexical items for EXIST in the lexicon: the nonsuppletive *iss* (80), the honorific suppletive root *kyey* (81), and the negative suppletive root *eps* (82).

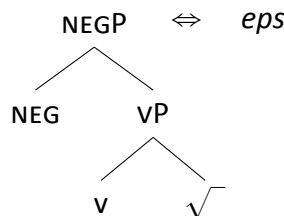
(80)



(81)

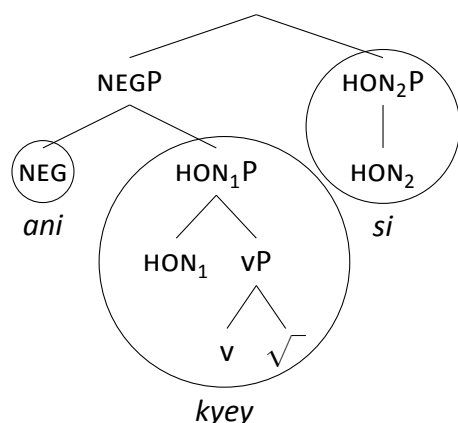


(82)



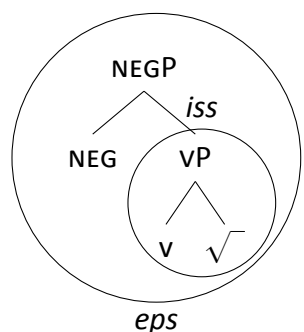
The structure in (83) shows that at the level of *vP*, (80) will be the perfect match. When honorific structure is built on top, (81) is the perfect match, and since (80) and (81) share conceptual information (81) can override (80). When at the next level *NEGP* is merged, the regular sentential negation will kick in, because the lexical item in (82) is not a candidate, allowing only lexicalisation of a structure without honorification. At the level of *HON2P* lexicalisation-driven movement will apply to find a lexicalisation for the feature *HON2*. This will result in the insertion of *-(u)si* in accordance with the lexical item in (70c).

(83)



When there is no honorification, *kyey* will not be a candidate for insertion and only *eps* or *iss* are possible, depending on whether negation is present or not.

(84)



Crucial in solving the paradox is the structure of the functional sequence on the one hand, and the structure of the two negative portmanteaus on the other hand: while *molu* ‘not know’ consists of a pointer to *al* ‘know’, hence allowing insertion of *molu* both in honorific and non-honorific environments, *eps* ‘not exist’ contains the structure of the root *iss* ‘exist’, but not the structure of *kyey* ‘be.HON’, disallowing it to occur in a honorific context. An important ingredient in solving the paradox has to do with how suppletive forms are handled in Nanosyntax, i.e. by decomposition. Suppletive forms are portmanteaus of hierarchically structured features that are lexicalized as one chunk. While these forms come across as a F-to-Mⁿ, once we cut open the form and look at its inside structure, it makes sense why the different meanings arise. What looks like one form at the surface underlyingly corresponds to different abstract meaning components, which explains the form-meaning mismatch.

1.2.2.3 Bambara

This section is based on unpublished work, that I presented at the Morphology as Syntax Workshop at UCLA in 2021 (De Clercq 2022), and of which some parts were discussed in De Clercq (2020a) (chapter 4 in Part II). De Clercq (2020a) also lies at the origin of my current ANR funded project *Sentential Negation Across the Globe (SNAG)*. *The finegrained structure of a linguistic*

universal. In the current section I discuss a prototypical case of fusion or polyexponence, i.e. TAM marking in an isolating language (**BickelNichols2013a**), Bambara (Mande). The approach taken here will be similar to the what we discussed before: the fusion we see in isolating languages can be explained by adopting the idea that one form phrasally lexicalizes multiple meaning components or syntactico-semantic features.

Bambara has two ways to mark perfective aspect: the suffixal marker *-ra* (which becomes *-la* after /r/ or /l/ and *-na* after /m,n/ or nasal vowel; Hewson 2016: 53) and the free morpheme *ye* (Dumestre 2003). Unergative (85a) and unaccusative verbs (85b) take *-ra*, while transitive verbs with NP complements take *ye* (86). Transitive verbs with PP-complements on the other hand also take *-ra* (87) (data from Koopman 1992: 559-560; Dumestre 2003).¹⁵

- (85) a. A kasi- **ra**.
s/he cry PFV
'S/he cried.'
- b. A taa- **ra**.
s/he go PFV
'She went.'
- (86) Den **ye** cè ye.
child PFV man see
'The child saw the man.'
- (87) N bó- **ra** i ye.
I visit PFV you at
'I visited you.'

The transitivity split disappears under the negative perfective marker *ma* (88), which replaces the perfective markers and expresses negation (Ermisch 2013).

- (88) a. Fali **ma** sa.
donkey PFV.NEG die
'The donkey has not died.'
- b. Birama **ma** liburu kalan.
Birama PFV.NEG book read
'Birama has not read the book.'

In the imperfective, there is no transitivity split either, and the single marker *bε* is used (89). The same is true when negation and IPFV are combined, i.e. there is a single marker *tε* (not illustrated below).

¹⁵Bambara is a tonal language, distinguishing between high and low tones. They are not indicated in the data presented here, but we refer the reader to Creissels (1988; 1989; 1992) for elaborate discussion. See also Dumestre (2003: ch1)

- (89) Seku **bɛ** boli / tiga sɛnɛ
 Sékou IPFV run / nut cultivate
 ‘Sékou runs / cultivates groundnuts.’ (based on Creissels 2007)

The two negators, *tɛ* and *ma*, cannot be stacked in the same sentence (90) (p.c. Dieydi Sylla), which crucially underscores the fact that they are in complementary distribution, and strongly suggests that they have the same scope, which I take to be scope over the tensed predicate (cf. De Clercq 2020b).

- (90) *Birama **ma** **tɛ** liburu kalan.
 Birama PFV.NEG IPFV.NEG book read

The aspectual markers cannot be stacked on each other either (91), and neither can the negators (*tɛ*/*ma*) be stacked on any of the present tense TAM-markers (*bɛ*/*ye*) (92) (p.c. Dieydi Sylla).

- (91) a. *Sanu **bɛ** **ye** baara kE
 Sanu IPFV PFV work do
 b. *An **bɛ** taa-**ra** sugu la.
 1PL IPFV go PFV market to
- (92) a. *Sanu **tɛ** **bɛ** baara kE
 Sanu IPFV.NEG IPFV work do
 b. *Sanu **bɛ** **tɛ** baara kE
 Sanu IPFV IPFV.NEG work do
 c. *Birama **ma** **bɛ** liburu kalan.
 Birama PFV.NEG IPFV book read
 d. *Birama **ma** **ye** liburu kalan.
 Birama PFV.NEG PFV book read
 e. *Birama **ye** **ma** liburu kalan.
 Birama PFV PFV.NEG book read

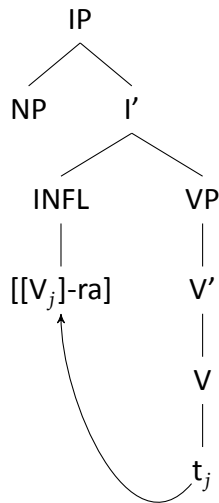
The stacking data show that all these markers are portmanteaus lexicalizing at least present tense, aspect and sometimes also negation. The data can be summarized as in (93).¹⁶

	PFV	NEG+PFV	IPFV	NEG+IPFV
INTRANS	-ra	ma	bɛ	tɛ
TRANS	ye	ma	bɛ	tɛ

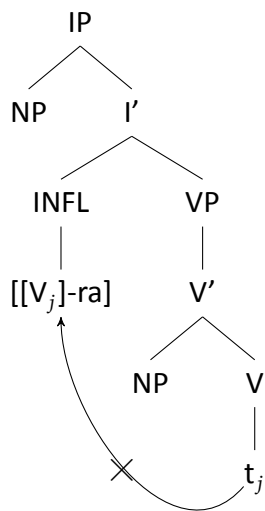
¹⁶We abstract away in this section from the past and future tenses, which come with additional morphemes for tense (see Dumestre 2003; De Clercq 2020a). We also do not discuss the fact that *tɛ* and *bɛ* can be used in different types of copular constructions, where they are syncretic with the imperfective marker. We hope to come back to this issue in future work, but for now, we refer the reader to Dumestre (2003: ch3) for discussion.

Koopman (1992) discusses the distribution of the perfective marking in Bambara and explains the data in (85)-(86) as a consequence of the Case Filter, and the absence of Case chains in Bambara, i.e. verbal traces cannot assign case, so that movement of the verb would leave the object unlicensed. The idea is that when the verb has no internal argument to assign case to, it can move to INFL and pick up *-ra*. The tree structures in (94) for intransitive structures, and (95)-(96) for transitive structures, based on Koopman (1992: 563), illustrate this.

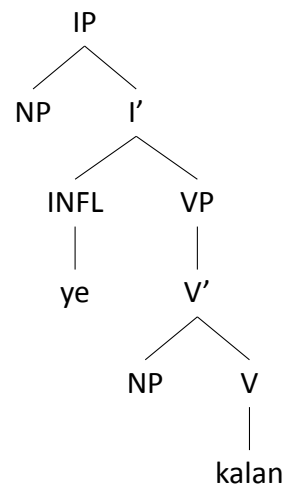
(94)



(95)



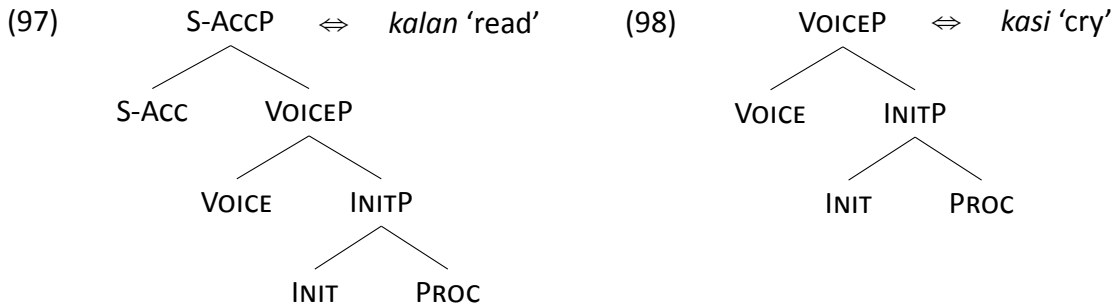
(96)



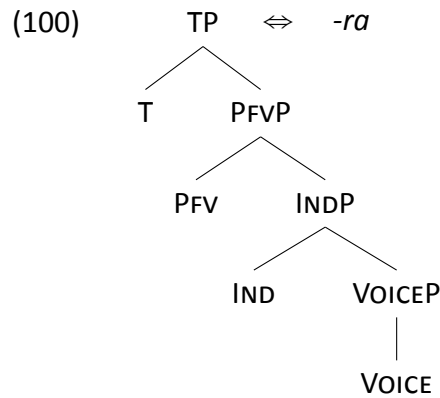
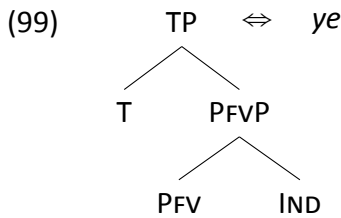
Verbs that do not need to assign structural accusative case raise to INFL, where they attach to the suffix *-ra* (94). Transitive verbs need to stay in situ to assign structural case to the object, with *ye* popping up in INFL as a dummy element (96). However, Koopman's account does not explain why there is neutralisation of the transitive-intransitive distinction under negation, and in the imperfective. If verbs that do not need to assign case can indeed raise to INFL, it remains unexplained why they do not do so with intransitive verbs under negation (88a).

A big part of the solution to this puzzle lies again in the structure of the lexicon in Bambara, and more in particular in the feature structure of the TAM markers and negative markers. In what follows we will focus on trying to solve the transitive-intransitive split in the perfective and the neutralisation under negation from this perspective.

A first crucial ingredient is that verbs in Bambara are either endowed with a S-Acc (Caha 2009) feature or not. The feature S-Acc captures one of the functions that are usually assigned to vP. When verbal roots are endowed with it, they select Accusatively marked objects, as illustrated for the verb *kalan* ‘read’ in (97). When they do not have this feature in their lexical structure, they are intransitive, as illustrated for the verb *kasi* ‘cry’ in (98). In addition to S-Acc, we assume that verbs are also specified for VOICE properties in Bambara. For now we keep it simple by adopting just one feature, but we assume that Voice will eventually also need to be decomposed. Finally, we adopt the Ramchandian structure to capture the argument structure of the verb, as also adopted in section 1.2.1, so verbal roots can be decomposed into RES, PROC and INIT depending on the lexical aspect they give rise to.

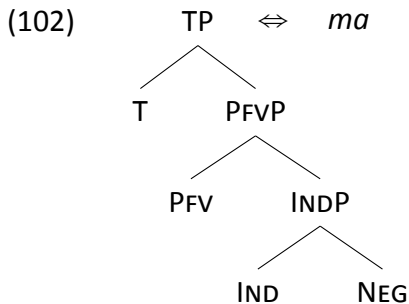


On the basis of the fact that the markers indicated in blue in (93) are insensitive to the transitive-intransitive split and precede the lexical verb, we deduce that they cannot realise S-Acc and realise features that sit higher than this feature in the structure. But let us first focus on the transitivity split in the perfective. For the transitive perfective marker *ye* we know that it is used in the indicative, hence we adopt the features IND (see Starke & Cortiula 2021; Cortiula 2023), PFV for perfective viewpoint aspect, and T for default tense. Support for this latter feature comes from the fact that *ye* is in complementary distribution with the marker *be*, which is used in present tense habituals. The lexical structure for *ye* is in (99). For the intransitive perfective marker *-ra*, we do not only need to make sure that it combines with intransitive verbs, but also that the marker *ye*, used with transitive verbs, cannot kick in when lexicalising the structure of intransitive verbs. The way to do that is to make sure that *-ra* lexicalizes a feature that sits below S-Acc in the functional sequence. We propose that this feature is Voice, as in the lexical entry in (100). This entry encodes the fact that *-ra* only occurs with intransitive verbs, i.e. verbs lacking the S-Acc feature between VOICE and PFV. A supporting argument for this is that *-ra* not only occurs with intransitives, but also in passive sentences, as in (101a) (Koopman 1992: 560).



- (101)
- a. Ji min- **na** sisan (den fè).
water drink PFV now child by
'The water has been drunk now (by the child).'
 - b. *Ji **ye** min sisan (den fè).
water PFV drink now child by

The negative perfective marker *ma* is insensitive to transitivity. It has a lexical structure that, just like *ye*, has its foot higher than the transitivity-related feature S-Acc. We assume that this foot consists of the features NEG and [IND], as in (102).

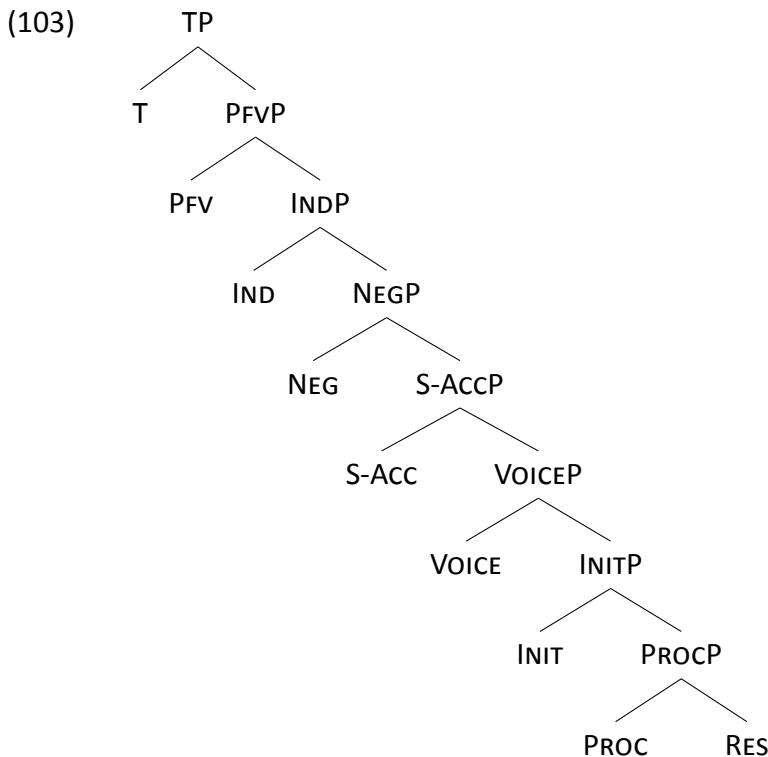


There is no strong argument to be made for the exact position of sentence negation in Bambara with respect to other TAM-related features. The reason for this is that all negators are fusional. For now we assume that there is a position for negation just above the vP domain, below IND, but nothing crucially hinges on where the negation exactly sits. In addition to NEG and IND, *ma* also lexicalizes PFV, as well as default Tense, T. This follows from the fact that the negator is incompatible with the marker used in present tense habituals or with any of the other TAM-related markers, as illustrated in (92) above.

Something we have not yet made explicit is that there is a crucial difference with respect to the bottom of the lexical structure of the markers in blue in table (93) and the one in red, i.e. *-ra*. *-ra* has a unary bottom, (100), while the other markers all have a binary bottom. Starke (2018) argues that the unary bottom of a suffix like *-ra* makes it a match for a syntactic structure in which the root, in this case the lexical verb, has moved to the left out of VOICEP in order to lexicalise the structure. The other markers are a consequence of first merge. They are created

in a separate workspace from the main derivation, and subsequently merged as (projecting) specifiers, in a position preceding the root. The consequence of this proposal by Starke (2018) is that the distinction between elements that precede, so called PRE elements, and elements that follow, so called POST elements, is lexically encoded.

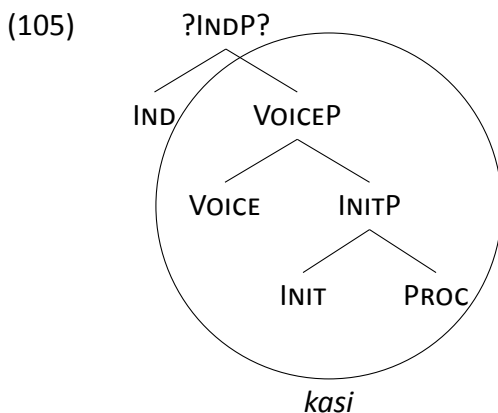
The complete functional sequence, of which the lexical items discussed above lexicalise different portions, is given in (103). Not all features are always required to be there: if imperfective is aimed for, the PFV feature is not going to be merged. If an intransitive structure is going to be realised, then S-ACC will not be merged. If an ongoing event is going to be realised, then there is not going to be [RES], and if the sentence is affirmative, then there will be no [NEG].



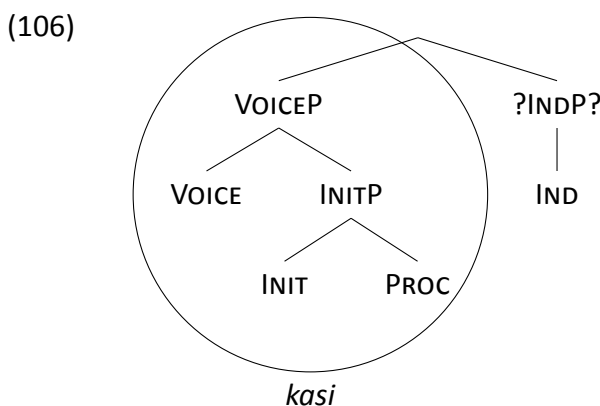
With all of this in place, we can proceed to the account of the transitive/intransitive distinction in the perfective, and its neutralisation under negation. We will first derive the split in transitivity in the perfective. The lexical items are set up to derive it and the crucial lexical element is *-ra*. Unlike the other TAM-markers, it is suffixal and it has its foot feature in VOICE, a feature below the transitivity feature S-ACC. In other words, this suffix eats into the lexical structure of Bambara verbs. Crucially, it will only be able to lexicalise intransitive verbs. The reason for this has to do with the lexicalisation algorithm in Nanosyntax, repeated in (104) (from (72) in section 1.2.2.2). It orders the different rescue operations that syntax resorts to if there is no immediate lexicalisation after a feature has been merged. These rescue movements are ranked from least costly to most costly.

- (104)
- Merge-F and Spell out FP
 - If (a) fails, move the spec of the complement of F, and retry (a)
 - If (b) fails, undo spec movement, move the complement of F, and retry (a)
 - If (c) also fails, attempt backtracking to the previous cycle and try next option for that cycle.
 - If merge-F has failed to spell out (even after backtracking), try to spawn a new derivation providing feature F and merge that with the current derivation, projecting feature F to the top node.

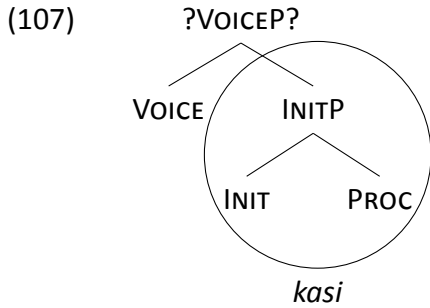
Up until now we did not really make use of the algorithm much, but the Bambara data point us to the fact that the order actually matters to derive particular facts. Let me explain why. If we are merging the structure of an intransitive predicate like *kasi* 'cry', and we come to the point where syntax merges an indicative feature IND, as shown in (110), then the lexical structure of a verb like *kasi* (given in (98) above), will no longer be able to lexicalise the structure.



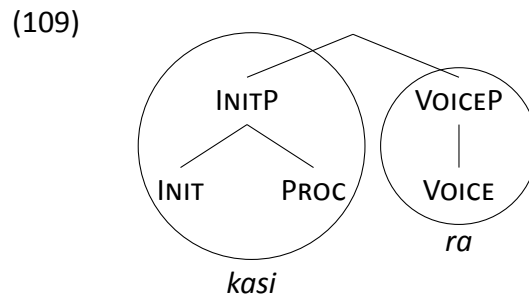
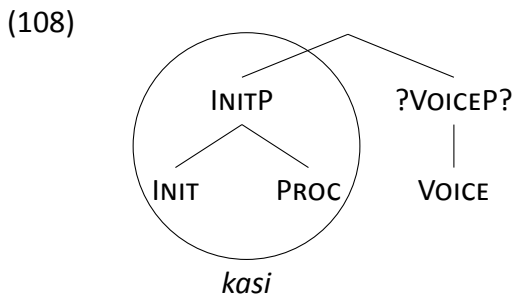
First, complement movement will be tried (since there is no spec available, step (104b) will be skipped), but this will also not lead to a lexicalisation, since there is no lexical item in the lexicon of Bambara that has a unary foot in IND that can lexicalise the feature on the right in branch in (106).



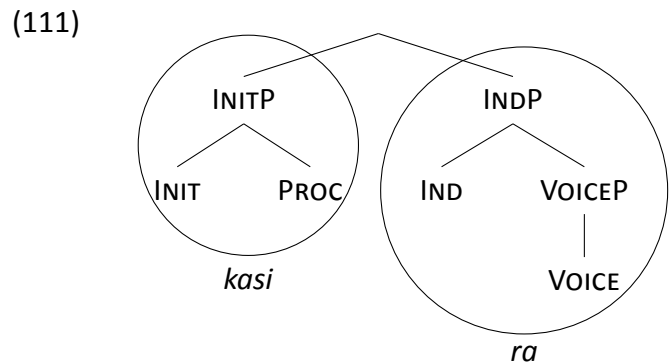
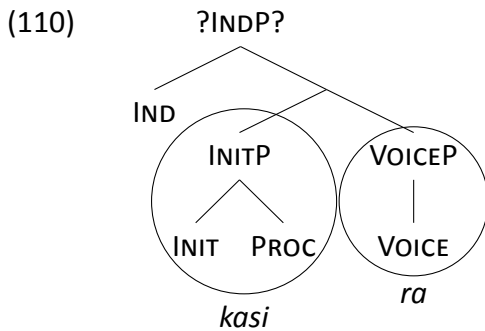
The next option in the algorithm is to backtrack (104d). This means that we go back to the previous cycle, i.e. to the point before VOICEP was lexicalized, as in (107).



At this point, syntax tries the next option for that cycle, i.e. move the complement, instead of immediate lexicalisation with the verb, resulting in the structure in (108). At this point, VOICEP can get a lexicalisation, thanks to the lexical structure of *-ra* in (100), which has VOICE as its bottom feature, resulting in the lexicalisation of the right branch with *ra*, as in (109), and leading to the suffixal order we wanted to derive for the perfective marker *-ra*.

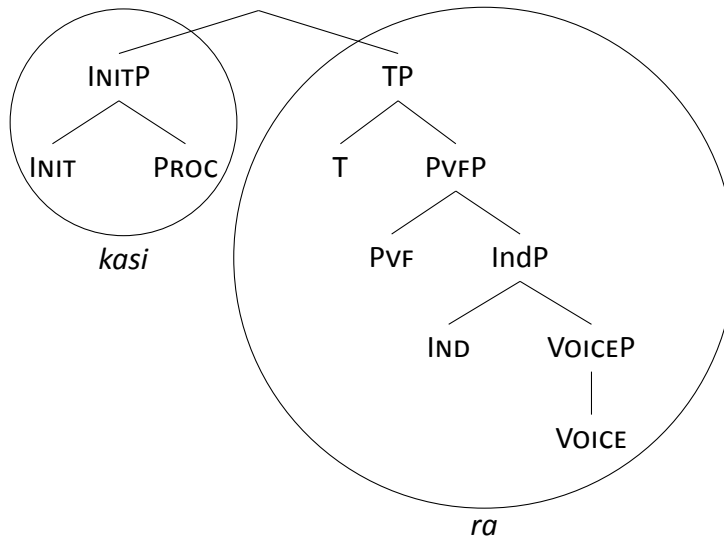


From here the structure will grow in its usual way, merging next the feature IND as in (110) and aiming for lexicalisation, which will be obtained after specifier movement, in accordance with the algorithm, and as illustrated in (111).



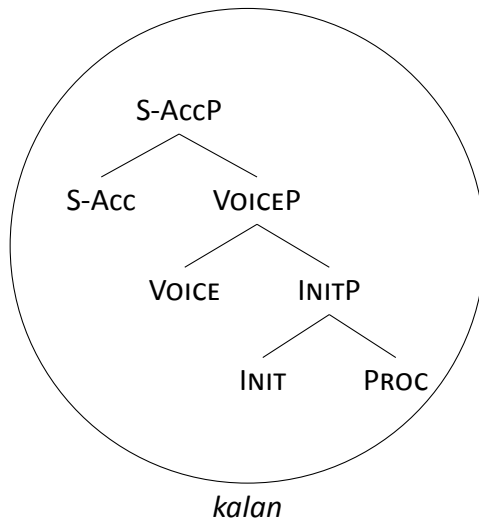
The same steps will apply after the merger of PFV and T, resulting in the structure in (112).

(112)



Next we turn to the derivation of a perfective transitive verb. The root first grows monotonously to the size of S-AccP, which can be lexicalised by the transitive verb *kalan* ‘read’.

(113)

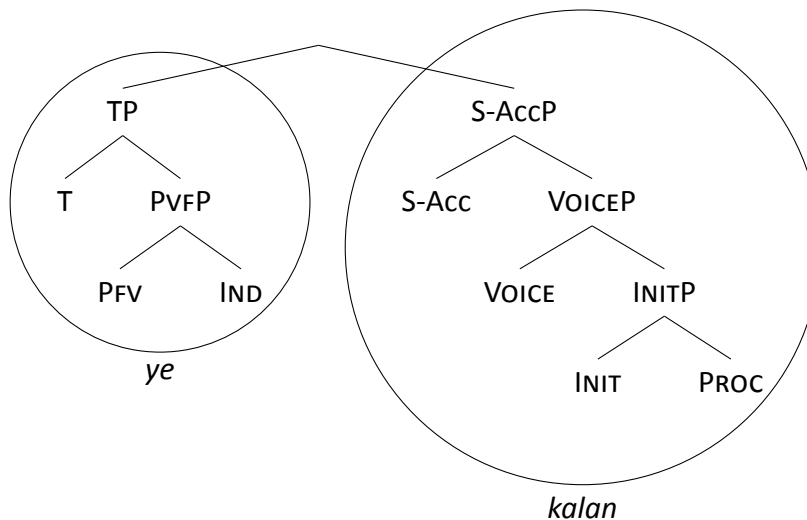


Then syntax merges IND, but at this point there will not be an immediate lexicalisation. Evacuation of the complement of IND will deliver a right branch [_{INDP} IND], but since this has a unary bottom, it does not match any lexical entry in our Bambara lexicon (recall from (99) above that *ye*, even though its lowest feature is IND, has a binary bottom). Backtracking will be tried, back to the previous cycle where S-Acc was merged, but trying other options for that cycle will not lead to any successful lexicalisation, because there is no lexical item with a unary bottom in S-Acc, which would be a match for the right branch created after lexicalisation-driven movement. At this point, backtracking goes back another cycle, to VOICE, and even further back if that is also unsuccessful (see Starke 2018; Caha 2019a; Janků 2022 for discussion and examples of derivations that backtrack over multiple cycles). These multiple cycles of backtracking, however, will only create lexical items with a unary bottom, but this will not help with the lexicalisation

of IND, since the only lexical entry that contains IND and that has a unary bottom is *ra* (100), but since (100) does not contain S-ACC, it cannot be a match in this case.

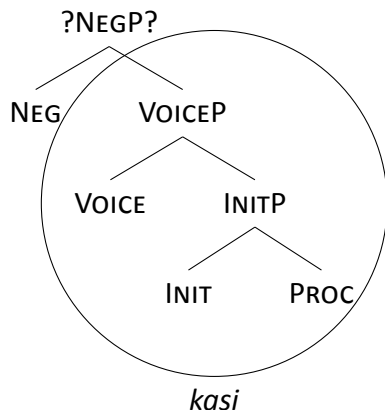
In order not to unduly complicate the discussion, I shall not go through all the cycles of backtracking, but take it as a given without further demonstration that these multiple cycles of backtracking do not result in a successful lexicalisation of IND. Consequently, all the backtracking is undone again, and the last step in the algorithm is tried: complex left branch formation. A new workspace with new derivation is created, in which IND will be merged with the next feature in the sequence, i.e. PVF. This structure will match a constituent in the lexical entry of *ye* in (99). After the subsequent merger of T and integration of the complex left branch into the main spine, we get (114).

(114)



We next turn to the transitivity neutralisation under the negative perfective *ma*, starting off with the intransitive verb *kasi* 'cry'. The derivation proceeds to the point where the verb has been lexicalized and NEG is merged, and needs to get lexicalised:

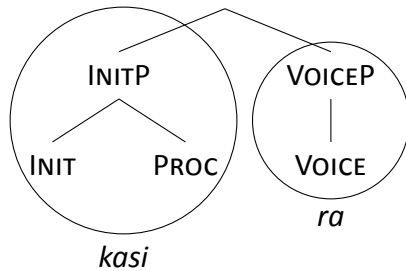
(115)



There is no lexical item in the lexicon of Bambara that can directly lexicalize the structure in (115), nor is there any after movement of the complement of NEG, since that creates a constituent [_{NEGP} NEG], with a unary foot, and there are no lexical items that have a unary foot NEG.

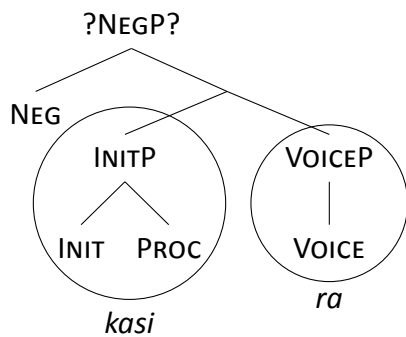
As a result, backtracking will be tried, and an other option will be tried for VOICE. Like before, this will lead to the structure in (109), repeated here as (116), resulting in the lexicalisation of VoiceP as *-ra*.

(116)



Even though there is a lexicalisation for VOICEP at this stage, this will not help for the lexicalisation of NEG in the next cycle:

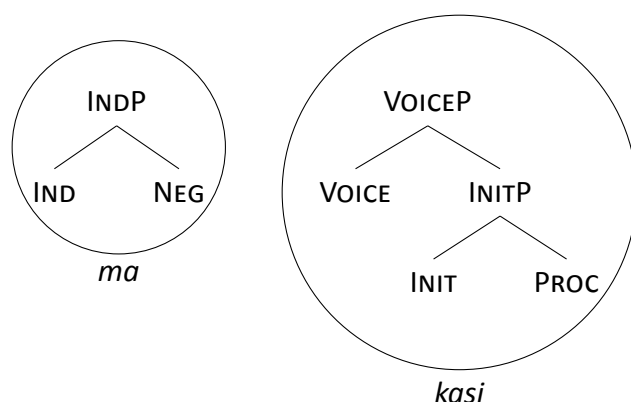
(117)



The reason is the same as before, namely that the Bambara lexicon has no lexical entries with NEG that have a unary foot. If we move the Spec INITP, we get a right branch with a unary foot in VOICE, but no negative marker has a unary foot. If we move the entire complement, the right branch has a unary foot in NEG, and the same problem arises. This problem persists across multiple cycles of backtracking, so that eventually all backtracking is undone again, and the derivation goes back to the initial state before backtracking, i.e. (115) above.

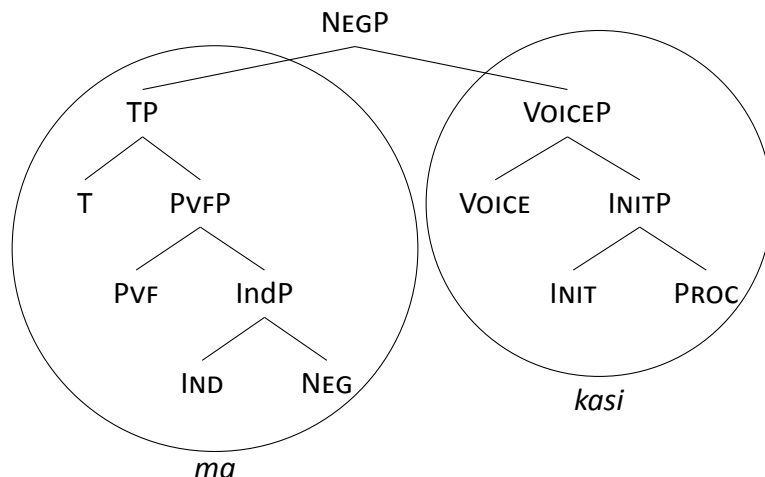
Backtracking having been unsuccessful, the derivation now moves to the final and most costly step in the algorithm, the creating of a complex left branch in a new workspace, using the feature NEG that needs lexicalisation in the current cycle. However, since complex left branches are new derivations, and not the result of movement, the bottom of this structure will be binary. Hence, NEG will combine with the next feature in line, i.e. IND. This will lead to a lexicalisation, as shown in (118).

(118)



At this point there are two options. Either the derivation will continue in the new workspace for as long as possible (see Starke 2018; De Clercq 2020b), or the complex left branch is closed again, and integrated into the main workspace, after which merge continues in the main spine (cf. Caha, De Clercq & Vanden Wyngaerd (2019) in chapter 6). The latter option may ultimately lead to reopening the complex left branch again and adding the new feature to the derivation of the complex left branch, a system which Caha (2019a) and Bergsma (2021) refer to as Multiple Merge. In both approaches, eventually the full structure of *ma* will be derived after the features PVF and T have been added to the complex left branch, as in (119).

(119)



Since the bottom feature of the complex left branch was NEG, we propose that NEG is the feature that is going to project in the main sequence. However, nothing crucial hinges on this at this point.¹⁷

The derivation of a transitive verb works analogously, except that the right branch grows

¹⁷One could also propose that all features in the CLB that have not yet been projected in the main spine, need to be merged in the main spine after the CLB has projected in the main spine, and the complex left branch gets attracted to the different specifiers thanks to an additional step in the lexicalisation algorithm, which includes feature-driven movement. This option has been explored in De Clercq (2019; 2020b); De Clercq & Reintges (2022), but we will not further dwell on this issue here.

until it reaches size S-AccP (see the discussion following (113) above). Then, NEG is merged, yielding a tree like (115) above, except that it adds S-AccP between NEG and VOICE. The derivation otherwise proceeds identically, thus accounting for the transitivity neutralisation under negation.

What the discussion of Bambara showed us is that prototypical cases of polyexponence can be well-captured in the system thanks to phrasal lexicalisation. As with the other *F-to-Mⁿ* mismatches, phrasal lexicalisation allows exponents to be stored in the lexicon with multiple syntactico-semantic features, accounting for why the same form can lexicalise these different meanings. The exploration into Bambara also showed us how differences between suffixes and prefixes can be captured in Nanosyntax by differentiating between them in terms of structural properties of the lexical items involved.

1.3 0-to-M

In section 1.2 we discussed syncretism, as one of the several *F-to-Mⁿ* mismatches dealt with in that section. One of the consequences of how we deal with syncretism in Nanosyntax is that syncretism does not only represent a *F-to-Mⁿ* mismatch, but also involves a *No Form-to-Meaning* mismatch, a *0-to-M*. More concretely, we looked at conversion of deadjectival verbs, and observed that a single form like *narrow* may express adjectival, inchoative or causative meanings. This phenomenon can be understood by adopting the idea that these forms are stored in the lexicon with multiple hierarchically organized syntactico-semantic features that can undergo phrasal lexicalization. In contrast, were one not to adopt phrasal lexicalisation, one would need to assume that the underlying features that give rise to these different readings are realized by zero morphemes (cf. the discussion in section 1.2.1). In other words, the mismatch *F-to-Mⁿ*, as far as it arises as a consequence of syncretism, can be argued to coincide with the mismatch *0-to-M*.

In this section, I will zoom in on yet another example, from an empirical domain that is known for having 0-to-M: the domain of imperatives (DFG Research Training Group 2021), and more in particular, imperatives in Latin. This work is relatively new, but a short paper has been published (Demonie & Clercq 2023) and is added to this volume in chapter 7 of Part II.

If we look at the formation of the present imperative in Classical Latin, then one can be struck straight away by the fact that 2SG.IMP does not make use of any additional morphology beyond the stem and the theme vowel. This is shown in Table 1.3, which shows the second person singular and plural of the imperative, indicative, and subjunctive in the four different verb classes of Latin.

The morphological poverty of the Latin 2SG.IMP is not rare from a typological point of view, since imperatives are known to be morphologically defective or poor (Zhang 1990). Nevertheless, imperatives have a very specific meaning, i.e. they put a requirement on the addressee

Table 1.3: Latin second person verbs in four verb classes

	am-ā-re	mon-ē-re	tēg-ē-re	aud-ī-re	cap-ē-re
2SG.IMP	am-ā	mon-ē	teg-ē	aud-ī	cap-ē
2PL.IMP	am-ā-te	mon-ē-te	teg-i-te	aud-ī-te	cap-i-te
2SG.IND.PRS	am-ā-s	mon-ē-s	teg-i-s	aud-ī-s	cap-i-s
2PL.IND.PRS	am-ā-tis	mon-ē-tis	teg-i-tis	aud-ī-tis	cap-i-tis
2SG.SUBJ.PRS	am-ē-s	mon-e-ā-s	teg-ā-s	aud-i-ā-s	cap-i-ā-s
2PL.SUBJ.PRS	am-ē-tis	mon-ē-ā-tis	teg-ā-tis	aud-i-ā-tis	cap-i-ā-tis

and hence can be regarded as a property that is restricted to the addressee (Portner 2005; Zanuttini 2008). Even though they are mainly used for commands (Kaufmann 2011), they can also be used for a variety of other speech acts, as illustrated in (120) (Isac 2015: 5).

- (120) a. Close the door! Open the window! (order/command)
 b. Stay away from me! (warning)
 c. Take the train if it's cheaper! (advice)
 d. Go to hell! (curse)
 e. Close the door, please! (request)

The combination of the poor morphology, and the dedicated meaning(s), seem to point to a mismatch: how is it possible that a requirement can be put on the addressee in the absence of any dedicated morphology? In what follows we have a look at Latin and argue that there is reason to assume that the 2SG.IMP is a portmanteau, lexicalising not only the relevant features needed to do aspectual and temporal work in the indicative and subjunctive, but also (at least) an addressee layer (Isac 2015; Zanuttini 2008), allowing it to give rise to imperative meaning.

The reason to believe that this is the case comes from what looks like ϕ endings in the 2PL.IMP. This form makes use of a dedicated ending *-te*, which we cannot see in any of the other 2PL forms in the verbal paradigm. This is shown in Table 1.4, which shows the second person endings in five tenses and two moods (indicative and subjunctive) for the verbs of the first conjugation (*amare* 'love').

Table 1.4: Latin second person verbs in five tenses and two moods

	PST.IND	FUT.IND	PFV.IND	PLUPFV.IND	FUT.PFV
2SG	amā-bā-s	amā-bi-s	amā-v-is-tī	amā-v-erā-s	amā-v-eri-s
2PL	amā-bā-tis	amā-bi-tis	amā-v-is-tis	amā-v-erā-tis	amā-v-eri-tis
	PST.SBJV		PFV.SBJV	PLUPFV.SBJV	
2SG	amā-rē-s		amā-v-eri-s	amā-vi-ssē-s	
2PL	amā-rē-tis		amā-v-eri-tis	amā-vi-ssē-tis	

Since there is a dedicated ending for 2PL imperative, we assume that the 2SG meaning is also present, but that it has no overt marking. If this assumption is correct, then the empirical picture looks as in (121).

(121)		PRS.IMP
	2SG	am-a- $\emptyset$
	2PL	am-a- <i>te</i>

Later on, we shall modify this idea and argue that there is underlyingly some feature present that in combination with the plural gives rise to *-te*, but without the plural can be lexicalized together with the theme vowel.

There is one mood, not often taken up in representations of the verbal paradigm of Latin, where we also see *-te* in 2PL: the future imperative (henceforth FUT.IMP). The future imperative was mainly used in Early Latin in plays, didactic texts and legal statutes. In this period, the future imperative indicated a time after the utterance time, and functioned a bit like a deferred imperative (Wurf 2007; Aikhenvald 2010; Schwager 2011; Banerjee & Kaur 2022). In Classical Latin, the future imperative is still attested in similar text types, especially in legal texts (cf. Decorte 2016), but it seems to have lost its temporal meaning. Characteristic marking of the FUT.IMP is with the suffix *-tō*, which has been identified as a tense marker (Pinkster 2015: 515).¹⁸

The FUT.IMP form cannot only be used in 2SG and 2PL (122), but also in 3SG and 3PL (123). The 2SG form has the ending *-tō* without any other marking, as illustrated with an example sentence in (122a), while in the 2PL *-tō* is used in combination with the plural ending that we know from PRS.IMP, i.e. *-te* (122b). In the 3P forms, 3SG has the same *-tō* ending, but 3PL has *-ntō*.

- (122) a. Si in Formiano non erimus, si nos amas, in Pompeianum
if in Formiae not be.IND.FUT.1PL if us love.IND.PRS.2SG to Pompeii
veni-to
come-FUT.IMP.2SG
'If we won't be in Formiae, please, come to Pompeii!' (Cic.Att.24.6)
- b. Vos huc decima **veni-to-te**!
you here at-four come-FUT.IMP-2PL
'Come here at four!' (Rh.Her.4.64.LI)
- (123) a. Uxor mea heres <ne> **es-to** ...
wife my heir NEG be.IMP.FUT.3SG
'Let my wife not be my heir ...' (Sen. Con. 2.7.9)

¹⁸It has been argued that this morpheme can be traced back to an old ablative of a demonstrative pronoun **-tod*, adding the meaning "from that point on" (cf. Barrios-Lech 2017, de Melo 2007, Fortson IV 2011).

- b. Primum verbi genus hoc “**conserva-nto**,” quo magis in legibus quam firstly of-word form this preserve.IMP.FUT.3PL which rather in laws than in foederibus uti solemus, imperantis est, non in treaties use.INF be-used-to.IND.PRS.1PL order.PART.PRS AUX.3SG NEG precantis.
ask.PART.PRS
‘In the first place, this form of words, ‘Let them uphold,’ which we are in the habit of using in laws rather than in treaties, implies a command, not an entreaty.’ (Cic, Pro Balbo, 36)

We can summarize the PRES.IMP and FUT.IMP for the Latin verb class *am-ā-re* ‘love’ as in (124).

(124)

	PRS.IMP	FUT.IMP
2SG	am-ā	am-ā-tō
2PL	am-ā-te	am-ā-tō-te
3SG		am-ā-tō
3PL		am-a-n-tō

When we take a look at the rest of the verbal paradigm in Latin in Table 1.5, which lists the third person forms of the first conjugation, we see that the regular ϕ -morphology for 3PL is *-nt* across the different tenses.

Table 1.5: Latin third person verbs in six tenses and two moods

	PRS.IND	PST.IND	FUT.IND	PFV.IND	PLUPFV.IND	FUT.PFV
3SG	ama-t	amā-ba-t	amā-bi-t	amā-v-i-t	amā-v-era-t	amā-v-eri-t
3PL	ama-nt	amā-ba-nt	amā-bu-nt	amā-v-ēru-nt	amā-v-era-nt	amā-v-eri-nt
	PRES.SBJV	PST.SBJV		PRF.SBJV	PLUPRF.SBJV	
3SG	am-e-t	amā-re-t		amā-v-eri-t	amā-vi-sse-t	
3PL	am-e-nt	amā-re-nt		amā-v-eru-nt	amā-vi-sse-nt	

It thus seems fairly straightforward that the 3PL.FUT.IMP *-ntō* can be argued to consist of the regular ϕ -morphology for 3PL *-nt*, followed by the marker of the FUT.IMP, i.e. *-tō*. The attested form *-ntō* can then be considered a consequence of degemination. Along the same lines we could also argue that 3SG underlyingly has the regular ϕ -marking *t* present, just like we see in the other tenses of the verbal paradigm. The suggested decomposition for the FUT.IMP endings is therefore as in (125), and finds support in Szexnerényi (1953).

- (125)
- | | | |
|-----------|---|------------|
| am-a-tō | < | am-a-t-tō |
| am-a-n-tō | < | am-a-nt-tō |

One could wonder whether there is any reason to assume that the ϕ -endings *-s* (and *-tis*) of 2P are also underlyingly present in the 2P.FUT.IMP forms, i.e. if *ama-tō* could reduce to *ama-s-tō*. However, even though there is an *s*-deletion rule in Latin (Cser 2016), it only applies when *-s* is followed by a voiced obstruent, which is not the case in the case of *-tō*. Moreover, even if one were to imagine that *-s* could get deleted, there is no way that *-tis* could get deleted by some phonological rule.

Taking everything until now into account, we arrive at the empirical picture in (126) for the FUT.IMP, which shows that the regular ϕ -endings in the 3P can be distinguished (marked in orange), which precede *-tō*, while the dedicated marker of the 2P *-te* follows *-tō* in 2PL.

(126)		FUT.IMP
	2SG	am-a-tō
	2PL	am-a-tō-te
	3SG	am-a- t -tō
	3PL	am-a- nt -tō

This means that we can now put these different endings in a sequence of imperative endings, (127). This sequence leads to a renewed insight into the regular endings that we see appearing in all different tenses (*t/nt*), and which also seem to be part of the underlying structure of the FUT.IMP, and the dedicated endings of 2PL.PRS.IMP.

(127) **-n(t)** < tō < te

The fact that these endings appear in different positions with respect to the FUT.IMP morpheme *-tō* strongly suggests that we cannot equate them. So the question is: how do they differ? We propose that what we see preceding *-tō* is the regular subject agreement morphology, the so called ϕ -features, while what we see following *-tō* is a realisation of the Addressee. In run-of-the-mill second person imperatives, the Subject and the Addressee are the same. However, 3P.FUT.IMP is used in situations in which the Addressee and the Subject are different. This explains why both subject agreement morphology and Addressee related morphology can be activated. A prototypical situation to imagine this type of imperative is one in which a parent tells a babysitter to ‘Make sure that they go to bed at 8pm!’, with *they* referring to the children the babysitter is going to take care of. In this case, the Addressee is the babysitter, and the subject morphology is related to the children. The upshot of the sequence in (127) is that 2P.IMP shows a gap in the structure: there are no ϕ morphemes (nor its underlying features) present in the PRS.IMP (see also Zanuttini (1991; 1994; 1997), Romanello & Repetti 2014, Isac 2015 for a similar claim related to the structural deficiency in imperatives). This gap corresponds to the morphemes marked in orange in (126). However, while the ϕ -features are missing in 2P, a Speech Act layer does get activated, and more in particular a layer for the Addressee (Ross 1970; Speas & Tenny 2003; Haegeman & Hill 2013).

There is a remaining issue with respect to the empirical picture sketched for FUT.IMP in (126). Since we see the dedicated Addressee morphology *-te* in 2PL, it seems quite straightforward to also adopt it as a zero in the 2SG, as we did for the 2SG.PRS.IMP. Can we do the same for 3P.FUT.IMP, i.e. can we assume that the Addressee layer gets activated and lexicalised by *-tō*? And why do we not get any forms with *-te* in 3P.FUT.IMP? We propose that, since this type of future imperative was usually used in legal texts in which the ADDRESSEE was non-specific, the default form, i.e. the singular, got used. As such, parallel with our proposal for 2SG.PRS.IMP, we adopt the idea that in 3P.FUT.IMP forms the Addressee feature is also underlyingly present, but zero-marked. In other words, the picture can be updated to (128), now sketching the situation for both PRS.IMP and FUT.IMP.

(128)

	PRES.IMP	FUT.IMP
2SG	am-ā- $\emptyset$	am-ā-tō- $\emptyset$
2PL	am-ā-te	am-ā-tō-te
3SG		am-ā-t-tō- $\emptyset$
3PL		am-a-nt-tō- $\emptyset$

We now have a rather full picture of what is going on with the morphology of Latin imperatives. While we will not go into all derivational details, we want to come back to the core point that we wanted to make with respect to the theme of this chapter. In spite of the fact that there is no dedicated morphology for the expression of 2SG.PRS.IMP, i.e. there is no form, we can argue, based on a comparison with the dedicated ending *-te* of 2PL.PRS.IMP, and the position of this *-te* in comparison to the underlying ϕ -morphology that we see in the FUT.IMP, that there is at least an underlying Addressee feature present, doing justice to the meaning that imperatives give rise to. This Addressee feature is in fact packed in the structure of the theme vowel in the present imperative, as we shall see below.

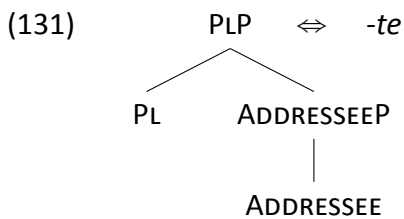
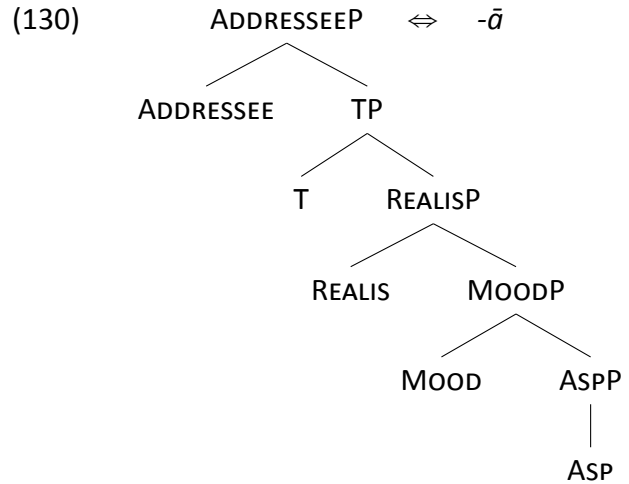
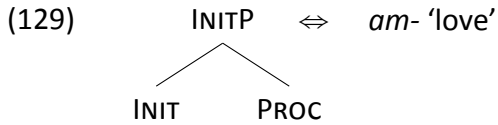
With respect to the other features present in the structure, we adopt the Ramchandian decomposition to analyse the internal structure of the verbal root (see sections 1.2.1 and 1.2.2.3). For the theme vowel, we propose that there is at least a default ASP feature present (Bermúdez-Otero 2013; Starke & Cortiula 2021; Cortiula 2023), and a MOOD feature. If one wants to build an indicative, one will need also a REALIS feature.¹⁹ MOOD and REALIS together give rise to an Indicative. Default tense, i.e. T is added as well, and then an ADDRESSEE feature.²⁰ Adding the

¹⁹I follow Starke & Cortiula (2021); Cortiula (2023) in the idea that the indicative is more complex than the subjunctive. While they capture this by adding a feature IND on top of MOOD, I replace IND with REALIS. The idea is that the combination of REALIS and MOOD gives rise to the indicative. If we add ADDRESSEE as well, we get an imperative.

²⁰While it has been proposed that tense may be missing from the structure of imperatives (Zanuttini 1991), a default future tense interpretation is inherent to imperatives and warrants the presence of a default tense feature T. A past tense interpretation on the other hand is entirely absent for imperatives, e.g. **Did go!*. So some impoverishment with respect to tense

ADDRESSEE feature gives rise to the imperative.²¹

With all of this in place, we propose the lexical structures for the verbal root *am-* 'love' in (129), for the theme vowel *-ā* in (130) and for the dedicated plural imperative ending *-te*.

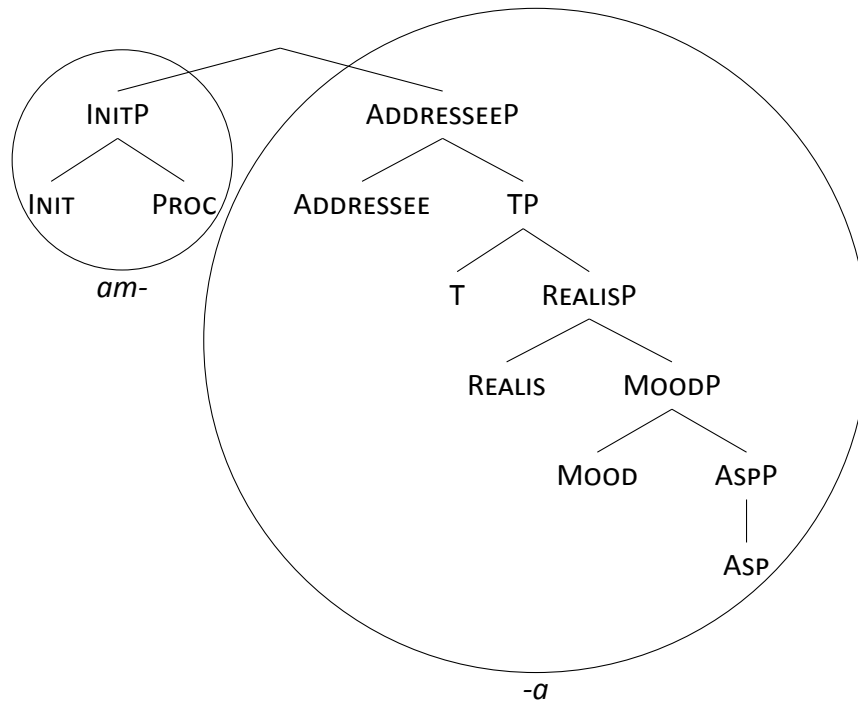


These lexical entries result in the structures in (132) for the 2SG present imperative and (133) for the 2PL imperative.

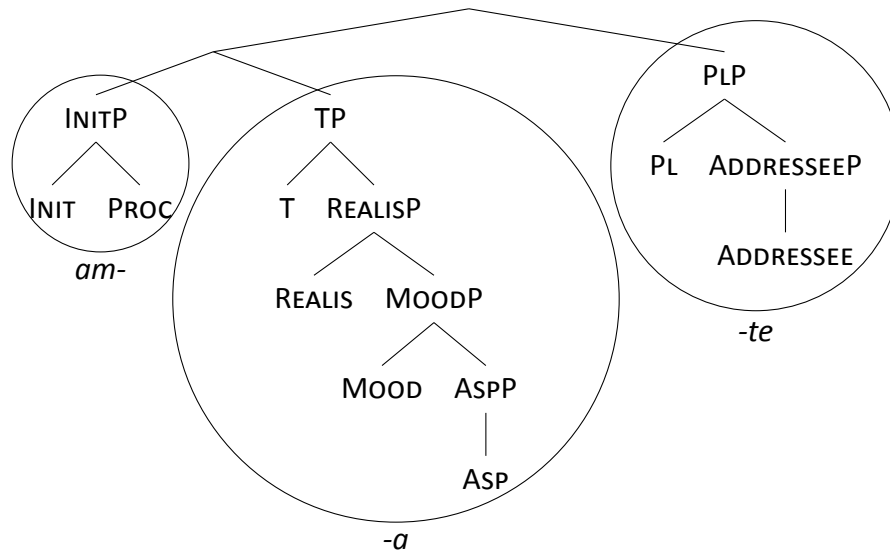
layers must be inherent in imperatives.

²¹While we assume that a Modality feature and/or a Jussive feature may also be present in the structure (Zanuttini 2008; Isac 2015), we keep the argumentation for such features for future work.

(132)



(133)



Observe that the ADDRESSEE feature is lexicalised by the theme vowel in the singular, but by the dedicated imperative ending *-te* in the plural. The fact that *-te* lexicalises ADDRESSEE explain why it is different from the 2PL ending that marks subject agreement elsewhere in the verbal paradigm (*-tis*).

We hope to have shown that even in cases where the morphology is extremely poor, as in the Latin 2SG.PRS.IMP, there may be reason to assume that in the absence of dedicated morphology, there are underlying features present that can be lexicalised with another form.

1.4 Many Forms to One Meaning (F^n -to- M)

In this section we discuss a mismatch in which several forms appear to give rise to only one meaning. We will discuss two subcases of this mismatch in this section. First, in section 1.4.1, we discuss a case of multiple exponence in Czech. Second, in section 1.4.2 we discuss two different types of doubling, namely allomorphy (section 1.4.2.1), and negative concord in section 1.4.2.2, both focused on Tamazight Berber.

1.4.1 Multiple exponence

In this section we discuss a case of multiple exponence in Czech degree morphology. The essence of what I discuss in this section is worked out in detail in Caha, De Clercq & Vanden Wyngaerd (2019), which is included in chapter 8 of Part II.

The productive way to form the comparative in Czech is by adding *-ejš* to the adjectival root, as illustrated in table (134). The fact that new words in Czech, like *benigný* in (134e) take *-ějš* is illustrative of this fact. We ignore the ‘hard’ agreement marker *-ý* here, as well as the soft agreement marker *-í* that is added to the comparative.

(134)	POS	CMPR	SPRL	
	a.	mil-ý	mil-ejš-í	‘nice/kind’
	b.	červen-ý	červen-ějš-í	‘red’
	c.	hloup-ý	hloup-ějš-í	‘stupid’
	d.	bujar-ý	bujar-ějš-í	‘wild’
	e.	benign-ý	benign-ějš-í	‘benign’

Interestingly, with some roots another allomorph of the comparative is used, i.e. *-š*, illustrated in (135).

(135)	POS	CMPR	gloss
	star-ý	star-š-í	‘old’
	slab-ý	slab-š-í	‘weak’

Additionally, the allomorph *-š* always pops up in the context of suppletive roots, as illustrated in (136).

(136)	ADJECTIVE		ADVERB		
	POS	CMPR	POS	CMPR	
	dobr-ý	lep-š-í	dobř-e	lép-e	‘good’
	špatn-ý	hor-š-í	špatn-ě	hůř-e	‘bad’
	mal-ý	men-š-í	mál-o	mén-ě	‘small’
	velk-ý	vět-š-í	hodn-ě	víc-e	‘big’
	dlouh-ý	del-š-í	dlouz-e	dél-e	‘long’

In some North-Eastern Bohemian dialects we can also observe what looks to be a zero allomorph. In the examples in (137) the soft agreement marker directly follows the root in the comparative, palatalizing the root-final *r* to *ř*.

(137)		St. Czech	N-E. Bohemian	
	POS	CMPR	CMPR	
	ostr-ý	ostř-ejš-í	ostř-ø-í	‘sharp’
	mokr-ý	mokř-ejš-í	mokř-ø-í	‘wet’

The data in (134) and (135) strongly suggest that *-ejš* can be decomposed into two parts, *-ěj-* and *-š*. If this decomposition is on the right track, then we are dealing with a situation where the comparative meaning is in some cases expressed by two morphemes, i.e. a prototypical case of multiple exponence.

Support for the decomposition comes from the fact that some adjectives miss the *-ěj-* part of the comparative, as in (135). Moreover, comparative adverbs systematically miss the *-š* part of the comparative adjective, as illustrated in (138).

(138)	CMPR ADJ	CMPR ADV	
	bujař-ej-š-í	bujař-ej- i	‘merrier’
	rychl-ej-š-í	rychl-ej- i	‘faster’
	červen-ěj-š-í	červen-ěj- i	‘redder’

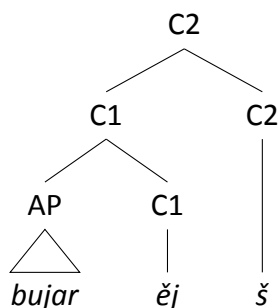
For these reasons, we decompose the comparative into (at least) two syntactico-semantic features, which we label C1 and C2, and which together give rise to the meaning of a comparative.²²

Under a terminal insertion approach a comparative with *-ějš* could be represented as in

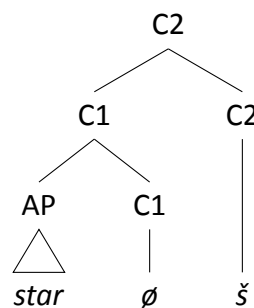
²²Without having any wish to provide a real semantics here, one could think of these two meaning components as providing a point and a direction. The point would be the standard of comparison, and the direction would indicate the direction of the difference on the scale. Varying the direction gives comparatives of superiority (*taller than*) or inferiority (*less tall than*); see section 1.5 for discussion of the meaning components POINT and DIR in the internal structure of the positive adjective. We hope to develop a matching semantics for the structures we propose in future work.

(139), with each comparative head lexicalised by an overt exponent. A comparative with just -š on the other hand would need to make use of a zero morpheme, as in (140), while roots that form the comparative with the zero allomorph would need to make use of two zeroes, as in (141).

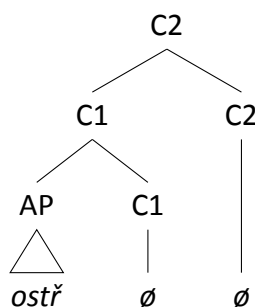
(139) The regular comparative



(140) The -š-comparative

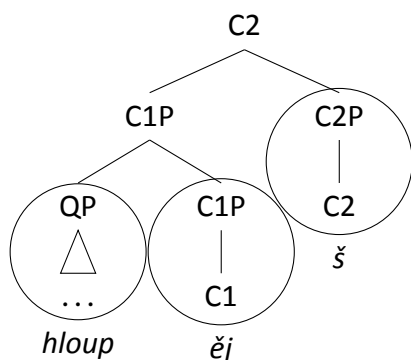


(141) The ø-comparative

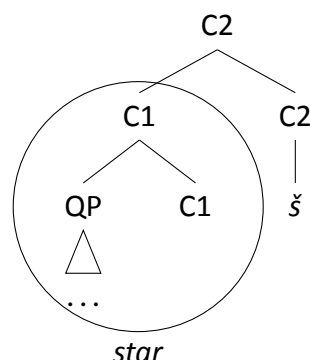


In Nanosyntax these apparent zero morphemes arise as a consequence of phrasal lexicalisation (see also our discussion in sections 1.1, 1.2.1 and 1.3). The different comparative allomorphs appear as a function of the size of the adjectival roots. If the root is big, there will be only ‘zeroes’, while if the root is of the medium type š will pop up. A root that is small will take both ěj and š. This idea of root size determining allomorphy is a very powerful tool and was first explored in Caha, De Clercq & Vanden Wyngaerd (2019). A nanosyntactic implementation of the structures in (139)-(141) is in (142) - (144).

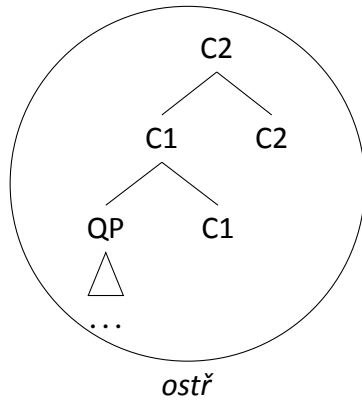
(142)



(143)



(144)



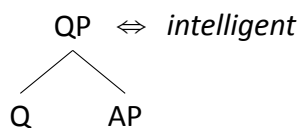
While we will not give detailed derivations here (I refer the reader to Caha, De Clercq & Vanden Wyngaerd (2019) in chapter 8), we hope to have shown that multiple exponence from a NanoSyntactic perspective boils down to decomposition of a morpheme, i.e. the comparative is decomposed into at least two syntactico-semantic features that together give rise to the comparative meaning, and which can either be lexicalized by dedicated comparative morphemes or by the adjectival root.

Before closing this section, we want to draw the parallel with English comparatives. The table in (145) shows that it does not involve a wild jump to see two different parts in the analytic comparative marker, *more*, comparable to what we just claimed for the Czech comparatives. The distinction between analytic and synthetic comparatives in English would under this approach follow from the size of the lexically stored structure for the root: if the root is structurally small, like the root of *intelligent* (146), then the analytic marker *more* will kick in (147). The syntax is in (148).

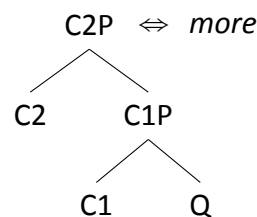
(145)

A	Q	C1	C2
rychl		ěj	š
lep			š
intelligent	mo	re	
old			er

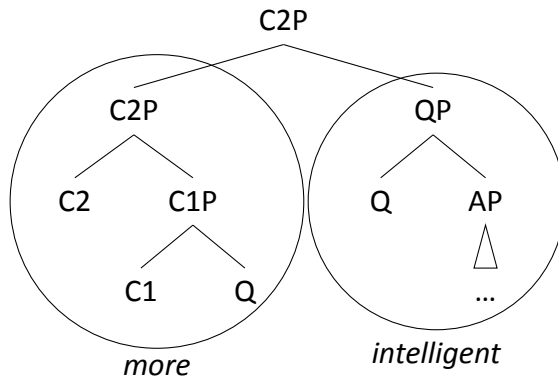
(146)



(147)

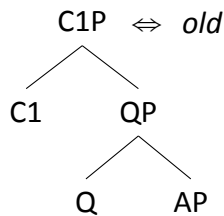


(148)

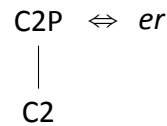


However, if the structure of the root is big, like for instance for *old* (151), then the synthetic marker *-er* (150) will be sufficient to lexicalize the second comparative head, as illustrated in the structure in (151).

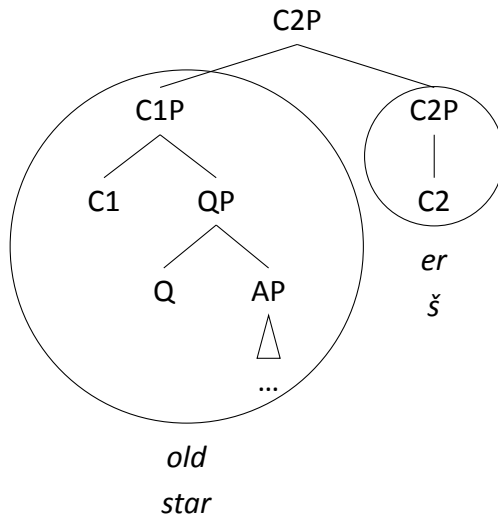
(149)



(150)



(151)



An interesting side-effect of the data discussed so far is the fact that it offers a natural account of root suppletion. Whenever an adjective has a suppletive root, one or both of the comparative markers will disappear. We refer to this as the *Comparative Suppletion Generalisation*, which is defined in (152) (Caha, De Clercq & Vanden Wyngaerd 2019: 507):

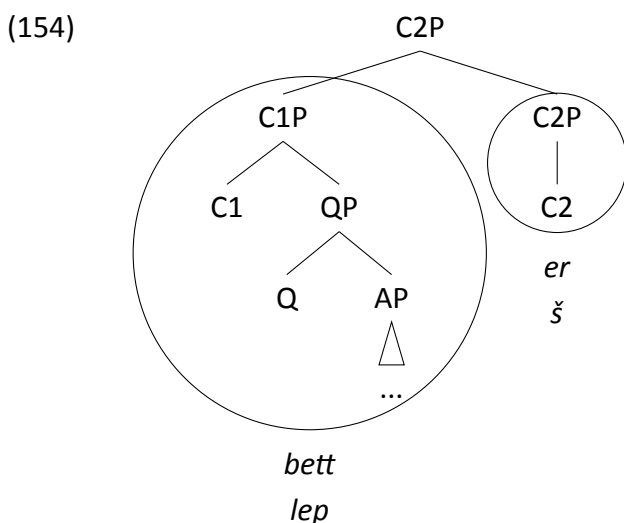
(152)

When the comparative is expressed by two overt markers in addition to the root (or when the comparative marker is a portmanteau that expresses the content of these two markers), there is no suppletion.

A consequence of this generalisation, when applied to English, is that analytic comparatives will never be combined with suppletive forms, since they lexicalize both C1 and C2. And indeed, there is no suppletion with analytic comparatives, an observation independently made by Bobaljik (2012) and referred to as the *Root Suppletion Generalisation*:

(153) Root suppletion is limited to synthetic (i.e. morphological) comparatives.

The *Comparative Suppletion Generalisation* follows naturally under the assumption that a suppletive root has to be of a different size than its nonsuppletive counterpart. For example, the positive degree *good* being of size QP, its suppletive counterpart *bett-* is of size C1P. Whenever there is suppletion, as for instance in the case of English *bett-er*, another typical example of multiple exponence, or its Czech counterpart *lep-š-í*, C1 is lexicalized by the suppletive root, as in (154), and the remaining C2 is lexicalized by *-er* or *-š-*, respectively.



Since *more*, like *-ější*, lexicalises both C1 and C2, it combines only with roots of size QP, which are nonsuppletive.

Summarizing, multiple exponence within the domain of degree morphology can be argued to follow from the proposed decomposition of the comparative into two heads. When the stored structure for adjectives is small, the two meaning components that together make up the comparative, i.e. C1 and C2, need to be realized by the lexical items the language has available in the lexicon. In Czech there are two different morphemes *ěj* and *š* that each lexicalize one of these layers when combined with an adjective that has a small lexical structure. However, when the size of the stored lexical adjective is bigger, only one of these morphemes will appear, and the illusion of multiple exponence will disappear, except of course in a case of root suppletion. Under this approach, suppletive comparative roots are at least one layer bigger than the positive root, hence predicting that whenever there is suppletion, and whenever a language makes use of more than one morpheme for the expression of the comparative, the

marking will be reduced under suppletion.

1.4.2 Doubling

This section draws on accepted work on negative concord in Quebliyeen Tamazight Berber, which is discussed in De Clercq (2025), and also included in chapter 9 of this work.²³ It builds on slightly older work on French negative concord (De Clercq 2019). De Clercq (2025) investigates the question whether there is any interaction between TAM and negative concord items (NCIs) of the type that we can observe between TAM and standard negators. This question has never been explicitly addressed in the literature, and the empirical work is taken as input for a Nanosyntactic approach to negative concord in De Clercq (2025). While this section will not discuss the typological angle of the paper, nor focus on its original research question, it will focus on two types of negative doubling in Quebliyeen Tamazight Berber, and how they can be modelled in Nanosyntax. The aim of this section is to show how the mismatch that we see is only apparent, and how it can be modelled making use of the Superset Effect and the way Nanosyntax stores lexical items. Most of the data in this section are from Quebliyeen Tamazight Berber (henceforth QTB), a Zemmour dialect spoken in the Middle Atlas Mountains in Central Morocco, as discussed in Ouali (2012: ch8).

1.4.2.1 Negative allomorphy

The first type of doubling we want to discuss involves a case of verbal allomorphy that arises in the perfective under negation in QTB.²⁴ (155a) illustrates an affirmative sentence in the perfective, and (155b) illustrates its negative counterpart. What is important in (155) is the fact that the positive form of the verb *idda* changes to *iddi* under negation, thus giving rise to a double realisation of negation (Ouali 2012: 160-161).

- (155) a. Idda ʕli.
 go.PFV.3.SG.M Ali
 Ali left.'
- b. **ur iddi.**
 NEG go.NEG.PFV.3.SG.M
 'He didn't leave.'

Standard negation in Berber is expressed by means of *ur*, and an optional morpheme *sha* (156).

²³The paper referred to is part of the output of the ANR-funded project *Sentential Negation Across the Globe (SNAG)*. *The fine-grained structure of a linguistic universal* (ANR-22CE28-0006).

²⁴Instead of referring to this type of polarity expression as *doubling* one could also refer to it as periphrastic (see Bonami (2015)).

- (156) **ur** swix (sha)
 NEG1 drink.PFV.1SG NEG2
 ‘I didn’t drink.’

In some Berber dialects negation is bipartite. However, since *sha* is optional in QTB, we will not take this to be the case (bipartite negation of the French *ne ... pas* type is discussed in De Clercq 2013; 2019; 2020b).

The type of allomorphy that we see in the pair in (155) presents a very typical type of verbal allomorphy under negation (Miestamo 2005). On the face of it, this is a doubling pattern, where both the negator and the verb are marked as negative, but there is only one negative meaning. However, I will argue that this doubling is only apparent, and that the *iddi* allomorph does not, in fact, contain a negation. Let me show why this is the case by means of the lexicalisation table in (157).

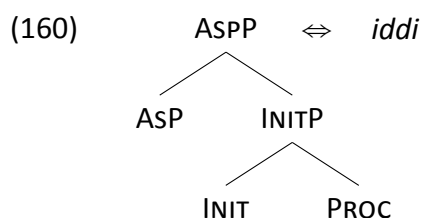
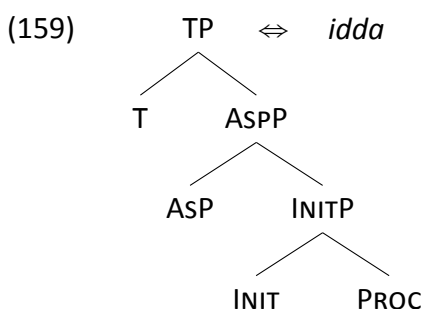
(157)

	NEG	T	ASP _{PFV}	INIT	PROC
a.				idda	
b.	ur			iddi	

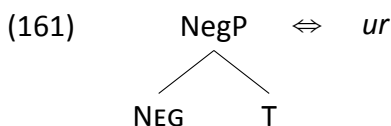
The top row shows the minimal set of features that one needs to generate the pair in (155). The features INIT and PROC are familiar from our earlier discussion of deadjectival verbs (sections 1.2.1 and 1.2.2.3), and refer to Ramchand’s (2008) decomposition of the verb phrase. In adopting the idea that sentential negation sits higher than T, we follow Ouali (2012) (and many of the references cited there). One of the crucial pieces of data presented by Ouali (2012) is that in the presence of isolating tense morphemes, as for instance in the future tense marker *dað*, *ur* precedes that tense particle, as in (158).

- (158) **ur** dað ayəx ibəttan
 NEG FUT buy.1SG clothes
 ‘I will not buy clothes’

The apparent doubling can be modelled by adopting the idea that there are two different verbal roots in the lexicon of Berber for the verb ‘leave’. One root is bigger and one smaller, as in the lexical structures for *idda* in (159) and *iddi* in (160), respectively.



If we furthermore adopt the idea that the standard negator *ur* lexicalizes T in addition to a NEG feature, as in (161), then we can understand, why the stem *idda* will not appear in the presence of negation: *iddi* will be a perfect match for all features up until ASP_{PVE} , and the standard negator *ur* will take care of the lexicalisation of tense and negation.



As such, we showed in this section how negative allomorphy on a verbal root can be captured in Nanosyntax as a consequence of root size, something we also discussed when we were tackling the Czech and English cases of multiple exponence. Nanosyntax thus allows us to unify these two phenomena, across the word boundary, and treat them both as cases of allomorphy that arise due to the fact that lexical items are stored with different structural sizes in the lexicon (Starke 2014b). In the next section we will tackle negative concord in Berber, and we will argue how it provides us with an important insight into the structure of negative concord items.

1.4.2.2 Negative concord

Unlike bipartite negation, which we briefly discussed in section 1.1, negative concord (henceforth NC) arises when argumental negative indefinites (like the equivalents of *nobody* or *nothing*) or negatively marked sentence adverbials (like the equivalents of *never*) co-occur with the standard sentential negator or with each other, while still giving rise to one negative meaning (Miestamo 2005). These negative indefinites can either appear in addition to the standard negator, as in the example from Czech (162), or in addition to one or more other negative indefinites (including those of the adverbial type), as in the example from Italian in (163) (examples from (Giannakidou & Zeijlstra 2017))

- (162) Dnes **nikdo** **ne**-volá **nikoho**.
 Today NEG.body NEG-calls NEG.body
 ‘Today nobody calls anybody.’
- (163) **Nessuno** ha telefonato a **nessuno**.
 NEG.body has called to NEG.body
 ‘Nobody has called anybody.’

Negative indefinites, also called n-words since Laka (1990) and more recently referred to as Negative Concord Items (henceforth NCIs) (Watanabe 2004: 578 for possibly the first use of the term), have been defined as in (164) by Giannakidou (2006).

- (164) N-word: An expression α is an n-word iff:
- α can be used in structures that contain sentential negation or another α -

- expression, yielding a reading equivalent to one logical negation; and
- b. α can provide a negative fragment answer (i.e., without the overt presence of negation).

The Czech and Italian data in (162)-(163) also bring us to another relevant distinction in the literature, namely the difference between Strict and Non-Strict NC (Giannakidou 1997): Czech is a strict NC language, because no matter whether the NCI appears pre- or postverbally, it always needs to be accompanied by the Sentential Negation (SN). In Italian on the other hand, it is only when the NCI appears in postverbal position that the SN is required (165b) (or another NCI, as in (163)). When the NCI is in preverbal position, the SN is not required (165b) (Zeijlstra 2004).

- (165)
- a. **Non** ha telefonato **nessuno**.
NEG has called NEG.body
'Nobody called'
 - b. **Nessuno** (*non) ha telefonato.
NEG.body NEG has called
'Nobody called'

In order to solve the mismatch that those multiple negative forms give rise to only one negative meaning, several different solutions have been proposed. Under one approach, the Negative Quantifier approach, all these NCIs are considered to be inherently semantically negative, hence negative quantifiers. At some point in the derivation, NEG-absorption takes place. This can happen thanks to the fact that all NCIs overtly or covertly move to the specifier of the position for sentential negation, i.e. NEGP, hence satisfying the NEG-Criterion.²⁵ This is the approach taken by Haegeman & Zanuttini (1991); Zanuttini (1991); Haegeman (1995). Another approach to NC, the Syntactic Agreement approach, considers NCIs to be semantically non-negative indefinites that are syntactically marked for negation. This approach finds its origins in Zeijlstra (2004; 2008), and has been highly influential ever since. Under this approach NCIs enter into an AGREE relation, either with a semantically negative (overt) negator, or with an abstract negative operator. A third approach, the Lexical Ambiguity approach, argues that NCIs are lexically ambiguous and stored twice in the lexicon, as negative and non-negative indefinites (Herburger 2001). A fourth approach, the Negative Polarity approach, treats NCIs as non-negative Negative Polarity Items (NPIs) (Laka 1990; Ladusaw 1992; Giannakidou 1997), i.e. on a par with words like *anything* and *anyone* in English.

In what follows I briefly point out some problems with each approach. The disadvantage of the Negative Quantifier approach is that it needs to rely on the principle of absorption, of which it is unclear how it works. Moreover, it relies heavily on covert movements (to satisfy

²⁵NEGP is a dedicated functional position in the syntax which finds its origin in Pollock (1989), and which has since then been argued to be either above or/and below TP.

the NEG-criterion) that are not all equally well-supported. The disadvantage of the Syntactic Agreement approach to negative concord is that it cannot explain why all NCIs can be negative in fragment answers, as in the definition in (164) (unless it takes recourse to ellipsis). Moreover, in order to explain why NCIs can be negative in pre-verbal position in non-strict negative concord languages like Italian, the theory needs to stipulate that each semantically non-negative NCI can trigger the appearance of a semantic negative operator to get licensed as a last resort. It goes without saying that this mechanism is very much a *deus ex machina* with little explanatory power. The disadvantage of the Lexical Ambiguity approach is that it requires all NCIs to be stored at least twice in the lexicon, as a negative indefinite and as a (non-negative) NPI. Finally, the disadvantage of the Negative Polarity approach is that it cannot capture the differences between NCIs and NPIs, in terms of licensing contexts, i.e. NPIs can appear in questions and conditionals but not (without negation) in fragment answers, while this is exactly the opposite for NCIs.

We will discuss negative concord data from QTB (already discussed above in section 1.4.2.1), which is considered to be a strict negative-concord language (Ouali 2012). QTB's morpho-syntax will allow us to understand the internal structure of NCIs better, and will point to the fact that NCIs must be really inherently negative. We will take this information as a starting point to develop a hypothesis about how negative concord may be captured in Nanosyntax.

The data in (166) show that QTB is a strict NC language. Both when *agid3* 'no-one' is in post-verbal position, as well as when it is in preverbal position, the SN *ur* is present. The morpheme *-n* on the verb in (166b) is an Anti-Agreement marker (glossed as AA), which appears whenever arguments are fronted to a position before the verb.

- (166) a. **ur** iddi **agid3**.
 NEG go.3SM.NEG.PFV no.one
 'No one left.'
- b. **agid3 ur** iddi-n yørskeela.
 no.one NEG go.3SM.NEG.PFV.PART-AA to.school.
 'Nobody went to school.'

The same negative perfective allomorph of the verb that we discussed in section 1.4.2.1 can be triggered in the context of NCIs. This is not unexpected since the negator *ur* is also present.

Interestingly, there is a construction in QTB, beyond the regular fragment answer context, illustrated in (167), in which the Strict NC pattern is broken, and *agid3* 'no-one' can give rise to negation on its own without the presence of the sentential negator *ur*. This is shown in (168).

- (167) a. Q: Ma ag idda-n?
 Who COMP come.3SM.PFV.PART-AA
 'Who came?'
- b. A: agid3
 no-one

‘No one’

- (168) **agid3** (ag) idda-n / *iddi-n
 no.one COMP go.3SM.PFV.PART-AA / go.3SM.NEG.PFV.PART-AA
 ‘Nobody left.’

Crucially, in (168), *agid3* ‘no-one’ precedes the optional complementizer *ag*. We also see in (168) that the negative allomorph of the perfective verb is unacceptable, which is perhaps not surprising since *ur*, the trigger of the allomorph, is also absent. We come back to the significance of this issue below. Moreover, we can be sure that we are not dealing with a biclausal structure (i.e. an elliptical negative cleft sentence), because of the presence of the Anti-Agreement marker *-n*, which indicates that *agid3* is the local subject of the verb *iddan*.

The table in (169) shows the two different patterns side by side: (169a) shows the pattern with the fronted NCI, while (169b) shows the regular strict NC pattern.

- (169) a. **agid3** (ag) idda-n
 no-one COMP go.3SM.PFV.PART-AA
 ‘Nobody left.’
 b. (ag) **agid3** ur iddi-n yərskeela
 COMP no.one NEG go.3SM.NEG.PFV.PART-AA to.school.
 ‘Nobody went to school.’

In order to represent the lexicalisation of the two different patterns in (169), we make use of the lexicalisation table in (170).

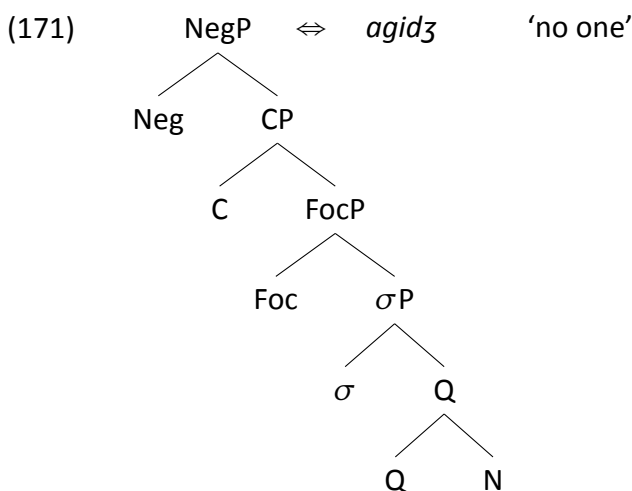
(170)

	NCI ₁	C	NCI ₂	NEG	T	ASP	INIT	PROC
a.	agid3	(ag)					idda	
b.		(ag)	(agid3)	ur			iddi	

Under the assumption that suppletion/allomorphy is conditioned in a strictly local configuration (see the discussion in section 1.2.2.2 on Korean), the data in (169) show that QTB NCIs cannot give rise to the negative allomorph on the verb. Moreover, while the sample is still small, it seems that no language in De Clercq (2025) has an NCI that conditions allomorphy on the verb. This warrants the (preliminary) conclusion that the negation that NCIs can give rise to, and the negation that the SN gives rise to, must be in a different position in the clause. This contrast between NCIs en SN is unexpected from the perspective of theories like the Negative Quantifier approach and the Syntactic Agreement approach. Moreover, what the data also show is that even NCIs in strict NC languages can, under certain circumstances, give rise to real negation on their own (168). This is unexpected under the Syntactic Agreement approach (though a hack is provided in the system to solve this problem) and the negative polarity approach.

The table in (170) captures the fact that only with the strict NC pattern can negative allo-

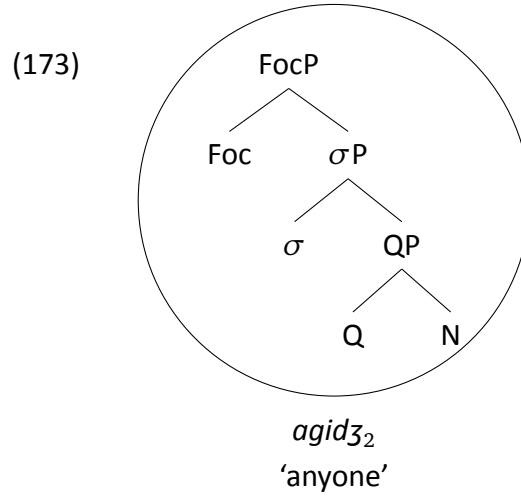
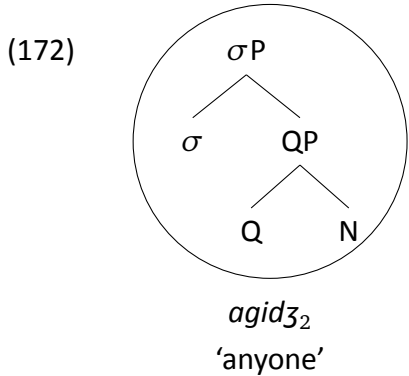
morphy arise in the perfective. However, the table does not tell us anything about how the negative meaning can pop up when the NCI precedes the complementizer. In order to understand this better, we need to consider the NCI *agid3* more closely and try to get closer to its internal structure. It is not really hard to argue that some nominal properties must be part of the NCI, so let us adopt the feature N, keeping in mind that it is a placeholder for some relevant nominal features. In order to capture the fact that this nominal is a type of quantifier we use the feature Q. However, we also need a feature to capture the NPI-like properties that NCIs have, and we adopt the feature σ for that (in line with work by Chierchia (2006) and the syntactic implementation by Labelle & Espinal (2014)).²⁶ Additionally, Ouali (2012: 157) states that *ag-id3* means ‘not.even-one’, suggesting that focus is morphologically contained inside the NCI. This is not unexpected since it has been claimed in the literature that Focus must be an inherent part of the internal structure of NCIs, forcing them to be in the scope of NEG (Watanabe 2004). What is interesting is that *ag—*, a part of the NCI, is syncretic with the complementizer *ag*. We take this syncretism to be meaningful in line with work by Baunaz & Lander (2018a) and propose that NCIs must consist of a property that they share with complementisers. For convenience sake, I will refer to this feature by means of [C]. Since the data in (168) show that the NCI can give rise to negation on its own when it is higher than C, I propose that the NCI also consists of a NEG at the top of its internal structure. The representation in (171) shows the lexical structure for *agid3*.



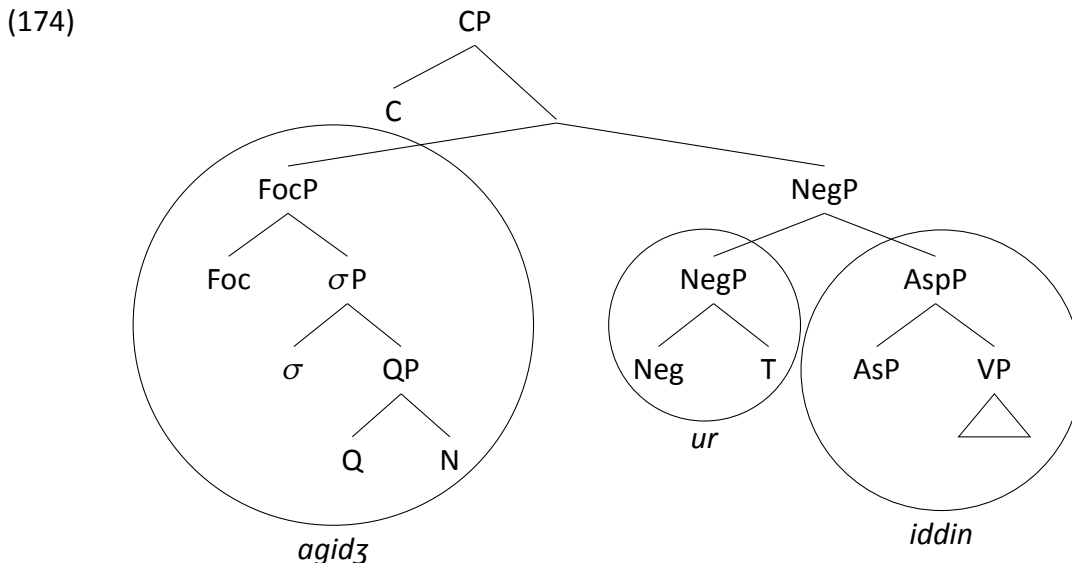
Thanks to the Superset Effect the structure in (171) can also lexicalize the syntactic structures in (172)-(173). In these cases, however, the lexical item in (171), while being a good match, is not lexicalising a NEG feature in the syntax, and hence the structure will lack semantic negation.

²⁶We mainly adopt this feature to capture the fact that positive indefinites are different from NPIs and to be able to regulate the competition between them. We are convinced that that σ (and the associated mechanism of exhaustification) is insufficient to capture the complex distribution NPIs (Giannakidou 2018).

As a result, the sentential negator will need to appear to give rise to negative force.

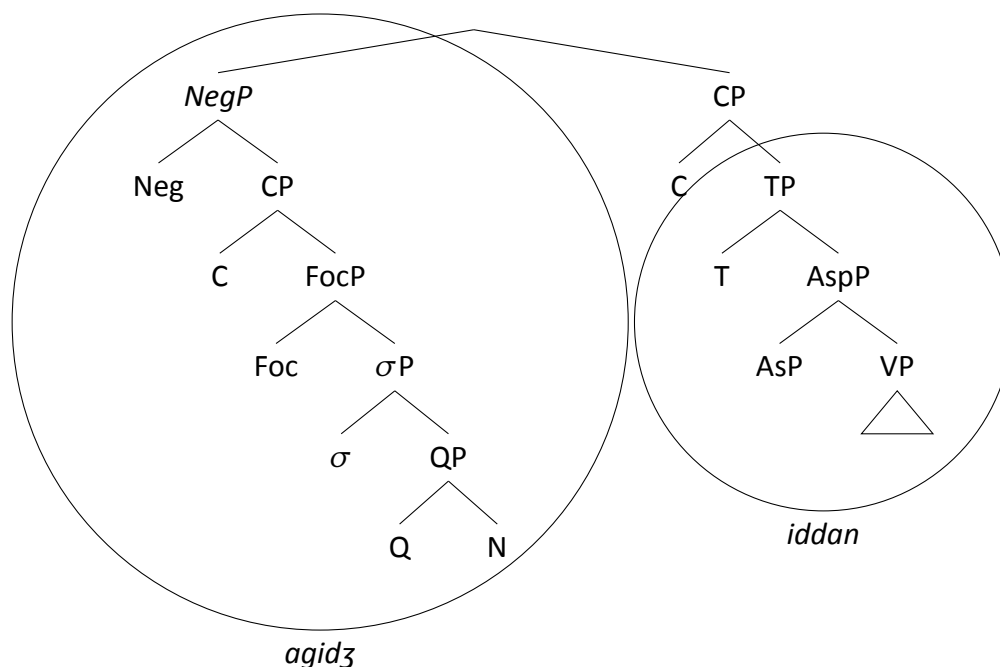


When NCIs are merged in postverbal position, they will have a syntax as in (172), since only a subset of the features of the lexical structure of the NCI will be present in the derivation at that point. FocusP will only be merged with the NCI when the NCI moves to a position above the vP-domain. This proposal is in line with syntactic approaches to Focus that argue for a FocusP for New Information Focus in a low left periphery, above vP (Belletti 2004), and one for Contrastive Focus in the high left periphery (Rizzi 1997). NEG will not be part of the internal syntax of the NCI when (172) is merged in syntax, accounting for why *ur* needs to be present to give rise to Sentential Negation. When the NCI appears in preverbal position, below the complementizer, as shown in (174), the syntax of the NCI will not reach up to the NCI-internal NEG either. Again, the SN marker *ur* will be needed to project a NEGP in the clausal syntax. The internal syntax of *agid3* may reach up to FocP in this case, but given the internal structure of NCIs in Berber, this structure will not be able to project a NEGP in the main spine.



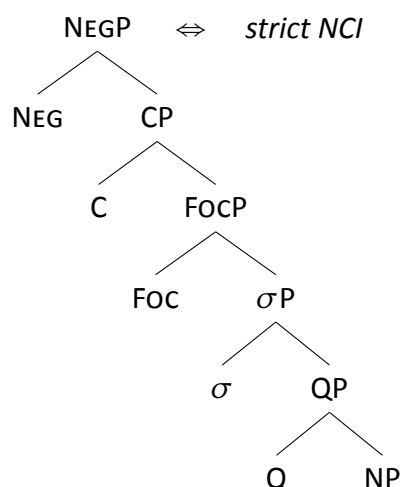
It is only when the NCI gets attracted to a high left peripheral position, above the complementizer *ag*, that the lexical item (171) can realize its full lexicalisation potential and project a NEGP in the syntax.

(175)

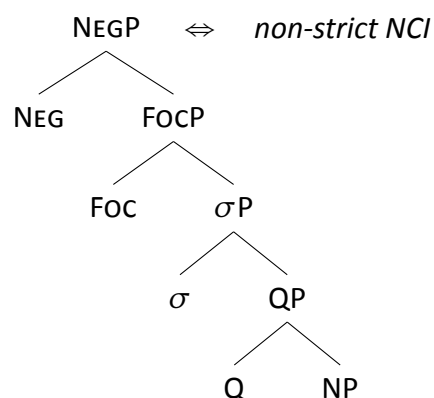


This type of approach, in which the position of NEG in the internal structure of negative NCIs or negative quantifiers interacts with the functional sequence, provides us with a potentially new way to look at negative concord. The situation in strict NC languages like QBT would correspond to (176). In a non-strict NC language like Italian, an NCI can give rise to real sentence negation only when it is in pre-verbal position, because the NEG feature sits lower in the NCI (below C), or because the NCI lacks the C-layer altogether, as illustrated in (177). A language like Standard English would under this approach have its NEG feature below FOC, allowing the negative quantifier to give rise to negation from a postverbal position, as in (178).

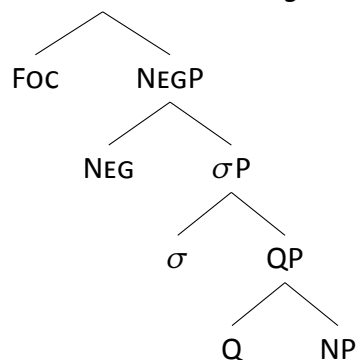
(176)



(177)



(178) FOC_P $\Leftrightarrow$ *Negative Quantifier*



While this hypothesis for a typology of concord is part of a bigger research programme, and lots more research needs to be done, we believe that this type of approach may provide a potential solution to a long-standing form-meaning mismatch involving negation. Thanks to the Superset Effect and the way lexical items are associated to hierarchically structured sets of features one form can be matched against different syntactic structures, accounting for why we perceive doubling phenomena as instances of form-meaning mismatches.

1.5 Form-to-No Meaning

This section is based on recent unpublished work that I presented with Pavel Caha (Masaryk University, Brno) and Guido Vanden Wyngaerd (CRISSP, KULeuven) at the 2023 NELS conference in Göttingen, at a Degree workshop organized by CRISSP in Leuven, and at Olinco 2023 in Olomouc. This research builds on work that we started in Caha, De Clercq & Vanden Wyngaerd (2019) and explored further in Vanden Wyngaerd, Starke et al. (2020); De Clercq, Caha et al. (2023), and it brings pieces of morphology in Czech adjectives to the fore that at first sight seem to have no meaning, but which upon closer consideration can each be given a clear-cut position in syntax, forcing us to consider whether they have some meaning after all. For this, we draw on the decomposition of the adjective that we proposed in Vanden Wyngaerd, Starke et al. (2020) when we considered the distribution of *k* in Slovak. Vanden Wyngaerd, Starke et al. (2020) is contained in chapter 10, but we will not discuss the Slovak data here. In the more recent research, we not only look at Czech *-k*, but also contrast it with a minimally different morpheme *-n*. Though these morphemes appear to be meaningless at first sight, and therefore to instantiate a form-meaning mismatch, I will argue that they in fact lexicalise semantically meaningful heads.

Czech adjectives can be divided into at least three classes based on their morphology. The first class of adjectives has the suffix *-n* in between the root and the agreement marker, e.g., *jem-n-ý* ‘smooth’ (179).

(179)

POS	GLOSS
jem- <i>n</i> -ý	‘smooth’
lev- <i>n</i> -ý	‘cheap’
hod- <i>n</i> -ý	‘kind’
šťast- <i>n</i> -ý	‘happy’
skrom- <i>n</i> -ý	‘modest’
něž- <i>n</i> -ý	‘gentle’
sluš- <i>n</i> -ý	‘kind’
pěk- <i>n</i> -ý	‘pretty’
mír- <i>n</i> -ý	‘peaceful’

The second class of adjectives has the suffix *-k*, e.g., *leh-k-ý*, see (180).

(180)

POS	GLOSS
leh- <i>k</i> -ý	‘easy’
heb- <i>k</i> -ý	‘smooth’
měk- <i>k</i> -ý	‘soft’
slad- <i>k</i> -ý	‘sweet’
hlad- <i>k</i> -ý	‘smooth’
níz- <i>k</i> -ý	‘low’
blíz- <i>k</i> -ý	‘near’
úz- <i>k</i> -ý	‘narrow’
krát- <i>k</i> -ý	‘short’

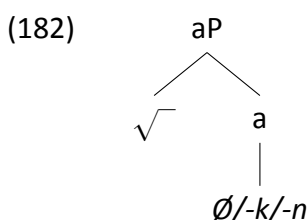
In the third class, the obligatory agreement marker attaches to the bare root, e.g., *slab-∅-ý* ‘weak.’ The absence of a marker equivalent to *-k* or *-n* is depicted by the *∅* in the first column of (181).

(181)

POS	GLOSS
slab- Ø -ý	'weak'
mil- Ø -ý	'lovely'
tich- Ø -ý	'silent'
mal- Ø -ý	'small'
dobr- Ø -ý	'good'
čist- Ø -ý	'clean'
hloup- Ø -ý	'stupid'
mlad- Ø -ý	'young'

In order to analyse these data, we rely on the notion of root size (already used in the analysis of Caha, De Clercq & Vanden Wyngaerd 2019 and discussed above in section 1.4.1).

In a Distributed Morphology type of architecture, the simplest analysis of the three classes would probably be as in (182):



One could then write insertion rules for arbitrary root classes to capture the distribution of the three allomorphs of *a*, with for instance *-k* as the default exponent, and the other markers sensitive to the context of a particular class.

- (183) Arbitrary root classes
- $a \rightarrow k$
 - $a \rightarrow n$ / Class-*n* __
 - $a \rightarrow \emptyset$ / Class- $\emptyset$ __

However, when we dig a bit deeper into Czech morphology we see that *-n* and *-k* are not allomorphs of a single head little *a*, but that they differ in their position with respect to the diminutive marker, as can be seen for the three different adjectival classes in the tables (184)-(186). The diminutive is very productive in Czech. It can intensify the meaning of negative scale adjectives (short → very short), and attenuate the meaning of positive scale adjectives (warm → warm-ish).

(184)

POS	DMV	GLOSS
jem- n -ý	jem- n -ouč- k -ý	‘smooth’
lev- n -ý	lev- n -ouč- k -ý	‘cheap’
hod- n -ý	hod- n -ouč- k -ý	‘kind’
šťast- n -ý	šťast- n -ouč- k -ý	‘happy’
skrom- n -ý	skrom- n -ouč- k -ý	‘modest’
něž- n -ý	něž- n -ouč- k -ý	‘gentle’
sluš- n -ý	sluš- n -ouč- k -ý	‘kind’
pěk- n -ý	pěk- n -ouč- k -ý	‘pretty’
mír- n -ý	mír- n -ouč- k -ý	‘peaceful’

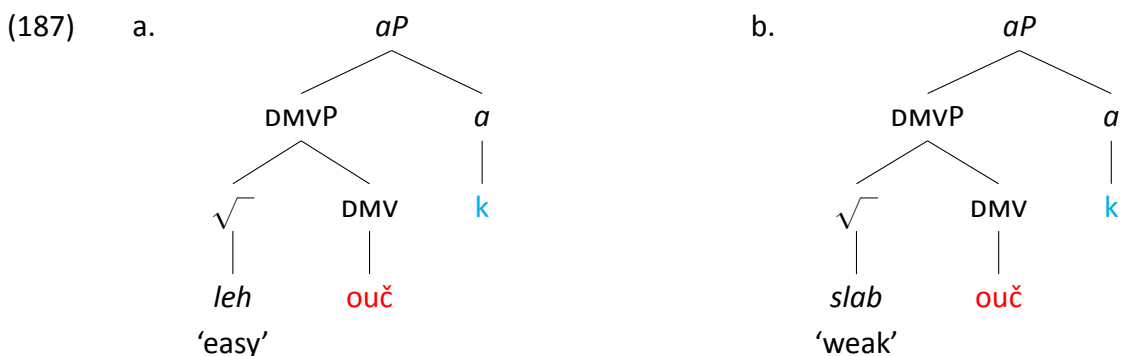
(185)

POS	DMV	GLOSS
leh- k -ý	leh-ouč- k -ý	‘easy’
heb- k -ý	heb-ouč- k -ý	‘smooth’
měk- k -ý	měk-ouč- k -ý	‘soft’
slad- k -ý	slad-ouč- k -ý	‘sweet’
hlad- k -ý	hlad-ouč- k -ý	‘smooth’
níz- k -ý	niz-ouč- k -ý	‘low’
blíz- k -ý	bliz-ouč- k -ý	‘near’
úz- k -ý	uz-ouč- k -ý	‘narrow’
krát- k -ý	krat-ouč- k -ý	‘short’

(186)

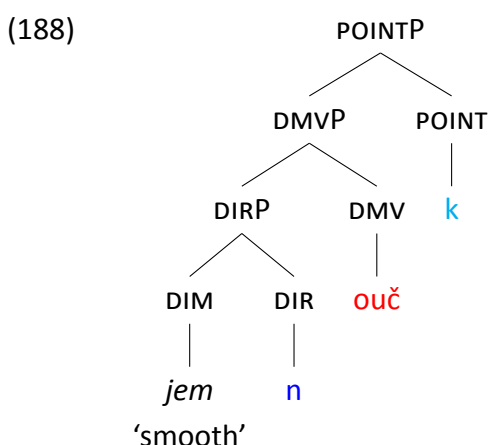
POS	DMV	GLOSS
slab- Ø -ý	slab-ouč- k -ý	‘weak’
mil- Ø -ý	mil-ouč- k -ý	‘lovely’
tich- Ø -ý	tich-ouč- k -ý	‘silent’
mal- Ø -ý	mal-ouč- k -ý	‘small’
dobr- Ø -ý	dobr-ouč- k -ý	‘good’
čist- Ø -ý	čist-ouč- k -ý	‘clean’
hloup- Ø -ý	hloup-ouč- k -ý	‘stupid’
mlad- Ø -ý	mlad-ouč- k -ý	‘young’

For the *k*-class we can see that the diminutive *ouč* appears in between the root and the adjectival augment *k* (e.g., *leh-ouč-k-ý* ‘very easy’). We must therefore posit the structure (187a), where DMV appears in between the root and the little *a*.



The position of DMV between the root and little *a* can also shed light on the diminutives of *-ø* adjectives, shown in (186), which acquire *k* in the diminutive (e.g., *slab-ouč-k-ý* ‘very weak’). This can be explained under the view in (183): since little *a* is no longer local to the root, the default exponent of little *a* (i.e., *k*) appears, as in (187b).

For the diminutives of *n* adjectives in (184), this type of explanation would not work: *n* precedes the diminutive, while *k* follows (e.g., *jem-ň-ouč-k-ý* ‘very smooth’). Once we realise that *n* and *k* can stack (when separated by DMV), we need to postulate two different projections: one for *n* (below DMV) and one for *k* (above DMV), as shown in (188).

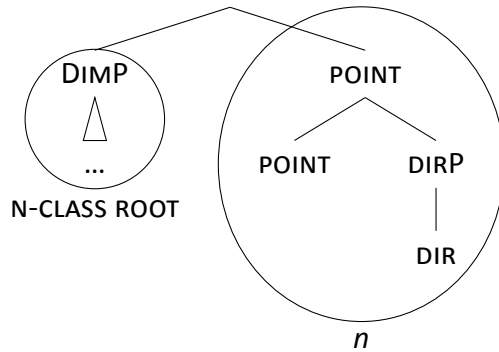


This also means we have to go beyond a system that has a root node and a little *a* head. The heads we assume in (188) are explained as follows. At the bottom there is a Dimension (DIM), topped by a DIRECTION (DIR). DIR determines the direction of the scale and helps distinguishing positive from negative adjectives. The additional projection POINT lexicalizes *-k*. POINT sets the standard on the scale. The decomposition of the adjective (and of the previous labels AP and QP in section 1.1) in DIM, DIR and POINT was also proposed in Vanden Wyngaerd, Starke et al. (2020).

Now that we have a layered structure for the adjective with three projections (excluding the diminutive head), we can provide an account of the three different adjective classes in terms of differences in root size. This solution relies on phrasal lexicalisation. Adjectives with the augment *-n* are the smallest: the root only lexicalises DIMP, and *-n* lexicalizes both DIR and

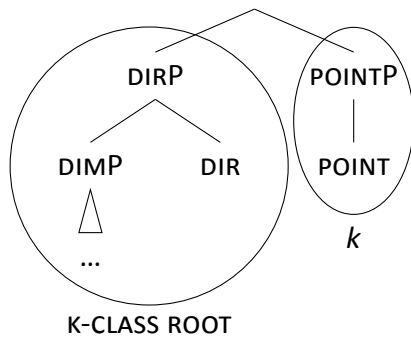
POINT:

(189)



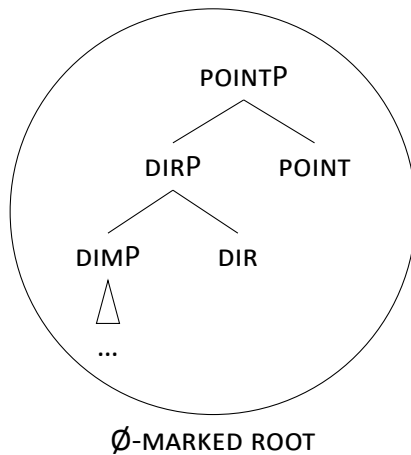
Adjectives with *-k* are slightly bigger: they lexicalize DIRP, while POINT is lexicalised by *k* (see (190)).

(190)



The $\emptyset$ adjectives, like *slab-y* 'weak', spell out the whole POINTP, obviating the need for any augment, as in (191).

(191)



The lexicalisation table in (192) illustrates the different root sizes.

(192)

	DIR	DIM	POINT	GLOSS
jem-n-ý	jem	-n		'smooth'
leh-k-ý		leh	-k	'light'
slab-ý		slab		'weak'

A crucial question is why *-k* appears in all diminutives. This can be explained in terms of intervention: when DMV is present, whatever precedes DMV can maximally lexicalise DIRP, and hence neither the root (for the $\emptyset$ -class adjectives) nor the *-n* augment can lexicalize POINT. Consequently, *-k* is needed for the lexicalisation of POINT in the diminutive. The table (193) shows this intervention effect of the diminutive.

(193)

	DIR	DIM	DMV	POINT	GLOSS
jem-n-ý	jem	-n	ouč	-k	'smooth'
leh-k-ý		leh	ouč	-k	'light'
slab-ý		slab	ouč	-k	'weak'

A closer look at the diminutive of Czech adjectives reveals that *-k/-n* are not allomorphs of the same adjectival head; nor are they expletive markers without any real meaning or role. They each realize a different projection, providing evidence for at least three heads in the structure of adjectival roots, and hence for a decomposition of positive degree adjectives into three semantically relevant meaning components.

1.6 Conclusion

In this chapter I discussed a set of empirical phenomena that I developed in work that was published or presented between 2019 and today. Lots of the empirical phenomena that I looked at (together with colleagues or alone) in the past 5 to 10 years can be captured under the central theme of form-meaning mismatches. The core idea in all case studies is that once we make the unit of study smaller, i.e. if we focus on submorphemic syntactico-semantic features, and once we adopt phrasal lexicalisation, phenomena like syncretism, fusion (in its different degrees and types), conversion, multiple exponence, contextual allomorphy, negative concord and expletive markers all get a natural explanation. The conclusion is that all these so-called mismatches stem from the deeply rooted assumption that there should be a one-to-one correspondence between form and meaning. However, in this chapter we showed that, from a Nanosyntactic perspective, form-meaning mismatches should not be regarded as exceptional: they are a natural consequence of how the functional sequence is packaged and cut up into portions that

are stored in the lexicon. Because that is what the lexicon is: a collection of stored portions of the functional sequence that are associated with exponents and conceptual information. Once this approach is adopted, it is only natural that form-meaning mismatches should be pervasive.

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Part II

Collection of papers

2 Nanosyntax: state of the art and recent developments

K. De Clercq, P. Caha, M. Starke and G. Vanden Wyngaerd

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Nanosyntax: state of the art and recent developments[♣]

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1. Introduction

One of the goals of current linguistic theory is to reduce grammatical variation to the properties of lexical items. This desideratum is known in the literature as the Borer-Chomsky conjecture (Borer 1984, Baker 2008: 353). The main idea behind the conjecture is that the operations of syntax are invariant across languages, variation arising solely from the interaction of those principles with language-specific lexical entries. This locates variation to a domain where the evidence for variation is indisputable -- the lexicon.

Nanosyntax shares this desideratum with current syntactic theories. However, the implementation of this idea in Nanosyntax differs from standard theories in one crucial respect. In Nanosyntax, the terminals of syntax are not language-specific feature bundles (i.e., pre-established assemblies of features), but individual features. Those features are furthermore hypothesised to be universal. There is thus no language-particular lexicon preceding the syntactic derivation. As a result, variation is restricted to the postsyntactic lexicon.

How can postsyntactic lexical entries trigger variation in the syntactic computation which precedes it? Starke (2014b; 2018) suggests that lexicalisation applies after every application of merge, and if it fails, the structure is handed back to syntax for last resort rescue operations — movements or complex left branch formation. In this model, the success of lexicalisation is thus dependent on the specific postsyntactic lexical items, and so are the last-

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resort movements. This resolves the timing tension of how postsyntactic entries influence what happens in syntax.

This mechanism proposed by Starke (2014b, 2018) is referred to as the *Lexicalisation Algorithm*. The Lexicalisation Algorithm regulates the process of externalization of syntactic structures. It takes the universal syntactic hierarchy or *functional sequence (fseq)* as an input and produces language-particular outputs due to its interaction with two other components of the grammar: the postsyntactic (language-particular) lexicon on the one hand and the (language-invariant) syntactic principles on the other hand. In effect, the algorithm steers the derivation in such a way that language-particular structures are formed as a function of language-particular (post-syntactic) lexicons. The prime source of variation has been identified as the size of a lexical entry, which refers to the number of features that can be lexicalised by a particular lexical item.

Work since 2018 has been marked by two important evolutions. The first of these is the realization that lexicalisation is not only affected by the size of lexical entries, but also by the structural arrangement of the features it contains. Two lexical entries of the same size may lead to different patterns of lexicalisation by organising the same features into different structures. The second innovation is more fundamental, in that it involves a change to the formulation of the Lexicalisation Algorithm itself. It is therefore natural for this volume to take the Lexicalisation Algorithm as its primary focus, and to explore its empirical effects across a wide range of phenomena. By assembling many different case studies under one roof, the volume showcases the wide applicability and the explanatory power of the algorithm.

In this chapter, we discuss the basics of phrasal lexicalisation in section 2, and relate it to Cinque's (2005) account of crosslinguistic differences in ordering. Section 3 explains how root size explains allomorph distribution. Section 4 introduces Starke's (2018) Lexicalisation Algorithm, and illustrates its operation using a concrete example. Section 5 discusses the shape

of lexical items, in preparation of section Section 6, which discusses the recent update of the algorithm involving pied-piping and subextraction.

2. The basics of phrasal lexicalisation

Since its beginnings, one of the tenets of Nanosyntax has been to provide a theory of the syntax-phonology interface that is direct in the sense that the output of syntax is directly translated onto the phonological representation, without any intermediate morphological structure. A crucial ingredient that allowed Nanosyntax to progress towards this goal was the concept of phrasal lexicalisation, explored by Starke since the early 2000s (Starke 2002; see also McCawley 1968, Weerman and Evers-Vermeul 2002, Neeleman and Szendroi 2007). This notion refers to a scenario where phonological interpretation is assigned to a phrasal node containing one or more terminals.

While exploring this notion, it is important to keep in mind that the expressive power of phrasal lexicalisation is inextricably linked to the constituent structure provided by syntax. We therefore inevitably introduce the notion of phrasal lexicalisation along with the specific Nanosyntactic assumptions about syntactic structures. These draw upon the work in the cartographic tradition (Cinque & Rizzi 2010a), which assumes the existence of highly structured syntactic hierarchies (referred to as *fseq*, the functional sequence, in the Nanosyntactic literature). See for instance the decomposition of NP (Abney 1987), CP (Rizzi 1997, 2004), IP (Cinque 1999), PP (Koopman 2000, Svenonius 2010), NegP (Zanuttini 1991, 1997; Poletto 2008), TopicP (Benincà and Poletto 2004; Frascarelli 2007; Frascarelli and Hinterhölzl 2007), as well as other domains.

A final important inspiration of the Nanosyntax approach is the theory of possible orders proposed in Cinque (2005). Let us illustrate all three components (the theory of phrasal

lexicalisation, the functional sequence, and the theory of ordering) by the example of Greenberg's Universal 39, given in (1).

(1) Greenberg's Universal 39 (Greenberg 1961: 95)

Where morphemes of both number and case are present and both follow or both precede the noun base, the expression of number almost always comes between the noun base and the expression of case.

This generalisation can be illustrated by languages such as Turkish in (2) or Tagalog in (3), where the number marker (in bold) intervenes between the root and the case, no matter whether we are looking at suffixes (as in (2)) or prefixes (as in (3)).

(2) Turkish

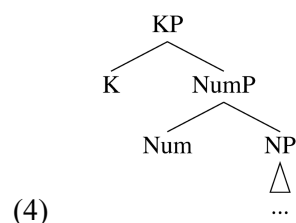
- | | |
|--|---|
| <p>a. el -de</p> <p> hand -LOC</p> <p> 'in hand'</p> | <p>b. el -ler -de</p> <p> hand -PL -LOC</p> <p> 'in hands'</p> |
|--|---|

(3) Tagalog

- | | |
|---|---|
| <p>a. sa =bata</p> <p> DAT =child</p> <p> 'to the child'</p> | <p>b. sa =mga =bata</p> <p> DAT =PL =child</p> <p> 'to the children'</p> |
|---|---|

The fact that languages tend to place number markers in between the noun and the case marker can be captured if we assume a crosslinguistically invariant hierarchy of functional projections

(the functional sequence, *fseq*), where number is lower than case, and therefore, by default, closer to the noun, see (4).



Assuming a universal hierarchy like this disallows the generation of orders such as Num>K>N (with case in between number and the noun), because such an order can only arise from a hierarchy where Num is merged higher than K, violating the universal hierarchy of merge order. This ingredient of the cartographic approach is fully embraced in Nanosyntax.

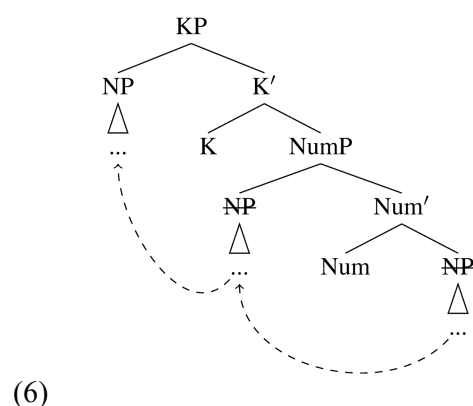
Consider now the fact that Greenberg (1961) states the generalization in (1) as a tendency, which means that there are counterexamples. As a case in point, consider the Classical Armenian example in (5).

- | | | | | |
|-----|----|-----------------|----|--------------------|
| (5) | a. | azg -aw | b. | azg -aw -k' |
| | | nation -INS | | nation -INS -PL |
| | | ‘with a nation’ | | ‘with nations’ |

Relevantly, as reported in a typological study by Kloudová (2020), examples like (5) only appear in a position after the noun, and never in a position before the noun (i.e., the order Num>K>N is not found). This observation falls in line with the approach pioneered by Cinque (2005) for the ordering of nouns, adjectives demonstratives and numerals (Greenberg’s

Universal 20). In this domain, we also observe one order before the noun, and various different orders after the noun.

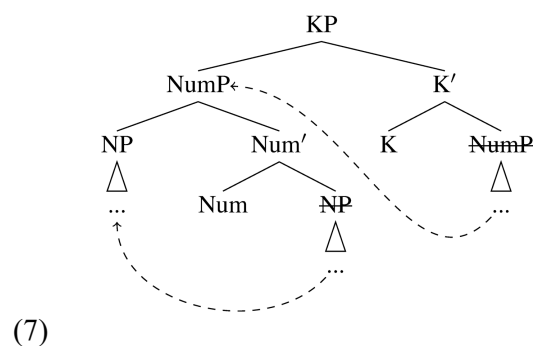
To account for this, Cinque (2005, 2009) adopts Kayne's proposal that the syntactic hierarchy is always mapped on a left-to-right linear order based on c-command. This produces the invariant order before the noun. For case and number morphology, the underlying structure in (4) thus maps onto the order $K > \text{Num} > N$. Any other order is derived by a leftward movement of a phrasal constituent containing the noun. In Cinque's approach, Classical Armenian thus has the structure as shown in (6).¹



In (6), instead of staying in the base position, the noun contained in the NP moves cyclically across the projections of number and case without inverting their order, yielding the surface order $N > K > \text{Num}$. The first movement that brings the noun across Num will be referred to as complement movement because the whole complement moves across Num. The second movement will be referred to as a ‘spec-to-spec’ movement because only the complex left branch inside the complement of K moves across it. As we shall shortly see, these two movement types (complement movement and spec-to-spec movement) form the backbone of the Lexicalisation Algorithm proposed by Starke (2018).

We must, however, mention at this point that while Nanosyntax adopts the geometric properties of the trees proposed in Cinque's approach, there are also some differences that become important later. Specifically, in Cinque (2005), specifiers are dominated by a label related to their sister, their movement is triggered by attracting features, and they enter in agree(ment) relationships. In Nanosyntax, phrases that sit in the same geometrical position lack these properties: their movement is triggered by the need to lexicalise the syntactic structure ('lexicalisation-driven movement', also referred to as 'evacuation movement'), a type of movement which doesn't project a label or leave a trace, contrary to feature-driven movements (see also Section 8). Furthermore, their mother node is not labelled. For now, this is merely a notational difference, but the unlabelled character of the nodes created by movement will become relevant in Section 6.

Consider now the Turkish order N>Num>K. This order would, under Cinque's (2005) approach, be derived by the sequence of two complement movements, whereby we first move the NP across Num, and then again move the whole complement across K, as in (7).



In sum, to derive the attested orders, Cinque assumes that for each contentful head that is merged into the tree, there are three different derivational options. First, it can be the case that the complement of the head stays in situ and there is no movement. Second, the specifier of the complement can move, as when the NP moves across K in Classical Armenian. Third, the whole complement of the head can move, as when the NumP moves across K in Turkish. Nanosyntax

adopts this general view on the derivation, even though it changes (as we shall discuss) some of the properties of the trees, e.g., the trigger for the movement and the labelling conventions.

With the basic assumptions concerning the functional sequence and the movements clarified, let us turn to phrasal lexicalisation. The role phrasal lexicalisation plays in Nanosyntax can be illustrated by means of a comparison between Latin and Turkish in Table .1.

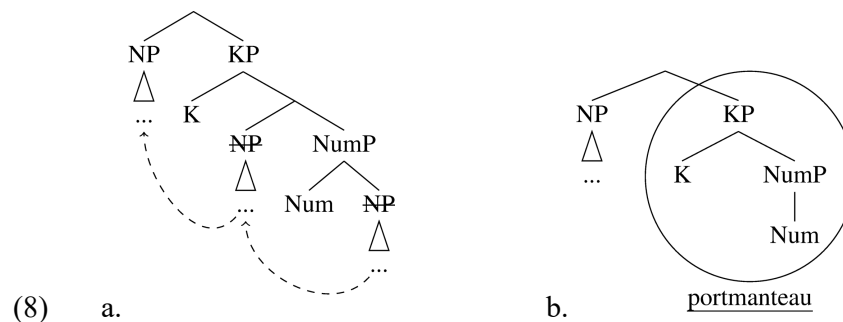
	TURKISH ‘HOUSE’		LATIN ‘GRANDFATHER’	
	SG	PL	SG	PL
NOM	ev	ev-ler	av-us	av-i
ACC	ev-i	ev-ler-i	av-um	av-os
GEN	ev-in	ev-ler-in	av-i	av-orum

Table 1 Latin and Turkish number and Case morphemes

In Turkish, the morphology is agglutinative: the morpheme *-lar* (with a harmonizing vowel) realizes PL, while *-i* realises accusative and *-in* genitive (see Türk and Caha 2021 for an analysis of the Turkish declension). In contrast, the Latin morphemes are fusional, i.e., they realise more than one grammatical feature. There is no way of segmenting the Latin morphemes into invariant plural or case markers.

In theories where lexicalisation only targets terminal nodes, the Latin facts present a paradox. One branch of the paradox stems from Greenberg’s Universal 39, which requires number and gender to be independent projections, universally ordered (e.g. by *fseq*). The other branch of the paradox stems from Latin: under terminal-node lexicalisation, Latin suggests that there are no two independent projections, but rather a single projection hosting the Latin morpheme combining case and number. Put differently, if there are two independent projections, the Latin morpheme is spread across several terminals, contradicting the terminal-based approach.

That paradox is resolved in Nanosyntax, without invoking any mechanism dedicated to it: the information spread across several terminals corresponds to a phrase, and phrasal lexicalisation will target that constituent – no paradox. Specifically, the idea is that the noun in Latin moves cyclically to the top of the tree, as shown in (8)(a). Note that since we now switch to a Nanosyntax account, we also change the labelling conventions, with unlabelled nodes dominating phrases which are similar to traditional "specifiers". After the movement, we obtain (8b), a structure where K and Num form a constituent (setting aside the issue of traces of the moved noun). This constituent is lexicalised by the portmanteau morpheme.



Phrasal lexicalisation extends the notion of constituency to a new domain: lexicalisation. When multiple terminals jointly move to a particular position (e.g., to the left of the verb in V2 languages), it is assumed that a whole constituent containing all the relevant terminals moves as one unit, i.e., as a constituent. When multiple terminals are not pronounced due to ellipsis, it is assumed that the ellipsis targets the full phrase containing these terminals (e.g., VP ellipsis). The idea in (8)(b) extends this rationale to the pronunciation of multiple terminals: when multiple terminals are realised by a single morpheme, this is because the morpheme lexicalises a constituent containing these terminals.

Summarising, the discussion above shows that phrasal lexicalisation in combination with the types of movements utilized in Cinque (2005) allows us to capture the differences between the number of terminals and the number of morphemes. It does so without altering the underlying syntactic structure (number and case remain separate terminals) and therefore allows

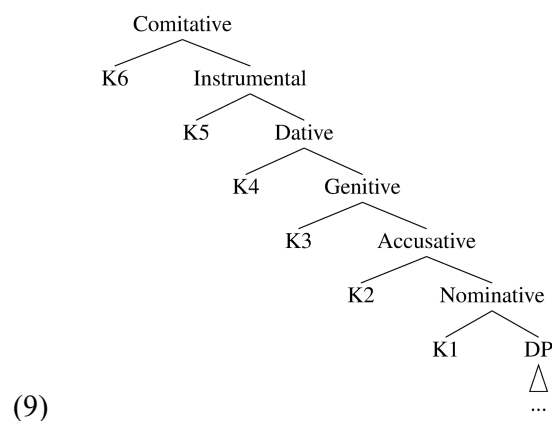
one to maintain the idea of a universal functional sequence. It also does not require any postsyntactic operation (like Fusion, an operation used in Distributed Morphology (Harley and Noyer 1999)), thereby also avoiding the need for a dedicated morphological structure, distinct from syntax.

Nanosyntax generalizes this logic beyond obvious examples of fusional and portmanteau morphology, such as the Latin genitive plural *-orum* above. To see this, consider agglutinative case markers, such as the Turkish genitive *-In* or the Armenian instrumental *-aw*: they also lexicalise with a single morpheme what other languages lexicalise with more than one morpheme. In Caha's (2010) containment data (Table) for instance, the genitive, which is lexicalised as a single morpheme in Turkish (as we saw in Table 1.1) is lexicalised by two suffixes in Vlakh Romani (the accusative suffix followed by *kor*), and by a preposition together with an accusative pronoun in English. Similarly for the instrumental, which is lexicalised as a single morpheme in Classical Armenian (see (5a)), but by two morphemes in Budukh and German.

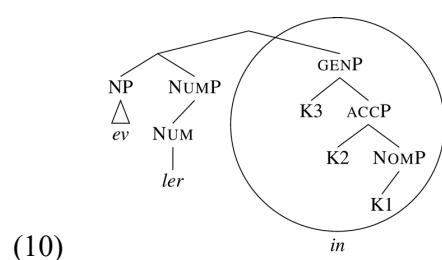
Case	Language	Expression		
		P	SUFF1	SUFF2
GEN	Vlakh Romani		ACC	<i>kor</i>
	English	<i>Of</i>	ACC	
DAT	Estonian		GEN	<i>le</i>
	Arabic	<i>Li</i>	GEN	
INS	Budukh		DAT	<i>Vn</i>
	German	<i>Mit</i>	DAT	
COM	Georgian		INS	<i>gan</i>
	Russian	<i>S</i>	INS	

Table 2 Case containment relations (from Caha 2010)

Based on this containment, syncretism patterns, and a number of other facts, Caha (2009, 2010) proposes a hierarchy of cases, such that every containment relation in Table is reflected by structural containment in the functional sequence: the comitative case is built by adding a feature (K6) to the instrumental, the instrumental is built by adding a feature (K5) to the dative, and so on, see (9).



Once this decomposition is adopted, agglutinative languages with no overt case containment, like Turkish, come out similar to Latin, in that they possess a case portmanteau that lexicalises all the relevant features. This is shown in (10), where it is assumed that NP first moves across Num, and then the entire NumP cyclically moves across the case features.



For more discussion, see Caha (2009), where it is argued that phrasal lexicalisation (based on a particular matching condition and incorporating the Elsewhere Condition) derives the fact that syncretism between case markers targets adjacent projections in the *fseq*, see (11). Such statements are referred to as the *ABA generalisation (cf. Baerman et al. 2005, Bobaljik 2012, Smith et al. 2019).

- (11) In the sequence *nominative – accusative – genitive – dative – instrumental – comitative*, syncretism targets only adjacent cases (Caha 2009: 10).

The study of syncretisms (as in (11)) and containment (as in Table 1.2) led to a rich body of work investigating the structure of the *fseq* in the nominal domain (Caha 2009; 2022; this volume; Taraldsen 2010; Starke 2017; Bergsma 2019; Kloudová 2020; Türk and Caha 2021; Janků 2022), directional expressions (Pantcheva 2011), negation (De Clercq 2013; 2020; Baunaz and Lander 2023), demonstratives (Lander 2016; Lander and Haegeman 2018), pronominals (Vanden Wyngaerd 2018), complementizers (Baunaz 2018; Baunaz and Lander 2018a; 2018c), cross-categorial syncretisms (Wiland 2019), the inflectional and derivational morphology of verbs (Taraldsen Medová and Wiland 2018; Baunaz and Lander 2019; Starke and Cortiula 2021; Cortiula 2023), degrees (Caha, De Clercq, Vanden Wyngaerd 2019, Vyshnevskaya 2022, Wiland 2023) or indefinites (Dekier 2021). The investigation of the *fseq* by means of the study of syncretisms and morphological and semantic containment is ongoing, and in this volume, the chapters by Dikmen and Demirok, and by Baunaz and Lander add to this investigation, too. We discuss the issue of syncretism and *ABA in more detail in sections 5 and 6.

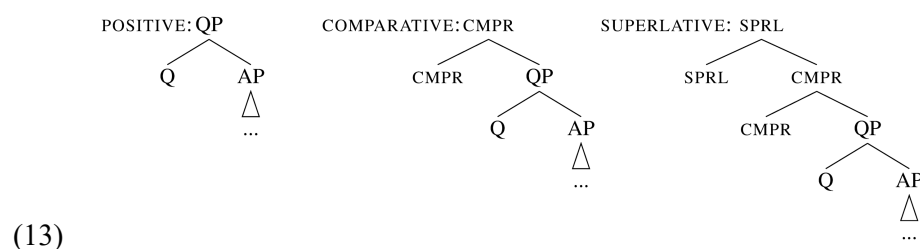
3. Root size as an allomorphy trigger

This section introduces the idea that not only functional morphemes, but also roots may differ in how many grammatical features they realise, and in doing so, give rise to allomorphic variation in suffixation patterns. For example, the roots *men* or *sheep* are lexically associated to plural, while the roots *boy* or *dog* are not. This is why the latter but not the former need a plural suffix. This section provides analogous examples from the comparison of Czech, English and Armenian degree morphology.

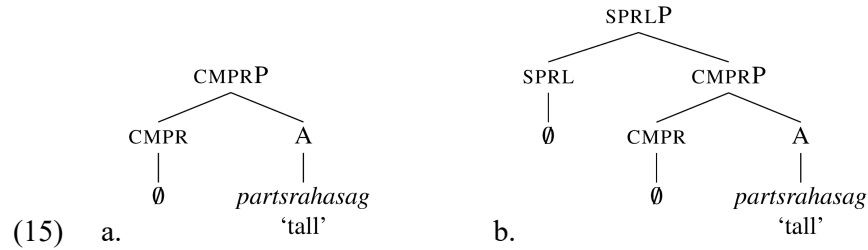
To introduce the basic idea, consider first Czech in (12). The examples show an increasingly complex morphology as we go from the positive (*intelligent-n-í* ‘intelligent’) to the comparative (*intelligent-n-ějš-í* ‘more intelligent’) to the superlative (*nej-intelligent-n-ějš-í* ‘most intelligent’). The extra morpheme is always in bold in (12).

- (12) a. Emm-a je intelligent-n-í. POSITIVE
 Emma-NOM be.3SG.PRS smart-N-FEM.SG
 ‘Emma is intelligent.’
- b. Emm-a je intelligent-n-**ějš**-í než Pavel. CMPR
 Emma-NOM be.3SG.PRS smart-N-CMPR-FEM.SG than Pavel
 ‘Emma is more intelligent than Pavel.’
- c. Emm-a je **nej**-intelligent-n-ějš-í ze všech. SPRL
 Emma-NOM be.3SG.PRS SPRL-smart-N-CMPR-FEM.SG from all
 ‘Emma is the most intelligent of all.’

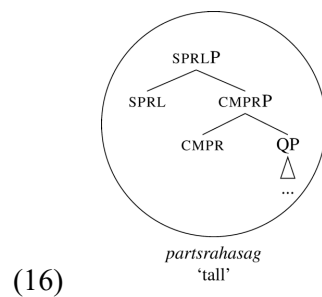
The increasing complexity of the morphology can be captured under the proposal (due to Bobaljik 2012) that the superlative degree contains the comparative, which is in turn based on the positive, see (13); recall also the discussion of containment in Section 2.



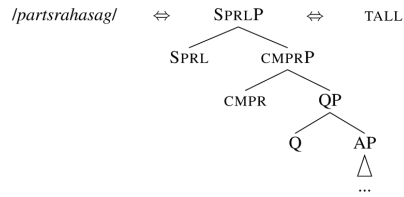
The positive degree as depicted in (13) is assumed to be bi-componential (enriching slightly Bobaljik’s original proposal). On top of the lexical AP, the positive contains a QP, a functional



However, if one adopts phrasal lexicalisation, then one could say that Armenian roots like *partsrahasag* ‘tall’ can lexicalise all features relevant to the adjectival root (QP), as well as the comparative and the superlative, as in (16).



At this point, we need to say something about how lexical items are stored in the lexicon. The general assumption is that lexical items are stored (memorised) links between representations belonging to different modules. For example, the depiction in (16) corresponds to the syntactic representation of the superlative. In the lexical entry (17), the syntactic tree is linked to a phonological representation (between slanted brackets) and conceptual information (in small capitals). In essence, the entry says that if syntax builds a structure like the one in (17), this structure can be linked to the phonological representation /*partsrahasag*/ and the concept TALL.



(17)

To determine whether a lexical entry matches a syntactic structure, Nanosyntax relies on identity. Specifically, if the constituent built by syntax is identical to a constituent stored in the lexicon, the entry of that constituent can be used to ‘lexicalise’ the syntactic representation; as a result, the relevant syntactic structure can be linked to a particular phonological and/or conceptual representation. This condition on matching is given in (18).

(18) *Condition on matching*

A lexically stored constituent L matches a syntactic phrase S iff S is identical to L.

The condition makes it possible for the lexical item in (17) to lexicalise either of the syntactic structures in (19), i.e. those of the comparative and the positive. This is so because any subconstituent of a lexically stored tree is itself stored in the lexicon, and therefore both the CMPRP constituent of (19)(a) and the QP constituent of (19)(b) find a matching constituent in the entry (17).



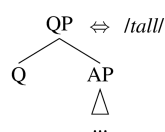
Phrasal lexicalisation, understood as identity matching, thus derives the syncretism between the three degrees of comparison that we observe in Armenian without any additional assumptions.

The matching condition (18) is an unpacked version of the matching condition used in Starke (2009): ‘A lexically stored tree matches a syntactic node iff the lexically stored tree contains the syntactic node’. To show that, let us first expand it to: ‘A lexically stored tree matches a syntactic node iff the lexically stored tree contains a constituent identical to the syntactic node’. This formulation further implicitly assumes that we look at the root constituent of the lexical tree, and because of that, the definition has to explicitly allow matching by constituents that it ‘contains’. Removing that implicit stipulation allows a simplified formulation: ‘A lexically stored tree matches a syntactic node iff the lexically stored tree is identical to the syntactic node’, i.e. (18). The goal of unpacking the original formulation is to make the identity requirement on matching explicit.

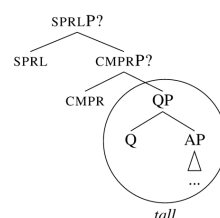
The matching condition (18) -- just like the original formulation -- derives the Superset effect, meaning that a lexical entry can lexicalise any phrase that it contains. Because of this, the matching condition is sometimes referred to as the Superset Principle (also in the subsequent chapters). However, it is important to realise that the Superset Principle is not an independent principle in addition to the matching condition (they are two different names for the same thing).

Let us now turn to English, which, unlike Armenian, requires additional morphology in the comparative and superlative (*tall* – *tall-er* – *tall-est*). If the lexical tree for *tall* is as in (20)(a) (omitting the conceptual information), we can understand why this is so: the lexical tree of *tall* is too small to lexicalize the degree features CMPR and SPRL, as (20)(b) shows. It can only realize the features of AP and QP. In effect, the root size of *tall* determines that the adjective will need help from other lexical items to lexicalize CMPR and SPRL.

(20) a. lexical entry

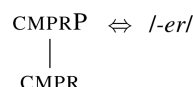


b. syntax

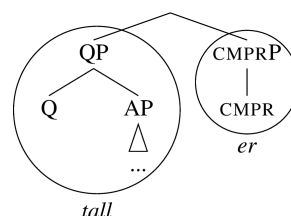


To capture the morphology of the forms *tall-er* and *tall-est*, we need two more assumptions. The first assumption concerns the morphosyntactic structure, namely, we need to assume that the QP in English moves across CMPR by complement movement, yielding the structure in (21)(b). In this structure, the CMPR feature is lexicalised as a part of the CMRP, using the entry in (21)(a).

(21) a. lexical tree



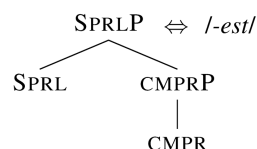
b. syntax



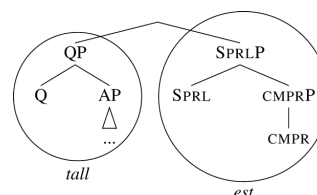
In the syntactic tree (21b), we assume the same types of lexicalisation-driven evacuation movements as introduced in Section 2, which leave no trace, and do not project a label (see the discussion in section 2).

In (22) we show an analogous analysis for the superlative.

(22) a. lexical tree



b. syntax

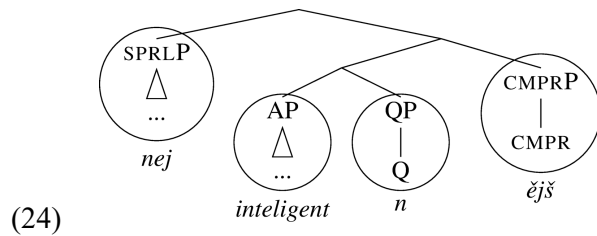


Note that as a consequence of the matching condition (18), the superlative *-est* with the lexical entry as in (22)(a) is a candidate for the lexicalisation of the comparative in (21)(b) because the CMPRP in (21b) is identical to a subtree of (22)(a). The reason why *-er* nevertheless appears in the comparative is that the lexical tree of *-er* in (21)(a) is a better match for the structure in (21)(b) as it has fewer superfluous features compared to (22)(a). In sum, when there are multiple candidates, they compete, and the winner is the lexical entry that is the best match with the syntactic structure. This competition principle is an instance of the Elsewhere Condition, going back to Kiparsky (1973):²

- (23) In case two rules, R1 and R2, can apply in an environment E, R1 takes precedence over R2 if it applies in a proper subset of environments compared to R2.

Let us now come back to the pattern of the Czech adjective *intelligent-n-í* ‘intelligent,’ with a comparative form *intelligent-n-ějš-í*, and a superlative *nej-intelligent-n-ějš-í*. Czech differs from English on two counts. First, unlike in English, the superlative marker does not replace (‘override’) the comparative, but rather attaches on top of the comparative. Intuitively, this is because the Czech morpheme *nej-* only lexicalises the feature SPRL, i.e. it is smaller than the English *-est*. This will require *-ějš-* to realize the CMPR feature in the superlative.

The second difference is that the positive in Czech is bimorphemic, consisting of the root and an adjectiviser *-n*. We may thus informally represent the situation in Czech as in (24).



Here the root only lexicalises the AP, which moves across Q by complement movement, and the adjectiviser lexicalises the QP. The whole QP then moves again across CMPR, and the CMPRP is lexicalised by the comparative *-ějš*. The lexical entry of *nej-* is unable to lexicalise the CMPR head realised by *-ějš*, and so we see the superlative prefix attaching to the comparative, which is unlike what happens with the English *-est*.

Summarising, this section so far showed that varying root size across languages (i.e., whether the root lexicalises AP, QP, or SPRLP) correlates with whether we find affixal material on the adjective in the positive, comparative and/or superlative. This can be captured if the root always lexicalises whichever features its lexical entry allows it to, and the remaining features are lexicalised by affixes or independent words.

It is important to observe that this type of variation is not only found across languages, but also within languages. For example, the English positive-comparative pair *bad* ~ *worse* has a suppletive root and a zero allomorph of the comparative. In Nanosyntax, the most natural analysis is one where there is no zero, rather *worse* has the same comparative structure as the Armenian *partsrahasag* ‘tall,’ i.e., containing no movement, recall (19)(a). This differs from *tall-er*, which requires complement movement, recall (21)(b). If that is so, we end up in a situation where different adjectives require different structures within the same language, i.e., English has both (19)(a) (no movement) and (21b) (complement movement), but for different roots.

Similarly, there are adjectives in Czech that do not require an adjectivizer, and their positive is monomorphemic, e.g., *chytr-ý* ‘smart’ with the comparative *chytr-ější* ‘smarter’.

Their structure must be therefore like the one of *tall ~ tall-er*, where the root lexicalizes the whole QP. Once again, this means that there is no movement of AP across Q with these adjectives, and we are once again led to conclude that different roots require different structures.

A compact way of representing and comparing the sizes of different lexical items is in the form of a lexicalisation table. The discussion of the degrees of comparison in Armenian, Czech, and English in this section, for example, can be summarised in a single overview in the form of a lexicalisation (Table). The top row of the table (following the language and the degree), represents the ingredients/features of the functional sequence, going from bottom to top as one reads from left to right. Actual morphemes that lexicalise these features are shown in the relevant cells of the table. Root size can be directly read off the table: the more cells a morpheme occupies, the larger its size. Black cells indicate features that are absent (e.g. the features CMPR and SPRL are not present in the positive).

language	degree	AP	Q	CMPR	SPRL
Czech	POS	intelligent	n		
Czech	CMPR	intelligent	n	ějš	
Czech	CMPR	chytr		ějš	
Czech	SPRL	intelligent	n	ějš	nej
English	POS	smart			
English	CMPR	smart		er	
English	CMPR	worse			
English	SPRL	smart		est	
Armenian	POS	partsrahasag			
Armenian	CMPR	partsrahasag			
Armenian	SPRL	partsrahasag			

Table 3: Lexicalisation table of the degrees of comparison in Czech, English, and Armenian

The lexicalisation table also shows how different root sizes yield different patterns of affixation. The concept of root (and affix) size therefore neatly captures the differences between classes of adjectives within a language, as well as the differences between different root classes across languages. In this volume, such differences play a crucial role in the contributions by Bergsma, Don, Merkuur and Smith, by Caha and Taraldsen Medová, by Natvig, Putnam and Wilson, by Taraldsen, and by Vyshnevskaya.

When there are multiple ways of lexicalising the same set of features (e.g., *worse* vs. **badd-er*), the system of lexicalisation is set up in such a way that a preference for one type of lexicalisation against another type follows from the system. More concretely, Starke (2018) proposes that the movements which give rise to different structures are not triggered by syntactic features, but by the need to create a structure that can be lexicalized, i.e., externalised at the interface. The theory of how movement (and structure building in general) interacts with lexicalisation is the main content of the Lexicalisation Algorithm, which is in turn the main focus of this book. The Lexicalisation Algorithm interacts with lexical items in a way that different lexical items give rise to different structures within the same language. We provide more detail in the next section.

4. The Lexicalisation Algorithm

4.1. Five steps

The first important assumption is that features are merged one by one, and that whenever a new feature is merged, lexicalisation must first succeed before the next feature can be merged. To this end, the structure created by Merge is matched against the existing lexical entries of the particular language. When a matching item is found for the newly formed structure, the derivation may continue by Merge F. However, when no matching item is found, syntax tries to rescue the phrase marker by performing left-branch movement (akin to spec-to-spec

movement in traditional theories) and lexicalising the result. If that fails, complement movement is attempted.

These steps are given in (25), and the first three of them (25)(a-b-c) quite straightforwardly correspond to Cinque's (2005) parameters of variation (no movement in (25)(a), spec movement in (25)(b), complement movement in (25)(c)) -- though in a different order of preference.

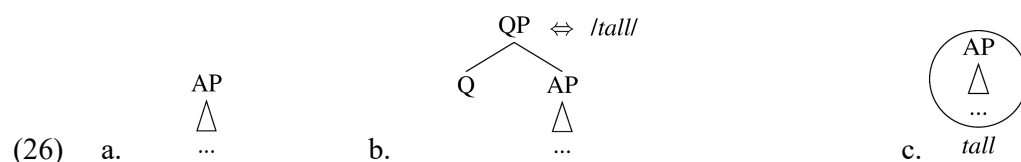
- (25) a. Merge F and lexicalize.
- b. If fail, try a spec-to-spec movement and lexicalize.
- c. If fail, try a movement of the complement of the newly inserted feature and lexicalize.
- d. If fail, go back to the previous cycle, and try the next option for that cycle.
- e. If fail, spawn a new derivation providing feature X and merge that with the current derivation, projecting feature X to the top node.
- (adapted from Starke 2018)

The ordering of the first option of the algorithm with respect to the others sets preferences for particular structures: lexicalisation without movement, whenever possible, is preferable to movements: this will ensure that *worse* is preferred to *badd-er*. In what follows, we discuss the movement options of the algorithm (25)(b-c) (section 4.2), backtracking (section 4.3), and complex specifiers (section 4.4).

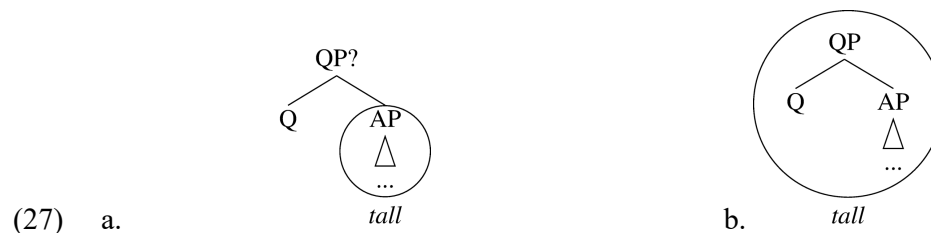
4.2. Movement

Turning first to the first three options of the algorithm (25)(a-b-c), let us see how English *tall* differs from the Czech *intelligent-n-i* 'intelligent' in derivational terms. The syntactic derivation starts by assembling AP, see (26)(a) (we assume that the AP is internally complex, as in Vanden

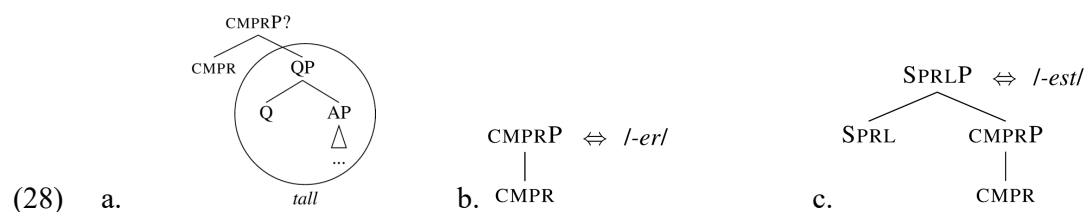
Wyngaerd et al. 2020). After this, the lexicon is checked, and since *tall* (repeated for convenience in (26)(b)) contains the AP, it is inserted, see (26)(c).³



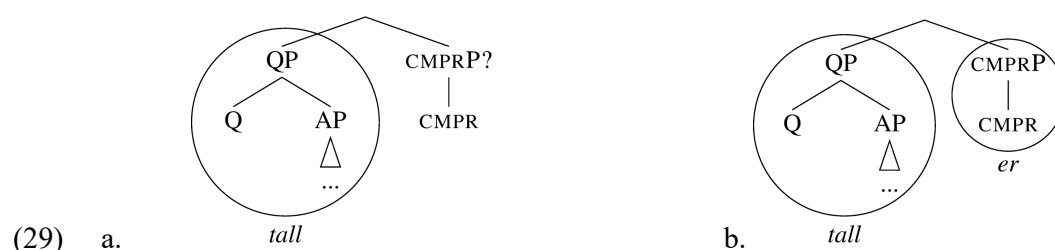
Since matching succeeded without any movement, the gradability feature Q can be merged, yielding (27)(a). (26)(b) is a perfect match and hence can lexicalize the structure, shown in (27)(b). This is the correct lexicalisation of the positive degree.



If deriving the positive degree was intended, the derivation would terminate (or merge the adjective with a noun). If the comparative is intended, syntax merges CMPR in the next step, see (28a). The lexicon is checked, but there are no lexical items matching this structure: the lexical tree for *tall* in (26)(b) does not contain the syntactic tree (28)(a). The lexical entries for *-er* and *-est* (repeated in (28)(b-c)) have CMPR, but neither of them matches the structure in (28a).



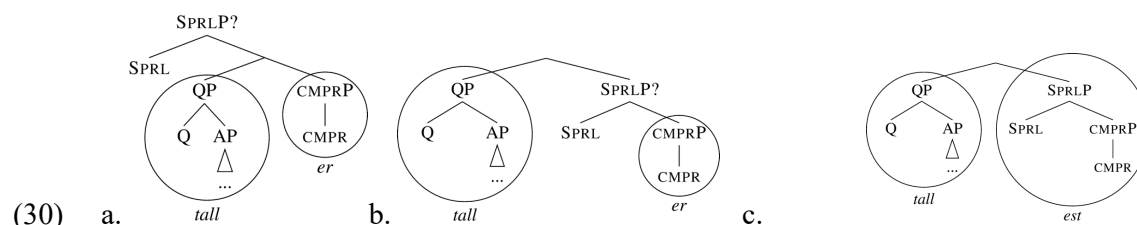
Since no match is found, the syntactic tree (28)(a) cannot be externalised. This leads to rescue movements, following the Lexicalisation Algorithm in (25). The first movement to be tried is moving the spec of the complement, recall (25)(b). However, the complement of CMPR has no spec in (28)(a), so syntax proceeds directly to complement movement (moving QP out of CMPRP), yielding (29)(a). In (29)(a), CMPRP can be lexicalized by the lexical item (28)(b), see (29)(b).



Summarising, movement is necessary if the constituent created by Merge does not find an immediate match in the lexicon. This movement is not a feature-driven, but it applies for the sake of lexicalisation: it is a lexicalisation-driven movement.

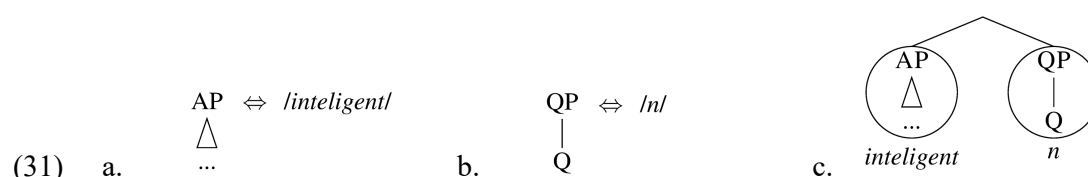
Because of this, the algorithm easily handles the fact that different roots interact differently with the derivation: for instance, if the root *worse* (capable of pronouncing (28)(a)) was compatible with the intended conceptual structure, (28)(a) would have been successfully lexicalised, and no movement would have been triggered.

Going back to our derivation of *tall*, consider now how the superlative is derived. First, syntax merges the feature SPRL to (29)(b), yielding (30)(a).



Since lexicalisation is rigidly cyclic, the lexicon is checked again, but there is no matching lexical item for the structure in (30)(a). Therefore, syntax tries moving the specifier (recall (25)(b)), yielding the tree (30)(b). After QP moves out, the SPRLP on the right branch can be lexicalized by the lexical item in (28)(c), overriding the previous lexicalisation, i.e. *-er*, and leading to the structure in (30)(c).

Let us now turn to Czech. In this language, some adjectives (e.g., *chytr-ý* ‘smart’) are rather similar to *tall* and do not require any extra morpheme in the positive. The morphology of such adjectives follows if they lexicalise QP, like the English *tall*. However, recall that the adjective *intelligent-n-í* ‘intelligent’ requires a special adjectiviser in the positive, namely *-n*. We can model this by attributing the size AP to the adjectival root *intelligent-* and letting *-n* lexicalise QP, as shown in the lexical entries in (31)(a-b).



Since (31)(a) cannot lexicalise the entire structure of the positive, AP has to move out of QP, yielding (31)(c), where the adjectiviser is inserted as the lexicalisation of the QP.

Summarising, we have now seen how the lexical items that a language has at its disposal ‘decide’ which of the three different movement scenarios in (25a-b-c) takes place. Given the appropriate lexical item, Merge can be followed straightaway by lexicalisation without movement as in (27)(b), but it can also be followed spec-to-spec movement (30)(b), or complement movement (31)(c). These are still the same movement types as proposed in Cinque (2005), but which of these applies is not dependent on ‘strong features,’ EPP, OCC and the like, but it depends on the lexical items available in each particular language – an explicit and detailed implementation of the Borer-Chomsky conjecture. Two additional steps of the

algorithm remain to be discussed now: backtracking, (25d) and complex left branch formation (25e).

4.3. Backtracking

Backtracking corresponds to step (25d) in the Lexicalisation algorithm. This step is tried when the movement steps that we have discussed so far do not lead to successful lexicalisation. When the derivation reaches a dead end in this way, it returns to the previous cycle, and attempts the next option of the algorithm. To give a feel for the kind of phenomena that give rise to a treatment in terms of backtracking, we turn to a new empirical case, the nominal declension in Czech, as discussed in Janků (2022). There are a bit less than 20 different ls, depending on the noun class. We discuss two of them here, and only three different cases, as shown in Table 1.4.

	‘bread’	‘tongue’
NOM	chleb-a	jazyk
ACC	chleb-a	jazyk
GEN	chleb-a	jazyk-a

Table 4: Partial declension paradigms for two Czech noun classes

The nominative and the accusative show a contrast between the *chleba* ‘bread’ and *jazyk* ‘tongue’ class that is straightforwardly amenable to a treatment in terms of varying root size. The fact that nouns of the *jazyk* class have no suffix in the nominative and the accusative suggests that they have a big enough root to lexicalise the entire functional sequence up to and including the accusative. The nouns of the *chleba* ‘bread’ class are smaller, and require a suffix *-a* in the nominative and the accusative (assuming, as argued in Caha 2009 and explained section 2, that the different cases involve features like K1, K2, etc., which stand in a

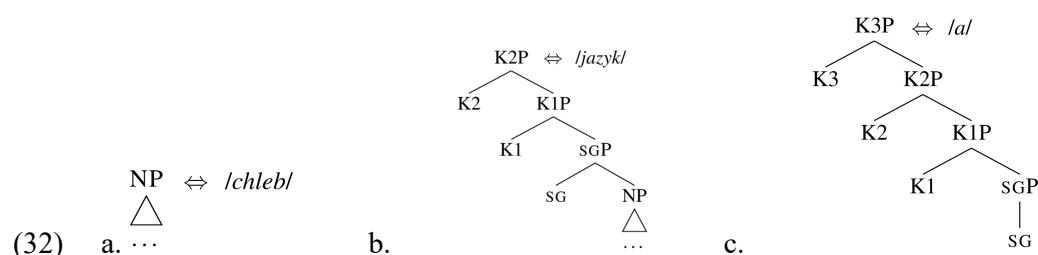
containment relation). Table 1.5 (somewhat simplified from Janků 2022) shows this in the form of a lexicalisation table.

	NP	SG	K1	K2	K3
NOM	chleb	a			
ACC	chleb	a			
GEN	chleb	a			
NOM	jazyk				
ACC	jazyk				
GEN	jazyk	a			

Table 5: Lexicalisation table for *chleba* 'bread' and *jazyk* 'tongue'

The surprise comes in the genitive, where the distinction between the two noun classes disappears, and both get the same suffix *-a*. This phenomenon of paradigm levelling is pervasive in Czech, where typically oblique cases show less noun class distinctions than structural cases. Informally put, it looks as if in the genitive, the noun *jazyk* shrinks to become the same size as that of the root *chleb*, requiring it to have the same suffix. This is shown schematically in Table 1.5 (see in particular the lines of GEN). The important point is that this shrinking of the root below its full lexicalisation potential is exactly what happens in a backtracking derivation.

Let us see in some more detail how this works. In (32) we show the lexical entries for the roots *chleb*-, *jazyk*-, and the suffix *-a*, respectively.



The derivation for *chleba* is straightforward and does not introduce anything new beyond what we discussed for the derivation of adjectives above. The same is true for the nominative and the accusative of the root *jazyk*, which allow direct lexicalisation of the structure by the root, and therefore do not trigger movement. However, when K3 is merged on top of K2P, the root cannot lexicalise K3P, and movement will be triggered. This will create a constituent [_{K3P} K3], for which there is no match in the lexicon, however. Although there is a suffix *-a* which can lexicalise K3, the lexical tree of the suffix, (32c), does not contain a constituent [_{K3P} K3]. At this point, the derivation reaches a dead end, and the backtracking option of the Lexicalisation Algorithm is activated, (25d). The derivation goes back to the point where SG was merged, and instead of taking the first option of the algorithm (direct lexicalisation), it now, on this second pass, takes the second option, and moves the complement, i.e. the NP. The result is a tree that has the constituent [_{SGP} SG] as its right-hand member, and since this constituent finds a match in (32)(c), it is lexicalised as *-a*, yielding *jazyk-a*. Subsequently, K1 is merged, and again movement of NP will be triggered, followed by lexicalisation of [_{K1P} K1 [_{SGP} SG]] by *-a*. The same happens when K2 and K3 are merged, ultimately yielding the genitive form *jazyk-a*, which will have an identical structure to the genitive *chleb-a*, effectively neutralising the distinction between these two different root classes. In this volume, backtracking features in Chapters 2 (Bergsma, Don, Merkuur and Smith), 7 (Demonie), 9 (Caha), 10 (Kasenov), 11 (Caha and Taraldsen Medová), 12 (Taraldsen), and 13 (Balsemin & Pinzin). In the literature, backtracking derivations are discussed, following Starke (2018), in Caha (2019; 2021; 2024), Wiland (2019) or Vanden Wyngaerd et al. (2020).

4.4. Complex specifiers

The final, last-resort option of the algorithm in (25), is the creation of a new workspace, where an entirely new, secondary, derivation is started, which is then subsequently merged with the primary derivation as a prefix or complex specifier, see (25)(e). Complex left branches merged in a new workspace surface as prefixes, or as other material that precedes the root (auxiliaries, complementisers, adverbs, etc). An example of this is the prefix *nej-* that marks the superlative in Czech, as in *nej-intelligent-n-ějš-í* ‘most intelligent’ (see (12)(c) above). Comparing Czech *nej-* to English *-est*, we see two differences. First, *nej-* stacks on top of the combination of the root and the comparative suffix, whereas *-est* attaches directly to the root. Second, *nej-* precedes the root, whereas *-est* follows it. The first of these differences arises because *-est* lexicalises both the CMPR and the SPRL feature, as is obvious from its lexical entry given in (28)(c) above. In contrast, Czech *nej-* lexicalises only SPRLP. If the lexical tree of *nej* had the shape $[\text{SPRLP SPRL}]$, it would be a suffix: since merge is binary, the only way to create a single-branching node like $[\text{SPRLP SPRL}]$ is to generate it as a binary branching one, and move one of the members out. Since movement is always upwards and linearized on the left, *nej*, would end up on the right, as a suffix. The theory thus predicts that suffixes have a single daughter in their lowest branching node (as shown for the Czech Case suffix *-a* in (32c) above), whereas prefixes (and roots) have binary branches in their lowest branching node (compare the root *chleb* ‘bread’ in (32a) above). This reasoning was proposed in Starke (2018) and is sometimes described as suffixes having a unary bottom, prefixes binary bottoms. Applied to the case of *nej-*, we will assume that it has a lexical tree of the shape $[\text{SPRLP SPRL CMPR}]$, with a binary bottom. When the feature SPRL is merged, it will not be directly lexicalisable by the root, and all the relevant movement options of the algorithm will produce a tree with the unary bottom $[\text{SPRLP SPRL}]$, but there is no match for that in the Czech lexicon. As a last resort, therefore, a new derivation will be spawned in a separate workspace to create a lexicalisable tree with the feature SPRL. In our example, the result

is [_{SPRLP} SPRL CMPR], which matches the lexical entry for *nej*.⁴ The main derivation may now merge the tree produced by the secondary workspace and proceed further. More discussion and examples of derivations with complex prefixes can be found in the papers by Wiland (this volume), Demonie (this volume), Gök and Demirok (this volume), Caha (2019) and De Clercq (2020).

As a final point in this section, we want to mention the issue of feature-driven movement, i.e. the syntactic movement of a phrase that has already been lexicalized in an earlier stage of the derivation. Such movements can be incorporated into the lexicalisation algorithm, by allowing the existing derivation to be searched for an already existing lexicalisation of F, either before merge-F or after a failure of merge-F to lexicalise. This is a largely unexplored area in the nanosyntactic literature, but see De Clercq (2019; 2020), De Clercq and Reintges (2022), Caha and Ziková (2023) and Starke (2024) for proposals incorporating feature-driven movement. Incorporation of feature-driven movement into the algorithm points into the direction that Nano is Syntax. We are confident that future work will explore this issue in much greater detail than our summary treatment in the present context.

5. The shape of lexical items

In this section, we focus on two apparent paradoxes for the theory introduced so far, namely *ABA violations and inverse marking systems. Systems of inverse marking have the property that a larger structure is lexicalized by less morphemes than a smaller one. A hypothetical case would be one where the positive degree of an adjective required a morpheme in addition to the root (as in the Czech *intelligent-n-i*), and the comparative of that same adjective would be lexicalized by the root alone (as in the case of Armenian is *partsrahasag* ‘taller’ discussed above). The lexicalisation algorithm would appear to rule out such a hypothetical situation: if the lexicon contains a root of size CMPRP, then it is never expected to give rise to movement

(and consequent suffixation) in the positive degree. We discuss such a pattern from the domain of singular and plural marking in Kipsigis in section 5.1 below. The other paradox is the existence of *ABA patterns in the domain of verbal inflection, which we turn to in section 5.2

We show that both these phenomena can be accounted for thanks to the fact that phrasal lexicalisation allows the lexicalisation of any shape of tree, including trees with "specifiers" (complex left branches). So far, the left branches of lexemes were always terminals, yielding trees of the shape [X [Y [Z]]]. However, if we moved YP across X, the resulting tree would be of the shape [[Y [Z]] X]. The latter is a well-formed syntactic object, and can therefore be stored lexically (Starke 2014a).

Lexically stored trees of this latter type can be referred to as trees with a Complex Left Branch (CLB). The existence of such trees introduces another type of variation in addition to root size. In the example just given, the size of the lexical entries is exactly the same, i.e. the exact same features X, Y, and Z are involved, but the way they are hierarchically organized in the lexical entries is different. The exploration of the possibilities of lexical items with CLBs constitutes the first of the two recent innovations we mentioned in the introduction (the other being the pied-piping-based algorithm, to which we turn in section 6). It turns out that next to root size, root shape can account for certain lexicalisation patterns which would otherwise be unexplained.

5.1. Inverse marking

This section shows that CLB lexical items provide a natural explanation for systems of inverse marking, which raise a paradox stemming from containment (Blix 2022). An example of such a system is the pattern of number marking in Kipsigis (Kalenjin). Table 1.6 summarizes the core of the system of nominal classification that Blix (2022) discusses.

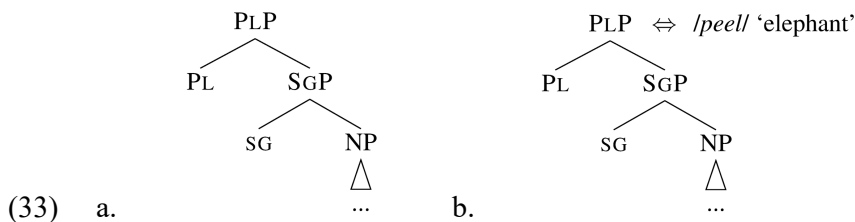
	root	SG	PL	
a.	kipaw	kipaw	kipaw- tiin	‘rhino’
b.	peel	peel- yaan	peel	‘elephant’
c.	pata	pata- yaan	pat- een	‘duck’

Table 6 Endo-Marakwet (Kalenjin), modified after Blix (2022) via Kouneli (2019, 2021)

The table shows that some nouns -- like *kipaw* ‘rhino’ on row (a) -- show a common pattern where the plural is marked with respect to the singular. Row (b) shows a case where the singular is marked. This type of marking is sometimes referred to as singulative, and the whole pattern can be summarised under the notion of an ‘inverse’: the idea is that each root has a default reference, which is either singular (*kipaw* ‘rhino’) or plural (*peel* ‘elephant’), and the inverse is the marking used when the actual number is the inverse of the default reference. Row (c) shows that there are also nouns where both the singular and plural are marked.

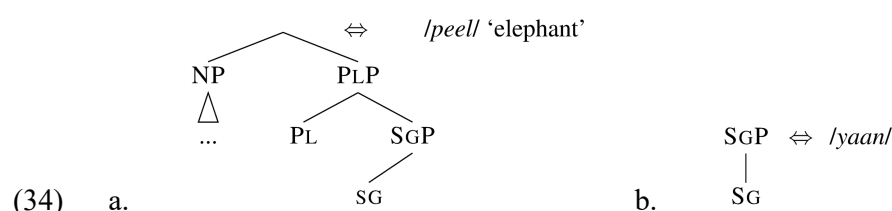
The potential paradox arises because it is impossible to represent both (a) and (b) patterns in terms of morphosyntactic containment because structural containment is asymmetric (i.e., if the structure of the plural properly contains the structure of the singular, it cannot be that at the same time, the singular structure properly contains the plural structure).

To state the problem in terms of the Lexicalisation Algorithm, let us adopt Blix’ functional sequence as in (33)(a) where the plural contains the singular (as in *chair* ~ *chair-s*, or as in row (a) of Table 1.6). Given that *peel* ‘elephant’ on row (b) has no marking in the plural, we would expect that it has an entry as in (33)(b).



But if that is so, then we would expect that *peel* can also lexicalise the singular number (corresponding to SGP), which is contained inside the lexical tree of *peel*, according to (33)(a).

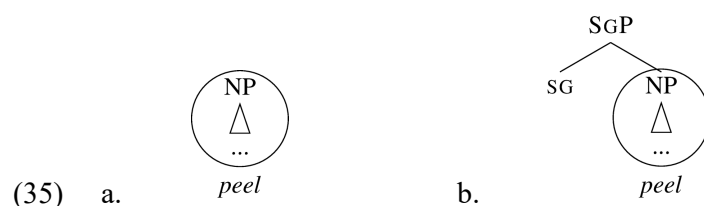
It turns out that the functional sequence (33)(a) is sufficient to account for the pattern in Table 1.6: If the lexical entry of *peel* contains a complex left branch (NP) arising from an evacuation movement, as in is **Fout! Verwijzingsbron niet gevonden.**(a), the inverse marking is derived. Such a tree arises in the course of the derivation when the NP cyclically moves from below SG (34(a)). Since **Fout! Verwijzingsbron niet gevonden.**(a) is a well-formed structure, and lexical items are links between well-formed syntactic structures and phonology, the theory expects such lexical items to exist. Observe that (33a) and (34a) contain exactly the same features, but arranged in a different shape.



The line connecting SGP with its daughter SG in **Fout! Verwijzingsbron niet gevonden.**(a) is oblique only to visually indicate the extraction site of the NP. It is notationally equivalent to the straight line in **Fout! Verwijzingsbron niet gevonden.**(b), and we shall use both notations in what follows. The lexical item in **Fout! Verwijzingsbron niet gevonden.**(b) represents the singular marker *yaan*. This entry is consistent with the pattern seen on row (c) of Table 1.6, where we see a root that has a marker both for the singular and the plural. Since this root always requires a number marker, we analyse it as lexicalising just an NP, and that is why *yaan* is needed in the singular.

Let us now observe how the markers in **Fout! Verwijzingsbron niet gevonden.** interact with the derivation of the noun *peel* ‘elephant.’ The first step in the derivation is to assemble the NP, which is lexicalised as *peel*, since NP is contained in the entry **Fout! Verwijzingsbron**

niet gevonden.(a), namely on the left branch. Since matching succeeded, the SG feature is merged, see **Fout! Verwijzingsbron niet gevonden.(b)**.

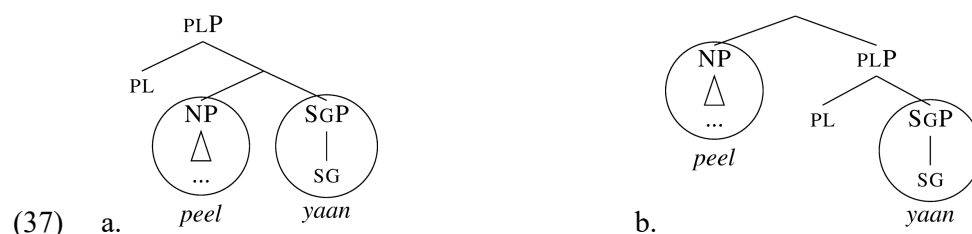


The initial problem with **Fout! Verwijzingsbron niet gevonden.(b)** was that when we postulated the entry for *peel* as in (33)(b), the expectation was that the structure **Fout! Verwijzingsbron niet gevonden.(b)** will be lexicalised by *peel*, yielding the incorrect unmarked singular. However, after we have modified the entry for *peel* to include a complex left branch, the tree in **Fout! Verwijzingsbron niet gevonden.(b)** is actually not contained in the lexical entry **Fout! Verwijzingsbron niet gevonden.(a)**: the entry contains no constituent that is identical to **Fout! Verwijzingsbron niet gevonden.(b)**. As a consequence, there is no match with **Fout! Verwijzingsbron niet gevonden.(b)** and the Lexicalisation Algorithm resorts to rescue movements.

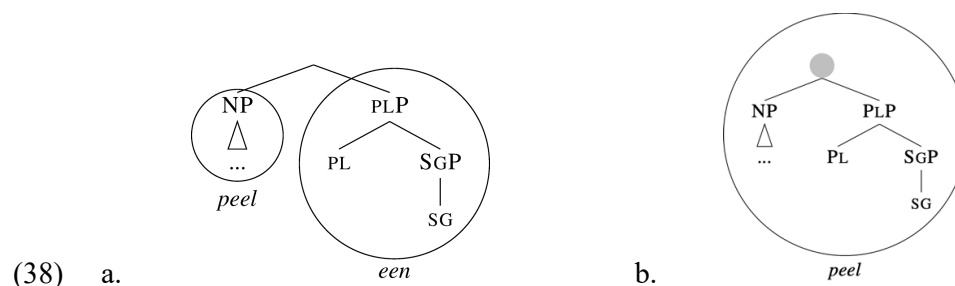
Spec-to-spec movement is inapplicable, complement movement yields **Fout! Verwijzingsbron niet gevonden.(a)**. This structure leads to the correct lexicalisation *peel-yaan* ‘elephant-SG’, as in **Fout! Verwijzingsbron niet gevonden.(b)**.



When PL is merged, as in 0(a), there is no match. Spec-to-Spec movement is therefore tried, yielding 0(b).



There are two ways to lexicalise 0(b). One option is to use the plural marker *een*, as in **Fout! Verwijzingsbron niet gevonden.(a)**. This marker appears in the plural of the noun *pata* ‘duck’ on row (c) of Table 1.6. Since *pata* ‘duck’ is just the lexicalisation of an NP, *een* is the realisation of the remaining features.



The other option is to use the entire entry of *peel*, as it matches the topmost constituent of the entire structure (i.e. the node marked by a grey circle in **Fout! Verwijzingsbron niet gevonden.(b)**). Since higher lexicalisations override previous ones, this lexicalisation will override the lexicalisation in **Fout! Verwijzingsbron niet gevonden.(a)**.

In sum, lexical items with a complex left branch capture ‘inverse’ patterns of marking, i.e. patterns that seem to give rise to a containment paradox.

5.2. ABA patterns

Another important feature of CLB entries is that they allow for the generation of a special type of ABA pattern, represented as A-Ax-A patterns by Blix (2022). Such a pattern is attested in a dialect of Friulian (spoken in Tualis) studied by Cortiula (2023). In this dialect, the first person is unmarked and syncretic with the third person (*bat* ‘I hit/he hits’), while the second adds a

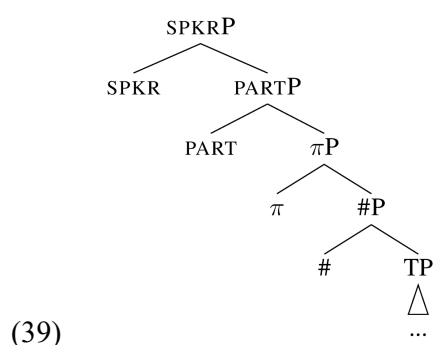
suffix to the root (*bat-s* ‘you hit’). When the persons are ordered 1-2-3, this gives rise to an A-Ax-A pattern, as shown in the Friulian column of Table 1.7.

		FRIULIAN <i>bati</i> ‘hit’	BULGARIAN <i>četa</i> ‘read’
SG	1	bat	čet-ox
	2	bat-s	čet-e
	3	bat	čet-e

Table 7 Present tense of *bati* ‘hit’ in Tualis Friulian and aorist of *četa* ‘read’ in Bulgarian

Before we show how CLB trees allow for the generation of this pattern, let us say why Cortiula adopts the ordering 1-2-3, rather than 2-1-3, which would make the Friulian paradigm trivially compatible with *ABA. The reason is that in other languages, of which Bulgarian in the second column of Table 1.7 is an instance, we find a syncretism between second and third person, which are incompatible with the 2-1-3 order. The syncretism between second and third person is in fact more common than the one between first and third person (even though the latter is far from being rare either; see Baerman et al. 2005; Baerman and Brown 2013; Ackema and Neeleman 2013). Cortiula (2023) therefore adopts the person ordering 1-2-3.

The specific decomposition of the agreement features she works with is given in (39) (see Béjar 2003, Starke 2020, Blix 2021a, Starke and Cortiula 2021).



Let us start unpacking this structure by pointing out that it is essentially an extended version of the proposal by Pollock (1988), who proposed that above the tense projection, there is a dedicated position for agreement features. In (39), each agreement feature is given an independent projection, in accordance with the general tenets of the Nanosyntax approach. Let us now introduce the ingredients.

Immediately above TP, we see the number projection #. Plural would require the addition of PL, as discussed in the previous section, but we ignore this here for simplicity. In the absence of PL, number is interpreted as the default number, which is singular. Above number are person features. If only the feature π was present, we would be looking at a default person, i.e., the third person. The second person has the additional feature PART(icipant), and first person is semantically most complex, also including the feature SPKR. With respect to this decomposition, the paradigm found in Friulian represents a *ABA paradigm.

The fact that the form *bat* can lexicalise the first-person structure suggests that it is able to lexicalise all the features present in (39). Given the matching condition (18), this allows it to lexicalise the third-person, which is structurally contained in the first person. What is unexpected is that the root *bat* needs the suffix *-s* in the second person: by standard logic of constituent matching, the root should be able to lexicalise PARTP on its own.

In the form of a lexicalisation table, the pattern looks as in Table 1.8.

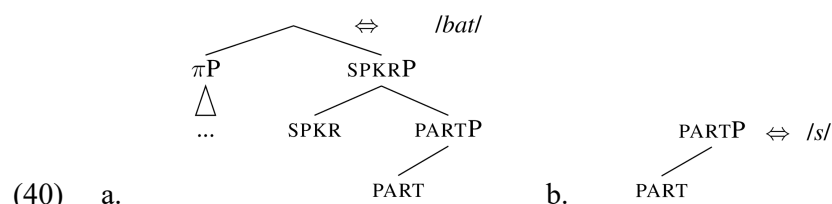
	INDP	T	#	π	PART	SPKR
1SG	bat					
2SG	bat				s	
3SG	bat					

Table 8. Lexicalisation table of the present tense singular in TF

This looks like 'selective shrinkability': a large lexical item A cannot shrink one notch (requiring help of another lexical item B instead), but can shrink two notches. We indicate this

nonshrinkability by means of the asterisk in Table 1.8. Applied to the case at hand, the root *bat* can lexicalise a constituent as large as SPKRP (1SG). It can also lexicalise the subconstituent π P of 3SG, but it cannot lexicalise the subconstituent PARTP (2SG), since it needs the suffix *-s* to lexicalise the feature PART. The inability of the root to lexicalise PARTP is what the asterisk in Table 1.8 indicates.

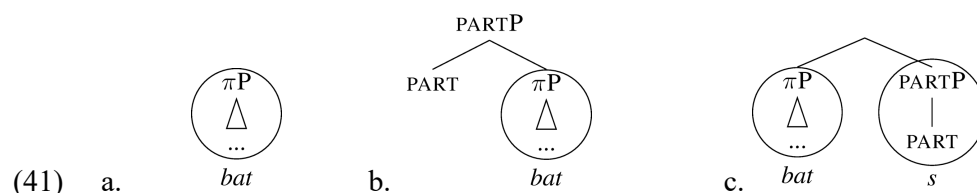
Lexicalising a structure which includes an evacuation movement however yields exactly that distribution. Cortiula proposes the lexical entry in **Fout! Verwijzingsbron niet gevonden.(a)** for the root *bat*, and the one in **Fout! Verwijzingsbron niet gevonden.(b)** for the 2nd person suffix *-s*.



This lexical entry with the CLB can lexicalise the most complex structure (SPKRP), but it does not contain a subconstituent that is identical to the structure of a 2SG: the PARTP in **Fout! Verwijzingsbron niet gevonden.(a)** only dominates PART, and not the rest of the features that would be needed for the root to lexicalise the full 2SG form of the root. On the other hand, the entry does contain a subconstituent as a CLB, namely π P, which corresponds with the least complex form (3SG). In the lexicalisation table (Table 1.8), the feature that is immediately to the left of the asterisk corresponds with the size of the CLB.

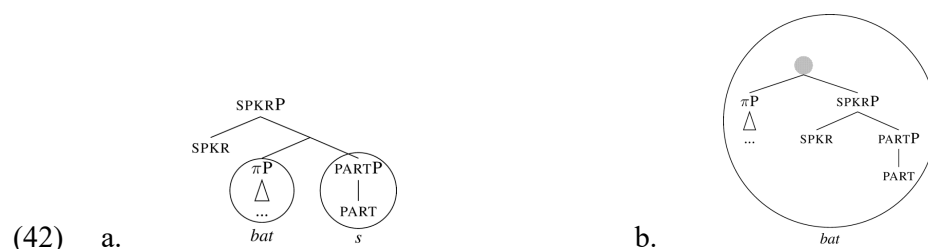
Let us now go through the derivation stepwise. The derivation first takes several steps to construct the π P, corresponding to the third person. Since π P is contained in the entry of *bat* in **Fout! Verwijzingsbron niet gevonden.(a)** as a CLB, the third person is lexicalised as *bat*, as in **Fout! Verwijzingsbron niet gevonden.(a)**. When the PART feature is merged, as in **Fout!**

Verwijzingsbron niet gevonden.(b), there is no match. Therefore, lexicalisation-driven movement is required. Spec-to-Spec movement does not apply as there is no such phrase to be moved, so that complement movement applies, yielding **Fout! Verwijzingsbron niet gevonden.**(c), which is lexicalised as *bat-s*.



Note that **Fout! Verwijzingsbron niet gevonden.**(c) cannot be lexicalised as *bat*, since there is no constituent in the lexical entry of *bat* in **Fout! Verwijzingsbron niet gevonden.**(a) that is identical to **Fout! Verwijzingsbron niet gevonden.**(c).

To derive the first person, the SPKR feature is merged, as in **Fout! Verwijzingsbron niet gevonden.**(a). Since there is no match, spec-to-spec movement is triggered, yielding **Fout! Verwijzingsbron niet gevonden.**(b). This constituent is identical to the lexical tree of *bat*, and the first person is therefore lexicalised as *bat*.



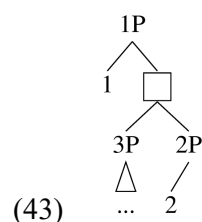
In sum, lexical entries with CLBs have been understudied until Blix's ground-breaking work (Blix 2021b), but once the spotlight is shone on them, they derive important morpho-syntactic patterns. Their empirical relevance has been demonstrated in the domain of Czech declension classes by Janků (2022), that of Friulian verb classes (Cortiula 2023), and the morphology of Czech adjectives (Caha, De Clercq and Vanden Wyngaerd to appear; Vanden Wyngaerd, De

Clercq and Caha to appear). In this volume, CLB entries are explored in the contributions by Bergsma, Don, Merkuur and Smith, by Caha, by Caha and Taraldsen Medová, by Kasenov, and by Vyshnevská.

6. A new logic for the lexicalisation algorithm

Starke (2022) suggests a novel way of thinking about rescue movements (evacuation movements), with the side-effect that a new type of evacuation becomes available: subextractions. The core of the idea is that rescue movements involve gradual pied-piping: Evacuation always targets the same constituent, but that constituent can pied-pipe gradually larger structures in the quest for a successful lexicalisation.

Let's unpack that by looking at a typical derivation such as **Fout! Verwijzingsbron niet gevonden..**



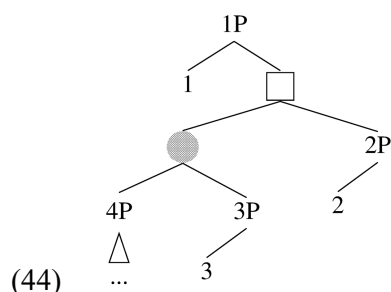
Here 3P has evacuated out of 2P, and the square node lexicalised successfully. Now feature 1 has been merged and the constituent cannot lexicalise as it is. Starke's reformulation of the lexicalisation algorithm always evacuates 3P in this configuration. That leads to a spec-to-spec evacuation. If this evacuation fails to yield a lexicalisation, 3P retries by pied-piping its mother node, i.e. the square constituent. This formulation derives both the existence of complement movement, without the need to stipulate that there is such an option and the fact that it comes after spec-to-spec evacuation, without having to explicitly list that order.

Why 3P though? Two answers have been given, one in Starke's original 2022 seminar and one in Caha's follow up 2023 seminar. In both formulations, only labelled nodes can be targeted by movement.

Starke's proposal is a generalisation of the insight in Cinque (2005) that only lexical categories move in these types of movements (in Cinque's case, only nouns move), together with the syntactic locality principle of moving the closest candidate. In the syntactic tradition (and in Cinque's system), "lexical categories" are the bottom of the clause, the bottom of *fseq*. The condition of moving only nouns, thus generalises to 'move a node containing the bottom of *fseq*'. If we combine that with 'move the candidate closest to the root of the tree', we obtain that in (43), the node 3P is the closest node (to the root of the tree) containing the bottom of *fseq*. Starke's proposal is thus that evacuation always targets the closest labelled node containing the bottom of the *fseq*, and if that fails to lexicalise, that same node pied-pipes its containing node.

In Caha's formulation, the same effect is obtained by restricting the target to the closest labelled non-remnant constituent. In (42) for instance, 3P is the only non-remnant node since 2P is a remnant in the sense that material has been extracted out of it (see Caha and Taraldsen Medová, this volume; Caha, this volume, for more details).

As a side-effect, this new way of thinking about evacuation movements allows for a new type of rescue movement: subextraction. Consider the tree in **Fout! Verwijzingsbron niet gevonden.**, a stage of the derivation where 4P has evacuated out of 3P, the resulting circled node has evacuated out of 2P, and the feature 1 is then merged.



Since only labelled nodes may be evacuation targets, 2P, 3P and 4P could in principle be evacuated, but the round or square nodes could not. In this situation, both Starke's and Caha's formulation have the effect that 4P ends up being the only candidate for evacuation. This then yields a new evacuation type: subextraction from within a "Spec". If the evacuation of 4P fails to create a lexicalisable structure, the mother node of 4P, marked by the gray circle, would be pied-piped, yielding a traditional spec-to-spec evacuation. If that in turn fails to create a lexicalisable structure, another degree of pied-piping is attempted, evacuating the entire square node – yielding a traditional complement movement. The new formulation thus deduces both the types of evacuation movements, and their relative ordering, while predicting new types of evacuation movement: sub-extraction, sub-sub-extraction, etc.⁵

By doing so, this new version of the algorithm preserves previous analyses, as it derives the same movements (spec-movement, the gray circle, and complement movement, the square), in the same order (always moving the spec before moving the complement). In addition to being more principled, preserving previous accounts, and predicting the existence of subextractions, this algorithm gives us new analytical options. Let's show this with a small case study of the Friulian conjugation, using Caha's formulation of the algorithm for concreteness:

- (45)
- (a) Merge F and lexicalise
 - (b) If fail, evacuate the closest labelled non-remnant constituent and lexicalise
 - (c) If fail, evacuate the immediately dominating constituent and lexicalise (recursive)

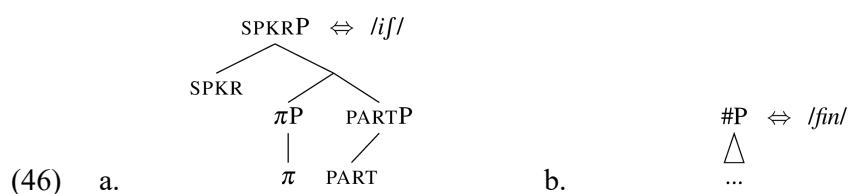
The Friulian conjugation of *bati* ‘to hit’ and *fini* ‘to finish’ are shown in Table 1.9.⁶

		FRIULIAN	
		<i>fini</i> ‘finish’	<i>bati</i> ‘hit’
SG	1	fin- <i>if</i>	bat
	2	fin- <i>if</i> -s	bat-s
	3	fin- <i>if</i>	bat

Table 9 Present tense singular of *bati* ‘hit’ and *fini* ‘finish’ in Tualis Friulian

The table shows that the two verb classes have the same overall pattern: there is no phi feature suffix in the first and third persons, while *-s* appears in the second person. The difference is that this happens after a single root in *bati* ‘hit’ whereas it happens after a multi-morphemic stem in *fini* ‘finish’. This difference makes the technical derivation for *bati* ‘hit’ inapplicable for *fini* ‘finish’. The new algorithm however allows a new derivational path, yielding R-a/R-a-x/R-a patterns (where R represents the root, and *-a* represents any number of suffixes).

The lexical entry for *if* contains a CLB, as shown in **Fout! Verwijzingsbron niet gevonden.(a)**, just like the lexical entry for *bat* in **Fout! Verwijzingsbron niet gevonden.(a)** did.



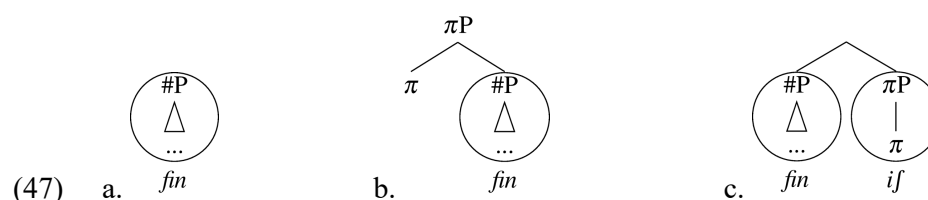
Notice an important difference: the entry for *if* does contain a constituent corresponding to the second person (π and participant), whereas the crux of the derivation of *bati* was that it does not contain any such constituent. Despite this, **Fout! Verwijzingsbron niet gevonden.(a)** will fail to lexicalise the second person, due to how the derivation is timed.

Simplifying Cortiula’s account, we posit the entry for the root *fin*- ‘finish’ as in (45)(b). This entry can only lexicalise features up to #P. Unlike *bat* ‘hit,’ *fin* cannot lexicalise π P, which is why *-if* has to appear. The lexicalisation table corresponding to this analysis is provided in Table 1.10. The table shows (with an asterisk on the line of 1SG) the special shape of the lexical entry for *-if*, which can shrink to the size of its left branch (π P), but not to the size of PART.

	TP	#	π	PART	SPKR
1SG	<i>fin</i>		<i>-if</i>	*	
2SG	<i>fin</i>		<i>-if</i>	s	
3SG	<i>fin</i>		<i>-if</i>		

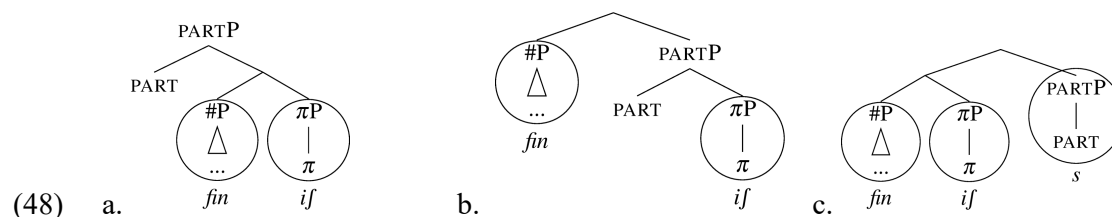
Table 10 Lexicalisation table of TF singular present tense of *if*-verbs

The derivation starts by assembling #P, see (47)(a). When π P is merged, as in (47)(b), *fin* is no longer a match. Following the Lexicalisation Algorithm, we must subextract the node which is closest to the root node, and which is at the same time labelled and non-remnant. In (47)(b), this is the #P node. After #P is evacuated from π P, we get the correct third person *fin-if*, see (47)(c).



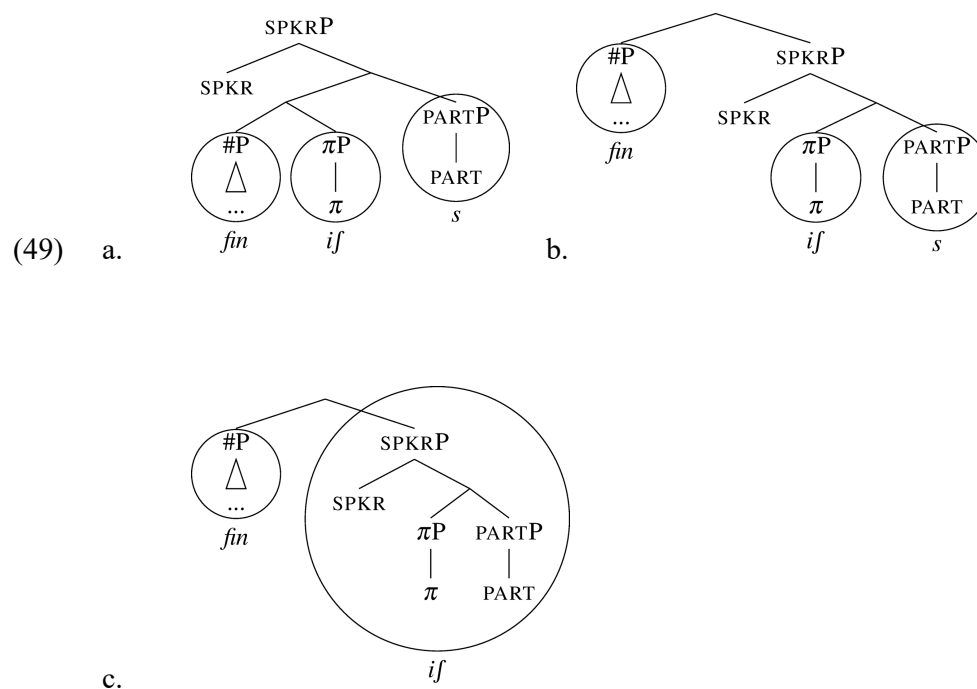
The derivation continues by merging PART, see (48)(a). Since there is no match, we must evacuate the closest non-remnant labelled node. This cannot be the complement of PART (since it is unlabelled), and it cannot be π P (since it is remnant), so it must be #P. Evacuating #P gives

us (48)(b), but there is no match for PARTP, since the entry for *-if* does not contain this constituent.



As a result, we revert to (48)(a), and we pied-pipe the node immediately dominating #P, which is the complement of PART. Evacuation of this node yields (48)(c), which can be successfully lexicalised, yielding the correct second person form.

Notice that so far, the pied-piping algorithm gives the exact same steps as the standard algorithm: we first moved the spec, and then the complement. The difference between the algorithms surfaces when we introduce the speaker feature, as in (49)(a).



Since there is no match for (49)(a), we need to move the closest non-remnant labelled constituent. Looking at (49)(a), this is not the spec of the complement, which has no label. Hence, at this point, the difference between the old and the new algorithm emerges, because the subextracting algorithm leads to the evacuation of #P. This is shown in (49)(b). We see that this movement strands the πP in between SPKR and PART, which is precisely the shape of the entry for *-i/*, which has a remnant πP stranded between these two features. As a result, the whole SPKRP can be lexicalised as *-i/* (49)(c). This derives the ABA-pattern displayed by the *-i/*-suffix in Tualis Friulian.

For further explorations of the revised algorithm, we refer the reader to Kasenov (2023), as well as Caha and Taraldsen Medová (this volume) and Caha (this volume).

7. Comparison with other approaches

It is beyond the scope of this chapter to provide a full and detailed comparison of the nanosyntactic approach with other approaches to syntax and morphology. We wish nevertheless to devote a few comments to this issue, in particular with respect to other approaches which we take to be particularly relevant to the current discussion. Keeping to the innovations in the Lexicalisation Algorithm, and the one involving their shape in particular, we can see an important difference with approaches that assume less restrictive formulations of the principles that determine when a lexical entry can lexicalise a given syntactic structure. We saw earlier (section 5) that the Matching Condition imposes not just a match in the number and type of features that lexical and syntactic representations must contain, but also a structural match in terms of constituent identity. This implies that a lexical tree of the shape $[[Y [Z]] X]$ is not a match for the syntactic structure $[X [Y [Z]]]$, even though the lexical tree and the syntactic tree have exactly the same set of features (namely, X, Y, and Z). A less restrictive alternative ignores the shape of the lexical entry, and just considers the question whether the set of features

stand in a subset-superset relationship, taking this to be a sufficient condition for lexicalisation. Such an alternative is proposed in Vanden Wyngaerd (2018), under the name of the Revised Superset Principle, which is used in the contribution by Baunaz and Lander (this volume). It is to be noted, however, that the types of phenomena discussed in section 5, like the Kipsigis inverse marking and the Friulian ABA patterns, cannot be accounted for under this alternative, less restrictive, approach in terms of the Revised Superset Principle. Crucially, a lexical entry that would contain the features X, Y, and Z could – under such an approach - always lexicalise a syntactic structure containing those features (or a subset of them), regardless of their structural arrangement.

Another approach uses spans rather than constituents (e.g. Ramchand 2008; Svenonius 2012, 2016; Merchant 2015). Spans are “a complement sequence of heads, normally in a single extended projection” (Svenonius 2012:1). Taking again the structure [X [Y [Z]]], the complement sequence of heads X-Y-Z (and any contiguous subpart thereof) is a span. In contrast, in the structure [[Y [Z]] X], with the (moved) constituent [Y [Z]], there is a span Y-Z and a (trivial) span X, but no span X-Y-Z, since the (moved) constituent is not the complement of X, nor the other way round. The spanning approach takes morphemes to be lexicalisations of spans: “[a] single morphological exponent (morpheme, for short) cannot spell out two heads (cannot “span” two heads) unless those heads are in a complement relation with each other” (Svenonius 2012: 2). This means that the results of section 5 cannot be derived in a spanning approach. To see this, consider the Kipsigis pattern of inverse number marking, where a noun like *peel* ‘elephant’ needs a suffix to mark singular number, but none to mark the plural. In our approach, this was done by assigning to such nouns a lexical entry with a CLB (see (33a) above), i.e. [[PL [SG]] N]. This allowed the root to lexicalise the features N, SG, and PL, while at the same time preventing it from realising the features N and SG in the singular, requiring a suffix to lexicalise SG. In the spanning approach, lexicalisation of the plural by the

root would require a span PL-SG-N (corresponding to a syntactic structure [PL [SG [N]]]), but a root with such a lexical entry would always be able to lexicalise the singular, and the need for a suffix in the singular would be unexplained.

Distributed Morphology (DM) is another framework using late-insertion (on top of early insertion of language-specific "bundles"). In contrast to Nanosyntax, it does not (usually) use phrasal lexicalisation, and instead all morphemes are inserted under terminal nodes. This means that an approach to allomorphy where root size conditions allomorphy, is unavailable since variable (syntactic) root size requires phrasal lexicalisation. Instead, DM takes extensive recourse to zero morphemes, and rules of contextual allomorphy to ensure their correct distribution. The combination of terminal spellout, zero morphemes, and rules of contextual allomorphy can also deliver the patterns of inverse marking, like the one we discussed for Kipsigis. However some of these patterns of distribution cannot be derived in DM without the postulation of accidentally homophonous morphemes, as argued in Caha (2024), Caha et al. (to appear) and Vanden Wyngaerd et al. (to appear).

ABA-patterns of the Friulian type are difficult to account for if the same decomposition of the agreement marker as in (39) above is adopted, with a set of features that stand in a containment relationship. A similar containment is crucial in Bobaljik's (2012) account for the absence of ABA-patterns in root suppletion in the triplet positive-comparative-superlative. Adopting such containment in the person-number sequence will make it consequently hard to allow the ABA-like distribution of the agreement marker. The fact that such patterns are widespread in verbal agreement marking (as noted above) may therefore lead the DM-theorist to abandon the containment hierarchy of (39), and in this manner develop an account for the occurrence of ABA-patterns (see e.g. Harbour 2008; 2016, and references cited there). Discussing the various avenues where this may lead is clearly beyond the scope of the present chapter. However, Taraldsen's chapter in the current volume compares Nanosyntax to

Distributed Morphology in his analysis of suppletive verbs in Romance. We also refer the reader to Caha (2018, to appear), and Barbiers, van der Wal and Vanden Wyngaerd (to appear), for more general discussion of the differences between DM and Nanosyntax.

8. Conclusion

This chapter has outlined the basic assumptions of Nanosyntax, focusing in particular on the Lexicalisation Algorithm. In addition to the basic lexicalisation-driven movements, the backtracking and complex specifier options of the Algorithm were shown to be a means of accounting for the phenomenon of paradigmatic levelling and prefixes, respectively. In addition, this chapter zoomed in on two recent evolutions within the framework. The first concerns the possibilities offered by lexical items with Complex Left Branches (CLBs), which considerably extend the empirical coverage of the nanosyntactic framework. The second evolution is a new logic for the Lexicalisation Algorithm proposed in Starke (2022), which simplifies it and as a side-effect adds subextractions as a type of rescue movement.

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Notes

¹ In Cinque (2005), the NP moves to the Spec of intermediate Agr projections. We ignore this here for simplicity.

² The chapter by Balsemin and Pinzin discusses an interesting case where the winner is determined by phonology, rather than by the Elsewhere Principle.

³ See Caha et al. (2019); Vanden Wyngaerd et al. (2021) for a discussion of how a particular root is selected among those that match. One idea is that roots do not compete in terms of the Elsewhere Condition, and one inserts the root that "one wants to talk about". This can be implemented by proposing that the conceptual representation restricts the candidate set of competing items, so that the root *tall* is the only root in the competition set (all other roots are irrelevant, since they are linked to the wrong concept).

⁴ There is some variation in the literature concerning which other feature combines with the newly required feature: some sources combine the last successfully lexicalized feature with the newly required feature, while other sources combine the newly required feature with the next in line. The latter option would require a decomposition of SPRL into two distinct features (as

proposed in De Clercq and Vanden Wyngaerd 2017). This issue is currently investigated and there is no final consensus on this topic.

⁵ Subextractions are a well-known syntactic type of movement, their use in nanosyntax continues the research programme of unifying syntax, morphology and beyond, by reusing and adapting standard operations and filters on structures. Within Nanosyntax, Wiland (2019) already proposed a subextraction step in the structure-building algorithm, ordering it after complement movement however. See also Pantcheva (2011) for a rather different derivational system, within a larger umbrella of Nanosyntactic investigations.

⁶ The table ignores certain phonological processes which convert the underlying form of the 2SG into the surface form *finis*. See Cortiula (2023) for discussion.

3 Zero morphology in change-of-state verbs

P. Caha, K. De Clercq, and G. Vanden Wyngaerd

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Research Article

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Zero morphology and change-of-state verbs

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Abstract: This paper discusses a theory of conversion (zero derivation) in terms of phrasal spellout. In this approach, there are no zero morphemes. Instead, the ‘silent’ meaning components are pronounced cumulatively within overt morphemes. As an empirical case, we discuss adjective/verb ambiguity as in *narrow*. As verbs, these roots have both an inchoative and a causative sense. Following Ramchand (2008), we assume that such deadjectival causatives contain three parts: the adjective denoting a state, a change-of-state component PROC, and a causative component INIT. Adopting a Nanosyntax approach, we propose that verbs like *narrow* spell out a complex node with all these abstract heads. The ambiguity between the inchoative, causative and adjective falls out as a consequence of the Superset Principle (Starke 2009), which states that a lexical entry can spell out any subtree it contains. Since both the inchoative sense and the adjective sense correspond to proper parts of the causative one, we derive these readings without the need to postulate zeroes. We show how these assumptions allow us to capture a broad range of patterns, focussing mainly on English and Czech.

Keywords: deadjectival verbs, change-of-state verbs, causative-inchoative alternation, nanosyntax, phrasal spellout.

1 Introduction

This article proposes an analysis of adjective-verb conversion. The phenomenon is illustrated in (1).

- (1) a. *The road is **narrow**.*
 b. *The road **narrow**-ed.*
 c. *The workers **narrow**-ed the road.*

Conversion refers to the fact that the adjective *narrow* in (1a) has the exact same form as the verb *narrow* in (1b-c). The verb can have either an inchoative (change-of-state) reading (1b), or a causative one (1c).

Alongside the conversion cases in (1), deadjectival verbs in English sometimes require an overt suffix, as in (2).

- (2) a. *The road is **wide**.*
 b. *The road **wide-n**-ed.*
 c. *The workers **wide-n**-ed the road.*

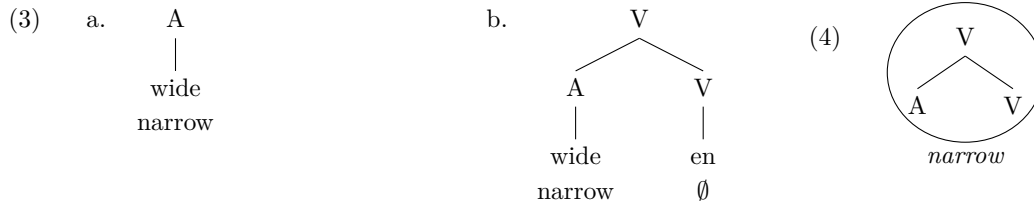
Interestingly, the verbs show the same ambiguity between an inchoative and a causative reading as the unsuffixed verbs.

The standard approach to deadjectival verbs attributes to the examples (1) and (2) the exact same bimorphemic structure, (3a,b), where the verb is derived from the adjective. The two classes differ only in whether the verbalising suffix is overt or not.

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The idea that we pursue in this paper is that in a case like *narrow*, the apparent zero marking of the verb arises as a consequence of phrasal spellout, where the relevant phonology is associated to a complex syntactic object, as indicated by the circle in (4). In developing the analysis, we also show how it accounts for the inchoative-causative ambiguity of the derived verbs, which we can observe in both types, the unsuffixed and the suffixed ones.

The paper is organised as follows. In section 2, we present some of the data that motivate the approach. Section 3 introduces the prerequisites for our analysis. The main proposal is described in section 4. Section 5 discusses suffixal marking, and ends with some predictions regarding the typology of morphological marking in the triplet adjective-inchoative-causative. Section 6 examines how these predicted patterns are realised in Czech deadjectival verbs. Section 7 briefly touches upon the topic of anticausatives.

2 The data

We start by presenting a representative selection of English deadjectival verbs in Table 1. In the first column, the table contains verbs that are zero-derived from adjectives. The other columns contain suffixed deadjectival verbs and one group of prefixed deadjectival verbs, as indicated in the column headings.

Tab. 1: Deadjectival verbs in English

	<i>-en</i>	<i>-ify</i>	<i>-ise</i>	<i>en-</i>
cool	tighten	solidify	generalise	enlarge
narrow	widen	prettify	formalise	enfeeble
open	shorten	simplify	americanise	enrich
thin	sharpen	humidify	sexualise	
dim	slacken	acidify	christianise	
tame	brighten	fluidify	commercialise	
blind	cheapen	falsify	conceptualise	
warm	coarsen	Frenchify	actualise	
clean	dampen	intensify	annualise	
empty	darken	uglify	grammaticalise	
clear	deaden	diversify	brutalise	
dry	deafen	greenify	centralise	

Our main point is that it is to a large extent an arbitrary property of the adjective whether it forms a zero-derived verb, or whether it requires a suffix, and if it requires an affix, which affix.

Let us start by noting that there are a couple of phonological restrictions to be observed. For instance, the suffix *-en* normally attaches to monosyllabic adjectives ending in a plosive, fricative or affricate (Plag 2003: 117-118, Carstairs-McCarthy 2002: 55-56). However, as Carstairs-McCarthy (2002: 87) noted, there are adjectives that have the right phonology and nevertheless do not take *-en*. The adjective *wet* is an example, because **wetten* does not exist, but the zero-derived *wet* does. In sum, while the phonological rule can explain why **greenen* will never be formed (*green* does not end in a plosive or fricative), additional factors are at play for adjectives that do satisfy the condition.

Something similar applies to the suffix *-ify*: this suffix attaches only to bases that are either monosyllabic, stressed on the final syllable or end in unstressed /ɪ/. However, this does not mean that *-ify* can attach to all adjectives that obey these phonological conditions. A case in point is the adjective *empty*, which allows for a zero-derived verb, contrasting with *pretty*, which does not.¹

In sum, even though there are some phonological conditions that restrict the formation of deadjectival verbs in English, it is impossible to predict which adjective allows for zero marking and which one requires an affix. As a result, we conclude that which class an adjective belongs to is to some extent arbitrary, and this information therefore needs to be learned and stored for each adjective in the lexicon.

Similarly, it is not predictable which suffix will appear on a suffixed deadjectival verb. For instance, it is not clear why *-ify* can derive *falsify*, while *-en* cannot derive **falsen*, even though the phonological conditions would allow for either of these suffixes (compare *coarsen*).²

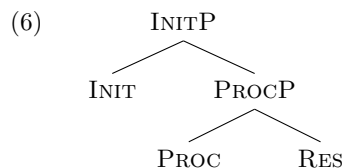
The next point we want to bring out is that all the affixes shown in the table show the inchoative-causative ambiguity. For reasons of space, we only illustrate this for the suffix *-en*, (5):

- (5) a. *Her stomach tightened.*
b. *She tightened the lid.*

In this paper, we present a Nanosyntactic account of how the difference between zero derivation and suffixation is encoded in the lexicon. Our main aim is to show that the Nanosyntactic framework, where zero-derivation is modeled by means of phrasal spellout, has the right properties to model this distinction in a way that allows us to provide an explanatory account not only for English, but also for other languages. Moreover, the very same type of account will allow us to explain the causative-inchoative ambiguity, which is found in English, but largely absent in Czech, as we discuss in Section 6.³

3 Prerequisites

In this section, we lay out the prerequisites for our analysis. Specifically, we adopt here the structure of deadjectival verbs as proposed in Ramchand (2008). In Ramchand's work, verbs in general are decomposed into a series of heads which she refers to as Initiation, Process, and Result. They are organised hierarchically as in (6):



The INIT head introduces the causation event. The PROC head specifies the nature of the change or process, whereas RES gives the result of the event. Each of these heads may license a specifier (not shown in (6)):

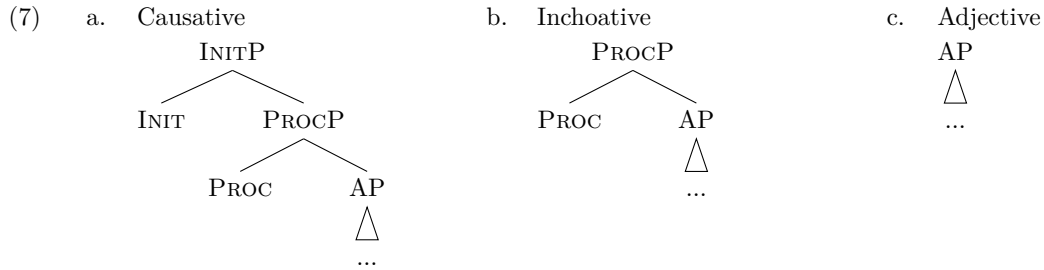
¹ The base alternations of adjectives to which *-ise* attaches are complex and beyond the space of this paper (see Plag 2003: 117–118, Plag 1999). There is also a fourth potential verbalising suffix *-ate*, which we leave out of consideration here. This suffix often attaches to truncated roots, especially if the base ends in two unstressed syllables (Carstairs-McCarthy 2002: 55, Plag 2003: 117).

² *-ise/-ify/en-* can also turn nouns into verbs, as in *terrorise*, *beautify* and *empower*, respectively (Carstairs-McCarthy 2002: 55). The conceptual basis involved in these verbs can, in certain cases at least, generalise over the noun/adjective distinction, e.g. both the noun *beauty* and the adjective *beautiful* involve the same scale. Francez and Koontz-Garboden (2015) refer to this shared conceptual basis as a *property concept*.

³ Since the main focus of this paper is on zero-derivation, we only want to capture the distinction between zero-derived adjectives and the rest. The allomorphy of derivational affixes won't be addressed here for reasons of space (see Caha, De Clercq, and Vanden Wyngaerd 2019 and De Clercq and Vanden Wyngaerd 2019 for two different types of approaches compatible with our proposal).

the specifier of the causation event is the initiator, the specifier of the process is the undergoer, and the specifier of the result is the resultee (Ramchand 2008: 40). For example, in the sentence *Mary gave the book to Bill*, *Mary* is the initiator who brings the event about, *the book* is the undergoer of the Process (it changes possession) and ‘Bill having the book’ is the Result. The INIT and RES heads are optional in that some verbs lack them. For instance, the verb *get* in *Bill got the book* lacks the INIT component.

Against this background, Ramchand (2008: 90,108) analyses deadjectival verbs as in (7). The trees depict the structure of the causative, the inchoative and the adjective, respectively. Recall that each of PROC and INIT brings along an argument position, which accounts for the argument-structure alternations.



Building on Hay, Kennedy, and McNally (1999), Ramchand (2008: 90) proposes that verbs like *dry* are “a special kind of process verb where the degree of verbal change is mapped onto a property scale of some sort (derived from a basic adjectival meaning). Thus, in their intransitive use, they are classic *proc* verbs, with the single argument being an UNDERGOER.” This is shown in (7b), which is just a special instance of the generally adopted structure (3b). Note that like Ramchand, we understand the AP in the complement of PROC as representing not directly the positive-degree adjective, but rather a scale on which both the inchoative and the positive adjective are based.⁴

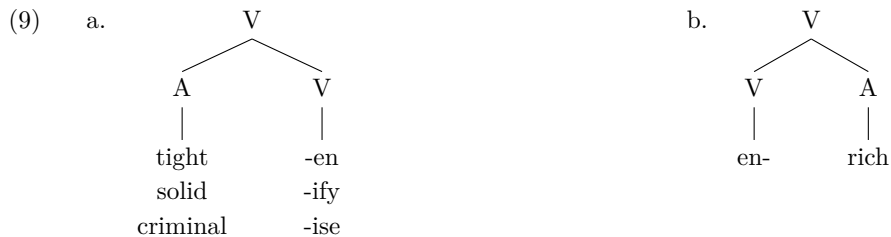
In Ramchand’s system, the causative in (7a) has an additional zero-marked INIT head, which introduces the Initiator argument. Verbs such as *narrow* in their causative use thus contain minimally two components of meaning that apparently receive no overt marking (i.e., INIT and PROC).

In sum, the syntactic structures in (7) are based on the two assumptions given in (8):

- (8) a. The verb contains the adjective.
b. The causative contains the inchoative.

The evidence in support of these assumptions is both morphological and semantic.

We start with morphological evidence in support of (8a). We have already discussed some of the evidence in Table 1, which shows how verbs are derived from adjectives by various affixes. The structural representation of these cases in (9) clearly shows how the verb is more complex than the adjective.



This type of evidence can be replicated for many languages, where verbs related to adjectives are typically morphologically more complex than the adjectives.

⁴ We refer the reader to Vanden Wyngaerd, Starke, De Clercq, and Caha (2020) for a discussion of some relevant facts concerning the relationship between scales and positive degree adjectives based on that scale.

Semantically as well, the meaning of the verb contains that of the adjective. For example, the verb ‘to open’ means ‘to become open’, or ‘to cause to become open.’ This type of paraphrase generally works quite well for the verbs in Table 1. In some cases, a paraphrase containing a comparative of the adjective is more appropriate. For instance, atelic sentences such *The soup cooled for hours* do not entail that the soup ultimately became cool, but they nevertheless entail a change along the relevant scale. As we have made it clear above, this is compatible with our proposal, since the AP at the bottom of the tree does not necessarily correspond to the positive degree, but rather to the scale along which the change proceeds.

Let us now turn to the second assumption in (8b). This is also supported by semantic evidence. Intuitively, *to open* in its causative sense means ‘to cause to open’, with the second occurrence of *open* being the inchoative one (but see Harley 2012 for a different view). Lundquist, Corley, Tungseth, Sorace, and Ramchand (2016: 2) put this more formally in what they call the *Causational Entailment*:⁵

- (10) *Causational Entailment*
 $\forall x \forall y [\text{CAUSE} (x, \text{INCH}(\text{Pred}(y))) \rightarrow \text{INCH}(\text{Pred}(y))]$

This entailment is what accounts for the (semantic) deviance of (11).

- (11) *#John broke the glass, but the glass didn't break.*

As far as the morphology is concerned, the picture is more complex. There are languages where the morphology supports the claim that the causative contains the inchoative. We reproduce some empirical evidence from Haspelmath (1993: 91) in (12). The table shows cases where the causative is morphologically more complex than the inchoative. We refer to this as the transparent pattern.

- (12)
- | | | | | |
|----|----------|----------------------------|---------|--------|
| a. | Georgian | <i>du-s</i> | ‘cook’ | (INCH) |
| | | <i>a-du-eb-s</i> | ‘cook’ | (CAUS) |
| b. | French | <i>fondre</i> | ‘melt’ | (INCH) |
| | | <i>faire fondre</i> | ‘melt’ | (CAUS) |
| c. | Arabic | <i>darasa</i> | ‘learn’ | (INCH) |
| | | <i>darrasa</i> | ‘teach’ | (CAUS) |

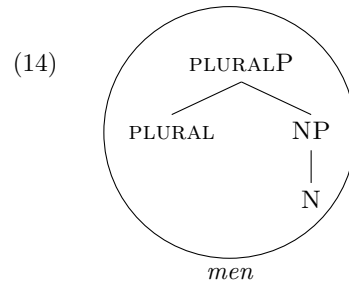
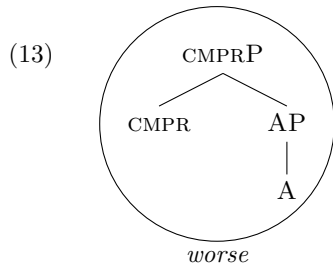
As the discussion unfolds, we will also get to see an inverse pattern, which is traditionally referred to as the *anticausative*, where the inchoative is morphologically more complex than the causative. We come back to anticausatives in section 7. For now, we focus on the causative-inchoative entailment and the transparent morphological pattern as support for the structures in (7).

4 Zero-marking as phrasal spellout

As highlighted in the introduction, our main goal is to explore the idea that there are no zero morphemes used in the derivation of deadjectival verbs. Our claim is that zero morphemes are an illusion, arising when grammatical meanings expressed by the purported zeroes are actually realized (in a portmanteau fashion) by the root and/or other morphemes.

The idea that roots can (in addition to the conceptual meaning) also express adjacent grammatical categories is perhaps best seen in cases of root suppletion (*bad–worse*; *man–men*). These cases can be understood as instances of a scenario where the suppletive root realizes both the lexical category (A or N) and the relevant grammatical category (CMPR or PLURAL). We show this in (13) and (14).

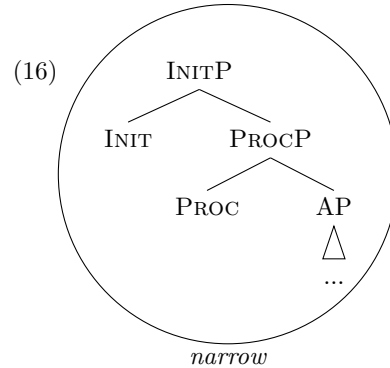
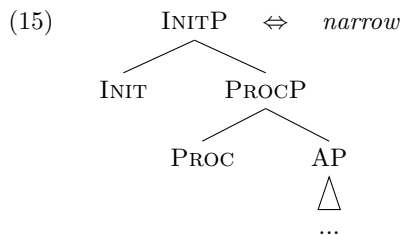
⁵ We slightly changed the formulation, with a universal rather than an existential quantifier taking scope over the entire conditional, to match what we take to be the intention of the entailment.



An alternative analysis could (of course) conclude that the relevant grammatical categories in these forms have a “zero allomorph.” While we are aware of the fact that there is no direct link between suppletion and the absence of regular morphology (e.g., *bett-er*), we think that the frequent absence of regular morphology with irregular roots such as *worse* or *men* is not accidental. Therefore, the intuition we shall follow is that whenever regular markers are absent with particular roots (such as in (13) and (14)), this is because the root already realises the relevant meanings, thereby blocking the appearance of regular morphology.⁶

In the remainder of this paper, we extend the type of analysis in (13) and (14) to deadjectival verbs. In the technical implementation of this idea, we follow the Nanosyntax approach (Starke 2009, 2018). Nanosyntax is a Late-Insertion theory of morphology, where the syntactic structure is assembled first, and then it is spelled out using lexical entries. In this conception, lexical entries are understood as stored links between syntactic representations on the one hand, and a phonology and/or a concept on the other hand. Crucially, the syntax part of the lexical entry consists of a full syntactic tree (rather than a terminal), similarly to what we saw in (13) and (14).

With this background in place, consider now the lexical entry for *narrow* in (15), which links the phonology /'næɹəʊ/ (represented by means of plain spelling) with a particular syntactic tree:⁷



When syntax builds the structure of a causative verb, as in (16), the lexical tree in (15) is identical to it, and spellout is successful. We represent this by placing a circle around the syntactic structure in (16). Phrasal spellout thus obviates the need for zero morphemes: the **INIT** and **PROC** heads do not dominate zero morphemes, but the lexical entry of *narrow* is such that it may cumulatively realise several heads, including **INIT** and **PROC**.

The next question is how the lexical entry of *narrow* can also realise the structure of an inchoative verb and an adjective. The phrasal-spellout model treats this as an instance of syncretism, i.e., different meanings expressed by the same form. Technically, this is achieved by proposing that lexical entries may

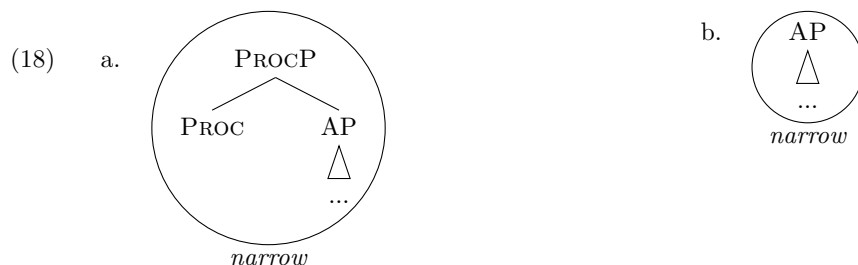
⁶ The issue of why suppletive roots are sometimes compatible with regular morphology, as in *better*, is addressed in Caha et al. (2019). We do not discuss this here for reasons of space, our main point being to motivate the idea that roots can (at least sometimes) realise grammatical meanings.

⁷ Our entries in this paper systematically ignore the concept associated with the entry for *narrow*, since these are not relevant to the current discussion.

lexicalise various different syntactic trees – under the condition that the syntactic tree is contained in the lexical entry. This condition is known as the Superset Principle:

- (17) Superset Principle (*Starke 2009*)
A lexically stored tree L matches a syntactic node S iff L contains the syntactic tree dominated by S as a subtree

The result of adopting the Superset Principle is that the entry for *narrow* in (15) can also spell out the inchoative structure (18a) and the adjective structure (18b), because these trees are contained in (15).



In sum, the mechanism of phrasal spellout gives us an elegant way of representing two related phenomena: on the one hand, the zero marking of certain types of deadjectival verbs, and on the other hand the syncretism between the inchoative and the causative verb.

5 Suffixal marking

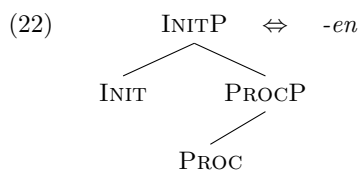
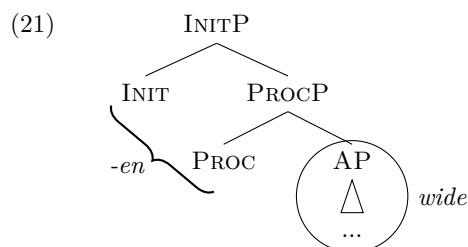
As we saw in Table 1, only a subset of adjectives allows for zero marking of the corresponding verb. A lexical item like *wide* contrasts with *narrow* in that *wide* is only an adjective and not a verb. The contrast is illustrated in (19).

- (19) a. *The workers narrowed/*wided the road.*
 b. *The road narrowed/*wided.*

Recall that whether an adjective behaves like *narrow* or *wide* is a matter of lexical idiosyncrasy, i.e., it has to be learned (and stored) for each relevant lexical item. In the phrasal-spellout model, this is achieved simply by associating an adjective like *wide* to an AP only:

- (20) AP $\Leftrightarrow$ *wide*

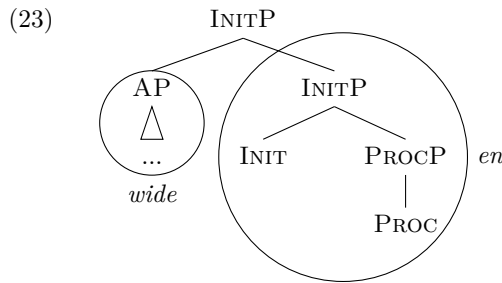

Since the lexical tree (20) does not contain the syntactic tree of a verb, such a lexical item will not be able to function as a verb: the root *wide* will need help from an additional morpheme to realise all meaning components, as informally shown in (21).



What is the lexical entry for *-en*? Recall that in Nanosyntax, lexical entries are understood as memorized links between syntactic structures (constituents) and phonology. Therefore, by proposing that the suffix *-en* spells out the heads PROC and INIT, as shown informally in (21), we are led to assign to it the lexical entry (22), which corresponds to a constituent containing these two features.

The geometry of the lexical entry reflects the structure (21): specifically, the entry (22) is identical to (21) minus the part of the structure that is spelled out as *wide*. In effect, the structure (21) literally divides into two parts, where one part is spelled out by *wide* and the other by *-en*.

But now a problem arises. Specifically, the lexical tree in (22) does not contain the syntactic tree (21), nor any subpart of it, as a subtree. Therefore, the Superset Principle is not met, and the heads PROC and INIT cannot be spelled out by (22) as long as the structure is as given in (21). This problem is resolved by moving the AP node to the specifier of the INITP, yielding (23). This movement is driven by the need to lexicalise the features INIT and PROC.



The INITP in (23) is now an exact match for the lexical entry of the suffix, and spellout is successful. Moreover, the affix *-en* now appears where we want it, namely as a suffix following the root.⁸

The inchoative version of *widen* works analogously, except that the syntactic derivation lacks the INIT head, (24a). Movement of the AP to the left yields (24b). Since the remnant PROCP is a subtree of (22), *-en* can spell out the PROCP in (24b).



It is relevant to note that if there was a suffix in the English lexicon that would only be specified for the circled PROCP, this suffix would also qualify as a match for (24b), and it would, in fact, take precedence over the ambiguous causative/inchoative *-en*, on the grounds that it is more specific. There is no such suffix in English, but we will be led to propose such a suffix in Czech in the next section.

To close off this section, we consider some implications of our theory for the typology of marking in the triplet adjective – inchoative – causative. So far, we have discussed two cases, instantiated by the English adjectives *narrow* and *wide*, respectively. However, the scenarios allowed for by the system just developed are wider. We show some of the predicted scenarios in Table 2.

This table should be read as follows. The top row represents the features found in various forms. The scenario where the root is a syntactic adjective is defined by the feature A. This scenario is represented on the lines marked by (a) (i.e. 1a, 2a, etc.). The grey cells stand for heads which are not present in the derivation, namely PROC and INIT. To derive an inchoative verb, we add PROC to the adjectival structure, as shown on lines (b). The causative verbs add INIT (lines (c)).

⁸ In Nanosyntax, such movements are enforced by the so-called spellout algorithm, i.e. they are spellout-driven. We presuppose the algorithm, but we can't go into its details for reasons of space, see Starke (2018), Caha et al. (2019).

Tab. 2: Five scenarios

	A	PROC	INIT	(typological pattern)
1a.				labile Eng. <i>wid-en</i>
1b.	root			
1c.		suffix		
2a.				equipollent
2b.	root	suffix1		
2c.		suffix2		
3a.				causative
3b.	root	suffix1		
3c.		suffix1	suffix2	
4a.				causative
4b.	root			
4c.			suffix	
5a.				labile Eng. <i>narrow</i>
5b.	root			
5c.				

The horizontal sections in the table (numbered 1-5) represent various root and suffix types predicted by our analysis, ordered according to root size, from the smallest (AP-size) roots in the top section to the largest (INITP-size) roots in the bottom one. A lexically small root like *wide* is shown in section 1. It can realise no more than the adjective feature A. When it functions as a verb, it needs a suffix, as shown on lines 1b and 1c, which correspond to the derivations we have just seen. The suffix is the same for the causative and for the inchoative verb, giving rise to syncretism between the causative and the inchoative.

We show an INITP-size root, like *narrow*, in section 5 of the table. As explained above, the Superset Principle ensures that such roots are three-way syncretic: they can realise each member of the triplet adjective-inchoative-causative, as shown on the lines 5a, 5b, and 5c, respectively.

In the final column, the table contains the names given in Haspelmath (1993) to different types of morphological marking of the causative-inchoative alternation. The two English patterns (given in section 1 and 5 of the table) are called ‘labile,’ showing syncretism between the causative and the inchoative. The reason why we have two different labile patterns is that Haspelmath does not consider the derivational relation to the adjective. As a result, his classification slightly deviates from ours, but we include Haspelmath’s terminology here to show that we are dealing with well-established typological patterns.

At least three additional patterns of marking are predicted by our system, shown in sections 2, 3, and 4 of the table. The first of these (section 2) is one where we have different suffixes for the inchoative and the causative, a possibility already pointed out in the discussion surrounding (24). This is a scenario where one suffix (suffix1) lexicalises just PROC, whereas a different suffix (suffix2) lexicalises both INIT and PROC, such that suffix1 (marking the inchoative) is replaced by suffix2 in the causative. This pattern is labelled the equipollent pattern in Haspelmath (1993), since it involves a common stem, to which different markers are added in the inchoative and the causative.

The second predicted pattern is in section 3 of the table. In this scenario, the causative suffix lexicalises only INIT and not PROC. Therefore, a causative verb is derived by stacking a causative suffix on top of an inchoative one, as shown on line 3c of the table.

The third additional pattern (section 4) is one where the root is of size PROCP, i.e. larger than a mere adjective, but smaller than a causative verb. Such roots are predicted to have a zero-derived inchoative, but a suffixed causative. Both patterns 3 and 4 are called causative in Haspelmath (1993), since they involve a causative that is derived from the inchoative.⁹

⁹ Haspelmath distinguishes two further patterns: an anticausative pattern, to which we return in section 7, and a suppletive pattern (e.g., *fall-drop* and *kill-die*), which we take to be a subtype of the *narrow* pattern.

In the remainder of this paper, we provide evidence for some of the additional patterns by discussing deadjectival verbs in Czech. Specifically, we argue that Czech instantiates Patterns 1, 2, and 4 of Table 2. Before we turn to this detailed discussion, we want to briefly consider evidence for the existence of all three of the predicted additional patterns in Turkish. The relevant patterns are shown in Table 3.¹⁰

Tab. 3: Turkish patterns of causative-inchoative marking

	adj	inch	caus	
Pattern 2	kir-li	kir-le- n -(mek)	kir-le- t -(mek)	'dirty'
Pattern 3	iyi	iyi- leş -(mek)	iyi- leş-tir -(mek)	'good'
Pattern 4	kuru	kuru-(mak)	kuru- t -(mak)	'dry'

The numbering of the patterns links them to the relevant sections in Table 2. The relevant suffixes (which we discuss below) are in bold; the bracketed suffix is the infinitival ending.

Starting with Pattern-2 roots, we can see that they have different suffixes for the inchoative and the causative, with the causative *-t* replacing the inchoative *-n*.¹¹ Pattern 3 roots have a causative suffix *-tir* that stacks on top of the inchoative *-leş*. Pattern 4 roots have a zero-marked inchoative and the causative suffix *-t*. The particular interest of Turkish resides in the evidence it provides for Pattern 3, which is one that Czech (to be discussed in detail in the next section) does not have. Pattern 3 is also interesting in that we consider it to be one that faithfully reflects the underlying meaning composition, with the inchoative derived from the adjective, and the causative further derived from the inchoative.

6 Three inch/caus scenarios in Czech

We start by describing causative verbs in Czech, because the picture here is rather simple: there is a single suffix *-i*, which derives causative verbs. We turn to inchoatives in Section 6.2.

6.1 The causative

Causatives are a relatively productive category in Czech. Using the Czech National Corpus (<https://www.korpus.cz>), where all our examples come from, we have collected causatives of about 250 different adjectival roots. An example of a causative is given in (25). (25a) is an adjective, (25b) a verb.

- | | | | | | |
|------|----|---|------|----|--|
| (25) | a. | <i>tup</i> - <i>ý</i>
blunt -AGR | (26) | a. | <i>tich</i> - <i>ý</i>
silent -AGR |
| | b. | <i>tup</i> - <i>i</i> - <i>l</i>
blunt -CAUSE -PST | | b. | <i>tiš</i> - <i>i</i> - <i>l</i>
silent -CAUSE -PST |

The causative suffix triggers palatalisations, as illustrated in (26), where (in IPA notation) *x* goes to *j*.¹²

¹⁰ We are grateful to Ömer Demirok for pointing this out to us. Turkish also has an anticausative pattern, which is not shown in the table (see section 7).

¹¹ The adjective 'dirt-y' is denominal, i.e. the root *kir* 'dirt' takes an adjectivising suffix *-li* in the ADJ column, and a general verbalising suffix *-le* in the two verbal columns. What is crucial is that the difference between the inchoative and the causative resides in the different suffixes *-n* vs. *-t*. We are grateful to Utku Türk for a useful discussion.

¹² Palatalisation is a regular process in Czech. For example, the NOM.PL.MASC.ANIM of the adjective *tich-ý* 'silent' is *tiš-í*, the comparative adverb is *tiš-eji* 'more silently'.

An interesting fact is that many adjectives require an aspectual (perfectivising) prefix in the causative. We illustrate this in (27). (27a) is the adjective *snadn-ý* ‘easy.’ (27b) is an unprefixated causative (which does not exist), (27c) is a prefixed causative. Building on previous work, Ramchand (2008: 140-1) proposes that such prefixes are analogous to English particles and occupy the (generally optional) RES head. We do not know why a prefix is obligatory with some verbs, but we agree with Ramchand that their presence (while sometimes required) is orthogonal to the patterns of causative/inchoative formation. In order for the prefixes not to influence our data, we illustrate each class with an unprefixated verb whenever possible. When we use prefixed verbs, we keep the prefix constant across the causative and the inchoative. That way, the role of unpredictable obligatory prefixes is kept to the necessary minimum.

- | | |
|--|--|
| <p>(27) a. <i>snadn -ý</i>
easy -AGR</p> <p>b. *<i>snadn -i -l</i>
easy CAUSE -PST</p> <p>c. <i>u- snadn -i -l</i>
UP easy -CAUSE -PST</p> | <p>(28) a. <i>tup -i -l</i>
blunt -CAUSE -PST</p> <p>b. <i>tup -í -m</i>
blunt -CAUSE -1.SG</p> <p>c. <i>tup -e -ní</i>
blunt -CAUSE -NMLZ</p> |
|--|--|

Another property of the causative suffix is that it shows allomorphy, as shown in (28). In the present tense, the vowel lengthens (28b), and in nominalisations, it is replaced by *-e* (28c). We note the variation for completeness’ sake, and we will only look at the past participle (28a) as the relevant form from now on. The reason is that the past participle is the most informative form from the perspective of verb-class identification. For example, some inchoative deadjectival verbs have an inchoative *-ě* in the past participle, so they clearly differ from the causatives which have an *-í*, see (28a). However, such inchoatives have the same present tense *-í* as causatives, see (28b). Therefore, looking at the past participle makes most sense.¹³

6.2 The inchoative

Having discussed the formation of causatives, we turn to their inchoative counterparts. Here, the situation is more complex, in that at least four different classes may be distinguished. In this section, we focus on three classes, which instantiate the predicted patterns in Table 2. We repeat the table here as Table 4, but with the three Czech verb classes added.

The Czech Class I is characterised by the fact that both the causative and the inchoative are marked by the same suffix (namely *-i-*), which is analogous to the English *wid-en* class. An example is the verb derived from the adjective *levn-ý* ‘cheap’. In (29a), we give a causative sentence based on this verb, and in (29b) we present the inchoative counterpart. The verb is the same in both sentences.

- (29) a. *Škoda Auto z- levn -i -la své dva hlavní modely.*
Škoda auto PFX- cheap -CAUS -PST its two main models
‘Škoda Auto has made its two main models cheaper.’
- b. *Vodka z- levn -i -la.*
vodka PFX- cheap -INCH -PST
‘Vodka got cheaper.’

Table 5 lists the roots in our database showing this inchoative-causative syncretism. This is a relatively minor class and we could therefore only illustrate it using a prefixed verb in (29).

The analysis of this class is analogous to English *wide*. The lexical entry of the root is of the smallest size (AP), see (30a). The suffix *-i* is then specified for the heads INIT and PROC as indicated in (30b). The (partial) derivation of the causative verb is shown in (31).

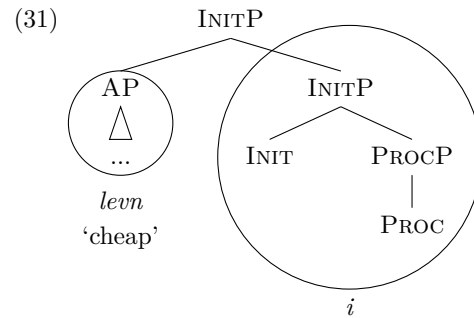
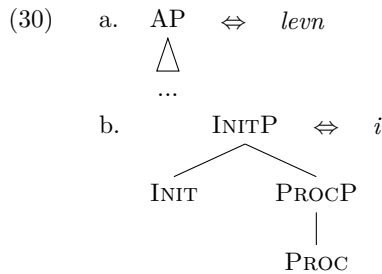
¹³ The allomorphy of the post-root causative/inchoative marker can be accounted for by assuming that in addition to the causative/inchoative meaning, these morphemes may also realise tense and aspect.

Tab. 4: Three verb classes in Czech

	A	PROC	INIT	
1a.				Eng. <i>wid-en</i> Cz Class I
1b.	root			
1c.		suffix		
2a.				Cz Class II
2b.	root	suffix1		
2c.		suffix2		
3a.				
3b.	root	suffix1		
3c.		suffix1	suffix2	
4a.				Cz Class III
4b.				
4c.	root		suffix	
5a.				Eng. <i>narrow</i>
5b.				
5c.	root			

Tab. 5: Class I verbs

gloss	adj	inch/caus
expensive	drah-ý	z-draž-i-l
cheap	levn-ý	z-levn-i-l
firm	pevn-ý	z-pevn-i-l
slow	pomal-ý	z-pomal-i-l
fast	rychl-ý	z-rychl-i-l
calm	mírn-ý	z-mírn-i-l
intensive	intenzivn-í	z-intenzívni-l



Class II differs from Class I in that it has a dedicated suffix *-ě* for the inchoative. A case in point is the adjective *hutn-ý* ‘dense,’ illustrated in (32). Specifically, (32a) contains the causative verb with *-i*. Interestingly, the inchoative in (32b) no longer has *-i*, but *-ě*:

- (32) a. *Půdu jsem hutn -i -l dostatečně.*
 soil.ACC AUX.1.SG dense -MAKE -PST sufficiently
 ‘I made the soil sufficiently dense.’
 b. *Zvuk hutn -ě -l a koncert ostře gradoval.*
 sound dense -BECOME -PST and concert sharply grew.in.intensity
 ‘The sound was getting denser and the concert was growing in intensity.’

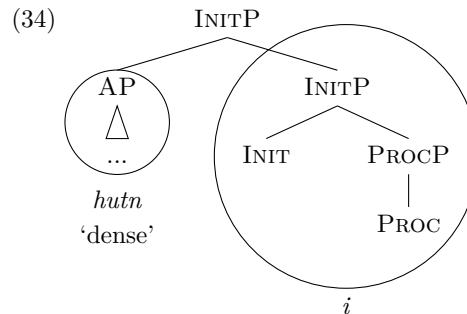
Table 6 lists all the roots that show the *-ě* suffix in the inchoative.

Tab. 6: Class II verbs

gloss	adj	inch	caus
coarse	drsn-ý	z-drsn-ě-l	z-drsn-i-l
clear	jasn-ý	z-jasn-ě-l	z-jasn-i-l
smooth	jemn-ý	z-jemn-ě-l	z-jemn-i-l
compact	hutn-ý	z-hutn-ě-l	z-hutn-i-l
liquid	kapaln-ý	z-kapaln-ě-l	z-kapaln-i-l
calm	klidn-ý	z-klidn-ě-l	z-klidn-i-l
beautiful	krásn-ý	z-krásn-ě-l	z-krásn-i-l
cultural	kulturn-í	z-kulturn-ě-l	z-kulturn-i-l
mighty	mohutn-ý	z-mohutn-ě-l	z-mohutn-i-l
dead	mrtv-ý	z-mrtv-ě-l	z-mrtv-i-l
national	národn-í	z-národn-ě-l	z-národn-i-l
nervous	nervózn-í	z-nervózn-ě-l	z-nervózn-i-l
affectionate	něžn-ý	z-něžn-ě-l	z-něžn-i-l
ugly	oškliv-ý	z-oškliv-ě-l	z-oškliv-i-l
transparent	průhledn-ý	z-průhledn-ě-l	z-průhledn-i-l
exact	přesn-ý	z-přesn-ě-l	z-přesn-i-l
strict	přísn-ý	z-přísn-ě-l	z-přísn-i-l
dark	temn-ý	z-temn-ě-l	z-temn-i-l
immobile	nehybn-ý	z-nehybn-ě-l	z-nehybn-i-l
unsure	nejist-ý	z-nejist-ě-l	z-nejist-i-l
uncalm	neklidn-ý	z-neklidn-ě-l	z-neklidn-i-l

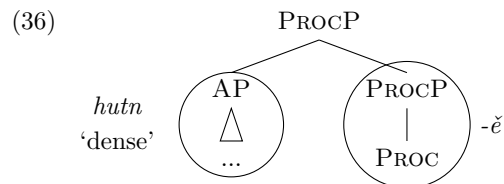
We analyse these roots as being of the same size as the Class I roots, i.e. they can realise an AP, see (33). As a result, they need the suffix *-i* to spell out the verbal heads PROC and INIT, see (34).

(33) AP $\Leftrightarrow$ hutn
 $\triangle$
 ...



Different from the Class I verbs, however, these verbs make use of a specialised inchoative marker spelling out just PROC. We show the lexical entry of this marker in (35).

(35) PROCP $\Leftrightarrow$ -ě
 |
 PROC



In the inchoative structure (see (36)), the marker *-ě* wins in competition against *-i*. The competition is regulated by the Elsewhere Principle, which favours the more specific marker over the more general one. Since *-ě* only applies in inchoative environments, and *-i* both in inchoative and causative ones, *ě* is more specific and wins the competition. The representation for the adjective-inchoative-causative triplet for the Class II roots is therefore as shown in section 2 of table 2.

Finally, the Czech Class III is illustrated in (37) by verbs based on the adjective *tich-ý* ‘silent.’ This class is characterised by the fact that the past participle has no marker for the inchoative at all, see (37b).

- (37) a. *Ššš, tiš -i -l mě otec.*
 psst silent CAUSE -PST me.ACC father.NOM
 ‘Psst, my father was making me silent.’
 b. *Posměšný křik skřetů tich -Ø -l.*
 mocking shouting goblins.GEN silent -BECOME -PST
 ‘The goblins’ mocking shouting was getting silent.’

Table 7 lists the roots that have an unmarked inchoative.¹⁴

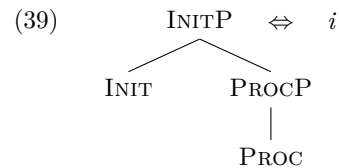
Tab. 7: Class III verbs

gloss	adj	inch	caus
deaf	hluch-ý	o-hluch-l	o-hluš-i-l
brown	hněd-ý	za-hněd-l	za-hněd-i-l
weak	chab-ý	o-chab-l	o-chab-i-l
lame	chrom-ý	z-chrom-l	z-chrom-i-l
blind	slep-ý	o-slep-l	o-slep-i-l
silent	tich-ý	z-tich-l	z-tiš-i-l
dark	tmav-ý	z-tmav-l	z-tmav-i-l
bitter	trpk-ý	z-trpk-l	z-trpč-i-l
tough	tuh-ý	z-tuh-l	z-tuž-i-l
alive	živ-ý	o-živ-l	o-živ-i-l

Given the zero marking in the inchoative, we analyze the roots in this class as capable of spelling out PROCP. The table (38) shows that roots of this size can realise both an adjective and an inchoative structure without any additional marker, see lines (38a) and (38b). The INIT projection of the causative needs an additional morpheme, see line (38c’).

(38)

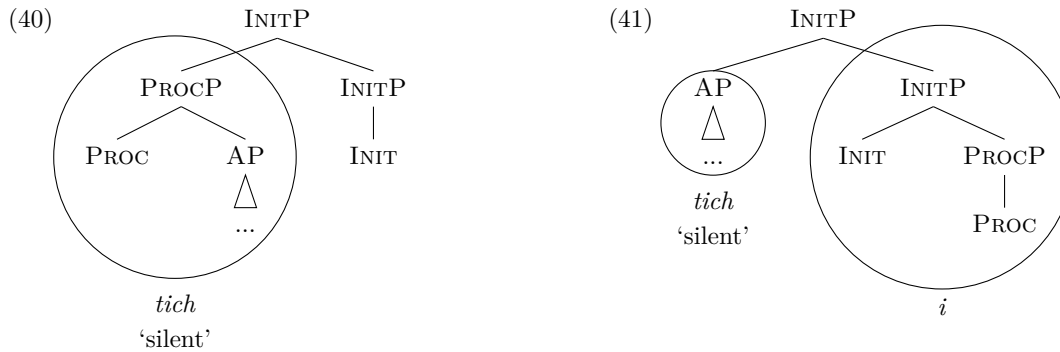
	A	PROC	INIT
a.	root		
b.		root	
c’.		root	-i
c”.	root		-i



However, recall that the causative marker *-i* can potentially realise both PROC and INIT features. This offers another possibility of lexicalising the causative structure, namely as (38c”).

We shall now argue that only the analysis in (38c”) is compatible with insertion based on the Superset Principle. Our starting point is precisely the observation that the causative suffix in Class III verbs (namely *-i*) is identical to the one in Classes I and II. This means that the same lexical item is involved, namely the one given in (30b) above, and repeated in (39). Such an entry cannot spell out a remnant INITP that only contains INIT but not PROC. Such a constituent is in (40), but it cannot be spelled out by *-i*, because the INITP is not a subtree in the lexical entry (39). This makes the analysis in (38c’) impossible.

¹⁴ In current Czech, there exist (diachronically newer) alternative forms of the past participle with *-nu-* in between the root and the past participle *-l*, see Taraldsen Medová and Wiland (2019).



Since the INIT feature fails to be lexicalised in a derivation like (40), the causative must be derived in such a manner that the root does not reach its full lexicalisation potential, stopping at AP, just as it does in the simple adjective. This is shown in (41). The suffix *-i* then lexicalises the same constituent INITP as with the Class I and Class II roots (see (31) above).

Technically, this derivation involves backtracking, the details of which we do not address here for lack of space. We refer the reader interested in the details of backtracking to Vanden Wyngaerd et al. (2020). What we do want to point out, however, is that there is some interesting empirical evidence for the analysis in (41). Note first that if (41) is correct, the root spells out the same features when used in the causative (namely A) as when used as an adjective. This is different from the inchoative, where it spells out a bigger constituent, A+PROC.

This has consequences for root suppletion. Following Caha et al. (2019), we assume that suppletive roots are differentiated by the number of features they realize. If that is so, we expect suppletive adjectival roots to differentiate between the inchoative environment on the one hand (the root spells out AP+PROC) and, on the other hand, the adjective and the causative environment (the root spells out AP). Interestingly, there is a set of roots in Czech with exactly this type of root distribution, see Table 8.¹⁵

Tab. 8: Adjectives with mildly suppletive forms in the inchoative

gloss	adj	inch	caus
dense	hust-ý	z-houst-l	z-hust-i-l
dry	such-ý	u-sch-l	u-suš-i-l
young	mlad-ý	o-mlád-l	o-mlad-i-l
weak	slab-ý	ze-sláb-l	ze-slab-i-l
golden	zlat-ý	zlát-l	zlat-i-l
yellow	žlut-ý	za-žlout-l	za-žlut-i-l

Summarising, Czech provides two reasons for analysing zero marking as an instance of phrasal spellout. The first reason is that Czech fills two gaps in the predicted typology of marking. Most notably, it features a set of roots, which are of PROCP size, with an overt causative, but a zero inchoative (Class III).

The second reason has to do with the pattern of root suppletion exemplified in Table 8. The pattern shows that the suppletive root is found where the regular marking is absent, much as in the *bad-worse* case. Specifically, what we see is that when PROC and INIT are spelled out by suffixes, we get no suppletion (The final column in Table 8). Suppletion only arises when the root spells out PROC. Such roots, then, clearly reveal that zero marking does not come for free, and the root must do a part of the job.

¹⁵ The roots are mildly suppletive. In some cases, the difference is only vowel length, as indicated by the accent sign.

7 The inverse pattern: anticausatives

As we have already noted, there is also a pattern where the inchoative is formed on the basis of the causative through the addition of a marker, often a reflexive.

(42) The inverse pattern

a.	caus:	V
b.	inch:	V+aff

Haspelmath calls this the anticausative pattern. This type of pattern is also found in Czech and it instantiates the fourth class of verbs, which was left undiscussed in section 6.2. In (43), we give an example of a causative derived from the adjective *tup-ý* ‘blunt.’ In order to express the inchoative reading (‘get blunt’), Czech must add a reflexive marker *se* to the causative, see (43b). There is no non-reflexive form that would be able to express this meaning.

- (43) a. *Já mu před zápasem vždycky tup -i -l brusle*
 I.NOM him.DAT before match always blunt -CAUSE -PST skates.ACC
 ‘I have always made his skates blunter before the match.’
 b. *Hráčům se tup -i -ly brusle*
 players.DAT REFL blunt -CAUSE -PST skates.NOM
 ‘The player’s skates were getting blunt.’

This pattern is apparently problematic in that a structure with fewer features (the inchoative) requires more markers than the semantically richer structure (the causative). In this section, we highlight two possible solutions. The choice between them is left for future research.

The first possible line of attack would capitalize on the fact that this type of anticausative marking involves a marker of reflexivity. Koontz-Garboden (2009: 80) observes that also cross-linguistically, reflexivisation and anticausativisation ‘seem almost always to be marked in a morphologically identical fashion’. Following Chierchia (2004), Koontz-Garboden (2009), Beavers and Koontz-Garboden (2011), one could then analyse the reflexive-marked verb, like *prodloužil se* ‘lengthened’ in (43b), as a reflexivized version of the causative verb. We represent this schematically as in (44).

$$(44) \quad V_{\text{ANTICAUS}} = \text{CAUSE} (x, \text{INCH}(\text{Pred}(x)))$$

This is different from a true inchoative verb, which we take to have a representation as in (45):

$$(45) \quad V_{\text{INCH}} = \text{INCH}(\text{Pred}(x))$$

Under this analysis, anticausatives like the one in (43b) are structurally different from English-type inchoatives like *widen*, in the sense that they are based on the causative structure. The reflexive marker *se* is, then, not a derivational affix on the verb, but a particular way in which the external or internal argument is realised. Koontz-Garboden (2009) supports this by pointing to potential meaning differences between anticausatives and true inchoatives. Specifically, the reflexivisation operation yields a predicate that is true if the single argument is both the Effector and the Theme. That is, the single argument is not simply undergoing some change (as a Theme), but it is at the same time also responsible for its own undergoing of the change. A consequence of this is that for anticausatives, the Causational Entailment (10) is predicted to not always be valid, as schematically shown in (46).

$$(46) \quad \text{No Causational Entailment with Anticausatives} \\
\neg(\forall x \forall y [\text{CAUSE} (x, \text{INCH}(\text{Pred}(y))) \rightarrow \text{CAUSE} (y, \text{INCH}(\text{Pred}(y)))])$$

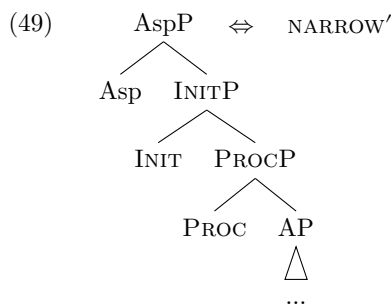
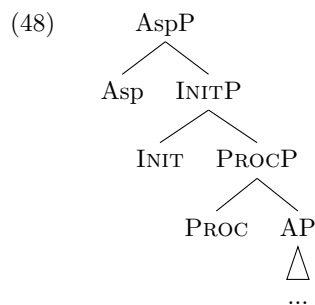
The example (47) (Koontz-Garboden 2009: 117) may clarify this. Spanish has both inchoative verbs like *empeorar* ‘worsen’, and anticausative ones like *romperse* ‘to break’ (with the reflexive *se*). With the inchoa-

tive, the Causational Entailment is valid, as shown by the deviance of (47a). But with the anticausative, speakers accept sentences like (47b), suggesting that the Causational Entailment is not always valid.

- (47) a. #*No empeoró ningún paciente. Los empeoró el tratamiento.*
 not worsened any patient them worsened the treatment
 ‘No patient worsened. The treatment worsened them.’
 b. *No se rompió ningún vaso; los rompió Andrés.*
 not REFL broke any glass them broke Andrew
 ‘No glass broke; Andrew broke them.’

However, there has been a debate as to whether this kind of approach is correct (see Schäfer and Vivanco 2015).¹⁶ Even if it were, the question remains how it can be extended to languages where the anticausative marker differs from the reflexive marker. This is notably the case in such languages as Turkish (Key 2013), Hungarian (Márkus 2015), and Korean (Jeong 2018). Let us therefore sketch here an alternative approach to the anticausative conundrum, explored in work by Márkus (2015).

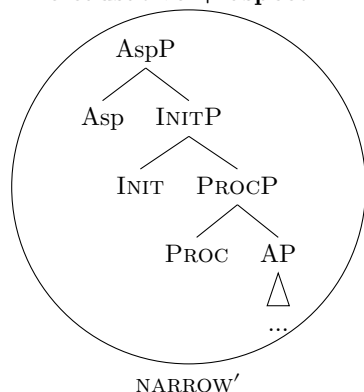
The main idea of the alternative approach is that causative and inchoative structures do not live in a vacuum, so to speak. In syntax, argument structure projections are always embedded under aspectual and temporal projections, as shown in a simplified form in (48).



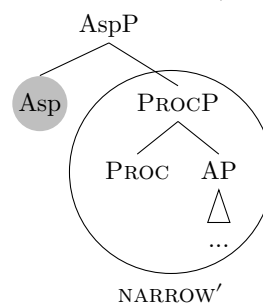
Suppose now that a language has a lexical entry like the one in (49). The lexically stored tree of this hypothetical entry does not only contain the argument structure projections, but also the aspectual projection ASP. This lexical entry can spell out all the projections in (48), see (50). However, the entry (49) cannot be used to spell out all the projections of the inchoative structure, as shown in (51). The reason is that the AspP given in (51) is not contained (as a constituent) inside the lexical entry (49). The lexical entry does contain the PROCP, so that PROCP can successfully lexicalise, as indicated by the circle in (51). But this leaves the ASP head without lexicalisation, which is marked by the grey circle in (51).

¹⁶ A reviewer points out that the English example in (11) is different from the ones in (47) in that the order of the causative and the negative sentence is reversed, and that this fact may be responsible for the judgments, rather than any difference between unmarked or reflexive-marked inchoatives.

(50) The causative + aspect



(51) The inchoative + aspect



As a result, an additional morpheme must be used to spell out Asp, and the inchoative ends up needing more morphemes than the causative. The reason for the extra morpheme is that the missing INIT head prevents constituent matching between the lexical tree of (49) and the syntactic structure.

The consequence of this proposal is that the reflexive is not associated to an argument-structure projection, but to an aspectual layer higher up in the structure. A piece of data suggesting that this may be correct is that in nominalisations (which presumably have less structure than verbs), the inchoative form loses the reflexive marker seen in (43b). We show this in (52), where the nominalisation in (52a) is derived from a causative verb (as the translation suggests), while (52b) is inchoative. What is remarkable is that including the reflexive marker in (52b) leads to ungrammaticality.¹⁷

- (52) a. ... o Llywelynově krutém tup-e-ní rohů případné konkurence
 about Llywelyn's cruel blunt-CAUS-ING corners.GEN potential competition
 '... about Llywelyn's cruel way of making the edges of his potential competition blunt'
 b. Při častém používání by docházelo k jejich tup-e-ní (??se).
 with frequent use would lead to their blunt-CAUS-ING REFL
 'A frequent use would lead to them getting blunt.'

It is worth pointing out that the corpus does contain the string *tupení se* 'blunting REFL,' but this is attested only in a reflexive/reciprocal meaning, not in the anticausative one. While we find this evidence suggestive, we shall not elaborate on any detail here, merely noting that the anticausative pattern of marking may receive an explanation even under the view (argued for here) that the structure of the causative always contains the structure of the inchoative.

8 Conclusion

The main idea of this paper is that derivational 'zero affixes' do not exist, and that conversion arises as an effect of cumulative exponence, where the root realizes the relevant meaning ingredients. We have implemented this idea in Nanosyntax, a framework with phrasal lexicalisation with matching governed by the Superset Principle. In addition, we adopted a relatively widespread idea that causatives contain inchoatives, and that inchoatives contain adjectives. Combining the Superset Principle and the two independent ideas allowed us to account for the following set of facts. (i) Not all adjectival roots function as verbs, and which roots may do so is largely unpredictable. We achieved this by relying on arbitrary lexical storage: only roots that spell out PROC and/or INIT function as verbs. (ii) Some zero-derived verbs are ambiguous between the causative and the inchoative reading. We encoded this by saying that these verbs spell out

¹⁷ The examples come from the Czech National Corpus, the judgement on the reflexive is that of the first author.

both PROC and INIT (as in English). (iii) Czech zero-derived verbs are only inchoative (but not causative). This is because they only spell out PROC, but not INIT. (iv) Zero-derived verbs in Czech may be (mildly) suppletive, while non-zero-derived causatives are not. This is because the adjectival root in zero-derived inchoatives pronounces the PROC component, which may be reflected via suppletion. (v) Finally, our assumptions allowed us to capture the different patterns of the inchoative/causative alternation that are known from the typological literature, including the superficially problematic anticausative pattern.

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4 Negation and the functional sequence

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Negation and the Functional Sequence

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Abstract There exists a general restriction on admissible functional sequences which prevents adjacent identical heads. We investigate a particular instantiation of this restriction in the domain of negation. Empirically, it manifests itself as a restriction on the stacking of multiple negative morphemes. We propose a principled account of this restriction in terms of the general ban on immediately consecutive identical heads in the functional sequence on the one hand, and the presence of a Neg feature inside negative morphemes on the other hand. The account predicts that the stacking of multiple negative morphemes should be possible provided they are separated by intervening levels of structure. We show that this prediction is borne out.

Keywords Negation · Adjectives · Decomposition · Nanosyntax

1 Introduction

The starting observation of this paper is a contrast between negative and positive adjectives: whereas positive adjectives can generally be prefixed with *un-*, negative ones systematically cannot. At the same time, negative adjectives are not resistant to negation *per se*, as they can be negated with *not*, as is shown in (1c).

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(1)	a.	unhappy	b.	*unsad	c.	not sad
		unwise		*unfoolish		not foolish
		unclean		*undirty		not dirty
		unfriendly		*unhostile		not hostile
		unhealthy		*unsick		not sick
		unkind		*unrude		not rude
		untrue		*unfalse		not false
		uneasy		*undifficult		not difficult

The observation of this pattern dates back to Jespersen (1942:466), and has been repeated since in the work of other authors (e.g. Zimmer 1964; Horn 1989:275; Horn 2005).

Data from the BNC corpus (100m words) and the COCA corpus (450m words), given in Table 1, confirm this pattern. The final two columns in the table in particular reveal the relevant contrast: the prefinal column gives the numbers for *un*-prefixed positive adjectives, and the final column shows the absence of the *un*-prefix with negative adjectives.¹

Table 1 *un*-prefixation with positive and negative adjectives (BNC–COCA)

	PosA	NegA	un-PosA	un-NegA
wise-foolish	wise 2,118–10,018	foolish 1,088–4,406	unwise 399–792	unfoolish 0–0
happy-sad	happy 11,166–55,400	sad 3,241–17,549	unhappy 1,822–5,763	unsad 1–0
kind-rude	kind 23,349–1,855,404	rude 942–3,386	unkind 257–512	unrude 0–0
true-false	true 17,577–90,165	false 3,529–14,944	untrue 277–1,195	unfalse 0–0
easy-difficult	easy 14,143–65,942	difficult 21,433–72,543	uneasy 915–3,386	undifficult 0–1

On closer scrutiny, the pattern turns out to be more general, in that multiple negative affixes are ruled out in general. For example, noun-derived adjectives with the negative suffix *-less* also resist *un*-prefixation, but not negation per se (see also Siegel 1974, Allen 1978):

(2)	breathless	*unbreathless	not breathless
	senseless	*unsenseless	not senseless
	merciless	*unmerciless	not merciless
	useless	*unuseless	not useless
	cheerless	*uncheerless	not cheerless

Here as well, corpus data confirm the judgments, as shown by Table 2.

¹ In the Tables 1, 2, and 3, the first figure in each cell gives the number of occurrences in the BNC, the second that in COCA.

Table 2 *-less* (BNC-COCA)

N-less	un-N-less	not N-less
breathless 459–1,505	unbreathless 0–0	not breathless 5–2
senseless 175–1,088	unsenseless 0–0	not senseless 0–0
merciless 122–611	unmerciless 0–0	not merciless 0–0
useless 1,244–4,529	unuseless 0–0	not useless 5–20

These cases contrast minimally with positive noun-derived adjectives ending in *-ful* (corpus data in Table 3):²

- (3)
- | | | |
|------------|--------------|----------------|
| successful | unsuccessful | not successful |
| lawful | unlawful | not lawful |
| eventful | uneventful | not eventful |
| helpful | unhelpful | not helpful |
| faithful | unfaithful | not faithful |

Table 3 *-ful* (BNC-COCA)

N-ful	un-N-ful	not N-ful
successful 10,564–40,400	unsuccessful 921–2,711	not successful 59–275
lawful 503–827	unlawful 896–892	not lawful 5–12
eventful 105–255	uneventful 118–429	not eventful 1–1

Another instantiation of the general restriction is found in the fact that combinations of *un-* and *dis-* are ruled out, as shown in (4) (Siegel 1977:190-191). The pattern extends more generally to combinations of *un+iN*, *un+ab*, *un+a*,

² The minimal contrast between *un-N-less* and *un-N-ful* is also noted by Allen (1978:30), who adduces it as an argument against Siegel's (1974) account of **un-N-less* in terms of level ordering.

as well as the repetition of identical negative prefixes (*un+un* and *dis+dis*), as is shown in (5):^{3,4}

- | | | |
|-----|-------------------|--------------------|
| (4) | *undishonest | not dishonest |
| | *undiscourteous | not discourteous |
| | *undisloyal | not disloyal |
| | *undiscomfortable | not discomfortable |
-
- | | | |
|-----|---------------|----------------|
| (5) | *unimpossible | not impossible |
| | *unillogical | not illogical |
| | *unabnormal | not abnormal |
| | *unatypical | not atypical |
| | *ununhappy | not unhappy |
| | *disdishonest | not dishonest |

In sum, the restrictions we observe on morphological negation suggest the following preliminary generalisation:

³ In principle, it should be possible to replicate the pattern in (1) with the *dis-* and *iN-* prefixes, i.e. they should not combine with negative adjectives. In practice, however, it turns out difficult to find relevant cases where the prediction could be tested. The reason is that the negative adjectives that would have to serve as the input to *dis-/iN-* prefixation for the most part either do not exist, or are not Latinate in origin. That is, next to *possible*, *proper*, *appropriate*, *coherent*, *honest*, etc. we have their polar opposites derived through *dis-/iN-* prefixation (*impossible*, *improper*, *inappropriate*, *incoherent*, *dishonest*, etc.). But we have hardly any underived adjectives that express the negative meanings of these adjectives, and that could serve as the input for *dis-/iN-* prefixation (Horn 2005). A possible candidate is the word *callous*, which could serve as a synonym of *indifferent*. Given its Latinate origin (< *callum* 'callus'), we would expect it to be prefixed with *iN-*. In line with the pattern in (1), *incallous* indeed does not exist in English, .

⁴ R. Kayne (p.c.) suggests to us a number of cases where *un-* attaches to a negative base, and which he finds acceptable:

- | | | |
|-----|----|---|
| (i) | a. | I'm happy to see that you've finally learned to act in an uncareless way. |
| | b. | He's finally made an unincoherent statement! |
| | c. | Why can't you learn how to make unsenseless remarks? |
| | d. | I'm trying to learn to wander around unaimlessly. |

However, these to us reveal the existence of a different *un-*, one that is necessarily stressed:

- | | | |
|------|----|---|
| (ii) | a. | *Im happy to see that you've finally learned to act in an unCAREless way. |
| | b. | I'm happy to see that you've finally learned to act in an UNcareless way. |

This type of stressed UN- has important characteristics of echo negation (Seuren, 1976) or metalinguistic negation Horn (1985, 1989). For example, a case like (iii) carries with it a speaker presupposition (Stalnaker, 1978) that the person in question's behaviour is usually careless, and the sentence denies that existing presupposition (Vanden Wyngaerd, 1999).

- | | |
|-------|---|
| (iii) | (We all know you usually act in a careless way, so) |
| | I'm happy to see that you've finally learned to act in an UNcareless way. |

Cases like these contradict the claim by Horn (1989, 1985:392) to the effect that metalinguistic negation does not incorporate prefixally, or the claim by Seuren (1976) that echo negation is only possible if the negation is in its canonical position, i.e. cliticised onto the auxiliary. See also Kayne (2017).

- (6) Negative morphemes cannot be stacked.

Although the restriction illustrated by (1) was noted a long time ago, it has defied a principled explanation so far. The literature does feature a more specific version of (6), independently formulated by Zimmer and Horn, which we shall call the Zimmer–Horn generalisation (or Z–H generalisation for short).

- (7) a. Negative affixes are not used with adjectival stems that have a ‘negative’ value (Zimmer, 1964:15)
 b. The stem to which a relatively nonproductive negative affix can attach tends to be an UNMARKED, WEAK POSITIVE scalar value (Horn, 1989:286)

Siegel (1977:192) accounts for the data in the left-hand column of (4) by means of an output filter on word formation rules:

- (8) Words in *un* are thrown out if the morpheme *dis* is uniquely contained in the cycle adjacent to *un*.

A somewhat more general formulation is given in Allen (1978:50), as well as Seuren and Jaspers (2014:632), respectively:

- (9) Condition on *un*-prefixation:
Un’s base may not have negative content.
 (10) A negative affix can only be attached to the positive member of a positive-negative pair.

Crucial to these generalisations and restrictions is that they formulate a constraint on *morphological* negation. In virtue of this fact, they capture the contrast between (1b) and (1c). At a theoretical level, they appear to provide an argument for the existence of the word (and hence morphology) as a separate domain: word-internal negation is subject to a restriction that syntactic negation is not subject to.

In this paper, we shall show that this conclusion is undermined by the existence of certain cases of syntactic negation, which show exactly the same restriction against stacking multiple negative morphemes. Concretely, the evidence we shall present concerns a negative syntactic modifier which is the cross-linguistic counterpart of *little*, and which can modify positive adjectives but not negative ones. This evidence is found in Dutch and French, amongst other languages; two relevant examples are given in (11):

- (11) a. Zijn houding is weinig actief/*passief
 his attitude is little active/passive
 ‘His attitude is not very active/passive.’
 b. Son comportement est peu actif/*passif
 her attitude is little active/passive
 ‘Her attitude is not very active/passive.’

At a theoretical level, we shall argue that the restriction that negative markers are excluded with negative morphemes follows from a general restriction on admissible functional sequences, formulated in (12).

(12) * $\langle X, X \rangle$

The functional sequence must not contain two immediately consecutive identical projections.

A constraint like (12) is commonly (if mostly tacitly) assumed in the (cartographic) literature, i.e. C does not select CP, T does not select TP, AgrS does not select AgrSP, D does not select DP, etc. An explicit formulation of it that comes closest to what we have in mind is the *Unlike Category Constraint* of Hoekstra (1984:85), a constraint which prevents a head of category X from taking a complement of the same category XP (see also Grimshaw 1997).⁵ We shall argue that the constraint in (12) explains the Z–H generalisation, as well as the other restrictions that have been proposed in the literature to capture the data discussed in this section.⁶

⁵ There is also an obvious parallel with the Obligatory Contour Principle in phonology, which prevents the adjacent occurrence of two identical elements on the melodic tier within the same lexical item. The OCP has been given a syntactic interpretation by e.g. van Riemsdijk (2008), Hiraiwa (2010) (and further references cited there). In contrast to these authors, however, we do not conceive of (12) as a haplology constraint, which is a constraint on form rather than meaning. Haplology is a restriction against two adjacent identical phonologies $/\alpha + \alpha/$. This will result in the removal one form, yielding $/\alpha/$, while the underlying features remain intact. This means that whatever it is that gave rise to the phonology $/\alpha + \alpha/$ is unaffected by the removal of one occurrence of $/\alpha/$. In contrast, the constraint we assume is a constraint on features, not phonologies, more in particular the syntactic structures underlying negation. These may or may not be realised by identical exponents. As a result, phonologically identical negative markers may in some cases be linearly adjacent (as in e.g. *You cannot not invite here*, to be discussed in section 5.2 below), whereas in others phonologically different ones may be precluded from appearing adjacently (as in **undis-honest*, discussed above). The restriction in (12) may ultimately be reducible to a labelling problem (Ott, 2011, Chomsky, 2013), or a linearisation problem (Richards 2010). Alternatively, it could be argued to follow from the requirement that the combination of features through Merge needs to observe the requirements imposed by the functional sequence, i.e. Merge (α, β) is subject to the requirement that α and β be distinct adjacent elements of the functional sequence. We shall not explore these connections any further here.

⁶ A reviewer points out a case potentially violating the principle in (12), namely topic recursion, as discussed in Rizzi (1997). There are a number of possibilities to bring such cases in line with our principle (12). A first possibility is that different heads are involved, i.e. different positions in the functional sequence. What looks like an X-X sequence would then actually be an X-Y sequence. Such an analysis has in fact been proposed for Italian by Benincà and Poletto (2004). They argue that only one Topic position exists, namely the one that linearly precedes Focus, and that it can contain only one topic, i.e. there is no recursion of topics, despite appearances to the contrary (see also Cinque 1977, 1983 on the distinction between hanging topics and CLLD). A single topic position has also been defended for Japanese sentences with multiple *wa*-marked phrases by Vermeulen (2007). Other languages that mark topics with a particle have been argued to admit of only one such topic (e.g. Gungbe, see Aboh 2004:311). A second possibility is that there is covert intervening structure between these topics, so that what superficially looks like an X-X sequence is in fact an X-Y-X sequence. An analysis of this general format is the ellipsis analysis of Left Dislocation of Ott (2014, 2015), which takes the topic to have been extracted from a sentence that subsequently undergoes ellipsis, and that is followed by a second sentence that contains

A second crucial ingredient of our proposal is the following:

- (13) Negative morphemes contain a Neg-feature.

This assumption presupposes a form of decomposition. The kind of decomposition that we shall assume is one where the syntax works with features, and the lexicon is postsyntactic. In a nutshell, we shall argue that negative morphemes like *sad* and *un-* contain a Neg-feature, and that a word like *unsad* instantiates an illicit stacking of Neg-features, in violation of (12). The concrete means by which a word like *sad* spells out a set of features is by assuming the nanosyntactic mechanism of *phrasal spellout*: every syntactic object created by Merge interfaces with the lexicon, and undergoes spellout if a matching lexical item is found. The organisation of the lexicon is subject to the following restriction (Starke, 2014a):

- (14) The lexicon contains nothing but well-formed syntactic expressions.

The consequence of this is that lexical items contain well-formed syntactic trees. More details on the mechanism of postsyntactic lexical insertion will be discussed in section 3.

The paper is structured as follows. In section 2 we discuss our reasons for believing that the contrasts discussed above are to be accounted for in syntactic rather than semantic terms. In section 3 we lay out the prerequisites for our analysis: we introduce the theoretical framework of nanosyntax, we discuss the feature structure of lexical adjectives, and we show how *un-*prefixed positive adjectives are derived. In section 4 we present the analysis that accounts for the data pattern above, i.e. the cases of morphological negation stacking, and we provide new data from low scope syntactic negation, arguing that the split is not between morphological and syntactic negation, but between high and low scope negation. Finally, in section 5 we provide an analysis for why both positive and negative adjectives can be combined with *not*, as shown in (1c).

2 Syntax or semantics?

Our decomposition analysis has obvious consequences for the semantic analysis of gradable adjectives. Providing such a fully worked out semantics is the topic of a different paper and therefore not a task we shall undertake here. What we would like to motivate in this section, however, is our approach to the facts discussed in the previous section in terms of a syntactic principle (i.e. (12)), rather than a semantic one.

the (fronted) resumptive pronoun. A third possibility is that multiple topics form a single complex constituent, and a single topic-comment structure. Under such a view, what looks like an X-X sequence would reduce to a single occurrence of X. Multiple topics would then be a case of ‘topic concord’, i.e. multiple topic-marked constituents interpreted as a single topic. A full investigation of these alternatives is beyond the scope of the present article, however.

It is obvious that a simple semantic constraint against the stacking of negations will not do, since there is nothing semantically wrong with the stacking of multiple negative operators:

- (15) $\neg\neg P(x)$

And as we saw in the previous section, double negation is also found in natural language:

- (16) a. That is **not impossible**.
b. He doesn't **not** like her.

A more sophisticated version of a semantic approach to the data discussed in the previous section tries to explain the restriction involved in terms of the distinction between contradiction and contrariety, as follows: contradictory negation can be stacked (onto contrary ones and onto each other), whereas contrary negation cannot.⁷ Two propositions are contradictory if they cannot be true together, nor be false together, such as the ones in (17):

- (17) a. Alexandra went to university.
b. Alexandra didn't go to university.

Two propositions are contrary if they cannot be true together but can be false together, such as (18a) and (18b), which are both false if Wendy is neither happy nor unhappy:

- (18) a. Wendy is happy.
b. Wendy is unhappy.

As (18b) shows, the negative prefix *un-* gives rise to contrary negation. Similarly, the relation between the adjectives *tall* and *short* is one that gives rise to contrary opposition, given that Rachel could be neither tall, nor short, i.e. the sentences of (19) could be false together:

- (19) a. Rachel is tall.
b. Rachel is short.

Returning to cases of double negation, the more sophisticated semantic approach to our data would imply that the cases in (16) are allowed because they stack a contradictory negator (*not*, *n't*) onto another contradictory one, or onto a contrary one (like *im-*). What is not allowed, however, is stacking a contrary negation onto a contrary one, and this is precisely what happens in cases like **unshort*.

Contradictory negation would under this approach be identical to logical negation (represented as $\neg$), whereas contrary negation would involve a differ-

⁷ This possible alternative was suggested to us by an anonymous reviewer. Though (Cinque, 1999:126) does not make this distinction between contradictory and contrary negation, his position is potentially compatible with this suggested alternative, in that he assumes that negation can occur freely on every adverb-related functional projection.

ent operator altogether (e.g. the **c**-operator proposed by Goddard 1960, the **R**-operator of McCall 1967, the κ -operator of Humberstone 2005, or Horn's 2014 **©**-operator). Representing contrary negation as κ for concreteness, the situation can be summarised as in (20):

- (20) a. $\neg\neg$
 b. $\neg\kappa$
 c. $*\kappa\kappa$
 d. $\kappa\neg$

Since affixal negation is for the most part contrary and syntactic negation contradictory (Horn, 1989), this semantic theory to all intents and purposes derives the equivalent results of the Z–H generalisation.

However, there are a number of reasons why we believe such a semantic approach is inadequate. The first of these is that it is empirically deficient, in the sense that there are many instances where contradictory negations cannot be stacked. Collins (2016) discusses a wide range of such cases, such as the following:

- (21) a. Not everybody was there.
 b. *Not not everybody was there.
- (22) a. Not many people were there.
 b. *Not not many people were there.
- (23) a. Not {even/only} John was there.
 b. *Not not {even/only} John was there.
- (24) a. Not {more than three/less than three/a lot of} people were there.
 b. *Not not {more than three/less than three/a lot of} people were there.
- (25) a. Not often do I manage to go on fall foliage tours.
 b. *Not not often (do) I manage to go on fall foliage tours.
- (26) a. I persuaded John not to like Clinton.
 b. *I persuaded John not not to like Clinton.

These all involve the illicit stacking of two contradictory negations, which therefore must be due to a different constraint, if indeed contradictory negators are semantically freely stackable. Collins argues that this is a syntactic constraint, which prevents the stacking of two negations in identical positions (*NEG NEG).⁸

The other reason why we believe the approach in terms of the semantic distinction between contradiction and contrariety is insufficient is that it is at best a description of the facts, not an explanation. Its potential theoretical

⁸ Collins' constraint is similar to ours, but does not extend to cases of morphological negation, nor to other types of X–X sequences. We believe his data could also be explained in terms of our general constraint in (12). We refrain from making a detailed comparison of his proposal with ours, as it would lead us too far afield.

interest rests on the assumption that contrary negation is somehow fundamentally different from contradictory negation, and involves a different operator altogether (as shown in (20)). But even if one adopts this view, there is nothing explanatory about (20): it is the mere observation of a pattern, with little generality, and no obvious explanation. The restriction that we propose in (12), in contrast, is quite general, covering not only all cases of negation, but extending beyond negation to many other cases.

More importantly, we believe that the assumption of a contrariety operator is not necessary. We believe that the contrary nature of the negation we see in the data discussed in section 1 can be derived from an underlying contradictory negation in the internal structure of negative gradable adjectives (whether lexically negative or with a negative prefix or suffix). In the remainder of this section, we shall briefly lay out the core of this analysis, which is largely based on Seuren (1978) (see also von Stechow 1984, 2008). We refer the reader to De Clercq and Vanden Wyngaerd (2018) for a more detailed presentation (and to Collins 2017 for a slightly different approach to the same issue).

We first define the distinction between contradiction and contrariety in terms of sets, as follows:

- (27) a. Contradiction
 $A \cup B = \mathcal{U}$
 $A \cap B = \emptyset$
 b. Contrariety
 $A \cup B \neq \mathcal{U}$
 $A \cap B = \emptyset$

The denotation of a proposition under this approach is not a truth value, but the set of situations in which it is true (Van Fraassen, 1971). Two propositions are contradictory if their union equals the universe of all possible situations, and contrary if their union does not denote the universe. In either case, their intersection is empty, which represents the fact that they cannot be true together.

The analysis that we propose of gradable adjectives is based on interval semantics, where gradable adjectives denote intervals or extents (Kennedy, 2001). A scale $\langle S, <_{DIM} \rangle$ is a set of linearly ordered points along a dimension DIM . An extent E is a nonempty subset of S with the following property (Landman, 1991:110):

$$(28) \quad \forall p_1, p_2 \in E, \forall p_3 \in S, [(p_1 < p_3 < p_2) \rightarrow (p_3 \in E)]$$

Assume further a degree function d_{DIM} , which maps any entity x which can be ordered along some dimension DIM onto a unique point on the scale $\langle S, <_{DIM} \rangle$. This unique point divides the scale into two intervals or extents, a positive and negative one. The positive extent of x with respect to $\langle S, <_{DIM} \rangle$ is defined as in (29a), and its negative extent as in (29b).

$$(29) \quad \text{a.} \quad POS_{DIM}(x) = \{p \in \langle S, <_{DIM} \rangle \mid p \leq d(x)\}$$

$$\text{b. } NEG_{DIM}(x) = \{p \in \langle S, <_{DIM} \rangle \mid \neg[p \leq d(x)]\}$$

The positive and the negative extents of x entertain a relation of contradictoriness as defined above: their union equals the entire scale, and their intersection is empty. We may therefore define a negative extent more concisely as follows:

$$(30) \quad NEG_{DIM}(x) = \neg POS_{DIM}(x)$$

A crucial assumption is that positive gradable adjectives denote a positive extent, and negative gradable adjectives a negative extent, as shown in (31) for the pair *tall-short* (see also Kennedy 2001, Heim 2006, Buring 2007, Heim 2008, Bobaljik 2012):

$$(31) \quad \begin{array}{ll} \text{a. } \llbracket \text{tall}(x) \rrbracket = POS_{HEIGHT}(x) \\ \text{b. } \llbracket \text{short}(x) \rrbracket = NEG_{HEIGHT}(x) \end{array}$$

That is, $\llbracket \text{tall}(x) \rrbracket$ is the set of degrees to which x is tall, whereas $\llbracket \text{short}(x) \rrbracket$ is the set of degrees to which x is not tall. Antonymic pairs like these therefore stand in a relationship of contradictoriness, for the reasons just explained. Given the equation in (30) above, we can now assume that negative scalar adjectives contain a logical negation in their internal structure.

$$(32) \quad \llbracket \text{short}(x) \rrbracket = \llbracket \neg \text{tall}(x) \rrbracket = \neg POS_{HEIGHT}(x) = NEG_{HEIGHT}(x)$$

So where does contrariety come from? The contrary nature of the opposition in an antonymic pair like *tall-short* derives from the way these extents interact with the contextual standard, which is a further essential part of the denotation of gradable adjectives (Wheeler 1972, Seuren 1978, Klein 1980, and much subsequent work). Following Seuren (1978), we take this contextual standard or average itself to be an extent, i.e. the set of degrees that counts as neither neither tall nor short. We now define the truth conditions for a sentence like *Linus is tall* as follows: the sentence is true if the positive extent of Linus's height includes the contextual average A_C .⁹ Negative adjectives work similarly, except that they are defined in terms of negative extents: *Kurt is short* is true in case the negative extent of Kurt's height includes A_C . Now suppose Eva is of average height, i.e. neither tall nor short. In this case, $d(Eva)$ yields a value that is included in the contextual average A_C ; in such a case, neither the positive nor the negative extent of Eva's height will include A_C . As a result, both the sentence *Eva is tall* and *Eva is short* will come out as false.

The above analysis is represented graphically in Figure 1, where the top line represents the height scale and the contextual average A_C . The second line represents Linus' tallness, with its positive extent as a solid line, and its negative extent as a dashed line. The representation shows that A_C is included in Linus' positive extent, as well as in Kurt's negative extent, but not included in either the positive or the negative extent of Eva's height.

⁹ For two extents X and Y , $X \subseteq Y \iff ((X \cap Y = X) \wedge (X \cup Y = Y))$.

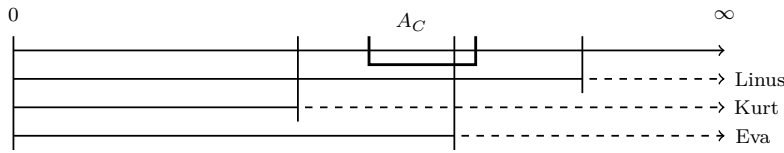


Fig. 1 Height scale with positive and negative extents

The result is that *Linus is tall* will come out as true, as will *Kurt is short*, as required. In contrast, the sentence *Eva is tall* will come out as false, as will the sentence *Eva is short*. This derives the contrary opposition of the latter two sentences, since they can both be false at the same time.

In sum, this analysis takes the fundamental semantic distinction between antonymic pairs of adjectives to be one of contradictory opposition, which derives from the presence of a Neg feature with the semantics of logical negation ($\neg$) in the internal makeup of negative gradable adjectives. Contrariety follows from the truth conditions of gradable adjectives, which are formulated in terms of a positive extent for positive ones, and a negative extent for negative ones, and the dependence on a context-dependent average A_C , which is itself an extent.

Given that cases of contrary opposition can be explained in terms of an underlying contradictory negation, the theoretical appeal of the semantic account of our data in terms of the semantic distinction between contradiction and contrariety is greatly diminished. Saying that contrary negations cannot be stacked and contradictory ones can is then ultimately unexplanatory, since in the end all types of negation can be explained in terms of an underlying contradictory negation, plus the effect of scales. More specifically, **un-short* would amount semantically to $\neg\neg POS_{HEIGHT}(x)$, via (32). Since this involves the stacking of two contradictory negations, it should not present a problem from a semantic point of view (see (20) above). So at this point, one would either have to dig deeper into the semantics of scales in order to explain **unshort*, or assume, as we do, that there is no syntax that generates the illicit double negation. In particular, the syntactic restriction that we propose explains the relevant data in terms of a general restriction on admissible functional sequences.

3 Prerequisites for the analysis

3.1 General background assumptions: nanosyntax

The analysis we propose is couched in the theoretical framework of nanosyntax, which finds its origins in cartography (Rizzi 1997, Cinque 1999, Haegeman 2012). In this section we present the most important background relevant to

understanding our proposal.¹⁰ Nanosyntax assumes that the lexicon is post-syntactic, and consequently, that the syntax manipulates features. Nanosyntax furthermore takes each feature to be a syntactic head (One Feature, One Head).¹¹ A lexical item consists of three parts: a phonology, a syntactic tree, and, optionally, a component of conceptual meaning, which is present in cases where the meaning of the lexical item in question is underdetermined by its features, i.e. with nonfunctional lexical items with rich lexical meanings.

Postsyntactic lexical insertion is subject to the *Superset Principle*:

(33) *Superset Principle*

A lexical entry may spell out a syntactic node iff the lexical tree is identical to the syntactic tree, or if it contains the syntactic tree as a constituent.

As a result of this principle, there may be a competition between several forms for insertion, for example if one lexical entry is a perfect match for a syntactic node, and another contains the syntactic node as a subtree. The winner of the competition is determined by the *Elsewhere Principle* (Kiparsky, 1973):

(34) *The Elsewhere Principle*

In case two rules, R_1 and R_2 , can apply in an environment E , R_1 takes precedence over R_2 if it applies in a proper subset of environments compared to R_2 .

An informal version of this principle states that the lexical item with the fewest superfluous features wins the competition.

As already mentioned above, a further important assumption is that of phrasal spellout. Except (possibly) in strictly agglutinative languages, features do not map onto phonological exponents in a one-to-one fashion. Rather, the relationship between features and exponents is typically many-to-one, or even many-to-many. In nanosyntax, the former situation is accounted for by assuming that lexical items contain stored syntactic trees, i.e. well-formed syntactic objects. These syntactic objects are then matched against the syntactic objects created by Merge in the syntax. If the matching satisfies the Superset Principle, spellout occurs. Every Merge step is a phase, i.e. after every Merge step, the process of matching the syntactic object against the lexicon is repeated. We shall discuss some derivations illustrating how this works in detail below.

¹⁰ We refer the reader to Starke (2009), Caha (2009), Pantcheva (2011), De Clercq (2013), Rocquet (2013), Lander (2016), Baunaz et al. (to appear) for a more detailed discussion of the nanosyntactic framework.

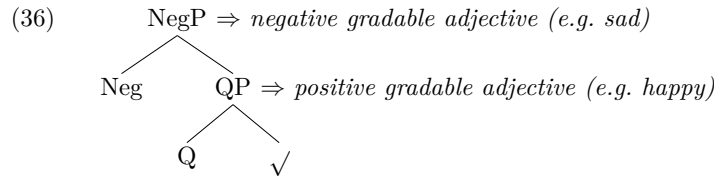
¹¹ See also Cinque and Rizzi (2008:50); Kayne (2007) adopts a similar principle, to the effect that UG imposes a maximum of one interpretable syntactic feature per lexical item.

3.2 The feature structure of adjectives

Since the syntax works with features, the first question we need to address is what are the features involved in adjectives. We assume that these involve a (partial) functional sequence or f_{seq} as in (35):

$$(35) \quad \langle \text{Neg}, Q, \surd \rangle$$

Starting at the bottom, the root feature is responsible for the rich lexical content that differentiates different adjectives from one another. The feature Q contributes gradability (an ordering of a set of degrees). Semantically, QP denotes a positive extent. Neg is a feature that maps the positive extent of QP into a negative extent, as explained above.¹² Putting these together in a tree, we get (36):¹³



The double arrows informally represent phrasal spellout. Lexical entries for three example adjectives are given in (37):¹⁴

- (37) a. $\langle /happy/, [_{QP} Q \surd], \text{HAPPY} \rangle$
 b. $\langle /sad/, [_{\text{NegP}} \text{Neg} [_{QP} Q \surd]], \text{SAD} \rangle$

The derivation of a positive gradable adjective like *happy* proceeds as follows (we omit irrelevant steps prior to the merging of Q): the syntax merges QP , consults the lexicon, and finds any gradable adjective. At this point, both positive and negative gradable adjectives are possible spellouts because of the *Superset Principle*: their features (i.e. (Neg) , Q , and $\surd$) are a superset of those of the syntactic tree QP (which dominates Q and $\surd$). However, negative

¹² For ease of exposition, we ignore the contextual average, which is introduced by another head at the top of the structure, in the manner of the POS head of Kennedy (2007). See De Clercq and Vanden Wyngaerd (2018) for discussion.

¹³ A reviewer asks if positive adjectives should not receive a Pos feature in place of the Neg feature of negative adjectives. This would in fact amount to a binary feature, like a Pol head that can take two mutually exclusive values $[\pm \text{Neg}]$. Starke (2004) offers a number of arguments against such an approach. The first is that $[\text{+Neg}]$ triggers weak islands, whereas $[\text{-Neg}]$ does not (or at least not always). The second is that certain types of movement are sensitive to $[\text{+Neg}]$, like negative inversion in English, but there are no similar movements that are triggered by $[\text{-Neg}]$. In this respect, $[\pm \text{Neg}]$ resembles $[\pm \text{Wh}]$ and $[\pm \text{Foc}]$: movement is often triggered by $[\text{+Wh}]$ and $[\text{+Foc}]$, but never by the corresponding negative values. Starke's conclusion is that the negative, unmarked, values of these features are syntactically absent. Since they are the unmarked values, they are recoverable in the interpretive component.

¹⁴ Small capitals represent the conceptual meaning component of semantically rich lexical items.

gradable adjectives will lose the competition against positive ones in virtue of the *Elsewhere Principle* because they have an extra Neg feature, i.e. they have more superfluous features than positive ones, which are an exact match.¹⁵ Positive gradable adjectives are all in a tie with respect to the *Elsewhere Principle*, which cannot designate any one of them as a winner. As a result, any one can be chosen to spell out QP.¹⁶ Optionally, syntax may proceed to merge NegP, and consult the lexicon again, as spellout is a cyclic process that applies after the merger of each XP. If this happens, spellout will find any negative gradable adjective in the lexicon, and since they are all in a tie, any one may get inserted. Note that positive adjectives are not candidates for spelling out NegP, since their features are not a superset of those of the syntactic tree, but a subset.

This analysis, and in particular the presence of a Q-feature in gradable adjectives, receives confirmation from the phenomenon of *much*-support (Corver, 1997:127):

- (38) a. John is fond of Mary. Maybe he is **too** *much* so.
 b. John is fond of Mary. Maybe he is **as** *much* so as Bill.
 c. The weather was hot in Cairo—**so** *much* so that we stayed indoors all day.

Much can, and in fact must, appear here, because pro-form *so* spells out less than QP, i.e. its lexical entry contains $\sqrt{P}$, not QP. As a result, *so* can spell out $\sqrt{P}$, but not QP, so that *much* is needed to spell out QP.¹⁷ We also explain the impossibility of *much* as an adjectival modifier (e.g. **much tall*). Since gradable adjectives already spell out QP, *much* is not needed to do so. Merging *much* on top of a gradable adjective would incur a violation of (12), as there would be an $f_{\text{seq}} \langle Q, Q, \sqrt{\rangle}$.

¹⁵ The spellout mechanism does not require the existence of a relation of antonymy between any one positive adjective and a negative counterpart, or vice versa. Even if a negative adjective lacks a positive antonym, it will lose the competition against other, nonantonymous, positive adjectives. Such lexical relations of antonymy could be incorporated into the analysis by means of the mechanism of pointers (Starke, 2014a), such that the lexical entry for *short* would contain a Neg-feature plus a pointer to the lexical item for *tall*. Since nothing hinges on this issue, we do not discuss it any further.

¹⁶ In principle, this situation could result in two different outcomes: a crash, or free choice of insertion of equally ranked candidates. In taking the latter option, we follow De Belder (2011).

¹⁷ Kayne (2002:96n39) argues that *so* is not an adjective because it follows, rather than precedes, the modifier *enough*:

- (i) John is fond of Mary, enough so to invite her to his party.

This is to be contrasted with *tall enough*, where *enough* follows the adjective. We take these facts to show that *so* is smaller than QP. One could think of this contrast in the manner of Starke (2014b), who suggests that a movement operation that targets the highest feature in an $f_{\text{seq}} \langle A, B, C \rangle$ will attract the spellout of AP but not that of BP. Applied to the case of *tall enough* vs *enough so*, the movement attracts *tall* but not *so*, since *so* is smaller than QP.

3.3 *Un*-prefixed positive gradable adjectives

We assume that the *un*-prefix spells out a Neg-feature and a Q-feature, as follows (De Clercq, 2017):

- (39) < /un/, [QP Q Neg] >

An argument for the presence of Q in *un*- concerns the fact that *un*- makes adjectives gradable. This appears most strikingly in the contrast with *non*-, as noted by Zimmer (1964:33):

- (40) a. non-christian: ‘(not) related to, pertaining to, characteristic of certain religious doctrines’
 b. un-christian: ‘a scale of conformity or opposition to certain norms’

Lieber (2004:121) observes the same correlation between the negative prefix *non*- and nongradability: ‘*non*- attaches to all kinds of adjectival bases, both gradable and ungradable [*sic*], and quite consistently forms negatives that are both nongradable and contradictory in meaning’. The systematic nature of this contrast between *un*- and *non*- is shown by the following cases:

(41)	A	non-A	un-A
	American	non-American	unamerican
	grammatical	nongrammatical	ungrammatical
	Cartesian	non-Cartesian	un-Cartesian
	maternal	nonmaternal	unmaternal
	motherly	??nonmotherly	unmotherly

A further example illustrating the contrast is given in (42):

- (42) a. The blood found in the closet was nonhuman/*inhuman.
 b. Their behaviour was inhuman/*nonhuman to the extreme.

In (42a), the denotation of *human* is a set of individuals, and the *non*-prefix selects the complement set.¹⁸ In (42b), in contrast, we are dealing with a scale, i.e. a set of degrees. Since only gradable adjectives can serve as the input for degree comparison, we expect only *un*-prefixed adjectives to allow such comparison, and this is indeed what we find:

- (43) a. This sentence is more ungrammatical than that one.
 b. *This sentence is more nongrammatical than that one.

¹⁸ A reviewer points out that some cases of *non*-prefixed adjectives can be graded, given the right context e.g. *Despite their perfectly human appearance, the androids’ behaviour was nonhuman in the extreme* or *It turned out that the androids were (even) more nonhuman than the aliens*. This reflects the fact that nonscalar adjectives (like *male*, *French*, *right*) can generally be coerced into a gradable meaning in the right context; see Matushansky (2013) for discussion.

In the same logic, Horn (1989:281) notes the following contrasts:¹⁹

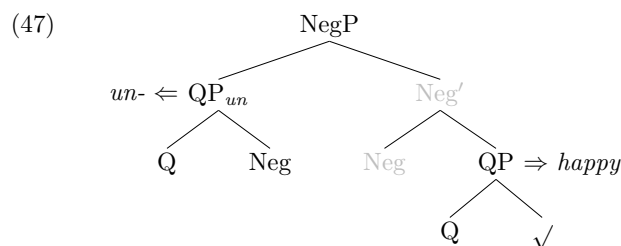
- (44) downright un-American/#non-American
 very un-Christian/#non-Christian
 extremely unnatural/#nonnatural
 somewhat immoral/#nonmoral
 awfully irrational/#nonrational
 rather unscientific/#nonscientific

Zwicky (1970) likewise observes a property of adverbs like *usually*, *typically*, *characteristically*, *probably* and a range of others.²⁰ These have a meaning as a sentence adverbial, illustrated by the fact that the adverb can be fronted without any change in meaning, as shown in (45b). However, the *un*-prefixed version of these adverbs cannot be fronted, as (46b) shows, and they only have an interpretation as a degree modifier:

- (45) a. The children are usually noisy.
 b. Usually, the children are noisy.
 (46) a. The children are unusually noisy.
 b. *Unusually, the children are noisy.

In sum, *un*- is a scalar negator with low scope; its scalarity derives from the presence of the feature Q in its lexical entry.

The tree of an *un*-prefixed positive gradable adjective is shown in (47):²¹



¹⁹ Horn (1989:281) further notes that with adjectives that have the *iN*-prefix, *un*- seems to take on the nongradable meaning of *non*-.

- (i) inhuman unhuman nonhuman
 irreligious unreligious nonreligious
 impious unpious nonpious

Observe, however, that in these cases, *un*- seems to be necessarily stressed, i.e. we have *inHUMAN* vs *UNhuman*. This is likely a different kind of *un*- than the one we are considering here. See note 4 above, as well as Selkirk (1982:100) for the claim, in the context of level ordering theory, that there are two different *un*-prefixes in English, one a Level I prefix, and another a Level II prefix.

²⁰ We are grateful to L. Haegeman for pointing this out to us.

²¹ We adopt the proposal by Starke (to appear) to the effect that prefixes, and functional material to the left of the lexical root in general, have a binary structure at the bottom (i.e. [X Y]), whereas suffixes have a unary bottom (i.e. [X [Y]]). The unary bottom is the result of moving the complement of Y across it. See the discussion of *-ful* and *-less* below.

Here, *happy* spells out QP in the usual way. In a parallel derivation, a complex specifier (QP_{un}) is created, which spells out as the negative marker *un-*. This complex specifier is then merged in the main spine as the Spec of NegP. An intuition shared between our proposal and more traditional proposals in terms of Spec-Head agreement (cf. Haegeman and Zanuttini 1991, Haegeman 1995) is that there is semantic concord between a head and a Spec with matching features, i.e. the relevant features are interpreted once, not twice.

The more radical implementation of this intuition in Starke (2004) is that two items are interpreted only once because there is only one item, i.e. that specifiers, like heads, can project, and if they do, the head is absent. Since our analysis does not hinge on the adoption of Starke's radical idea, we have trees compatible with the more traditional view that both a head and a specifier may be present, with the semantically vacuous heads shown in a lightgray to distinguish them from contentful heads, such as the ones found in e.g. (36). What is most important for our purposes is that the introduction of the negative marker *un-* introduces a Neg in the functional sequence of the main spine.²²

4 Analysis: low negation

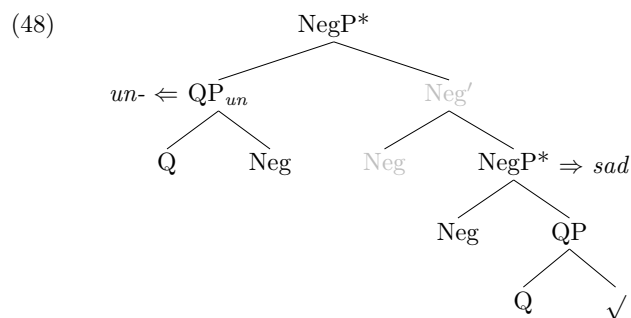
4.1 *Un*-prefixed negative gradable adjectives

We now have the elements available to account for the empirical generalisation in (6), which states that multiple negative morphemes cannot be stacked. We show that this generalisation can be explained in terms of the restriction on admissible f_{seqs} formulated in (12).

On the one hand, we have the structure of the *un*-prefix in (39), which involves a Neg feature; on the other hand, we have argued in section 3.2 that negative gradable adjectives also contain a Neg-feature in their internal makeup. Combining the two, as in *unsad*, yields the following tree:²³

²² An issue that we shall not address in any detail here is the relations between the triple *happy-unhappy-sad*, and others like it (see (1) above). Syntactically, the derivation of *unhappy* in (47) is different from that of *sad* in section 3.2 above, in that *un-* is a complex specifier, whereas *sad* only contains a projecting Neg-feature. We remain neutral on the question of the lexical relatedness of the *sad-unhappy* pair, i.e. whether they mean exactly the same or something slightly different. Elements of rich lexical content, if any, are contributed by Encyclopedia (McGinnis-Archibald, 2016).

²³ Observe that in the main projection line, Neg is higher than QP, whereas in the complex specifier it is the other way round. The reason for this is that Neg is a feature which may appear at various positions in the functional sequence: if it appears at the bottom, it results in a negative marker (e.g. *un-*), if it appears above Q it results in a negative gradable adjective (e.g. *sad*, or *weinig* 'little', to be discussed below). However, in both cases it projects as Neg in the main spine, in line with Starke (2004).



The lexical item *sad* is the phrasal spellout of a NegP that dominates an adjectival projection line $\langle \text{Neg}, \text{Q}, \checkmark \rangle$. In a parallel derivation, a complex specifier is created, which spells out as *un-*. After merging this specifier, a second NegP will project. The resulting tree in (48) violates the restriction on admissible functional sequences, since we now have two immediately consecutive NegP projections (we mark the offending consecutive NegPs with an asterisk).²⁴

One might ask why QP_{un} could not occur in the specifier of the lowest NegP, eliminating the higher NegP, and potentially avoiding the violation of (12). Such a derivation is ruled out, however, in a rather classical approach to negation, because a single NegP would contain two interpretable Neg features. In the complex specifier/head framework of Starke (2004) that we have assumed, the f_{seq} determines that there can be a NegP in the main projection line on top of QP. This NegP may result from merging either a head (as in the case of *sad*, see (36) above), or a complex specifier (as in the case of *unhappy*, see (47) above). But crucially one cannot merge both a head and a specifier, as this will necessarily introduce a second contentful negation in the f_{seq} , and therefore lead to a violation of (12).²⁵

4.2 *Un*-prefixed derived negative gradable adjectives

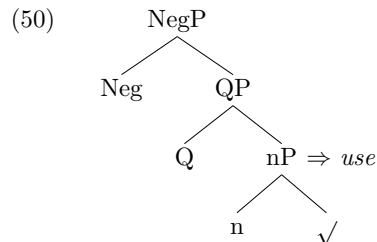
As we saw in section 1, the restriction against prefixing negative adjectives with *un-* extends to derived negative adjectives, such as noun-derived adjectives suffixed with *-less*. The lexical entries for the suffixes *-ful* and *-less* are given in (49):

- (49) a. $\langle \text{ful}, [\text{QP Q}] \rangle$
 b. $\langle \text{less}, [\text{NegP Neg} [\text{QP Q}]] \rangle$

²⁴ P. Caha (p.c.) points out a parallel with a proposal by Starke (2014b), who discusses how differences in the size of indefinites like *someone*, *something*, *somewhere* influence their modifiability by *or other*, e.g. *someone or other*. Such modification is not possible in French (**quelqu'un ou un(e) autre* 'someone or other'), which Starke accounts for by assuming that the French indefinites spell out a larger structure than the English ones. In the same spirit, *sad* spells out a larger structure than *happy*, and is therefore not modifiable by *un-*.

²⁵ See also the Generalized Doubly Filled Comp Filter of Koopman (2000), which prohibits lexical material in both the specifier and the head of a given X-bar projection.

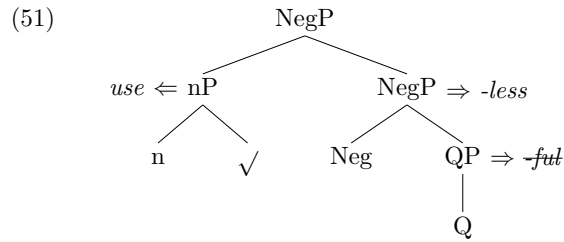
The feature Q reflects the fact that both suffixes derive gradable adjectives; their functional nature derives from the fact that the lexical entries do not contain a root feature. The negative suffix *-less* contains a Neg feature, which makes it minimally different from *-ful*.²⁶ We start out with the derivation of *useful*.



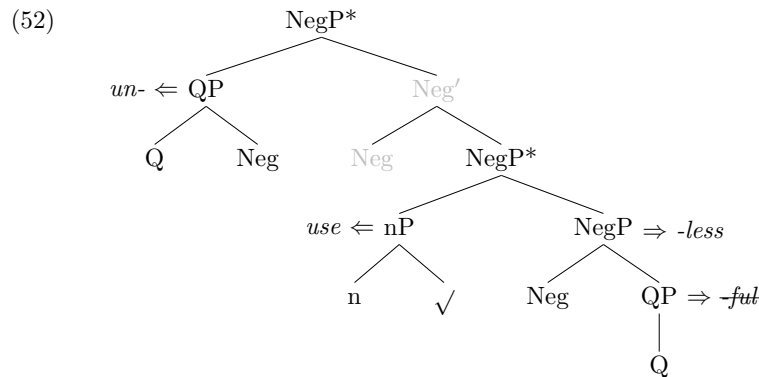
The syntax first merges the nominal structure of the noun. For concreteness, we represent this as an nP, which spells out as any contentful underived noun. Then the Q head is merged, creating QP. The lexicon, however, contains no items that spell out a QP with an nP in it. In order to be able to spell out Q, spellout-driven movement applies, adjoining nP to QP. The lower segment of QP now dominates only its head Q; QP can now be spelled out by the suffix *-ful*, since it is an exact match. The lexical item *-less* is also a candidate for insertion at QP, as its lexical entry contains QP as a subtree, but since it contains more superfluous structure it will lose the competition against *-ful*. We now have the adjective *useful*. The derivation of the adjective may stop here, or continue to merge NegP. Spellout-driven movement of nP applies again; at NegP, *-less* is the only candidate, since the features of *-ful* are not a superset of those of the syntactic tree, but a subset. In contrast, *-less* is a perfect match and gets inserted. The insertion of *-less* at NegP overwrites the earlier spellout of QP as *-ful*. The resulting tree after movements is given below (we assume that movement does not leave traces; cf. e.g. Lasnik 1999):²⁷

²⁶ It is also minimally different from *un-* in having a unary bottom, which accounts for its suffixal nature; see note 21 above.

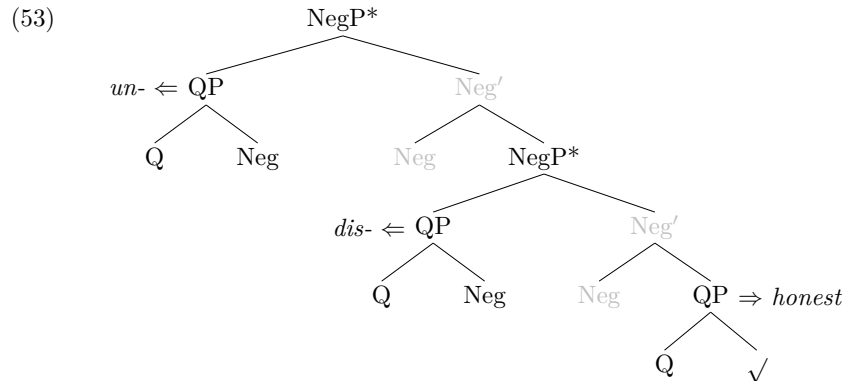
²⁷ A way of looking at the relation between *-less* and *-ful* is in terms of suppletion, i.e. *-less* suppletes for *-ful* in a grammatically negative environment. In other words, *-less* equals *-ful*+NEG, in a manner that is similar to the way *went* suppletes for *go*+PST. The overwriting of spellouts of earlier cycles by later spellouts in contexts of suppletion is restricted by a recoverability condition. Assuming that the lexical item for *-less* contains that of *-ful* as a subpart, *-ful* is recoverable once it gets overwritten by *-less*. See Caha et al. (2017) for discussion.



Let us now proceed to show why *unuseless* is impossible. We start out from the tree in (51), and add a complex specifier (merged in a parallel derivation), which spells out *un-*. This specifier needs to be merged into a SpecNegP, but doing so will result in the projection of a second NegP in the main spine, leading to the familiar <Neg, Neg> violation in the functional sequence.



Although we shall not go through all the derivations in detail, it will be clear that examples like *unhappy* and *undishonest* and others discussed in section 1 will all incur the same violation of the principle in (12). A sample derivation for *undishonest* is given in (53):



Like the earlier trees involving stacked negative morphemes, this tree features an illegitimate $f_{\text{seq}} < \text{Neg}, \text{Neg}, \text{Q}, \sqrt{} >$.

In sum, the restrictions on morphological negation discussed in section 1 are accounted for in a principled manner by the restriction on the f_{seq} in (12).

4.3 Additional support

4.3.1 Low syntactic negation

We now present evidence showing that the data pattern in section 1 is not restricted to morphological negation, but extends to cases of syntactic negation. The Dutch adjectival modifier *weinig* ‘little’ shows exactly the same restriction as the negative prefix *un-* in not combining with negative adjectives:

- (54)
- a. *weinig actief*/**passief*
little active/passive
 - b. *weinig gezond*/**ziek*
little healthy/sick
 - c. *weinig correct*/**fout*
little correct/wrong
 - d. *weinig verstandig*/**dom*
little clear/confused
 - e. *weinig interessant*/**saaï*
little interesting/boring
 - f. *weinig duidelijk*/**verward*
little clear/confused

This restriction extends to *weinig* + derived negative adjective, i.e. adjectives derived with the negative prefix *on-* ‘un’ or the negative suffix *-loos* ‘less’.²⁸ The former of these cases is illustrated in (55):

- (55)
- a. *weinig geloofwaardig*/**ongeloofwaardig*
little credible/unbelievable
 - b. *weinig verstandig*/**onverstandig*
little intelligent/unintelligent
 - c. *weinig aantrekkelijk*/**onaantrekkelijk*
little attractive/unattractive
 - d. *weinig duidelijk*/**onduidelijk*
little clear/unclear
 - e. *weinig zichtbaar*/**onzichtbaar*
little visible/invisible
 - f. *weinig geduldig*/**ongeduldig*
‘little patient/impatient’

²⁸ Some speakers find the contrasts in (54) less sharp than those in (55). We found no reflection of this fact in the corpus data that we are going to discuss shortly, however.

For the negative suffix *-loos* ‘-less’, we see the same pattern. The noun-derived adjectives with the positive suffixes *-vol* ‘-ful’ or *-rijk* ‘-rich’ in (56) contrast minimally with the negative adjectives in (57): the positive ones are modifiable by *weinig* ‘little’, the negative ones are not.

- (56) a. *weinig berouwvol*
 little remorseful
 b. *weinig begripvol*
 little understanding
 c. *weinig hoopvol*
 little hopeful
 d. *weinig succesvol*
 little successful
 e. *weinig belangrijk*
 little important
- (57) a. **weinig ademloos*
 little breathless
 b. **weinig zinloos*
 little senseless
 c. **weinig genadeloos*
 little merciless
 d. **weinig nutteloos*
 little useless
 e. **weinig belangeloos*
 little disinterested

These judgments are confirmed by corpus data from the *Corpus Hedendaags Nederlands* (Corpus of Contemporary Dutch).²⁹ Table 4 gives the results for *weinig* followed by three types of adjectives: positive adjectives (first column), *on*-prefixed adjectives (second column), and negative adjectives (third column). The contrast to note is between the first column (*weinig* + positive adjective) on the one hand, and the second and third columns, which in general show zero hits.

²⁹ The *Corpus Hedendaags Nederlands* is a collection of more than 800,000 texts taken from newspapers, magazines, news broadcasts and legal writings (1814-2013).

Table 4 *weinig + adjective* (CHN)

weinig + PosA	weinig + onPosA	weinig + NegA
aangenaam (11) 'pleasant'	onaangenaam (0) 'unpleasant'	vervelend (1) 'annoying'
vriendelijk (9) 'friendly'	onvriendelijk (0) 'unfriendly'	vijandig (0) 'hostile'
duidelijk (47) 'clear'	onduidelijk (0) 'unclear'	verward (0) 'confused'
interessant (71) 'interesting'	oninteressant (0) 'uninteresting'	saai (0) 'boring'
geloofwaardig (103) 'credible'	ongeloofwaardig (0) 'unbelievable'	
verstandig (7) 'intelligent'	onverstandig (0) 'unintelligent'	dom (1) 'stupid'
aantrekkelijk (137) 'attractive'	onaantrekkelijk (0) 'unattractive'	afstotelijk (0) 'hideous'
zichtbaar (110) 'visible'	onzichtbaar (0) 'invisible'	

Table 5 gives the results for derived adjectives with *-vol/-rijk* and *-loos*. Here, too, we see a marked contrast between the first and the second column in the number of hits:

Table 5 *weinig + N-loos/-vol/-rijk* (CHN)

weinig + N-vol/-rijk	weinig + N-loos
waardevol (6) 'valuable'	ademloos (0) 'breathless'
begripvol (5) 'understanding'	zinloos (0) 'useless'
hoopvol (70) 'hopeful'	genadeloos (0) 'merciless'
succesvol (127) 'successful'	nutteloos (0) 'useless'
sfeervol (2) 'cosy'	sfeerloos (0) 'cheerless'
belangrijk (11) 'important'	belangeloos (0) 'disinterested'

Although we shall not undertake to demonstrate it here, the same restriction can be shown to hold for the French counterpart of *weinig*, namely *peu* 'little' (see De Clercq and Vanden Wyngaerd 2017). Ducrot (1973) notes that most French speakers will avoid using *peu* as a modifier of negative adjectives, an 'avoidance' which will become 'actual refusal if these adjectives have a neg-

ative prefix', e.g. **peu désagréable* 'little unpleasant', **peu maladroit* 'little unwieldy', **peu inintéressant* 'little uninteresting'. Another piece of evidence underscoring the parallel between the negative prefix and *peu* is the fact that French takes recourse to modification by *peu* 'little' where prefixation with a negative prefix is impossible, e.g. *peu important* 'little important' rather than **inimportant*. In other words, there seems to be a semantic equivalence between *iN*-prefixation and modification by *peu*, exactly like in Dutch.³⁰

The Z–H generalisation, formulated as a restriction on morphological negation, cannot be extended to these cases. However, we will show that such data can be subsumed under the restriction on admissible functional sequences formulated in (12) above, which gives our theory a larger descriptive and explanatory power. Under our account, there is no need to postulate additional principles to account for the ungrammaticality of *weinig A-loos* 'little A-less', as its degraded status falls under the same theory as the one of *on-A-loos* 'un-A-less'. This theory has the effect of preventing the stacking of multiple negative morphemes if they are too close to one another structurally. Whether the negative marker is morphological or syntactic is immaterial. The relevant distinction is whether the relevant negative heads are adjacent in the functional sequence or not.³¹

Our analysis derives exactly this result. Assume that *weinig* 'little' spells out the features Neg, Q, and $\sqrt{}$.³²

$$(58) \quad < /weinig/, [\text{NegP Neg [QP Q } \sqrt{}]] >$$

³⁰ We owe this observation to M. van Oostendorp. Also see Ruytenbeek et al. (2017) for similar differences in acceptability between *pas très actif* 'not very active' vs *pas très passif* 'not very passive'.

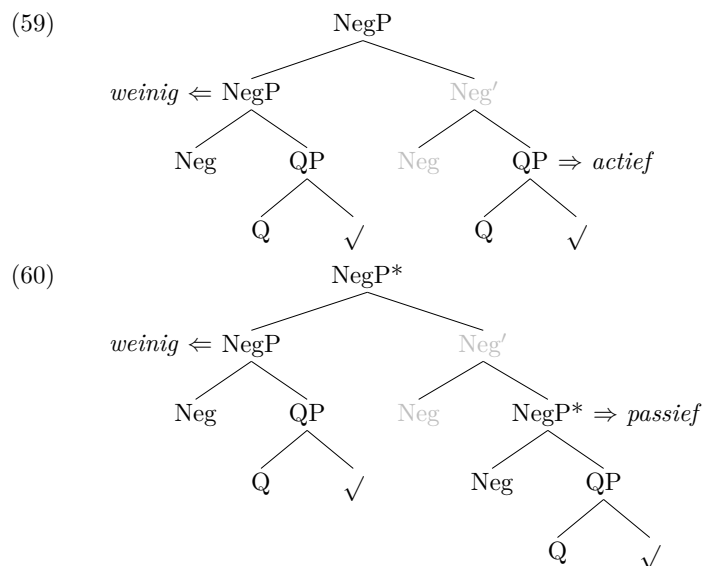
³¹ A reviewer suggests that the restriction we see here might be unified with or reduced to the following well-known contrast with measure phrases: *The table is 80cm long/*short*. However, neither *weinig* 'little' nor its polar opposite *veel* 'much' are able to occur as measure phrases:

- (i) a. **De tafel is veel/weinig lang.*
the table is much/little long
- b. **De tafel is veel/weinig kort.*
the table is much/little short

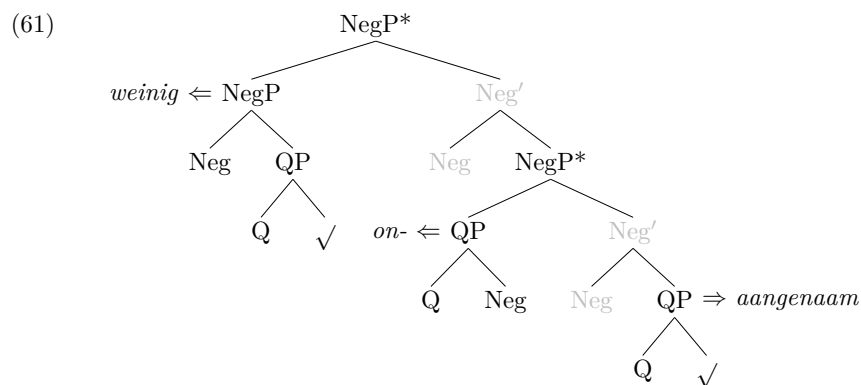
Also, if indeed *weinig* were a measure phrase and its inability to co-occur with negative adjectives followed from this fact, then we would expect to find this property in the entire system of the Q-words (Solt, 2015, Rett, 2016) to which *weinig* 'little' belongs, but this is not the case. For one thing, the polar opposite of *weinig*, *veel* 'much' cannot appear preadjectivally at all, e.g. **veel intelligent* 'much intelligent'; the same is true for its English and French analogues (**much intelligent*, **beaucoup intelligent*). With morphological comparatives, the polarity restriction is reversed: *veel/*weinig langer* 'much/little longer'. For another, the comparative degree of *weinig* 'little' (i.e. *minder* 'less') can modify positive and negative adjectives alike (e.g. *minder verstandig/onverstandig* 'less intelligent/unintelligent'). This set of facts suggests to us that the system of Q-words of which *weinig* 'little' is a part behaves differently from measure phrases.

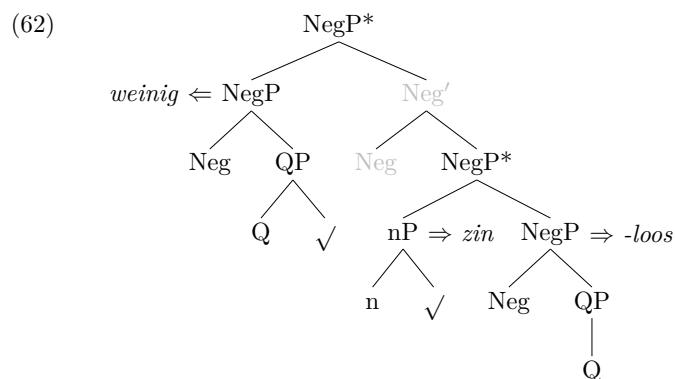
³² Note that the internal structure of *un-* and *weinig* is slightly different, not only in terms of where Neg appears (cf. footnote 23), but also with respect to the presence of a $\sqrt{}$ in *weinig*. We set these differences aside here.

Combining *weinig* ‘little’ with a positive adjective is unproblematic, as (59) shows, but with a negative adjective a violation of (12) will ensue, as shown in (60):



The fact that the negative adjective modifier *weinig* ‘little’ shows the same polarity restriction as the *un*-prefix is accounted for by the constraint formulated in (12), and assuming that both negative adjectives and *weinig* ‘little’ contain a Neg feature. The same violation of (12) occurs with derived negative adjectives (e.g. **weinig onaangenaam* ‘little unpleasant’; **weinig zinloos* ‘little useless’), and for the same reason. The trees for these are given below:





In (62), spellout-driven movement has applied and raised nP from a position below QP to a position left-adjoined to NegP in the manner described earlier (see (50) and surrounding discussion).

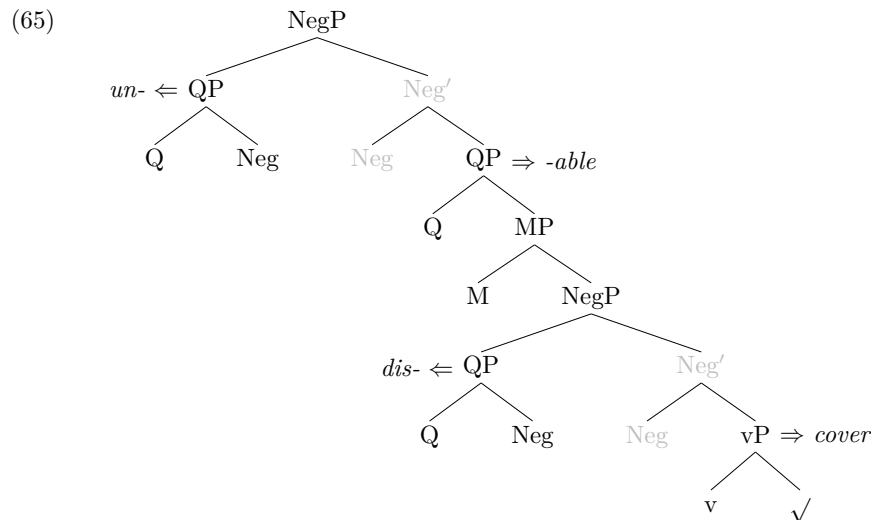
4.3.2 Structural nonadjacency

We now turn to some apparent counterexamples to the general pattern preventing the stacking of multiple negative morphemes, which actually turn out to support the general analysis developed in this paper. These are listed below (the English examples are from Siegel 1977, Zimmer 1964, Horn 1989, 2005):

- (63)
- | | |
|----------------|-----------------|
| undisheartened | unscathed |
| undisputed | undefeated |
| undiscoverable | unblamable |
| unharmmed | unobjectionable |
| invulnerable | irreproachable |
- (64)
- | | | |
|-----------------|-----------------|----------------------------|
| ongecompliceerd | 'uncomplicated' | (compliceren 'complicate') |
| ongevaarlijk | 'safe' | (gevaar 'danger') |
| onschadelijk | 'harmless' | (schade 'harm' n.) |
| onschuldig | 'innocent' | (schuld 'guilt') |
| ondogmatisch | 'undogmatic' | (dogma 'dogma') |

These adjectives are all derived from (negative) verbs or nouns. This means that the negative meaning of *disheartened*, for example, derives from the verbal prefix *dis-* in the verb *dishearten*; in a case like *defeated*, the negative meaning comes from the verb *defeat*; if there is a Neg head in that verb (an issue we remain agnostic upon at this point), it will be contained in the verb itself. Subsequently, a (nonnegative) morpheme is attached that derives adjectives (e.g. *-able*, *-ed* in English; *-lijk*, *-ig*, *-isch* in Dutch), after which the negative prefix *un-/on-* is added. The two negative projections are now structurally separated by the nonnegative, adjective-deriving, suffix. There is consequently no violation of (12), as the two Neg projections are not adjacent in the func-

tional sequence (see also Siegel 1977; Horn 1989:277). The tree in (65) shows this for *undiscoverable*.³³



The two NegPs are separated by other projections in the f_{seq} , so that the tree does not violate (12).³⁴ We leave it to the reader to verify that the other cases in (63) and (64) work in the same way.

³³ The tree ignores the spellout-driven movement of the verb *discover* to the Spec of QP needed to derive the suffixal word order in *discoverable*. Another issue concerns the featural makeup of *dis-*. In view of its reversative or privative semantics when used with verbs (Horn, 1989, Plag, 2003), it is likely that its feature composition is slightly different from the adjectival *dis-*. At the same time, it is striking that both *un-* and *dis-* have this double adjectival and verbal use (e.g. *untie*, *undo*). Also note that both *un-* and *dis-* can occur with nouns (e.g. *unrest*, *disquiet*). The suffix *-able* contains an element of modality that we represent by M. We shall not investigate these issues any further at this point, however, since they are orthogonal to the point we wish to make.

³⁴ The grammaticality of *unafraid*, with a negative adjective prefixed with *un-*, also seems an exception to the observed pattern at first. However, its grammaticality presumably also follows from the presence of more underlying structure. *Afraid* is part of a class of a-adjectives, for which it has been observed that they do not occur in attributive position (Bolinger 1971, Beard 1995, Huddleston and Pullum 2002, Larson and Marušić 2004, Cinque 2010, Boyd and Goldberg 2011, Yang 2015) and which have been argued to have developed historically from prepositional phrases (Long 1969). Following Rauh (1993), Bruening (2011a,b), Yang (2015), we assume that prepositional structure can be assumed in these APs. If this proposal is on the right track, then a derivation with additional structure between the negative adjective and the prefix (as in (65)) would explain why *unafraid* is possible.

5 Analysis: high negation

5.1 Sentential negation

At the outset, we noted that positive and negative adjectives alike can be negated with *not*, (1). In this section, we show how this follows from our analysis.

In a nutshell, *not* is a negative marker that takes scope in a higher position in the sentence, as a result of which its insertion will not lead to a violation of (12), due to the presence of intervening levels of structure. The higher scope position of *not* is itself a result of its internal makeup, i.e. its feature composition. We shall in addition argue that the clause contains two positions where *not* may appear. This claim is supported by the fact that cases exist where two occurrences of *not* may be stacked onto one another, in apparent violation of (12). This violation is only apparent, however, since *not* may occupy two different positions.

Languages often have a variety of negative markers. English, for example, has the negative markers *un-*, *in-*, *dis-*, *-less*, which above we argued all involve (at least) a Neg and a Q-feature. The evidence in section 1 above moreover suggests that these Q-markers form a natural class, given their complementary distribution, i.e. the fact that they cannot co-occur. For this reason, we call them Q^{Neg} markers. It is equally clear that *not* is in a different class, given that it can be stacked on top of the elements of the Q^{Neg} class, as also demonstrated in section 1. The class difference also correlates with a scope difference, in that *not* is a negative marker with sentential scope, whereas the other markers have constituent scope. The scopal difference can be demonstrated using the question tag diagnostic, which is a classical test for sentential negativity (Klima, 1964, Brasoveanu et al., 2014).

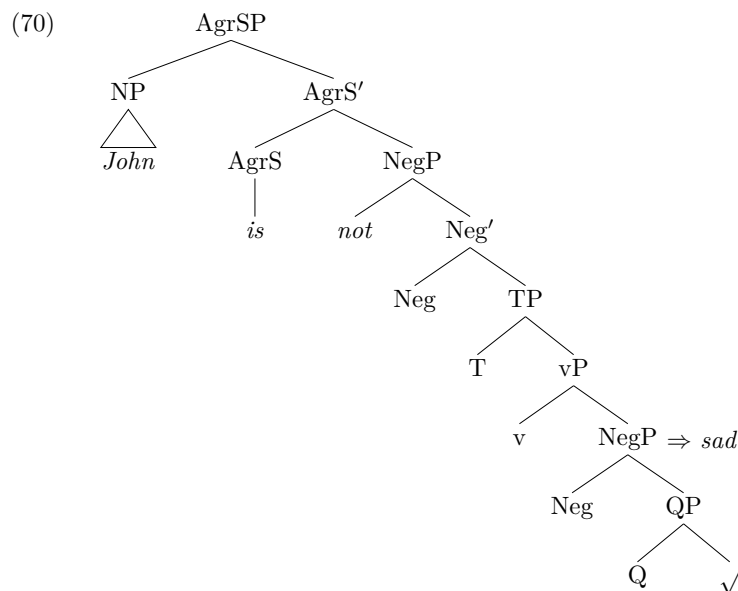
- (66) a. She isn't happy, is she?
 b. She is not happy, is she?
- (67) a. She is unhappy, isn't she?
 b. His behaviour was dishonest, wasn't it?
 c. The tea is sugarless, isn't it?

Sentences with (sentential) *not* make the sentence syntactically negative, as witnessed by the positive tags in (66). Sentences with Q^{Neg} markers are syntactically affirmative at the sentential level, as shown by the negative tags in (67). Further diagnostics introduced by Klima (1964) include the *either/too*-test, the *neither/so*-test, and the *not even*-test (see also McCawley 1998:604-612). All of these tests reveal a difference between the sentential negative marker *not* and the Q^{Neg} markers.

- (68) a. John didn't vote for Clinton, and Mary didn't vote for him *too/either.
 b. John didn't vote for Clinton, and *so/neither did Mary.
 c. John will not accept suggestions, *(not) even reasonable ones.

- (69) a. Mary was unfriendly, and John was unfriendly too/*either.
 b. Mary was unfriendly, and so/*neither was John.
 c. Mary was unfriendly, (*not) even to her own children.

We take these facts to indicate that the sentential negator *not* takes scope at a higher position in the clause, as shown in the following tree (we ignore the internal structure of *not* for now and simply represent it as *not* in the Spec of a high NegP in the clausal spine):³⁵



The negative marker *not* is generated in the SpecNegP that sits in the clausal spine at the TP-level. This clausal NegP is separated by (at least) two intervening heads (T and v) from the NegP that spells out *sad*. There is consequently no violation of the restriction against immediately successive identical heads in the f_{seq} , given in (12) above.

5.2 Focus negation

There is evidence for at least one more Neg head in the clausal spine, in addition to the position for sentential negation and the one for the Q^{Neg} markers. There is in fact a wide consensus in the syntactic literature on negation that

³⁵ Regarding the relative positions of NegP and TP, it has been claimed that (one position for) negation is higher than TP in English (or universally) by Haegeman (1995), Holmberg (2003), Moscati (2006, 2010, 2012), Temmerman (2012), Holmberg (2013). The same claim has also been made for French (Rowlett, 1998), Italian and English (Zanuttini, 1996, 1997) and Spanish (Laka, 1994). Zanuttini (1997) distinguishes four different positions for negation within the TP-domain, one of which is the position above TP.

there are (at least) two syntactic positions for negation in the clause (e.g. Lasnik 1972, Zwicky and Pullum 1983, Zanuttini 1997, Haegeman 1995, van Kemenade 2000, Cormack and Smith 2002, Haegeman 2002, Holmberg 2003, Schwarz and Bhatt 2006, Biberauer 2008, Tubau 2008). Adding morphological negation (largely ignored by the syntactic literature), we arrive at three distinct positions for negation. Our position in fact agrees with that of Horn (1989:517), who distinguishes inflected negation (sentence-level) from particle negation (VP-level) and incorporated negation (AP-level: *unhappy*). The following sentences provide evidence for this third position for negation, which is in between the high position of sentence negation and the low position of Q-negation.

- (71) a. ALL the students weren't happy, were they? $Neg > \forall$
 b. All the students were NOT happy, weren't they? $\forall > Neg$

These examples (based on Horn 1989:490) show that sentential negation (as in (71a)) correlates with positive tags and negation scoping over the quantifier. Focus negation (as in (71b)) correlates with negative tags and the negation scoping below the quantifier. The question tag test also identifies the *not* that occurs in small clauses as being nonsentential:

- (72) They consider it not likely, don't they?

The negative tag indicates that the sentence in question is syntactically affirmative, and consequently that *not* does not have sentential scope. At the same time its scope cannot be as low as that of *un-* and the other Q^{Neg} markers, since this kind of focal negation stacks onto the Q^{Neg} markers:

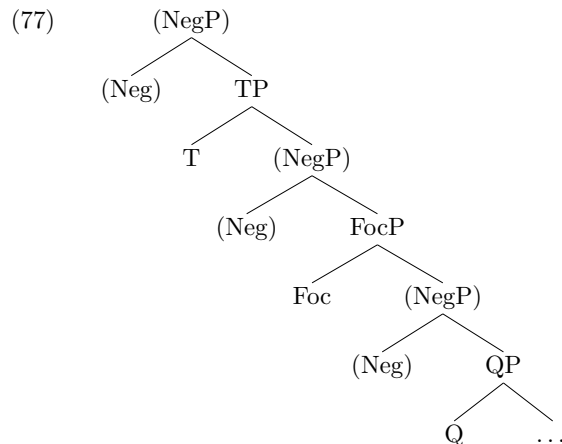
- (73) a. All the students were NOT unhappy/sad.
 b. They consider it not unlikely.

If focal negation were in the same position as the Q^{Neg} markers, we would expect it to show complementary distribution with them; since this is not what we find, we must conclude that focal negation occupies a different position, separated from the other two positions for negation by intervening heads.

In the same line of reasoning, sentential and focal *not* stack onto each other, suggesting that they, too, occupy different positions (see Horn 2014:21ff for some attested examples).

- (74) A: You should talk to your grandfather more when he comes to visit us.
 B: Well, I haven't NOT talked to him, have I?
 (75) A: Did you enjoy the movie?
 B: Well, I didn't NOT enjoy it, but I did look at my watch a couple of times.
 (76) You cannot NOT invite your aunt to your wedding.

We take the low scope or focus *not* to sit in a NegP immediately dominating so-called low FocP (Belletti, 2001, 2004, Jayaseelan, 2001, 2008, Kandybowicz, 2013). This yields the following (simplified) picture of the main clausal spine, with the three positions for negation:³⁶



The three different positions for negation in the main clause functional sequence have been put between brackets because we assume, following Starke (2004), that negation is an optional element and does not project in the syntax in the absence of overt negative markers in the clause.

5.3 The internal structure of negative markers

The clausal functional sequence in (77) provides three possible positions where negative markers can appear: these are the three specifier positions of NegP, where a negative marker may be merged as a complex specifier. Which Spec-NegP a negative marker appears in, i.e. its scope position, is determined by its internal makeup. A similar correlation between internal structure and scope-taking position in the clause has been claimed by Endo and Haegeman (2014) with regard to the position and internal structure of adverbial clauses. The same idea is present in the Level Embedding Regime of Williams (2003, 2009). In this section, we consider the question of the internal structure of negative markers in greater detail.

³⁶ De Clercq (2013) argues that there is a fourth position for negation, which she calls DegP (or ClassP in later work, De Clercq 2017), and which is situated between QP and FocP. The English negative marker that spells out DegP is *non-*, which seems to be in a different class from the Q^{Neg} markers. For the sake of simplicity, we ignore this fourth type of negation here. See also Zanuttini (1997), Poletto (2017), who likewise argue in favour of four distinct positions for negation on the basis of detailed study of Italian dialects. The positions they distinguish are restricted to the TP-domain and do not involve positions for affixal or lexical negation.

The basic idea we wish to pursue is that the features and the functional sequence $\langle T, \text{Foc}, Q \rangle$ of the clausal spine (77) are replicated in the internal structure of negative markers (see also Poletto 2017). Concretely, negative markers have a single Neg-feature at the bottom, which defines them as negative markers, but which is packaged with different sets of features from the $\langle T, \text{Foc}, Q \rangle$ sequence, depending on the type of negative marker.

Let us illustrate this with the English negative markers *not* and *un-*. The internal structure for sentential *not* is richer than for the Q^{Neg} markers discussed above, and contains the three scope-related features $\langle T, \text{Foc}, Q \rangle$, as well as negation proper (De Clercq, 2013). The relevant lexical entries are given in (78) ((78b) being repeated from (39) above):

- (78) a. $\langle \text{/not/}, [\text{TP } T [\text{FocP } \text{Foc } [\text{QP } Q \text{ Neg }]]] \rangle$
 b. $\langle \text{/un/}, [\text{QP } Q \text{ Neg }] \rangle$

As we saw, *not* is both a marker for sentential negation and focus negation, i.e. it may spell out two different kinds of syntactic trees, and, as a result of this, take scope in two different positions.³⁷ This fact is accounted for by the lexical item in (78a). The Superset Principle as formulated in (33) above will ensure that *not* can spell out both a sentential negative marker (TP) and a focus negation marker (FocP). In the case of sentential negation, the lexical tree in (78a) will be an exact match for the syntactic tree, whereas in the case of focus negation, the lexical tree (TP) contains the syntactic tree (FocP) as a subtree. The lexical item in (78a) cannot spell out negation at the QP-level because of the *Elsewhere Principle*: although *not* is a candidate for insertion, it will lose the competition against the Q^{Neg} markers because they contain less superfluous structure.

An important argument in support of this decomposition analysis of negative markers is provided by crosslinguistic syncretism patterns. De Clercq (2013, 2017) shows how crosslinguistically the hierarchy in (77), which is based on scope and stacking facts, is confirmed by syncretism patterns found in Greek, English, French, Czech, Mandarin Chinese, Modern Standard Arabic, Hungarian, and Persian, which represent four different language families.³⁸ Negative markers are always syncretic in a way that respects the sequence in (77), i.e. there are no ABA-patterns. English, for example, has an AAB-pattern: the T-negator and the Foc-negator are syncretic (*not*), and different from the Q-negators. Greek has an ABC-pattern, i.e. three different negative markers, thus confirming the existence of these three types. Czech has an AAA-pattern, with the three markers fully syncretic. But there appears

³⁷ We leave aside the possibility of phonological reduction or phonological stress.

³⁸ De Clercq (2018) has extended the sample of languages to 21 languages, which belong to 10 different language families, with two languages with an affiliation that is under debate: Indo-European (English, Swedish, Dutch, French, Greek, Czech, Macedonian, Russian, Persian), Sino-Tibetan (Chinese), Austronesian (Malagasy), Semitic (Hebrew, MS Arabic, Moroccan Arabic), Finno-Ugric (Hungarian), Turkic (Turkish), Khoe (Khwe), Carib (Hixkaryana), Uto-Aztecan (Tümpisa Shoshone), Dravidian (Malayalam), Korean (family debated), Japanese (family debated)

to be no language with an ABA pattern, i.e. with a T-negator that would be syncretic with a Q-negator across a Foc-negator.

(79)

	T ^{Neg}	Foc ^{Neg}	Q ^{Neg}
English	not	not	un-
Greek	dhen	oxi	a-
Czech	ne	ne	ne
unattested	A	B	A

The syncretisms between negative markers show that scopally different negative markers are structurally related, i.e. share certain features (cf. Caha 2009 on Case). They furthermore provide evidence as to how these features can be ordered, i.e. they yield the functional sequence in (77). The lexical entries that we assumed for *not* and *un-* in (78) show the structural relatedness of the two English negative markers: the tree for *un-* is structurally contained in that for *not*.³⁹

As we said above, where a Neg marker takes scope in the clause is determined by its internal structure. A Neg marker with a T feature takes scope at TP: it will be inserted as a complex specifier in the SpecNegP that immediately dominates TP (see (77) above). Similarly, a negative marker that spells out FocP is inserted in the SpecNegP that dominates FocP in the clausal spine, etc.

Summarising the results of section 5, we have seen that the restriction on stacking multiple negative morphemes onto one another that we started out with (see (6) above) turns out not to be a restriction on morphological vs syntactic negation. Neither is it a restriction that can be formulated in terms of phonological identity: although certain phonologically identical negative markers are not stackable (e.g. **unun-*, **disdis-*, etc.), others are (e.g. *not not*). The relevant criterion is where a negative marker takes scope: if two markers take scope in identical positions, they are not stackable, because this will create an inadmissible functional sequence. This is so even if they are phonologically different, or not linearly adjacent, as the discussion in section 4 has shown. If two negative markers take different scopes, they can be stacked, even if they are phonologically identical. This is notably the case with English *not*, which may either be a sentential or a focus negator. As a result, two occurrences of *not*—each with different internal structure—may appear in the same clause, because they take scope in NegPs that are separated by intervening levels of structure, thus avoiding a violation of (12).

³⁹ The idea of an internally complex negative marker is also present in Poletto (2008, 2017). However, her approach differs from De Clercq (2013, 2017) in many ways and most crucially in that so-called lexical negation is not taken into account.

6 Conclusion

We have proposed a restriction on admissible functional sequences that forbids sequences of immediately successive identical heads. We have shown how it allows an account of a range of facts that have hitherto defied a principled explanation. These facts involve a restriction on the stacking of multiple negative morphemes, both within and across words, i.e. the restriction is not confined to the word-internal domain, but manifests itself across word boundaries. The proper analysis of the facts is in terms of a distinction between the scope position of different types of negative markers, such that negative markers with different scope positions may be stacked onto each other, but negative markers with identical scope positions may not. The analysis supports an architecture of the grammar where there is no separate domain of morphology, given that the design principles and restrictions apply identically across the domains traditionally labelled as syntax and morphology. Our account crucially relied on the presence of a Neg feature in negative markers and negative adjectives, and on a late insertion model of grammar. The mapping of syntactic feature bundles onto a phonology happens in a maximally straightforward manner under the nanosyntactic mechanism of phrasal spellout.

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5 Late Insertion and Root Suppletion

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LATE INSERTION AND ROOT SUPPLETION¹

INSERÇÃO TARDIA E SUPLEÇÃO DE RAIZ

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RESUMO: Este artigo propõe uma abordagem de inserção tardia Nanossintática para a supleção de raízes. Nós mostramos que essa teoria nos permite explicar a supleção de raízes dentro de uma teoria da gramática estritamente modular, que não faz nenhuma diferenciação sintática entre raízes distintas. Como ponto de partida, focamos primeiramente nas dificuldades arquitetônicas que surgem em uma teoria modular das raízes na abordagem da Morfologia Distribuída. Em seguida, demonstramos como a Nanossintaxe contorna esses problemas e tratamos de duas questões empíricas potenciais para o tratamento Nanossintático (exponência múltipla e localidade), mostrando como, na verdade, elas fornecem suporte para a abordagem proposta.

PALAVRAS-CHAVE: Inserção tardia; Nanossintaxe; Morfologia Distribuída.

ABSTRACT: This article proposes a Nanosyntactic Late-Insertion approach to root suppletion. We show that this theory allows us to account for root suppletion within a strictly modular theory of grammar, which makes no syntactic distinction between different roots. As a starting point, we first focus on the architectural difficulties that arise for a modular theory of roots in the Distributed Morphology approach. We then show how Nanosyntax circumvents these problems, and address two potential empirical issues for the Nanosyntactic treatment (multiple exponence and locality), showing that they in fact provide support for the approach proposed.

KEYWORDS: Late insertion; Suppletion; Nanosyntax; Distributed morphology.

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INTRODUCTION

There are several good reasons to adopt a syntax-based Late Insertion model as an approach to morphology (see Halle & Marantz, 1993; Marantz, 1994; and Embick & Noyer, 2007 for a discussion). We start by briefly discussing what we take to be the two most important ones, namely universality and modularity.

Universality refers to the idea that the atoms of syntactic trees (its terminals) are not language specific, but they are drawn from a universally-available set of features. In Late-Insertion models (which place the lexicon after the syntactic derivation), the syntax no longer has to deal with the arbitrary and language-specific objects that lexical items are. Instead, the atoms of syntax correspond to a universal set of features that refer to syntactically relevant semantic distinctions, like ANIMATE, COUNTABLE, PLURAL, etc. Language-particular elements are introduced late through lexical items, understood as unpredictable and language-specific linking between the universal features and a language-particular phonology, and, in some cases, also (encyclopaedic/conceptual) meaning, that is not syntactically relevant. In Late-insertion models, these language-particular aspects of individual languages become available only after syntax, in the process of the ‘externalisation’ of syntax through lexical insertion.

The second major advantage of Late Insertion is *modularity*. By this we mean the following:

(1) *Strong Modularity Thesis* (SMT)

Syntactic representations only contain entities that are relevant for the application of syntactic principles and operations.

This excludes from the syntactic module everything that is not syntactically relevant, specifically, the phonological, conceptual and encyclopaedic information that is associated with lexical items. To take a concrete example, the difference between *dog* (singular) and *dogs* (plural) is syntactically relevant, as seen in various kinds of agreement processes triggered by nouns, but the difference between *cat* and *dog* is purely conceptual, and is not relevant to any syntactic operation. Late Insertion offers a way to separate the syntactically relevant from the syntactically irrelevant. It provides a type of architecture in which syntax is in principle unable to refer to syntactically irrelevant properties of lexical items like their phonology or conceptual information. On the other hand, if the atoms (terminals) of syntactic derivations were the traditional lexical items, then their phonology (and conceptual meaning) would necessarily be present in syntax as well. This clashes with the insight that neither their phonology nor conceptual information play any role in syntax.

Within Distributed Morphology (DM), this view has been uncontroversial and universally adopted for functional lexical items like the comparative *-er* etc. It has been

a common stance to extend Late Insertion also to open-class lexical items, so-called roots. As Marantz (1996, p. 16) puts it “[n]o phonological properties of roots interact with the principles or computations of syntax, nor do idiosyncratic Encyclopaedic facts about roots show any such interactions” (see also the Principle of Phonology-Free Syntax of Zwicky, 1969; Zwicky & Pullum, 1986; and Miller, Pullum and Zwicky, 1997). If Late Insertion is applied also to lexical items like *cat* and *dog*, we derive the fact that syntax cannot distinguish between them either in terms of their phonology or in terms of their conceptual and/or encyclopaedic meaning.

The syntactic irrelevance of the distinction between roots like *cat* and *dog* has been taken to its logical conclusion in models such as Halle and Marantz (1993), Marantz (1996) and De Belder and Van Craenenbroeck (2015), where syntax contains just a single root symbol $\sqrt{}$, which is an object devoid of syntactic, phonological or semantic properties. $\sqrt{}$ works as a pure placeholder for the insertion of the morphological root. Concrete root morphemes are inserted late into such a non-discriminate $\sqrt{}$ terminal based on a free choice (Harley & Noyer, 1999).

In this context, the issue of suppletion is relevant. Since suppletive roots have, by definition, several phonologically unrelated forms depending on the context, it must be the case that roots enter the derivation without any phonology, and acquire it only once the appropriate context has been determined. As Haugen and Siddiqi (2013), Harley (2014) and Arregi and Nevins (2014) have argued, Late Insertion of roots also allows the theory to deal effectively with root suppletion, while simultaneously maintaining the two advantages alluded to above.

However, within a classical DM architecture (Halle & Marantz, 1993; Harley & Noyer, 1999), this fully modular approach to roots meets two challenges. One is related to competition among roots. The other concerns the proper pairing of conceptual and encyclopaedic information with phonological information (as we discuss in section 1). Because of these two issues, some recent approaches adopt the view that individual roots are, after all, differentiated in syntax. For instance, Harley (2014) (following Pfau, 2000, 2009) uses arbitrary numerical indexes to this effect. While this move solves the two issues noted above, it no longer satisfies the Strong Modularity Thesis as formulated in (1), i.e. it fails to deliver a fully modular architecture, for reasons that we discuss in the body of the paper.

In this context, the main goal of this article is to show that it is possible to account for root suppletion in a universalist, fully modular approach to roots, i.e., without the need to introduce arbitrary diacritics on roots in syntax. In developing a theory along these lines, we will be drawing on the Nanosyntax theory of spellout (Starke, 2009, 2018).

The organisation of the paper is as follows. In Section 1, we discuss some of the main findings emerging from the treatment of root suppletion in DM. We also highlight here two challenges that arise for these approaches. We introduce the architecture of

Nanosyntax (Starke, 2009, 2018) in sections 2 and 3. We show that this theory allows for an approach to root suppletion where neither of the two problems mentioned above arises. By demonstrating this, we show that Nanosyntax allows for an approach to root suppletion where syntactic terminals are both modular and universal. In sections 4 and 5, we defend the approach against two kinds of possible objections. The first objection is that phrasal lexicalisation, on which the theory relies, is not the right tool for suppletion, because it does not allow for multiple exponence. Section 4 argues that it does. The second objection is that it cannot handle non-local conditioning of suppletive roots. Section 5 argues that this is, in fact, a desirable result, despite apparent challenges.

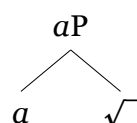
1 ROOTS AND SUPPLETION IN DM

In the current section, we discuss two kinds of approaches to roots in DM. In Section 1.1, we discuss an approach that is both universalist and modular. However, this approach faces two challenges: one related to suppletion and another one related to the overall architecture of grammar in DM. In Section 1.2, we discuss an approach that addresses these challenges by differentiating among roots in syntax. Our point is that while root-differentiation is successful in resolving the issues, it does not deliver a fully modular architecture. These considerations form the background to our own proposal that we put forth in Section 2.

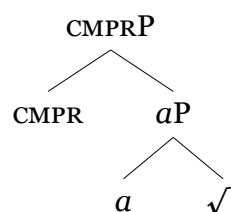
1.1 TWO CHALLENGES FOR SINGLE-ROOT APPROACHES

This section highlights two challenges facing a fully modular, universalist theory of roots. To make a number of points about root suppletion and beyond, we will be often using the positive and the comparative degree as an example (for a number of reasons, one of them being the fact that this is a well-researched topic, thanks to the work by Bobaljik, 2012). The structures we will be initially assuming are as depicted in (2). More specifically, we will be assuming that the positive degree (in (2a)) is contained in the comparative degree (2b), which adds the *CMPR* head on top (Bobaljik, 2012). We will decompose the positive degree into a $\sqrt{}$ node and a little *a* node, to maintain easy comparison with existing proposals in the DM literature (but see Vanden Wyngaerd et al., 2020 and De Clercq et al., to appear for a different, root-free type of approach to the bottom of the functional hierarchy).

(2) a. positive



b. comparative



We want to make two points about (2), which relate to the issues of universality and modularity that we discussed in the previous section. The first point is modularity. The trees in (2) show a syntactic representation that is in agreement with the Strong Modularity Thesis of the introduction. To see this, consider the fact that it is impossible to tell from the structures (2a,b) alone what phonology or meaning they represent (as opposed to the roots *nice* or *kind*). This is a good thing, because syntax cannot distinguish these two roots either (in a similar way in which it cannot differentiate between the root *cat* and *dog*). The inability to capture in syntactic representations such irrelevant differences would be a failure by design, because this is what modularity is all about.

Another way to show that the syntax we assume for the positive and the comparative degrees respects modularity is to imagine that we were told to draw the syntactic structures for the positive degree of the adjectives *kind* and *nice*. With just a single $\sqrt{}$ node at our disposal, both structures would look exactly the same, namely as (2a). This shows that the trees in (2) respect the Strong Modularity Thesis, where information that is irrelevant to syntax is not represented in syntax.

Note that things would be different if we allowed ourselves to use multiple root symbols, one for each root. In such case we would draw different structures for *nice* and *kind*, one with the symbol $\sqrt{\text{NICE}}$ another one with the symbol $\sqrt{\text{KIND}}$. Our point is that when we are able to tell from the syntactic trees which root they represent – without that distinction being relevant for syntax – the Strong Modularity Thesis is violated. This is the case no matter what specific information we use to differentiate $\sqrt{\text{NICE}}$ from $\sqrt{\text{KIND}}$: it could be an arbitrary index, a concept, a phonological string or a combination thereof; the problem remains the same.

The second point about the structures in (2) is that they are universal. Specifically, if all languages have the same ingredients for the positive and the comparative (as proposed in Bobaljik, 2012), it is impossible to say whether the structures in (2) depict the structure of English or Latin roots. This is a good thing too: this is what universality is all about. In contrast, if the trees had symbols such as $\sqrt{\text{NICE}}$ $\sqrt{\text{PULCHER}}$ (Latin for ‘beautiful’), we would be able to tell which language we are dealing with.

Despite its good features, some technical issues arise for this view when we try to implement it in the DM model as given in Harley and Noyer (1999). We present their model (with slightly updated labels) in Figure 1.

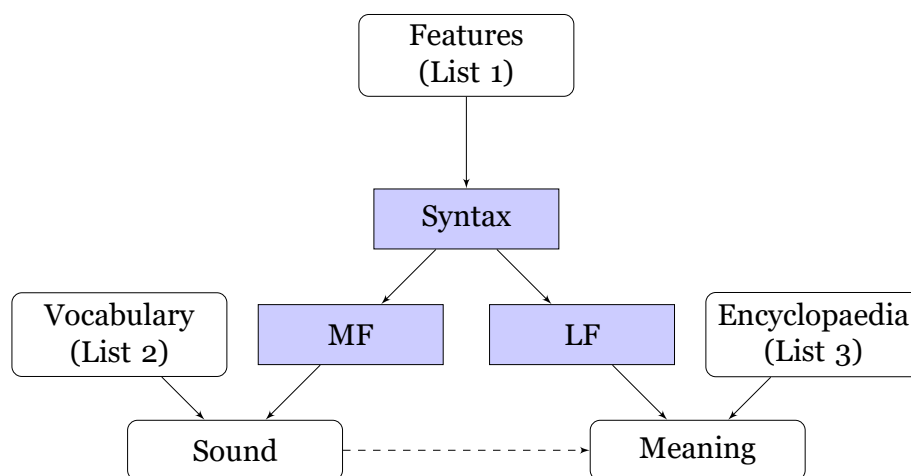


Figure 1: A version of Distributed Morphology (based on Harley & Noyer, 1999)

Figure 1 shows that the syntactic computation begins from features (or feature bundles) drawn from a list that is referred to as List A in Harley and Noyer (1999) and as List 1 in Arregi and Nevins (2014). These features enter the narrow syntactic derivation, where they are assembled into syntactic trees. For reasons that are not directly relevant for the current concerns, DM then splits the derivation of the sentence into two different branches: we have the so-called PF branch (leading from syntax to sound) and the LF branch (leading from syntax to meaning). On the PF branch, various purely morphological operations take place. These may displace, add or remove nodes and features, enriching or impoverishing the structure provided by the narrow syntax. The placement of these operations on the PF branch is motivated by the fact that they do not affect the meaning. The output of these operations is a structure that we refer to as the morphological form (MF), which at this point still consists of abstract features. On the basis of the MF, the sound representation is constructed by consulting the list of exponents (Vocabulary Items, List 2). The two arrows going into the sound representation thus indicate the fact that the sound is determined jointly by the two components, namely by the features that are found at MF, and the list of exponents selected to match the features at MF.

The meaning representation is formed in an analogous fashion. First we construct the so-called Logical Form, LF. Harley and Noyer (1999) are not explicit about how the LF is constructed, but we assume that as in standard minimalism, LF in their conception arises through the application of covert movements such as quantifier raising and the like. In any event, LF is still an abstract syntax-internal representation. All such syntax-internal representations (composed of abstract features and their structures) are marked by the (violet) shading in Figure 1.

As a part of mapping the LF to meaning, the Encyclopaedia is consulted and contributes encyclopedic and/or conceptual aspects of meaning in a fashion analogous to

the list of exponents in Vocabulary.

With the basics in place, let us turn to the first challenge. The observation is that the separation of the phonological information (List 2) from the encyclopaedic/conceptual information (List 3) has consequences for the representation of roots in syntax and hence, also for modularity. To see that, recall the fact that the trees in (2) contain just a single root symbol $\sqrt{}$. In Harley and Noyer (1999), all root morphemes such as *cat* and *dog* reside in the Vocabulary, and they are inserted into such a non-discriminate $\sqrt{}$ terminal based on a free choice. Given this setup, the question arises how LF knows what kind of exponent has been inserted at the PF branch of the derivation, so that if *kæt* is inserted from the Vocabulary at the PF branch, the LF learns about this and the Encyclopaedia contributes the right encyclopaedic/conceptual information at the meaning side. What must be prevented is that the non-discriminate $\sqrt{}$ symbol arrives at LF and Encyclopaedia has the power to independently select a particular lexical item, so that, in effect, /kæt/ means DOG.

In order to avoid this, Harley and Noyer (1999) suggest that a direct communication line (depicted by the dashed arrow) is established between the sound and the meaning, which makes sure that the Encyclopaedia knows what information it should provide. The architectural consequence is that sound and meaning are linked both via the syntactic derivation (as is standardly assumed), but also outside of it for the sole purpose of root insertion. The direct communication line between sound and meaning is the first complication that arises in DM when a fully modular approach to roots is adopted.

As we shall see in Section 1.2, there are versions of DM that do not need such a direct communication line between sound and meaning. However, we will also see that the price one has to pay for eliminating it is that roots must be differentiated within syntax, which is something that we avoid in our own proposal in Section 2.

Let us now turn to the second challenge that a theory based on non-discriminate roots faces. It comes from root suppletion and the related notion of root competition. In the DM framework, root suppletion, as in the pair *good-better*, is accounted for by contextual specification of Vocabulary Items (VIs), which insert phonology under the terminals, in this case the $\sqrt{}$ node. We need two distinct Vocabulary Items for the suppletive pair *good-bett*; they are as in (3), which are slightly adapted from Bobaljik (2012) to fit the trees in (2) above:

- (3) a. $\sqrt{} \Leftrightarrow \text{bett-} / ___] a] \text{CMPR}]$
 b. $\sqrt{} \Leftrightarrow \text{good}$

In the positive degree, these VIs are not in competition with each other, as there is no CMPR head there, so that only (3b) meets the structural description, and *good* will be inserted. In the comparative, given in (2b), however, a competition between (3a) and (3b) will arise, since the structure generated in syntax meets the structural description

who argues that suppletive verbs in Hiaki have rich lexical meanings, for which an analysis in terms of functional heads is unlikely (cf. Haugen & Siddiqi, 2013). To deal with this issue, Harley (2014) has argued that $\sqrt{\text{ }}$ s are individuated in the syntax, i.e., prior to vocabulary insertion, by means of a numerical index. We shall describe this approach in the next section, noting that while it successfully addresses the challenges identified, it does so at the cost of proposing different syntactic representations for different roots.⁶

1.2 ROOT DIFFERENTIATION

According to Harley (2014), the pre-syntactic lexicon contains an infinity of different, individuated, $\sqrt{\text{ }}$ s (see also Pfau, 2000, 2009). Free choice of a root is then not exercised at the point of insertion (as in Marantz' approach), but at an earlier point, namely in the selection from List 1, i.e., when the elements that will serve as the input to the syntactic computation are selected. At the point of insertion, the competition is consequently restricted to the VIs that can spell out the particular root symbol selected, e.g., $\sqrt{153}$.

The Vocabulary Items that we shall need under Harley's proposal would then be as in (6):

- (6) a. $\sqrt{153} \Leftrightarrow \text{bett-} / ______] a] \text{CMPR}]$
 b. $\sqrt{153} \Leftrightarrow \text{good}$

The idea here is that both *good* and *bett-* are two different forms capable of spelling out the symbol $\sqrt{153}$. What this proposal achieves is that *bett-* is no longer a comparative of *just any* $\sqrt{}$ (as it has been in (3)), but a comparative of *one particular* $\sqrt{}$ with a unique index. Once this is the case, the problem with root competition disappears. This is because *bett-* will not be a candidate for any other root than $\sqrt{153}$, and so it will never outcompete *nice*, which spells out (say) $\sqrt{154}$.

This proposal has as an additional benefit in that direct communication between the PF branch and the LF branch of the grammar (i.e. the dotted line in Figure 1) is no longer needed, since the index will be present from the start of the derivation, and be carried through to both MF and LF.

However, the proposal achieves these results as a consequence of adopting indexed roots, so *cat* and *dog* have a different index. In one sense, this proposal is elegant in that syntax still does not operate over roots that are specified for language-particular phonology and/or encyclopaedic meaning: it still has the potential of being universalist (although it would appear to predict that all languages have the same number of roots,

⁶ Other solutions are thinkable, but have been shown to be less viable. For example, one could suggest that VIs for $\sqrt{}$ nodes do not compete with each other at all (something which is suggested by the free choice rule in (4)), but such a generalised lack of competition at the $\sqrt{}$ node would lead to the problem that both *gooder* and *better* are generated (Harley & Noyer, 1998).

a prediction that is not obviously true). In other words, the symbol $\sqrt{153}$ is not specific to English, but it is found in all languages (which will, of course, associate a different phonology and/or encyclopaedic content to it). At the same time, the differentiation via numerical indexes is a case that violates the Strong Modularity Thesis, as explained in the discussion below (3). By containing symbols such as $\sqrt{153}$, syntactic representations contain information that syntax cannot process. The index passes intact through the syntactic computation with a single purpose, namely to be used by Lists 2 and 3. So even though the numerical index is more abstract than a root with a concrete phonology, the problem is still the same: we are representing a distinction in the syntax that the syntactic computation itself cannot make use of, in violation of the Strong Modularity Thesis.

To summarise the content of this whole section, let us repeat what is going to be important as we move to our own proposal. The most important observation is that in a system with non-discriminate roots, root suppletion leads to a problem with competition: once *bett-* outcompetes *good*, it also outcompetes *nice*, because *good* and *nice* have exactly the same lexical entry (spelling out $\sqrt{}$).

To solve the competition problem, we must be able to uniquely identify the lexical item that undergoes suppletion, and limit the competition to those VIs which stand in a suppletive relation to this particular item. Marantz (who works with just a single $\sqrt{}$) makes suppletive items unique by placing them in the class of ‘functional’ heads, which allows one to identify the unique grammatical feature whose realisation shows suppletion. Harley (2014) suggests that this view is empirically problematic and proposes that $\sqrt{}$ s are individuated by an index.

Our main objection against individuating roots in syntax is that by doing so, one in fact abandons strict modularity and allows for a theory where *cat* and *dog* have different syntax. This is because *cat* and *dog* have a different index, and syntax could be sensitive to this property (since it is present inside it). On the other hand, if we stipulate that syntax is not sensitive to such indexes, then modularity issues arise.

An additional issue is that if $\sqrt{}$ s really lack any constant *substantive* property, i.e. something more contentful than a mere index, one needs to seriously wonder why they should be differentiated in narrow syntax at all.⁷ The indexation of roots looks like a technical solution to a technical problem, rather than an advance in the understanding of the nature of roots.

In the following sections, we will argue that there is a way to handle root suppletion without the need to differentiate roots in syntax by an arbitrary index, thereby adhering to strict modularity. In order to avoid the competition problem, we too will have to find a way to identify the unique root which undergoes suppletion. The device we are going

⁷ In fact, one may wonder why they should be present in the syntax at all to begin with. Since this question is orthogonal to our concerns, we shall leave it aside here.

to use to this effect is called a *pointer*, and we introduce it in the next section.

2 CYCLICITY AND PHRASAL LEXICALISATION

In this section, we describe the main features of an account that allows for root suppletion with just a single $\sqrt{}$ in syntax (or without any $\sqrt{}$ at all, if $\sqrt{}$ s are to be eliminated, as in Ramchand, 2008 or Vanden Wyngaerd et al., 2020). What makes such a theory possible is cyclic phrasal lexicalisation, where suppletive items stand in a containment relationship.

In order to present this idea in an accessible way, we will begin with the suppletive pair *bad*–*worse*, which has been treated by nonterminal lexicalisation also in Bobaljik (2012). We shall then return to *good*–*bett-er* in the following section. The relevant lexical entries (with the required containment relation) are given in (7). Regardless of the treatment of *bad* (to which we return), the important point here is that *worse* spells out a nonterminal node properly containing the structure that *bad* spells out.



Independent support for (7b) comes from the fact that *worse* lacks the regular CMPR marker *-er*. This is accounted for if its lexical entry pronounces the terminal where *-er* gets usually inserted, as is the case in (7b). Similarly, the reason why *bad* spells out a full phrase is that it shows no overt *a*, differing from adjectives like *risk-y*, *crapp-y*, *tin-y* etc.

We will get to the technical details of nonterminal insertion shortly, but the main intuition is this: when syntax builds just the *aP* (corresponding to the positive degree), only *bad* will be inserted, because its lexical entry provides an exact match for the syntactic tree. The lexical item for *worse*, in contrast, is not an exact match: it is too big. ‘Too big’ may be understood either in an absolute sense (it is not a candidate for insertion at all), or in a relative sense (it is a candidate, but it is too big relative to *bad*, with which it is in competition). When syntax builds *CMPRP*, only *worse* is an exact match and will be inserted, this time because *bad* is too small (either in the absolute or in the relative sense).

There are several ways of formalising the phenomenon that an exact match gets inserted, and not a lexical item which is either too big or too small. For instance, Bobaljik (2012) relies on the Subset Principle, augmented by Radkevich’s (2010) Vocabulary Insertion Principle (VIP), which states that the phonological exponent of a vocabulary

item is inserted at the minimal node dominating all the features for which the exponent is specified. On this account, the Subset Principle makes sure that *worse* is too big for the positive, and the VIP makes sure that *bad* is too small for CMPRP. Another available option, which we develop and explain later, adopts the Superset Principle (Starke, 2009). For now, the main point is that no matter how the ‘too big/too small’ difference gets encoded, we initially run up against the same conundrum as the terminal-based proposal in section 1. In order to see that, let us once again turn to the fact that there are a number of roots in free competition with *bad*:

- (8) aP $\Leftrightarrow$ *good, nice, kind, small, intelligent, bad, ...*
-

Again, the problem is that once syntax builds the CMPRP, all of these are going to be ‘too small’ compared to *worse*. The problem resides in the fact that the lexical entry in (7b) says that whenever the syntax combines the $\sqrt{}$ node with *a* and CMPR, *worse* will be an exact match for such a constituent. Other lexical entries (like *nice*) might be candidates for insertion as well, but since they are not an exact match for the comparative structure, *worse* will win, independently of how the competition is to be implemented.⁸

However, in the new setting based on phrasal lexicalisation, a new type of solution to this problem becomes available, if one more ingredient is added into the mix. The addition that is needed is that the lexicalisation process, which associates a particular phonology with a syntactic structure, proceeds bottom-up, as in Bobaljik (2000, 2002) and Embick (2010) or Starke (2009, 2018). We phrase this as (9), noting that (9) need not be seen as an axiom, but rather the consequence of two proposals, which are given in (10).

- (9) *Bottom-up Lexicalisation*
 If AP dominates BP, spell out BP before AP.
- (10) a. Merge proceeds bottom up.
 b. Lexicalisation applies after every Merge step.

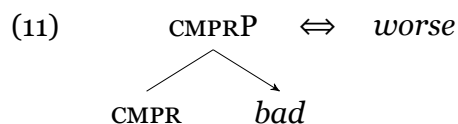
The bottom-up nature of lexicalisation, and the fact that it targets non-terminals is what makes it possible to propose a single- $\sqrt{}$ syntax that can accommodate root suppletion. In order to see this, consider the fact that lexicalisation (as it proceeds to higher and higher nodes) must keep track of what it has done at lower nodes, so that it can ship this information to PF at some relevant point.⁹ In this type of architecture, the problem is solved if we require that the phrasal lexical item (7b) can apply at CMPRP *only if* the

⁸ Our discussion of this issue is indebted to Michal Starke (p.c.).

⁹ This could happen either at the very end of the derivation (as we currently assume), or such shipments can be sensitive to phases, see, e.g. Embick (2010), Merchant (2015) and Moskal and Smith (2016).

lower *aP* node has been lexicalised by *bad*. Equivalently, *worse* is inapplicable if (by free choice of root) we have lexicalised *aP* by a different lexical entry than *bad*.

In order to encode this proposal, let us rewrite the lexical entry for *worse* as in (11), where instead of the *aP* node, we write *bad*. Following Starke (2014), we refer to this device as a pointer. It is an object that is present inside one lexical entry and refers to another lexical entry.



The entry (11) reads as follows: lexicalise CMPRP as *worse*, if one daughter is the CMPR feature, and the other daughter (i.e., *aP*) has been lexicalised as *bad* at the previous cycle.¹⁰

The idea behind (11) presupposes that the process of lexicalisation has at least two parts: matching (lexical search) and pronunciation (shipment to PF). The crucial point is that the lexicalisation procedure may make multiple searches of the lexicon (leading to the selection of various matching items) before the ultimate pronunciation. Crucially, when a matching lexical entry is found for a given node (say *aP*), this does not mean that this lexical entry is immediately shipped to PF for actual realisation. The match is remembered, and it will eventually be sent to PF; but if later on, a lexical item matching a higher node (say CMPRP) is found, then the first (lower) candidate is not sent to PF at all: only the higher lexicalisation survives. In Nanosyntax, the replacement of a lower match (*bad*) by a higher match (*worse*) is called ‘overriding.’

(12) *Cyclic override* (Starke, 2009)

Each successful spellout overrides previous successful spellouts.

As said, overriding means that a matching item at a node XP (*worse*) prevents that any item matching a node contained inside XP (*bad*) is shipped to PF. Overriding is a general property of cyclic bottom-up lexicalisation.

Recall now that from the perspective of a single- $\sqrt{\quad}$ theory, the problem with suppletive lexical items like (7b) was that they could override just any root. The pointer device introduced in (11) is here to restrict unlimited overriding: *worse* can only override *bad*. Caha, De Clercq, and Vanden Wyngaerd (2019) encode this by the so-called Faithfulness Restriction:

¹⁰ Pointers are used within Nanosyntax also beyond root suppletion. The reason that led Starke to introduce pointers (in unpublished work) are idioms like *shoot the breeze*. In this case, intuitively, we have a lexical entry that introduces a particular concept (CHAT) when the VP contains the relevant lexical items *shoot*, *the* and *breeze*. In other words, the idiom is to be stored with pointers to these lexical items as: CHAT $\Leftrightarrow$ [$\rightarrow$ shoot [$\rightarrow$ the $\rightarrow$ breeze]]. Pointers have also been used to model syncretism in multi-dimensional paradigms (Caha & Pantcheva, 2012; Vanden Wyngaerd, 2018; Blix, 2021), and to deal with issues in nonproductive morphology (De Clercq and Vanden Wyngaerd, 2019).

(13) *Faithfulness Restriction (FR, preliminary)*

A lexicalisation α may override an earlier lexicalisation β iff α contains a pointer to β

To conclude, let us stress the crucial point, which is that we now have a way to account for root suppletion with just a single $\sqrt{}$ (or a single A, or, potentially, just functional heads all the way down). To achieve this, we have introduced a bottom-up phrasal lexicalisation procedure. In this kind of system, insertion at the $\sqrt{}$ node is free. But once the choice has been made, the Faithfulness Restriction limits the overriding of the initial choice only to lexical items whose lexical structure consists of a pointer to this initial choice. This way, we can restrict *worse* to be the comparative of *bad* using a pointer, rather than an arbitrary index on $\sqrt{}$ in the syntax.

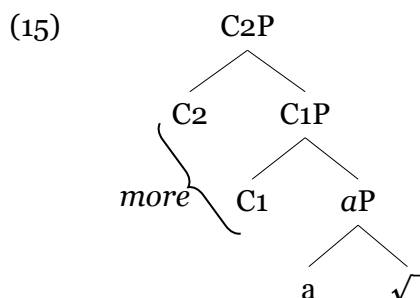
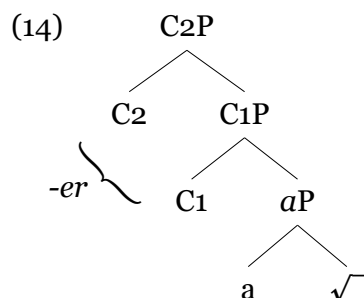
3 PHRASAL LEXICALISATION AND MULTIPLE EXPONENCE

From the perspective of a modular and universal syntax, the zero theory of $\sqrt{}$ s is that there is only a single $\sqrt{}$ (or perhaps no $\sqrt{}$ at all, if the bottom of the functional sequence is simply a feature like all the others). In the previous section, we have argued that the problems posed by suppletion can be reconciled with such a modular syntax if cyclic phrasal lexicalisation is adopted. However, an objection that is sometimes raised against the principle of phrasal lexicalisation is that of multiple exponence or double marking, i.e. the phenomenon where suppletion in the root is accompanied by regular marking.¹¹ This is, for example, the case in a form like *bett-er*, which multiply expounds the comparative: once in the root, and once in the suffix. Here our theory faces a conundrum: how can *bett-* spell out CMRP (as required by the phrasal lexicalisation theory), while at the same time leaving CMRP available for the insertion of *-er*? No such issue arises in a theory with terminal lexicalisation: the suppletive root is an allomorph which is inserted in the context of the CMRP head, and the CMRP head may itself be realised by a separate suffix.

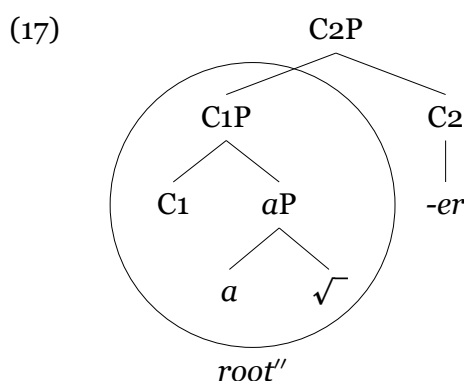
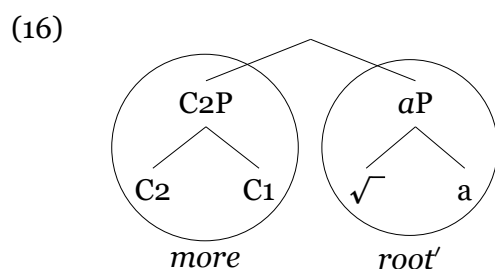
In this section, we suggest a solution to this problem. The solution is based on the observation that in cases where suppletion co-occurs with overt marking, the overt marking tends to be ‘reduced’, often a substring of a different, nonreduced marker. To see this on an example, let us turn back to English. Here we have *-er* and *more* for the

¹¹ One solution to this issue, suggested for instance in Haugen and Siddiqi (2016), would be to say that decomposing suppletive forms (like *bett-er*) into two pieces is actually doubtful. Such an approach could be supported by the fact that in degree achievements like *to better something*, the ‘comparative’ *-er* (if it is one) is retained (unlike in, say, *to cool something*), which may suggest that *-er* could have actually been re-analysed as a part of the root. If that is so, the form *better* would be a nondecomposable comparative form in trivial conformity with the nonterminal-spell-out hypothesis. We do not want to dismiss this approach in its entirety: there are clearly cases where suppletive forms are nondecomposable, and we think that these are suggestive of a solution in terms of nonterminal lexicalisation. However, we do not think that such a solution is universally applicable for reasons that become clear as we proceed.

comparative, and *-est* and *most* for the superlative. Clearly, *-er* and *-est* are morphologically reduced compared to *more/most*, if only because they are affixes while *more* and *most* are free-standing items. Further, there are morphological and semantic reasons to think that *mo-re/mo-st* actually contain *-er/-est* as a proper part. Such a containment relation between the two comparative markers can be captured if we decompose the single CMPR node into two heads, C1 and C2, as shown in (14) (cf. Caha, De Clercq & Vanden Wyngaerd, 2019). Reduced comparative marking can now be analysed as expressing only C2, as in (14), while full marking spells out both C1 and C2, as in (15).



We leave it open as to how exactly lexicalisation applies in the case of *more*, as the main focus is on its complement, i.e. that part of the structure that is lexicalised by the root (but see De Clercq & Vanden Wyngaerd, 2018 for discussion of this issue). We only note that phrasal lexicalisation requires C1 and C2 to form a constituent: this could be achieved by head-movement (Matushansky, 2013), Local Dislocation (Embick, 2007) or by Complex-Spec formation (Caha, De Clercq & Vanden Wyngaerd, 2019), as shown in (16). What is crucial is that this type of marking occurs on top of roots which spell out only the *aP* constituent, as shown by the constituent on the right hand side in (16).

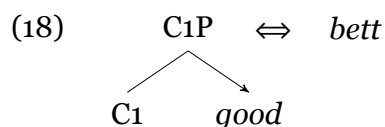


In (17), we show that the reduced marker appears on top of roots which spell out C1P, leaving it again aside how the surface order is derived, as this would take us too far afield (see Caha, De Clercq & Vanden Wyngaerd, 2019 for a worked-out proposal). The crucial point here is the size of the constituent spelled out by the root and by the functional marker. In particular, given that the number of features is constant in (16) and

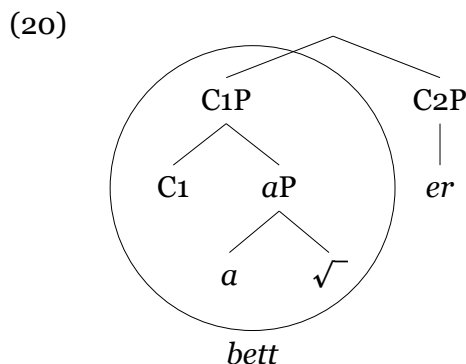
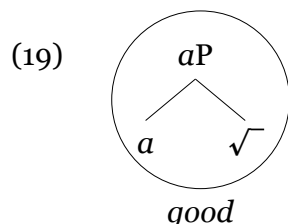
(17), we observe a trade-off between the size of the root and the size of the comparative marker. In particular, we can distinguish between large roots, which spell out C1, and combine with reduced markers. Smaller (*aP*-sized) roots must combine with *more*. The difference between the two classes of roots can be easily encoded in the lexicon: some roots will be specified for C1, others will not.

With the background in place, let us now show how a form like *bett-er* can be derived. As a starting point, consider the observation that suppletive adjectives like *bett-er* only occur with the reduced markers (i.e., *-er/-est*) and never with the full markers (i.e., there is no case like **more/most bett*), as observed by Bobaljik (2012). In a theory with a single $\sqrt{}$ using pointers like the one we have sketched above, this observation follows. In particular, the tree in (16) (with full marking) is correctly predicted to be incompatible with suppletion. That is because the root in (16) pronounces a constituent (*aP*) that exactly corresponds to the positive. Under the theory with single $\sqrt{}$ and pointers, suppletive roots must stand in a containment relation, one overriding the other. Therefore, the comparative root must spell out at least one extra feature compared to the positive, but such a feature is not available in (16), making it incompatible with suppletion.

Turning now to (17), this scenario allows for root suppletion on our account, although it does not require it. We first show how root suppletion works, and then we turn to nonsuppletive roots that combine with the reduced marker. Suppletive roots like *bett* will have an entry like (18), with a pointer to a different root.



In this case, *good* first spells out the *aP*, as shown in (19), which is a stage of the derivation that corresponds to the positive. If C1 is added, *bett-* is inserted at C1P. This C1P is subsequently merged with C2, yielding the full comparative structure in (20). For concreteness, we place C1P to the left of C2P, reflecting a leftward movement operation of C1P, which we do not discuss in detail here.



We now turn to nonsuppletive roots that combine with *-er*. In order to show how

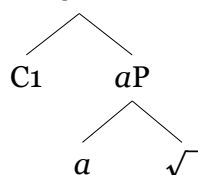
they are accounted for, we shall diverge from our reliance on a broad spectrum of conceivable approaches to phrasal lexicalisation, and focus on one particular version, due to Starke (2009, 2018). The specific component of this theory which we now need, is a matching procedure based on the Superset Principle.

(21) *The Superset Principle* (Starke, 2009)

A lexically stored tree L matches a syntactic node S iff L contains the syntactic tree dominated by S as a subtree

The principle says that if there is an entry like (22), then it can spell out a C1P, as well as aP (because aP is contained in it).

(22) C1P $\Leftrightarrow$ *old, nice, smart, great, ...*



If a root has such an entry, it can be used both in the positive (i.e., as an aP), and, at the same time, appear with reduced marking in the comparative. In English, the adjectives *old* or *nice* would be examples of such roots. The possibility of entries like (22) is what leads us to say that if a root spells out C1P (and thus occurs with reduced marking in the comparative), it does not have to be necessarily suppletive.

To sum up, the theory sketched up to now has two parameters of variation. The first parameter is related to the absolute size of the (morphological) root: it either spells out aP or C1P. At the level of data, this parameter distinguishes between roots that combine with *more* and those that take *-er*. The second parameter distinguishes two classes of roots of the size C1P, i.e., two classes of roots that combine with *-er*. The difference is whether the entry for the root has a pointer in it or not: suppletive roots like *bett-* do (overriding *good*), nonsuppletive roots like *old* do not.

Before we develop this concept further, we need to refine the Faithfulness Restriction slightly. Notice first that the entry for adjectives like *old* in (22) is very similar to the entry we have originally considered for *worse*, recall (7b). The problem with (7b) was that it could spell out the comparative form of just about any root, which is why we introduced the Faithfulness Restriction in (13). The FR states that overriding at C1P only happens if the overridee has a pointer to the overridee. As a result, the entries of suppletive adjectives will always contain a pointer to another entry. The entry for the adjective *old* in (22) does not contain a pointer, so it is not allowed to override other roots.

However, such roots do raise an issue related to overriding and faithfulness. In a bottom-up cyclic system, the $\sqrt{}$ is always spelled out first. Here all lexical items that

contain the $\sqrt{}$ node are candidates thanks to the Superset Principle, and we let free choice decide. Suppose we choose an entry like *old*. The next step is to merge little *a* with the $\sqrt{}$, forming *aP*, and we again try to spell it out. What we need to achieve is that *old* is inserted at *aP*, forming the positive-degree form *old*.

Strictly speaking at this point, the lexicalisation of *aP* as *old* must override the lexicalisation of the $\sqrt{}$ node (also *old*), which (due to the Faithfulness Restriction) requires a pointer that *old* lacks. At the same time, we are not literally overwriting one entry by another, since we want to insert at *aP* the very same entry that we inserted at the $\sqrt{}$ node. This must be legal, otherwise an entry such as (22) would never get to use its lexicalisation potential. In order to allow this, we augment the FR in the following way:

(23) *Faithfulness Restriction (FR)*

A lexicalisation α may override an earlier lexicalisation β iff

- a. α contains a pointer to β
- b. $\alpha = \beta$

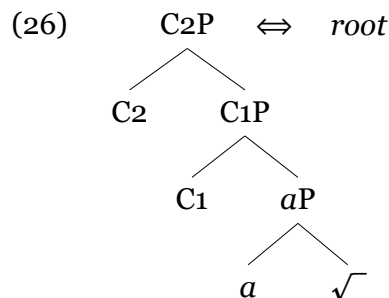
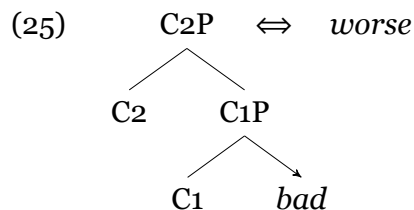
The clause (23b) now allows the entry (22) to keep overriding itself all the way to C1P. When C2 is merged, however, C2P cannot be spelled out by (22), and C2 is lexicalised by *-er*.

Finally, in order to capture the full spectrum of adjectival roots in English, we must introduce roots of two more sizes. To see that, consider again *aP* sized roots:

$$(24) \quad \begin{array}{c} aP \\ \swarrow \quad \searrow \\ a \quad \quad \sqrt{} \end{array} \Leftrightarrow \textit{good, intelligent, ...}$$

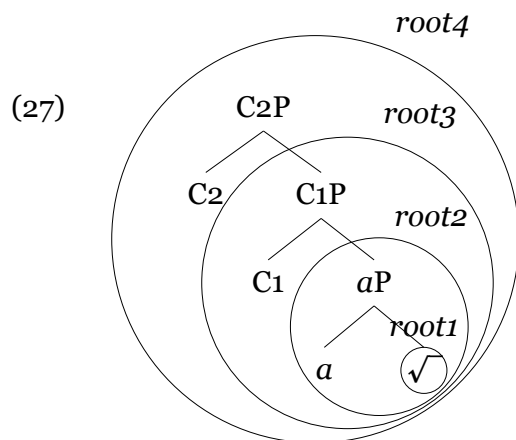
The reason for claiming that these roots spell out the entire *aP* (as opposed to spelling out just the $\sqrt{}$) is the existence of morphologically complex positive-degree adjectives, like *slim-y*, *happ-y*, *cheek-y*, etc., where arguably, *-y* spells out little *a*. Since the *aP*-sized roots are not further decomposable, but distribute like positive degree adjectives, we treated them as spelling out the *aP*. But for the morphologically complex adjectives, where *-y* spells out the little *a*, we must specify the root only for $\sqrt{}$.

Another possible type of root is a root that spells out the whole C2P. This root spells out both C1 and C2, and hence it appears with no comparative marking whatsoever. Such roots come again (in principle) in two flavours. One type of such roots has a pointer to a different root, as in (25), and then the root works as a suppletive counterpart of a positive root. A case in point is the entry for *worse*, which contains a pointer to *bad*.



The other type is as in (26), without a pointer. English has no such adjectives, but we find cases like this in certain varieties of Czech, to be discussed in section 4.1 below.

In sum, the approach sketched in this section distinguished the $\sqrt{}$ (a syntactic node) from the morphological root, which spells out the $\sqrt{}$ node (or, equivalently, whatever is at the bottom of the functional sequence) and potentially other nodes. This allows for a variety of roots in the morphological sense, while still maintaining a single $\sqrt{}$ in syntax. The variety of roots that our theory makes available can be visualised as a set of concentric circles, encompassing various sizes of structure, as shown in (27):



The various types of roots correspond with different types of morphological marking. A size 1 root (*root1* in (27)) appears with an overt little *a* in the positive, and the comparative marking comes on top of the little *a* (*happ-i-er*). A size 2 root (*root2*) has no overt marker corresponding to little *a*, and full comparative marking. A size 3 has no overt little *a*, and reduced comparative marking, while a size 4 root has no overt little *a* and no comparative marking

From the perspective of suppletion, we note that roots that reach up to the comparative zone (namely size 3 and 4) may work as suppletive comparatives of positive roots (those of size 2). The crucial theoretical possibility allowed by the split CMPR system is the existence of suppletive roots of size 3, corresponding to *bett-*, since these show the property of multiple exponence. Size 3 roots can both work as suppletive counterparts to positive-degree roots of size 2, and, at the same time, combine with an overt com-

parative marker, namely *-er*. This extends the reach of our theory to examples where suppletive roots combine with overt markers, i.e. cases of multiple exponence. Note, however, that the impression of multiple exponence is only apparent in our proposal, since the root and the ending expone different features, C1 and C2. It also follows from this analysis that in cases of multiple exponence, we will observe a certain type of ‘reduction’ of the relevant marker. In the following section, we present two case studies which further illustrate and refine the reduction effect under suppletion.

4 EMPIRICAL SUPPORT

4.1 CZECH

The first case study concerns the interaction between comparative marking and suppletion in Czech. We start from the fact that the traditional descriptions recognise three different allomorphs of the comparative (see Dokulil et al., 1986; Karlík, Nekula & Rusínova, 1995; Osolsobě, 2016). We give them on the first three rows of (28). Each row starts by the relevant allomorph, followed by the positive, comparative and the superlative. The final morpheme in each form is the agreement marker, which we ignore in extracting the comparative allomorph. Following this approach, the allomorph in (28c,d) is zero (no overt marker). We address the *k/č* alternation in (28c) shortly.

(28)	allomorph	POS	CMPR	SPRL	GLOSS
a.	ějš	chab-ý	chab- ějš -í	nej-chab- ějš -í	‘weak/poor’
b.	š	slab-ý	slab- š -í	nej-slab- š -í	‘weak’
c.		hez-k-ý	hez-č- -í	nej-hez-č- -í	‘pretty’
d.		ostr-ý	ostř- -í	nej-ostř- -í	‘sharp’

On the first two lines, we illustrate the *ějš-í* and *-š-í* allomorphs with two adjectives that are semantically and phonologically similar. We do so to show that the allomorphy is not driven by phonology or semantics. Rather, the distribution is governed by arbitrary root class: *-ějš* is the productive allomorph, while *-š-í* is restricted (occurring with 72 out of 5440 adjectives sampled in Křivan, 2012).

On the third and fourth line, we illustrate the zero allomorph, and two facts should be noted. First of all, the positive and the comparative are not homophonous: their morphological identity is obscured by phonological interactions with the agreement markers. Specifically, the agreement marker *-í*, found in the comparative, triggers the palatalisation of the base (*k* goes to *č*), while the elsewhere agreement marker *-ý* does not palatalise the base (see Caha, De Clercq & Vanden Wyngaerd, 2019 for a discussion of the palatalisations). As a result, the forms are distinct. The second fact to be noted is that in the standard language, zero marking only occurs after a particular adjectival marker, namely *-k*. This morpheme is similar to the English *-y* in that it sometimes

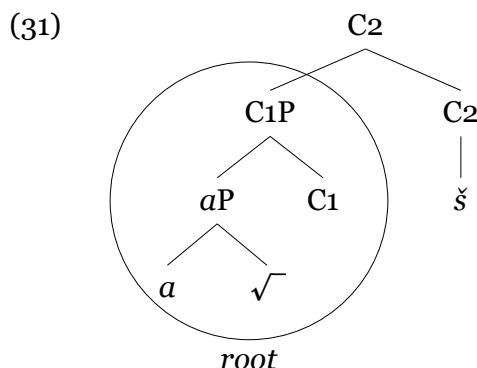
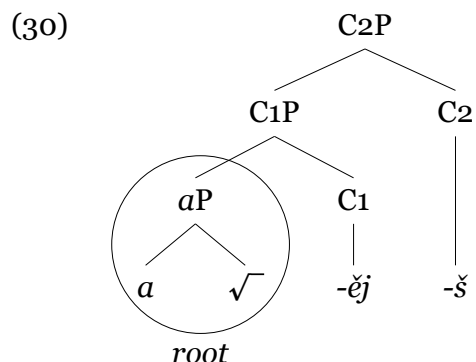
occurs after nominal roots (e.g., *sliz-k-ý* = ‘slim-y’) and sometimes after cranberry type of morphemes (e.g., *hez-k-ý* = ‘prett-y’). Because of its limited distribution, it is not clear whether the $\emptyset$ allomorph needs to be recognised as a separate marker, or perhaps dismissed as a special realisation of *-š* after *-k*. We do, however, recognise the zero as a relevant allomorph to consider, because in the dialects of North Eastern Bohemia (Bachmannová, 2007), one finds it also after nonderived adjectives, as shown on the last row (28d). We note, however, that much of our reasoning is valid even if it turns out that the zero allomorph is an effect of phonology, rather than morphology.

Taking the traditional descriptions at face value, an interesting generalisation is that going from the first to the third line, in (28), we see an increasingly ‘reduced’ realisation of the full marker *-ějš-*. (28b) suggests that *-š* is a substring of *-ějš*. This makes it tempting to decompose *-ějš* into two morphemes, *-ěj* and *-š*, as suggested by Caha, De Clercq, and Vanden Wyngaerd (2019).

Independent evidence for decomposing *-ějš* comes from comparative adverbs, seen in the second column of (29). Here the *-š*-part of the comparative adjective is systematically missing, while *-ěj* is preserved. This confirms that *-ěj* and *-š* are independent morphemes.

(29)	CMPR ADJ	CMPR ADV	
	chab-ěj-š-í	chab-ěj-i	‘weak’
	rychl-ej-š-í	rychl-ej-i	‘fast’
	červen-ěj-š-í	červen-ěj-i	‘red’

Given our model with two comparative heads, the facts are easily captured if *-ěj* and *-š* spell out C1 and C2 respectively. With *aP*-sized roots, both markers surface, see (30). With roots of the size C1P, only *-š* appears, as in (31).



Zero marking arises when the root spells out all of the projections, as in (26) above. Recall that (26) was presented as a logical option allowed by our system, and though it was not attested in English, we need it to account for *ostr-ý* ‘sharp’ in (28). This concludes our discussion of ‘regular’ comparatives, i.e., those based on the same base as found in the positive, and we now turn to suppletive comparatives.

Given our theory of suppletion where suppletive roots override the base, comparative suppletion requires a root that spells out a different node than the positive. Since the positive spells out *aP*, a suppletive comparative root must be at least of the size *C1P*. This idea interacts with our account of comparative allomorphy. Specifically, since roots of the size *C1P* cannot combine with *-ěj-š* (recall (31)), we now predict that suppletive roots should be incompatible with *-ěj-š*. To verify this, the table (32) presents an exhaustive list of suppletive adjectives based on Dokulil et al. (1986, p. 379) and Osol-sobě (2016). The table shows that the prediction is borne out: all suppletive adjectives require the ‘reduced’ *-š* allomorph.¹²

(32)	POS	CMPR	GLOSS	POS	CMPR	GLOSS
	dobr-ý	lep-š-í	‘good’	špatn-ý	hor-š-í	‘bad’
	velk-ý	vět-š-í	‘big’	mal-ý	men-š-í	‘little, small’
	dlouh-ý	del-š-í	‘long’			

We submit these facts here as an important confirmation of the current model, which predicts that when there are two or more ways of marking the comparative, suppletion is incompatible with the full marker. With reduced markers, we find both suppletive and regular cases, depending on whether the entry of the size *C1P* has a pointer or not.

It is thanks to phrasal lexicalisation, the mechanism of pointers, and the post-syntactic lexicon that the single $\sqrt{\quad}$ approach can be maintained against the surface diversity of morphological roots. Roots can be stored in the lexicon without functional structure, with (more or less) functional structure, and with or without a pointer, resulting in the different types of roots that we observe. Crucially, suppletive forms can be linked to their base form without the need to code the identity of the base on the syntactic $\sqrt{\quad}$ node.

4.2 LATIN

Latin provides further evidence for the correlation between reduced marking and suppletion predicted by our theory, but in contrast to Czech, it shows the effect in the superlative. The regular marking of comparative and superlative is shown in (33a).

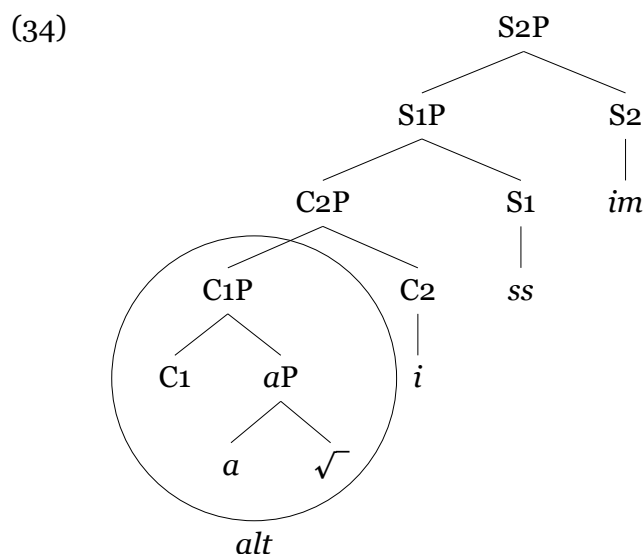
(33)	POS	CMPR	SPRL	GLOSS	marking in SPRL
a.	alt-us	alt-i-or	alt- i-ss-im -us	‘tall’	full marking
b.	mal-us	pe- or	pe- ss-im -us	‘bad’	SPRL lacks <i>-i</i>
c.	bon-us	mel-i-or	opt- im -us	‘good’	SPRL lacks <i>-i-ss</i>
d.	magn-us	ma-i-or	max- im -us	‘big’	SPRL lacks <i>-i-ss</i>
e.	parv-us	min- or	min- im -us	‘small’	SPRL lacks <i>-i-ss</i>
f.	mult-us	plūs	plūr- im -us	‘much’	SPRL lacks <i>-i-ss</i>

¹² See Caha, De Clercq, and Vanden Wyngaerd (2019) for the discussion of potential counterexamples.

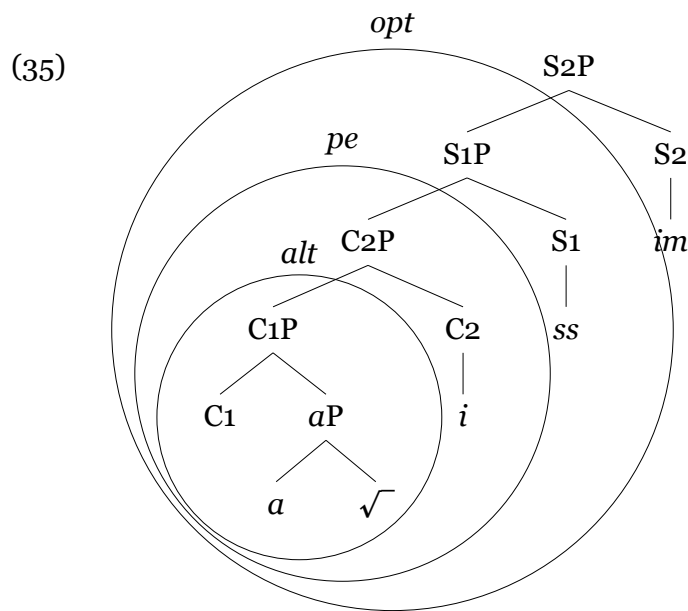
We segment the regular superlative into five morphemes (following De Clercq & Vanden Wyngaerd, 2017). The first morpheme is the root (*alt*), and the last one (*-us*) an agreement marker. The reason for treating the three middle markers *-i*, *-ss* and *-im* as separate morphemes is that they can be missing in the irregular forms shown in (33b-f). These represent an exhaustive list of the suppletive cases given by Gildersleeve and Lodge (1903, p. 46).

We analyse *-i* (the first of the post-root superlative morphemes) as a comparative marker, i.e., as a morpheme identical to the *-i* of the comparative *alt-i-or*. We treat *-i* in the same way as the English *-er*, namely as the lexicalisation of C2. Consequently, we analyse *-or*, which follows *-i* in the comparative, as an agreement marker. We do so because the masculine form *alt-i-or* ‘taller, M.SG’ alternates with the neuter *alt-i-us*. As a C2 marker, *-i* is compatible with suppletion. In (33c), for instance, the positive degree root *bon-* realises aP, the suppletive comparative root *mel-* realises C1P, and *-i-* is the marker of C2.

The remaining two morphemes mark the superlative, which we split into S1 and S2, analogously to CMPR. The structure of *alt-i-ss-im(-us)* thus looks as follows:



Against this background, consider the fact that the superlative marking with suppletive roots is *always* reduced, see (33b-f). There is not a single suppletive root in Latin which keeps all the three pieces in place, as indicated in the final column of (33). Specifically, we see two classes of suppletive roots. The majority of suppletive roots lacks the C2 *-i* as well as the S1 *-ss*, and we would thus analyse them as spelling out S1P. However, *pe-* lacks only the *-i*, which, on the assumption that *-i* is C2, leads to the proposal in (35).



This picture has implications for the analysis of the comparative. Specifically, all suppletive roots which spell out a projection larger than C1P should make *-i* disappear not only in the superlative, but also in the comparative. This is true for the adjectives *minor* ‘smaller’ and *plus* ‘more’, as well as, arguably, *pe-or* ‘worse,’ where the glide in the comparative *pe[j]or* results, on our analysis, from phonological factors (hiatus filling). Note that *plus* lacks the agreement marker *-or*, and Gildersleeve and Lodge (1903, p. 46) analyse it as a neuter form, with the masculine cell left blank. Here we treat *plus* as spelling out minimally S1P, lacking *-i* in the comparative, and in the superlative also *-ss*. We leave the reasons for the lack of agreement in the comparative open to interpretation.¹³

The (c) and (d) cases of (33) warrant some further comment, since they have *-i-* in the comparative but lack it in the superlative. This is because they instantiate an ABC-pattern, with two different suppletive roots: one of size C1P (explaining the presence of *-i-* in the comparative), and another of size S1P (explaining the absence of both *-i-* and *-ss-* in the superlative). These suppletive roots successively point to one another, e.g. the lexical entry for *opt-* contains a pointer to *mel-*, which itself contains a pointer to *bon-*.¹⁴

¹³ Note that *plus* and *plur* are two shapes of a single root, with *s* undergoing rhotacism in intervocalic positions, which happens also in the comparative when inflected, cf. *plur-is* ‘more, GEN.SG.’

¹⁴ We take *s* in the superlative *maks-im-us* ‘biggest’ to be a part of the root, given that it is not a geminate like the superlative S1 marker. The comparative *ma-i-or* ‘bigger’ could arise from the root *mag-*, as suggested in Bobaljik (2012), with the root final *g* first assimilating to *j* (yielding *maj-j-or*), which is then reduced due to degemination. Bobaljik (2012) concludes from this that this adjective has a regular AAA pattern, and hence, that it is irrelevant for suppletion. However, this move requires the parsing of the positive as *mag-nus*, which we see little evidence for. We therefore treat this as an ABC pattern (*magn-ma(g)-maks*).

Related to this we note that the literature sometimes makes a distinction between weak or mild suppletion, where the related forms are phonologically similar (e.g. *sing-sang*), and strong suppletion,

In sum, this case study also illustrates how $\sqrt{\quad}$ and roots should be treated differently. Whilst there is only one $\sqrt{\quad}$ in syntax, there are many different types of roots in the lexicon, which store $\sqrt{\quad}$ with (or without) other pieces of structure. The trade-off between the size of roots and the superlative degree morphology in Latin shows that suppletion follows from the size of lexical entries of morphological roots and not from the nature of the syntactic $\sqrt{\quad}$.

5 NON-LOCAL ALLOMORPHY

In this section, we turn to a case of suppletion in Korean discussed by Chung (2009). Choi and Harley (2019) have argued that this represents a case where allomorphy may be conditioned nonlocally, possibly skipping intervening morphemes. The relevant situation of nonlocal allomorphy is abstractly depicted in (36).

- (36) a. $\sqrt{153(A)}$ -AFF1
 b. $\sqrt{\mathbf{153(B)}}$ -AFF1-AFF2

What the formulas in (36) depict is that insertion at the root $\sqrt{153}$ is suppletive and varies between A ($\sqrt{153(A)}$) and B ($\sqrt{153(B)}$). The suppletion is conditioned by the presence/absence of AFF2. The trigger is nonlocal because it is separated from the root by an intervening affix, as (36b) shows.

Choi and Harley (2019) operate in a framework where lexical insertion happens under terminal nodes only, and suppletion is a case of contextual allomorphy, as illustrated in (3) above. Therefore, they capture the facts by saying that the $\sqrt{153}$ is realised as B if somewhere higher up in the structure the triggering feature [F] is found. [F] is then independently realised by AFF2.

For our analysis, such cases represent a challenge. The reason is that in order not to discriminate among roots in syntax, our system relies on pointers and overriding, thereby eliminating the possibility of standard rules of contextual allomorphy, which make reference to a wider context. In our system, the ‘trigger’ for allomorphy needs to be strictly local, in the sense of ‘having to be a part of the set of features expounded by the suppletive root.’ A situation like the one depicted in (36), where the presence of a suppletive root is apparently caused by a higher head across an intervening affix therefore cannot arise in our system.

What we want to do in this section is illustrate the viability of an approach to apparent non-local cases that goes along the following lines: what appears to be nonlocal is

where such similarity is absent, as in the pair *bad-worse* (see e.g. Pomino & Remberger (2019), and references cited there). The former type is often treated as morphologically regular, and subject to post-morphology phonological readjustment in DM (see Bobaljik, 2017). We make no difference in the treatment of cases of weak or strong suppletion, in that we take both weakly and strongly suppletive allomorphs to be lexically stored in terms of an entry containing a pointer. We thus do not rely on phonological readjustment rules (see Harley & Siddiqi, 2013 for the same approach).

only so under certain assumptions about the structure, but if we enrich the structure, what looked like a case of nonlocal allomorphy starts looking like local allomorphy, in the sense of the previous paragraph. We will argue that nonlocal conditioning of allomorphy is both unnecessary and undesirable. It is unnecessary once we enrich the structure involved in negation and honorification in Korean. It is also undesirable because it predicts the wrong results once the interaction between negation, causation, and honorification is taken into account.

In addition, the Korean data that we shall discuss in this section also present a case of double exponence in the domain of honorific marking. We shall propose to deal with this case of double exponence by analogy with the one of the English comparative form *bett-er*, i.e. by means of, on the one hand, an enrichment of the functional hierarchy with an additional head, and the principle of phrasal lexicalisation on the other.

5.1 A KOREAN PARADOX

Korean shows a paradox with respect to the conditioning of allomorphy in the domain of negation and honorification, which was first discussed by Chung (2009), and then taken up again by Choi and Harley (2019). The paradox is summarised in the table in (37).

(37)

	regular pattern	$\sqrt{\text{EXIST}}$	$\sqrt{\text{KNOW}}$
a.	$\sqrt{X}$	iss	al
b.	NEG $\sqrt{X}$	eps	molu
c.	$\sqrt{X}$ HON	kyey-si	al-si
d.	NEG $\sqrt{X}$ HON	ani/mos kyey-si	molu-si

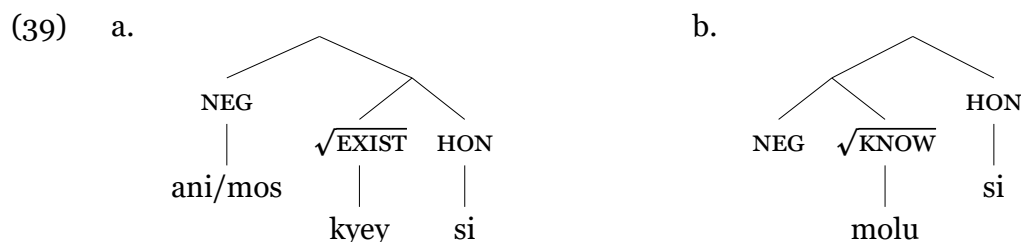
The first column of (37) abstractly shows the regular way of forming honorific and negative forms by affixes. The regular markers of negation and honorification occur each on a different side of the root, so that it is not *a priori* obvious what their hierarchical relation is with respect to the root. Their hierarchy could be, however, inferred indirectly by looking at suppletive forms, which are given in the last two columns of the table.

The general form of the paradox is this. Assuming (as we do) that suppletion is always local, the hierarchy of negation and honorification could be determined by looking specifically at the cell where both are present. This is the last row of the table (37). But the facts from suppletion point either way, paradoxically: one verb ($\sqrt{\text{EXIST}}$) suggests that the honorific head is closer to the root, because the root shows the ‘honorific’ allomorph where both triggers are available (as indicated by the shading). On the other hand, the verb $\sqrt{\text{KNOW}}$ suggests that negation is closer to the root, because it has a suppletive negative form even in the presence of an overt honorific suffix. Let us now look

at the patterns in more detail.

The first verb, *iss* ‘exist’, has a negative suppletive form *eps* ‘not exist’ (37b), which is a portmanteau expressing negation (Chung, 2009).¹⁵ It also has a suppletive honorific form *kyey-si* (37c). When negation and honorification co-occur, as in (37d), *kyey-si* is used, suggesting that honorification takes precedence over negation in determining the allomorph of the root (*ani* and *mos* ‘not’ are analytic negative markers). Assuming that this precedence is a function of greater structural closeness, this suggests the functional hierarchy in (38a), represented treewise in (39a). The second verb in (37) is *al* ‘know’, which also has a suppletive negative form, the portmanteau *molu* ‘not know’ (37b). There is no allomorph in the presence of honorification, and the regular honorific marker (*u*)*si* is expressed on the verb (37c). When honorification and negation co-occur, as in (37d), *molu* appears again, suggesting that negation takes precedence over honorification in determining root allomorphy. In terms of structural closeness to the root, this suggests the opposite conclusion from the one reached earlier, namely (38b)/(39b).

- (38) a. NEG > HON > $\sqrt{\text{EXIST}}$
 b. HON > NEG > $\sqrt{\text{KNOW}}$



The paradox exists in virtue of the following two assumptions:

- (i) there is only a single functional hierarchy (for Korean), i.e. (38a) and (38b) cannot both be correct
- (ii) allomorphy is conditioned strictly locally (see Bobaljik, 2012; Moskal (2013); Moskal & Merchant, 2015, and many others).

The first of these assumptions is uncontroversial. Opinions diverge, however, as to which of the two hierarchies shown in (39) is the correct one. Choi and Harley (2019)

¹⁵ *iss-* has three meanings (Martin, 1992, p. 319), one of which is ‘exist’, another is ‘stay intentionally’, and a third meaning is ‘have’. The negation of *iss-* ‘stay’ is *an(i) iss* (Chung, 2009, p. 539; Chung, 2007, p. 124–127), while the negation of *iss-* ‘exist’ and ‘have’ is *eps*. Chung (2007) argues convincingly that *iss-* ‘exist’ is adjectival, while *iss-* ‘stay’ is verbal, showing additional functional morphology on the root in the present tense. We hypothesize that the absence of negative suppletion with *iss-* ‘stay’ follows from intervening structure between *ani* and the root *iss-* ‘stay’, violating locality. However, we focus on *iss-* ‘exist’ in this paper, deferring a detailed account of how the different readings for *iss-* arise to future research.

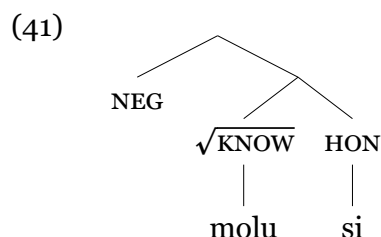
assume that the functional hierarchy illustrated in (40a) is the correct one, whereas Chung (2009) argues in favour of (40b):¹⁶



Regardless of which of these two options is taken, it seems clear that a strict version of (ii) cannot be maintained. In the next section, we discuss the proposal by Choi and Harley (2019) which abandons (ii), and some of the problems that it faces.

5.2 NON-LOCAL ALLOMORPHY AND CAUSATIVE INTERVENTION

We start out by taking a closer look at the specifics of the account by Choi and Harley (2019). Recall first that they adopt the structure in (40a), where the HON head is local to the root. This takes care of the allomorphy on *iss-* ‘be’ in a straightforward and local fashion, the HON head taking precedence over negation when both are present. In order to account for the pattern of suppletion found with *al* ‘to know’, Choi and Harley (2019) argue that if no suppletive form has been inserted locally, root allomorphy can be conditioned from a distance, as long as the conditioning head is within the complex X° head (Bobaljik, 2012). This condition is satisfied for $\sqrt{\text{KNOW}}$, where no suppletive VI exists that is conditioned by HON, so that NEG can condition allomorphy of the root across HON. This is shown in (41), where the suppletive portmanteau *molu* ‘not know’ is inserted under the root terminal, conditioned by a NEG head separated from it by an intervening HON head. This account cannot be replicated under constituent lexicalisation.



In what follows we shall discuss two cases (not discussed by Choi & Harley, 2019) that are structurally analogous to (41), but that show different behaviour, in that a

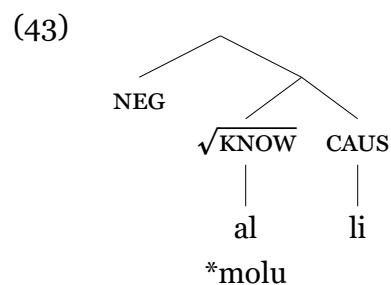
¹⁶ Choi and Harley’s (2019) hierarchy is not (only) the consequence of an assumption they make about the functional sequence. Instead, it results from their assumption that honorification is a form of agreement with an honorific subject, which is realised on *v* as a result of a rule of HON-sprouting that applies in the morphological component if a specific configuration is realised, i.e. if the verb is c-commanded by an NP with [+hon] (cf. Marantz, 1991, 1993). The result of this node sprouting rule makes HON lower in the structure than NEG. See Kim and Sells (2007) for arguments to the effect that Korean honorific marking is not to be considered as agreement.

higher head is not able to trigger suppletion of the root across an intervening head. We shall refer to these two cases as instances of ‘causative intervention’, as they involve the intervention of a CAUS head between the root and a higher functional head (HON or NEG). This intervening causative head appears to block the suppletive realisation of the root, unlike what happens in (41). The facts of causative intervention suggest to us that suppletion has to be strictly local after all, since the presence of a causative intervener appears to block it. We shall then proceed to develop an alternative account for the case of (41) in section 5.3.

The first case of causative intervention involves negation. Again taking the verb *al* ‘know’, we see the following pattern (Chung, 2007):

- (42)
- | | | | |
|----|-------------------------------|----------------------|--------------------|
| a. | $\sqrt{\text{KNOW}}$ | al | ‘know’ |
| b. | NEG $\sqrt{\text{KNOW}}$ | molu | ‘not know’ |
| c. | $\sqrt{\text{KNOW}}$ CAUS | al-li | ‘let know, inform’ |
| d. | NEG $\sqrt{\text{KNOW}}$ CAUS | ani/mos al-li | ‘not inform’ |

The case is similar to the one in (37), with negation occurring to the left of the verb and the other marker (honorific or causative) to its right, thus yielding a potential ambiguity as to the hierarchical relations. However, in this case the relative scope of the negation and causative marker can be deduced from the meaning. Specifically, in (42d), the meaning is ‘not inform’ (‘not let know’) rather than ‘cause to not know,’ suggesting a scopal hierarchy NEG > CAUS > $\sqrt{\text{KNOW}}$. Given that hierarchy, the structure of (42d) looks as in (43), which is exactly as in (41), modulo the substitution of CAUS for HON. Yet in this case, the NEG head is apparently unable to trigger the insertion of the suppletive negative portmanteau *molu* across the intervening CAUS head, since only the nonsuppletive root *al* ‘know’ is possible.



This is unexpected under Choi and Harley’s (2019) proposal.

The second case of causative intervention in Korean involves honorific suppletion, which is found with a small number of Korean verbs, given in (44) (Kim & Sells, 2007, p. 312).¹⁷

¹⁷ The regular *iss-usi-ta* means ‘have’, while the suppletive *kyey-si-ta* means ‘be/exist’ or ‘stay’ (Martin, 1992, p. 217). As already mentioned in footnote 15, this paper focuses on *iss-* ‘exist’ and the morphological patterns it triggers, and leaves the other readings and its associated morphological patterns out

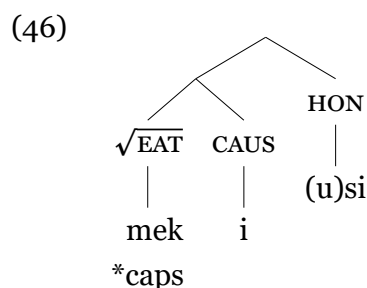
(44)	ROOT-DECL	ROOT-HON-DECL	ROOT-HON-DECL	
a.	mek-ta	*mek-usi-ta	caps-usi-ta	‘eat’
b.	ca-ta	*ca-si-ta	cwum-usi-ta	‘sleep’
c.	iss-ta	iss-usi-ta	kyey-si-ta	‘be, exist, have’

These show the following pattern in the interaction with causation:¹⁸

(45)

a.	$\sqrt{\text{EAT}}$	mek	‘eat’
b.	$\sqrt{\text{EAT}}$ CAUS	mek-i	‘let eat’
c.	$\sqrt{\text{EAT}}$ HON	caps-usi	‘eat’
d.	$\sqrt{\text{EAT}}$ CAUS HON	mek-i-si	‘let eat’

Since the causative and the honorific markers are on the same side of the root, it is easy to see their hierarchy, which we take to be the mirror image of the linear order, i.e. HON > CAUS > $\sqrt{\cdot}$.¹⁹ This translates into the following tree structure:



We see the same pattern as in (43) above: HON is not able to trigger the insertion of the suppletive honorific form *caps* ‘eat’ under the root node across an intervening CAUS head. This is a second case, then, that is unexpected under the approach of Choi and Harley (2019). In these two cases of causative intervention’, a CAUS head intervenes between the root and a higher functional head (HON or NEG), and blocks the suppletive realisation of the root. We take the data of causative intervention to cast serious doubt on Choi and Harley’s (2019) claim that triggers for suppletion can be nonlocal.²⁰

5.3 TOWARDS AN ALTERNATIVE: DECOMPOSING HON

In the light of the preceding discussion, we shall now proceed to develop an alternative analysis of the Korean paradox in terms of phrasal lexicalisation and the strictly local allomorphy requirement that this approach entails. The fundamental ingredient of our

of consideration.

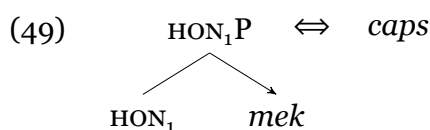
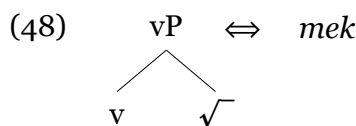
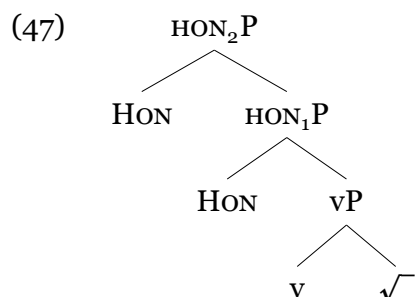
¹⁸ The verb *iss* 'be, exist' does not permit causation, so that the interaction of honorification with causation cannot be illustrated for this verb.

¹⁹ The alternation between *-usi-* and *-si-* is phonologically conditioned: *-u-* is an epenthetic vowel that appears between two heteromorphemic consonants (Chung, 2009, p. 543).

²⁰ A possible way out for Choi and Harley (2019) to account for this problem would be to argue that CAUS is a cyclic node, which blocks suppletion. It will be clear that such arbitrary marking of heads as interveners, while deriving the facts, seriously undermines the conceptual appeal of the proposal.

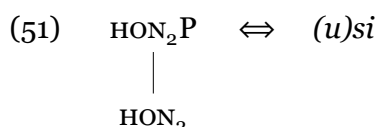
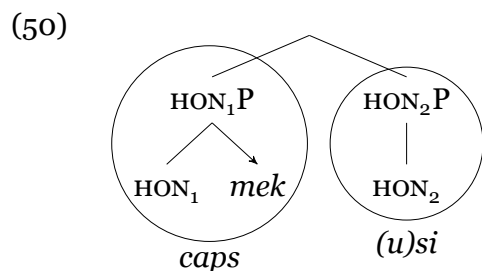
alternative is the idea that honorification (like comparative/superlative formation) involves two HON heads. In this section, we explain how assuming two heads yields the relevant honorific forms. In Section 5.4 we add the causative head, and show how it interacts with honorification. In Section 5.5, we pinpoint the NEG head in a particular syntactic position and show how it interacts with causative formation. Finally, with all the ingredients in place, Section 5.6 explains how the two HON heads allow us to resolve the Korean paradox noted in Choi and Harley (2019).

In order to see why assuming two honorific heads is needed, consider the example of the honorific suppletive form *caps-usi* ‘eat-HON’. On the one hand, the root has a special honorific shape (the non-honorific shape of the root is *mek* ‘eat’); on the other hand, the root is accompanied by an overt honorific marker *-usi* ‘HON.’ Therefore, just like in the case of *bett-er*, we hypothesise that two honorification-related heads are involved, as in (47). For the lack of a better term, we call them HON₁ and HON₂. The verbal structure these heads come on top of is abbreviated as [v √] in (47).



In this setting, the co-occurrence of a suppletive root and an overt honorific suffix is easily captured. Specifically, we associate the string *mek* ‘eat’ with a constituent of the size vP, as in (48), while *caps* ‘eat.HON’ is a realisation of HON₁P plus whatever features are contained in the verb *mek* ‘eat’. The lexical entry of *caps* ‘eat.HON’ therefore looks as in (49); it basically says that *caps* is the honorific form of *mek* ‘eat.’

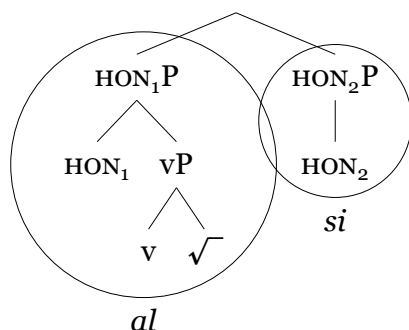
The structure of the full honorific form *caps-usi* is then as given in (50). We can see that the honorific root *caps* ‘eat.HON’ applies at HON₁P, overriding the nonsuppletive *mek* ‘eat’ in the process. The structure further presupposes that the constituent spelled out by *caps* ‘eat.HON’ moves out of HON₂P, and the honorific marker is inserted as the spellout of the remnant HON₂P. The lexical entry for the honorific *-(u)si* is as in (51).



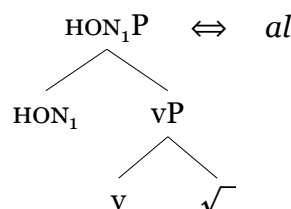
The reader will notice that this account follows the same logic as developed for *bett-er* in Section 3 (*bett-* spells out C1P, *-er* spells out C2.)

Moving further along the same path, non-suppletive honorifics will be captured in the same way as English non-suppletive forms like *smart-er*. Recall that a root like *smart* spells out the whole C1P (like *bett*), but it lacks a pointer. Quite parallel to this treatment, we are assuming that Korean non-suppletive roots spell out the whole HON₁P, as shown in (52). This structure shows the honorific form of the (non-suppletive) verb *al* ‘to know.’

(52)



(53)



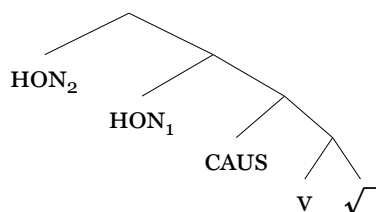
This leads us to posit for *al* ‘know’ a lexical entry like the one in (53). Because of the Superset Principle, this makes the verb also usable in non-honorific environments, spelling out just the vP. The ambiguity of non-suppletive roots (spelling out either vP or HON₁P) will be important later on.

5.4 ADDING CAUSATIVES

This section introduces the causative head CAUS in the structure. It starts by providing a structure for regular verbs, and then turns to suppletive verbs.

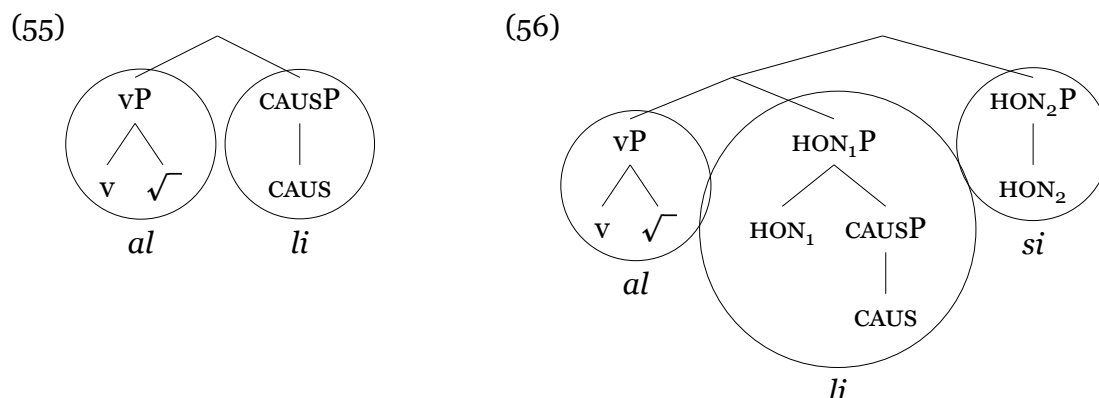
To begin with, we shall place the CAUS head in the tree relative to HON₁ and HON₂. The empirical facts to be discussed suggest that CAUS is below HON₁. The relevant hierarchy is shown in (54).

(54)



How does the presence of CAUS influence the derivation? The main consequence is that regular verb roots with an entry like (53) (spelling out $\sqrt{}$, v and HON₁) will only be able to spell out $\sqrt{}$ and v, since CAUS makes it impossible for such a root to also spell out HON₁. As is obvious from (54), CAUS intervenes. The root will therefore spell out just

the vP and move to the left; after the movement, the causative suffix *-li* spells out CAUS, as in (55). We are using the root *al* ‘to know’, whose entry is in (53).



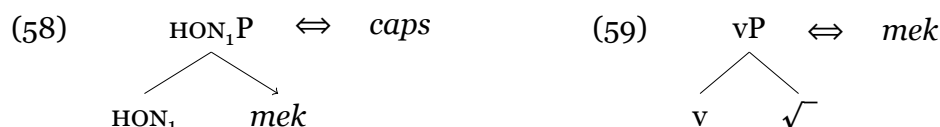
The tree in (55) represents the structure of the non-honorific causative. The honorific causative is derived by adding HON₁ and HON₂ on top of the non-honorific causative. On the surface, this leads to the addition of (u)*si* to the causative, yielding *al-li-si* ‘to let know, honorific.’

Let us next address the question of what happens to HON₁ in the honorific causative. We know that it is not spelled out by (u)*si*, which can realise only HON₂. Since the root cannot spell out HON₁ either (since CAUS intervenes), we conclude that HON₁ is spelled out along with the causative *li*. The full structure we are assuming is therefore as in (56). Note that the causative *li* spells out different structures in (55) and (56) and alternates (in a manner reminiscent of regular roots) between a non-honorific use in (55) and an honorific use in (56). In particular, its lexical entry is specified as containing both HON₁ and CAUS, which allows it to spell out either both of these features, or just CAUS.

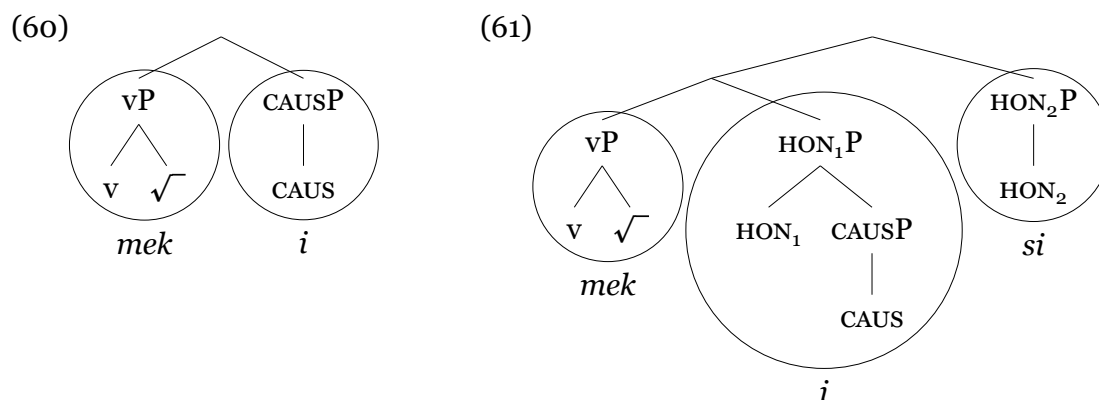
The interest of this derivation is that it allows us to explain why causativisation blocks honorific suppletion. The relevant facts are as repeated in (57) (originally (45)), and the relevant observation is that the honorific causative in (57d) uses the non-honorific root.

- (57)
- | | | | |
|----|------------------------------|-----------------|-----------|
| a. | $\sqrt{\text{EAT}}$ | mek | ‘eat’ |
| b. | $\sqrt{\text{EAT}}$ CAUS | mek-i | ‘let eat’ |
| c. | $\sqrt{\text{EAT}}$ HON | caps-usi | ‘eat’ |
| d. | $\sqrt{\text{EAT}}$ CAUS HON | mek-i-si | ‘let eat’ |

This pattern can be captured by using the entries for ‘eat’ as proposed in (48) and (49) above. We repeat them in (58) and (59).



With these lexical entries, both types of causatives (i.e., both the honorific and the non-honorific causative) are correctly expected to be based on the nonsuppletive root *mek* ‘eat.’ To show that, we give in (60) and (61) the structures for the non-honorific and the honorific causative respectively. These structures are the same as those in (55) and (56), only the root is different (‘eat’).

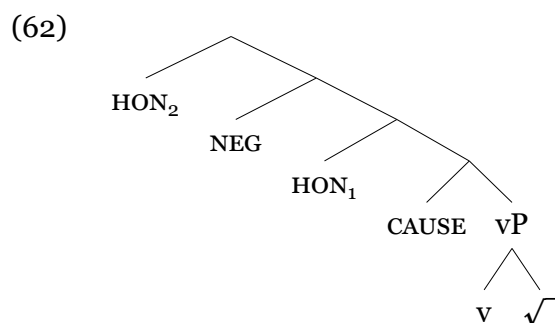


The most relevant thing to look at is the honorific causative in (61). In this structure, the HON₁ head is separated from the vP by the CAUS head. Therefore, just like with regular verbs, HON₁ cannot be spelled out by *caps* ‘eat.HON.’ This explains why the root only spells out vP and surfaces as *mek* in the honorific causative.

5.5 ADDING NEGATION

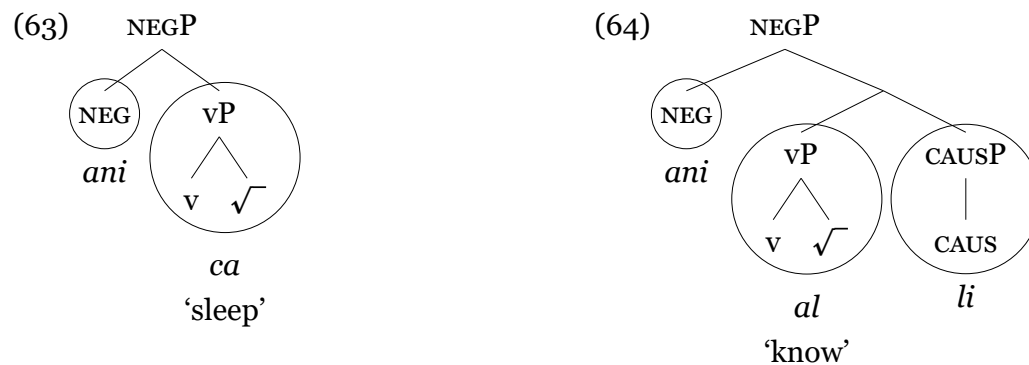
This section puts forth a proposal for negative suppletion. We also propose a specific position for the NEG in the tree, paving the way for the solution to the paradoxical interaction between negation and honorification.

Let us begin by proposing a particular position for negation in the tree. Recall first that the existing literature contains diverging opinions on the position of NEG. Some authors place it higher than HON (Choi & Harley, 2019), others place it lower than HON (Chung, 2009). Our account with split HON allows us to take the good aspects of each of these proposals by placing the NEG head in between the two honorific heads. The full structure we shall be assuming is therefore as in (62).



All the heads above vP are assumed to be optional in the sense that they can be either present or absent. The only caveat applies to the two HON heads, which are either both present, or both absent.

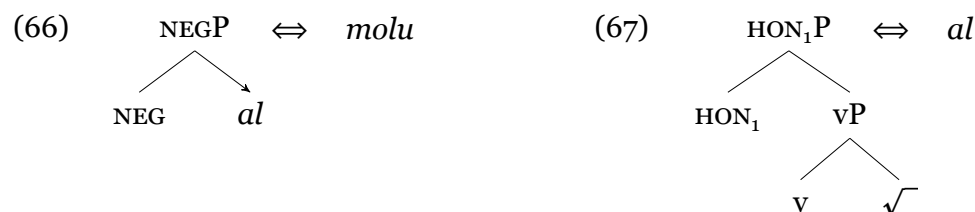
As a default, the NEG head is spelled out by one of the two negative markers *ani* or *mos*. The tree (63) shows the negation of a simple verb root *ca* ‘sleep.’ The tree in (64) shows the negation of a causative.



Let us now turn to the verb *al* ‘know’, which has a negative suppletive form *molu*. The pattern of this verb is repeated in (65) (repeated from (42)). The most remarkable datapoint is on line (d), showing that the causative blocks the use of *molu*- despite the presence of NEG in the structure, and leading to an analytic marking of negation as *ani* or *mos*.

- (65)
- | | | | |
|----|-------------------------------|----------------------|--------------------|
| a. | $\sqrt{\text{KNOW}}$ | al | ‘know’ |
| b. | NEG $\sqrt{\text{KNOW}}$ | molu | ‘not know’ |
| c. | $\sqrt{\text{KNOW}}$ CAUS | al-li | ‘let know, inform’ |
| d. | NEG $\sqrt{\text{KNOW}}$ CAUS | <i>ani/mos al-li</i> | ‘not inform’ |

To see how our system accounts for this, let us first provide the lexical entry for the suppletive negative form *molu* ‘not know,’ see (66). The entry contains a pointer to the non-negative verb *al* ‘know.’ That is, it realises the NEG head and whatever features are realised by *al* ‘know.’ The entry of *al* is repeated in (67) (recall (53)).



Our account correctly predicts that the lexical item in (66) cannot be used in the causative structure as the spellout of NEG and the root *al*. The reason is that CAUS intervenes between NEG and the root. Instead, the vP, the CAUS head, and NEG have to be each spelled out by a distinct morpheme; the relevant structure is shown in (64) above. It correctly

predicts that even though NEG is present in the structure of the negated causative, the suppletive negative form *molu* ‘not know’ cannot not be used.

In sum, the CAUS head is closest to the root. Like all the other functional heads in the above sequence, it is an optional element. When present, it triggers ‘causative intervention’ both for suppletive negative verbs (as discussed in this section) and suppletive honorific verbs (as discussed in the previous section). The reason for this is the presence of CAUS blocks the lexicalisation of the the bottom-most vP along with HON₁ or NEG.

5.6 EXPLAINING THE PARADOX

So far, we discussed the interaction between causativisation and negation, and between causativisation and honorification. What we still have not discussed is the interaction between negation and honorification, which is precisely the domain where the paradox discussed in section 5.1 arises. For convenience, we repeat the relevant data here.

(37)

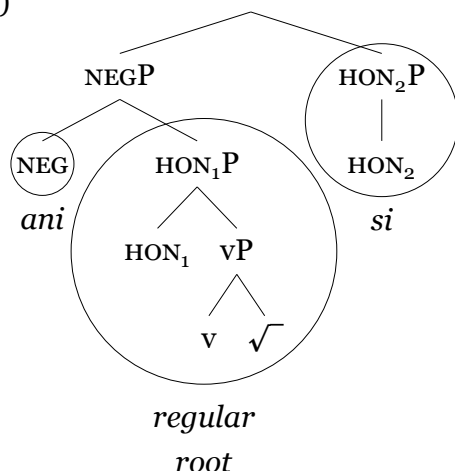
	regular pattern	$\sqrt{\text{EXIST}}$	$\sqrt{\text{KNOW}}$
a.	$\sqrt{X}$	iss	al
b.	NEG $\sqrt{X}$	eps	molu
c.	$\sqrt{X}$ HON	kyey-si	al-si
d.	NEG $\sqrt{X}$ HON	ani/mos kyey-si	molu-si

The paradox is that with the root $\sqrt{\text{EXIST}}$, the presence of the honorific blocks negative suppletion, as shown on line (d). This suggests that HON is closer to the root than NEG. With $\sqrt{\text{KNOW}}$, the honorific does not block negative suppletion, suggesting NEG is closer to the root.

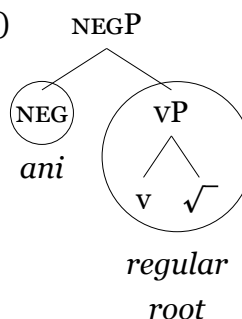
The point of this section is to show that with the two HON heads in the structure, this paradox can be resolved. We first focus on deriving the regular pattern as given in the first column on the left. Recall that for such regular roots, we assume that they spell out HON₁P.

As a starting point, let us draw the structures for a negated honorific and a negated non-honorific. The relevant structures are then as given in (68) (negated honorific) and (69) (negated non-honorific).

(68)



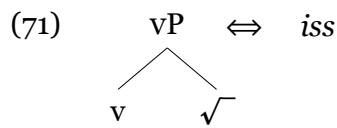
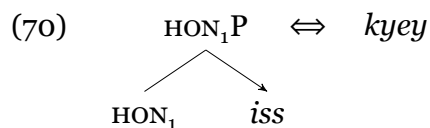
(69)



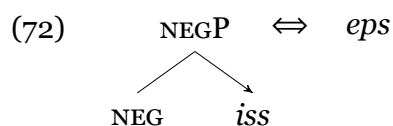
What do these structures predict for the negative suppletive form *molu* ‘not know’? Recall that the lexical entry for *molu*, given in (66) above, contains NEG and a pointer to *al*. First of all, they predict that *molu*- will be able to spell out the whole structure (69), provided vP is lexicalised by *al*.

The structures further predict that honorification will not block negative suppletion for *al* ‘know.’ The reason is that the root *al*- ‘know’ is associated to a constituent of the size HON₁P, recall (67) above. Therefore, *al* can spell out either the HON₁P in (68), or just the vP in (69). Since *molu* ‘not know’ has a pointer to *al*, it will apply always when the NEG_P contains NEG and *al*, regardless of the size of the structure spelled out by *al*. Therefore, whenever a NEG head is merged to a constituent that has been spelled out as *al* on the previous cycle, *molu* is going to be inserted. Therefore, we correctly predict that we find *molu* both in (68) and (69). The consequence is that with *molu* ‘not know,’ honorification does **not** block negative suppletion.

Next we turn to the verb *iss* ‘be, exist.’ This verb has three different suppletive allomorphs. It has a suppletive honorific form *kyey*, which spells out HON₁P and contains a pointer to *iss*, see (70). The non-honorific form *iss* then spells out just vP, see (71).

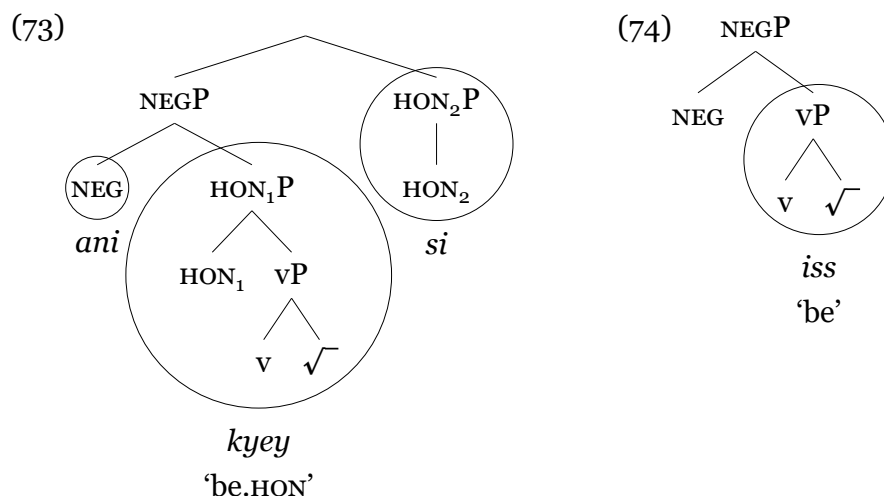


Then, *iss* ‘be, exist’ also has a suppletive negative form *eps* ‘not exist’, whose lexical entry is as given in (72):



What do these entries predict about the interaction between honorification and nega-

tion? In order to see that, consider the fact that the applicability of a negative suppletive form is always evaluated at NEG_P. The suppletive item applies when NEG has a sister that has been spelled out by a particular lexical item. Therefore, in order to determine the applicability of (72), we need to look at whether the sister to NEG is spelled out by *iss* ‘be’ or not. The examples in (73) and (74) show the relevant structures.



What we see here is that in the honorific structure (73), the HON₁P is spelled out by the honorific suppletive form *kyey*, not *iss*. Therefore, the negative suppletive *eps* does not match the NEG_P in (73), and we correctly get the spellout *ani kyey* ‘not be.HON.’ The consequence is that in this case, the presence of HON does block negative suppletion. This is different from the non-honorific structure, where *iss* is the only candidate for spellout. Therefore, (74) is correctly predicted to be realised as *eps* ‘not be’.

The solution to the paradox relies on the most basic essence of the pointer hypothesis: namely that suppletive lexical items point one to the other, and each spells out a structure of a different size. Since *al* is ambiguous between spelling out HON₁P and vP, the form *molu* ‘NEG.know’ inherits this ambiguity and we find it both in the honorific and non-honorific environments. However, ‘be, exist’ is not ambiguous in the same way: we find *iss* in non-honorific environments, and *kyey* in honorific environments. The negative suppletive *eps* ‘not be’ has a pointer to *iss*, and it therefore inherits the distribution of *iss*, and both are only found in non-honorific contexts.

Importantly, this solution maintains the idea that suppletive roots spell out structures of different sizes, and therefore, they never compete with each other: there is no point in the derivation where both *eps* ‘not be’ and *iss* ‘be’ can be inserted. Since we do not need competition among suppletive lexical items, we can maintain the idea that whenever multiple roots match, the choice between them is free.

CONCLUSION

This paper has proposed an approach that reaches an important theoretical goal, namely to allow for root suppletion within a theory of narrow syntax that adopts the Strong Modularity Thesis, which holds that syntax is both phonology-free and concept-free. By dispensing with indexed $\sqrt{\text{ }}$ s, the theory is also compatible with approaches where $\sqrt{\text{ }}$ s are dispensed with altogether (Ramchand, 2008). What makes this type of theory available is a bottom-up phrasal lexicalisation, where $\sqrt{\text{ }}$ s are kept distinct from morphological roots. The latter come in a variety of classes, each class associated with a particular amount of functional structure.

We have further explained why and how such a theory is compatible with the fact that suppletion often co-occurs with overt marking. In order to test the predictions, we looked at the details of comparative/superlative suppletion in English, Czech and Latin. What we found is that suppletion in these languages is inevitably correlated with the reduction of overt morphology, which supports the empirical predictions of the model.

We finally discussed a paradox involving suppletion in Korean involving the interaction of negation and honorification in suppletion. Depending on the verb, different patterns are observed, which lead Choi and Harley (2019) to argue that allomorphy can be conditioned non-locally, across an intervening head. We discussed two cases which represent a similar type of configuration, but that does not give rise to allomorphy, and concluded that allomorphy must be conditioned strictly locally after all. Our solution to the Korean paradox consisted in a postulation of two different HON heads, one below and one above negation. The different behaviour of different verbs when negation and honorification interact could then simply be attributed to the different lexical entries for the relevant suppletive roots. Our theory of suppletive roots as involving pointers in lexical entries moreover ensured the correct distribution of suppletive roots, without having to take recourse to indexed $\sqrt{\text{ }}$ nodes. We could thus maintain the Strong Modularity Thesis, with a phonology- and concept-free syntax, and with free choice of the root at the first cycle of insertion.

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6 Tense and sentential negation: a typological perspective

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Tense and sentential negation

A typological perspective

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The association between sentential negation and tense has solid foundations in the literature. It has even been argued that sentential negative markers consist of a Tense feature (De Clercq 2018, 2020). This paper adduces the first results from a typological study and data from Bambara in support of this claim. In addition, the Bambara data also point to the morphological realisation of present or default tense, to a hierarchy for tense and to a position for sentential negation in that hierarchy.

Keywords: negation, TAM, Bambara, typology, nanosyntax

1. Introduction

It is a fairly common assumption in the generative literature that sentential negation is associated with tense.¹ This idea can be traced back to Zanuttini (1996) and was generally accepted in much subsequent work, taking the shape of a NegP that dominates TP (Zanuttini 1991; Haegeman 1995; Zanuttini 1997 and many others). This proposal is based on a careful consideration of the distribution and properties of different negative markers within one language, their position with respect to the hierarchy for adverbs proposed by Cinque (1999), the position with respect to tense morphemes and the scopal properties of sentential negative markers with respect to tensed predicates. In more recent work, De Clercq (2018, 2020) did not only adopt the idea that sentential negation sits in a position associated with tense, but she also argued on the basis of syncretism patterns between scopally different negative markers that the internal featural structure of sentential negative markers is richer than previously assumed and that in addition to [Neg], a sentential neg-

1. I want to thank two anonymous reviewers for their useful comments and Guido Vanden Wyngaerd and Michal Starke for discussion of some of the issues raised in this paper. All errors are of course mine.

ative marker also consists of a default tense feature (henceforth [T]). Even though the evidence from syncretisms indeed supports the idea of an additional feature, very little direct support was provided for the fact that this additional feature is or needs to be [T]. The present paper will provide some support for this claim and will go beyond that, by showing that negative markers – from a cross-linguistic perspective – can not only interact with tense, but also with mood and aspect, the so-called TAM-domain, and that hence it is possible that negative markers spell out even more features of the TAM-domain than just a [T]-feature.

In the next section I will introduce the notion of TAM-conditioned negative allomorphy and the hypothesis that fed into the typological investigation. The results of the typological investigation will be presented in Section 3. Finally in Section 4 I will zoom in on Bambara to get more direct support for [T] inside a sentential negative marker.

2. Negation and TAM

In many languages around the world sentential negative markers remain constant, regardless of the Tense, Aspect, Mood or Modality (henceforth TAM) expressed on the predicate, as is the case in English for instance, illustrated in (1).

- (1) a. John is **not** playing.
- b. John was **not** playing.
- c. John has **not** played.
- d. John would **not** play with you.
- e. ...

However, unlike English there are languages in the world that show Allomorphy In Sentential Negative Markers conditioned by TAM (henceforth AINT), as mentioned by Payne (1985); Dryer (2011) and Horn (2001). Horn (2001: 449) mentions that Classic Arabic made use of *maa* to express negation in the context of past tense and of *laa* to express negation in the context of nonpast, while Payne (1985: 209) mentions that Maori uses *kiihai* in the context of completed aspect and *kaahore* for non-completed aspect. Most discussions of TAM-conditioned allomorphy are restricted to the domain of speech acts though, and more in particular to imperatives, as in Sadock & Zwicky (1985).

A thorough and systematic investigation of sentential negative markers from the perspective of AINT in general is lacking, but would be helpful to get a better understanding of the internal structure of sentential negative markers, not in the least the question whether negative markers consist of a [T] feature, which is one of the research questions in this paper.

Before we can start a small-scale typological investigation into this domain we need to have a closer look at the different types of languages there are with respect to sentential negative markers. A first type is the type that is represented by English in (1): a type that does not make use of TAM-conditioned negative allomorphs. A second type can be illustrated by a language like Bafut, a Niger-Congo language spoken in Cameroon, that makes use of different *suppletive* negative markers conditioned by declarative, (2), imperative, (3), or conditional mood, (4).²

- (2) a. *Bó tù'ù mə ŋkì.*
 they carry PST_{imm} water
 'They have just fetched water.'
- b. *Kāā bó sì ŋkì tū'ù.*
 NEG they NEG water carry
 'They have not fetched water.' (Chumbow & Tamanji 1994: 215)
- (3) a. *Fá ŋkī wá.*
 give water that
 'Give that water.'
- b. *Tsūū ŋkī wá fá.*
 NEG water that give
 'Do not give that water.' (Chumbow & Tamanji 1994: 225–226)
- (4) a. *Ngwín káló tú'ú ŋkì*
 Ngwin F2 carry water
 'Ngwin will fetch water (tomorrow).'
- b. *Mbá Ngwìn lōn túú ŋkì tú'ú*
 If Ngwin F2 NEG water carry
 'If Ngwin fails to fetch water.'

A third case that is worthwhile taking a closer look at is Bengali (Ramchand 2004: 40–41). While present and past tenses are negated by means of *na*, (5–a), the perfective in Bengali, expressed as in (5–b), can only be negated by *ni*, (5–c), a form that is *partially identical* to *na* and hence partially suppletive, illustrating yet another type of TAM-conditioned allomorphy, that is similar to the second type in that it shows partial suppletion.

- (5) a. *Ami amṭa kha- cch- i na.*
 NOM mango.CLF eat- PROG/PRS- 1SG NEG
 'I am not eating the mango.' [Bengali]

2. There are tense conditioned allomorphs for negation as well in Bafut, but it goes beyond the confines of this paper to discuss this. I refer the reader to Chumbow & Tamanji (1994) for a detailed discussion of Bafut.

- b. *Ami amṭa kheyē- ch- i*
 NOM mango.CLF eat.PFV- PRS 1SG
 ‘I have eaten the mango.’
- c. *Ami amṭa kha- i ni*
 NOM mango.CLF eat 1SG NEG
 ‘I didn’t eat the mango.’

Interestingly, both in Bafut and Bengali there are not only changes in the form of the negative marker conditioned by TAM, but there are also changes in the TAM-marking conditioned by negation, a phenomenon studied by Miestamo (2005).³ In Bafut, in the immediate past (PST_{imm}), which is interpreted as a completed event, *mə* disappears under negation, as in (2),⁴ while in Bengali the perfective form *kheyē* changes back to the root form *kha* under the perfective negator *ni*.⁵ The interaction between the AINT and the TAM-changes under negation strongly suggest that the features of sentential negative markers and the features related to TAM are somehow part of the same continuum. Even though the cases discussed here do not reflect on tense, the discussion of English, Bafut and Bengali is instructive to set up a typology of sentential negative markers, i.e. a hypothesis concerning the type of languages that we expect to find in the languages of the world.

We hypothesize that there are 6 logical possibilities, illustrated in Table 1. Languages that display *suppletive* AINT and TAM-changes under negation (TYPE 1A) or no TAM-changes under negation (TYPE 2A); languages that have *partially identical* AINT and TAM-changes under negation (TYPE 2A) or no TAM-changes under negation (TYPE 2B); languages that do not have NEG allomorphy but that have TAM-changes under negation (TYPE 3) or no TAM-changes under negation (TYPE 4); languages that have a negative allomorph, but one that is not conditioned by TAM, and that show TAM-change (TYPE 5) or no TAM-change (TYPE 6).

3. This paper is restricted to a discussion of TAM-conditioned negative allomorphy and negation-conditioned TAM-allomorphy. However, the role of Φ -features on the shape of the negative marker and the role of negation on the shape of Φ should not be underestimated. The latter of the two is also discussed in Miestamo (2005).

4. It should be mentioned that in all negative clauses in Bafut, regardless of the mood involved, the position of the verb changes under negation. Even though this does not qualify as TAM-allomorphy, it is most definitely an alternation, i.e. asymmetry in Miestamo’s terms, conditioned by negation.

5. These allomorphic changes sometimes trigger changes in the interpretation of the tenses as well, as for instance in (5–c). I refer the reader to Ramchand (2004) for a more detailed discussion of the semantics of both negative markers.

Table 1. Typology of sentential negative markers

	TAM-change	No TAM-change
Suppletive AINT	TYPE 1A	TYPE 2A
Partially identical AINT	TYPE 1B	TYPE 2B
No NEG allomorphy	TYPE 3	TYPE 4
Other NEG allomorph	TYPE 5	TYPE 6

In the next section I will test this typology with logical options for NEG-TAM interactions against a sample of 50 languages to see whether the logically possible languages are also attested in our sample.

3. Typological investigation

The set of languages in Table 2 are 50 languages discussed in Miestamo (2005). Miestamo (2005) investigated a typological sample of 297 languages for the asymmetries they display under negation. One of the main asymmetries that languages display is TAM-change. However, Miestamo did not focus in his discussion on the negative markers themselves, nor on the interaction between the negative markers and TAM-change. In what follows, I will scrutinize the first 50 languages in Miestamo's Appendix for their negative markers and the interactions with tense/aspect-marking on the verb, i.e. I will check them against the typological hypothesis in Table 1.

It is clear that this way of sampling does not make the current sample typologically representative for the languages of the world and hence all correlations are tentative at this point and should be treated as first results in a bigger ongoing project. The phylum, genus and macroarea of the languages taken up in the sample are listed in the Appendix and are based on Miestamo (2005). Important to mention is that the presented results focus on negative markers conditioned by tense and aspect and on tense and aspect changes under negation. Most mood and modality related categories, apart from irrealis, are not considered in the present count. This is an immediate consequence of the fact that Miestamo (2005)'s investigation, which was the main source for this typological investigation, was only concerned with changes under negation in declarative main clauses and most mood-related changes are relevant in embedded domains. Future research will include more mood and modality related categories and hence lead to an update of the results and classification presented here. I will nevertheless refer to the changes under negation as TAM-changes, since sometimes there are changes with respect to irrealis mood under negation.

Table 2. Typological sample

Abipón	Abkhaz	Achumawi	Acoma	Alamblak	Albanian	Ainu
Amele	Andoke	Apalaí	Apurinā	Arabic(Egyptian)	Araona	Comanche
Arapesh	Armenian (East)	Asmat	Awa Pit	Aymara	Bafut	
Bagimi	Bambara	Barasano	Baré	Basque	Batak (Karo)	
Bawm	Beja	Bella Coola	Berber (Middle Atlas)	Birom	Boko	
Bororo	Brahui	Burarra	Burmese	Burushaski	Canamarí	
Candoshi	Canela- Krahô	Cantonese	Carib	Cayuvava	Chamorro	
Chehalis (Upper)	Chinantec (Lealao)	Chinook (Lower)	Chorote	Chukchi	Chumash	

The results of the investigation are numerically summarized in Table 3 and listed per type and language in Table 4. The reasoning behind the current classification is that if a language has at least one TAM-conditioned negative allomorph (AINT) in the language as a whole it belongs to TYPE 1A, 1B, 2A or 2B. If it also has at least one negation-conditioned TAM-change, it belongs to TYPE 1A or 1B. Only if there is no negation conditioned TAM-change at all, but at least one AINT, will the language be classified as TYPE 2A AND 2B. Languages that are categorized as Type 3 do not have any negative allomorphs at all, but they do have negation-conditioned TAM-allomorphs. Type 4 languages do not show any negative allomorphs nor any TAM-changes under negation in the language as a whole. Some languages belonging to TYPE 1A or 1B could be argued to belong to both types, since a language can have both fully suppletive and partially identical negative markers conditioned by other TAM-categories. The same is true for TYPE 2A and 2B.

What is striking in Table 3 is the strong correlation between the presence of AINT and some or other TAM-change under negation, i.e. while there are quite some instances in the sample of TYPE 1A and 1B languages, there are no TYPE 2A and 2B languages. For languages without negative allomorphs, i.e. TYPE 3 and 4, absence of negative allomorphs seems to correlate more often with absence of TAM-change, but the correlation is less radical. There are a couple of languages that have negative allomorphs that are not conditioned by TAM, i.e. TYPE 5 and TYPE 6. It seems that these negative allomorphs are either conditioned by ϕ -features (as is most probably the case for Candoshi and Chorote) or by phono-

logical conditions (Burushaski, Abipón). A more detailed investigation of TYPE 5 and 6 is required, but remains out of the scope of this paper.

Table 3. Typology of sentential negative markers

	TAM-change	No TAM-change
Suppletive AINT	TYPE 1A: 7	TYPE 2A: 0
Partially identical AINT	TYPE 1B: 9	TYPE 2B: 0
No NEG allomorphy	TYPE 3: 9	TYPE 4: 21
Other NEG allomorphy	TYPE 5: 0	TYPE 6: 4

Even though the picture in Table 3 could well be skewed by the relatively small set of languages considered for this discussion, it turns out that not all logically possible types are attested in the current sample. Type 2a and 2b and Type 5 are not attested. If this is confirmed by further research and by a more balanced typological sample, then these results may lay bare a new language universal about the interaction of allomorphy between negation and TAM.

The most intriguing results of the typological investigation for the present paper are on the one hand what we see left of the vertical line in Table 4, i.e. the fact that all languages in the sample with AINT also show TAM-changes under negation, and on the other hand what we see right of the vertical line, i.e. that the absence of AINT does not imply an absence of TAM-change under negation, i.e. the existence of TYPE 3.

In the next section we will interpret these results and zoom in on Bambara to shed more light on the initial question this paper started out with: is there an empirical reason to argue that sentential negative markers may consist of a [T] feature?

4. Towards an analysis

4.1 Interpreting the typological data

The data from our typological survey in Section 3 point to the fact that when the form of the sentential negative marker in a language changes due to TAM this brings about changes in the TAM-domain as well. Crucially, when nothing changes in the form of the sentential negative marker due to TAM, the TAM-domain can nevertheless still be affected due to the mere presence of the sentential negative marker, i.e. the pattern displayed by TYPE 3. This pattern is important since this suggests that the mere presence of a sentential negative marker activates the TAM

Table 4. Typological sample: Division per types

TYPE 1A	TYPE 1B	TYPE 3	TYPE 4	TYPE 5	TYPE 6
Alamblak	Amele	Abkhaz	Achumawi	Burushaski	Abipón
Bukiyip	Egyp Arabic	E. Armenian	Acoma	Candoshi	Chorote
Albanian	Awa Pit	Aymara	Ainu		
Bambara	Bafut	Bagirmi	Andoke		
Bafut	Bororo	Berber	Apalai		
Beja	Brahui	Boko	Apurina		
Chukchi	Cantonese	Burrara	Araona		
	Upper Chehalis	Burmese	Asmat		
	Chinantec (Lealao)	Comanche	Barasano		
			Baré		
			Basque		
			Batak		
			Bawm		
			Bella Coola		
			Birom		
			Canela-Krahô		
			Carib		
			Cayuvava		
			Chamorro		
			Chumash		
			Chinook Lower		
TYPE 2A	TYPE 2B				
–	–				

domain in a non-trivial way and could provide support for the hypothesis that there is a [T]-feature inside negative markers. As mentioned before, more data are required to confirm these results and to test their significance with respect to the languages of the world.

If we assume that language variation boils down to differences in the structural size of lexical items (Starke 2011; Caha et al. 2019; Vanden Wyngaerd et al. 2020), i.e. the number of features that lexical items spell out according to the Nanosyntactic spellout principles (Starke 2009; Caha 2009; Baunaz et al. 2018),⁶ then the variation we get to see in the languages of the world with respect to AINT

6. Nanosyntactic spellout is rigidly cyclic and phrasal, i.e. after every step of Merge syntax tries to spellout.

and the TAM-change under negation could be regarded as a trade-off between features spelled out by markers pertaining to the TAM-domain and features inherently spelled out by negative markers. Table 5 illustrates this idea informally.

The first row provides a set of arbitrary abstract features. In the following rows it is hypothetically and informally illustrated how the relevant markers (predicates, negative markers, TAM-markers) can spell out different sets of features (A, B, C, D) depending on the language one is dealing with. The table shows that there can be gaps for TAM-related categories and negation (row 2 and 3): both categories can be optional and can therefore be absent in certain configurations; furthermore the table shows that all morphemes can grow or shrink featurally, depending on the language we are dealing with: the predicate (row 4); the negative marker (row 5 and 6) or a TAM-morpheme (row 7). I added ϕ (person, gender, number) to this table as well, but I will remain agnostic about it for now, even though interactions between ϕ , negation and the TAM-domain have been observed (Miestamo 2005 and see Section 3).

Table 5. Lexicalisation table

	A	B	C	D	E
1.	PRED(-)	TAM	NEG-m	TAM	ϕ
2.	PRED(-)		NEG-m	TAM	ϕ
3.	PRED(-)	TAM		TAM	ϕ
4.	PRED(-)		NEG-m	TAM	ϕ
5.	PRED(-)	TAM	NEG-m		ϕ
6.	PRED(-)	NEG-m			ϕ
7.	PRED(-)	TAM		NEG-m	ϕ

What Table 5 does not show is which types in Table 4 correlate with which row. In order to pinpoint that, we need to do more typological research, understand better what the abstract features in the toprow stand for and how many features each morpheme can consist of, i.e. know the precise functional sequence (FSEQ) of these morphemes. It is the aim of the bigger project behind this paper to investigate this, but for now the aim is to focus solely on the internal structure of the sentential negative morpheme (Neg-m in Table 5) and more in particular on whether it makes sense to argue for a default [T]-feature.

The typological investigation in Section 3 already suggested that the presence of a negative marker can bring about allomorphy in the TAM-domain, regardless whether the negative marker has an allomorph or not that is conditioned by TAM. However, what was not made explicit is what the role is of present tense. When

we referred to AINT up until now, the allomorphs conditioned by tense are never conditioned by present tense. The main reason for this is that the standard negator, which is by definition not represented as an AINT, is the negator we find in the context of the present tense indicative. If the informal trade-off of features between TAM-markers and AINT is on the right track, then it could very well be the case that standard negators, which typically appear in the context of present tense spell out a (default) tense feature. This would also explain why present tense is very often morphologically zero coded (Bybee et al. 1994). However, this hypothesis also presumes that we expect to see present or default tense morphologically realised in at least some languages in the world, possibly also contained within the negative marker.

It seems that there is such a language in the small sample discussed in this paper. Bambara does not only provide support for the idea that present tense can be morphologically realized, but also for the idea that it can be part of the feature composition of negative markers. In addition, Bambara also sheds light on the hierarchy of tense features. In Section 4.2 we discuss the data from Bambara.

4.2 Negative markers and tense: Support from Bambara

Bambara is a Niger-Congo language spoken in Mali.⁷ The present tense marker *bε* becomes *tε* in a negative sentence, cf. (6). It thus seems as if *b-* is replaced by the negative element *t-*. The same is true in the present progressive, where progressive aspect is additionally marked by means of *ka*, (7).⁸

7. Bambara is grouped with Type 1a and should thus have a fully suppletive AINT. This is indeed the case. To negate an affirmative sentence with a transitive predicate expressing perfective aspect, as in (i-a) the *fully suppletive* negative marker *ma* is used, (i-b).

- (i) a. *Birama ye liburu kalan.*
 Birama PRF book read
 ‘Birama has read the book.’
 b. *Birama ma liburu kalan.*
 Birama PRF.NEG book read
 ‘Birama has not read the book.’

Due to a lack of space it is not possible to discuss perfective tense with intransitive predicates and past perfect tense. However, the negator remains the same (cf. (Ermisch 2013)).

8. Bambara examples use a simplified orthography without tone marking and follow Dumestre (2003) in spelling the vowel of the marker under discussion /ε/, not /E/ as in Ermisch (2013).

- (6) a. *Sanu bε baara kε*
 Sanu PRS work do
 ‘Sanu works.’
 b. *Sanu tε baara kε*
 Sanu PRS.NEG work do
 ‘Sanu does not work.’ (Ermisch 2013: 2)
- (7) a. *N bε ka liburu kalan.*
 1SG PRS PROG book read
 ‘I am reading the book.’
 b. *N tε ka liburu kalan.*
 1SG PRS.NEG PROG book read
 ‘I am not reading the book.’ (Ermisch 2013: 3)

Based on these data we can safely say that negation sits higher in the functional sequence than progressive aspect. However, it is less clear whether negation sits higher than the morpheme that gives rise to present tense. One could argue that the data in (6) suggest that present tense consists of two features, as illustrated in Table 6. However, under this approach one would assume there is an independent morpheme, ϵ , in Bambara, that can be spelled out separately. Since this is not the case, it seems more likely that $b\epsilon$ is (phrasally) spelled out by the feature responsible for the expression of present tense/default tense, represented here by PRS and illustrated in Table 7.⁹ Under this assumption $t\epsilon$ is the phrasal spellout of that same PRS and NEG. However, it is not possible to decide whether NEG sits below or above PRS on the basis of the data in (6)–(7). I will assume that it sits above PRS for now, as in the table in (7).¹⁰

Table 6. Bambara: Present tense 1

(NEG)	PRS1	PRS2
	b	ϵ
t	ϵ	

9. It is not the aim of this paper to discern the right featural properties of present, past or future tense nor to discuss the semantics of tense or the relationship between grammatical tense and semantic tense. The tense-related featural labels used in this paper are placeholders for whatever features responsible for the expression of present tense, past or future tense.

10. It has been argued by Dumestre (2003); Ermisch (2013) that $b\epsilon$ should be glossed PRES.AFF. However, there is no more reason to gloss ‘be’ as AFF, than there would be to gloss English *is* that way. Moreover, there is a focus particle in Bambara, *dé* (Prokhorov 2014), that is dedicated to functions similar to emphatic *do* in English. It seems to me that this particle would be a better candidate to be glossed AFF.

Table 7. Bambara: Present tense 2

(NEG)	PRS
	bɛ
tɛ	

It thus seems that Bambara provides morphological support for the morphological realisation of present tense and the idea that sentential negative markers spell out a tense feature in addition to a NEG feature. However, Bambara's morphology has more to offer, both with respect to the FSEQ of tense and with respect to the position of negation in that FSEQ.

The past tense morpheme in Bambara is *tun bɛ*, which literally contains the present tense marker *bɛ*. The negative past form, *tun tɛ*, thus shows that the negative past tense consists of the marker used in the negated present tense. It thus seems fair to say that the marker used in the present tense is some sort of default tense marker, which can be part of the morphological make-up of other tenses.

- (8) a. *N tun bɛ liburu kalan.*
 1SG PAST PRES book read
 'I was reading the book.' / 'I used to read the book.'
- b. *N tun tɛ liburu kalan.*
 1SG PAST PRES.NEG book read
 'I was not reading the book.' (Ermisch 2013: 4)

Assuming, as we mentioned before, that the feature NEG sits above PRES, Bambara gives us the following hierarchy.

- (9) (PROG) < PRS < NEG < PST

To get a more complete picture we also need information on future tense. Near future tense in Bambara is expressed as in (10), while distant future tense is expressed as in (11). Especially the distant future tense is informative, since it shows that the present or default tense marker is present in the future tense. The negative marker *tɛ*, which as we said consists of present tense, is present in both distant and near future tenses. The examples are from Ermisch (2013), but I modified the glosses in accordance with the analysis in this paper.

- (10) a. *Birama na mobili ko.*
 Birama FUT car wash
 'Birama is going to wash the car.'
- b. *Birama tɛ- na mobili ko.*
 Birama PRS.NEG- FUT car wash
 'Birama will not wash the car.'

- (11) a. *Muso be-na den sOrO.*
 Muso PRS-FUT baby have
 ‘The woman will have a baby.’
 b. *Muso te-na den sOrO.*
 Muso PRS.NEG-FUT baby have
 ‘The woman will not have a baby.’

Assuming the mirror principle (Baker 1985), the Bambara future tense data indicate that future tense *na* sits hierarchically higher than negation and present tense. Moreover, the fact that the past tense marker is a free morpheme in a negative context, while the future tense marker *na* is a bound morpheme under negation, leads us to propose that *na* is closer to the present tense than the free morpheme *tun*. This gives us an order for Past, Future and Present tense.

The order for tense, negation and progressive aspect in Bambara is visualized by means of a lexicalisation table, cf. Table 8: the shorthand for the tense features and the negation is in the top row, while the fact that some morphemes span several cells captures nanosyntactic phrasal spellout in an informal way.

Table 8. Bambara

PST	FUT	NEG	PRS
			be
		te	
tun			be
tun		te	
	na		be
	na	te	

At this point – and under the assumption of Chomsky (2001:2)’s Uniformity Principle – we have actually completed the mission of this paper, i.e. we provided support from Bambara and from a typological investigation for the idea that sentential negative markers may consist of a [T]-feature.

5. Conclusion

This paper set out to provide support for the well-established claim in the literature that sentential negative markers are associated with tense and more radically, that there is [T] feature inside sentential negative markers (De Clercq 2018; 2020). First, the paper discussed data from a small-scale typological investigation

that showed that the presence of TAM-allomorphy under negation does not necessarily coincide with TAM-conditioned negative allomorphy, while the presence of TAM-conditioned negative allomorphy always correlates with TAM-allomorphy under negation in the current sample. Crucially, the study also showed that the mere presence of a standard negator may impact TAM-marking, providing strong support to the idea that negation interacts with the TAM domain in non-trivial ways. Finally, the investigation suggests that some logically possible types of languages may not exist in the languages of the world. More research is required to confirm this. Second, we speculated on the role of the standard negator with respect to present tense. We suggested that if we were to assume that there is a default tense feature inside standard negative markers, this could explain why present tense is so often zero coded and why standard negators typically appear in the context of present tense. Third, we discussed Bambara, whose morphology does not only tell us more about the FSEQ for tense and negation, but it crucially also shows that a sentential negative marker indeed may consist of a default/present tense feature.

Abbreviations

The glosses and abbreviations used in this paper are:

AFF	(affirmation)
AINT	(Allomorphy in Sentential Negative Markers conditioned by TAM)
CLF	(classifier) F2 (tomorrow future)
FUT	(future)
NEG	(sentential negation)
NOM	(nominative)
PRF	(perfective)
PRS	(present)
PROG	(progressive)
PST	(past)
PST	<i>imm</i> (immediate past)
SG	(singular)
TAM	(Tense, Aspect, Mood/Modality).

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Appendix

Table 9. Sample

Language	Phylum	Genus	Macroarea
Abipón	Mataco-Guaicuru	Guaicuruan	South America
Abkhaz	Caucasian	Northwest Caucasian	Eurasia
Achumawi	Hokan	Palaihnihan	North America
Acoma	Keresan	Keresan	North America
Alamblak	Sepik-Ramu	Sepik	Australia/New Guinea
Albanian	indo-European	Albanian	Eurasia
Ainu	Ainu	Ainu	Eurasia
Amele	Trans-New-Guinea	Madang	New Guinea
Andoke	Andoke	Andoke	South America
Apalaí	Carib	Carib	South America
Apurinā	Maipuran	Maipuran	South America
Arabic(Egyptian)	Afro-Asiatic	Semitic	Africa
Araona	Tacanan	Tacanan	South America
Arapesh	Torricelli	Torricelli	Australia /New Guinea

Language	Phylum	Genus	Macroarea
Armenian (East)	indo-European	Armenian	Eurasia
Asmat	Trans-New-Guinea	Central& South New Guinea Kutubuan	Australia/New Guinea
Awa Pit	Barbacoan	Barbacoan	South America
Aymara	Aymaran	Aymara	South America
Bafut	Niger-Congo	Bantoid	Africa
Bagirmi	Nilo-Saharan	Bongo-Bagirmi	Africa
Bambara	Niger-Congo	Western Mande	Africa
Barasano	Tucanoan	Tucanoan	South America
Baré	Maipuran	Maipuran	South America
Basque	Basque	Basque	Eurasia
Batak (Karo)	Austronesian	Batak	Southeast Asia/ Oceania
Bawm	Sino-Tibetan	Kuki-Chin-Naga	Southeast Asia/ Oceania
Beja	Afro-Asiatic	Northern Cushitic	Africa
Bella Coola	Salishan	Bella Coola	North America

Table 10. Sample

Language	Phylum	Genus	Macroarea
Berber (Middle Atlas)	Afro-Asiatic	Berber	Africa
Birom	Niger-Congo	Platoid	Africa
Boko	Niger-Congo	Eastern Mande	Africa
Bororo	Macro-Ge	Bororoan	South America
Brahui	Dravidian	Northwest Dravidian	Eurasia
Burarra	Australian	Burarran	Australia/New Guinea
Burmese	Sino-Tibetan	Burmese	Eurasia
Burushaski	Burushaski	Burushaski	Eurasia
Canamarí	Katukinan	Katukinan	South America
Candoshi	Candoshi	Candoshi	South America
Canela-Krahô	Macro-Ge	Ge-Kaingang	South America
Cantonese	Sino-Tibetan	Sinitic	Southeast Asia/Oceania
Carib	Carib	Carib	South America
Cayuvava	Cayuvava	Cajuvava	South America
Chamorro	Austronesian	Chamorro	Southeast Asia/Oceania
Chehalis (Upper)	Salishan	Tsamosan	North America
Chinantec (Lealao)	Oto-Manguean	Chinantecan	North America
Chinook (Lower)	Penutian	Chinookan	North America
Chorote	Guaicuru	Mataco	South America
Chukchi	Chuckotko-Kamchatkan	Chuckotko-Kamchatkan	Eurasia
Chumash	Chumash	Chumash	North America
Comanche	Uto-Aztecan	Numic	North America

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7 The Addressee in the present imperative in Latin

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The Addressee in the present imperative in Latin

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Abstract. In this paper we adduce morphological support for the activation of a Speech Act layer, more in particular an Addressee layer, in the syntax of Latin present imperatives (see Isac 2015 and Zanuttini 2008 for similar proposals). Evidence comes from a comparison between present and future imperatives. While the regular ϕ -morphology is generally absent and substituted for a different set of endings in these paradigms, it does resurface when the subject and Addressee of the imperative do not correspond, as is the case for the third person forms of the future imperative. We argue that this suggests that the ϕ -morphology, dedicated to the lexicalisation of the subject needs to be distinguished from the Addressee morphology in Latin, and that, accordingly, they must also be distinguished in syntax. We provide an analysis, couched in Nanosyntax (Starke 2009 et seq.) for the present imperative forms of the first verb class following *amāre* ('love').

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1 Introduction

In this paper, we investigate the imperative in Classical Latin (1st century BCE - 3rd century CE). We focus, in particular, on what looks like the agreement morphology of this mood, because it is markedly different from that of the indicative and the subjunctive. The table in (1) illustrates these differences by means of the present tense forms of the first class verb, *amāre* ('love'). The focus is on the second person singular and plural, because these are the only forms available for the present imperative in Latin. The relevant endings are marked in bold.¹

		PRES.IMP	PRES.IND	PRES.SBJV
(1)	2SG	am-ā- Ø	am-ā- s	am-ē- s
	2PL	am-ā- te	am-ā- tis	am-ē- tis

As we can see from the data in (1), the singular imperative only consists of the root *am-*, and the theme vowel *-ā*, whereas the other singular forms mark more grammatical information about person/number (ϕ for short) by the ending *-s*. And while the plural imperative does have an ending, *-te*, that could pass for ϕ , it still only bears a slight resemblance to the ending, *-tis*, of the other plural forms. So, this brief comparison between the moods in the present tense seems to indicate that the indicative and subjunctive share the same endings, whereas the imperative seems to have its own unique set.

To show that the indicative and subjunctive pairing of ϕ -endings is not characteristic of the present tense, and that these endings could therefore be viewed as the 'regular' ones, in opposition to the imperative ones, we provide table 1. The table gives an overview of all the available active verb forms in both moods. As we can see, the singular ending *-s* and plural ending *-tis* are omnipresent, (except for the PRF.IND.2SG, *-ti*, but see Baldi (2002) for a discussion on this) and we cannot find another zero ending or *-te* anywhere else.

¹The table in (1) only illustrates the active verb forms. The reason for this is that passive and deponent imperatives are exceptionally rare, despite being attested (Pinkster 2015: 512; 520-512). This is most probably because passivity and controllability are conflicting properties. We thus do not opt to include them in the current discussion, rendering comparisons with other passive present tense forms irrelevant. In addition, we will only account for the first verb class for now. The other verb classes largely exhibit the same imperative morphology, so the analysis we propose is expected to be transferable.

	PST.IND	FUT.IND	PRF.IND	PLUPRF.IND	FUT.PRF
2SG	amā-bā-s	amā-bi-s	amā-v-is-tī	amā-v-erā-s	amā-v-eri-s
2PL	amā-bā-tis	amā-bi-tis	amā-v-is-tis	amā-v-erā-tis	amā-v-eri-tis
	PST.SBJV		PRF.SBJV	PLUPRF.SBJV	
2SG	amā-rē-s		amā-v-eri-s	amā-vi-ssē-s	
2PL	amā-rē-tis		amā-v-eri-tis	amā-vi-ssē-tis	

Table 1

The question which arises from these observations is why such a morphological split should exist in Latin between the endings of the indicative/subjunctive and the imperative. This is, as far as we know, an unaddressed question in the literature, as the verb endings have mainly been described from a diachronic point of view, often with the goal of reconstructing their origins and developments.

With regards to the singular forms, for instance, Baldi (2002) notes that the imperatives of thematic verbs (i.e. verbs which put a vowel between the stem and the personal ending) already lacked endings in PIE, and that the *-s* we now see uniformly in the indicative and subjunctive, originally stems from two different endings, **-si* and **-s* respectively. With regards to the plural forms, Baldi (2002) and W.Fortson IV (2011) note that both endings *-te* and *-tis* are derived from the indicative PIE ending, **-te*. The final *-s* of *-tis* emerged most probably through analogy with the regular first person plural *-mus*.

Now, while these reconstructions may be able to tell us something about how the ‘current’ forms came about, they do not really shed light on the reason why the imperative endings turned out differently from the indicative and subjunctive ones. In this paper we aim to provide an account which explains the distinction between the endings. Building on the well-established idea that the imperative does not only differ morphologically from indicatives and subjunctives, but also structurally (see for instance Isac 2015, Romanello & Repetti 2014, Zhang 1990, and Zanuttini 1994, 1997 amongst others), we will show on the one hand that the Latin imperative lacks certain features that are present in the other moods, and on the other hand that there is evidence to assume that imperatives contain an Addressee-layer (Zanuttini 2008) which is absent in the other moods. In a nutshell, we will argue that the fact that there are two sets of endings is a direct reflection of the different underlying structures. The analysis which we propose is couched in Nanosyntax (Starke 2009 et seq.).

The structure of the paper is as follows: In section 2, we first explore the contrast that is most relevant to the current discussion, namely the ϕ -distinction between the imperative and the indicative and subjunct-

ive that was addressed above. For this, we will draw on a comparison of the present imperative and future imperative. Through this comparison, we adduce support for a special Addressee layer in the syntax of the imperative which is absent in the other moods. On the basis of this discussion, we propose a first structure for the present imperative endings. After that, in section 3, we move on to discussing how the imperative also differs from the other moods in that it is impoverished for both tense and ϕ . We propose the functional sequence for present imperatives and demonstrate how it deviates at least from the structure of the indicative. A comparison with the subjunctive is postponed till future research. Finally, in section 4, we provide some necessary prerequisites about Nanosyntax, and proceed with a detailed nanosyntactic derivation of the singular and plural present imperative forms. In the Conclusion, we summarise and conclude the findings of this paper.

2 Evidence for a speaker-addressee split

So far, we have assumed that the endings of the imperative, i.e. the $\emptyset$ in the singular and *-te* in the plural, correspond to ϕ . In comparison with the indicative and subjunctive, which had *-s* and *-tis* in the same positions (cf. (1)), this seemed a logical assumption. However, Latin also has a future imperative paradigm, shown alongside the present one in (2), which seems to call this hypothesis into question because of its ordering of morphemes.²

	pres.imp	fut.imp
(2) 2sg	am-ā	am-ā-tō
2pl	am-ā-te	am-ā-tō-te
3sg		am-ā-tō
3pl		am-a-n-tō

Firstly, note that the future imperative is marked throughout by the morpheme *-to*. This morpheme has been identified as a tense marker (Pinkster 2015: 515).³ Secondly, note that the endings in the second person, which again resemble ϕ -endings, are identical to those of the

²The use of the future imperative forms is restricted to certain genres (e.g. plays, legal/didactic texts). Usually, however, these imperatives refer to a time further away than the present time, or to a time when a certain condition is met. For that reason, they often co-occur with time adverbs which further specify the time frame within which the command applies (see Allen & Greenough 1903, Barrios-Lech 2017, Decorte 2016, Pinkster 2015, and Rosén 1999 for more details on the future imperative).

³Originally, this morpheme stems from an archaic ablative of a demonstrative pronoun **-tod*. An imperative like *amātō* would thus have meant something like ‘love, from

present imperative: no marker follows *-to* in the singular, but *-te* follows *-to* in the plural. These two facts are still within expectations. If we recall, for instance, the future indicative forms *amā-bi-s* and *amā-bi-tis* in table 1, we also found the same order of morphemes: stem > fut.tense > ϕ . What complicates the picture, however, is the third person of the future imperative.

Unlike the second person imperative, the third person is a type of imperative in which the subject and the person or people addressed by the command do not necessarily overlap (Pinkster 2015: 512). As is illustrated in (3), the sentence in (3a) is uttered in a kind of letter-correspondence, and in the one in (3b) it is clear that the verb form would be used in a legislative text. In both circumstances, the command is directed to a certain target audience (e.g. the person receiving the letter, or the people listening/reading the law), which does not overlap with the person or people which are affected by the command (e.g. the wife, or the people to whom the law applies).

- (3) a. Uxor mea heres <ne> **es-to** . . .
 wife my heir NEG be.IMP.FUT.3SG
 ‘Let my wife not be my heir . . .’
 (Sen. Con. 2.7.9)
- b. Primum verbi genus hoc “**conserva-nto**,” quo
 firstly of-word form this preserve.IMP.FUT.3PL which
 magis in legibus quam in foederibus uti
 rather in laws than in treaties use.INF
 solemus, imperantis est, non
 be-used-to.IND.PRES.1PL order.PART.PRES AUX.3SG NEG
 precantis.
 ask.PART.PRES
 ‘In the first place, this form of words, “Let them uphold,”
 which we are in the habit of using in laws rather than in
 treaties, implies a command, not an entreaty.’
 (Cic, Pro Balbo, 36)

Crucially, the third person forms also show two morphological twists: the singular form seems syncretic with the second person singular, and the plural has an unexpected morpheme *-n* preceding the tense marker *-to*. However, it has been argued by Szexnerényi (1953) that the traditional segmentation in (3) of these forms is incorrect. Instead of *amā-tō* and *ama-ntō*, these third person forms are rather degeminated

that point on’ in the spatial sense (cf. Barrios-Lech 2017, de Melo 2007, W.Fortson IV 2011).

forms whose underlying form corresponds to *amā-t-tō* and *ama-nt-tō*. In these original forms, we can clearly recognise the regular third person ϕ -morphemes *-t* and *-nt* that are found throughout the active verbal paradigm, as can be verified in table 2 below.

	PRES.IND	PST.IND	FUT.IND	PRF.IND	PLUPRF.IND	FUT.PRF
2SG	ama- t	amā-ba- t	amā-bi- t	amā-v-i- t	amā-v-era- t	amā-v-eri- t
2PL	ama- nt	amā-ba- nt	amā-bu- nt	amā-v-ēru- nt	amā-v-era- nt	amā-v-eri- nt
	PRES.SBJV	PST.SBJV		PRF.SBJV	PLUPRF.SBJV	
2SG	am-e- t	amā-re- t		amā-v-eri- t	amā-vi-sse- t	
2PL	am-e- nt	amā-re- nt		amā-v-eru- nt	amā-vi-sse- nt	

Table 2

If correct, it would thus seem that the 3SG is actually not syncretic with the 2SG, because it has an additional ϕ morpheme *-t*, and that the morpheme *-n* is a residue of the ϕ -morpheme, *-nt*. With this in mind, we can see that the regular ϕ -markers precede the future tense imperative marker *-to* in the third person, whereas the special morpheme of the second person, *-te*, follows it. This suggests that what we assumed to be ϕ in the second person imperatives, i.e. the zero marking in the singular and *-te* in the plural, is likely marking something else. If *-te* had really been a ϕ -morpheme, then any other piece of ϕ -morphology should have followed *-to* as well. This means that we would have expected to encounter forms like **ama-to-t* or **ama-to-nt* for the third person. Contrary to fact. Similarly, we can also wonder whether *-to* is only a future tense marker, like for instance *bi-* in the indicative. If it was, then we would also have expected all ϕ -morphology to follow it, but this is also not the case, given *ama-nt-to*.

If our reasoning is on the right track, then we are now one step closer to understanding the hierarchy of special imperative markers in Latin. While the future imperative morpheme *-to* can be stacked on top of the regular ϕ -endings, represented by *-n(t)-*, the indicative tense morphemes, like the future indicative *bi/bu*, sit below them. In addition, the second person plural *-te* (and the second person singular $\emptyset$) can be stacked on top of the future imperative morpheme. This leads us to the hierarchy in (4), which confirms the intuition that the endings we see in the present (and future) imperative are not regular ϕ -endings.

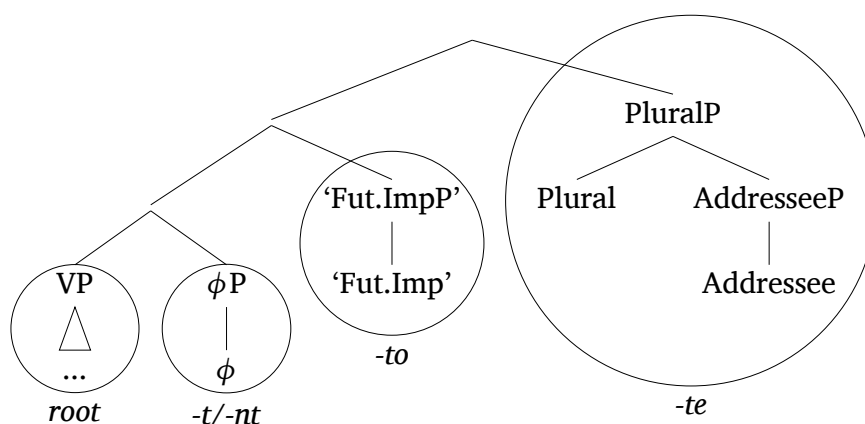
- (4) tense > -n(t)- > fut.imp > te/ $\emptyset$

Furthermore, we already noted that the special 2SG and 2PL imperative endings, $\emptyset$ and *-te* respectively, are completely absent anywhere else (cf.

table 1), but now, we see that the situation for 3SG and 3PL is rather the opposite; any dedicated imperative ending is absent, and the regular ϕ -endings appear instead. This is an interesting difference, because it tells us something about the morphosyntax of two pragmatically different situations. A second person imperative is used in a situation in which the subject of the event, i.e. the one who is executing a particular action, and the Addressee, i.e. the person addressed by the command, at least partially overlap. A comparable example in English would be a command as ‘Go to school, boys!’ in which the boys are also the ones who need to go to school. In a third person imperative this requirement of overlap between Subject and Addressee seems absent. The type of situation we are dealing with is typically one in which the person executing the event is different from the Addressee to whom the command is uttered. There is no straightforward way to express this in English, but consider a command such as ‘Let her go to bed by 8pm!’ in which the person going to bed is not the person who needs to make sure she does.

The morphology of the imperatives in Latin thus shows that a distinction should be made between the morphology used to refer to the subject of an event (cf. the regular ϕ markers), and the morphology used to refer to the Addressee, i.e. to the one who is meant to act upon the command they receive (cf. the special *-te* in IMP.PRES.2PL). To put it in more technical terms, the morphology of both imperatives in Latin suggests that a Speech Act layer (Ross 1970, Speas & Tenny 2003, Haegeman & Hill 2013) may get activated in the imperative when the subject of the event and Addressee overlap, an idea also proposed by Zanuttini (2008) and Isac (2015). The tree in (5) informally shows how the various morphemes of the future imperative would maximally lexicalise all the parts of the structure.

(5)



To summarise, what the morphology of the future imperative clearly shows is that there are Speech Act features, which sit higher in the structure than the features of regular ϕ , and that they are activated when the subject and the Addressee necessarily overlap, a situation that arises solely in the second person imperative. When the subject and the Addressee deviate, only regular ϕ morphemes are present. While not as morphologically conspicuous, we will propose that this Speech Act layer, which sits high in the left periphery, also gets activated in the second person forms of the present imperative. In 2PL this Speech Act layer will be lexicalised by *-te*, while in 2SG a zero morpheme will lexicalise this layer - or as is common in Nanosyntax, the theme vowel will lexicalise this layer thanks to phrasal lexicalisation.

Before we work out the details for our proposal, we will first discuss the remainder of the structure of present imperatives in the next section.

3 The structure of the FSEQ in present imperatives

From our detour to the future imperative in section 2, we deduced the structural hierarchy in (4), here repeated in the table in (6) with the corresponding morphemes.

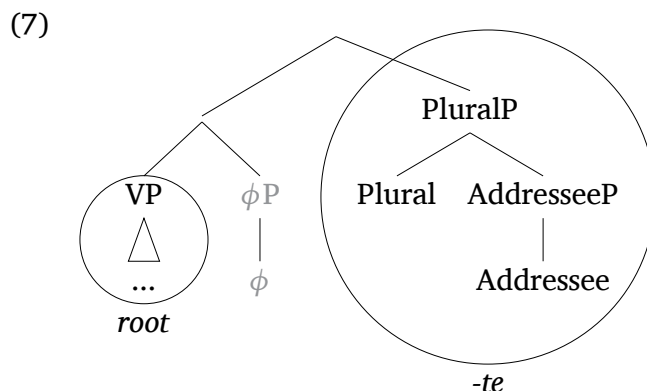
	VP	ϕ	'Fut'	Addr
(6) 2sg	amā	∅	-to	∅
2pl	amā	∅	-tō	-te
3sg	amā	-t	-tō	∅
3pl	amā	-nt	-tō	∅

From the overall picture in (6), we concluded that the future imperative shows that the endings of the second person imperatives, i.e. $\emptyset$ and *-te*, cannot be treated as regular ϕ -morphology, since they sit in a different position from the third person ϕ -morphology, i.e. *-t* and *-nt*. We subsequently argued that $\emptyset$ and *-te* are lexicalising a Speech Act layer, in particular an Addressee layer.

However, another important point which the table in (6) shows is that the realisation of this Speech Act layer and ϕ are in complementary distribution: when the subject and the addressee of the imperative overlap, which is the case for the second person, the Speech Act layer can be lexicalised but not the ϕ -layer, and when the subject and the addressee do not overlap, the ϕ -layer can be lexicalised but not the Speech Act layer.

This point brings us to another important ingredient of our proposal of the present imperative, namely impoverishment. What we seem to be able to conclude from the complementary distribution is that, when the subject and addressee overlap - which is the default situation for the present imperatives - the features realising ϕ are absent from the imperative structure, i.e. they will not be merged in syntax. This idea is not so far-fetched in that it stays with the rather long-standing intuition that so-called ‘true’ imperatives, i.e. unique verb forms that are specifically used to express commands, are morphologically defective or impoverished (see for instance Zanuttini 1991, 1994, 1997, Romanello & Repetti 2014, Isac 2015).

Considering present imperatives do not contain the future imperative morpheme *-to*, the structure in (5) is thus updated as follows, (7). The absent feature is marked in light grey.



However, in accordance with the literature, we also follow the claim that these ‘true’ imperatives are impoverished for Tense (see for instance Zanuttini 1991, 1994, Platzack & Rosengren 1998), because imperatives in general do not have a temporal (past) interpretation. The time of evaluation in imperatives is the speech time, as is illustrated by the example in (8).⁴

⁴It is debatable whether this general claim holds for future imperatives, even though they could technically be considered as ‘true’ imperatives, following (Romanello & Repetti 2014)’s criteria. While a near future interpretation is intrinsic in all imperatives, the future imperative interprets the command as having to be met in the further future, hence really adding a temporal interpretation to the command. We would like to argue that the future imperative does not lack Tense, and hence differs from the prototypical present imperative in this sense. This, however, may raise some questions with respect to the nature of *-to*. For if we assume that Tense sits below ϕ , as in other tensed forms as the third person future *ama-bi-t* or past *ama-ba-t*, then it seems unlikely that *-to* is the morpheme which will lexicalise Tense, seeing it sits higher than ϕ (cf. *ama-t-to*). A way out of this issue may be to consider *-to* the lexicalisation of some type of modal

- (8) a. Do try this!
 b. *Did try this!
 (Isac 2015: 51)

The idea that present imperative structures in Latin do not contain a Tense feature is also supported by the behaviour of negation when it interacts with the imperative. To begin with, the standard negator *non* cannot be combined with the present imperative, as is shown in (9). Since many languages have negative markers that sit in a position above tense, and for which it has been argued that its position is parasitic on the TP it selects (Zanuttini 1991, 1994), the absence of Tense could easily explain the concomitant absence of the standard negator.

- (9) 2SG : (**nōn*) *amā*!
 2PL : (**nōn*) *amāte*!

Additionally, the standard negator is compatible with the present indicative, (10a), giving rise to ‘neutral’ sentential negation, and with the present subjunctive, (10b), giving rise to modal negation such as negative potentialis constructions or counterfactuals. This again points to the fact that the present imperative lacks Tense, whereas the other moods do not, thus enabling the negator to select them, but not the imperative.

- (10) a. *Leporem et gallinam et anserem gustare fas*
 hare and hen and goose taste divine-law
 non putant
 NEG consider.PRES.3PL
 ‘The tasting of hare, hen and goose they do not consider the law.’
 (Caes, Gall. 5.12)
- b. *Hunc tu non ames?*
 this you NEG love.SUBJ.PRES.2SG
 ‘Would you not love such a man?’
 (Cic. Att. 4.19)

Another important point to mention here is that the other Latin negator, *nē*, which has been identified as a modal negator (cf. Lakey 2015, Baunaz & Lander 2023), is also incompatible with the present imperative, (11).

necessity, as was pointed out to us by an anonymous reviewer. We hope to return to this issue in future research.

- (11) 2SG : (**nē*) *amā*!
 2PL : (**nē*) *amāte*!

However, this negator is mainly used to form prohibitive constructions, with the subjunctive (in either the present or the perfect tense) not with the imperative, as is illustrated in (12). Just as with *nōn*, this indicates the presence of a Tense feature in the subjunctive, which is absent from the present imperative.

- (12) a. *Quam ob rem . . . solem alterum nē metueritis.*
 what to thing . . . sun second NEG fear.SUBJ.PERF.2PL
 ‘For which reason ... do not fear the second sun’
 (Cic. Rep. 1.32, in Pinkster 2015: 501)
- b. *Hanc pete: ne metuās fastus*
 this search NEG fear.SUBJ.PRES.2SG arrogance
limenque superbum
 threshold = and haughty
 ‘Do not fear arrogance and a haughty threshold!’
 (Mart. Ep. 1.70.13)

Moreover, the future imperative is compatible with *nē*, unlike the present imperative, (13).⁵

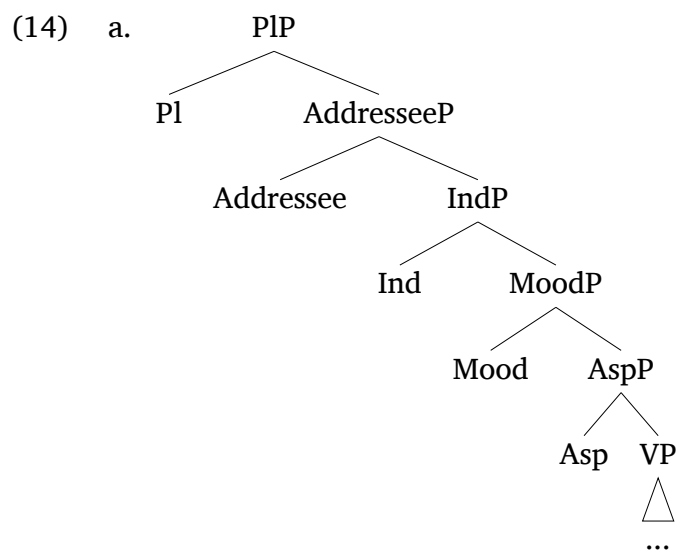
- (13) *Boreā flante, nē arātō,*
 north-wind blow.PART.PRES.ABLE NEG plough.IMP.FUT.2SG
sēmen nē iacitō
 seed NEG sow.IMP.FUT.2SG
 ‘When the north wind blows, plough not nor sow your seed.’
 (Plin. H. N. 18.334)

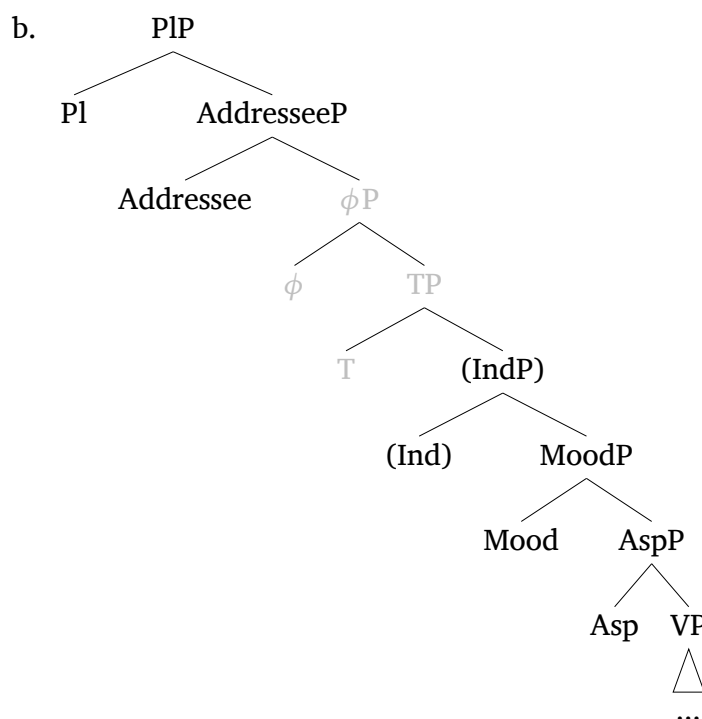
What the data in this section thus underline is that there is a fundamental structural difference between the present imperative and the indicative/subjunctive on the one hand, but also between the present and future imperative. This difference is interpreted in terms of deficiency: present imperatives are morphologically deficient in such a way that (i)

⁵It should be pointed out that the standard negator *nōn* is also incompatible with the future imperative, even though we assume this is a tensed imperative form. The reason for this incompatibility is still unclear at the moment, but we assume it must be due to the presence or absence of at least one other feature which disrupts the selection between the negator and the verb. A more fine-grained analysis of both elements is required to solve this issue, but since the main focus of this article is neither on the future imperative nor on negation, we leave this to future research.

they cannot lexicalise ϕ -features and (ii) that neither of the negators is compatible with them because of a lack of Tense, i.e. ϕ and Tense features are not merged, and there is no space for a NEGP in the inflectional or discourse related part of the clausal structure. Other moods and the future imperative clearly can project both.

In accordance with this discussion, we propose that the final functional sequence for the present imperative is as in (14a). The structure in (14b) shows the layers, again in light gray, that we assume to be present in the indicative and the subjunctive, but which are missing from the imperative. However, there is not only impoverishment in the imperative. There is also enrichment, as the Speech Act layer gets activated (i.e. Addressee with an additional, privative plural), a layer that is absent in indicatives or subjunctives. Note that, following the verbal structures in Starke & Cortiula (2021), we have added a few features which have not been mentioned previously: default Aspect, default Mood and Ind(icative). In line with their proposal, we also consider the indicative feature to be the element which minimally distinguishes the indicative mood from the subjunctive mood (hence why it is put between brackets in 14b).





Now that we have introduced the main elements which separate present imperatives from present indicatives/subjunctives, i.e. the addition of the Speech Act layer and the omission of ϕ and Tense, we proceed in section 4 with the derivation of the 2SG and 2PL of the present imperative within the Nanosyntactic framework.

4 Analysis

4.1 Nanosyntax

In this section, we will present a Nanosyntactic analysis (cf. Caha 2009, Starke 2009) for the present imperative forms of the first class verb *amāre*. Before we proceed to the derivations, we need to briefly say something about Nanosyntax, the theoretical framework we are using.

Nanosyntax is a Late-Insertion theory, which lexicalises syntactic structures after each step of merge by means of Phrasal Lexicalisation. After each merge, syntax interfaces with the lexicon to check whether the merged structure can be matched against an item in the lexicon. The lexicon contains lexical items that consist of a lexical tree structure, phonological and conceptual information. If there is a lexical item in the lexicon that matches the syntactic structure, then lexicalisation is successful and syntax proceeds to merge other features. If lexicalisation

is unsuccessful, syntax has to perform some rescue operations, which are referred to as lexicalisation-driven movements. These movements happen in a specific order, according to an algorithm. We rely on Starke (2018)'s definition of the algorithm, reproduced here in (15).

- (15) Merge-F and
- a. Spell out FP
 - b. If (a) fails, move the spec of the complement of F, and retry (a)
 - c. If (b) fails, undo spec movement, move the complement of F, and retry (a)
 - d. If (c) also fails, attempt backtracking to the previous cycle and try next option for that cycle.
 - e. If merge-F has failed to spell out (even after backtracking), try to spawn a new derivation providing feature X and merge that with the current derivation, projecting feature X to the top node.

The matching between the lexical structure and the syntax happens in one of two ways: either there is a 1:1 relationship between the syntactic and lexical structure, or the match occurs in accordance with two principles, the Superset Principle, (16) and the Elsewhere Principle, (17) Baunaz & Lander (2018).

(16) *Superset Principle*

A lexical tree L can match a syntactic tree S if L is a superset (proper or not) of S. L matches S if L contains a node that is identical to a node in S and all the nodes below are also identical

(17) *Elsewhere Principle*

If more than one L- tree can lexicalize the same S- tree (by the Superset Principle), then the L- tree with the least amount of superfluous material is chosen.

To put it informally, the Superset Principle in (16) states that a lexical structure forms a match with a syntactic structure as long as the latter is a subpart of it. However, this principle on its own is not sufficiently constrained for situations where multiple lexical items of varying sizes 'compete' to lexicalise the same tree. For this purpose, the Elsewhere Principle in (17) exists. It states that when there are several candidates for lexicalisation, the smallest one will take precedence over the others.

With these basics about Nanosyntax in mind, we can move on to a more detailed derivation of the present imperative forms, in which the Nanosyntactic principles discussed here will be illustrated.

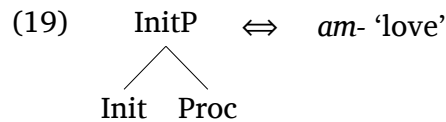
4.2 Analysis of the present imperative

4.2.1 Lexical items

If we recall the present imperative paradigm, (18), there are essentially three morphemes involved: the root *am-*, the theme vowel *-ā*, and the plural Addressee morpheme *-te*.

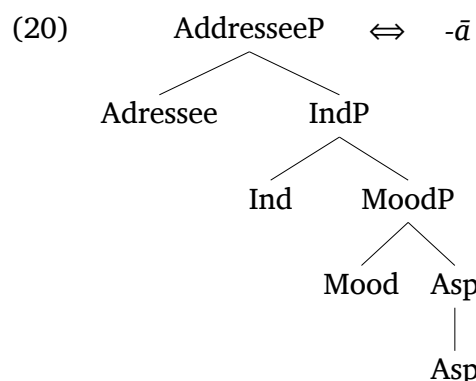
		PRES.IMP
(18)	2SG	<i>am-ā-∅</i>
	2PL	<i>am-ā-te</i>

Starting with the lexical structure of the root of the verb *amāre*, we assume that it has the shape in (19).⁶ For the features of the verbal root, we adopt Ramchand (2008)’s decomposition of verbal predicates into INIT (Initiation), i.e. the causation event, PROC (Process), i.e. the event or process itself, and RES (Result), i.e. the result state of the event. As can be seen in (19), we assume that the root *am-* lexicalises PROC and INIT, but not RES.



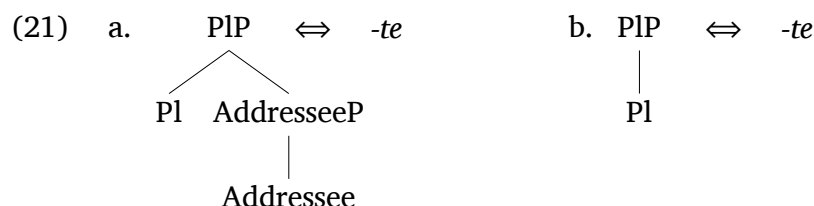
Following the root *am-* is the theme vowel of the first verb class, i.e. *-ā*. We propose that it has a lexical structure as in (20), and is capable of phrasally lexicalising the default viewpoint Aspect feature (ASP), the default Mood feature (MOOD), the Indicative feature (IND) and the Speech Act related feature, ADDRESSEE (Ross 1970, Speas & Tenny 2003, Hill 2013, Haegeman & Hill 2013). This way we can already account for 2SG *am-ā*.

⁶As mentioned before, lexical items in Nanosyntax are stored with their phonology, conceptual information and lexical structure. In this paper we represent the lexical structure and the corresponding orthography. We leave out conceptual information and phonological representations.



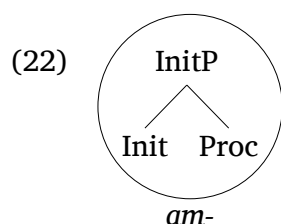
The lexical structure of the theme vowel $-\bar{a}$ as it is drawn in (20), seems to suggest that we claim that the indicative is contained in the imperative. This may sound like a controversial claim because imperatives are usually considered to be a different Mood, hence making it unlikely that the indicative mood is a feature of the imperative mood. However, in line with McGinn (1977), we would like to argue that the core of imperatives and indicatives is the same. The way in which they are distinct is that indicatives always need a Tense layer and a ϕ layer, while true imperatives do not have such a requirement. For this reason, the INDICATIVE feature, taken from Starke & Cortiula (2021), should actually be viewed as a placeholder for a component of the indicative mood rather than a sole feature constituting it. The indicative is thus a mood realised by several features amongst which we need at least default Mood, Indicative, and default Tense. Since this Tense layer is absent in the structure of the present imperative, the indicative mood will not arise, and vice versa. The present imperative requires the features illustrated for the theme vowel in (20): default Mood, Indicative, and Addressee.

The last lexical item we need to determine the structure of is the plural morpheme $-te$. In theory, there are two options regarding its lexical structure: it can either have either have the structure in (21a) or the one in (21b). However, given that Plural distinguishes a singular Addressee from a plural one, we will assume that Plural and Addressee are lexicalised together by the same morpheme, and adopt the structure in (21a).

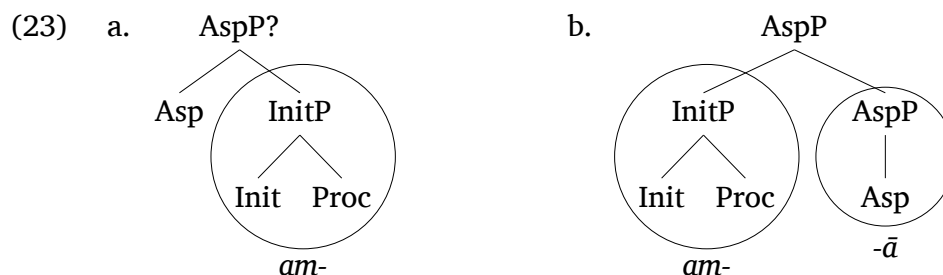


4.2.2 Derivation

Turning now to the derivation, syntax will start by merging the features relevant for the verbal lexical structure, i.e. INIT and PROC (Ramchand (2008)). At the level of INITP, the lexicon will have many possible candidates to be inserted, and as we are dealing with *amāre* ('love'), the item *am-* will be chosen, (22). From then on, this is the root that will be maintained in line with the principles of Free Choice and Faithfulness discussed in Caha, De Clercq & Vanden Wyngaerd (2019). The principle of Free Choice allows for free selection of one particular lexical item for insertion when a lot of different items are in competition to lexicalise a particular node, as is the case for roots. The principle of Faithfulness imposes to stay faithful to that choice (i.e. override with another conceptually unrelated lexical item is disallowed) until a movement happens and a new lexical item needs to be inserted.

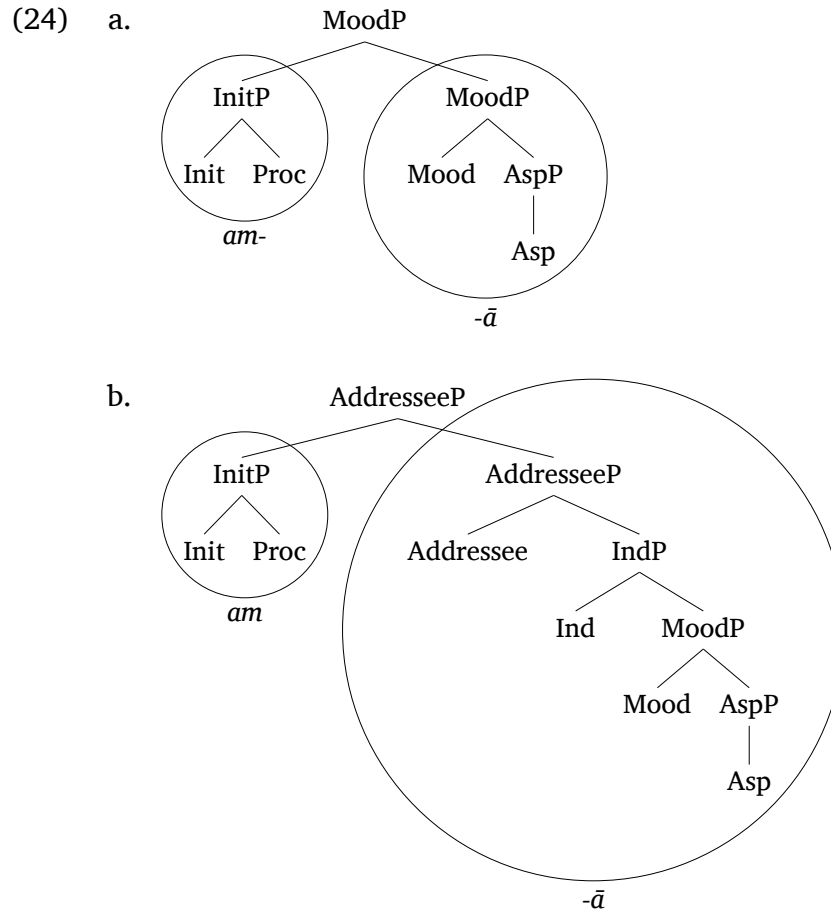


In a next step, syntax will merge default ASPECT, ASP, (23a). Since the root has the lexical structure in (19), it cannot lexicalise the structure syntax has constructed, compelling syntax to execute rescue movements in the order that is prescribed by the algorithm (cf. (15)). As there is no specifier yet, (cf. the first movement step in the algorithm), the complement will be moved out, and ASP will be lexicalised by the theme vowel, as shown in (23b).



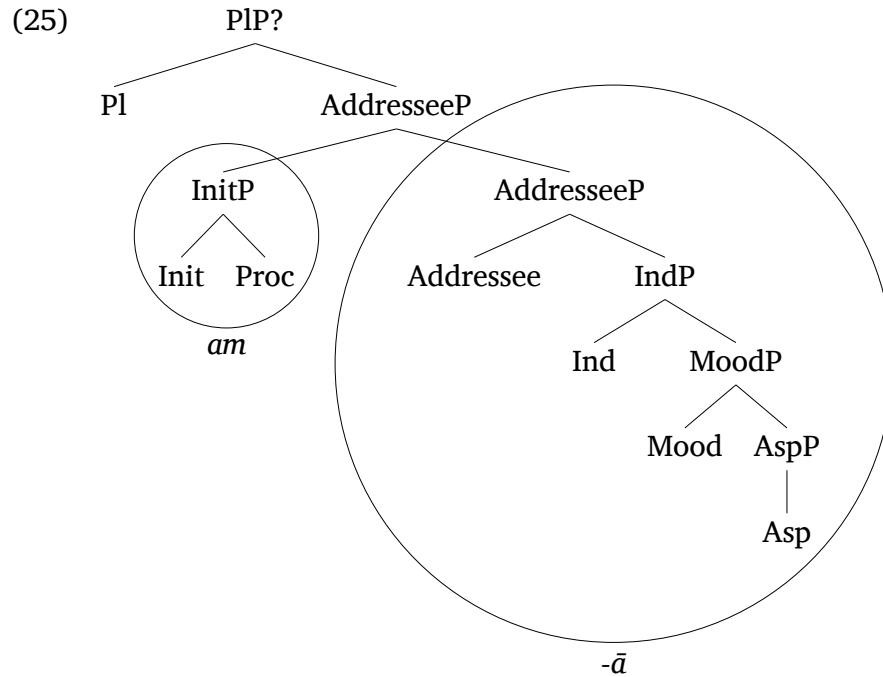
The next feature in line, default MOOD, will then be merged, resulting again in the lexicalisation (and override) of the theme vowel after performing spec movement (15b), (24a). After MOOD, the INDICATIVE fea-

ture will be merged and the same rescue operation will lead to the lexicalisation by means of *-a*. When syntax derives a present imperative, the feature following the Indicative will not be Tense, but will be ADDRESSEE. By merging this feature, the Tense and ϕ (and negation) layers are thus skipped, as was illustrated in (14b). ADDRESSEE can be realised by the same lexical item in (20), leading to the output *amā* for 2SG and to the structure in (24b). One important thing to mention here is that thanks to phrasal lexicalisation, there is no need not adopt a $\emptyset$ -ending for the 2SG imperative as we have done so far for illustrative purposes: the lexical item of the theme vowel can be argued to take care of lexicalising ADDRESSEE.



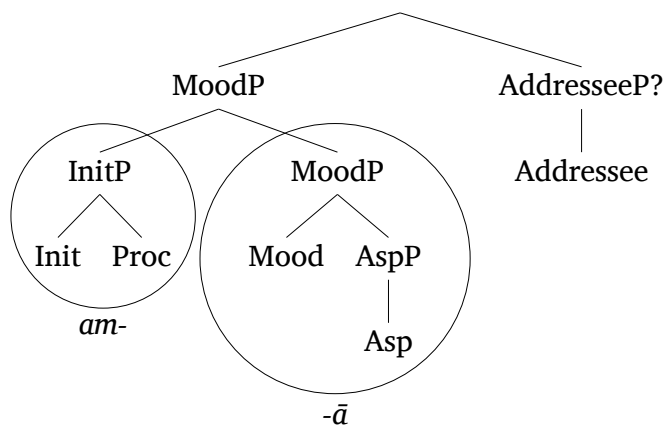
In order to lexicalise the plural imperative, which contains the plural Addressee feature, the derivation will involve a procedure called backtracking (cf. Starke 2018). This step is mentioned in step (15d) of the algorithm which we introduced in section 4.1. At the point when the structure in (24b) is merged, syntax will continue merging the feature

PLURAL, when a plural addressee is aimed for, leading to the derivation in (25).



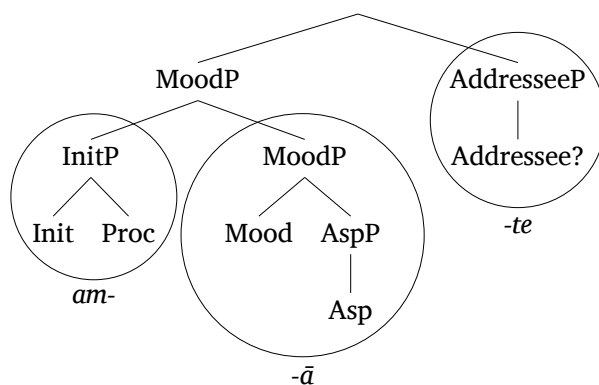
However, at this point, no lexical item will be found that can lexicalise the derived structure, which means that different movements must be tried again. Yet, neither spec movement, (15b) nor complement movement, (15c) will lead to a successful lexicalisation. The next step in the algorithm is backtracking, (15d). By backtracking the last merge of Plural and the previous lexicalisation of ADDRESSEE with the theme vowel are undone first, upon which a different step in the algorithm is tried for ADDRESSEE. Since the realisation of *am-ā* involved spec movement, this time complement movement will be tried, leading to the structure in (26).

(26)

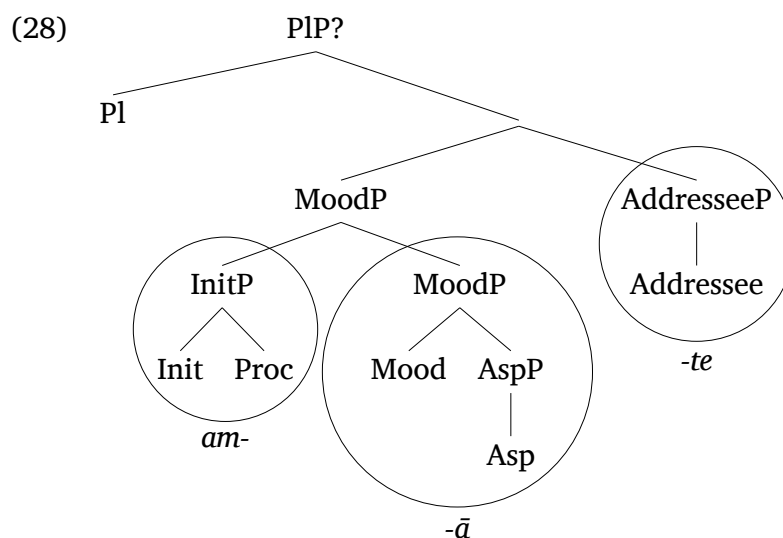


For the current structure, only (21a) is a possible candidate for lexicalisation of ADDRESSEE, (27).

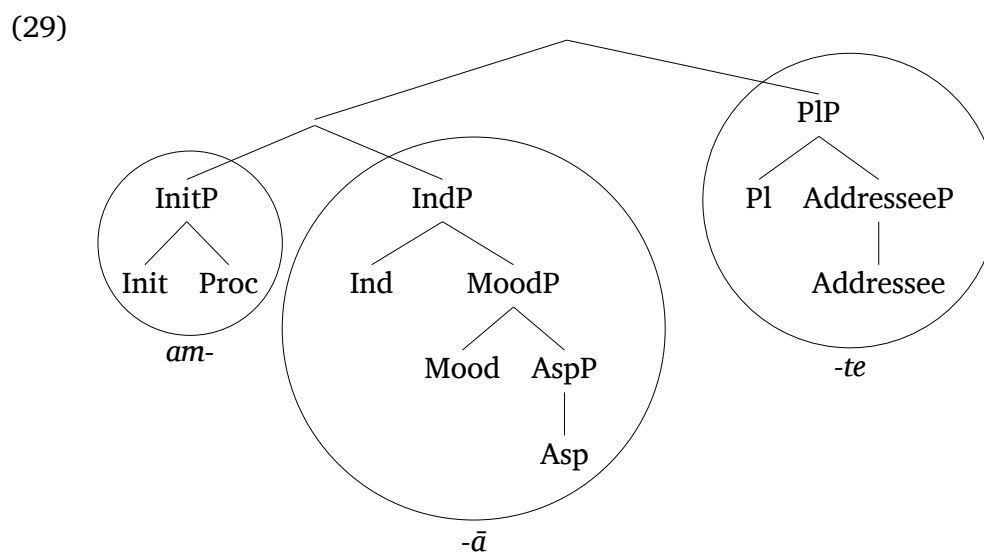
(27)



Now that ADDRESSEE is lexicalised, the next feature in line, PLURAL, is merged again. The structure looks as in (28).



PLURAL cannot be lexicalised in this position. Spec movement is tried, leading to the structure in (29). PLURAL can now be lexicalised together with ADDRESSEE by means of the lexical item in (21a), yielding the output *am-ā-te*, the second person plural of the present imperative.



We have now shown how we can derive both the singular and the plural form of the imperative within Nanosyntax.

5 Conclusion

In this paper we had a closer look at the present imperative endings, *-te*, and $\emptyset$ in Latin. On the basis of a comparison with the future imperative, we argued that the dedicated imperative endings are in a different position from the regular ϕ -morphology, and that they only show up when the subject of the event and the Addressee overlap. We proposed that the present imperative in Latin is structurally impoverished in lacking ϕ and Tense layers, but enriched in projecting a Speech Act layer, Addressee (and Plural). We provided Nanosyntactic derivations for the 2SG and 2PL of the present imperative of the first verb class *amāre*.

Acknowledgements

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8 The Fine structure of the comparative

P. Caha, K. De Clercq and G. Vanden Wyngaerd

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THE FINE STRUCTURE OF THE COMPARATIVE

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Abstract The paper provides evidence for a more articulated structure of the comparative as compared with the one in [Bobaljik \(2012\)](#). We propose to split up Bobaljik's CMPR head into two distinct heads, C1 and C2. Looking at Czech, Old Church Slavonic and English, we show that this proposal explains a range of facts about suppletion and allomorphy. A crucial ingredient of our analysis is the claim that ad-

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jectival roots are not a-categorial, but spell out adjectival functional structure. Specifically, we argue that adjectival roots come in various types, differing in the amount of functional structure they spell out. In order to correctly model the competition between roots, we further introduce a Faithfulness Restriction on Cyclic Override, which allows us to dispense with the Elsewhere Principle.

1. Introduction

The seminal study of Bobaljik (2012) has put the morphosyntax of degree comparison firmly on the research agenda. Central to his claims is the Containment Hypothesis (Bobaljik 2012:4):

(1) *Containment Hypothesis*

The representation of the superlative properly contains that of the comparative.

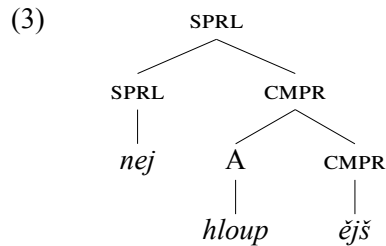
A language like Czech provides overt morphological evidence for this hypothesis, in that the marker of the comparative (–ějš) also shows up in the superlative, which adds the prefix *nej–* to the comparative form:¹

¹ The final vowels in these forms are concord markers. We ignore them in what follows.

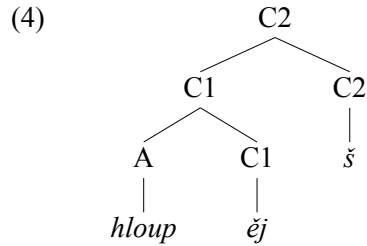
(2)

POS	CMPR	SPRL	
mil-ý	mil-ejš-í	nej-mil-ejš-í	‘nice/kind’
červen-ý	červen-ějš-í	nej-červen-ějš-í	‘red’
hloup-ý	hloup-ějš-í	nej-hloup-ějš-í	‘stupid’
bujar-ý	bujar-ějš-í	nej-bujar-ějš-í	‘merry’

The syntax Bobaljik proposes is accordingly as in (3). The crucial part of the tree is the containment relation between the degrees, such that the superlative contains the comparative, which in turn contains the positive adjective (A). The direction of branching (and the corresponding order of the heads) is subject to cross-linguistic variation, adjusted below to fit the Czech pattern of *nej-hloup-ějš-í* ‘most stupid’:



In this paper, we want to take this proposal a step further and propose (following in part Bopp 1833, as well as Caha 2017a;b and De Clercq & Vanden Wyngaerd 2017) that the CMPR head in the above tree needs to be split up into two distinct heads, as shown in (4), with *-ěj* spelling out C1 and *-š* spelling out C2:



In section 2, we present evidence for this structure coming from the allomorphy of the comparative markers in Czech. In section 3, we present the details of our theoretical framework (Nanosyntax). In sections 4 and 5, we extend the analysis developed on the basis of the Czech to data from Old Church Slavonic and English. In section 6, we turn to suppletion in Czech and English, and show how the bi-partite structure explains an interaction between the allomorphs of the comparative and root suppletion. In section 7, we argue that the Elsewhere Principle should be abandoned, and its effects captured by the Faithfulness Restriction on spellout.

2. Allomorphy of the Czech comparative

2.1. The three allomorphs

As we have shown in (2) above, the Czech comparative can be formed by suffixing *–ějš* to the root. We give a couple more examples below.

(5)	POS	CMPR	palatalisation		gloss
	stroh-ý	strož-ejš-í	[fɪ → ʒ]	(velar)	‘austere’
	divok-ý	divoč-ejš-í	[k → tʃ]	(velar)	‘wild’
	kulat-ý	kulat-ějš-í	[t → c]	(alveolar)	‘round’
	bujar-ý	bujař-ejš-í	[r → ɹ]	(alveolar)	‘merry’
	benign-í	benign-ějš-í	[n → ɲ]	(alveolar)	‘benign’
	hloup-ý	hloup-ějš-í	[p → pj]	(labial)	‘stupid’
	blb-ý	blb-ějš-í	[b → bj]	(labial)	‘stupid’

The suffix *-ějš* is the only productive comparative marker and newly coined gradable adjectives follow this pattern (see ‘benign’ for an example).

To get a full picture, it is important to take note of the consonant mutations (in effect, palatalisations) that affect the base to which *-ějš* attaches. Palatalisation affects the final consonant in a way that is shown in the last column of the table. The palatalisations in (5) are not an automatic response of the base to the following phoneme [ɛ], as there are [ɛ]-initial suffixes which do not trigger palatalisation. Hence, it is an arbitrary property of the vowel/suffix that it triggers palatalisation, and so this needs to be somehow encoded in the lexicon. The traditional spelling of the *-ějš* suffix reflects this by placing a diacritic over the vowel: *ě* is an *e* which triggers the palatalisation of the preceding con-

sonant. As we shall see below, there are different types of palatalisation triggers in Czech, with different properties. For the purposes of the discussion below, we will note the palatalisation trigger in the *–ějš* suffix as PAL_{VAL} , i.e. $\text{ějš} = \text{PAL}_{\text{VAL}}\text{ějš}$; the subscript indicates that palatalisation affects velars, alveolars, and labials. Such a notation serves the purpose that it transforms the non-segmental marker into a segmental-like object that we can work with in the morphological analysis. At the same time, PAL is admittedly nothing more than a stand-in for an actual phonological analysis of how palatalisation works, which we deliberately leave aside. In the orthography, when *–ějš* is attached to the base, palatalisations are sometimes marked on the preceding consonant (as in *že, če, ře*), and sometimes on the vowel (in *tě, ně, pě, bě*), as appears from the table (5).

The next point in our description is the allomorphy of the comparative marker. Specifically, as the traditional descriptions recognise (Dokulil et al. 1986; Karlík et al. 1995; Osolsobě 2016), there are two more ways of forming comparatives. The first of these is to attach only *–š*, as in the forms below:

(6)	POS	CMPR	palatalisation		gloss
	drah-ý	draž-š-í	[ɸ → ʒ]	(velar)	‘expensive’
	tich-ý	tiš-š-í ²	[x → ʃ]	(velar)	‘silent’
	bohat-ý	bohat-š-í	–	(alveolar)	‘rich’
	star-ý	star-š-í	–	(alveolar)	‘old’
	slab-ý	slab-š-í	–	(labial)	‘weak’

It can be observed that –š also triggers palatalisations, but this process affects only velars, while alveolars and labials are unaffected. This suggests that if palatalisation is to be seen as an effect of juxtaposing the base and the comparative –š, then –š (like –ějš) must be accompanied by a PAL diacritic too. Importantly, since the palatalisations triggered by –š are of a different kind than those associated to –ějš, we will introduce a different palatalisation trigger for this case, namely PAL_V (for the palataliser of velars). As we shall show below, there is also a third type of palataliser in Czech, PAL_{VA}, which is found with the soft declension marker –í, and which palatalises velars and alveolars but not labials. This situation is summarised in (7).

- (7) a. PAL_{VAL} palatalises preceding velars, alveolars, and labials
(e.g. –ěj)

² Orthography requires that *tišší* ‘more silent’ is written with double š, but it is pronounced with a single š due to degemination.

- b. PAL_{VA} palatalises preceding velars and alveolars (e.g. $-i$)
- c. PAL_{V} palatalises preceding velars (e.g. $-\check{s}$, $-\check{s}t$)

This has also consequences for the morphological analysis of the palatalisations: given that the palatalisations are different for different allomorphs, we will not segment out ‘palatalisation’ as something that marks all comparatives, since there is no unique palatalisation. Rather, the two types of palatalisation that we see in the comparative suffixes are each firmly connected to the particular allomorph that gets attached to the base.

Note finally that a purely phonological account of the distribution of the allomorphs $-\check{e}j\check{s}$ and $-\check{s}$ seems unlikely. For example, the adjectival root *kulat-* ‘round’ from (6) is similar to *bohat-* ‘rich’ in (7), but they pattern differently. Similarly, *star-* ‘old’ in (7) ends in the same segments as *bujar-* ‘merry’ in (6), and yet they differ in allomorph selection. For this reason, we will treat $-\check{s}$ here as a morphologically conditioned allomorph of $-\check{e}j\check{s}$.

The third (and final) allomorph recognised in the traditional descriptions is a zero marker. Some examples are given in (8).

(8)	POS	CMPR	palatalisation	gloss
			[k → tʃ]	
	kluz-k-ý	kluz-k-ø-í	kluz-č-ø-í	‘slippery’
	leh-k-ý	leh-k-ø-í	leh-č-ø-í	‘light/easy’
	hez-k-ý	hez-k-ø-í	hez-č-ø-í	‘pretty’
	měk-k-ý	měk-k-ø-í	měk-č-ø-í	‘soft’
	ten-k-ý	ten-k-ø-í	ten-č-ø-í	‘thin’
	vlh-k-ý	vlh-k-ø-í	vlh-č-ø-í	‘wet’

In these representations, which represent the traditional analysis of these forms, the comparative marker is taken to be zero (indicated by the $\emptyset$ sign). The adjectives in question all have a final $-k$, a derivational suffix, which we will come back to in Section 2.4 below. In the comparative, $-k$ is followed by the soft agreement marker $-í$. The soft agreement marker is independently known to trigger palatalisation of velars ([k → tʃ]; see (7b) above). As a result, in the comparative $-k$ appears as $-č$.

The existence of a third, zero, allomorph (while not crucial) does have a role to play in our analysis, and so we need to consider a potential alternative to the traditional analysis. The alternative would be to reduce the pattern in (8) to suffixation by $-š$, i.e. the second allomorph (illustrated in (6) above). This alternative analysis is represented in (9).

(9)

POS	CMPR	palatalisation [k → tʃ]	degemination [tʃʃ → tʃ]
kluz-k-ý	kluz-k-š-í	kluz-č-š-í	kluz-č-í
leh-k-ý	leh-k-š-í	leh-č-š-í	leh-č-í
hez-k-ý	hez-k-š-í	hez-č-š-í	hez-č-í
měk-k-ý	měk-k-š-í	měk-č-š-í	měk-č-í
ten-k-ý	ten-k-š-í	ten-č-š-í	ten-č-í
vlh-k-ý	vlh-k-š-í	vlh-č-š-í	vlh-č-í

The idea is that the sequence *k-š* (in the second column) is subjected to phonological processes, ultimately producing the same result as the analysis where *-í* is attached directly to the adjective. Specifically, *-š* would first trigger the palatalisation of *k* as shown in the third column of (9). After that, the [tʃʃ] cluster is simplified to [tʃ] by degemination (see also footnote 2), yielding the surface forms as in the final column of (9). These are identical to the forms in (8), i.e. the two analyses yield the same output.

The phonological derivation in (9) is not purely hypothetical. Its existence is supported by a parallel derivation found in the case of locative adjectives. In the NOM.PL, these adjectives have the suffix *-št-í*. This suffix triggers no change in alveolar-final stems, as shown in the ‘Norwegian’ column of (10). This fact shows that the *-št* suffix be-

longs to the same class as the comparative suffix $-\check{s}$, in that it does not trigger palatalisation of alveolars. Like all palatalisation triggers, $-\check{s}t$ triggers the palatalisation of velars ($[\text{h}] \rightarrow \text{ʒ}$, $k \rightarrow \widehat{\text{tʃ}}$), as shown in the two rightmost columns of (10). When $-\check{s}t$ follows a $-k$ -final base, as in the ‘Turk(ish)’ column, palatalisation ($[k \rightarrow \widehat{\text{tʃ}}]$) is followed by degemination ($[\widehat{\text{tʃtʃ}} \rightarrow \widehat{\text{tʃ}}]$), yielding the $[\widehat{\text{tʃ}}]$ output from the underlying $k-\check{s}$.

(10)		‘Norwegian’	‘Prague’	‘Turk(ish)’
	NOUN	nor	Prah-a	turek
	ADJECTIVE	nor- $\check{s}t-i$	prah- $\check{s}t-i$	turek- $\check{s}t-i$
	palatalisation	–	pra $\check{z}$ - $\check{s}t-i$	ture $\check{c}$ - $\check{s}t-i$
	degemination	–	–	ture $\check{c}$ -t- i
	output	no[r ʃ]tí	pra[ʒ]tí	ture[$\widehat{\text{tʃ}}$]tí

The sequence of phonological processes we see at work in the final column of (10) is the same as the one that are postulated in the alternative analysis of the (superficially) zero-marked comparatives in (9).

Summing up, we see that zero marking of the comparative and $-\check{s}$ affixation yield the same surface forms for $-k$ final stems. The issue of the proper analysis of the comparative is therefore hard to decide (see [Scheer 2001:34](#), who also remains agnostic about which option should be chosen).

However, there are dialects with additional zero-marked forms, and these allow us to tease these analyses apart, supporting the zero-marking account. In order to understand the facts, let us first show what happens to a base to which the soft adjectival *-i* attaches. For instance, relational ‘kind-possessive’ adjectives are often derived from the underlying nouns by simply placing the soft declension marker *-i* after the root of the noun, as in (11).

(11)	NOUN	ADJECTIVE	palatalisation	gloss
	vlk	vlč- <i>i</i>	$[k \rightarrow \widehat{tʃ}]$ (velar)	‘wolf’
	vydr-a	vydř- <i>i</i>	$[r \rightarrow \text{ř}]$ (alveolar)	‘otter’
	sup	sup- <i>i</i>	– (labial)	‘vulture’

These adjectives are most naturally used in context where English would have compounds, like ‘a wolf track’ or ‘a vulture nest.’ In these contexts, they seem semantically not very different from nouns, which gives us some grounds to believe that we are actually looking at examples where a nominal root is directly followed by a soft agreement marker. Certainly, there is no suspicion here that the comparative *-š* may be involved. What is relevant for us are the palatalisations we observe in these examples. As the last column of the table (11) brings out, these are of the type PAL_{VA} : velars and alveolars are palatalised, while labials remain unaffected.

With this background in mind, let us now turn to the dialectal comparative forms, which (as we propose) feature a zero comparative marker. They come from the dialect of North-East Bohemia (Bachmannová 2007), and they are given in the table below, along with the standard forms. The difference between the two sets of forms is not a matter of a difference in the application of phonological processes, but a consequence of the roots belonging to different morphological classes. The standard form takes the *-ějš* allomorph, the dialectal forms take a different one.

(12)	St. Czech	N-E Bohemia	palatalisation	gloss
	POS	CMPR	CMPR	
	ostr-ý	ostř-ejš-í	ostř-ø-í	[r → ʀ] ‘sharp’
	mokr-ý	mokř-ejš-í	mokř-ø-í	[r → ʀ] ‘wet’

These examples are unproblematic for the idea that there is no overt comparative marker. As we saw in (11), the soft-declension marker *-í* belongs to the PAL_{VA} class, i.e. it triggers the palatalisation of both velars and alveolars, which is what we observe in (12).

On the other hand, the *-š* based analysis of zero marking cannot be applied to the dialectal forms. This is because the comparative suffix *-š* belongs to the PAL_{V} class, which does not trigger the palatalisation of alveolars, as shown in (6) above. A minimal contrast can be observed between a form like *star-š-í* ‘older’ (no palatalisation or *r* when followed

by $-\check{s}$), and the forms in (12), where r is palatalised. PAL_V is also attested with the $-\check{s}t$ suffix, discussed in (10) above. This suffix likewise does not trigger palatalisation of r , i.e. we get $[rʃ]$ in the output, as shown in (13).

(13)	NOUN	ADJECTIVE	palatalisation	gloss
	bratr	bratr- $\check{s}t$ - $\acute{i}$	–	‘brother’

If the underlying sequence of (12) were $ostr$ - $\check{s}$ - $\acute{i}$, we would expect $ostr$ - $\check{s}$ - $\acute{i}$, with non-palatalised r , to surface here. The dialectal forms thus seem best captured as a case where the zero suffix has a wider distribution as compared to the standard language, and we shall therefore assume that the traditional descriptions are correct in proposing the existence of a zero allomorph for the Standard Czech k -adjectives as well. At the same time, we also wish to point out here that the analysis of the k -adjectives that we develop only needs a minor modification if they are taken to instantiate the second allomorph ($-\check{s}$).

Summing up the traditional descriptions, the two non-productive allomorphs of the productive comparative marker $-\check{e}j\check{s}$ - $\acute{i}$ correspond to various degrees of its reduction, going from left to right, as (14) brings out:

- (14) a. $\check{e}j\check{s}$ - $\acute{i}$

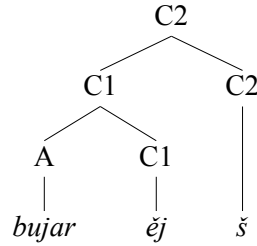
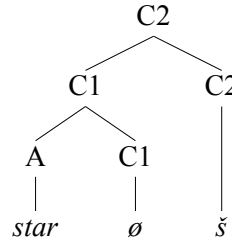
- b. *š-í*
- c. *-í*

One of the main goals of this paper is to present a theory that explains how such a reduction works and that makes some non-trivial predictions about how this type of reduction interacts with root suppletion.

2.2. *Decomposing the comparative*

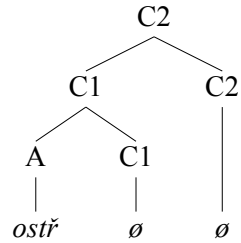
Moving now to the actual account, it is trivial to provide an analysis for the difference between the $-\emptyset$ class and the $-\check{s}$ class. This difference is perfectly analogous to pairs such as *sheep—sheep- $\emptyset$* vs. *heap—heap- s* : one (minor) class has a null marker where the other class has an overt marker. This simple logic can be extended to the difference between $-\check{s}-\acute{i}$ and $-\check{e}j\check{s}-\acute{i}$, provided we assume that the suffix $-\check{e}j\check{s}$ actually splits into $-\check{e}j$ and $-\check{s}$. If that is so, the $-\check{e}j\check{s}$ class becomes an $-\check{e}j-\check{s}$ class, and the $-\check{s}$ class then becomes a $-\emptyset-\check{s}$ class, with the morpheme $-\check{s}$ shared by both forms. In structural terms, this means that comparatives have two heads, C1 and C2, where the $-\check{e}j$ exponent spells out C1, and $-\check{s}$ spells out C2, in the manner indicated in (15). For comparatives which lack the $-\check{e}j$ exponent, we assume, for now, that there is simply a zero allomorph of $-\check{e}j$, as indicated in the tree in (16).³

³ As before, the structures are adjusted to fit the surface order of Czech, which is the mirror image ordering of the underlying functional sequence (see Cinque 2005). As

(15) The $-\check{e}j\check{s}$ comparative(16) The $-\check{s}$ comparative

If only one position of exponence were assumed, we would need to store both $-\check{e}j\check{s}$ and $-\check{s}$ as separate allomorphs (in competition for a single position), and their partial identity would be purely accidental.

The third class of comparatives can still be captured as a reduced version of (16), with both markers silent, as shown in (17).

(17) The $\emptyset$ -comparative

Decomposing the comparative $-\check{e}j\check{s}$ into two morphemes $-\check{e}j$ and $-\check{s}$ thus allows for a ‘principled’ approach to the gradual reduction of the full marker, an account which cannot be stated without such a decomposi-

in Cinque’s work, we analyse mirror orders as arising through a series of standard movement operations, which affect phrasal nodes and move them either cyclically (i.e. spec-to-spec), or roll-up style. We highlight the mechanics of these movements in the next section.

tion. By ‘principled approach’ we mean an approach where the similarity between $-\check{e}j-\check{s}$ and $-\check{s}$ is not accidental: they share an identical morpheme.

Interestingly, it turns out that there is independent evidence that $-\check{e}j\check{s}$ should be split into two parts indeed. The evidence comes from comparative adverbs, seen in the second column of (18). Here the $-\check{s}$ part of the comparative adjective is systematically missing.

(18)

CMPR ADJ	CMPR ADV
bujař-ej-š-í	bujař-ej- i ‘merrier’
rychl-ej-š-í	rychl-ej- i ‘faster’
červen-ěj-š-í	červen-ěj- i ‘redder’
hloup-ěj-š-í	hloup-ěj- i ‘sillier’
bujař-ej-š-í	bujař-ej- i ‘merrier’

The absence of $-\check{s}$ in the adverbs (highlighted by a gap between $\check{e}j$ and the adverbial marker) is hard to attribute to phonology, because the adverbial markers are vowels like the agreement marker $-\acute{i}$, and they differ only in length. But if $-\check{s}$ is not deleted due to a contact with the following vowel, it must correspond to a separate morpheme that is simply missing in all the adverbs, a conclusion which independently supports the split analysis. We will come back briefly to the adverbial forms in section 4.

2.3. Root types and the distribution of zeroes

With the ‘split-CMPR’ proposal in place, we turn to an interesting asymmetry in the way zero morphology is distributed in Czech comparative adjectives depending on the stem chosen. The stem may be morphologically simplex, corresponding to the root only, or it may be complex, as with $-k$ derived adjectives. In this section, we are interested in how roots control zero morphology in the positions corresponding to C1 and C2, and we turn to stems in the next section.

The pattern we shall explore is that we find roots that require a zero marker next to them, with the outer marker overt, as in (19b) (overt exponence is represented by α). Furthermore, if the analysis in (17) is on the right track, the outer morpheme can also be silent, but only if the inner one is, so that zero morphology spreads from the root outwards, as in (19c). What we do not find is a case where a particular root requires a pattern in which the inner marker is overt and the outer zero, as in (19d).

(19)		C1	C2	
a.	ATTESTED:	$\sqrt{\text{Type 1}}$	$-\alpha$	$-\alpha$ (<i>hloup-ěj-š-i</i>)
b.	ATTESTED:	$\sqrt{\text{Type 2}}$	$-\emptyset$	$-\alpha$ (<i>star-ø-š-i</i>)
c.	ATTESTED:	$\sqrt{\text{Type 3}}$	$-\emptyset$	$-\emptyset$ (<i>ostř-ø-ø-i</i>)
d.	NOT ATTESTED:	$\sqrt{\text{Type } *}$	$-\alpha$	$-\emptyset$

This pattern leads to the following generalisation:

- (20) Root-determined zeroes spread from the inside out, and cannot skip intervening heads.

Before we go on to discuss the explanation for (20), we must address some empirical issues (relevant for the generalisation), which have to do with the proper treatment of palatalisations. So far, we have analysed them as an indivisible (even if floating) part of each of the suffixes, such that in effect $\check{e}j = \text{PAL}_{\text{VAL}}ej$, and $\check{s} = \text{PAL}_V\check{s}$, etc. One might go a step further and argue that the PAL element is not part of the suffix to begin with.

For instance, a reviewer builds on the (diachronically motivated) idea that soft declension does not *trigger* palatalisation, but rather reflects the fact that the stem ends in a palatal consonant to begin with. If that is so, we have to grant the palatalisation in the third pattern an independent status, and treat it as an exponent of some comparative head. Once palatalisations are granted morphemic status, we move towards a system where we could distinguish three different components of the comparative marker instead of two, with PAL as the third element. As suggested by the reviewer, one possible segmentation of the three different patterns of comparative formation would then look as in (21), with PAL realised as *j* after a vowel and labials (i.e., in the first pattern):

(21)		C1	C2	C3	our analysis
a.	$\sqrt{\text{Type 1}}$	PAL _{VAL} e	PAL	š	(<i>hloup-ěj-š-i</i>)
b.	$\sqrt{\text{Type 2}}$	Ø	PAL	š	(<i>star-ø-š-i</i>)
c.	$\sqrt{\text{Type 3}}$	Ø	PAL	Ø	(<i>kluz-k-ø-ø-i, ostr-ø-ø-i</i>)

Here the palatalisation of $\sqrt{\text{Type 3}}$ would be caused, not by the soft declension marker, but by the palatal element in C2. Such analysis would then contradict our claim that the root-determined zeroes can only spread from the inside out, since C3 would be silent where C2 is realised by PAL.

While we find the alternative worth considering, one problem we see in this move is that there is no independent evidence for decomposition. Recall that our argument for splitting *-ěj* and *-š* rests in part on the fact that *-š* is missing in comparative adverbs, where only *-ěj* is found. Analogously, if PAL and [i:] were independent in the soft declension marker, we would want to see PAL independently of the following [i:]. But there are no forms (or parts of forms) like that; these only arise when *-í* or other palatalisation triggers follow.

The second reason why the analysis in (21) is problematic is that the proposed decomposition works only for the standard language, because it makes the palatalisation triggers that occupy C2 identical. But as we have seen when looking the dialectal forms in (12), the palatalisations are actually different. In (21b), we are looking at PAL_V, while in (21c),

we are dealing with PAL_{VA} . This casts doubt on the particular segmentation in (21), where an identical PAL is shared across the (b) and the (c) example.

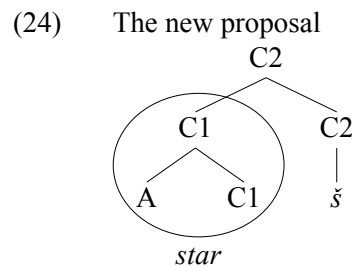
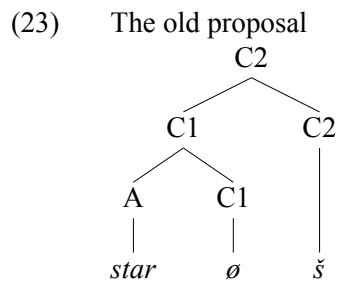
If we wanted to make the proposal in (21) descriptively accurate, we would be rather led to structure the facts in a way as shown in (22). This proposal preserves the idea that PAL_{VA} is an independent comparative marker (in C3 in (22)), syntactically distinct from the following agreement marker, and at the same time captures the fact that palatalisation is different in the (b) and (c) cases.

(22)		C1	C2	C3	our analysis
a.	$\sqrt{\text{Type 1}}$	$\text{PAL}_{\text{VAL}}\text{e}$	$\text{PAL}_{\text{V}}\check{\text{s}}$	PAL_{VA}	(<i>hloup-ěj-š-í</i>)
b.	$\sqrt{\text{Type 2}}$	$\emptyset$	$\text{PAL}_{\text{V}}\check{\text{s}}$	PAL_{VA}	(<i>star-ø-š-í</i>)
c.	$\sqrt{\text{Type 3}}$	$\emptyset$	$\emptyset$	PAL_{VA}	(<i>kluz-k-ø-ø-í</i> , <i>ostř-ø-ø-í</i>)

However, once we structure the facts like this, the very same generalisation concerning zero distribution emerges again: zeroes spread from the inside out. In sum, even if PAL is treated as an exponent of the comparative, the details of how palatalisations work lead us to an analysis which observes the generalisation (20). In the account that follows, we return to our original bi-componential analysis. The reason (as highlighted above) is that it is harder to defend the independence between

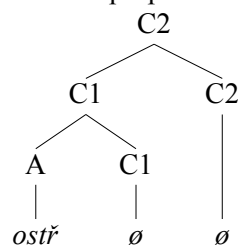
PAL_{VA} and the following agreement marker. However, if it turns out that the facts ought to be structured as in (22), this will have no theoretical consequences for the type of proposal we shall put forth.⁴

Let us now then turn to the explanation we propose for the asymmetrical distribution of zeroes. We argue that this follows from a theory where zero exponents arise as a consequence of non-terminal spell out, where a single lexical item may realise multiple positions in the syntactic/morphological structure (see e.g. Williams 2003; Siddiqi 2006; Starke 2009; Radkevich 2010; Haugen & Siddiqi 2016). In such a theory, the root may spell out a unit that corresponds to multiple terminals, including one or more comparative heads. If such an approach is adopted, the original structures with zeroes (on the left) are updated as depicted on the right.

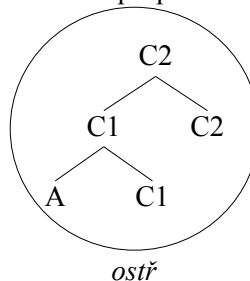


⁴ The advantage of introducing PAL under C3 is that it allows for a purely phonological approach to the issue of hard/soft declension: soft declension is what one gets after soft stems, i.e. stems ending in PAL_{VA} . However, we think that the issue is more complex than this, and that there is a more substantial (morphosyntactic) difference between the two declensions. We refrain from discussing this issue in full here, as it would lead us too far afield.

(25) The old proposal



(26) The new proposal



It is clear from the diagrams that if root-controlled zeroes are the product of non-terminal insertion, then they will make the two comparative markers disappear inside out, rather than in any other fashion. This derives the absence of the fourth pattern in (19), repeated (for convenience) below:

(27)			C1	C2	
a.	ATTESTED:	$\sqrt{\text{Type 1}}$	-α	-α	(hloup-ěj-š-í)
b.	ATTESTED:	$\sqrt{\text{Type 2}}$	-ø	-α	(star-ø-š-í)
c.	ATTESTED:	$\sqrt{\text{Type 3}}$	-ø	-ø	(ostř-ø-ø-í)
d.	NOT ATTESTED:	$\sqrt{\text{Type}^*}$	-α	-ø	

Given the type of explanation proposed, we expect the absence of (27d) to be valid for all cases where a particular root requires a certain pattern of overt/zero exponence. However, the role of the root in triggering the pattern is crucial. Consider, for instance, the adverbs presented in (18) above, a representative instance of which is repeated in (28).

(28)	CMPR ADJ	CMPR ADV	
	bujar- <i>ej</i> -š- <i>í</i>	bujar- <i>ej</i> - <i>i</i>	'merrier'

In the adverbial form of (28), the C2 marker *š* is missing, while the C1 *ěj* exponent is present. This might on the face of it appear to instantiate (27d). But the crucial point is that the absence of C2 *š* in the presence of C1 *-ěj* is triggered here by 'being an adverb,' and not by 'having a particular root'. We will have more to say about the analysis of adverbs in section 4.

To sum up, we are proposing an account of the allomorphic variation in terms of a bi-partite structure, which explains why one allomorph is a reduced version of the other, and non-terminal spell out, which explains why zeroes first appear close to the root and expand outwards.

The correct pairing of the three comparative allomorphs with individual roots is accounted for by proposing different lexical entries for different classes of adjectival roots. The first class is the regular one, which merely spells out A, as shown in (29a). Adjectives that take only *-š* in the comparative spell out a larger structure, consisting of A and C1 (see (29b)). Finally, adjectives that spell out the entire A+C1+C2 show no CMPR marker, and their entry is as in (29c). Analysing the allomorphy this way, we dispense with diacritics that pair the right allomorph with

the right root. The correct comparative allomorph simply corresponds to a residue that is not spelled out by a given root.⁵

- (29) a. Root that combines with $\check{e}\check{j}\check{s}-(i)$ $\Leftrightarrow$ [A]
 b. Root that combines with $\check{s}-(i)$ $\Leftrightarrow$ [[A] C1]
 c. Root that combines with $-(i)$ $\Leftrightarrow$ [[[A] C1] C2]

The last ingredient we need to introduce in this section is the Superset Principle (we slightly modify the original formulation of [Starke 2009](#) so as to avoid any misunderstanding).

(30) *The Superset Principle:*

A lexically stored tree L matches a syntactic node S iff L contains the syntactic tree dominated by S as a subtree

The Superset Principle makes sure that these entries can also appear in the positive degree. This is particularly relevant for entries of the type in (29b) and (29c), as these are larger than the syntactic structure of the positive degree, and still they can occur in the positive degree. The Superset Principle ensures this: in the syntax, the positive degree corresponds only to the feature [A], which is contained in all the lexical entries in (29), and so any of these entries can be inserted in the positive degree. The Superset Principle also accounts for the adjectives of the

⁵ We will update the entries slightly in section 2.4, where we further decompose A.

third category, which show a syncretism between the positive and the comparative degree.⁶

2.4. *Splitting up A*

Now recall from section 2.1 that there is a particular class of adjectives in Czech that has the suffix *-k* in the positive and apparently a zero-marked comparative. These are repeated in the table below.

(31)	POS	CMPR	gloss of A	BASE	gloss
	kluz-k-ý	kluz-č-í	‘slippery’	s-kluz	‘a slide’
	břit-k-ý	břit-č-í	‘sharp’	břit	‘edge’ (of a knife)
	hoř-k-ý	hoř-č-í	‘bitter’	hoř-e	‘sorrow’
	sliz-k-ý	sliz-č-í	‘slimy’	sliz	‘slime’
	vlh-k-ý	vlh-č-í	‘wet’	vláh-a	‘dew’
	ten-k-ý	ten-č-(í)	‘thin’	*ten	—

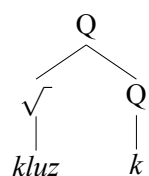
In the table, the stem final *-k* is factored out as a derivational suffix, in line with traditional accounts such as Dokulil et al. (1986), Karlík et al. (1995) or Janda & Townsend (2000:24). This segmentation is motivated by the fact that the roots to which the adjectiviser *-k* attaches lead an

⁶ At this point, we need to acknowledge a degree of variation in comparative formation. For instance, the adjective *bohat-ý* ‘rich’ allows for two comparative forms, namely *bohat-š-í* (ca. 786.000 hits on Google) and *bohat-ěj-š-í* (ca. 21.000 hits). We attribute such variation to the possibility that different speakers may assign different lexical entries to particular roots, so that some speakers store the root *bohat* ‘rich’ as C1P, others as AP (more specifically QP, cf. section 2.4), others (maybe) as both.

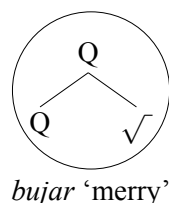
independent life, as shown in the column marked BASE. Note that in some cases (e.g. ‘thin’), the root is just a cranberry morpheme and does not exist independently of the adjectival context.

Such morphologically complex positives require that we split up the positive-degree A into a root feature and a functional head Q, which we take to be a head that contributes gradability (Bresnan 1973; Corver 1997; De Clercq 2013; 2017; De Clercq & Vanden Wyngaerd 2017). This makes enough space in the positive degree (corresponding now to a constituent dominated by Q) for both the root and the *-k*, as shown in (32). Here we assume that the (verbal) root *kluz* ‘to slide’ spells out $\sqrt{}$, and the *-k* suffix spells out the Q projection.⁷

(32)



(33)



Given the semantic contribution which we have invested into the Q head (gradability), we expect that this head is present in all gradable adjectives. However, many gradable adjectives have no adjectival suffix and

⁷ We use the notation $\sqrt{}$ to indicate the bottom of the functional sequence for convenience only. We consider it likely that $\sqrt{}$ in fact needs to be further decomposed into functional heads all the way down (comparable to the way V is decomposed into a series of functional heads in Ramchand 2008). Under this perspective, $\sqrt{}$ should be read as a mere shorthand for a series of functional heads, the exact nature of which is yet to be determined. This issue does not affect the arguments made in this paper.

their stem (the base to which the inflectional agreement is added) corresponds to the root only. We propose that such adjectives (which we discussed in the previous section) spell out the whole QP, as shown in (33).

Consequently, we must update all other types of roots (those that spell out also comparative heads) accordingly. The final lexical entries of selected roots, each belonging to one of the classes identified above, are given below.

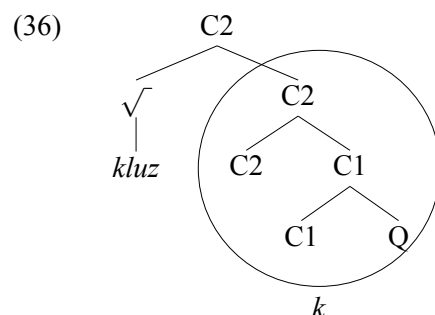
- (34)
- | | | | |
|----|---|-------------------|---------|
| a. | $\sqrt{}$ | $\Leftrightarrow$ | /kluz/ |
| b. | $[\text{QP Q } \sqrt{}]$ | $\Leftrightarrow$ | /bujar/ |
| c. | $[\text{C}_{1\text{P}} \text{C}_1 [\text{QP Q } \sqrt{}]]$ | $\Leftrightarrow$ | /star/ |
| d. | $[\text{C}_{2\text{P}} \text{C}_2 [\text{C}_{1\text{P}} \text{C}_1 [\text{QP Q } \sqrt{}]]]$ | $\Leftrightarrow$ | /ostř/ |

The entry (35a) is the new type of root, which we have introduced in this section: it spells out only the $\sqrt{}$ node, and cannot spell out the whole positive degree (i.e., QP). Hence, it requires additional morphology in the positive (i.e., $-k$). All other root types (i.e. those in (35b-d)) spell out QP at least, and as a result, they have no suffix in the positive (save the agreement). They correspond to the three classes of roots identified in the preceding section. *Bujar* ‘merry’ type of roots (see (35b)) leave both C1 and C2 without spellout, combining with $-\check{e}j-\check{s}$ in the comparative. *Star* ‘old’ can spell out C1 in addition, combining with $-\check{s}$ in the

comparative. Finally the dialectal *ostř* ‘sharp’ needs no additional morphology either in the positive or in the comparative. The lexical entries for the comparative suffixes are given in (35).

- (35) a. [C_{1P} C₁] ⇔ /ěj/
 b. [C_{2P} C₂] ⇔ /š/

Finally, let us turn to the lexical entry of *-k*. This suffix can appear either as a Q suffix in the positive, as in (34), or it can spell out the whole complex consisting of Q+C₁+C₂ (in the comparative), as in (36).



The mechanism by which *-k* can spell out both Q (as in (32)) and the larger structure dominated by C₂ (as in (36)) is again the Superset Principle: taking the lexical entry of *-k* to be the larger structure, as shown

in (37), it can ‘shrink’ to spell out any node that it contains as a constituent.^{8,9}

$$(37) \quad [C2 [C1 Q]] \Leftrightarrow /k/$$

A question that arises at this point is how these structures are created. In particular, looking at the structure in (36) and comparing it to earlier structures as in (34), we see that they contain quite different constituents: (36) has a constituent consisting of Q+C1+C2 to the exclusion of $\sqrt{}$, where $\sqrt{}$ precedes the rest of the terminals, whereas (34) has the $\sqrt{}$ node at the bottom of the hierarchy, and there is no constituent Q+C1+C2 to the exclusion of $\sqrt{}$. In the following section we turn to the details of how these structures are derived from a common underlying functional sequence.¹⁰

⁸ Observe that the lexical item in (37) spells out Q, C1, and C2. A possible alternative consists in assuming that it spells out only Q and C1, and that $\check{s}$ is needed on top of k to spell out C2. That would amount to the alternative analysis discussed in section 2.1 above, where Standard Czech in fact only has only two allomorphs, and the third (zero marking) reduces to the second (marking by $\check{s}$), modulo phonetically triggered $\check{s}$ -deletion. Our proposal is therefore compatible with either analysis.

⁹ A reviewer asks if we ever get *root*+Q as a synthetic rather than a portmanteau adjective in English. One could think in this respect of a derivational affix like *-y*, which can derive adjectives from nouns (e.g. *arty*, *crappy*, *skinny*, etc.). But there are also quite number of adjectives ending in *-y*, which on the face of it are underived (e.g. *nifty*, *sloppy*, *silly*, *pretty*, *happy*, *kinky*, *bonny*, *busy*, *canny*, *phoney*). An alternative analysis of these would take them to consist of a root plus *-y* suffix, in the manner of the Czech *-k*-adjectives.

¹⁰ An anonymous reviewer observes that with some adjectives, like *slad-k-y* ‘sweet,’ the *-k* drops in the comparative, yielding *slad-š-i* ‘sweeter.’ We see two possibilities for capturing these adjectives. One option is to loosen the containment hypothesis

3. Getting technical: the spellout algorithm

In this section, we make explicit our assumptions concerning the details of how lexical insertion works. While our proposal for splitting the comparative into two heads is to a large extent independent of these assumptions, they will be crucial as the discussion gets more involved. They are also useful for readers to fully understand how in our approach the universal hierarchy of heads is transformed into a language-specific structure with a particular order of the elements. We adopt here the general Nanosyntax approach as developed in recent years by Michal Starke (Starke 2009 et seq.), and specifically we will draw upon the latest additions to the framework as described in Starke (2018). In section 7 we shall propose a nontrivial modification to the framework by eliminating the Elsewhere Principle.

and admit that the comparative does not fully contain the positive (since with these adjectives the comparative form does not contain the positive form). Another option is to rely on pointers, a device which we introduce later on. The behaviour of these adjectives could be captured with the pointer technology by means of a lexical item for *slad* that consists of C1P (which makes it incompatible with $-\acute{e}j$), but with a moved root inside the lexical item and a pointer to the $-k$ morpheme at QP-level:

$$(i) \quad [_{C1P} C1 [_{QP} [\sqrt{}] [_{QP} \rightarrow k]]] \Leftrightarrow \textit{slad}$$

This entry contains the $\sqrt{}$ node, so it can spell out $\sqrt{}$. It cannot spell out $[Q+\sqrt{}]$, so in the positive, it needs $-k$. It includes a pointer to this $-k$, so it can spell out a C1P that contains C1 and a constituent with a root in it and a $-k$ suffix, eliminating both by over-ride, i.e. we get the spellout *slad* in the comparative. An entry like this, however, raises some technical and empirical issues which we cannot deal in the current paper, so that we leave this issue for future work.

3.1. *Merge F, Move complement of F*

The first assumption is that structures are built by merging one feature at a time. Every time a new constituent is created by merging a feature, the structure must be lexicalised. A ‘lexicalised’ structure is one where all its meaningful elements are realised (cf. [Fábregas 2007](#)). Still following Starke’s work, we adopt here the proposal that only non-terminal nodes can be spelled out (i.e. there is no insertion at a terminal).

Suppose, for instance, that syntax has produced an XP and merged it with the feature F, as in (38a). The lexicon is then consulted, and if it contains a lexical entry which contains the newly formed FP, then the structure is successfully lexicalised. (38b) is an example of such an entry.

- (38) a. $[_{FP} F [_{XP}]]$
 b. $[_{FP} F [_{XP}]] \Leftrightarrow / \alpha /$

The spellout at FP ‘overrides’ any prior the spellout of XP; in other words, whatever lexical item was selected for insertion after syntax had constructed XP is no longer considered. This is called ‘Cyclic Override’ in the Nanosyntactic literature.

Suppose now that the lexicon does not contain (38b), but instead two separate entries, one for XP, and another one for F and its projection:

- (39) a. XP $\Leftrightarrow$ /β/
 b. [FP F] $\Leftrightarrow$ /γ/

When the lexicon is accessed to see how (38a) can be spelled out, there is no way to spell out FP, since neither (39a) nor (39b) contain the whole FP (including the XP that FP contains), which is required by the Superset Principle. However, since every feature must be lexicalised at every cycle, a repair mechanism is set in motion. Repair mechanisms correspond to various types of movement operations which alter the structure, hoping that lexicalisation will succeed. We will now introduce the various types of movement one by one, and then explain the logic behind their choice.

The first possibility is to move the complement of F out of FP, as shown in (40). This movement option is called rollup or snowball movement, since it takes along anything that is contained in XP.

- (40) [FP XP [FP F t]]

In this new ‘repaired’ configuration, all the features may be spelled out. XP is realised by (39a), and the lower FP can be lexicalised using (39b) (we assume that traces do not count).¹¹

¹¹ Various proposals have been made in the literature as to the precise mechanism from which this follows. The by far simplest assumption is that the movement in (40) does not leave a trace. For concreteness, we keep the trace in the representation for now.

These two abstract scenarios (direct lexicalisation in (38) and complement movement in (40)) correspond to two derivational options that we need to account for the various classes of Czech comparatives. Direct lexicalisation takes place when we merge C1 with the positive adjective (QP), as in (41a), and we lexicalise them at once, using an adjective of the size C1P, see (41b).

- (41) a. $[_{C1P} C1 [_{QP} Q \sqrt{\quad}]]$
 b. $[_{C1P} C1 [_{QP} Q \sqrt{\quad}]] \Leftrightarrow /star/$

However, suppose that we want to use a QP-size root; then we cannot lexicalise (41a) directly. We must move QP across C1, as in (42):

- (42) $[_{C1P} QP [_{C1P} C1 t]]$

In the configuration (42), all features can be lexicalised. QP is spelled out by the root (with the entry in (43a)), while the lower C1P is spelled out by *-ěj* (its entry is in (43b)). Note as well that the movement of QP across C1 also derives the correct suffixal order of exponents, which is the mirror image of the underlying functional hierarchy.

- (43) a. $[_{QP} Q \sqrt{\quad}] \Leftrightarrow /bujar/$
 b. $[_{C1P} C1] \Leftrightarrow /ěj/$

Suppose now that we add C2 on top of C1. Once again, two derivational options arise. Direct lexicalisation (by a root that can spell out C2P) derives zero comparatives:

- (44) a. [C2P C2 [C1P C1 [QP Q $\sqrt{\quad}$]]]
 b. [C2P C2 [C1P C1 [QP Q $\sqrt{\quad}$]]] $\Leftrightarrow$ /ostř/

If a root is not large enough, snowball movement of C1P across C2 is triggered, and C2P is spelled out as –š:

- (45) a. [C2P [C1P ...] [C2P C2 *t*]]
 b. [C2P C2] $\Leftrightarrow$ /š/

The spellout of C1P is marked by three dots in (45a), because it is variable (as has already been discussed). Specifically, C1P may either be lexicalised directly by a root like *star* ‘old,’ in which case we derive the –š comparative. Alternatively, there may be movement inside C1P (see (42)), in which case we derive the –ěj-š comparative.

Notice finally that if a given root has a lexical entry that allows it to spell out C1P directly, then it will always do so. This suggests a hierarchy of operations, where movement is a last resort: only if direct lexicalisation fails does movement apply. Without such a hierarchy, the system would be able to spell out C1P as **star-ěj* instead of *star* ‘old’,

which is clearly an undesirable result. Therefore, the two spellout options discussed so far must be ranked in a spell-out algorithm as follows:

(46) *Spell-out Algorithm (v1, preliminary)*, *Starke (2018)*

- a. Merge F and spell out.
- b. If (a) fails, move the complement of F, and spell out.

The spell-out algorithm, combined with the lexical entries in (47), allows us to derive three of the patterns observed, namely *bujar-ěj-š* ‘merrier’, *star-š* ‘older’, and *ostř-* ‘sharper,’ with the lexical entries specified as follows:

- (47)
- a. $[QP\ Q\ \sqrt{\quad}] \Leftrightarrow /bujar/$
 - b. $[C1P\ C1\ [QP\ Q\ \sqrt{\quad}]] \Leftrightarrow /star/$
 - c. $[C2P\ C2\ [C1P\ C1\ [QP\ Q\ \sqrt{\quad}]]] \Leftrightarrow /ostř/$
 - d. $[C1P\ C1] \Leftrightarrow /ěj/$
 - e. $[C2P\ C2] \Leftrightarrow /š/$

Now we turn to how complex adjectives like *kluz-k* ‘slippery’ are derived.

3.2. *Move spec-to-spec*

In order to obtain the correct forms of these adjectives, we must introduce yet another type of repair strategy: spec-to-spec movement. This repair strategy does not remove the whole complement of the newly added F, but only the Spec of the complement. In [Starke \(2018\)](#), the operation occupies a fixed place in the hierarchy of repairs as given in (48):

(48) *Spell-out Algorithm (v2, preliminary)*

- a. Merge F and spell out.
- b. If (a) fails, try spec-to-spec movement of the node inserted at the previous cycle, and spell out.
- c. If (b) fails, move the complement of F, and spell out.

We will comment on the ordering of the operations shortly, but first we want to show how the new derivational option works on its own. Let us then go back to (40), which represents the stage of the derivation where XP extracts out of FP. Suppose we now continue the derivation by adding a new feature, F2 (for clarity, we rename F to F1):

(49) $[_{F2P} F2 [_{F1P} XP [_{F1P} F1 t]]]$

Suppose now that we have a lexical entry like in (50), i.e. one that spells out both F1 and F2:

$$(50) \quad [_{F2P} F2 [_{F1P} F1]] \Leftrightarrow /δ/$$

This lexical entry cannot spell out (49), since (50) does not contain (49). Hence, movement must take place. In this particular case, what is needed in (49) is to move the XP (i.e. the highest Spec of the complement of F2) out of F2P, as shown in (51). Here the XP has been cyclically moved from the complement of F1 to its Spec, and then further on to Spec,F2P:

$$(51) \quad [_{F2P} XP [_{F2P} F2 [_{F1P} t [_{F1P} F1 t]]]]$$

F2P can now be spelled out as /δ/ using (50), since the structure is identical to the lexical entry (ignoring traces).

A specific example of this strategy is the derivation of *–k* adjectives. Recall that the lexical entry for *–k* is as follows:

$$(52) \quad [_{C2P} C2 [_{C1P} C1 [_{QP} Q]]] \Leftrightarrow /k/$$

When deriving the QP (the positive degree of such adjectives), we combine the Q head with the $\sqrt{\quad}$, see (53a). For roots (like *kluz*) that are not adjectival, i.e. do not spell out QP, we cannot spell out (53a) without

movement. We thus move the $\sqrt{}$ across Q, as shown in (53b), and spell out QP by $-k$. The derivation now continues by merging C1, as in (52c). This structure cannot be spelled out without movement, and hence, $\sqrt{}$ moves on by spec-to-spec movement, yielding (52d).

- (53) a. $[_{QP} Q \sqrt{}]$
 b. $[_{QP} \sqrt{} [_{QP} Q t]]$
 c. $[_{C1P} C1 [_{QP} \sqrt{} [_{QP} Q t]]]$
 d. $[_{C1P} \sqrt{} [_{C1P} C1 [_{QP} t [_{QP} Q t]]]]$

Here the remnant C1P is spelled out by $-k$, which has thereby pronounced the projection of C1 that would, on its own, be pronounced as $-ěj$. This is the correct result, since the comparative of the $-k$ derived adjectives definitely lack this marker, regardless of whether they do or do not have the comparative $-š$.

This derivation also gives us reasons to rank the spec-to-spec movement higher than the movement of the whole complement. Suppose that instead of moving spec-to-spec as shown in (53d), we first moved the complement of C1, producing (54) as the first attempt, after failing to spell out (53c):

- (54) $[[_{QP} \sqrt{} [_{QP} Q t]] [_{C1P} C1 t]]$

In this structure, CIP gets spelled out by $-ěj$, which would follow in the string after the root and the $-k$ suffix, deriving the incorrect *kluz-k-ěj*. If this was the first attempt at a repair structure, all the relevant features would be spelled out, and there would be no motivation for trying another repair strategy. In other words, *kluz-č-ěj* (after palatalisation) would block the correct output *kluz-k*. In order to avoid this, we have to rank spec-to-spec movement before complement movement. This result provides an empirical support for the proposal in Starke (2018), who considers spec-to-spec movement more economical on the grounds that it carries less material.

The spellout algorithm in (48) applies recursively, i.e. after each successful spellout, the entire procedure in (48) is repeated. We shall return to the formulation of the spellout algorithm in section 5 below, where we discuss markers of the comparative that precede the adjective. These are not accommodated under the current formulation in (48) above, and will require a further extension of it. However, before discussing such cases, we turn to some new data.

4. Comparatives in Old Church Slavonic

This section turns to comparatives in Old Church Slavonic (OCS). OCS is the oldest Slavic language attested in writing (9th century). Even though it is not a direct ancestor of Czech, it clearly exhibits much of

the same patterns, demonstrating their diachronic stability. More importantly, it also features facts that may look problematic for our approach to zeroes, as pointed out to us by an anonymous reviewer, with reference to [Bopp \(1833\)](#). However, we think that if these facts are properly analysed, there is no counterexample to our analysis coming from OCS.

Let us start by looking at the GEN.SG form of the comparative, since it is more characteristic of the entire paradigm than the nominative and the accusative, which show some special properties; we discuss the latter case forms next. OCS is like Czech in that it has root-controlled allomorphy. The table in (55) shows two such allomorphs: one is found with the adjective *bol-* ‘big’ (in the first row), the other with *nov-* ‘new’ (in the second row). According to [Aitzetmüller \(1978:131-2\)](#) and [Lunt \(2001:77-8\)](#), these are the only two types of comparatives in OCS.¹²

(55) Two types of adjectives in OCS ([Aitzetmüller 1978:131-2](#))

	M	N	F	gloss
GEN	bol- jьš-a	bol- jьš-a	bol- jьš-ę	‘bigger’
GEN	nov-ě-jьš-a	nov-ě-jьš-a	nov-ě-jьš-ę	‘newer’

These data are amenable to an analysis that is virtually identical to modern Czech: the marker *ě* spells out C1, *jьš* spells out C2. The root *bol-* is specified as C1P, which leads to the disappearance of *-ě* on the surface.

¹² We shall see immediately that [Bopp \(1833\)](#) distinguishes a third allomorph as well. The grapheme ѣ corresponds to a ‘reduced’ vowel *i*, sometimes written as *ī*.

However, things get more complicated when we look at full paradigms which include the nominative and the accusative. We start from the adjective ‘newer.’ This adjective has a bi-morphemic comparative in (55), with separate markers for C1 and C2 present. However, in the following table, the C2 marker *-jbš* remains silent in the shaded cells.¹³

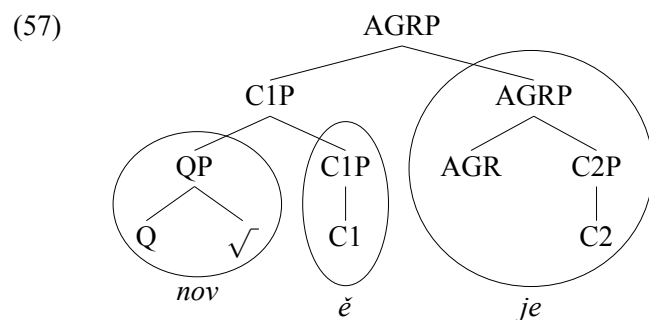
(56) ‘Newer’ in Old Church Slavonic (Aitzetmüller 1978:132)

	M		N		F
NOM	nov-ě-	jь	nov-ě-	je	nov-ě-jьš-i
ACC	nov-ě-jьš-bь		nov-ě-	je	nov-ě-jьš-q
GEN	nov-ě-jьš-a		nov-ě-jьš-a		nov-ě-jьš-e
LOC	nov-ě-jьš-i		nov-ě-jьš-i		nov-ě-jьš-i
DAT	nov-ě-jьš-u		nov-ě-jьš-u		nov-ě-jьš-i
INS	nov-ě-jьš-emь		nov-ě-jьš-emь		nov-ě-jьš-ejъ

This is apparently problematic for our generalisation in (20), which claims that zero markers spread from the root out. However, as with the comparative adverbs, the crucial point is that our generalisation in (20) regulates root-controlled zeroes, and the C2 zeroes in (56) do not belong to this category. Rather than a specific root, it is a particular combina-

¹³ There is an alternative segmentation/analysis of these forms, namely one where the inflectional endings in the shaded cells are just *-b/-e*, in which case the preceding *j* would be a leftover of the C2 comparative marker. In such case, we would need to split C2 *-jbš* into C2a *-j* and C2b *-bš*, and propose an analysis according to which only C2b goes missing. This analysis is viable, but we shall not develop it further in this paper.

tion of ϕ -features (whose markers immediately follow C2) which is to be blamed for the absence of an overt C2 exponent. We therefore propose that in these cases, the absence of C2 is not due to phrasal spellout by the root (which clearly does not control this effect), but due to the fact that the ending, which expresses a particular ϕ -feature combination, spells out [C2+agr]. This is shown in (57), with roll-up movement of QP across C1, and then cyclic spec-movement of C1P to the left of AGR.¹⁴



This shows that our analysis in fact allows for a ‘zero’ C2, even if C1 is non-zero, but only in case the zero spellout of C2 is conditioned by outside morphology, and not by the root.¹⁵

¹⁴ AGR is in fact internally complex, *-je* being a portmanteau for gender, number and case. We abstract away from this complication since it is orthogonal to our concerns.

¹⁵ It is worth noting that the Czech adverbs, which realise C1 *-ěj* but not C2 *-š* (see (18) above), share this property of an affix-controlled zero. Diachronically, they are the continuation of the neuter forms shown in (56). Their analysis is perfectly analogous to (57), substituting ADV for AGR.

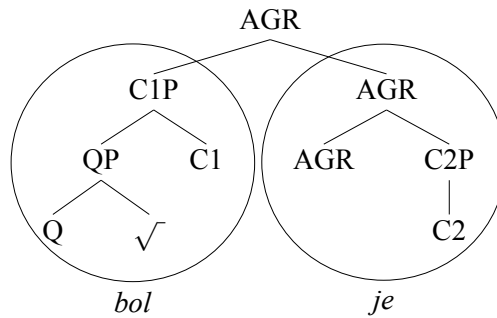
This analysis is confirmed by the behaviour of the adjective ‘bigger.’ In (58), we give its full paradigm. The most telling contrast is between the ACC of the masculine and neuter paradigms. Here one can clearly see that the neuter has neither C1 *ě* (as this gets realised by the root), nor C2 *-jěš*, since this morpheme gets eaten by agr *-je*.

(58) ‘Bigger’ in Old Church Slavonic ([Aitzetmüller 1978:131](#))

	M		N		F
NOM	bol-	jь-jь	bol-	je	bol- jьš-i
ACC	bol-	jьš-jь	bol-	je	bol- jьš-q
GEN	bol-	jьš-a	bol-	jьš-a	bol- jьš-ę
LOC	bol-	jьš-i	bol-	jьš-i	bol- jьš-i
DAT	bol-	jьš-u	bol-	jьš-u	bol- jьš-i
INS	bol-	jьš-емь	bol-	jьš-емь	bol- jьš-ejq

The analysis of the NOM/ACC neuter forms is shown in (59).

(59)



This form then confirms that the disappearance of C2 is a process that is completely independent of the particular root, because all roots lose C2 before the relevant endings, exactly as predicted by the current account.¹⁶

The second issue we want to address here is the development from the OCS type of system to the Czech type of system. A part of this change can be understood as a relatively straightforward case of paradigm levelling. In particular, during the Early Old Czech, “under the influence of the feminine and the oblique cases, there appeared [in the shaded cells of (58)] an analogical suffix *–š’*, which was in complementary distribution with *ějš’*” (Lamprecht et al. 1986:313, translation from Czech is ours). In our system, this change corresponds to a change in the lexical entries of the agreement markers, which now fail to spell out C2, leaving C2 to be spelled out by *š*, as in the other cases and genders. This devel-

¹⁶ The only remaining issue is the form of the NOM.SG.M which has the ending *–jb-jb*, and not the expected *–jb*. As far as we could determine, this is an effect of declension type. Specifically, OCS has two different adjectival declensions, an indefinite one, which we present in the declension tables, and a definite one. The definite declension is derived from the indefinite one by the addition of a pronoun. For instance, from the indefinite *bol-je* ‘bigger, N’ one derives *bol-je-je* ‘bigger, N-DEF’. However, in the comparative only, the definite masculine form is the same as the indefinite one, namely *bol-jb-jb*, a form which morphologically resembles the definite paradigm by the ‘doubling’ of the inflection. In sum, we think that the M.SG. cell shows a definite form for reasons that are unclear to us, yet this seems to be already remarked in Bopp (1885:409), who claims that “comparatives always follow, in the masculine and neuter, the definite declension.” In accordance with this statement, Bopp also gives the neuter in *je-je*, which the later grammars give as the indefinite with a single *–je*, while keeping the double *jb-jb* in the masculine.

opment led to the present day paradigms with *–ější* and *–ší* throughout the paradigm, as illustrated in the first two columns of (60).

(60) Levelled paradigms in Czech, neuter forms

	‘newer’	‘better’	‘thinner’
NOM	nov-ěj-š-í	lep-š-í	ten-č-í
ACC	nov-ěj-š-í	lep-š-í	ten-č-í
GEN	nov-ěj-š-ího	lep-š-ího	ten-č-ího
LOC	nov-ěj-š-ím	lep-š-ím	ten-č-ím
INS	nov-ěj-š-ím	lep-š-ím	ten-č-ím
DAT	nov-ěj-š-ímu	lep-š-ímu	ten-č-ímu

However, this leaves it unexplained why there are three comparative allomorphs in Czech. In particular, how did the *–ø* allomorph, an example of which is seen in the third column, arise? According to Kosek (2014:131-2), this paradigm is to be interpreted as a case where paradigm levelling proceeded in the opposite direction, and the oblique forms were attracted to the nominative. He writes: “during the development of Old Czech in the middle period, an a-historical way of comparative formation by the suffix *–í* came to existence. This new type of comparative marking represents the emancipation of the original M.NOM.SG form” (translation ours).

What happened, then, is that in some cases, all the case forms levelled in the direction of the NOM.SG (nonfeminine). However, the NOM.SG had no comparative marker, and since the endings could not have been analysed as spelling out the comparative C2 any longer (as this was impossible in the other classes, which had the same agreement markers plus the C2 marker *š*), the only option was to analyse the paradigm in a way that it was the root which spelled out both C1 and C2. In sum, different directions of paradigm levelling led to the result that a new class of comparatives was born.¹⁷

The final point we want to turn to is a comment on the analysis of OCS presented in Bopp (1833), brought to our attention by an anonymous SL reviewer. The first thing to note is that Bopp (1833:421-6) (unlike the later grammarians) distinguishes not two, but four different ways of marking the comparative. The masculine singular forms of Bopp's four allomorphs are given in the table (61). The rows correspond to our well-known distinction between two types of roots: those that spell out C1 are on the first row, those that fail to spell out C1 are on the second row. In Bopp's work, this two-way distinction is cross-cut by an additional distinction, namely whether C2 is overt or not. In the left column (labelled 'C2 zero'), we find the forms that we are already

¹⁷ It is worth noting here that in the genetically closely related language Polish there was an evolution toward a system where the distribution of the two allomorphs (*-ejsz(y)* and *-sz(y)*) is phonologically conditioned (Rubach 1986:272; Bethin 1987; Szpyra 1992:286).

familiar with from Aitzetmüller (1978). Here, as we have proposed, the agreement markers spell out C2. In the second column (labelled ‘C2 overt’), we find two new forms where $-(j_b)\check{s}$ appears to the immediate left of the agreement marker. The segmentation of the forms is ours, agreement in brackets.

(61) OCS NOM.SG.M according to Bopp (1833:421-6)¹⁸

	C2 zero	C2 overt
C1 zero ($\sqrt{bol-}$)	$(-j_b-j_b)$	$-\check{s}(-j_b-j_b)$
C1 overt ($\sqrt{nov-}$)	$-\check{e}(-j_b)$	$-\check{e}-j_b\check{s}(-j_b-j_b)$

When we look at the new forms in the final column with the knowledge of the later development of the OCS comparative, it seems that Bopp’s allomorphs with $-\check{s}$ are not special independent allomorphs of the comparative, but rather the harbingers of paradigm levelling. This interpretation of Bopp’s forms with $-\check{s}$ is supported by the fact that the

¹⁸ Bopp explicitly distinguishes only three ways of marking the comparative, and introduces the fourth way ($\check{e}-j_b$) as a sub-paradigm of j_b-j_b , even though the two markers do not appear on the same roots. He also uses slightly different transcription, shown in (i).

(i)	M	F	N
a.	$-je\check{i}\check{s}i\check{i}$	$-je\check{i}\check{s}aja$	$-je\check{i}\check{s}ee$
b.	$-\check{s}i\check{i}$	$-\check{s}aja$	$-\check{s}ee$
c.	$-i\check{i}(-je\check{i})$	$-\check{s}i$	$-jee$

Here Bopp’s *je* corresponds to Aitzetmüller’s $\check{e}$, Bopp’s $\check{i}$ and *i* correspond to Aitzetmüller’s j_b . Bopp’s $-\check{s}$ initial suffixes in (ib) are rendered as $-j_b\check{s}$ in Aitzetmüller ($-b\check{s}$ in Lunt), which unifies the two distinct $-\check{s}$ forms in (61).

forms on the first row are illustrated by one root (Bopp gives *un(-jb-jb)* well as *un-š(-jb-jb)* ‘better, M.SG’), while the forms on the second row are illustrated by another one, namely by *jun-ě(-jb)* and *jun-ě-jbš(-jb-jb)* ‘younger, M.SG.’

Being primarily interested in the origin of these forms, Bopp hypothesised that the form in the bottom right corner (*-ějbšjbjb*) could be segmented as *-ějb-š-jb*, and analysed as combining the *-ějb* of the left-bottom cell, and the *-š-jb-jb* of the top right cell. If this analysis was correct, it would entail that the *-ějb* form in its entirety is C1, with C2 radically missing. As an anonymous reviewer points out, this would be problematic for our claims.

For reasons that were made clear above, we do not think that this problem is real. The reason is that Bopp’s analysis of the forms is rather superficial. Firstly, he ignores the fact that the variation in the table is split along two dimensions. As a consequence, he assumes that the innovative form (*jun-ě-jbš(-jb)* ‘younger’) arose as a result of augmenting its old form *jun-ě(-jb)* with a marker of the innovative shape belonging to a different root class. This completely misses the logic of a parallel development where both root classes simply adjusted their nominatives to the rest of the paradigm (a fact that Bopp fails to see, for he only looks at the nominative forms). In addition, Bopp in effect suggests that the whole old ending *-ě-jb* corresponds to an invariant C1 in all gen-

ders and numbers, which seems unrealistic, given that *-jb* in *jun-ě-jb* is a M.SG ending (as Bopp himself admits). For all these reasons, we do not think that Bopp's reasoning represents a valid base for rejecting our claim that there are roots which require the pattern with C1 overt and C2 silent. This conclusion seems to have been also adopted in the later descriptive tradition (Aitzetmüller 1978:131-2 and Lunt 2001:77-8), who rejected the innovative forms as such, and removed them from the descriptions.

An important take-home message is that many of the same patterns that we find in present-day Czech have been part of the language for more than a thousand years, demonstrating the robustness and stability of the “double comparative” marking. Also, we have seen that there are circumstances under which C2 may be silent, even when C1 is present, which we predict to arise only in case C2's zero exponence is triggered by an inflectional marker higher up in the structure, as is the case in OCS. Finally, we acknowledge here Bopp's thinking as a precursor to our ideas, though we are not sure that the reasons that led him to his “double comparative” analysis are correct.

5. English comparatives

5.1. *The issue*

The theory outlined so far leads to an expectation that more languages will exhibit a difference between two types of adjectival roots, which differ in their size. In this section, we argue that the difference between *–er* and *more* comparatives in English can be captured along these lines, which leads to an interesting prediction about suppletion, to be explored in the next section. Specifically, following [Corver \(1997\)](#); [Bobaljik \(2012\)](#), we will analyse *more* as a complex marker, since it is the comparative form of *much*. Building on this observation, we can establish the following (tentative) Czech-English correspondence table, where we draw a parallel between the Czech complex marker *–ěj-š* and the English *more*, as well as a parallel between the simplex comparatives:

(62)

√	Q	C1	C2
bujar		ěj	š
star			š
intelligent	mo		re
old			er

In the table, we have decomposed *more* into two pieces, but it may also be considered an opaque piece of morphology realising both C1 and C2. The proposal we are about to develop is independent of the surface

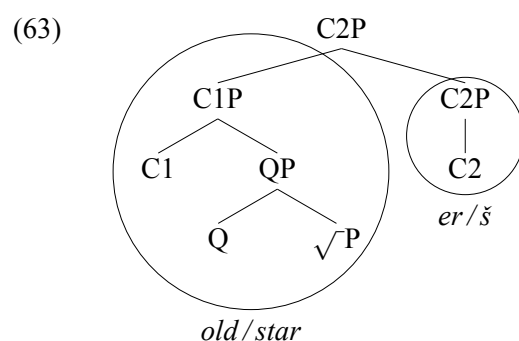
decomposability of *more*, and we will in fact adopt a proposal where *more* is a portmanteau for C1 and C2.

As a result of attributing different complexity to the two different comparative markers (C2 for *-er*, C1+C2 for *more*), and assuming that the underlying structure of the two classes of comparatives is the same, we automatically arrive at different lexical specifications of the roots which combine with these markers. Specifically, roots like *intelligent* are specified as a QP, while roots like *old* are specified as C1P. The result is a theory where the selection between a root and a particular comparative marker is a consequence of different root size.¹⁹

Based on the theory we have laid out in section 3, we can straightforwardly express the parallel between Czech and English in the form of tree representations. What we are proposing here is that English comparatives also involve two comparative heads. In comparatives expressed by *-er*, the suffix spells out a single feature (C2), and the whole class is thus analogous to *š-i* comparatives in Czech. In order to bring out the

¹⁹ A lexical approach to the distinction between morphological and periphrastic comparatives in English is also taken by Bobaljik (2012); Gouskova & Ahn (2016), among others. This is not to deny that there might be certain phonological or other regularities governing the choice for either option. There is in fact an extensive literature on this issue, which has identified a range of factors relevant to the choice, such as the frequency of the adjective, as well as a variety of phonological factors (see e.g. Jespersen 1954; Aronoff 1976; Quirk et al. 1985; Culpeper & Leech 1997; Graziano-King 1999; Mondorf 2003; Graziano-King & Smith Cairns 2005; LaFave 2005; Hilpert 2008; Mondorf 2009; Bobaljik 2012; Matushansky 2013; Enzina 2017).

parallel, we have simply placed the Czech lexical items for the comparative ‘old-er’ alongside the English lexical items.



For the English analytic comparatives, however, a different structure is needed than for the Czech *–ěj-š-i* comparatives, despite the fact that both *–ěj-š-i* and *more* combine with roots of the QP size. If we simply specified *more* for the features C1 and C2, similarly to the Czech suffix *-k* in (37), we would expect that the root would have to move to the left of *more* (deriving **intelligentmore*), just like in Czech the root moves to the left of *-k*. In order to satisfactorily address this issue, we need to return to the spellout algorithm, and extend it to account for comparative markers preceding the adjectival root.

5.2. Merge (X,Y)

Recall from section 3 above that in our system, all suffixes arise due to complement extraction. This is shown in (64), where the complement

of F has moved up, landed at a place indicated by the dots, and left just a single branch inside FP.²⁰

(64) [... [FP F]]

Lexical items in our system are simply links between a well-formed syntactic structure and its pronunciation. Therefore, lexical entries for suffixes will naturally copy this tree-shape, and always contain a non-branching FP as their lowest ingredient. However, this option does not exhaust the logical possibilities of what a syntactic structure may look like, and by the same token, what a lexical entry may look like. In particular, a lexical entry can also correspond to a syntactic object created by the merger of two features, as in (65). Ultimately, this will be a type of entry which we will use for prefixes, and we will now work towards clarifying the reasoning behind this conclusion.

(65) [F_{2P} F₂ F₁]

We start from the observation that the entry (65) spells out a node (F_{2P}) out of which nothing has been extracted. Such structures may correspond to various objects in the ‘real world,’ depending on the overall structure of the syntactic theory. The least controversial use of such lex-

²⁰ From here on, we shall omit traces. Since their presence may be inferred from the nonbranching projecting structure [FP F], that structure serves as the equivalent of one with a trace.

ical entries is when they correspond to roots. For instance, in Bobaljik (2012), an example of such an entry is when $[A + \text{CMPR}]$ are spelled out together, which corresponds for instance to a suppletive comparative root like *worse*. In our system, a root like *bujar* ‘merry’ realises $[Q + \sqrt{\quad}]$, which also corresponds to a syntactic object of the type in (65).

Such (internally) complex roots may of course move (by spell-out-driven movement, for instance), and then they end up in the specifier of a functional head, as in (66), where F2P is the phrase corresponding to (65), and F3 is the functional head in whose Spec it resides after movement from its complement.

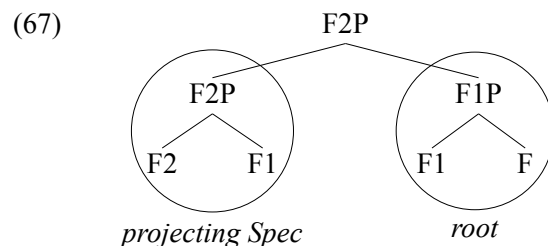
(66) $[_{F3P} \text{ F2P } [_{F3P} \text{ F3 }]]$

The placement of F2P into the Spec of F3P does not contribute anything to the meaning composition, which is reflected by the fact that the specifier does not project, and F3 remains the head of the whole phrase. However, Starke (2004) proposes that in addition to such ‘non-projecting’ specifiers, there are also ‘projecting’ specifiers. Projecting specifiers are like heads in that they contain features inside them that contribute to the building of the main projection line, but which at the same time are internally complex. Starke (2004) discusses *wh*-phrases as a case in point, and such structures have recently been used in Caha &

Pantcheva (to appear) or Taraldsen et al. (2018) to deal with classifiers in Bantu.

To approximate the idea on Indo-European data, consider, for instance, the Norwegian word *kopp* ‘cup.’ It has a use as an ordinary noun, referring to an object. However, it also has a classifier use. Specifically, in Norwegian, one may say *en kopp kaffe*, literally ‘a cup coffee’ (without *of*), where *kopp* resembles a functional head, which partitions the mass denotation of coffee into units containing the quantity defined by the classifier. However, even when used as a classifier, *kopp* retains nominal characteristics. First of all, it defines the relevant QUANTITY by reference to the prototypical size of a THING, namely *a cup*. This can be captured if *cup* has a lexical entry [QUANT N]. The presence of N inside the classifier (which overlaps with that of *coffee*) is confirmed by its formal behaviour: it controls agreement on adjectives that are placed higher up in the structure.

A projecting Spec is thus a phrasal object, which contributes to the meaning composition and provides its label to the whole structure, as in (67). Such a structure may also allow for a (partial) overlap in features between the Spec and the root (both have F1).



We further adopt the proposal by [Starke \(2018\)](#), which says that such projecting Specs correspond not only to separate words, but also to pre-fixes.

With the background clarified, let us now turn to the issue of how the formation of a projecting Spec fits into the overall architecture of a theory based on spellout-driven movement. Obviously, constructing a complex Spec presupposes that we create the Spec in a separate derivational workspace, and only then add it to the main projection line.

Let us consider an abstract example of a derivation with such a projecting Spec in the form of a prefix. Suppose that we have added the feature F3 to a phrase containing F2 and F1, as in (68a). Suppose that F2P spelled out successfully as β given the lexical item in (68b), and further that the only entry containing F3 is the one in (68c).

- (68)
- a. $[_{F3P} F3 [_{F2P} F2 F1]]$
 - b. $[_{F2P} F2 F1] \Leftrightarrow \beta$
 - c. $[_{F3P} F3 F2] \Leftrightarrow \alpha$

In such a scenario, the spellout of F3P would fail no matter what movement we try. If we extracted the complement of F3, we would need to spell out the boldfaced constituent in (69a). However, this syntactic constituent is not identical to any node contained in the entry (68c).

- (69) a. $[_{F3P} [_{F2P} F2 F1] [_{F3P} \mathbf{F3}]]$
 b. $[_{F3P} [F1] [_{F3P} \mathbf{F3} [_{F2P} \mathbf{F2}]]]$

Similarly, if we extracted the complement of F2, F1, raising it to adjoin to F2P, and subsequently applied spec-to-spec movement to yield (69b), we would have to spell out the boldfaced constituent in (69b). This constituent does contain only the features F3 and F2 (like the entry (68c)), but what is required for spellout in Starke (2009) is that the lexical entry contains a constituent that is fully identical to the syntactic structure. In (69b), the non-branching node above F2 makes the structure different from (68c).

What we then need to do is to add F3 not as a single feature, but bake it into a complex Spec by starting a new derivation that will merge F3 with some other feature. Which feature is this going to be? The two most salient options are either the next feature higher up (F4), or the next feature lower down (F2). If we choose the former option, we end up with a Spec consisting of F4 and F3. This is sub-optimal, since this way we may be forced to merge more features than we actually

want, like F4 in the above example. A more concrete case would be one where, for instance, we want to merge the gradability feature Q, and the only way to do it is to add it as a part of a complex Spec. If we defined things in a way that Q has to merge with the next feature higher up, we would have to add C1 to Q. But such a result is unusable, since we would derive a (semi-)comparative instead of a positive. Formulated more generally, if a Spec spawned by the need to spell out F always adds F and F^{+1} , unwanted results arise. Therefore, we suggest that a new derivation spawned by the need to spell out F always merges F with F^{-1} , and that it is this combination that is the first thing the system constructs after opening the new workspace.

We further assume that before F is used to start a new derivation, it is removed from the main derivation, since it cannot get spelled out there. We follow [Starke \(2018\)](#) in considering complex Spec formation as the absolute last resort, which is tried only after everything else fails. All of this is integrated into the spellout algorithm in (70).

(70) *Spellout Algorithm (v3, preliminary)*

- a. Merge F and spell out.
- b. If (a) fails, try spec-to-spec movement of the node inserted at the previous cycle, and spell out.
- c. If (b) fails, move the complement of F, and spell out.

- d. If (c) fails, remove F from the main workspace, merge it with F^{-1} and spell out.

The result of the algorithm is that if it comes to (70d), we end up with two workspaces, one old, the main projection line, and the other freshly created. Both can be spelled out, but they are not integrated together. As to when exactly should the two workspaces be joined again, is a topic of ongoing research. Nevertheless, a simple implementation which works well for the purpose of this paper is that the second workspace remains open and subject to further merge F operations only as long as the Spec spells out as one piece. Once we would need to do movements within the Spec, the workspace is closed and the Spec integrated with the main workspace. This is the content of (71b), which we add as a new and final step in the formulation of the spellout algorithm.

(71) *Spellout Algorithm (final version)*

- a. Merge F and spell out
- b. If (a) fails in the second workspace, undo merge F , and close the workspace by merging it with the main derivation
- c. If (a) fails in the first workspace, try spec-to-spec movement of the node inserted at the previous cycle, and spell out.
- d. If (b) fails, move the complement of F , and spell out.

- e. If (c) fails, remove F from the first workspace, merge it with F^{-1} in a second workspace, and spell out.

With the theory in place, we turn back to comparatives with *more*.

5.3. Analytic comparatives

Recall that in our analysis, adjectives which combine with the comparative marker *more* are simple gradable adjectives, i.e. they spell out QP. They differ from adjectives which mark the comparative suffixally, which spell out C1P:

(72) a. [QP Q $\sqrt{\quad}$] $\Leftrightarrow$ /intelligent/

b. [C1P C1 [QP Q $\sqrt{\quad}$]] $\Leftrightarrow$ /old/

(73) [C2P C2] $\Leftrightarrow$ /-er/

Given that *-er* only spells out C2, (72) correctly encodes the fact that the root *intelligent* cannot combine with *-er* to form a comparative, since neither the root nor the suffix spell out C1. In the analysis we propose, the C1 feature is present in the lexical entry for *more*. What does this entry look like beyond having C1?

Recall that in the theory of spellout-driven movement that we have developed above, PRE-markers have to have a specific shape of their lexical entry, which at its bottom contains both the feature to be spelled

out (i.e. C1), but also an overlapping feature to which C1 is added in order to form a left-branch Spec.²¹ This has to be the Q feature, which is the topmost feature of an adjective like *intelligent*. In addition, *more* also spells out C2, and its lexical entry must then be as follows:

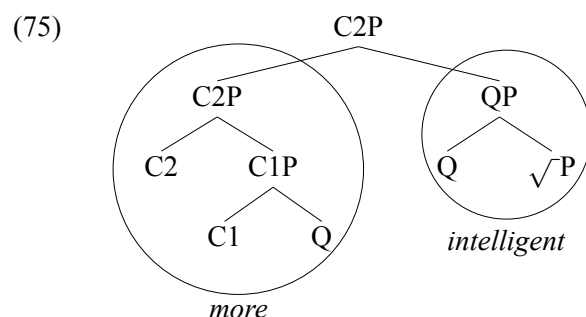
$$(74) \quad [C2P \ C2 \ [C1P \ C1 \ Q]] \Leftrightarrow /more/$$

Including the Q in the lexical entry is not only theoretically required, but it is also necessary in order to express the suppletion relation between *more* and *much*, where *much* is a Q marker (rather than a C1 marker), as argued in De Clercq & Vanden Wyngaerd (2017). We will say more about suppletion in the next section.

With the lexical entries in place, let us turn to how the derivation proceeds. Suppose we construct QP and spell it out by *intelligent*. We then go on to merge C1. However, *intelligent* cannot spell out C1, since its lexical entry does not contain it. Therefore, repair operations are triggered. These fail to deliver the right configuration for *more*, essentially because they fail to construct a constituent with a binary bottom, which is needed for *more*. The derivation therefore ultimately has to continue by creating a complex Spec in a separate workspace, merging C1 with

²¹ We adopt here the process of complex spec formation as outlined in Starke (2018). De Clercq & Vanden Wyngaerd (to appear) explore the potential empirical benefits that arise from assuming that no feature overlap is possible between a complex spec and the main derivation. Since this issue is orthogonal to our concerns, we do not explore it any further here.

Q. Since the Spec spells out directly, the feature C2 is merged with the specifier in the separate workspace. Once the Spec is integrated with the main projection line, the final product looks as follows:



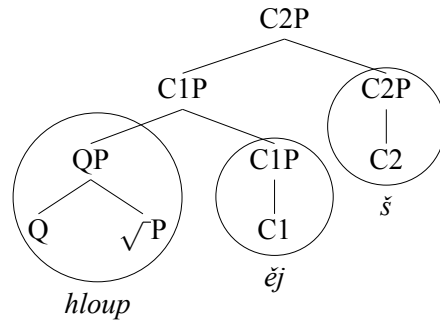
This is (obviously) a different derivation than the one found with Czech QP-sized roots. In Czech, when C1 is added, the derivation is steered in a different direction since the lexicon contains the suffixal C1 marker *-ěj*:

- (76) a. $[_{C1P} C1] \Leftrightarrow /ěj/$
 b. $[_{C2P} C2] \Leftrightarrow /š/$

Due to the existence of *-ěj*, a snowball movement of QP succeeds in producing a remnant C1P that can be lexicalised (see the C1P in (77)). This feeds into the next cycle, where C2 is added, first as a feature to the left of the structure spelled out as *root+ěj*. But then again, due to

the lexical entry for $-\check{s}$ in (76b), another snowball movement leads to a successful spellout. The result of the derivation is shown in (77):

(77) The $-\check{e}j\check{s}$ comparative



Ultimately, the relevant difference between the different kinds of structures is triggered by the fact that roots and affixes vary in their size and shape. Because of the way spellout proceeds, the roots always spell out as much structure as possible, and the rest is left for post-markers (suffixes) or pre-markers (*more*). Notice that on this approach, we do not need any *ad hoc* lexical diacritics to express how a root selects the right allomorph. The correct structures simply arise as the result of an interaction between the spell-out algorithm and the size and shape of the lexical items.

6. Suppletion

6.1. Portmanteau suppletion and contextual suppletion

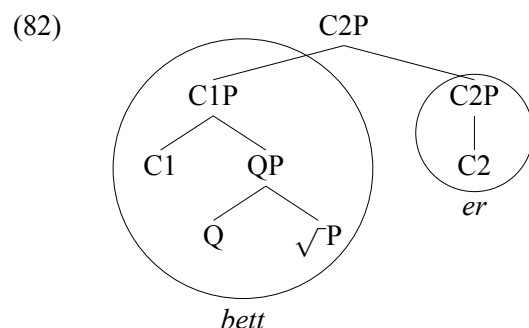
The approach to allomorphy of the comparative in Czech provides us with the possibility to explore a new—more restrictive—approach to suppletion in comparative formation than standardly assumed. In order to show that, we first consider the Distributed Morphology (DM) approach to this phenomenon, in particular the one developed by Bobaljik (2012).

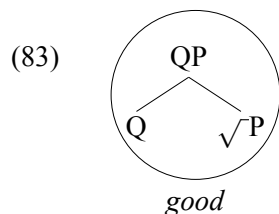
In this approach, there are two different kinds of suppletion, (which we will call) portmanteau suppletion and contextual suppletion. The first kind is instantiated by such pairs as *bad-worse*. Bobaljik proposes that the suppletive form is a portmanteau for a complex node containing the A node and the CMPR feature.²² The Vocabulary Items for this case are shown in (78). They are an intuitive way of encoding that *worse* in English conveys the meaning of both the root meaning BAD and the meaning of CMPR.

- (78) a. $\sqrt{\text{BAD}} \oplus \text{CMPR} \Leftrightarrow \text{worse}$
 b. $\sqrt{\text{BAD}} \Leftrightarrow \text{bad}$

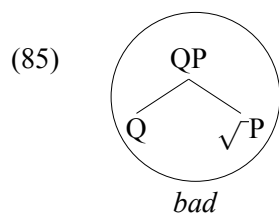
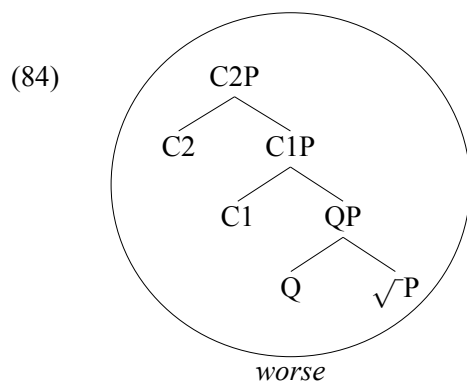
²² Bobaljik is noncommittal about the question how this complex node arises, either through fusion of two heads under a new (complex) terminal, or through spellout of a nonterminal node that dominates both heads.

sometimes, suppletion apparently co-occurs with overt marking. For instance, Embick (2016:275) observes that phrasal spellout theories predict that ‘[s]pecial stem allomorphs of a Root triggered by a feature [X] should not co-occur with an independent realization of [X].’ But this prediction is incorrect, as in the case of the comparative *bett-er*, since suppletion triggered by comparative morphology co-occurs with an overt comparative marker. However, once *cmpr* is decomposed into C1 and C2, this objection vanishes, and the portmanteau (or phrasal spellout) analysis captures the facts straightforwardly. Specifically, we can say that *bett-* spells out the root plus C1, and that *-er* spells out C2. This is shown in (82), where the circles indicate how the syntactic constituents map onto spellouts. This way, *bett-* differs from the positive *good* given in (82), and still leaves space for the comparative *-er*.



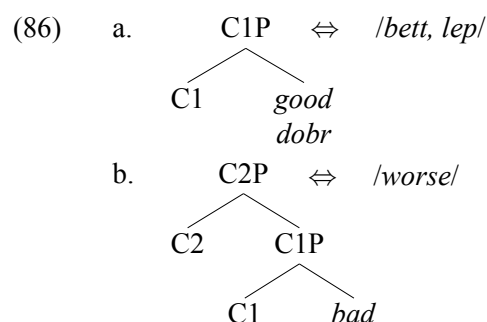


This approach also allows us to capture cases like *bad—worse*, where there is no apparent double marking, see (84) and (85).



In our theory, suppletive roots have very much the same entries like non-suppletive roots. For instance, *bett–* spells out **C1P**, just like for instance *old* (recall (63) above). Similarly, a root like *worse* is similar to a root like *ostr–* (see recall (26) above). Where suppletive roots differ is that

they entertain a lexical relation with another lexical item. We represent the lexical relatedness of the *good*–*bett* pair by assuming that the lexical item *bett*– contains a reference (a pointer) to an existing lexical item, *good*, as in (86a). Similarly, *worse* contains a pointer to *bad*.



Items with pointers like *bett*– are read as follows: C1P can be spelled out as *bett*– only if in the previous cycle QP has been spelled out as *good*. This means that if the syntax builds QP, only *good* is a candidate for insertion. *Bett*– only becomes a candidate after C1 is merged with *good*. The insertion of *bett*– at C1P will prevent the pronunciation of *good* in the comparative because of Cyclic Override: a successful spellout of C1P will override any earlier spellout of material contained inside C1P.

Non-suppletive roots with a morphological comparative (like *old*) spell out C1P exactly like *bett*– in (82), but lack a lexically related counterpart which they point to; as a result, they will be inserted both in the positive degree (as QPs) and as the comparative degree (as C1Ps).

Note that pointers in our system point to lexical items. This can be interpreted in two ways. First, we can assume that lexical entries can be uniquely identified by their phonological representation.²³ This might be an issue under the following circumstance: suppose the pointer points to a phonology / α /, and there are two lexical entries with the same phonology / α /, i.e. homophones, but only one of these is suppletive. Under this scenario, phonology is insufficient to uniquely pick out the lexical item that undergoes suppletion. An anonymous reviewer brings up English *bad* as a relevant case (with reference to Bobaljik's work). This has the usual, negative, meaning, but also a positive one ('awesome, cool'), which has a non-suppletive comparative and superlative (*badd-er*, *badd-est*).²⁴ The system could deal with this case, minimally, by distinguishing homophones with an index on their phonologies, so that there would be / α_1 / and / α_2 /.

The second potential approach to pointers would be to assume that they point to the full lexical entry, which is a triplet of the type <PHON, SYNT, SEM>. Under this approach, the morphologically regular *bad* 'cool, hip' would count as a different entry from the morphologically irregular *bad*, since their SEM is different. For practical purposes (to

²³ The objection raised in Harley (2014) against identifying roots by their phonologies concerns suppletion, but in our system this objection does not hold because of the existence of the pointer in the suppletive entry.

²⁴ We assume here, for argument's sake, that the two senses of *bad* correspond to two different lexical items, but see Arregi & Nevins (2014) for a different view.

avoid clumsy notation), lexical items in the postsyntactic lexicon can be assigned a unique (numerical) index, and then pointers can be written using these indices.²⁵ Since our system can deal with the problem mentioned in two different ways, i.e. it is compatible with both of the approaches just sketched, we shall not attempt to resolve the issue here.

To round up, at this point, we have rephrased Bobaljik's contextual allomorphy rule in terms of portmanteau suppletion, while at the same time avoiding the problem of apparent double marking. Such a reduction of two mechanisms for suppletion in favour of a single mechanism is theoretically attractive. In addition, it makes our theory also more empirically predictive, as we argue below.

6.3. *The Comparative Suppletion Generalization*

In order to see the prediction which arises when contextual suppletion is abandoned, consider the following reasoning. We start from the fact that under the portmanteau-based approach to suppletion, we get a suppletive root pair only when we have two entries for a root. The positive degree entry spells out a QP, and the suppletive comparative entry adds either C1 (*bett-*) or C1 and C2 (*worse*). From this setup, it follows that suppletive roots cannot leave both comparative heads C1 and C2 available for insertion. This predicts that a bi-partite comparative marker (which

²⁵ Note that this solution is different from distinguishing roots by an index in the presyntactic lexicon.

necessarily spells out both C1 and C2) is incompatible with suppletion (which spells out minimally QP+C1). We state the prediction below.

(87) *The Comparative Suppletion Generalisation (CSG)*

When the comparative is expressed by two overt markers in addition to the root (or when the comparative marker is a portmanteau that expresses the content of these two markers), there is no suppletion.

The CSG, when applied to English, predicts that syntactic comparatives with *more* never trigger suppletion: since *more* spells out both C1 and C2, it is predicted to be incompatible with suppletive roots.²⁶ This prediction is borne out, and has been observed independently in Bobaljik's study. However, he proposes to subsume it under a different generalisation, namely the *Root Suppletion Generalisation* (RSG) (Bobaljik 2012:3):

(88) *Root Suppletion Generalisation*

Root suppletion is limited to synthetic (i.e. morphological) comparatives.

²⁶ Empirically, it seems to be the case that prefixes are incompatible with suppletion in general. De Clercq & Vanden Wyngaerd (to appear) attempt to derive this result within the current framework. See also Moskal (2013) on prefix-suffix asymmetries.

Bobaljik proposes to derive the RSG as a consequence of a locality restriction, which prevents contextual suppletion between items separated by a word boundary (or, more specifically, by a phrasal projection of the head). What we have seen, however, is that the facts which the RSG derives for English can also be derived by the CSG. In order to see where the predictions of RSG and CSG differ, we would like to turn back to the Czech data, which provide an interesting test case for the two approaches.

6.4. *Suppletion in Czech*

Recall that in Czech, there is a productive bi-morphemic comparative marker *–ěj-š-(i)*, and two reduced comparative markers (*–š-(i)* and the zero marker). The important difference with English is that all the markers, including *–ěj-š*, are word-internal, and so the approach based on Bobaljik's RSG and contextual suppletion makes no predictions about this case. In particular, there is no reason to expect that *–ěj-š*-comparatives should be incompatible with suppletion. However, if we restrict suppletion to portmanteau suppletion (to the exclusion of contextual allomorphy), we expect that there should be no suppletion in *–ěj-š*-comparatives, because these spell out two different heads, and should therefore block suppletion by the CSG (87). This section explores this prediction and argues that it is in fact borne out.

Czech descriptive grammars vary as to how many forms are considered suppletive, since the phenomenon is known to be gradual and there is a fuzzy boundary between ‘pure’ suppletion, irregularity (e.g., irregular lengthening or reduction of the root) and full concatenativity. With this in mind, let us turn to the actual descriptions. Dokulil et al. (1986:379) and Ziková (2016) list five suppletive adjectives which are considered ‘fully irregular’; these are given in (89). Osolsobě (2016) considers the first four suppletive, while the last one of them is ‘irregular, but not suppletive’.²⁷

(89)	ADJECTIVE		ADVERB		
	POS	CMPR	POS	CMPR	
a.	dobr-ý	lep-š-í	dobř-e	lép-e	‘good’
	špatn-ý	hor-š-í	špatn-ě	hůř-e	‘bad’
	mal-ý	men-š-í	mál-o	mén-ě	‘small’
	velk-ý	vět-š-í	hodn-ě	víc-e	‘big’
b.	dlouh-ý	del-š-í	dlouz-e	dél-e	‘long’

For completeness, (89) also lists the forms of the corresponding adverbs, in the final two columns. These show the same suppletive root as the adjective (though subject to palatalisation and vowel lengthening), but cru-

²⁷ We consider both of these cases suppletive, as we endorse a theory without morphologically triggered readjustment rules.

cially lack the *-ěj*, just like their adjectival counterparts. Non-suppletive forms mostly have *-ěj* both in adjectives and adverbs, as in (90).

(90)

ADJECTIVE		ADVERB		
POS	CMPR	POS	CMPR	
rychl-ý	rychl- ej -š-í	rychl-e	rychl- ej -i	‘fast’
červen-ý	červen- ěj -š-í	červen-ě	červen- ěj -i	‘red’
hloup-ý	hloup- ěj -š-í	hloup-ě	hloup- ěj -i	‘silly’
bujar-ý	bujar- ej -š-í	bujar-e	bujar- ej -i	‘merry’

As we saw earlier, the implication goes only one way: it is possible for a root to spell out C1P (and hence lack *-ěj*) without being suppletive, i.e. without having a lexically related competitor for the spellout of the positive degree (QP) (see (6) above). Therefore, the theory predicts that some non-suppletive roots may also lack *-ěj* both in the adjective and the adverb. One case confirming it is given in (91).

(91)

ADJECTIVE		ADVERB		
POS	CMPR	POS	CMPR	
drah-ý	draž-š-í	draz-e	dráž-e	‘expensive’

All facts considered, we conclude that all the suppletive adjectives listed in Dokulil et al. (1986:379) and Ziková (2016) belong in the class that

lacks the C1 marker *-ěj*. This fact can be seen as nothing but an instance of the CSG, repeated below for convenience:

(87) *The Comparative Suppletion Generalisation (CSG)*

When the comparative is expressed by two overt markers in addition to the root (or when the comparative marker is a port-manteau that expresses the content of these two markers), there is no suppletion.

6.5. *Some potential counterexamples*

In this section we take a closer look at two potential counterexamples to the CSG as formulated in (87) above. The most serious challenge to the CSG is posed by the adjective *brz-k-ý* ‘early’, which has the suppletive comparative *dřív-ěj-š-l* ‘former/earlier’, with the two comparative suffixes, in apparent contradiction of the CSG. The pair is treated as suppletive in Křivan (2012:26).

However, the overview article Osolsobě (2016) designates the suppletive relation as ‘controversial’. The reason for this is that there is a plausible alternative analysis of the comparative *dřív-ěj-š-l*, namely as an adjective derived from the comparative adverb *dřív-e* ‘earlier’. This analysis is adopted also in Dokulil et al. (1986:382).

The basic idea of these approaches is that the meaning ‘early’ is encoded in the Czech grammar primarily through temporal adverbials. This holds both for the positive, where the adverb *brz-o* ‘early’ seems to be the basic form, and the adjective *brz-k-ý* is derived from this adverb by the adjectival *-k*, which we have already discussed above (section 2.4). This is similar to English, where the adjective *early* (at least historically) also contains an adverb, specifically the adverbial suffix *-ly* attached on top of a cranberry type of root *ere-*.

The same derivational relationship carries over (for Czech) to the comparative. The idea is that the comparative adverb is again the base from which the adjective is derived. The derivational relations between adverbs and adjectives are depicted in (92), using simple arrows.

(92)

	POS		CMPR	‘early’
ADV	brz-o	⇒	dřív-e	
	↓		↓	
ADJ	brz-k-ý		dřív-ější	

In addition, the double arrow indicates the suppletive gradation of the adverb. Under this view, there is thus indeed a suppletive relation between the roots *brz-* and *dřív-*, but only in the adverbs. And crucially, here the CSG (87) is obeyed: the form *dřív-e* indeed lacks the overt C1 marker *-ěj*, and behaves like all suppletive adverbs, recall (89). No direct derivational relation holds between the two adjectives.

The only suspicious part of the analysis (92) is the fact that the marker which derives the adjective *dřív-ějš-í* from the adverb is the same as the comparative marker. This gives the impression that the form is actually composed of a suppletive root and the regular bi-morphemic comparative marker on top. However, the *-ějš-(i)* suffix independently turns adverbs into adjectives, as in the following cases, treated as analogous to the problematic form *dřív-ějš-(i)* in Dokulil et al. (1986:382):

(93)

	ADVERB		ADJECTIVE
	<i>zítr-a</i> ‘tomorrow’	<i>zítř-ějš-(í)</i>	‘tomorrow’s’
	<i>včer-a</i> ‘yesterday’	<i>včer-ějš-(í)</i>	‘yesterday’s’
	<i>zd-e</i> ‘here’	<i>zd-ějš-(í)</i>	‘local’
	<i>vn-ě</i> ‘outside’	<i>vn-ějš-(í)</i>	‘outside’
	<i>vedl-e</i> ‘beside’	<i>vedl-ějš-(í)</i>	‘neighbouring’
	<i>tam</i> ‘there’	<i>tam-ějš-(í)</i>	‘the N there’
	<i>nyn-í</i> ‘now’	<i>nyn-ějš-(í)</i>	‘present-day’

The table shows that there is indeed a suffix *ějš-(i)*, homophonous with the comparative marker, which derives adjectives from temporal and locative adverbs. The resulting forms are incompatible with any regular comparative modifier, like measure phrases, *than*-phrases, etc. Summing up, we propose that the adjective *dřív-ějš-í* ‘earlier’ is to be analysed as an adjective derived from a suppletive adverb (for reasons inde-

pendent of CSG). Once this analysis is adopted, the form is fully consistent with CSG; in fact, the suppletive adverb *dřív-e* adds another piece of evidence in its favour.

The second and final potential counterexample is the adjective ‘white’ (Janda & Townsend 2000:25). The comparative is either completely regular, or shows a change of the root vowel. The latter pattern is problematic, since this irregular shape is accompanied by the *–ěj-š-(i)* marker.

(94)	POS	CMPR	SPRL	
	bíl-ý	bíl-ejš-í	nej-bíl-ejš-í	‘white’
	bíl-ý	běl-ejš-í	nej-běl-ejš-í	‘white’

The pattern could be captured by saying that *bíl* is an elsewhere form, and that *běl–* is specifically comparative/superlative. Our theory would then predict that *běl* is incompatible with *–ějš*, contrary to fact. However, there is an independent problem with this analysis, namely that *běl* also appears in contexts which reject comparative roots. To see this, let us turn to property denoting nouns like *stupid-ity*. Such nouns are derived in Czech by the suffix *–ost* ‘-ness/-ity’, which attaches to the positive roots:

(95)

ADJECTIVE		NOUN		
POS	CMPR	POS-BASE	CMPR-BASE	
špatn-ý	hor-š-í	špatn-ost	*hor-ost	‘bad(ness)’
mal-ý	men-š-í	mal-ost	*men-ost	‘small(ness)’
nízk-ý	niž-š-í	nízk-ost	*niž-ost	‘low(ness)’

With this in mind, it is surprising that the noun for ‘whiteness’ can be both *běl-ost* (the preferred choice) and (more marginally) *bíl-ost*. We expect only the latter, as *-ost* otherwise never attaches to comparative stems. This makes it difficult to analyse *běl-* as a comparative form.

Another argument to the same effect comes from the phenomenon of A-o-N compounding. This involves a sequence of an adjective and a noun, which normally shows up as a sequence of phrases, as in the first column of (96). This sequence may be subject to adjectivising suffixation with *ští*. In such circumstances, the adjective of the A-N sequence loses its agreement, and appears followed by an invariant ‘connecting morpheme’ *-o* instead. This is shown in the second column of (96).

(96)

	A+N		A-o-N compound
a.	velk-á	Morav-a	velk-o-morav-ští
	great-AGR	Moravia-F	great-o-morav-ian
b.	bíl-á	hor-a	běl-o-hor-ští
	white-AGR	mountain-F	white-o-mountain-ian

What is relevant for us is that in the course of this process, the adjective ‘white,’ shown in (97b), switches from the root shape *bíl-* to the root shape *běl-*. In this context, however, the root shape *běl-* must be interpreted as a positive degree. The reason is firstly interpretation, and also the fact that suppletive adjectives like *velk-ý* ‘great’ show unambiguously positive in the compound. This leads us to suggest that there are actually two independent roots, *běl* and *bíl*, each specified as a regular positive degree adjective, i.e. QP, and each with its own (almost) full paradigm. This idea, where each of the relevant forms is expected to have two shapes, is depicted below (POS1 is the root that appears in the positive degree, POS2 the one in incorporated adjectives like (96)):

(97)	ADJECTIVE			NOUN	
	POS1	POS2	CMPR	POS-BASE	
ROOT 1	bílý	bíl-	bílejší	bíl-ost	‘white’
ROOT 2	bělý	běl-	bělejší	běl-ost	‘white’

The distribution of the roots in the positive degree is still to be explained, but this cannot be done so by reference to degree, because of the appearance of *běl-* in the nominalisation and in the incorporated positive. In sum, we conclude that there are good reasons to assume that the CSG

holds without exception, and constitutes a strong generalisation that tells us something about how suppletion works.²⁸

7. Faithfulness

At this point we need to address two different, but related, problems with our assumptions so far. Both of these problems will be solved by introducing a Faithfulness Restriction on the principle of Cyclic Override. To see the issue, recall that we have assumed the existence of adjectives of varying sizes, as in (34) above, repeated here, which are represented arboreally in (98). The two comparative suffixes of Czech are repeated in (35):

²⁸ Janda & Townsend (2000:25) also give the forms for the adjective ‘holy’ as in (ia):

(i)	POS	CMPR	SPRL	
a.	svat-ý	(??)svět-ější	nej-svět-ější	‘holy’
b.	svat-ý	svat-ější	nej-svat-ější	‘holy’

These would lead to similar problems as discussed for ‘white.’ However, according to native-speaker judgements, the comparative is marginal at best, as indicated by the question marks. The superlative is well-formed, but it only has an ‘elative’ meaning (‘holy to a very high degree’) and it lacks the relative meaning ‘holier than all others’. To express the latter meaning, the ‘regular’ series would have to be used, as given in (ib). Elative superlatives can be independently shown not to conform to the principles that govern the form of regular comparatives and superlatives. This has been argued by Bobaljik (2012), who shows that elatives display ABA patterns, as in Italian *buono-migliore-buonissimo* ‘good-better-excellent’. Therefore, we set this example aside.

- (98)
-
- ostr
- star
- bujar
- C2P
- C2
- C1P
- C1
- QP
- Q
- √

²⁹ To be consistent with our earlier assumptions about the first Merge operation in the derivation, this would require that $\sqrt{}$ be binary, i.e. composed of at least two even smaller heads. Since the argument we develop is independent of this issue, we ignore this complication for now.

be free, depending on what one wants to express. But this intuition is incompatible with the Elsewhere Principle, which in the case of multiple candidates for spellout selects the one that is the closest match with the syntactic structure, and discards all others. This would in effect allow only the insertion of adjectives of the *kluz*-type at $\sqrt{}$, i.e. the ones with the smallest lexical trees, and rule out insertion of all the other types of adjective. If we go on to merge Q, creating QP, consulting the lexicon finds adjectives of that size and will insert one of them, to the exclusion of all adjectives of different sizes. As a result, a root like *star* ‘old’ (lexically specified as C1P) will not be allowed to surface in the positive degree, corresponding to a QP. We shall call this problem the problem of Free Choice: free lexical choice is severely restricted by the system set up so far, in a way that is undesirable.

The free choice problem was noted, in a different but related context, by Marantz (1995; 1996; 1997). Marantz observes that allowing competition at the level of root insertion will have the effect that a suppletive root will win the competition from all others (since it is more specific), preventing their insertion, and thus Free Choice. Marantz (1997) proposes to solve the problem by claiming that root suppletion does not exist (and that apparent counterexamples like *go-went*, *bad-worse* instantiate the functional vocabulary; see also Embick & Halle 2005). Harley (2014) argues against Marantz’ position, claiming instead that

(syntactic) roots are individuated in the syntax, i.e. prior to vocabulary insertion, by means of a numerical index (following [Pfau 2000; 2009; Acquaviva 2009](#)). This implies that the presyntactic lexicon contains a multitude of different $\sqrt{}$ s, as many as there are vocabulary items. Insertion rules then eliminate the competition between roots by making reference to these indices (except the suppletive ones, which have the same index).

Under this conception, the Free-Choice problem does not arise, as the syntactic computation is free to select a particular $\sqrt{}$ (say $\sqrt{539}$) to start the derivation, and post-syntactic insertion is restricted to the lexical item corresponding to that root. We believe this solution goes against the spirit of modularity, in particular, the separation of the syntax and the (postsyntactic) lexicon. As pointed out in [Marantz \(1996\)](#), the difference between *cat* and *dog* is irrelevant to syntax, and therefore syntax should not encode the difference between them. This is an important piece of motivation for late insertion models. Numerical indices on roots reintroduce the difference between *cat* and *dog* in the syntax. Therefore, we do not consider numerical indexes to be a solution the Free-Choice problem.³⁰

³⁰ Note also that this solution is unusable if syntax contains functional heads all the way down, as in [Ramchand \(2008\)](#), i.e. if the presyntactic lexicon does not contain a $\sqrt{}$ at all. See note 7 above.

The second problem is even more serious, and is in a way the mirror image of the first. We shall call it the problem of Faithfulness. Suppose we found a way to spell out QP as *bujar* ‘merry’, and proceeded to merge C1. To derive the correct *bujar-ej*, spellout driven movement has to apply (as explained in section 3 above). But given (98), there is not just one but two different types of lexical items that could spell out C1P without movement, namely adjectives of size C1P, like *star* ‘old’, and adjectives of size C2P, like *ostř* ‘sharp’. Given the fact that the spellout algorithm favours nonmovement over movement derivations, and given the principle of Cyclic Override, at C1P the ‘unfaithful’ spellouts *star* ‘old’ and *ostř* ‘sharp’ could override the earlier spellout of *bujar* ‘merry’, and the form *bujar-ej* would never be derivable. We call these spellouts unfaithful because they do not preserve the lexical choice made at the lowest level of insertion. This forced spellout of items which are unfaithful to the original lexical choice results from the following three factors: (i) the spellout algorithm (which favours spellout without movement), (ii) the principle of Cyclic Override (which allows unfaithful spellouts), and (iii) the way we have set up the structure of the lexical items.

The solution to both of our problems (i.e. the Free Choice and the faithfulness problem) lies, we believe, in restricting the applicability of the mechanism of Cyclic Override by imposing a Faithfulness Restriction. A first formulation of the restriction is given in (99):

(99) *Faithfulness Restriction (FR)* (to be modified)

A spellout / α / may override an earlier spellout / β / iff / α / = / β /

The FR entails that a lexical item may be overridden at a higher level only if the ‘overrider’ is the same lexical item. In the example we just discussed, *bujar* ‘merry’ could never get overridden by *star* ‘old’ or *ostř* ‘sharp’ at C1P, since they are not the same lexical items. In contrast, if $\sqrt{P}$ were spelled out as *star*, this spellout can (and will) be overridden by *star* both at QP and C1P, since the overrider is the same lexical item.

As it stands, the FR is overly restrictive, however. If we think back of the way suppletion works, an adjective like *good* is of size QP and gets overridden by *bett* at C1P. This is a kind of an override that we want to allow. However, the FR as given in (99) blocks it, since *good* and *bett* are not the same lexical item. But there is a sense in which this case is different from the ones where overriding should be blocked. Specifically, there exists a suppletive relationship between *good* and *bett*–, which is expressed by the pointer to *good* in the lexical entry of *bett*–. Building on this, we can now bring in the suppletive cases by allowing override by lexical items which are either identical, or lexically related by means of a pointer:

(100) *Faithfulness Restriction (FR)*

A spellout / α / may override an earlier spellout / β / iff

- a. $/\alpha/ = / \beta /$
- b. the lexical item $/\alpha/$ contains a pointer to $/\beta/$

In the case where $/\alpha/ = \textit{bett}$ and $/\beta/ = \textit{good}$, the former will be allowed to override the latter, and so for suppletive spellouts in general. In sum, the FR requires us to be faithful, i.e. to stick to a particular spellout as long as we can, once we have made a Free Choice to insert it at the lowest level of spellout.

Let us now turn to the second problem, namely how to allow for a Free Choice among lexical items. Currently, Free Choice at $\sqrt{}$ is unavailable because of the Elsewhere Principle (EP), which governs competition between lexical items, and restricts the choice to roots of the *kluz* class, to the exclusion of adjectives that spell out QP, C1P, or C2P. With the FR in place, we have even limited the options further, since once a lexical entry of the size $\sqrt{}$ (like *kluz*) has been selected (and the EP would effectively make this the only option), the FR prevents us from overriding this later by any other adjective than the one originally inserted. The solution to this conundrum, we propose, is to give up the EP: at $\sqrt{}$, Free Choice applies (see also Harley 2014), and the subsequent derivation must be faithful to the original choice. This means that any of the four types of adjectives listed in (34) can be inserted at $\sqrt{}$. This will steer the derivation in the right direction, with (last resort) move-

ment derivations for smaller roots (like *kluz*), or nonmovement ones for larger types of roots.

Does the EP have a role to play in the rest of the derivation? We will argue that the answer is no. To see where the EP has traditionally been relevant, consider the competition between lexical items like *good* and *bett*. Imagine, for instance, that *good* was specified as QP, and *bett*– as C1P, but without a pointer. In such case, *good* and *bett*– would both be candidates for the spellout of QP, and we would need the EP to select *good* and rule out *bett*–. The EP does that by choosing the candidate which is more specific, i.e. has fewer superfluous features; this prefers *good* and rules out *bett*. Can we do without the EP here as well? The answer to this question is already in place. Recall that earlier we suggested an interpretation of pointers which implies that a lexical item with a pointer is only a candidate for spellout if the item pointed to has been spelled out at the lower cycle. This will in fact eliminate *bett*– from the set of candidates for spellout at QP, since *good* needs to be spelled out first. Only after C1 is merged, can *bett*– be inserted, overriding *good*.

A second empirical domain where the EP has been used is that of prefixes and suffixes of varying sizes, such as the Case suffixes discussed in [Caha \(2009\)](#). Suppose we have a paradigm as in (101), with three distinct Case endings, indicated by the Greek letters.

- (101) NOM N- α
 ACC N- β
 GEN N- γ

In Caha (2009), the lexical items for such endings would look as in (102).

- (102) a. $[_{K1P} K1] \Leftrightarrow / \alpha /$
 b. $[_{K2P} K2 [_{K1P} K1]] \Leftrightarrow / \beta /$
 c. $[_{K3P} K3 [_{K2P} K2 [_{K1P} K1]]] \Leftrightarrow / \gamma /$

Now when K1 is merged, spellout driven movement will raise the noun to the left of the suffix, leaving the structure $[_{K1P} K1]$ to be spelled out. At that point, the three lexical items of (102) are all candidates for insertion, since they all contain the syntactic tree. The EP is then needed to ensure that the correct one (the most specific one) is chosen, which would be (102a). By selecting the right candidate, the EP is also important in deriving the *ABA restriction (see Caha 2009 for discussion). So if we want to eliminate the EP altogether, we shall need an alternative take on these cases.

What we propose is to extend the account of suppletion presented above to this case. That is, just as *bett-* is a suppletive form of *good*, we claim that the accusative suffix $/\beta/$ is a suppletive form of the nominative

/α/ (in the same way, the genitive /γ/ suppletes for the accusative β). This means that the lexical entry for the accusative suffix /β/ contains a pointer to the nominative suffix /α/, as shown in (103a). In the same way, the genitive /γ/ contains a pointer to the accusative /β/, as shown in (103b):



Suppose now the syntax creates [_{K1P} K1]: at this point, only /α/ is a candidate for insertion, since /β/ is only a candidate if /α/ has been spelled out in the previous cycle (and /γ/ only if /β/ has been spelled out previously). If subsequently K2 is merged (and after spec-to-spec raising of the noun to the left of K2), the only candidate is /β/, by the same logic, and /β/ gets inserted, as required. The spellout /β/ overrides the earlier spellout /α/ without violating the FR in virtue of its second clause, which allows suppletive items to override items to which they point in their lexical entries.

In sum, we have put in place a system where the EP is dispensed with. This allows Free Choice for roots at the bottom of the derivation, depending on ‘free will’, i.e. what one wants to talk about. Further derivations are restricted by the syntactic shape of the root one has cho-

sen: if it is big, it can grow larger than if it is small. In either case, when the structure available in the root is all ‘used up’ by the syntactic structure, a number of things can happen as the syntactic tree grows further. The first possibility is that the current spellout is overridden by a suppletive form. The second is that snowball movement applies, and the newly merged feature is spelled out by an additional lexical item (i.e. a suffix). This will now be allowed despite the FR, since snowball movement does not lead to Cyclic Override, but to the agglutination of suffixes. The third possibility is that a new derivation is started in a separate workspace, and merged as a prefix later on. This will also respect the FR in that no Cyclic Override takes place (except internal to the prefix, but this is then subject to the FR).

8. Conclusion

In this paper, we proposed an articulated structure of the comparative in terms of two distinct heads, C1 and C2. We presented morphological evidence from Czech and Old Church Slavonic in support of these heads, and developed an analysis of root suppletion that makes crucial use of them. In particular, we proposed to reduce all forms of suppletion to portmanteau suppletion, and discussed a generalisation (the *Comparative Suppletion Generalisation*) which this move gives rise to. Our analysis heavily relied on the idea that adjectival roots come in varying

sizes. This required us to introduce a Faithfulness Restriction, which takes over most of the work done by the Elsewhere Principle, which can consequently be dispensed with. We were able to achieve all these results relying on the decomposed comparative structure and its interaction with the language particular shapes of lexical entries, which may influence the way derivations unfold via the mechanism of spellout-driven movement.

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9 Negative Concord and TAM: a new perspective

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Negative Concord and TAM: a new perspective*

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Abstract

It has been proposed in the syntactic literature on Negative Concord (NC) that Negative Concord Item(s) (NCIs) move to a particular position in syntax, i.e. SpecNegP (Haegeman 1995, Zanuttini 1997), to give rise to concord. SpecNegP is also the position for (some) Standard Negators (SN). Under another theory, the syntactic agreement approach to NC (Zeijlstra 2004), NCIs are licensed by an abstract negative operator that also licenses the standard negator. In other words, in both accounts NCIs and SNs seem to be related, either because they end up in the same position, or because they share the same licenser. This raises the question whether the allomorphy that arises due to the conditioning between TAM (tense, aspect, mood/modality) and the standard negator (SN), also arises between TAM and NCIs. This paper explores this question by focusing on Quebliyeen Tamazight (Ouali 2012) and Egyptian Arabic (Benmamoun et al. 2013), two Semitic languages from the Afro-Asiatic phylum. The investigation shows that, while both languages display allomorphy conditioned by the interaction between TAM and SN, NCIs are not conditioned by TAM or do not condition TAM. On the basis of this, I conclude that theoretical accounts that make use of the same mechanism to capture the sentential negativity of NCIs and SNs may be in need of revision. I show how Nanosyntax can account for why the interaction between TAM and NCIs differs from the interaction between TAM and SNs, and how NC is a byproduct of the size of lexically stored trees, phrasal lexicalisation and the Superset Principle.

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1 Introduction

1.1 NEG-TAM allomorphy

There¹ are at least four logical ways in which TAM-morphology and Standard Negators² (henceforth SN) may morphologically interact, summarised in (1).

- | | | | | | |
|-----|---------------|-----|-------------------|----|----------|
| (1) | Type A | TAM | $\Leftrightarrow$ | SN | Bengali |
| | Type B | TAM | $\Rightarrow$ | SN | Albanian |
| | Type C | TAM | $\Leftarrow$ | SN | Abkhaz |
| | Type D | TAM | | SN | English |

A first possible situation is one in which TAM and the SN mutually condition each other (Type A). This is the situation in Bengali (Ramchand 2004, Ullah 2007, Banerjee 2021). The Standard Negator (SN) *na* (2), becomes *ni* in the context of the perfective (3).

- | | | | | | | |
|-----|------------------------------|-----------|------|----------|-----------|-----------|
| (2) | Ami | amṭa | kha- | cch- | i | na |
| | I.NOM | mango.CLF | eat | PROG/PRS | 1SG | NEG |
| | 'I am not eating the mango.' | | | | | |
| (3) | Ami | amṭa | kha- | i | ni | |
| | I.NOM | mango.CLF | eat | 1SG | NEG | |
| | 'I didn't eat the mango.' | | | | | |

However, not only the negator changes, the verbal morphology also changes. While the root in the negated perfective is the default root *kha*, without negation, the root in the perfective is *kheye* (4), showing that there is a bidirectional conditioning between SN and TAM.

- | | | | | | | |
|-----|---------------------------|-----------|---------|-----|-----|--|
| (4) | Ami | amṭa | kheye- | ch- | i | |
| | I.NOM | mango.CLF | eat.PFV | PRS | 1SG | |
| | 'I have eaten the mango.' | | | | | |

Type B is the type of languages where TAM-morphology conditions the SN (Dahl 1979, Kahrel 1996, De Clercq 2020b), (Horn 2001:367/446-452). This is

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²I adopt the term from Miestamo (2005), referring to the main negator a language has, and if there are several negators, it refers to the negator used in simple present declarative clauses.

the case in Albanian, which has *'s/nuk* as a SN (5)³, but the suppletive marker *mos* in the context of the subjunctive, optative and imperative mood (Turano 2000).⁴ I refer to this type of negator, as well as to the negative allomorph *ni* that appears in the perfective in Bengali, as **Allomorphs In Sentential Negation** conditioned by Tense, Aspect, and Mood/Modality, henceforth AINTs.

- (5) **'S/Nuk** vajta (më) në bibliotekë.
 NEG went-1s (anymore) in library
 'I didn't go to the library (anymore)'
- (6) a. Merr çadrën që të **mos** lagesh
 Take.2S umbrella that MOOD NEG wet.2S.SUBJ
 'Take the umbrella, so that you don't get wet'
 b. **Mos** lexo këtë libër.
 NEG READ.2S.IMP this book
 'Don't read this book!'
 c. **Mos** vdeksh kurrë.
 NEG die.2S.OPT never
 'May you never die'

In Type C languages, the presence of the SN conditions the TAM-marking on the verbal predicate, a topic studied extensively in Miestamo (2005). This is the case in Abkhaz (Hewitt 1979), where the SN *-m* has an impact on how the different tenses and aspects are morphologically expressed. For instance in the 1SG.PRS.AFF of the verb 'stand up' *s'gəlojt* becomes *s'gələm*, with the negative marker *m* not only adding negation but also replacing the finite ending *-(y)t* (Miestamo 2005:120)

A last logical option is one in which neither TAM nor the SN seem to interact morphologically, as is the case in English (Type D): (7) shows clearly how regardless of TAM the same negator can be used in Standard English. Moreover, the TAM-marking is the same with or without negation.⁵

- (7) a. He is not talking to her father. PROG
 b. He has not talked to her father. PFV

³Both *'s/nuk* are treated in Miestamo (2005), Turano (2000) as if they are interchangeable, with no meaning differences or other distributional differences associated to them. This was confirmed to me by Bujar Rushiti.

⁴*Mos* also negates gerunds and infinitives. See Turano (2000) for more details on Albanian negation.

⁵The example here is about Standard English and does not include substandard varieties which definitely include all kinds of allomorphy conditioned by NEG or by TAM, like the use of *dontcha* (Van der Auwera 2006), the use of *aint* (Yang 2016) and *amn't* (Bresnan 2001) to name just a few.

- | | | |
|----|----------------------------------|-----|
| c. | He did not talk to her father. | PST |
| d. | He would not talk to her father. | MOD |
| e. | ... | |

The typological hypothesis in (1) has been checked for 50 languages picked from Miestamo (2005) sample (see De Clercq 2020a for more explanation), yielding—after some recent updating and fine-tuning—the tentative result in (8) for the different types.⁶

(8)

	TAM	NEG _{AIN} T	
Type A	+	+	14
Type B	–	+	5
Type C	+	–	12
Type D	–	–	19

The question that arises at this point is whether this typology also holds when negative concord items (henceforth NCIs) are involved.⁷ Do we expect NCIs to interact with TAM, i.e. can their presence condition the TAM-domain, or can they be conditioned by the TAM-domain? As far as I know, this question has not yet been explicitly addressed in the literature. One can come up with multiple reasons for this: this connection has not been mentioned in the typological literature on negative indefinites that looked for any parameter that may impact negative indefinites (Alsenoy 2014, Van der Auwera & Van Alsenoy 2016, Van der Auwera 2017, Van der Auwera & Alsenoy 2018); argumental NCIs and SNs have a very different merge position; the allomorphy between SN and TAM is in general understudied; the semantic problems related to NCIs are considered far more important than the morphosyntactic issues, etc...

In spite of this, it is not unthinkable that there is conditioning between NCIs and the TAM domain. Both under the negative quantifier-approach to NCIs (Zanuttini 1991, Haegeman & Zanuttini 1991, Haegeman 1995, Zanuttini 1997) and the syntactic agreement approach to NCIs (Zeijlstra 2004, 2008, Penka 2011, and many others), the SN and NCIs are at some point in the derivation either in the same position or treated equally. In order to investigate this question in more detail we will first discuss the aforementioned syntactic theories for NC in more

⁶While De Clercq (2020a) still distinguished between 6 types, the current paper makes use of a simplified typological hypothesis to get a less fine-grained, but clearer picture of the available patterns in the languages of the world.

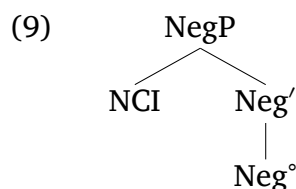
⁷For an introduction to negative concord items I refer the readers to the introduction of this volume, and to the references cited there.

detail, and the two core types of NC, i.e. Strict and Non-Strict Negative Concord (Giannakidou 1997).⁸ Next, we will move on to the empirical study of the question that we just raised.⁹

1.2 Syntactic Theories of Negative Concord

Before we move on to the data that are crucial to the investigation at hand, we need to zoom in on the two syntactic theories mentioned in the previous section, and discuss the issue relevant for the current paper in a bit more detail.

The first approach that we zoom in on is the NEG-criterion approach (Haegeman & Zanuttini 1991, Zanuttini 1991). Under this approach, NCIs are negative quantifiers, i.e. they are inherently negative. They need to move (overtly or covertly) to the specifier of a Neg° for concord to arise. The idea is that thanks to the movement to this position NEG-absorption can take place, leading to concord. The absorption is a consequence of the NEG-criterion: a head has only one specifier (in line with Kayne 1993) and hence movement to this one specifier will bring all NCIs/SNs in a position in which the head Neg° absorbs or factors out all other NEGs. If this movement does not take place, then the NCIs, which are negative quantifiers, each contribute their own negation. Crucially for our present purposes, the SN may also sit in Spec,NegP, like for instance French *pas* or West Flemish *nie*.¹⁰



Given that NCIs and SNs can sit in the same position under the NEG-criterion approach, to be licensed by the same [NEG] feature, and given the idea that NCIs are intrinsically negative quantifiers, it seems a valid question to ask whether

⁸We will not discuss less well-known patterns of non-strict negative concord, as described for Persian in Van der Auwera & Koohkan (2022). In Persian the non-strict pattern is different in the sense that the negator always needs to be present, but the indefinite can vary from being a negatively polar indefinite to a neutral indefinite.

⁹While this study focuses only on NCIs due to the scope of the volume, it would be relevant to investigate this question also on the basis of languages that have negative quantifiers of the English type, i.e. *no* and *nothing*. I leave this issue for future research.

¹⁰Italian *non* would be in Neg°, as well as Flemish *en*, and French *ne*. I refer the reader to the aforementioned works for more details.

NCIs can condition TAM or be conditioned by TAM, and if they can, whether this conditioning happens in the same way as with SN.

The other syntactic theory that we want to consider is the syntactic agreement approach (Zeijlstra 2004, 2008). Under this approach NCIs are negative indefinites, i.e. they are semantically non-negative, but syntactically marked for negation by means of a feature [uNeg]. In order to explain how concord is derived in this system, we need to make a distinction between two types of Negative Concord: Strict and Non-Strict NC (Giannakidou 1997, 2001).

If a language has Strict Negative Concord (henceforth SNC), like Czech for instance, NCIs need to be accompanied by the SN regardless of whether the NCI is in pre- or post-verbal position, as illustrated for Czech in (10).

- (10) a. **Nikdo** ženě peníze **ne**-dal
 nobody woman.DAT money.ACC NEG-gave
 ‘Nobody gave the money to the woman’
 b. Ženě **ne**-dal **nic**
 woman.DAT NEG-gave nothing
 He gave nothing to the woman. [Czech] (p.c. Lucie Janků)

Zeijlstra (2004) captures the SNC pattern by proposing that all NCIs and the SN are marked as [uNeg] and a covert (semantically) negative operator (OP in table (11)) licenses all [uNeg]-marked constituents in its scope. This is visualized in line (a) of the table in (11).

(11)		OP	SN	NCI
	a. SNC	iNeg	uNeg	uNeg
	b. Non-SNC	—	iNeg	uNeg
	c. Non-SNC	iNeg	—	uNeg

If a language has non-strict negative concord, like Standard Italian for instance, a NCI in post-verbal position requires the presence of the SN (12a), but cannot be combined with it when in preverbal position (12b) (unless double negation is aimed for).¹¹

- (12) a. **Non** ha telefonato **nessuno**
 NEG has called nobody
 ‘Nobody called’

¹¹The fact that there is the requirement to have SN present with the NCI in postverbal position in Italian is not equally strict in all substandard varieties of Italian. See the introduction to this volume by Van der Auwera and Gianollo.

- b. **Nessuno** (***non**) ha telefonato
 nobody NEG has called
 ‘Nobody called’

For this pattern Zeijlstra (2004) proposes that the SN comes with an interpretable [NEG], i.e. [iNeg], thus licensing the [uNeg] on the NCI in post-verbal position. This is illustrated in line (b) of the table in (11). However, when the NCI is in preverbal position, Zeijlstra (2004) argues that the presence of [uNeg] on an NCI allows for a self-licensing mechanism, i.e. the uninterpretable syntactic negative feature allows for the appearance of a negative operator that licenses the [uNeg] on the NCI (see also Penka 2011). This is shown in line (c) of the table in (11).

Crucial for our purpose is the fact that the NCI and the SN are licensed by the same negative operator in a SNC language, raising the issue of whether these elements share other properties, like the way they interact with the TAM-domain. For non-SNC languages, the syntactic agreement approach suggests that NCIs and SNs are different: SNs are self-licensing and really negative, while NCIs are not negative.

In the next section, we will dive into the question of how NCIs interact with TAM on the basis of two languages, Berber and Egyptian Arabic.

2 The data

De Clercq’s (2020b) 50 language sample consists of 31 languages which are of type A/B/C with respect to the SN marker. The 19 languages of type D were not considered in this paper, since there is no clear NEG-TAM interaction with the SN. Eleven languages of the A/B/C type seem to have some type of Negative Concord (NC), as shown in table 1. For some languages in the sample the data at my disposal were insufficient to establish whether there is NC, or what type the NC is. If there is *NC?* in the table, then this means that I am not sure about the type of concord, but that it looks as if there is negative concord. For the full overview of the 50 languages with the type and concord pattern, see section 4.

Most languages for which I have access to good data are SNC languages. In SNC languages, as outlined in section 1.2, the SN is argued to be always present when a NCI is used, making these languages—at first sight—less suitable to investigate the research question at hand, i.e. investigate the interaction between NCIs and TAM. If the interaction or effect of the NCI on TAM would be the same as the effect of SN, then these languages would not be revealing anything, since the SN would also always be present and prevent us from knowing what the effect of the

	Genus	Type	NC
Abkhaz	West caucasian	C	SNC
Albanian	Indo-European	B	SNC
Armenian (Eastern)	Indo-European	A	SNC
Awa Pit	Barbacoan	A	SNC
Aymara	Aymara	C	(optional) NC
Bagirmi	Nilo-Saharan	C	NC, type?
Bambara	Mande (Niger-Congo)	A	SNC
Berber (Middle Atlas)	Berber	C	SNC
Burmese	Burmese (Sino-Tibetan)	C	NC
Comanche	Numic	C	NC?
Egyptian Arabic	Semitic	B	Non-SNC

Table 1: Negative concord in the sample

NCI is. The effect can therefore be best studied in Non-SNC languages. However, if TAM were able to condition allomorphy on the NCI, then both SNC and Non-SNC languages should be equally good to show this effect. Therefore, we will look at one language of the SNC type and one language of the Non-SNC type. For SNC languages, we will look at Berber, and for Non-SNC we will look at Egyptian Arabic. We will start the discussion with the SNC language.

2.1 Berber

Berber is a continuum of dialects or languages, some not mutually intelligible, which are spoken in Morocco, Mauritania, Algeria, Tunisia, Libya, and the Oasis of Siwa in Egypt. This paper focuses on one dialect group, Tamazight, and more in particular on Quebliyeen Tamazight Berber (QTB), a Zemmour dialect spoken in the Middle Atlas Mountains in Central Morocco. The data are mainly drawn from Ouali (2012:ch. 8).

QTB is a Type C language (cf. table (1)), which means that the SN conditions TAM: under negation the perfective form of the verb, which denotes past tense, changes from what we see in a positive environment. While *idda* ‘he left’ is the form in a positive clause (13a), *iddi* is the form used under negation (13b).

- (13) a. **Idda** ʔli.
 go.PFV.3.SG.M Ali
 Ali left.’
 b. **ur iddi**
 NEG go.NEG.PFV.3.SG.M
 ‘He didn’t leave.’ (Ouali 2012:160-161)

QTB uses *ur* as a SN, but the SN often co-occurs with an optional additional marker, as illustrated in (14). Because of this marker, Berber is often described as having a bipartite negation system. However, for the dialect under investigation this marker is optional (cf. Ouali 2012:146), and so we will not discuss the optional bipartite status of QTB further in this paper.

- (14) **ur** swix (sha)
 NEG1 drink-pfv.1s NEG2
 ‘I didn’t drink.’

QBT is traditionally considered a SNC language, since the SN *ur* needs to be present with NCIs (15), regardless of whether the NCI is in postverbal (15a), or preverbal position (15b).¹²

- (15) a. **ur** iddi **agidʒ**
 NEG go.3SM.NEG.PFV no.one
 ‘No one left.’
 b. **agidʒ ur** iddin yərskeela
 no.one NEG go.3SM.NEG.PFV.PART to.school.
 ‘Nobody went to school.’

These SNC data do not seem to show us any interesting changes compared to what we see under SN: just as with SN, the verb takes the negative perfective form *iddi(n)*.

Before we move on, I want to point out that *agidʒ* can be used in fragment answers as in (16) (from Ouali 2012:152), supporting the idea that *agidʒ* really is a NCI and not a NPI. For more tests in support of this claim I refer the reader to Ouali (2012:150-155)).

- (16) a. Q: Ma ag didan?
 Who COMP come.PFV.PART
 ‘Who came?’
 b. A: *agidʒ*
 no-one
 ‘No one’

¹²When the subject is extracted across the verb, the marking on the verbal form is different, taking an *-n*. Ouali (2012) considers this an Anti-Agreement effect, and glosses this as Part, referring to Participle, in line with the traditional grammars on Berber. I follow his example, but I do not analyse the Anti-Agreement Effect in this paper.

Concerning the interaction between NCIs and TAM, an important piece of data can be observed in QTB. While (15b) is described as the neutral pattern in Ouali (2012), it is also possible to drop the SN when *agidʒ* is in preverbal position. Crucially, in that case the regular TAM-allomorphy on the verb is not visible (17).

- (17) **agidʒ** (ag) iddan / *iddin
 no.one COMP go.3SM.PFV.PART / go.3SM.NEG.PFV.PART
 ‘Nobody left.’

While the examples in (15a) and (15b) do not allow us to see what the role is of the NCI, because the SN is also present, the example in (17) shows that when the SN is no longer present there is no negative allomorphy on the verb. In other words, QTB shows that it is not just any [NEG] that can condition verbs in the perfective, but that only the [NEG] present in the SN can condition the perfective form of the verb. Moreover, QTB also shows that *agidʒ* ‘no-one’ remains unaltered regardless of when the SN is present or not.

2.2 Egyptian Arabic

We will now take a closer look at the only Non-SNC language in the sample, Egyptian Arabic. Egyptian Arabic (also discussed in Miestamo 2005, Benmamoun et al. 2013, Ouali & Soltan 2014, Alqassas 2021) is a type B language ((1)), which means that in the context of particular TAM the SN changes to an allomorph, a so-called AINT (cf. sect. 1.1 above). For Egyptian Arabic this AINT is *mif*, which is the only possible negator in the context of the future tense (also called the *ha*-imperfect) (19), instead of the SN *ma... f* (18b), which is the SN used for the past tense (also referred to as the perfect) (18a).¹³

- (18) a. katab
 write.PST.3SG.M
 ‘He wrote.’
 b. **ma**-katab-*f*
 NEG-write.PST.3SG.M-NEG
 ‘He didn’t write.’
- (19) Howa **mif** ha-yroh el welayat el motaheda
 he NEG FUT-go the states the united

¹³I am simplifying the picture here, because *mif* is also used in questions and is optionally used in the imperfect. However, due to the optional nature of its appearance under imperfect aspect I decided to focus on its obligatory appearance in the future tense.

‘He will not go to the United States.’ (p.c. Dia Awaad and Amgad Farrag)

Egyptian Arabic is Non-SNC, which means that the NCI needs to be accompanied by the SN when the NCI is in postverbal position (20a), but cannot be accompanied when the NCI is in preverbal position (20b).

- (20) a. **ma-šuf-t-i-ŷ** **walaa waahid.**
 NEG-saw-1SG-EV-NEG no one
 ‘I didn’t see anyone.’
 b. **walaa waahid** gih.
 no one came.3SG.M
 ‘Nobody came.’ (Ouali & Soltan 2014:162)

When the AINT is used with the future tense, the exact same pattern can be observed: the same AINT is present in the context of postverbal NCIs, (21) and the AINT disappears in the context of preverbal NCIs (22).

- (21) Howa **miŷ** ha-yeŷzem **walaa waahid.**
 he NEG FUT-invite no one
 ‘He will not invite anybody.’
 (22) **Walaa waahid** ha-yroh el welayat el motaheda.
 no one FUT-go the states the united
 ‘Nobody will go to the United States.’ (p.c. Dia Awaad and Amgad Farrag)

Crucial for our present purposes is the observation that the absence of *miŷ*, the AINT that was needed in the context of the future tense, does not need to be compensated for in any possible way by any marking on the NCI or anywhere else in the sentence. This suggests that the allomorphy that arises in the context of future tense does not affect all types of NEG, but only the NEG associated with the SN. The Egyptian data hence show that the NCIs themselves are not conditioned by the future tense, unlike the SN. Conversely, the NCI also do not condition tense.

2.3 Discussion

Both QTB and Egyptian Arabic suggest that the interactions that we see between the SN and the TAM-domain disappear in the context of NCIs: NCIs are not conditioned by TAM nor do they condition TAM. It thus seems that on the basis of the morphosyntactic properties it is very important to keep SNs and NCIs apart and

to treat them differently, unlike what is done under the NEG-criterion approach or the syntactic agreement approach (with respect to the treatment of SNC). The Berber data also suggested that even in SNC languages NCIs can give rise to semantic negation, outside the context of fragment answers ((16)), as in (17). For the latter it could be argued that the NCI can get licensed due to the presence of a negative operator in the ellipsis site, but this option is not available for the fronted NCIs in the Berber data. One could of course propose—in line with Zeijlstra (2004, 2008)—that this type of fronted NCIs are self-licensing, i.e. that the presence of a [uNeg] presupposes the presence of a semantic negative operator that licenses it. If there is no overt one, a covert one will appear. But would this not boil down to saying that NCIs are semantically negative after all? How else could the presence of a semantically non-negative indefinite (a [uNeg]) give rise to the appearance of a covert, semantically negative operator ([iNeg])?

Summarizing, the data from Berber and Egyptian Arabic support the idea (i) that NCIs are really intrinsically negative (as under the Neg-criterion approach), and (ii) that the [NEG] of NCIs and SNs cannot be the same [NEG]. The question that we want to address next is whether it is feasible to set up a proposal that captures (i) the allomorphy that we see between SN and TAM, (ii) the absence of allomorphy between NCIs and TAM, and (iii) the schizophrenic behaviour of NCIs in Strict and Non-Strict Negative Concord Languages. We will argue in the next section that this is feasible and that this can be done within Nanosyntax. We will illustrate this for the Berber data that we discussed in section 2.1.

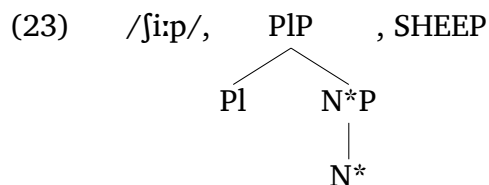
3 A Nanosyntactic view on allomorphy and NC

In this section we will explain how the conditioning between SN and the TAM-domain in Berber can be captured in Nanosyntax, and why we do not see this interaction with NCIs. In the meantime, the proposal will also account for the ambiguity NCIs give rise to in different positions in the clause, more concretely for the concord puzzle that we described in section 1.2 in general, and in section 4 with respect to Berber and Egyptian Arabic. Before we embark on this, we will briefly explain Nanosyntax.

3.1 Nanosyntax: the basics

Nanosyntax (Caha 2009, Starke 2009) is a late-insertion theory that lexicalizes the output of Syntax after each step of Merge, making use of phrasal lexicalisation.

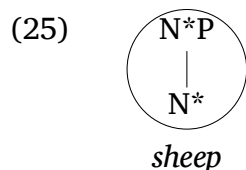
The consequence of the extremely cyclic nature of lexicalisation is that Syntax interfaces with the lexicon at every Merge step, allowing for an immediate interaction between the two. Thanks to this, there is no need for post-syntactic operations, nor for an independent morphological component. The lexicon consists of lexical items that not only come with phonology (between slanted brackets) and conceptual information (in capitals), but also with stored lexical tree structures, as illustrated for the noun *sheep* in (23).¹⁴



After each Merge step, the syntax is checked against the lexicon and a matching lexical item is searched for. The matching condition¹⁵ that mediates between the Syntax and the Lexicon is informally referred to as the Superset Principle, and its definition is in (24).

- (24) A lexically stored tree matches a syntactic node iff the lexically stored tree contains the syntactic node. (Starke 2009:3)

To understand what (24) can do, consider the lexical item in (23) again. Thanks to (24), (23) is a candidate for insertion both for the syntax in (25), without the plural, and for the syntax in (26), when syntax merged the [Pl] feature.

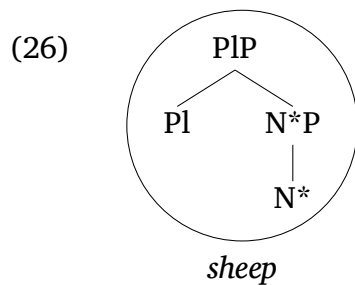


¹⁴The * in the lexical structure indicates that we assume that the nominal structure is internally more fine-grained. However, for the purpose of this explanation, we do not go deeper into that.

¹⁵The matching condition can be defined as in (i).

- (i) A lexically stored constituent L matches a syntactic phrase S iff S is identical to L.

The Superset Principle in (24) is actually a theorem that follows from the matching condition. If a syntactic tree is identical to a stored L tree, then the consequence of this is that the S tree may also be contained in the L tree, with the L tree being bigger in structure. This is explained in detail in Caha et al. (to appear).



With this much in place about NS, we continue our discussion of the Berber data under consideration. More technicalities about Nanosyntax will be explained as we go along. For a more detailed introduction to Nanosyntax I refer the reader to Baunaz & Lander (2018), Starke (2018), Caha et al. (2019, to appear).

3.2 Berber

The first thing we want to take a look at is how we can understand the fact that SN conditions the TAM categories on the verb, and why this is not the case with NCIs. Second we want to have a closer look at the structure of NCIs.

In order to capture the fact that SNs can condition TAM and vice versa, a functional category NEG and the relevant TAM category must be in an extremely local relation in Nanosyntax, i.e. they must be contiguous in the structure.¹⁶ The table in (27) is a lexicalisation table illustrating in an informal way phrasal lexicalisation for the lexical items relevant to our discussion of Berber. The top line is a simplification of what we assume are the relevant parts of the functional sequence (henceforth fseq): what is most to the left in the table is the top of the syntactic tree. For the discussion of the contrast between *idda* ‘he left’ and *ur iddin* ‘he didn’t leave’ it suffices to assume a simplified fseq.

(27)

	NEG	T	ASP _{PFV}	INIT	PROC
a.			idda		
b.	ur		iddi		

We follow Ouhalla (1991) and Ouali (2012:67-68) in proposing that the position for the SN must be higher than tense in the syntax of Berber. One of the reasons

¹⁶Nanosyntax differs in this way from other approaches to allomorphy and suppletion, in which strict contiguity is not required and looser implementations of locality are proposed. I refer the reader to Choi & Harley (2019) for an overview of different morphosyntactic approaches to allomorphy and suppletion.

for this is the fact that the SN always precedes the auxiliary, as illustrated in (28).

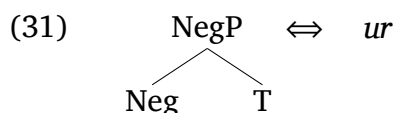
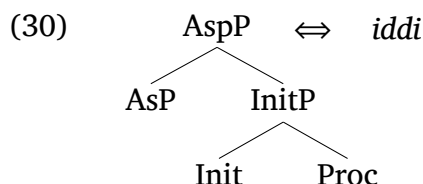
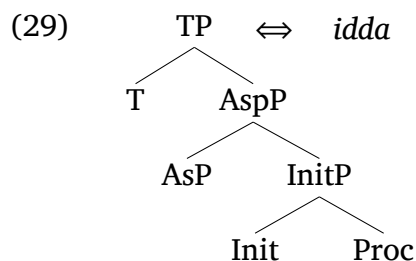
- (28) **ur** dað ayəx ibəttan
 NEG FUT buy.1SG clothes
 ‘I will not buy clothes’

The idea that NEG is higher than Tense is also present in the work by Holmberg & Platzack (1988), Haegeman (1995), Zanuttini (1996), De Clercq (2020b) and many others. TP then is higher than the features related to viewpoint Aspect, in this case the perfect, and viewpoint aspect is higher than the features relevant to the lexical aspect of the verb, captured here by means of Init, which stands for the Initiator of the Process, which is abbreviated by PROC (Ramchand 2008).

What the table shows is that we assume that the form *idda* cannot only lexicalize features relevant to the verbal domain, but also features relevant to the aspectual and the tense domain. The reason for adopting this idea is not far-fetched: *idda* gives rise to a past tense reading, as was illustrated in section 4. When it comes to line (b) in table (27), it is shown how we can capture the allomorphy that we see under the SN if we adopt the idea that there is another lexical item stored in the lexicon, *iddi*, which is smaller in structure, and a negator which is not only lexicalizing [NEG], but also T. The reason that *iddi* will only be used in the context of the negator is that if there is no negation present and you want to lexicalize *he left*, *iddi* will not be able to lexicalize the Tense layer, but will only reach up to perfect aspect, it being insufficiently big to lexicalize a fully inflected positive verb. Consequently, *idda* will be called for. However, in the context of negation, *iddi* will be the best possible match at the level of perfect aspect, since the SN *ur* will lexicalize the relevant T-layer together with NEG. In other words, if we adopt the idea that *ur* is a portmanteau lexicalizing tense and negation, then root conditioning under perfect aspect follows naturally. While the idea to look at *ur* as a portmanteau is new, the idea that Negation interacts with Tense (and Agreement) in Arabic and Berber is not new (see Ouali 2012).

What is intuitively captured by the lexicalisation table boils down to having the lexical items (29)-(31) in the lexicon of Berber (phonology and conceptual information are left out for simplicity’s sake):¹⁷

¹⁷One idea that is present in the way the lexical items are set-up is that we—for now at least—assume that roots are not lexicalized independently, but together with their vowels. So instead of having the root *k-t-b* in the lexicon of for instance Arabic, the idea would be that there are different lexical items for *kitab* ‘book’ and *kataba* ‘he wrote’. This idea is not new for Semitic nor for Berber and is present in the work of Moktadir 1989, Dell & Elmedlaoui 1991, Bensoukas 2001. However, this proposal is by no means a final stance on how I think root and pattern morphology



Our next task is to explain why NCIs in Berber do not interact with TAM. However, we can only understand this by addressing another issue first. We first need to find an answer to the question how it is possible that an NCI in a SNC language can give rise to what looks like real negation when it is in clause-initial position (see (17) above).

In order to answer this question, we will have another look at a lexicalisation table. This time the table we started out with in (27) is slightly modified to also host the NCI in preverbal position. In (32) this position is uninformatively labelled NCI (the brackets around *agidʒ* ‘no-one’ indicate that it can occur both pre- and post-verbally). The table shows the two patterns that we were confronted with in (15)-(17). Even though we understand now why the presence of *ur* leads to the presence of *iddi*, it still remains a mystery why *ur* needs to be present in line (b), but can be absent in line (a).

(32)

	NCI	T	NEG	ASP	INIT	PROC
a.	agidʒ				idda	
b.	(agidʒ)		ur		iddi	

This issue takes us to the heart of the NC-puzzle. As we explained before, it is not only in this particular context for QTB that NCIs can give rise to real negation: they can also give rise to negation in the context of an elliptical answer, as in (16), repeated here in (33) (Ouali 2012:152).¹⁸

- (33) a. Q: Ma ag didan?
 Who COMP come.PFV.PART

(McCarthy 1979) should be dealt with in NS.

¹⁸See also the introduction for discussion of NCIs in elliptical answers.

- ‘Who came?’
 b. A: *agidʒ*
 no-one
 ‘No one’

In order to understand why NCIs in QTB can give rise real sentence negation in these environments, we need to take a closer look at example (17) again, repeated here in (34).

- (34) ***agidʒ*** (ag) iddan / *iddin
 no-one COMP go.3SM.PFV.PART / go.3SM.NEG.PFV.PART
 ‘Nobody left.’

What we did not pay attention to in section 2.1 is the presence of the complementizer *ag*. Upon careful consideration of Ouali’s (2012)’s data it seems that the pattern without *ur*, shown on line (a) of (32), is only possible when the (optional) complementizer follows the NCI, while with the regular concord patterns, the NCI is below the complementizer, (35).¹⁹

- (35) a. ***agidʒ*** (ag) iddan
 no-one COMP go.PFV.PART
 ‘Nobody left.’
 b. (ag) ***agidʒ*** ur iddin *yərskeela*
 COMP no.one NEG go.PFV.PART to.school.
 ‘Nobody went to school.’

This means that we can now update the table in (32) to the table in (36), where we see that the disappearance of the SN is linked to the appearance of the NCI in a position above a complementizer.

(36)

	NCI ₁	C	NCI ₂	NEG	T	ASP	INIT	PROC
a.	<i>agidʒ</i>	(ag)					<i>idda</i>	
b.		(ag)	(<i>agidʒ</i>)	<i>ur</i>			<i>iddi</i>	

Since (i) it must be clear that NCIs have more internal features than just [N], and since (ii) we have already seen when we determined the inner structure of *ur* ‘not’ that the position of a morpheme in the functional sequence is indicative of the feature structure of this morpheme, we are now at a point to propose that the

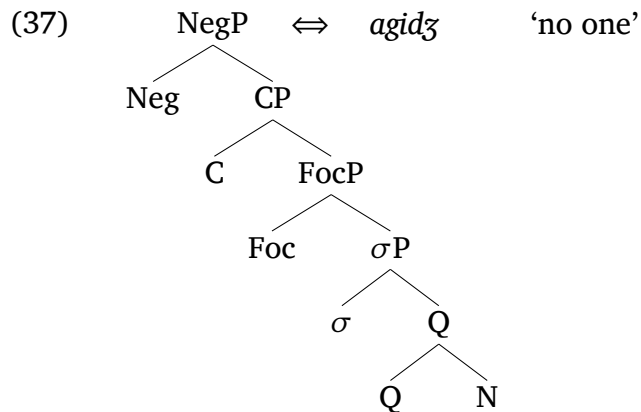
¹⁹One cannot argue that *agidʒ* in (34) is part of a cleft sentence, with the negative reading of the NCI following from the deletion of a regular SN, because the Anti-Agreement effect (see footnote 12) on the main verb shows that *agidʒ* must have been fronted.

different positions of *agidz* go hand in hand with the fact that the NCI is lexicalizing different amounts of structure in different positions in the fseq.

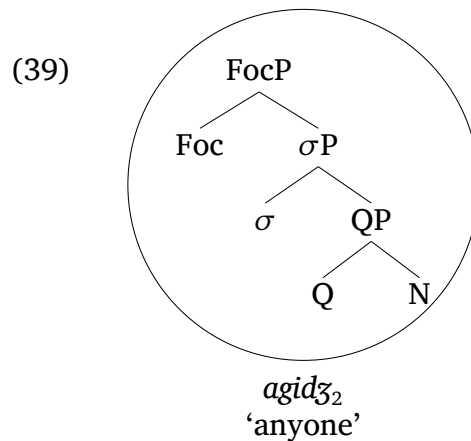
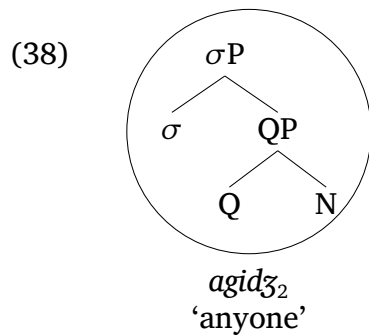
To make this idea more explicit we first take a look at what the feature structure of NCIs could be like. A first component is, in the case of *agidz*, definitely a nominal element, N. In order to capture the idea that this nominal is a quantifier of some sort, we assume that the feature Q will also be needed. This feature structure will allow for the lexicalisation of a positive indefinite (like for instance *some(one)*). However, we also need an ingredient that will allow the lexicalisation of a negative polarity item (like for instance *any*). We will use the feature σ for that (in line with work by Chierchia (2006) and the syntactic implementation by Labelle & Espinal (2014)).²⁰ Next, we want to propose, in line with work by Watanabe (2004), that a Focus feature is an inherent and necessary part of the internal structure of NCIs (as opposed to NPIs), forcing it to be in the scope of a [NEG]. Ouali (2012:157) shows for Berber—in line with Watanabe (2004)—that *ag-idz* literally means ‘not even-one’, and that hence focus is morphologically contained inside the NCI. Finally, I want to propose that the morpheme responsible for Focus, i.e. *ag-*, is also syncretic with the complementiser *ag*. We know from Baunaz & Lander (2018) that cross-categorical syncretisms between nominal elements and complementizers is typologically widespread. We take this syncretism to be meaningful and propose that NCIs must consist of a property shared with complementisers. For convenience sake, I will refer to this feature by means of [C]. Since the data in (34) clearly show that the NCI can give rise to negation on its own, in spite of the fact that the language is SNC, I propose that the NCI also consists of a NEG at the top of its internal structure. All these ingredients, that are all based on fairly regular assumptions in the literature and the empirical properties of NCIs in Berber, yield the lexical structure in (37).²¹

²⁰We realize that σ (and the associated mechanism of exhaustification) can probably not do justice to the complex distribution and behaviour of NPIs (Giannakidou 2018). However, we adopt σ to capture the fact that positive indefinites are different from NPIs and to be able to regulate the competition between them. We hope to come back to this in future work.

²¹One could also imagine decomposing *agidz* in *idge* ‘one’, which lexicalizes the structure up until QP, and *ag-* which lexicalizes the rest of the feature structure.

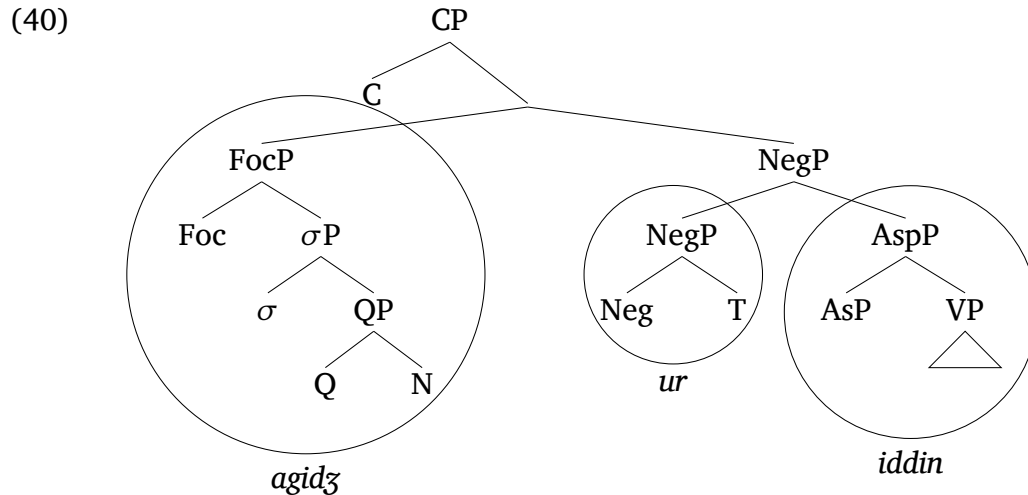


Thanks to the Superset Principle (24), the lexical item (37) can also lexicalize the structures in (38)-(39):



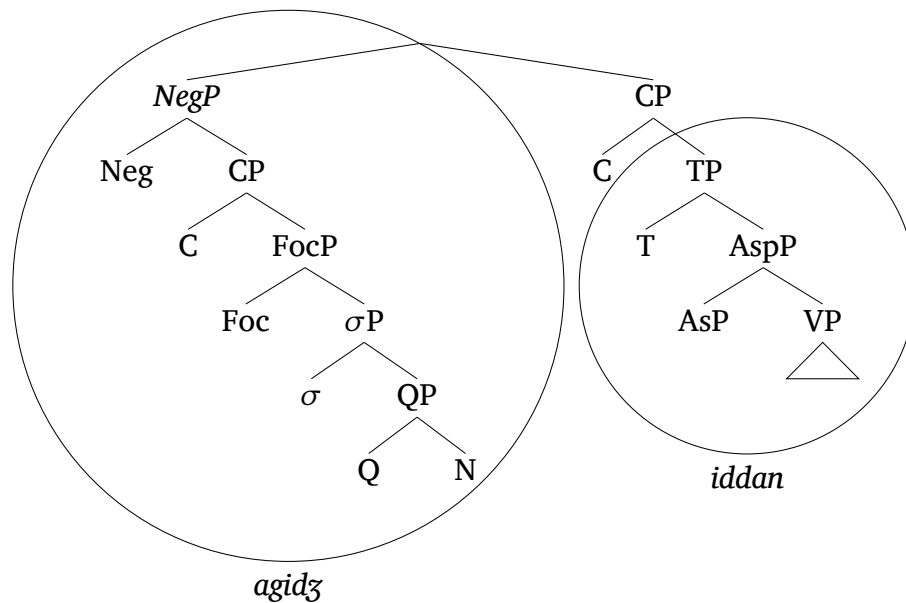
Crucially, when NCIs are merged in postverbal position, they will have a syntax as in (38): only a subset of the features of the lexical structure of the NCI will be present in the derivation at that point. FocusP will only be merged with the NCI when the NCI moves to a position above the vP-domain. This proposal is in line with syntactic approaches to Focus that argue for a FocusP for New Information Focus in a low left periphery, above vP (Belletti 2004), and one for Contrastive Focus in the high left periphery (Rizzi 1997). NEG will not be part of the internal syntax of the NCI when (38) is merged in syntax, accounting for why *ur* needs to be present to give rise to Sentential Negation. However, thanks to the Superset Principle, the lexical item in (37) will be able to lexicalize this structure. Also when the NCI appears in preverbal position, below the complementizer, as shown

in (40), the syntax of the NCI will not reach up to the NCI-internal NEG and hence again the SN will be needed to project a NegP in the clausal syntax. The internal syntax may reach up to FocP in this case, but given the internal structure of NCIs in Berber, this structure will not be able to project a NegP in the main spine.

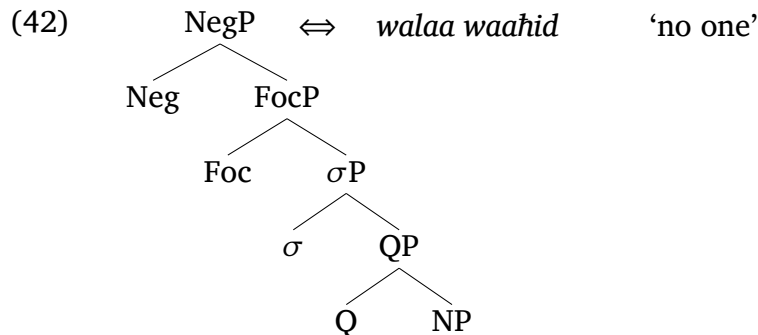


However, when the NCI appears in a position before the complementizer, the syntax of the NCI can grow to its full potential, as illustrated in (41), and hence the LI in (37) will be a perfect match. This time however the NCI can also project a Neg, higher in the structure, and hence the SN is not needed to give rise to Sentential Negation, giving rise to the unexpected pattern that we discussed for QTB.

(41)

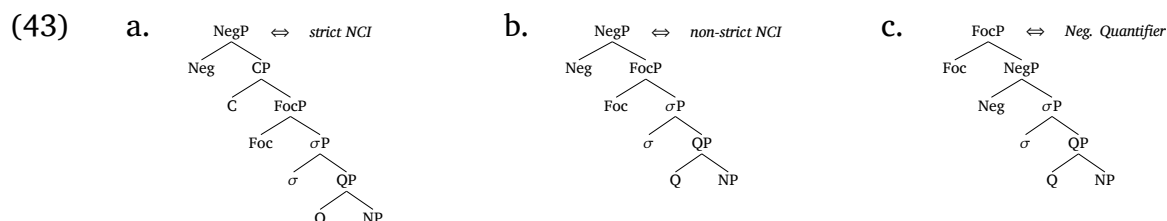


The underlying idea in the current proposal is that for a language like Egyptian Arabic, which is Non-SNC, the internal structure of the NCI differs in the sense that the NCI-internal NEG feature takes a different position, below the C-element, just above the Foc-element, accounting for why Egyptian Arabic does not need a SN as soon as the NCI is in a position above vP. The structure for Egyptian Arabic *walaa waahid* is in (42).²²



²²As with *agidz* in Berber it would be an option to argue that the NCI is not stored as one chunk in the lexicon, but that the sequence in (42) is lexicalised by multiple lexical items. One could namely further decompose *walaa waahid* into the positive indefinite *waahid*, that could possibly lexicalize the structure up until QP, and then *walaa*, which literally consists of a coordinator *wa* and a negator from Classical Arabic *laa*, which could be argued to lexicalize the upper part of the structure. We will not decide on this, since nothing crucially hinges on it.

In other words, the current proposal suggests that NC is a side-effect of an interaction between the Superset Principle and how lexical items are stored in the lexicon of a particular language. If the NCI-internal NEG sits high in the lexical structure of an NCI, as we saw for Berber, then it is only in the context like (34) or the elliptical contexts, typical of SNC languages, that the NEG can project in the main spine (tree structure (43a)). However, if the NCI-internal NEG sits lower, below the C-layer, then the NCI will be able to project its own NEG as soon as it moves to a position higher than vP (43b). Finally, when a language has quantifiers that have a structure in which NEG sits just above QP, as in (43c), then the system will not be Negative Concord, but will be of the Negative Quantifier type, like English.



The advantage of this hypothesis for concord is that it does not require us to postulate two different lexical items for the NCI in the lexicon of a particular language to capture the ambiguity that NCIs give rise to (see amongst others (Herburger 2001)).²³ Under the current proposal, the ambiguity arises as a natural consequence of the Superset Theorem, a matching principle that is independently needed to guarantee lexicalisation. The current proposal for concord also brings us back to the issue this paper started out with, i.e. the question whether NCIs can condition TAM and be conditioned by it. If allomorphy in SNs and TAM arises due to what we proposed for Berber, i.e. as a consequence of SN “eating” into the structure of the TAM domain, and if the structure that we just proposed for NCIs is on the right track, then it follows why NCIs and the TAM domain do not condition each other: their functional sequences simply do not share features, apart from the feature NEG. The only interaction that we see therefore, is that the NCI can replace the SN, because both can satisfy the need of projecting a NEG in the main spine.

²³See also the introduction of this volume for an elaborate discussion of ambiguity in various proposals for NCIs.

4 Conclusion

This paper started from two mechanisms - proposed in the syntactic literature - to capture NC: 1) moving the NCI to the same position as the position for SN in syntax (NEG-criterion approach), or 2) licensing NCIs and SNs by the same semantic operator (syntactic agreement approach for SNC). The observation related to the discussion of these mechanisms was that in both approaches SN and NCIs are - in terms of negativity - treated similarly and associated with the same position. This led to the question whether there are also morphosyntactic reasons from the domain of allomorphy to treat NCIs and SNs in the same way. On the basis of a small crosslinguistic investigation, and a more detailed discussion of Egyptian Arabic and Berber we concluded that it does not seem to be the case that NCIs condition TAM or are conditioned by TAM in the way SNs do this. This conclusion is in line with the typological work by Alsenoy (2014), Van der Auwera & Van Alsenoy (2016), Van der Auwera (2017), Van der Auwera & Alsenoy (2018) that looked at any parameter that could be related to (negative) indefinites, and didn't find any. Since the SNC language Berber provided us with a strong argument to consider NCIs as intrinsically negative (in line with the NEG-criterion approach), we concluded that the [NEG] in SNs and the [NEG] in NCIs must be a different [NEG]. We developed a Nanosyntactic account to capture (i) the allomorphy that arises between SN and TAM, (ii) the absence of allomorphy in the presence of NCIs, and (iii) the fact that NCIs in SNC languages can give rise to negation. The current paper gave rise to a new perspective on NC: under this approach NC is a side-effect that arises due the interaction of the Superset Principle with stored lexical items. It goes without saying that this new perspective is also a research hypothesis that needs to be tested against more languages.

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Appendix

The table below shows the first 50 languages from Miestamo (2005), which De Clercq (2020a) used for her pilot study. The ‘type’ label is based on the hypothesis developed in the introductory section of this paper, and deviates somewhat from the pilot study. Only the negative concord patterns for types A/B/C are provided (see discussion in section). For the determination of the types and the negative concord patterns the main resources are: Miestamo (2005), reference grammars of the relevant languages and native speakers if available. To determine the concord pattern for Aymara I want to thank Roger Gonzalo Segura, my Aymara informant, whom I was able to consult via Gabriel Martinez Vera, whom I also want to thank. The variety of Aymara in the table is Southern Aymara, as spoken in the Peruvian side of Lake Titicaca.

Concerning the abbreviations in table 2: ? means that I am not sure whether there is concord or not, **No NC** means that the language does not have Negative Concord, **SNC** means that the language has Strict Negative Concord, **Non-SNC** stands for Non-Strict Negative Concord, **NQ** stands for Negative Quantifier. If the cell is **empty** I did not look into the concord pattern, because the language is of Type D . If I added ? to one of the abbreviations, then it means that I am not entirely sure.

	Genus	Type	NC
Abipon	Guaicuruan	D?	
Abkhaz	West caucasian	C	SNC
Achumawi	Palaihnihan	A	No NC?
Acoma	Keresan	D	
Ainu	Ainu	D	
Alamblak	Sepik	A?	?
Albanian	Indo-European	B	SNC
Amele	Trans New Guinea	A	No NC
Andoke	Andoke	D	
Apala\`i	Cariban	C	No NC
Apurina	Maipuran	D	
Araona	Pano-Tacanan	D	
Armenian (Eastern)	Indo-European	A	SNC
Asmat	Central and South New Guinea	C	?
Awa Pit	Barbacoan	A	SNC
Aymara	Aymara	C	NC
Bafut	Niger-Congo (Eastern Grassfield)	A	No NC
Bagirmi	Nilo-Saharan	C	NC, type?
Bambara	Mande (Niger-Congo)	A	SNC
Barè	Tucanoan	B	NQ
Barasano	Arawakan	D	
Basque		C	No NC
Batak	Sundic	D	
Bawn	Sino-Tibetan	B	No NC
Beja	Northern Cushitic	A	No NC?
Bella Coola	Bella Coola (Salish)	D	
Berber (Middle Atlas)	Berber	C	SNC
Birom	Niger Congo	D	
Boko	Mander (Niger-Congo)	C	?
Bororo	Macro-Ge/Bororoan	A	No NC?
Brahui	Dravidian	A	No NC
Bukiyip (Arapesh)	Torrrelli	A	NQ
Burara	Burarran	C	NQ
Burmese	Burmese (Sino-Tibetan)	C	NC
Burushaski	Burushaski	D	
Canamari	Katukinan	?	
Candoshi	Candoshi	B	?
Canela-Krah	Carib	D	
Cantonese	Sino-Tibetan	A	No NC
Carib	Carib	D	
Cayuvava	Cayuvava	D	
Chamorro	Chamorro	D	Non-SNC
Chechalis (Upper)	Salishan (Coast)	A	NQ
Chinantec (Lealao)	Oto-Mangue	A?	
Chinook (Lower)	Chinookan	D	
Chorote	Mataco	C	?
Chukchi	Chukoto-Kamchatkan	A	No NC
Chumash	Chumash	D	
Comanche	Numic	C	NC?
Egyptian Arabic	Semitic	B	Non-SNC

Table 2: The sample

10 How to be positive

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Glossa: a journal of general linguistics 5(1): 23. doi:<https://doi.org/10.5334/gjgl.1114>

RESEARCH

How to be positive

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In this paper, we discuss a cross-linguistically rare pattern of comparative formation found in Slovak. This pattern is theoretically interesting, because it violates a candidate universal on the relationship between the positive and the comparative degree. The universal, discussed in Grano & Davis (2018), says that the comparative is always either identical to, or derived from, the positive degree. This universal is violated by a number of adjectives in Slovak. These adjectives have a suffix *-k* in the positive degree, which is absent in the comparative. We capture this pattern in terms of a non-containment structure of the positive and the comparative degrees and the nanosyntax model of spellout (Starke 2009 et seq.).

Keywords: adjectives; gradability; positive; comparative; containment

1 Introduction

We start by presenting some background on comparatives in Slovak. Consider first the data in (1):¹

(1)	POSITIVE	COMPARATIVE	
	star-ý	star- š-í	‘old’
	múdr-y	múdr-ejš-í	‘wise’
	sliz-k-ý	sliz-k-ejš-í	‘slim-y’

The first thing to note is that all the forms can be segmented into an adjectival base followed by an agreement marker (*-y/-ý* in the positive and *-í* in the comparative).² We are setting these agreement markers aside in our analysis, since the main focus of our paper rests on how the comparative is formed. In the table (1), the comparative is always formed by taking the positive degree (minus agreement) and suffixing it with *(ej)š* (followed again by agreement). The comparative marker shows an allomorphy between the markers *š* and *ejš*, which we shall disregard for now.³

¹ Accents over vowels indicate length. Wedges over consonants indicate palatalisation. For instance, *š* is IPA [ʃ], *ž* (to be encountered later) is IPA [ʒ]. The difference between orthographic *i* and *y* in the agreement ending represents the difference between soft and hard declension.

² The agreement marker tracks the number, gender and case of the head noun. All tables show masculine singular nominative forms.

³ Traditional grammars describe this allomorphy as phonologically governed. The long allomorph (*ejš*) appears after consonant clusters and after sibilants, the short one (*š*) elsewhere (Dvoňč et al. 1966: 210–12). We adopt this analysis here, but we do not rule out the possibility that there may be a morphosyntactic principle underlying the distribution of these allomorphs, as is the case in Czech (Caha et al. 2019).

It will become highly relevant that the same procedure is observed also by adjectives that are morphologically complex in the positive degree. An example of such an adjective is on the bottom line of (1). The adjective consists of a root, followed by a suffix *-k*, followed by the agreement marker. We shall refer to this marker *-k* as an *augment*. In the adjective *sliz-k-ý* ‘slimy,’ the augment is preserved in the comparative, giving rise to *sliz-k-ejš-í* ‘slimier.’ We label such a pattern of comparative formation the CONTAINMENT pattern, because of the fact that the positive degree is used as the basis to which the comparative marker *-(ej)š* attaches.

Against this background, consider the fact that adjectives with augments sometimes show a different pattern of comparative formation, given on the bottom line of the table below:

(2)

POSITIVE	COMPARATIVE	
star-ý	star- š-í	‘old’ CONTAINMENT
múdr-y	múdr-ejš-í	‘wise’
sliz-k-ý	sliz-k-ejš-í	‘slim-y’
ťaž-k-ý	ťaž- š-í	‘heav-y’ TRUNCATION

What we see here is that for the adjective *ťaž-k-ý* ‘heavy,’ the augment must be dropped before the comparative marker. The expected form **ťaž-k-ejš-í* is ungrammatical. We call this the TRUNCATION pattern: the comparative contains something which is less than the form of the positive degree, even when discounting the agreement marker.

The broader relevance of these data is linked to the truncation pattern in particular, because it contradicts a candidate universal proposed by Grano & Davis (2018: 133).

- (3) *Candidate Universal*
Universally, the comparative form of a gradable adjective is derived from or identical to its positive form.⁴

The way Grano & Davis (2018) understand this universal is to explicitly rule out a situation like the one found with the Slovak adjectives of the *ťaž-k-ý* ‘heavy’ class. As they put it, ‘it should [...] be impossible to find a language in which both the positive and comparative forms are independently derived from a common base’ (their pattern D; Grano & Davis 2018: 134). This is exactly what happens with *ťaž-k-ý* ‘heavy’, where the common base is the root *ťaž-*, to which the positive adds the suffix *-k*, whereas the comparative adds *-š*, both followed by an agreement marker.

Summarising the pattern illustrated by the adjectives in (2) schematically, we can distinguish the following three classes:

(4)

root	POS	CMPR
old	—	—
heav-y	AUG	—
slim-y	AUG	AUG

The adjectives of the *old* class have no augment, either in the positive or the comparative. The *heavy* class adjectives have an augment in the positive, but lose it in the compara-

⁴ It is not clear to us what Grano & Davis (2018) think about cases of suppletion like *good-bett-er*, which do not seem to be covered by their universal. We suspect that they implicitly disregard suppletion, i.e., that their universal only applies to cases of regular comparative formation. We discuss root suppletion in Slovak in section 7.

tive. The adjectives of the *slimy* class, finally, have an augment both in the positive and the comparative.

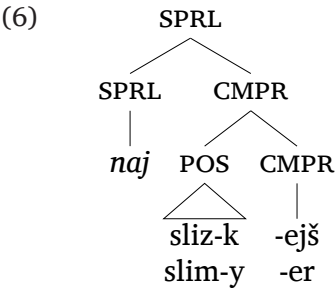
In what follows, we develop an analysis for this pattern and its distribution in the Slovak adjective system. In section 2 we discuss the problem from the perspective of the relation between morphological and structural containment, and present a global outline of the analysis. In section 3 we present a detailed empirical overview of the Slovak evidence. Section 4 considers the question whether the distribution of *-k* in the positive and comparative can be reduced to independently established semantic or phonological properties of the relevant roots. Section 5 provides our hypothesis concerning the morphosyntactic structure of the positive and the comparative. Building on these structures, Section 6 gives a fully formal and algorithmic treatment of the patterns within nanosyntax. Section 7 discusses the relationship between the relevant patterns and suppletion with special reference to Bobaljik (2012). Section 8 addresses some issues concerning the meaning of the positive and the comparative heads in the proposed framework.

2 Morphological and structural containment

There is a nontrivial relationship between structural and morphological containment. For example, Bobaljik (2012) has argued at length that the structure of the superlative universally contains the structure of the comparative. Supporting evidence for this claim is that there are languages where the structural containment is reflected in morphological containment as well. A case in point is again Slovak, where the superlative is formed by prefixing the comparative form with *naj-*. This is shown in (5) for the four adjectives discussed earlier in (2), with a Bobaljik-style structure in (6).

(5)

CMPR	SPRL	
star-š-í	naj-star-š-í	‘old’
múdr-ejš-í	naj-múdr-ejš-í	‘wise’
sliz-k-ejš-í	naj-sliz-k-ejš-í	‘slimy’
ťaž-š-í	naj-ťaž-š-í	‘heavy’



Yet at the same time, not all languages reflect this structural containment relationship in terms of morphological containment, whereby the superlative is formed by adding a morpheme to the comparative. In English, for example, a superlative like *short-est* does not contain the comparative form *short-er*. In such languages, as Bobaljik suggests, the superlative marker (*-est*) either pronounces both CMPR and SPRL as a portmanteau marker, or CMPR is silent in the context of SPRL.

Our concern here is the structural relationship between the positive and the comparative, and what the Slovak data can tell us about it. Bobaljik does not put forth an explicit argument to the effect that the structure of the comparative contains the structure of the positive, but he understands that to be the null assumption (see Bobaljik 2012: 32, ex. (37)). If that is indeed the case, we have no trouble whatsoever explaining the containment

pattern using the structures that Bobaljik proposes. In (7a), the positive is taken to have a certain structure, and the comparative structure is derived by merging a comparative head on top of this structure (as shown in (7b)). The containment pattern, illustrated here by the adjective *slim-y-er*, is straightforwardly predicted by such structures.



However, if we put a truncation-class adjective in the same structure, we inevitably end up with a wrong result, as shown in (8b).



The containment structure leads us to expect the form in (8b), which is, however, ungrammatical, the correct comparative form being *ťaž-ší* 'heavier', without the augment. The conclusion we come to is that in order to account for the truncation pattern, we need to say something special.

One logical option would be to pursue an analogy to the analysis of *-est* and say that the Slovak comparative marker is a portmanteau for CMPR and *-k*. However, in Slovak, this analysis is implausible. Specifically, in English, *-er* never precedes *-est*: this is because *-est* spells out *-er* plus SPRL. In Slovak, this is different. Specifically, in the containment class, *-k* does precede the comparative marker. This makes it obvious that the comparative marker does not spell out *-k*, and hence, the disappearance of *-k* in the truncation class must have some other reason.

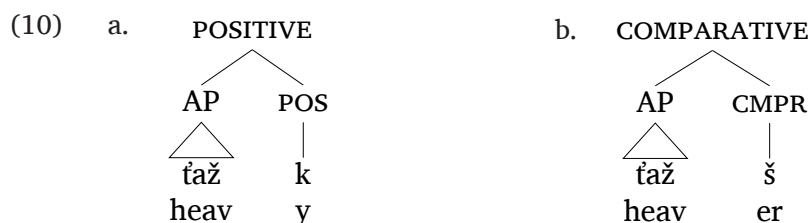
Another strategy would be to rely on context-sensitive rules. A possible analysis of the augment using this type of device is depicted below:

- (9) a. $a \Leftrightarrow \emptyset$
 b. $a \Leftrightarrow -k / \text{ ______ } \{ \sqrt{\text{HEAVY}}, \dots, \sqrt{\text{SLIMY}}, \dots \}$

What the rules say is that there is a particular functional head in the extended projection of all Slovak adjectives, which is called little *a* in (9). This head is generally left unexpressed (see (9a)), but receives an overt realisation (as *-k*) in the context of the roots which actually have *-k* in the positive (this is what (9b) says).

The challenge for such rules is how one should distinguish between the containment and the truncation class. In particular, in order to achieve the disappearance of the augment *-k* with the truncating adjectives in the comparative, the rule in (9b) will somehow have to be deactivated for the roots of the 'heavy' class in the context of a higher CMPR head. While this can certainly be achieved (e.g., by a rule of impoverishment, or an additional zero exponent), we do not find such a style of solution attractive.

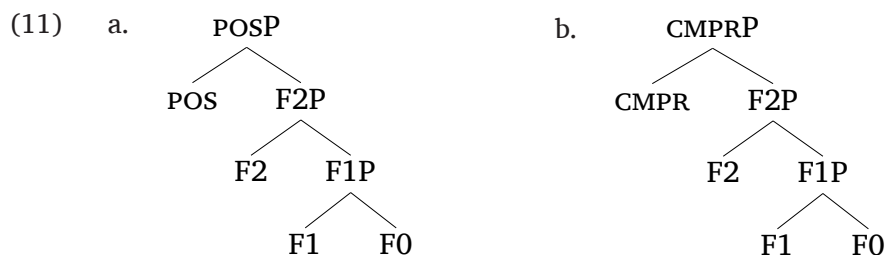
We will therefore explore here an alternative which derives the existence of the Slovak truncation pattern from a particular type of structure. The specific type of approach we propose here is given in (10), and it relies on non-containment structures. These assume a decomposition of the positive degree into two parts (each potentially complex): an adjectival base (called AP for now) and a head POS, as shown in (10a). POS is a label that does not necessarily designate a specific head (we shall in fact change the label later on); its goal for now is simply to say that the positive degree has some structure that is not a part of the base to which the comparative attaches, see (10b).



We will call the structures *symmetric* because the two degrees share a common base (AP), and each develop this base in a different direction. These different directions are labelled POS and CMPR for now. They may each correspond to a single head, but they may each correspond to a series of heads. We make a specific proposal about this in section 5.⁵

It is easy to see that these structures straightforwardly handle the truncation pattern. The augment is dropped in the comparative, because the POS head, which dominates the augment, is absent in the comparative.

However, it seems harder to deal with the containment pattern. With nothing else said, the augment should *always* be absent in the comparative. To handle containment, we need two additional proposals. The first one is a further decomposition of the AP. Specifically, we argue that the AP given in (10) decomposes into three projections, which (for now) we label simply F0, F1, and F2; as we shall show in section 5, these features correspond to independently needed ingredients of gradable adjectives. Combining this decomposition of the AP with the idea of a symmetric relationship between the degrees, we end up with the hierarchies in (11) as the more refined version of the structures in (10) (ordering aside).



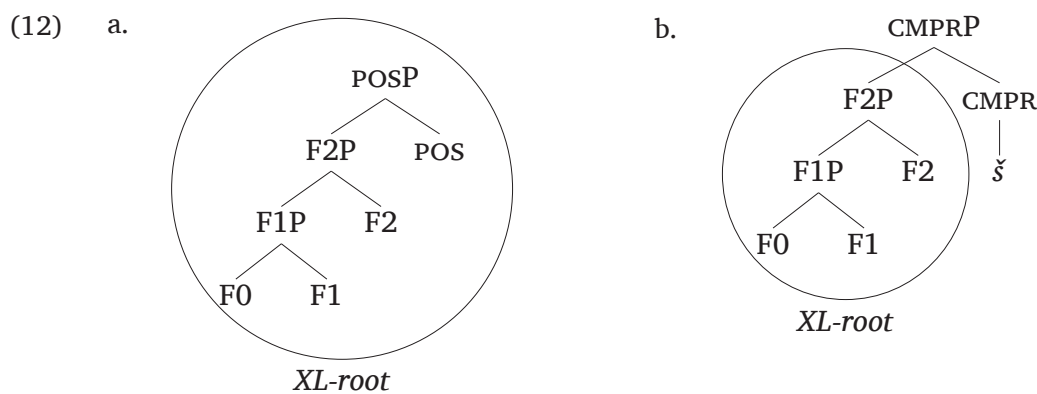
The second thing we need is phrasal spellout (Starke 2009). According to this proposal, roots as well as affixes may spell out phrasal constituents containing multiple heads. For example, an adjective that has no augment in the positive degree will be able to spell

⁵ An anonymous reviewer points out that these structures recall the proposal that there is a single DEG head in the projection line of the adjective, whose value is either positive or comparative (see, e.g., Corver 1997). This is certainly one way our proposal can be interpreted. However, we explore a different avenue in section 5.

out the full POSP in (11a) with all the heads it contains. This is an analytical option which, in a phrasal spellout model, replaces the traditional idea that POS is pronounced by a null marker, or that POS corresponds to a type-shifting rule (Neeleman et al. 2004; Kennedy 2007).

In this setting, the specific factor controlling the appearance and the disappearance of the augment is going to be *variable root size* (cf. Caha et al. 2019). This means that different roots can be associated to different sizes of structure in the decomposed AP, because of the arbitrary nature of lexical storage.

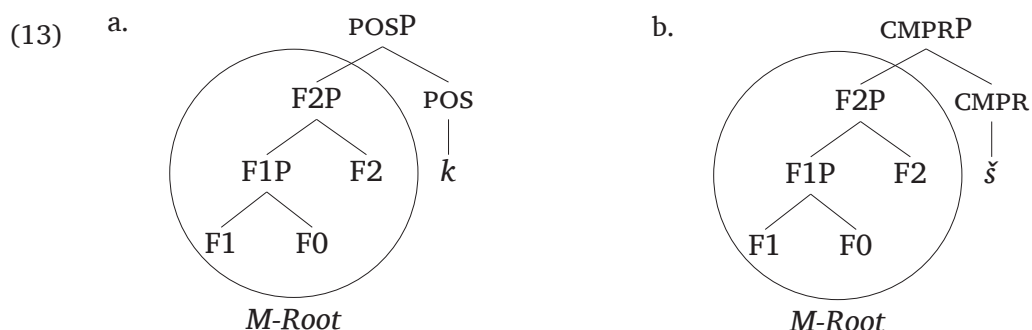
Specifically, in order to capture the three classes of adjectives in Table (4), we will distinguish three classes of roots that will differ by the number of heads they are able to spell out. Extra Large roots (henceforth XL-roots) are those of the *old* class: their lexical entry includes the complete structure of the positive degree, as shown in (12a). The circle represents the structure that the XL-root realises.



In the comparative (12b), the XL-root lexicalises the F2P. This structure corresponds to a subtree of its lexical entry, which is the full POSP. The possibility to spell out also proper subparts of the full lexical specification follows from an insertion principle that we shall adopt, namely the Superset Principle (Starke 2009). We will introduce this principle in much greater detail in section 6.

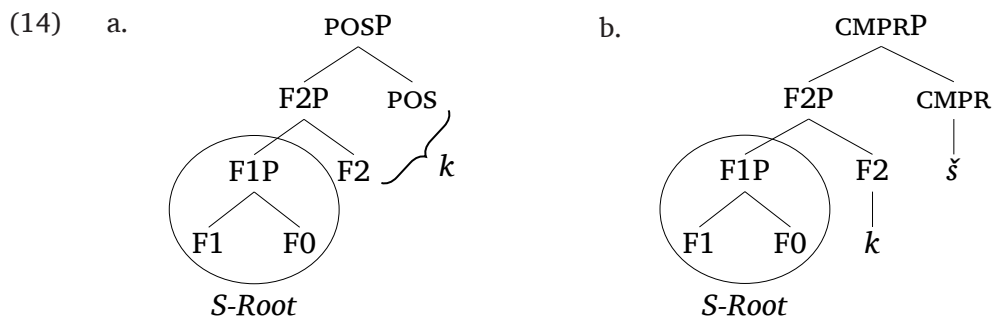
Since XL-roots are big, they do not need an augment, neither in the positive, nor in the comparative. These roots therefore give rise to a containment pattern without any augment.

Medium-sized roots (M-roots) are those of the *heavy* class, which featured the problematic truncation pattern in the comparative. Their lexical entry differs from XL-roots in that it lacks the POS head, i.e. their size corresponds to the F2P node. This means that they will not be able to spell out the POS head, and they will therefore need the augment in the positive to spell out POS, see (13a).



There is no need for an augment in the comparative, because the comparative does not include POS; the comparative marker will appear directly on top of the root spelling out F2P. This results in the truncation pattern.

Finally, adjectives of the *slimy* class have Small roots (S-roots). Their lexical entry corresponds to F1P, as shown in (14). In the positive (14a), F2 and POS will need lexicalisation, and this will be done by the augment.⁶



In the comparative (14b), the root alone is likewise insufficient to realise all the features. While the CMPR head is lexicalised by the comparative marker *-š*, and the root realises F1P, this still leaves F2 in need of lexicalisation. Because of this, *-k* is found also in the comparative with these roots. Therefore, S-roots exhibit a containment pattern, with an augment both in the positive and the comparative.

3 The Slovak data

This section discusses the Slovak patterns in more detail. We focus here in particular on those Slovak adjectives that have an augment in the positive degree. As we intend to show, they fall into two subclasses: one drops the augment in the comparative, giving rise to the truncation pattern, whereas the other keeps it, showing the containment pattern. We argue that the distinction between the truncation class and the containment class is an arbitrary property of the root. We do so by showing that the bifurcation cannot be captured by other (phonological, morphological or semantic) means.

To demonstrate this, we give here an exhaustive record of adjectives with the augment *-k* in the positive. Those that show truncation are shown in Table 1, while Table 2 lists those that follow the containment pattern. We shall introduce the tables in turn.⁷

The first column of both tables lists the positive degree, segmented into the root and the augment *-k*. That *-k* is indeed an independent morpheme (in both classes) can be demonstrated using diminutive adjectives. The diminutive forms are in the second column. They (usually) convey a positive attitude towards the addressee or the object with the relevant property. One example of their use would be parent-child interactions.

What is relevant for us is that the various diminutive markers *-un/-uč/-ulin* attach directly after the root (not after *-k*). This shows that the augment is a morpheme separate from the root (to which *-un* attaches). Not all adjectives have such a diminutive version;

⁶ To this end, F2 and POS need to form a constituent. How this happens is a matter that we address in section 6. We shall also argue that with M-roots in the positive, *-k* in facts spells out both F2 and POS. This issue is immaterial to the argument being made here.

⁷ The source of the positive/comparative forms is the Slovak Academy of Science webpage <http://slovníky.juls.savba.sk>, where the most authoritative dictionaries of the language are available in an electronically searchable format. The diminutive adjectives listed in the table are the ones we found in the Slovak National Corpus (<https://korpus.sk>). Since we consider it unlikely that the corpus contains all the possible diminutives, the gaps in the table should not necessarily be taken to indicate the absence of the diminutive forms.

Table 1: Slovak complex adjectives which drop *-k* in the comparative.

POSITIVE			DIMINUTIVE				COMPARATIVE			gloss
root	AUG	AGR	root	DIMIN	AUG	AGR	root	CMPR	AGR	
ľah	k	ý	ľah	un	k	ý	ľah	š	í	‘light’
mäk	k	ý	mäk	un	k	ý	mäk	š	í	‘soft’
ten	k	ý	ten	un	k	ý	ten	š	í	‘thin’
blíz	k	y	bliz	un	k	ý	bliž	š	í	‘close’
úz	k	y	uz	un	k	ý	už	š	í	‘narrow’
níz	k	y	niz	un	k	ý	niž	š	í	‘low’
krát	k	y	krat	un	k	ý	krat	š	í	‘short’
hlad	k	ý	hlad	un	k	ý	hlad	š	í	‘smooth’
slad	k	ý	slad	un	k	ý	slad	š	í	‘sweet’
vlh	k	ý	vlh	uč	k	ý	vlh	š	í	‘wet’
ried	k	y	ried	ulin	k	ý	red	š	í	‘thin’
ťaž	k	ý					ťaž	š	í	‘heavy’
krot	k	ý					krot	š	í	‘tame’
prud	k	ý					prud	š	í	‘steep’

Table 2: Slovak complex adjectives which keep *-k* in the comparative.

POSITIVE			DIMINUTIVE				COMPARATIVE				gloss
root	AUG	AGR	root	DIMIN	AUG	AGR	root	AUG	CMPR	AGR	
heb	k	ý	heb	un	k	ý	heb	k	ejš	í	‘smooth’
sliz	k	ý	sliz	un	k	ý	sliz	k	ejš	í	‘slimy’
kreh	k	ý	kreh	un	k	ý	kreh	k	ejš	í	‘fragile’
brit	k	ý					brit	k	ejš	iacute;	‘scathing’
hor	k	ý					hor	k	ejš	í	‘bitter’
hyb	k	ý					hyb	k	ejš	í	‘nimble’
krep	k	ý					krep	k	ejš	í	‘energetic’
mrz	k	ý					mrz	k	ejš	í	‘ugly’
trp	k	ý					trp	k	ejš	í	‘bitter’
syp	k	ý					syp	k	ejš	í	‘loose’
váz	k	ý					váz	k	ejš	í	‘sticky’
vrt	k	ý					vrt	k	ejš	í	‘nimble’
vlh	k	ý	vlh	uč	k	ý	vlh	k	ejš	í	‘wet’ ⁹

but crucially, in all cases where a diminutive adjective exists, the diminutive marker comes directly after the root.⁸

Let us now turn to the containment class in Table 2.

⁸ Note that in Table 1, we consider the *-k* that follows the various diminutive markers *-un/-uč/-ulin* to be the very same augment that is found with the non-diminutive adjective. This is not crucial for our point: the two *-ks* could also each be a different morpheme, and this would not affect our argument here. The only important thing at this point is that the non-diminutive adjectives in Table 1 are morphologically complex. We return to the derivation of the diminutive adjectives in section 6.4.

⁹ Observe that the adjective *vlh-k-ý* ‘wet’ occurs in both tables, forming the comparative with either truncation or containment. Other adjectives like this (according to dictionary searches) are *krot-k-ý* ‘tame’ and *prud-k-ý* ‘steep.’ We treat this as a case where either multiple roots must be stored, or where different speakers have different entries. Our goal here is not to capture the occasional variation, but to describe the patterns.

Here as well we see that the diminutive *-un* (where available) attaches directly after the root. This fact is crucial; it provides evidence that the base of the positive degree is to be segmented in two parts in this class as well. It is therefore impossible to say that, for instance, the comparative form *heb-k-ejš-í* ‘smoother’ on the first line in Table 2 exhibits *-k* in the comparative because the *-k* is not a suffix to begin with, but a part of the root. The diminutive *heb-un-k-ý* ‘smooth’ shows that *-k* is a suffix, yet with this root, it is preserved in the comparative.

4 Potential analyses

Let us now turn to the issue of what determines whether a particular adjective exhibits the containment or the truncation pattern. The first option to consider is that the distinction is to be explained in phonological terms, e.g., that *-k* drops after roots with a particular phonological shape. The second possibility we need to consider is that the difference between truncation and containment is semantic in nature, e.g., that adjectives with particular semantic properties keep the augment, while other types drop it. We discuss these in turn.

4.1 Phonology

A phonological analysis is made unlikely by the existence of near minimal pairs such as the ones in Table 3.

Here the members of each pair present similar phonological environments, yet behave differently with respect to the choice for truncation or containment in the comparative. Thus the first two roots end in *h*, whereas the latter two end in *t*, the final consonant in either case being preceded by a vowel. Yet the former member of each pair shows truncation, the latter containment, suggesting that the distribution of the augment in the comparative is not determined by phonological factors. Admittedly, the quality of the vowel preceding the consonant is different, yet this has generally no influence on the truncation/containment difference, as Tables 1 and 2 show.

As an anonymous reviewer points out, another phonological issue to consider is the following. In principle, it could be the case that the two classes of adjectives do not differ in the presence/absence of *-k* in the morphosyntactic structure, but only differ in what allomorph of the comparative they require. In particular, the suggestion is that the truncation class requires the comparative marker *-š*, whereas the containment class selects *-ejš*, as shown in (15a). In the truncation class, a CVC root is followed by the augment *-k* and the comparative *-š*, yielding a triconsonantal cluster *Ckš*. In the containment class, now characterised by the presence of the *-ejš* allomorph in the comparative, no such cluster arises. In the case of the truncation class, so the reviewer suggests, this would give rise to a phonologically triggered process of consonant cluster simplification through deletion of *k*, yielding the surface contrast in (15b).

Table 3: Similar phonological environments, different comparatives.

POSITIVE			DIMINUTIVE				COMPARATIVE				gloss
root	AUG	AGR	root	DIMIN	AUG	AGR	root	AUG	CMPR	AGR	
l'ah	k	ý	l'ah	un	k	ý	l'ah		š	í	‘light’
kreh	k	ý	kreh	un	k	ý	kreh	k	ejš	í	‘fragile’
krot	k	ý					krot		š	í	‘tame’
brit	k	ý					brit	k	ejš	í	‘sharp’ (humor)

- (15)
- | | | | |
|----|-------------------------|------------|-------------|
| | | TRUNCATION | CONTAINMENT |
| a. | underlying comparative: | CVC-k-š | CVC-k-ejš |
| b. | k-deletion: | CVC-š | CVC-k-ejš |

Under this analysis, the truncation of the positive would be a purely phonological, and not a morphosyntactic, effect.

There are two arguments against this approach. First of all, it is incompatible with the phonological account of the distribution of the -š and -ejš allomorphs, which states that -ejš occurs after clusters, and -š elsewhere (see note 3 above). Suppose now that we abandoned that account in order to allow for (15a), we expect there to be examples of simplex adjectives that end in a consonant cluster, and select the -š allomorph, as shown in (16a). Assuming a process of triconsonantal cluster simplification, we further expect to find this process at work in CCš clusters, as shown in (16b).

- (16)
- | | | | |
|----|-------------------------|------------|-------------|
| | | TRUNCATION | CONTAINMENT |
| a. | underlying comparative: | CVCC-š | CVC-k-ejš |
| b. | C-deletion: | CVC-š | CVC-k-ejš |

However, there are no cases that fit this description, as far as we are aware. While there are many adjectival roots which end in two consonants (e.g., *príkr-y* ‘steep’, *hust-ý* ‘dense’, *čistý* ‘clean’, *rýchl-y* ‘fast’), these all select the -ejš allomorph. Roots ending in three-consonant clusters also exist (e.g., *bystr-ý* ‘witty/smart’, *ostr-ý* ‘sharp’), but none of these feature the consonant cluster simplification that is allegedly at work with the truncating *k*-adjectives. Finally, we note that there is at least one case in the Slovak National Corpus that features the sequence *Ckš*, namely *jorkšírsky teriér* ‘Yorkshire terrier’, a fact which casts further doubt on the existence of the consonant cluster simplification rule supposedly responsible for the truncation pattern.

The second argument against (15) is that Slovak has also an augment with the phonological shape VC, namely -*ok*. This augment is found with four adjectives, three of which show a truncation pattern, while the fourth shows a containment pattern, as shown in Table 4.

In this case, it becomes difficult to derive the difference between the truncation and the containment class from a phonological truncation rule, since neither class has a tri-consonantal cluster created by the addition of the comparative -š after -*ok*. One would moreover not only have to assume the deletion of -*k* but also of the vowel preceding it, a kind of deletion which lacks any plausible phonological motivation. Therefore, we do not consider the phonological account sketched in (15) as a plausible alternative to a morphosyntactic account that we pursue here.

For reasons of space, we will not pursue here the analysis of -*ok* in any detail. The conclusion we want to draw from Table 4 is that the truncation pattern is not a quirk of a single marker, or a single phonological environment; it looks like a structural property

Table 4: Complex adjectives in Slovak: the suffix -*ok*-.

	POSITIVE			COMPARATIVE				gloss	
	root	AUG	AGR	root	AUG	CMPR	AGR		
a.	hl’b	ok	ý	hl’b		š	í	‘deep’	TRUNCATION
b.	šir	ok	ý	šir		š	í	‘wide’	
c.	vys	ok	ý	vyš		š	í	‘tall’	
d.	div	ok	ý	div	ok	ejš	í	‘wild’	CONTAINMENT

of the Slovak grammar that shows at multiple places in the language. Moreover, the pattern is diachronically stable, going back at least a thousand years. Specifically, it has been present already in Old Church Slavonic, the language of the oldest Slavic manuscripts. As Lunt (2001: 78) notes, already back then “[t]he positive [showed] a derivational suffix in *-k* which is lost in the comparative.” A morphosyntactic truncation analysis has been also adopted in the reference grammar of Slovak (Dvonč et al. 1966: 211).

4.2 Semantics

Let us now turn to the possibility that the distribution of *-k* (including the distinction between the containment and the truncation pattern) is determined not by phonology, but by the abstract meaning of the adjective. Following this avenue, we find that there is indeed a semantic regularity that unifies all adjectives with *-k*. In particular, as we shall discuss in more detail below, all adjectives with *-k* are scalar. What we mean by this is that adjectives with *-k* are gradable, tend to have antonyms and their interpretation is context-dependent. For example, the adjective *ľah-k-ý* ‘light,’ seen in Table 3, has a synthetic comparative, it has an antonym (*ťaž-k-ý* ‘heavy’), and its meaning is context-dependent: *a light/heavy handbag* has a different comparison class (and consequently a different meaning) than *a light/heavy truck*. These facts confirm our initial decision to treat *-k* as a marker that spells out POS and other projections of gradable adjectives.

At the same time, the relationship between scalarity and morphology is imperfect. To show that, we list a representative sample of adjectives with the augment *-k* in the first column of Table 5. The first thing we want to focus on are their antonyms (horizontally in the same row).

What we see is that most of the time, the antonyms lack the augment despite the fact that they too are gradable, context-dependent and have an antonym. Hence we must conclude that the correlation between *-k* and scalarity is only a one way implication: *-k* entails scalarity, but scalarity does not necessarily lead to *-k*.¹⁰

Table 5: Semantic category does not predict distribution of *-k*.

	K-ADJECTIVE				ANTONYM				
1.	ťaž	k	ý	‘heavy’	ľah	k	ý	‘light’	truncation
2.	krát	k	y	‘short’	dlh		ý	‘long’	
3.	mäk	k	ý	‘soft’	tvrd		ý	‘hard’	
4.	vlh	k	ý	‘wet’	such		ý	‘dry’	
5.	slad	k	ý	‘sweet’	—				
6.	kreh	k	ý	‘fragile’	pev	n	ý	‘firm’	containment
7.	brit	k	ý	‘sharp’	tup		ý	‘blunt’	
8.	heb	k	ý	‘smooth’	hrub		ý	‘rough’	
9.	sliz	k	ý	‘slimy’	—				
10.	hor	k	ý	‘bitter’	—				

¹⁰ The adjective *pev-n-ý* ‘firm’ on line 6 of Table 5 has yet another augment in the antonym. This augment (*-n*) is a common adjectivizing morpheme in Slovak (e.g., *lac-n-ý* ‘cheap,’ *chut-n-ý* ‘tasty,’ *bezpeč-n-ý* ‘safe,’ *sil-n-ý* ‘strong,’ *pôvab-n-ý* ‘charming,’ *hovor-n-ý* ‘talkative,’ *čier-n-y* ‘black,’ etc.). Perhaps it also doubles up as a passive participle morpheme (*unave-n-ý* ‘tired,’ *otvore-n-ý* ‘open,’ etc.) Unlike *-k* and *-ok*, this augment never drops in the comparative as far as we are aware. We do not discuss the augment *-n* in detail here (since this is orthogonal to our concerns), while nevertheless noting that such behaviour could be captured by attributing to this marker not only the high functional layers of the adjectival projection (POS and F2), but also its lower reaches (including F1 and possibly also F0).

Let us now turn to the division between truncation and containment, represented in the table by the horizontal line in the middle. The specific question we address is whether scale structure can be made responsible for the bifurcation between the two patterns. It turns out that it cannot. To show that, we have selected the adjectives in the first column from different sub-classes of scalar adjectives. Consider, for instance, the adjectives ‘heavy’ (line 1) and ‘short’ (line 2), both belonging in the truncation class. They represent the so-called open-scale adjectives in the terminology of Kennedy & McNally (2005). Both have antonyms and both are context-dependent (a short giraffe is a different kind of short than a short grasshopper). However, they differ in that ‘heavy’ is associated with the positive extent on the scale, while ‘short’ is a negative adjective (cf. *How heavy/#short is he?*). This shows that the augment *-k* is not associated with adjectives of a particular direction (positive or negative), and that both positive and negative adjectives may exhibit the truncation pattern.

A similar state of affairs is found in the containment class (lines 6–10). The adjective *kreh-k-ý* ‘fragile’ is a negative adjective with an open scale, while *brit-k-ý* ‘sharp’ is a positive open-scale adjective. The important conclusion is that both positive and negative open-scale adjectives may exhibit or fail to exhibit *k* in the positive degree, and may belong to either the truncation or the containment class in the comparative.

The adjective *vlh-k-ý* ‘wet’ on line 4 is a so-called minimum standard adjective (in the terminology by Kennedy & McNally 2005): it suffices that an object (like a chair) has a minimum amount of wetness for it to count as (a) *wet (chair)*. Another distinguishing property is that the comparative entails the positive: if *A is wetter than B*, it follows that *A is wet*. This is not the case for the open scale adjectives on line 1 and 2: for example, *A is heavier than B* does not entail that *A is heavy*. Despite the difference in scalar structure between ‘wet’ and ‘heavy,’ they both belong in the truncation class.

Moreover, we find minimum standard adjectives (associated to partially closed scales) in the containment class as well. To see that, consider the adjective *sliz-k-ý* ‘slim-y,’ which behaves semantically like ‘wet’: if *A is slimier than B*, it entails that *A is slimy*. The main difference with *wet* seems to be the viscosity of the material rather than the type of scale. Despite the fact that ‘slimy’ has abstractly the same type of scale as ‘wet,’ they pattern differently in the comparative.

Finally, lines 5 and 10 show two instances of taste adjectives. These adjectives do not have antonyms: being sweet does not, for instance, entail not being bitter (consider caramelised sugar). However, they are context dependent: *a sweet beer* is a different type of *sweet* than *a sweet cake*; a cake with the same degree of sweetness as beer would not count as sweet. The main point is, however, that one of the taste adjectives (‘sweet’) belongs in the truncation class, the other (‘bitter’) belongs in the containment class. So once again, it seems that the type of scale or even the type of an adjective do not, as far as we can see, allow us to distinguish what kind of morphological pattern the relevant adjective will show. The bifurcation between containment and truncation thus seems to be then best understood as an arbitrary property of the relevant root. This fact is reflected in our analysis sketched above, which relies on a lexically arbitrary property of the root, namely its size in the lexicon.

Summarising this section, we have seen that Slovak has complex adjectives in the positive, which are built up according to the template in (17):

(17) root-K-AGR

Such adjectives either drop the augment *-k* in the comparative, or they keep it. We have shown that this bifurcation is in large part arbitrary, and cannot be predicted on the basis

of phonology or semantics. While we did note that the adjectives with the augment are scalar, their scalar properties do not determine the particular pattern they belong in. It should be further noted that the appearance of an augment with a particular root is also in part arbitrary, simply because not all scalar adjectives have an augment, as becomes clear from observing the antonyms in Table 5.

Our goal in the next section will be to model this behaviour in nanosyntax. In this approach, lexical items are associated to syntactic constituents of various sizes, differing as to what features are contained in their lexical entry. In the next section, we present such an analysis and argue that it leads to interesting predictions for root suppletion, to be discussed in section 7.

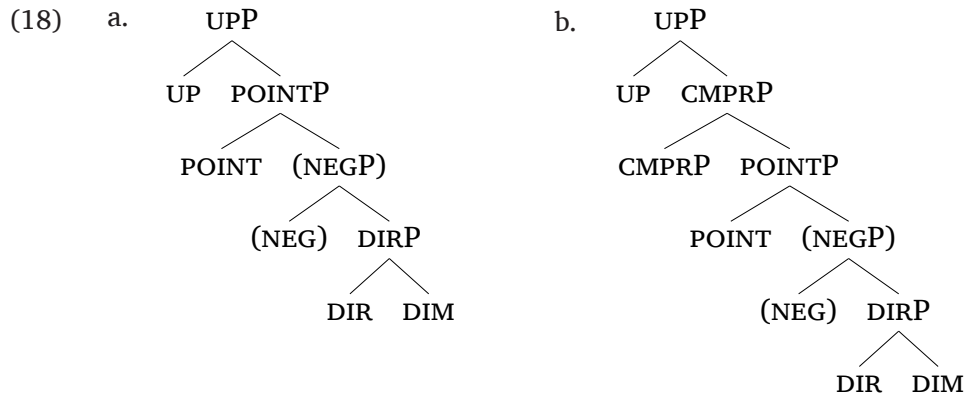
5 Decomposing the adjective

As the first step in our analysis, we provide labels for the features F0, F1, and F2, which we have been assuming in section 2 as an important part of our preliminary analysis. In general terms, these features correspond to independently needed ingredients of gradable adjectives. At the bottom of the hierarchy, there is a dimension, which could be size, velocity, height, color, etc. This means that $F0 = \text{DIM}$. Scalar adjectives come with an ordering on top of a dimension, which we represent by the feature DIR, i.e. $F1 = \text{DIR}$. Directions may be reversed by means of an optional reversal operator (called NEG in De Clercq & Vanden Wyngaerd 2019). The presence or absence of NEG distinguishes antonymic pairs of adjectives from one another (like *tall-short*, *wet-dry*, *heavy-light*, etc.). Such pairs of adjectives thus involve the same dimension, but the direction of the scale is different: positive for *tall*, negative for *short* (see De Clercq & Vanden Wyngaerd 2018 for a semantic analysis of this contrast).¹¹

This leaves us with F2 to provide a label for. We shall assume that F2 basically introduces a point on the scale, and we therefore label it as POINT. The precise position of this POINT on the scale may (in principle) be further specified by additional functional material. However, in the positive degree, the projection POINT is ‘underspecified;’ it does not introduce any specific point, but rather the fact that there is a point to begin with. As a result, the position of the point is determined by the context: it corresponds to the contextual standard.

The relevant point introduced on the scale by POINT will divide the scale into two parts, one which exceeds this point and another one, which does not. The argument that the adjective is predicated of will have a degree of the relevant property that falls within the part of the scale that exceeds the relevant POINT. Therefore, the final projection above POINT is the projection UP, which replaces our original POS (see Neeleman et al. 2004 for the UP component). Putting all of this together, we arrive at the structure in (18a) for the positive degree, with the optional NEG feature in brackets.

¹¹ Two anonymous reviewers ask about the status of non-gradable adjectives in our system. Our general take is that these adjectives correspond to a structural subset of the gradable adjectives, because they lack a scale. In the decomposition proposed here, this means that they have only the DIM projection (and possibly others lower down). A potential difficulty (brought to our attention by an anonymous reviewer) is that for some adjectives, e.g., *biological* or *presumed*, it does not seem to be clear as to what ‘dimension’ they apply to. One possibility for dealing with this would be to change the label of our DIM head into PROPERTY, so that non-gradable adjectives simply denote properties. Some of these properties can be arranged according to a single dimension, and these can become gradable by adding the DIR head on top of the PROPERTY, yielding an alternation between a non-gradable use (when DIR is absent) and a gradable use (when DIR is present). We do not deal here with these issues, as doing so would get us too far from the main narrative. See Kennedy & McNally (2010) for discussion of the gradable/non-gradable alternations in color adjectives.



In the comparative, the composition proceeds similarly, with the difference that the relevant POINT on the scale is not left unspecified, but it is provided by the standard of comparison in the *than*-phrase. As a reflection of this fact, we shall maintain the head CMPR in the structure as a way of saying that the value of the POINT will be provided by the *than*-phrase, possibly (but not necessarily) located in the Spec of this projection. After the CMPR head specifies the precise position of the relevant point on the scale, we still need to add the information that the argument of the adjective has a degree of the property that exceeds the CMPR point. This means that we again place the head UP above the CMPR head. The structure of the comparative then looks like (18b).

Note that what has happened now is that the POS head of our original *symmetric* structures has turned into UP, while our original CMPR head corresponds to two heads, namely CMPR and UP. The consequence is that we no longer have a structure where POS and CMPR are two fully disjoint – and mutually incompatible – continuations of the shared POINTP. Rather, the relationship between the positive and the comparative is one that we shall label *gapped*. This is because the positive has a ‘gap’ in between POINT and UP, i.e., in a place where the comparative has the extra projection CMPR.

The gapped relation shown in (18) is one of the possible interpretations of our symmetric structure introduced in (10). (The other interpretation is that POS and CMPR of (10) are two values of a single head, recall footnote 5.) We consider the ‘gapped’ interpretation of the relation more interesting in that it allows us to consider all features privative and rigidly ordered, but beyond this conceptual motivation, we do not provide here any empirical facts that force the gapped interpretation over the symmetric one.¹²

As we shall see, the gapped structure will allow for the informal analysis given in section 2 to remain in essence the same. The reason for this is that we shall propose that the two projections above POINT in the comparative (i.e., CMPR and UP) are actually spelled out by the comparative marker *-(ej)š*. Once this proposal is in place, the root (and any potential augments) still need to spell out just POINTP in the comparative, while they must spell out the full UPP in the positive, which yields the truncation pattern in exactly the same way as highlighted in section 2.

In particular, the distribution of the augment *-k* will still be explained in terms of different root sizes of the three different classes of adjectives. XL-roots can pronounce the entire structure in (18a), as well as the POINTP subtree that is found in the comparative below CMPR and UP, the two projections spelled out by the comparative *-(ej)š* in

¹² This view on the relationship between the positive and the comparative degree is highly similar to what is proposed by Kennedy & Levin (2008) on semantic grounds, with a POS head (comparable to our UP) both in the positive and the comparative.

Slovak. With this XL class of adjectives, no augment is present in either the positive or the comparative.

At the other extreme, S-roots are of size DIRP (or NEGP, if they are negative). As a consequence, they need the augment *-k* both in the positive and the comparative. The augment helps them spell out the part of the sequence which they themselves are unable to spell out. In the positive, these are the heads POINT + UP, two heads which we take to be the lexical specification of the augment. In the comparative of S-roots, *-š* spells out CMPR + UP, the S-root spells out all the projections up to DIR/NEG, and *-k* is still needed to spell out the POINT projection.

Finally, M-roots are of size POINTP. This means that the augment is needed in the positive, but will disappear in the comparative, since the comparative morpheme comes directly on top of POINTP.

An informal schematic overview of the difference between the different types of roots is given in (19):

(19)

DIM	DIR	(NEG)	POINT	UP
XL-root				
M-root				k
S-root			k	

We provide a more detailed derivation of both the positive and comparative degrees of these three adjective classes in the next section. At this point, we want to point out two predictions made by our approach. The first is that the lexical specification of the augment as the spellout of POINT + UP correctly predicts that *-k* only appears with gradable adjectives, because it spells out projections that are irrelevant for non-gradable adjectives (these have only DIM, recall footnote 11). The second prediction is that all root sizes should include both positive and negative adjectives, as they can all optionally spell out NEG. Both of these predictions are confirmed (see the discussion of the semantic classes of the *k*-adjectives in section 4.2 above).

Finally, it is worth mentioning explicitly that there is no $\checkmark$ node at the bottom of the tree, which thus consists of functional heads ‘all the way down.’ This means that on our approach, roots like *nice*, *kind*, *good*, *prett-*, etc. are not acategorical, but spell out functional heads (an approach pioneered for verbs by Ramchand 2008). On this approach, the way roots differ from functional morphemes is not in the type of a head that they spell out. Rather, roots are understood as lexical items that are associated to a concept in their lexical entry, while functional morphemes are not. The meaning of the latter consists solely of the grammatically relevant meaning (functional heads) that they spell out. When a root is inserted during spellout as the realisation of a particular set of functional heads, the associated concept is activated in the conceptual representation. The model of spellout we shall be assuming is the topic of the next section.

6 The formal derivation

6.1 Prerequisites

The analysis we shall present is couched in the framework of nanosyntax, a late-insertion model that has a postsyntactic lexicon, and a syntax where each feature is a syntactic head (Caha 2009; Starke 2009). The model is represented schematically in Figure 1.

The syntax creates syntactically well-formed objects, which are then matched against the lexicon for pronunciation. The lexicon plays the role of a translator between the syntactic representation on the one hand and phonology and concepts on the other. For

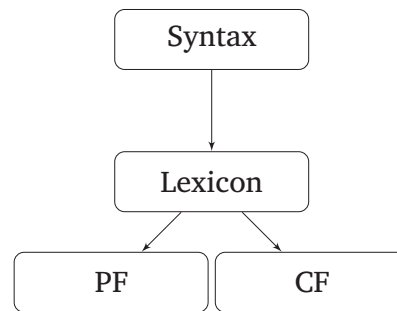
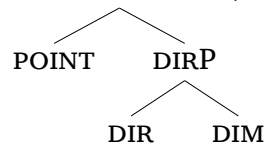


Figure 1: Nanosyntactic model of grammar.

content words, a lexical item links a syntactically well-formed tree with a well-formed phonological object at PF and a concept at CF (Starke 2014). Functional words only map syntax onto pronunciation at PF, but do not contribute any additional conceptual meaning beyond the grammatical meaning contributed by the set of functional heads the functional morpheme spells out. A model of spellout based on these assumptions leads to a restricted lexicon in the sense that only well-formed syntactic trees (rather than any haphazard set of features) can be linked to a phonology and/or concept inside a lexical entry.

A crucial part of this model is the idea that linking syntactically well-formed objects to phonology is not restricted to terminals. Since phrases containing multiple terminals are also well-formed syntactic objects, they may be easily pronounced by a single piece of phonology (a single morpheme in the traditional sense), provided these terminals form a constituent (i.e., they correspond to a syntactic object). An example is provided in (20) by the entry for an M-root. It will become clear in the course of the discussion that in our model, spellout in fact has to target phrasal nodes and cannot target terminals.

(20) POINTP $\Leftrightarrow$ /M-Root/

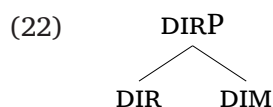


Whether a particular syntactic tree is matched by a lexical item is determined by the Superset Principle (Starke 2009):

(21) *The Superset Principle*

A lexically stored tree L matches a syntactic node S iff L contains the syntactic tree dominated by S as a subtree.

As a consequence of the Superset Principle, the M-root entry given in (20) is also applicable as the spellout of a DIRP in (22), since such a DIRP is contained inside the lexically stored tree.

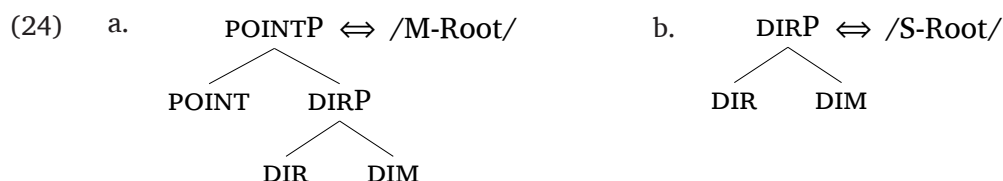


Recall now that roots (which in our system means ‘morphemes associated to conceptual information’) come in various sizes. Therefore, it will sometimes be the case that multiple roots of different sizes match a particular structure. We shall be assuming that in such cases, there is Free Choice, and we may insert the lexical item that we ‘want to talk about’:

- (23) *Free Choice of the root*
When several roots match, the choice among them is free.

This statement is compatible with the idea that normal competition occurs between roots, but their choice is restricted by the concept they are attached to.

To see how this works, consider the lexical items in (24), where (24a) represents an M-root and (24b) an S-root:



If syntax produces a DIRP like the one in (22), then both of these roots are candidates per the Superset Principle (21), because DIRP is contained in both entries. What we do not want to happen is that some version of the Elsewhere Condition forces the insertion of the S-root (e.g., ‘slimy’) on the grounds that it is a ‘perfect match,’ since that would in effect prevent M-roots (e.g., ‘heavy’) from being usable as the spellout of DIRP at all. The problem with this would be that a set of concepts (associated to M-roots) would not be available at all in the given context, an unwanted result. In order not to run into a problem like this, we may simply avoid postulating the Elsewhere Condition for roots. Alternatively, we may say that the Elsewhere Condition does apply to roots, but that the set of competing roots is restricted only to those that are associated to the relevant concept in a way that ‘slimy’ and ‘heavy’ never compete. In either case, (23) will emerge as a consequence of such a setup (i.e., it is not an axiom).

Moving on to additional assumptions, we adopt here the cyclic spellout algorithm described in Starke (2018) (cf. Baunaz & Lander 2018; Caha et al. 2019). The gist of the algorithm is that structure building (Merge F) is intertwined with spellout, such that spellout applies after each application of Merge F. The way spellout applies is by targeting the FP (i.e. the topmost node) that has just been created by Merge F. The algorithm is phrased in such a way that the spellout of FP must actually succeed for Merge F to continue further. The simplest way for spellout to succeed is by finding a matching item for the whole FP. However, if that fails, the structure is rejected at the interface and returned to syntax. In order to save the structure, syntax must perform certain rescue operations, movement in particular. The movements happen in a pre-defined order, first trying to remove the Spec of FP out of FP, and then checking again if this helps with finding a spellout for the FP. The last step is moving the complement. These successive steps in the spellout algorithm are formulated in (25).

- (25) Merge F and
- Spell out FP
 - If (a) fails, attempt movement of the spec of the complement of F, and retry (a)
 - If (b) fails, move the complement of F, and retry (a)

It is important to stress that ‘Spell out FP’ in (25a) does not equal ‘Pronounce FP.’ Spellout can be intuitively understood as ‘finding a match’ in the lexicon. When a matching item is found, this is enough for Merge F to proceed, with actual pronunciation postponed.

Keeping this in mind, consider how cyclic derivations proceed. Suppose that syntax constructs FP and spells it out, i.e. it finds a match in the lexicon. The derivation then

continues by another step of Merge F, e.g., by merging F2 with FP, producing the constituent $[_{F2P} F2 FP]$. At this point, spellout applies again. Suppose that a matching item for the full F2P is found. This will lead to the previous match (at FP) being forgotten, and only the highest match survives. This is known as Cyclic Override:

(26) *Cyclic Override*

Lexicalisation at a node XP overrides any previous match at a phrase contained in XP.

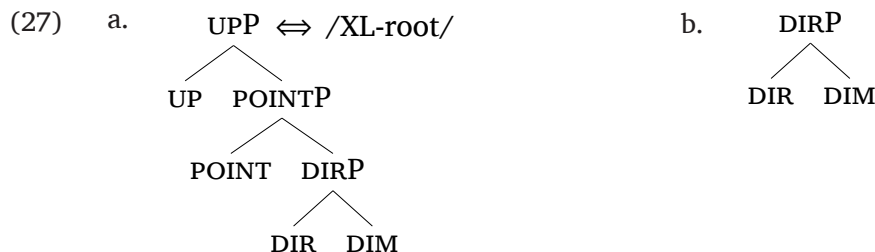
Note that (26) should not be considered a principle on its own, but rather a consequence of the architecture where spellout cyclically targets higher and higher nodes. Note, however, that roots associated to different concepts cannot override each other, recall (23) (see Caha et al. 2019 for discussion). For example, an XL-root like ‘old’ cannot override an M-root like ‘heavy.’

The final tool in our spellout toolbox is backtracking, which is a last resort operation that is activated when the derivation gets stuck following the algorithm described above. We will say more about backtracking at the relevant place in the discussion.

We shall illustrate the workings of the above mechanisms and principles as we proceed. While doing so, we demonstrate that the correct pairing of roots and augments, as well as their correct ordering, can be derived using the tools described above.

6.2 The positive degree

We start by considering the derivation of the positive degree of an XL-root (e.g., ‘old’). These lack augments both in the positive and in the comparative (recall (4)). An XL-root has a lexical entry like (27a). In the first step, the syntax merges DIRP as in (27b).

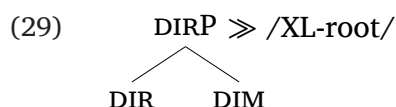


Recall now that the way the syntax interfaces with the lexicon is via the Spellout Algorithm, which we repeat in (28).

(28) Merge F and

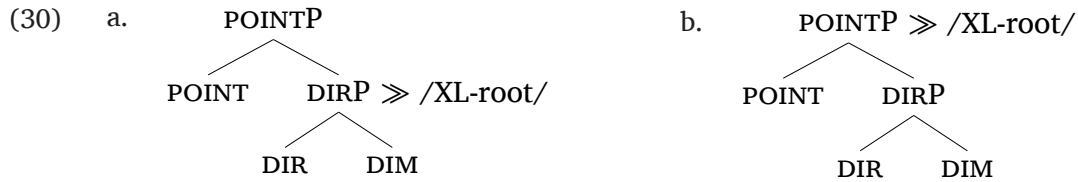
- a. Spell out FP
- b. If (a) fails, attempt movement of the Spec of the complement of F, and retry (a)
- c. If (b) fails, move the complement of F, and retry (a)

According to (28), the first thing we should do is try the spellout of DIRP without any movement (28a). Since DIRP is contained in the lexical tree of the XL-root, the XL-root can spell out DIRP. We indicate successful spellout with the $\gg$ sign:

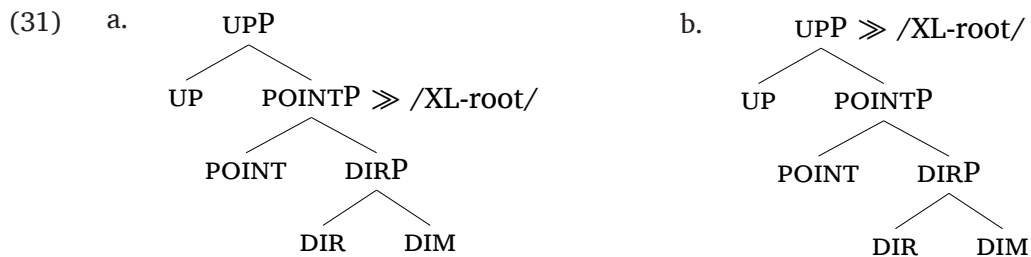


In the next step of the derivation, POINT will be merged on top of DIRP, producing POINTP as in (30a). The spellout algorithm is cyclic, and the lexicon will therefore be consulted

again in an attempt to spell out the newly formed POINTP. A match will be found in the same XL-root, so that spellout at POINTP is successful, and it will override the earlier spellout at DIRP, as shown in (30b).

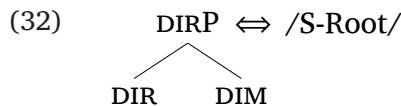


The same procedure will be repeated one more time, producing the whole UPP (corresponding to the positive degree), which is still spelled out as the XL-root.

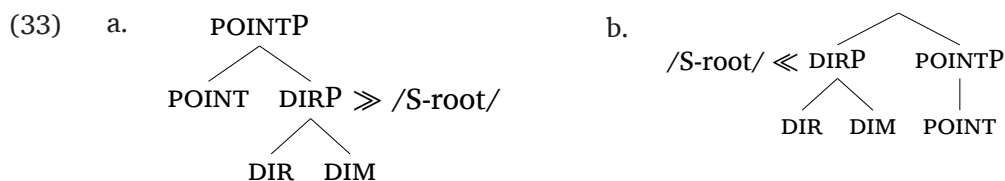


Derivations with augments in the positive work similarly in the initial stages, but they differ in that at some intermediate point in the derivation, the application of Merge F will result in a tree that is no longer contained in the lexical tree of the relevant root, i.e., the syntactic tree will become too big to be realised by the root. As a result, spellout will fail, and rescue strategies will be applied.

Let us see in more detail how this works with an S-root like *sliz*- ‘slime’ in *sliz-k-ý* ‘slimy’. An S-root has a lexical entry of the size DIRP, as shown in (24b) above, repeated below for convenience.



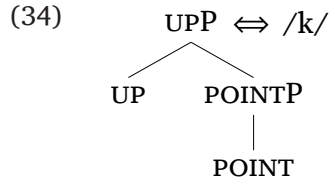
On the first step of the derivation, when DIR and DIM are merged, an S-root can spell out this constituent. However, if POINT is merged to DIRP, creating a POINTP shown in (33a), direct spellout fails. This is because the lexical tree of an S-root like *sliz*- no longer contains the syntactic tree.



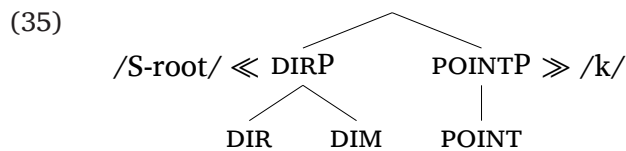
Note in addition that the POINT head cannot be spelled out as a terminal (without movement), because spellout (as per the spellout algorithm) only targets phrases. Since spellout fails, the structure is rejected at the interface and returned to syntax, which has to rescue the structure in accordance with the spellout algorithm. The first rescue strategy that should be tried is the movement of the Spec of the complement of POINT (see (28b)), but since the complement has no Spec, this

option is undefined. That leads to complement movement, as mandated by (28c), yielding (33b).

At this point, the lexicon will again be consulted, and it will find a match for POINTP in the augment *k*, whose lexical entry is given in (34). Recall that the specification of *-k* has already been introduced informally in (14a).



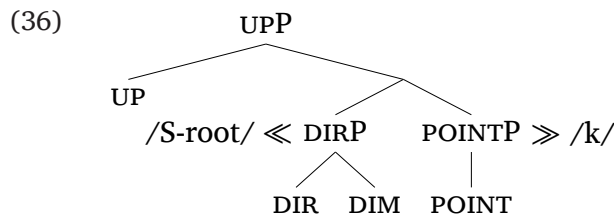
Observe that the syntactic tree POINTP of (33b) is contained in (34) as a subtree, so that spellout can occur:



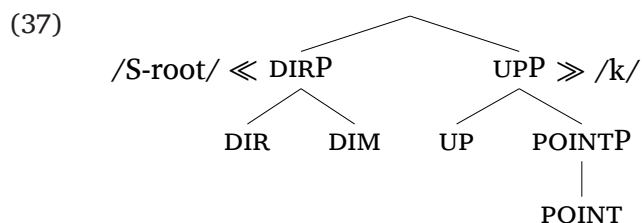
As a side effect, the underlying structure has been rearranged in a way that the correct linear order of the root and the suffix is derived.

Note also that spellout movement has simply removed the DIRP out of the POINTP, leaving no trace behind. In this property, spellout movement differs from standard feature-driven movement, which does leave a trace. The reason why spellout movement leaves no trace is the fact that unlike feature-driven movement, it does not lead to two interpretive positions, see Starke (2018) and Caha (2019: Chapter 4) for discussion. This also allows us to keep the spellout procedure matching syntactic trees with lexical ones maximally simple, since the trace does not enter into the computation of matching. We adopt this ‘traceless’ convention here, noting that it is not crucial for the system to work; the alternative would be for spellout to ignore such traces.

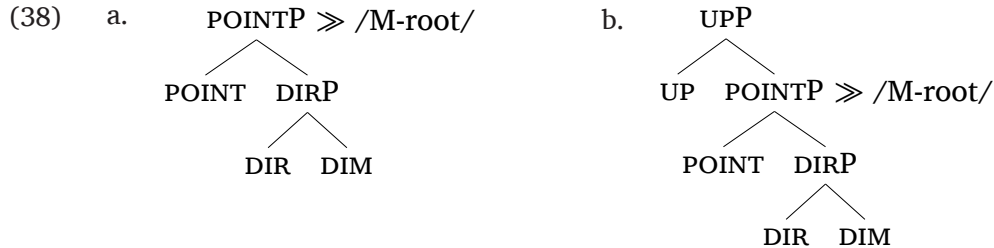
The derivation now continues by merging UP to (35), producing (36):



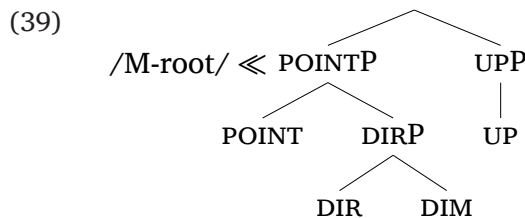
No lexical entry matches the UPP in (36), and movement of the Spec of the complement of UP applies. This moves the DIRP out of UPP, yielding (37). This derives, as the right hand member of the tree, an UPP that is identical to the lexical entry of *-k* in (34). The marker *-k* can therefore spell out the UPP in (37), overriding the earlier spellout of POINTP by the same *-k*.



Let us now turn to the derivation of M-roots, which is the most technically complex. To see why, consider first their lexical entry given in (24a). This entry will initially allow for a derivation where we merge features and spell out without movement, ultimately reaching the stage where POINTP is spelled out by the root, which exhausts its lexicalisation potential; see (38a). Once POINTP is spelled out by the root, the next feature (UP) is merged, yielding (38b).



Since the M-root cannot spell out (38b), rescue movements take place. The complement of UP has no Spec, so the movement of the complement of UP will apply, yielding (39):

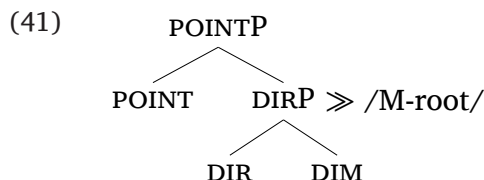


But now the right hand part of the tree (39), i.e., the UPP, cannot be realised by the augment, since the UPP of (39) is not a subtree of the augment's entry (34). In fact, there is no lexical entry in the Slovak lexicon that can apply to this structure, which leads to spellout getting stuck: no rescue operation produces a lexicalisable output, and Merge F cannot continue.

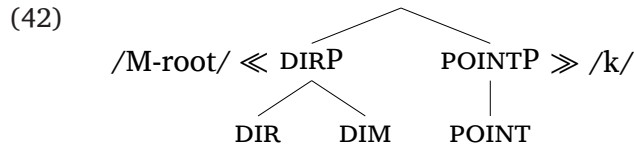
When a derivation gets stuck in this way, backtracking is set in motion. What backtracking does is that it undoes one Merge F operation and goes back to the previous stage in the derivation. Once there, it tries a different derivational option than the one which had previously applied – and ultimately led nowhere. This is formulated in (40):

- (40) *Backtracking*
When spellout fails, go back to the previous cycle, and try the next option for that cycle.

Reverting to the previous cycle brings us to POINTP, as in (41).



At the first pass through this cycle (38a), we have directly spelled out POINTP by the M-root (the first option of the spellout algorithm (28)). However, this has led to a dead end, so (40) instructs us to try the next option. This would be movement of the Spec of the complement of POINT. However, since this is undefined, we end up with the third option of (28), namely complement movement. This leads to the structure in (42). Here we POINTP can be spelled out by -k, since it is contained in the lexical item (34).

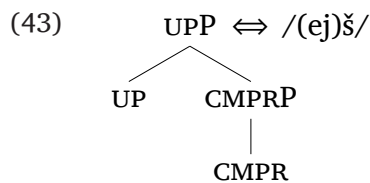


At this point, the derivation of M-roots runs exactly in parallel to the derivation of S-roots, see (35). Just like with S-roots, we merge UP to (42). The structure then looks as in (36), and fails to spell out without movement. We therefore move the DIRP out of UPP, by Spec movement, yielding a structure identical to (37) above, except that DIRP is spelled out by the M-root. The positive degree of an S-root and an M-root therefore end up identical (both have an augment), even though their root sizes are different.

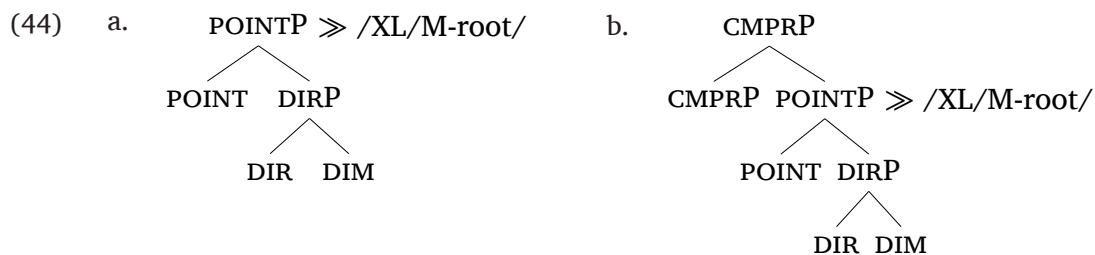
To sum up, this section showed how the derivation of the positive degree proceeds such that only XL-roots lack AUG. M-roots and S-roots need one.

6.3 The comparative

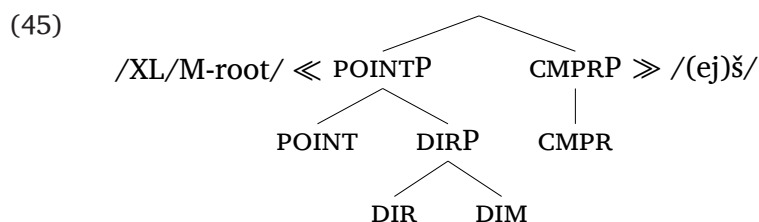
Let us start the discussion of comparatives by making explicit the lexical entry of the comparative morpheme we propose here, see (43).



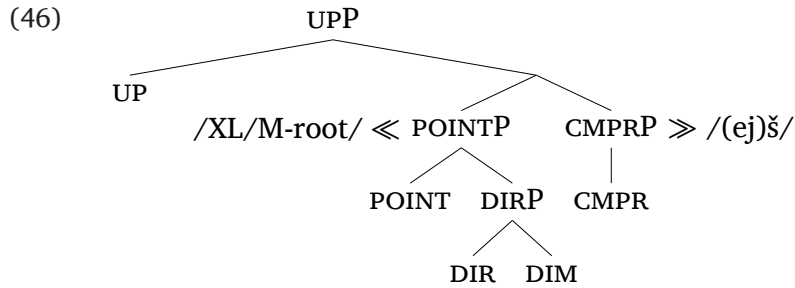
Let us now turn to the comparatives of XL-roots and M-roots, which have identical derivations. They both recursively Merge F and spell out without movement up to POINTP, see (44a), after which CMPRP is merged, see (44b).



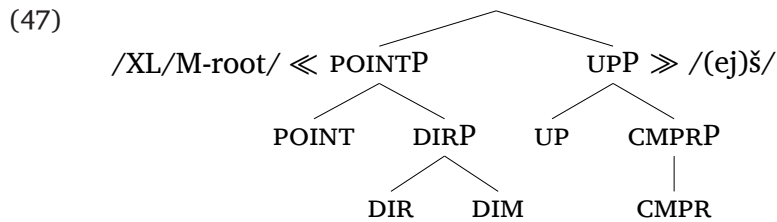
Since the lexicon contains no roots of this size, rescue movement must be applied, and this will move the complement of CMPR to the left, giving rise to (45). The right hand branch of this tree matches with the lexical entry for the comparative suffix, so that spellout is successful, and the correct linear order is derived.



Subsequently, the feature UP is merged to (45), producing (46).

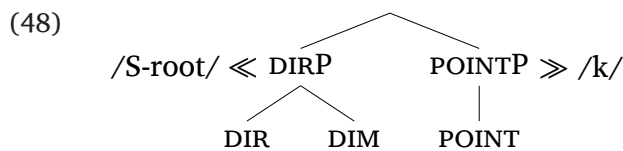


This structure fails to spell out, leading to rescue movements. Spec movement is tried first, producing (47), which spells out successfully.

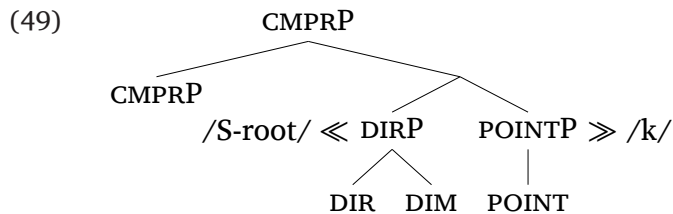


This is how the problematic truncation pattern is derived: M-roots are not large enough to spell out the positive degree (since they lack UP in their lexical entry), and so they need an augment in the positive. But since the comparative is built by adding CMPR on top of POINTP, and since M-roots can spell out POINTP, no augment is needed in the comparative.

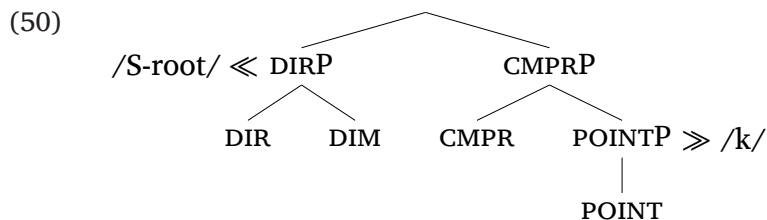
With S-roots, the derivation proceeds exactly as in the positive degree, up to the level of POINTP. This means that rescue movement will have to be applied at the merger of POINT, causing DIRP to be raised across POINT, leading to the structure in (48) (repeated from (35)).



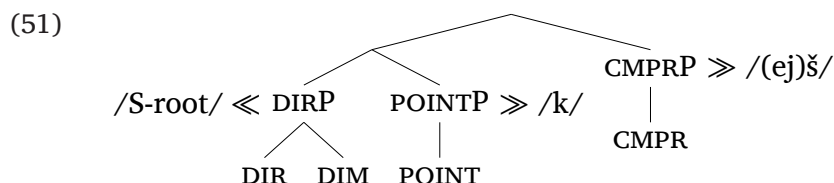
Next, CMPRP will be merged, yielding (49).



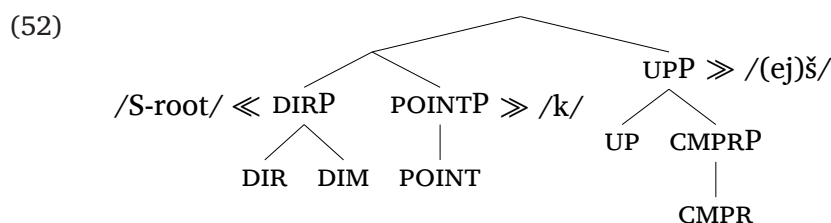
(49) does not spell out, so the Spec of its complement (i.e. DIRP) is raised across it, creating (50).



But spellout for CMPRP will fail because no lexical item contains this syntactic tree in the lexically stored tree. We therefore need to undo this movement, revert to (49), and apply the next rescue option of the spellout algorithm, which is movement of the complement of CMPR. This will result in the structure in (51), where CMPR can now spell out successfully as the comparative suffix after the evacuation of its complement.



The final step of the derivation is to add the UP element to the structure (51). This structure fails to be spelled out, and Spec movement is therefore tried. The output of Spec movement is in (52), where UPP correctly spells out as *-(ej)š*. This leads to the correct result, where the comparative form of S-roots contains the augment. We moreover derive the correct linear order of the suffixes, with the augment appearing between the root and the comparative marker.

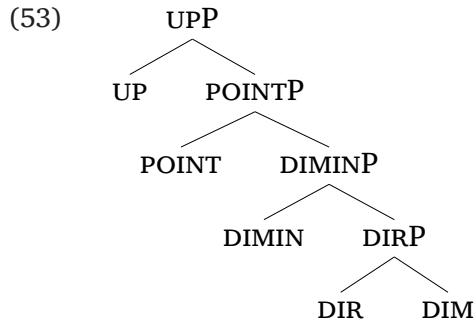


This concludes the discussion of the formal derivation of the positive and comparative degrees with the three different root types. We have shown how the derivations unfold on the basis of the different root sizes and the spellout algorithm, deriving the pattern of data discussed in the introduction: with XL-roots there is never any augment, with M-roots there is an augment in the positive but not in the comparative, whereas with S-roots, there is an augment both in the positive and the comparative.

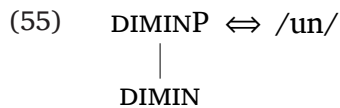
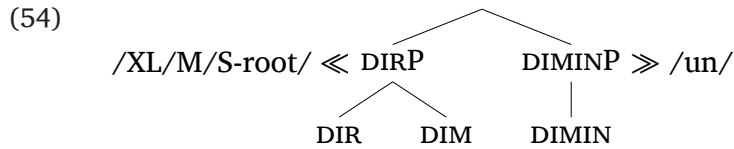
An essential aspect of our analysis are the non-containment/gapped structures we are proposing for the relation between the positive and the comparative. We believe that these are an improvement over any type of approach that would rely on containment structures. It is hard to see how a containment view on the positive and the comparative would be capable of delivering both the truncation pattern and the containment pattern, a result that our approach does achieve.

6.4 The diminutive

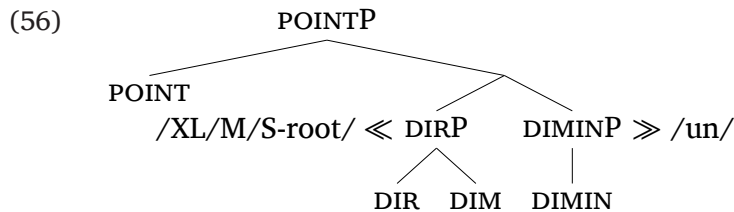
Finally, let us briefly address the issue of the derivation of the diminutive forms. Our goal is to show how they are derived for M-/S-roots, and what prediction this makes for the diminutives of XL-roots. Recall first from Tables 1 and 2 that all adjectives with augments pattern alike in that the diminutive marker *-un* always attaches to the root, and it is always followed by the augment *-k*. We can capture this pattern by proposing that the diminutive marker spells out an (optional) features DIMIN, which is just below POINT, i.e., on top of DIRP in the case of positive adjectives:



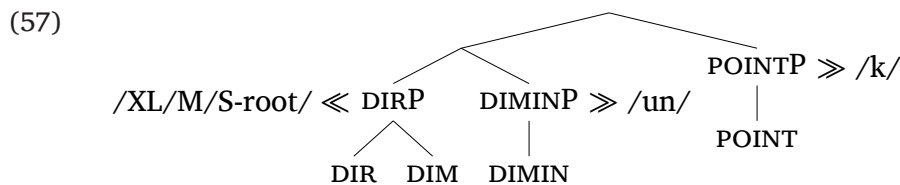
In this structure, we expect that all root sizes (including XL-roots) will maximally spell out DIRP. They cannot spell out the DIMINP, because none of the roots contains the DIMIN feature. Therefore, with all kinds of roots, the DIRP will have to move to the Spec of the DIMIN projection, as in (54). We are assuming that the lexical entry for the diminutive is as in (55).



When POINT is merged on top of (54), we get (56):



There is again no way to spell out without movement. Spec movement (of DIRP) also fails, because the resulting constituent containing POINT and DIMINP cannot be spelled out (there is no lexical entry that matches). Therefore, complement movement produces the structure in (57), where POINTP is spelled out by the augment, and the correct linear order is derived.



This account of diminutives predicts that all adjectives with a DIMIN morpheme must have the augment, including XL-roots. This prediction is borne out, and the diminutives of XL roots must indeed have the augment -k. This is shown in Table 6 for two XL-roots. A diminutive without the augment is ungrammatical.

Table 6: Diminutives of Slovak simplex adjectives.

POSITIVE		DIMINUTIVE				COMPARATIVE			gloss
root	AGR	root	DIMIN	AUG	AGR	root	CMPR	AGR	
mlad	ý	mlad	un	k	ý	mlad	š	í	‘young’
čist	ý	čist	un	k	ý	čist	ejš	í	‘clean’

7 Suppletion

We believe that our approach based on symmetric/gapped structures can also accommodate earlier results achieved by Bobaljik (2012) pertaining to the absence of ABA patterns in root suppletion in the triplet positive—comparative—superlative. This may come as a surprise, since Bobaljik’s observations have traditionally been interpreted as evidence for containment between the positive and the comparative.¹³ Upon closer examination, however, it turns out that containment between the positive and the comparative is not needed. We discuss the issues relating to suppletion and symmetric structures in the current section.

7.1 *ABA

Let us first introduce the basic facts. As Bobaljik observes, root suppletion in the triplet positive—comparative—superlative is restricted. In particular, if the comparative is suppletive with respect to the positive degree (as in *good—bett-er*), then the superlative is suppletive as well (i.e., there is no triplet where *good—bett-er* would be followed by **good-est*). Bobaljik states this as in (58).

- (58) *The Comparative-Superlative Generalization, part I (CSG1):*
If the comparative is suppletive, then the superlative is also suppletive.

This generalisation also holds in Slovak, as the following table illustrates. First, in order to get some background on superlatives in Slovak, the topmost line shows a regular adjective where the superlative adds the prefix *naj-* to the comparative. This is a typical pattern, found with all comparative—superlative pairs in the language.

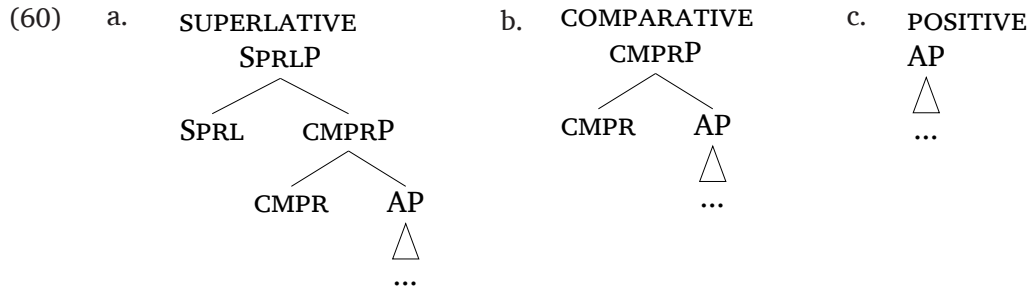
- (59) Superlatives and the *ABA in Slovak (Dvonč et al. 1966: 212)

POS	CMPR	SPRL	gloss
star-ý	star-š-í	naj-star-š-í	‘old’
dobr-ý	lep-š-í	naj-lep-š-í	‘good’
zl-ý	hor-š-í	naj-hor-š-í	‘bad’
mal-ý	men-š-í	naj-men-š-í	‘small’
A-ý	B-š-í	naj-A-š-í	not attested

Against this background, the middle part of the table illustrates the pattern found with suppletive adjectives. We see here first of all that the comparative has a suppletive root compared to the positive. The very same suppletive root is found in the superlative, which, as in the regular case, is still derived from the comparative by the prefix *naj-*. Such a pattern of root suppletion is called the ABB pattern by Bobaljik. The final line in the table makes it clear that there is no adjective in the language that would show the ABA pattern, with one root (A) found in the positive and in the superlative, and a different root (B) in the comparative.

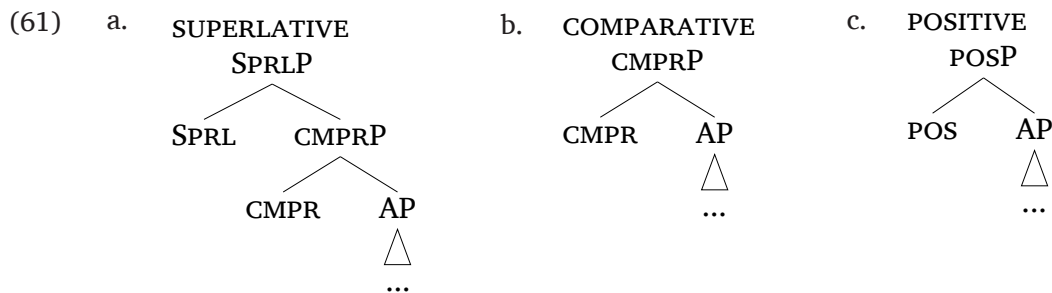
¹³ As far as we could determine, Bobaljik (2012) remains neutral on this issue, though he does draw trees where the comparative contains the positive, see Bobaljik (2012: 32).

These facts are usually interpreted in terms of structures shown in (60) (as suggested in Bobaljik 2012: 32).



Given these structures, Bobaljik excludes the ABA as follows. Suppose that we have an adjective with one root in the positive (A) and a different one in the comparative (B). For such an adjective, the comparative root B will be made sensitive to the presence of CMPR. The positive root A (with no particular specification) does not get to surface in the comparative, because it is less specific than the dedicated comparative root B. In this setup, it is impossible to insert the non-specific root A in the superlative, since the structure of the superlative contains the very same element that triggers the presence of the suppletive root B in the comparative (namely CMPR). This makes it impossible to fall back on the default (or elsewhere) root A. The root B will be a better match in the superlative than the non-specific root A for exactly the same reasons for which it is a better match in the comparative.¹⁴

However, note that the very same reasoning can be replicated for the non-containment structures we have been proposing here, as long as we stick to the hypothesis that the superlative contains the comparative. A simplified version of such a proposal is shown in (61). We are switching back to our original symmetric structures here, noting that they can be interpreted in multiple ways, one of which is that POS and CMPR are two different values of a single head, and another possible interpretation is the ‘gapped’ type of structure we have been investigating in the previous section.



To see how Bobaljik’s original reasoning applies to these structures, consider it once again. Suppose first that we have a pair of suppletive roots, one in the positive (A) and a different one in the comparative (B). One analysis says that the positive root (A) is the elsewhere case, and that the comparative root (B) is sensitive to CMPR. This derives the ABB pattern, and rules out the ABA pattern, for the same reason as before: the structure of the superlative contains the element CMPR, which triggers the appearance of B. This is so even if CMPR decomposes into CMPR + UP, as we have suggested in the previous section. As long as the superlative contains CMPR and the positive lacks it, the *ABA follows.

¹⁴ For an approach to the *ABA generalisation in terms of the phrasal spellout model adopted here, see De Clercq & Vanden Wyngaerd (2017).

It is interesting to note, in addition, that the non-containment structures open some new analytical possibilities. Specifically, if the trees are as in (61), we can consider the root in the comparative (B) to be an elsewhere form, while the root in the positive (A) is triggered by the POS head. Under such a scenario, the POS-specific root A will only appear in the positive, giving (again) rise to an ABB pattern, and not ABA. Under the structures in (61), this is because A is triggered by POS and neither the comparative or the superlative contain POS. Under the ‘gapped’ interpretation considered in (18) above, the positive and the comparative share a common feature (UP) at the top. In order to get an ABB in such a system, A (as the special form) must be triggered due to adjacency to UP. Such adjacency is available in the positive, but fails to obtain in the comparative, as well as the superlative. This makes A unavailable both in the comparative and the superlative. In other words, under both interpretations of the symmetric structures, *ABA follows. In sum, this means that both the scenario in (60) (suggested in Bobaljik’s work) and the one in (61) (which we are exploring here) are equal in their ability to derive the *ABA pattern.

We also wish to note that our proposal does not in fact contradict the core of what Bobaljik proposes. Specifically, he clearly states that what is at stake for *ABA to follow is the containment relation between the comparative and the superlative:

- (62) *The Containment Hypothesis* (Bobaljik 2012: 4):
The representation of the superlative properly contains that of the comparative.

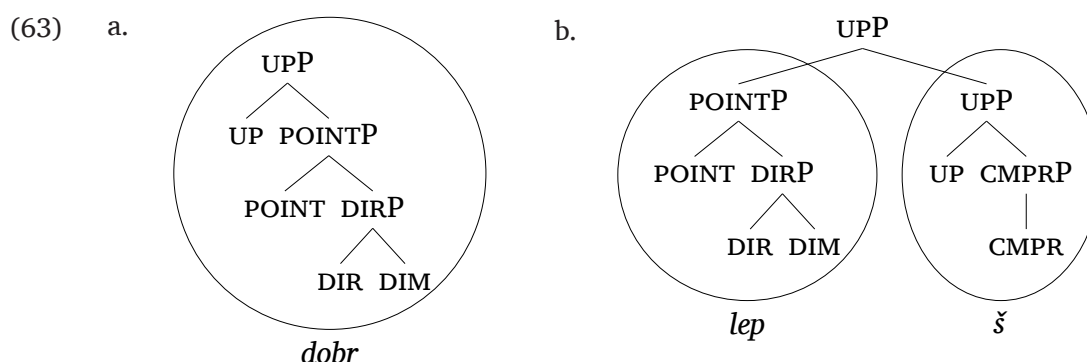
This statement is obviously as much in line with our (symmetric/gapped) proposal of how to be positive (depicted in (61)) as with the Russian-doll containment structures in (60).

In conclusion, we note that there seem to be a number of ways how *ABA patterns may arise in the grammar, as a growing number of works acknowledge (see, e.g., Caha 2017; Andersson 2018; Bobaljik & Sauerland 2018, for how the so-called ‘overlapping’ structures yield *ABA; for an approach to *ABA based on symmetric relation between the first two members like the one suggested here, see Christopoulos & Zompì 2019).

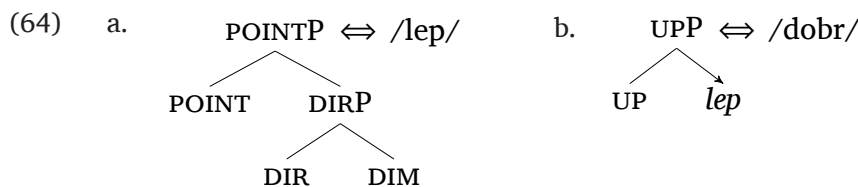
7.2 Suppletion as evidence for backtracking

In this section, we would like to elaborate in more detail on how suppletion is implemented in Nanosyntax, and we would like to use this as an additional evidence for the postulation of backtracking.

To have a specific example to work with, consider, first the suppletive relation between *dobr-ý* ‘good’ and *lep-š-í* ‘better.’ The starting point of our discussion is the fact that the positive degree *dobr-ý* ‘good’ has no augment, and therefore, the root *dobr* must spell out a structure of the size UPP. This is shown in (63a). The root *lep* has no augment either, and it therefore spells out the whole constituent that augment-less roots in the comparative spell out, i.e. POINTP. This is shown in (63b).

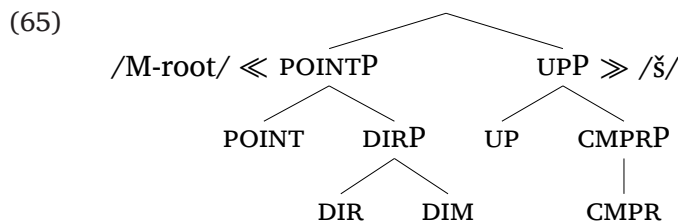


The interesting fact that can be observed in these structures is that we do not need context specification to express the distribution of these roots, because they are already differentiated by the size of the structure they spell out. Specifically, we may easily express their difference by storing the root *lep* ‘bett’ as a lexical item of the size POINTP (an M-root), as in (64a). The root *dobr* ‘good’ is of a different size, namely UPP (an XL-root). However, *dobr* differs from standard XL-roots in that it is the positive degree version of *lep*. An ordinary XL-root like ‘old’ does not stand in a suppletive relationship to another root. The way this special relationship of *dobr* to *lep* is expressed in nanosyntax is by using the so-called pointer (Starke 2014; Caha et al. 2019). In our particular case, the entry of the root *dobr* ‘good’ says that it is the spellout of a structure that includes the head UP as one of its daughters, and where the other daughter corresponds to a structure that has been spelled out by the lexical item *lep* ‘bett’ at the previous cycle. This is what the lexical entry (64b) says.

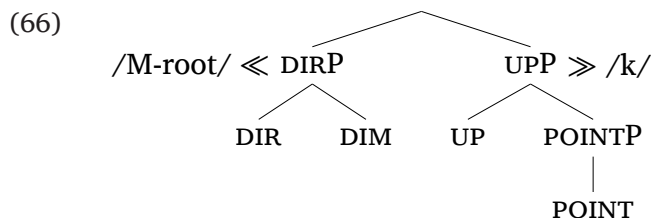


A crucial consequence of this approach is that suppletive lexical items always spell out constituents of different size. It is impossible to have one and the same constituent expressed by two roots with one suppletive for the other.

With the basic understanding of suppletion in place, we want to present a piece of empirical support for the backtracking analysis of the M-roots. Recall that these lack the augment in the comparative, but have it in the positive degree. Under the derivation that we developed earlier, M-roots (which are specified for POINTP) spell out the whole POINTP in the comparative. The relevant structure is given in (65) for convenience, repeated from (47).



This contrasts with the positive. Here (due to Backtracking) M-roots spell out only DIRP. This is shown in the structure (66), repeated from (37).



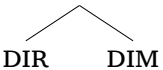
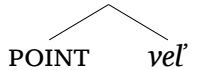
In the context of the theory of suppletion sketched above, this leads to a prediction. Specifically, the difference in the size of structure spelled out by the root may be reflected in a

different shape of the root. Specifically, we allow a situation where an M-sized root spells out POINTP in the comparative, and stands in a suppletive relation to a positive degree root (of the size DIRP), which must combine with an augment.

Slovak provides some evidence that bears out the prediction of the Backtracking derivation. The relevant adjective is the adjective *veľ-k-ý* ‘big’, which has an augment in the positive, loses it in the comparative and simultaneously changes to a suppletive root, as shown in (67) (data from Dvonč et al. 1966: 212).

(67)	POS	CMPR	SPRL	gloss
	<i>veľ-k-ý</i>	<i>väč-š-í</i>	<i>naj-väč-š-í</i>	‘big’

These adjectives thus support our claim that with truncating adjectives, the root spells out constituents of different root sizes in the positive and the comparative, which is a consequence of the backtracking derivation. The lexical items for *veľ*- ‘big, POS’ and *väč*- ‘big, CMPR’ are given below for completeness.

(68)	a.	DIRP $\Leftrightarrow$ /veľ/	b.	POINTP $\Leftrightarrow$ /väch/
				

A further prediction that we make is that such suppletion will not arise with S-roots, which have an augment both in the positive and the comparative. These adjectives spell out DIRP in both cases, which makes it impossible to characterise such roots as differing in size. This prediction is also borne out, and we find no suppletion in the containment class of augmented adjectives.

8 The semantics of POS

In the semantics literature, a (silent) POS head is often assumed in the positive but not the comparative degree (e.g., Kennedy 2007). Our symmetric/gapped structures discussed in the previous sections make a rather similar assumption, albeit argued for on morphological grounds. The main commonality between the proposals is that the CMPR head does not apply to the full positive degree, but to something that is less than the positive. The question we wish to address in this section is whether *-k* can be taken to be a realisation of this semantically motivated POS head. We shall argue that the relationship between the augment and the POS head is not one-to-one, and that consequently the augment *-k* cannot be taken as an instantiation of POS.¹⁵

Let us say what the POS head does. The semantic POS head is responsible for the context-sensitivity of gradable adjectives, i.e. the fact that their interpretation depends on a contextual standard or comparison class (cf. Wheeler 1972; Klein 1980, and much subsequent literature). For example, Grano & Davis (2018: 133) provide the following semantics for POS (where *g* is a measure function, and *d_c* the degree of the contextual standard):

$$(69) \quad \llbracket \text{POS} \rrbracket = \lambda g_{\langle d_c, \langle e, t \rangle \rangle} \cdot \lambda x. \exists d [g(d)(x) \wedge d > d_c]$$

¹⁵ Chinese *hen* has been argued to be an overt positive degree morpheme (e.g., Sybesma 1999; Liu 2010), but see Huang (2006); Grano (2012); Zhang (2015) for some alternative views. Grano & Davis (2018) discuss a potential candidate for the realisation of POS in Arabic, but dismiss it as inconclusive.

Applied to an adjective like *tall*, this yields (70):

- (70) $[[\text{POS}]] ([[\text{tall}]]) ([[\text{John}]]) = \exists d[\text{height}(j) \geq d \wedge d > d_c]$
 ‘There is some degree d such that John’s height meets or exceeds d and d exceeds a contextually determined threshold d_c .’

The fact that the comparative lacks this reference to a contextual standard in its semantics is the main argument for assuming that this null degree head POS is restricted to the positive degree, and does not occur in the comparative. As stated earlier, such a semantic analysis agrees to some extent with our symmetric/gapped structures given above, where the CMPR head does not attach to the full positive.

However, it is clear from the above that the semantic contribution of *-k* cannot simply be equated with that of the POS head of the semantics literature. The first reason is that the semantics of the POS head encodes meaning that is distributed across two heads in our proposal: first, POS in (69) introduces a contextual standard, and second, it also says that the argument of the adjective has a degree that exceeds this standard. In our gapped proposal, these ingredients are split across two heads, where the lower head POINT introduces the contextual standard, and the higher head UP is the ingredient responsible for the ‘exceed’ component of meaning.

Yet another way in which *-k* is different from the traditional POS is that in the containment class of adjectives, *-k* also shows up in the comparative, where it clearly lacks the POS semantics of (69). This property of *-k* can easily be accounted for under the phrasal spellout theory that we assume. In particular, we have claimed that the lexical entry of *-k* contains both POINT and UP. What that means is that *-k* is a marker that spells out either POINT alone (as in the comparative), or POINT and UP (in the positive). There is, in the phrasal spellout theory, no one-to-one relation between syntactic heads and exponents. The UP/POS meaning is carried by the syntactic head, not by the marker, which may be pronouncing different heads in different environments.

Further support for the claim that *-k* does not correspond in a one-to-one fashion with the semantic POS comes from measure phrases. These may specify an extent, as in *Radek is 1.5m tall*, or ask for one (e.g., *How tall is Radek?*). Adjectives with such measure phrases do not carry positive degree semantics as in (69), yet in adjectives that have an augment, measure phrases do co-occur with *-k*. This is shown in (71):

- (71) a. Tank T-72 je ťaž-k-ý 42 ton.
 tank T-72 is heav-y 42 tons.
 ‘The T-72 tank is 42 tons heavy.’ [Slovak National Corpus]
 b. Aký ťaž-k-ý je tank T-72?
 how heav-y is tank T-72?
 ‘How heavy is the T-72 tank?’

This suggests that *-k* has further internal structure, and is able to realise extents of various types: a contextual extent (as in (69)), or an extent which is overtly specified, as in sentences without a measure phrase, or questioned, as in (71b). This raises intricate questions of the semantics and the distribution of measure phrases, which we cannot do full justice to in the present context, and which we therefore refer to future research. The take home message from this section is that in the phrasal spellout theory, syntactic heads (or features) provide the semantic atoms, whereas the exponents typically map onto these heads in a one-to-many fashion, i.e. a single exponent typically realises multiple semantic

atoms. The quest for a morphological instantiation of the semantic POS head may therefore well turn out to be illusory in the end.

9 Conclusion

We have shown that the distribution of the augment *-k* in Slovak adjectives shows a peculiar pattern, which challenges the candidate universal in Grano & Davis (2018) to the effect that the comparative is either identical to or contains the positive degree. We have argued that this pattern can be fruitfully analysed under the root size approach, where allomorph selection, and more specifically the distribution of zeroes (i.e. zero augments in our case), is a function of root size. Under this approach, there are no zero markers under terminal nodes, but instead roots may realise constituents of variable sizes, thus creating the impression of zero realisation of certain heads or features.

The distribution of the augment *-k* in Slovak also required that we postulate symmetric structures, where a common adjectival base is elaborated on in different directions in the positive and the comparative. We also showed how the root size model provides an elegant account of root suppletion, which maintains the *ABA generalisation of Bobaljik (2012), and which receives some confirmation from suppletive patterns found with certain *k*-adjectives. Finally, we showed how the phrasal spellout model explains the fact that the correlation between the augment *-k* and the semantic atoms of the positive degree is not one-to-one but one-to-many.

Abbreviations

ADJ = adjective, AGR = agreement, AUG = augment, CMPR = comparative, DIM = dimension, DIMIN = diminutive, DIR = direction, POS = positive, SPRL = superlative

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Competing Interests

The authors have no competing interests to declare.

Author Contributions

The four authors have made an equal contribution to this paper. They are listed in the reverse alphabetical order.

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Part III

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